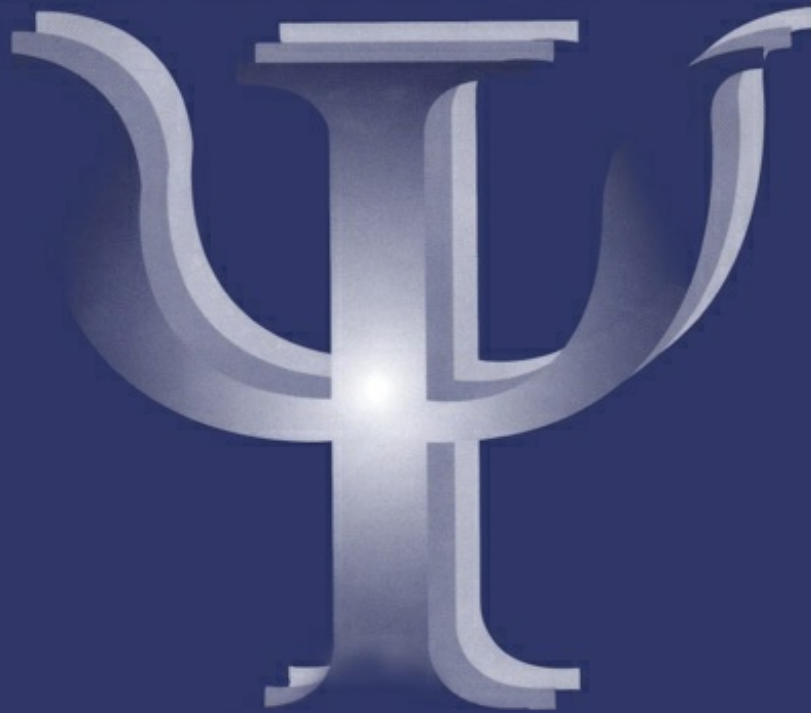


V.M. Bekhterev

COLLECTIVE REFLEXOLOGY



The Complete Edition

Lloyd H. Strickland, editor

Eugenia Lockwood and Alisa Lockwood, translators

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Acknowledgments—1999: The Complete Edition

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Lloyd H. Strickland
Carleton University
December 1999

Acknowledgments—1992: Part 1

Thanks are due to many people for the appearance at long last of this book. Eugenia Lockwood obviously deserves primary credit for the translation (I must claim subsequent interpretive and editorial errors as my own), but it may not be otherwise apparent how much else I owe her for other help and support over the years of my work in the area of Russian and Soviet social psychology. The Social Science and Humanities Research Council of Canada (Grant # 410-87-1056) provided funds for Ms. Lockwood's salary during the two-year translation process, and Carleton University provided invaluable aid in a number of ways: first through "start-up" funds in the form of Faculty Research Grants, application for which was supported by the then Deans of Graduate Studies (Syd Wise and John ApSimon) and Social Science (Dennis Forcese); second, through the prompt replacement of our stolen Macintosh computer by the latter's successor, Dean Marilyn Marshall; and third, through the sustaining support of a number of colleagues who accepted my insistence, at a time when it was not politically stylish to do so, that a relatively unknown, long-dead Russian physiologist/physician was really as important a figure in the history of my field as I claimed he was. Chief among them are Will Webster and Bill Jones, successive Chairs of Psychology, Larry Black, former Director of Carleton's Institute of Soviet and East European Studies, and fellow social psychologists Warren Thorngate, Fran Cherry, Betty Bayer, and Miho Hotta. Finally, to scholars like Michael Cole, James Wertsch, Jaan Valsiner, and Vladimir Trusov, who did not need to take my word for Bekhterev's contribution and who offered the kind of support that only consistent interest can contribute, I am particularly grateful. Putting Bekhterev's thoughts in a form to which my many colleagues could relate has

been a long pull, and I hope that for them the product has been worth the wait.

Clearly, if the reader construes the appearance of this book as the product of a collective effort, he/she will be entirely correct.

Lloyd H. Strickland
Carleton University
December 1999

Contents

Formative Years of Bekhterev's Ideas

Bekhterev and Mind/Body Dualism

The "Hidden Energy," its Accumulation, and Constant Transformation

From Evolutionary Ideas to a Theory of Activity

Bekhterev's Interest in Hypnosis, and the Foundations of Social Life

Bekhterev's Methodological Imperatives: The "Objective" Study of
Developing Phenomena

Part 1

Editor 's Foreword—1992

In Place of a Foreword

The necessity of applying a strictly objective method of studying the composite personality. Subjectivism in social sciences was borrowed from psychology and, as a remnant of anthropomorphism, it is now being expelled from all biological sciences; social sciences and humanities should follow suit. Collective reflexology represents an attempt to base one of the most important areas of social studies on a strictly objective foundation. Sociology, which has so far been based on biology and psychology, should from now on draw sustenance from biology and reflexology. The relationship between reflexology, biology, and psychology. The human personality is a phenomenon of biosocial origins. The significance of associative reflex theory as proof that the human personality is not an exclusively social product. Understanding social phenomena as collective reflexes. The misrepresentation by some subjectivist sociologists of the evolution of social phenomena as being dominated by a single principle such as imitation

(Tarde) or instinct (W. McDougall). The application of a strictly objective, or reflexological, method rules out such one-sidedness. The laws of the manifestation of collective activity are basically the same as those of individual activity.

1 Introduction

The unit of public events in the phenomena of social life is not the individual person but the whole society or a collective. Hence, individual psychology is not suitable for understanding social events. The attempts of Steinthal, Lazarus, W. Wundt, and others to create "folk psychology:" Studies of the special characteristics of the crowd by Tarde, LeBon, Sighele, Mikhailovskii, and others. The latest publications on this subject in Russia and abroad. Their common fault is in the subjective method of approaching the problem.

Collective reflexology is a purely objective science in the strictest sense of the word. The role of higher-order elements, or higher reflexes, in such social phenomena as war, collective work, exchange, the economy, the financial sphere, etc. Manifestations of associative activity, or the sum total of higher reflexes, in the collective as the subject matter of collective reflexology.

Collective reflexes. Regarding the characteristics of masses there should exist a certain law of their development. The basic laws of associative activity of the composite personality are the same as the laws of all animate and inanimate nature. The world is ruled by the same principal laws common to inorganic, organic and supraorganic, or social, phenomena.

2 Definition of Collective Reflexology

Collective reflexology should not be identified with so-called folk psychology since the former is concerned with all manifestations of associative activity of both nations and small social groups. Although myths, traditions, and language provide valuable material for the characterology of a people, they by no means exhaust the subject. Science, art, literature, etc., are also original products of folk creativity. Collective reflexology, unlike individual reflexology, is aimed at elucidating the ways in which relations between individuals in social groups produce the social results of their associative activity. Members of a collective often reveal certain personality traits that do not find manifestation in their private lives. Collective reflexology is an intermediate link between two sciences: reflexology and

sociology. Sighele, De la Grasserie and Rossi on collective psychology. The boundary between collective reflexology and sociology. The goals of collective reflexology. Collective characterology or the characterology of social groups. Sociology cannot exist without collective reflexology. The two sciences should not be identical. Common features and differences between history and collective reflexology. The key to the collective reflexes in prehistoric times is to be found in studies of prehistory. Individual evolution of human reflexes repeats, as it were, the evolution of reflexes in prehistoric man.

3 The Subject Matter and Method of Collective Reflexology

The subject matter of collective reflexology is the study of the emergence, evolution, and activity of collective groups united by a certain "cementing material" irrespective of their nature and goals. Collective activity develops along the same lines as individual activity. Manifestations of social activity or social movement as a social reflex. Subjective terminology is unacceptable when evaluating social reactions. A collective is based on common interests and goals. Organized and unorganized collectives. Collectives differ in their composition, goals and nature of interrelations between the members. The size of collectives. Application of energy theory to the collective. The wrong paths taken by this school. Subjectivism and objectivism in sociology. The subjectivism of contemporary psychology and its relationship to sociology. Reflexology and its significance in this respect. Failed attempts to approach sociology using the physiological method. Can objectivism and subjectivism coexist in sociology? The method of collective reflexology as well as sociology should be purely objective. As far as its method is concerned, collective reflexology should follow individual reflexology. The value of the statistical method for collective reflexology. Statistics as mass observation that provides material for collective reflexology. A collective experiment in the form of a questionnaire. Direct experiment with a whole collective. Experimentation under natural conditions and its importance. The method, application, and tasks of the collective experiment.

4 The Interrelation of Personality and Society

A neopositivist view of personality as a product of society. Objections. The biosocial theory of personality. The regulation of individual behavior by

society. Importance of the professions, occupations, environmental conditions and climate for individual associative activity. The latter is largely social rather than individual. The joint associative activity of society is the resultant force, as it were, of individual activities pursued by its members. A collective ensures the prevalence of quantity, while a personality reveals individual creativity. The difference in individual behavior in isolation and in a crowd. Subjection of personality to social conventions. Progression and evolution of social life is reflected in the establishment of certain relations between personality and society. Traditions and social norms limit the personality in those types of activity that are more or less mechanical. The importance of individuality in social life. Uniqueness of individual development is the basis for the social progress of nations. The role of outstanding persons in social life. A good social order is not an enemy of the personality. Deification of popular idols. The ways and means of managing and ruling a composite personality. The role of suggestion and "psychological" contagion. The importance of organization in controlling a crowd. The clarity of the goals to be achieved by a crowd and the importance of faith in the cause and in the leader. The growing significance of the qualitative aspect of personality in the process of its gradual development. The quality of a particular collective depends on its members. The meaning of unity and organization in the functioning of a collective. Dependence of its actions on its organization. The importance of the past for the activity of a collective.

5 The Collective as a Composite Personality

A definition of the collective as a composite personality. Various authors on this question. The concept of the crowd as an unorganized collective. Organized collectives and their varieties. A unified collective as a composite organism. Dimensions of the composite personality. A collective does not destroy the role of each participant who in one way or another affects the resultant force that determines the movement and functioning of the collective. The individuality of a collective. Conditions that lead to the development and establishment of a collective. Every society as a collective survives as long as its existence is justified by general conditions. A street crowd and its characteristics. The biased views of the crowd. While there are regularities in the behavior of a collective, it also depends on a number of circumstances.

6 On Collectivity

The role of natural and social selection in social evolution. Social instinct theory. The theory of the associative reflex in social evolution. Socialization as a result of individual experience. The importance of similarities between individuals. The importance of differences between mutually complementing individuals. Mutual assistance as a unifying factor. Interaction theory as an explanation of sociality. Its incompleteness. Conditions that encourage collectivity. The role of speech, coordinated movement, facial expression, gestures, symbolic objects, money and music as mediators of interaction in a collective.

7 Mutual Suggestion, Mutual Imitation, and Mutual Induction as Unifying Factors

The role of suggestion and induction in a collective. Imitation and its significance under social conditions of existence such as association and equalization. The causes of imitation. Examples from literature. The primary source of imitation is rivalry and confrontation. The place of collective experience. The direct influence of one individual on another separated by a distance via transmission of nervous energy. My own and my school's experiments in this field. Mediated and direct interaction between members of a collective.

8 Language as a Unifying Factor

The unifying role of language. Verbal suggestion and persuasion by oral speech and printed matter. The importance of language in the development of culture, science and art. The role of language in propaganda and manipulation of public opinion. Linguistic or vernacular differences as a factor opposing unification. Linguistic evolution.

9 On Collective Reflexes in General

Common collective reflexes. Panic as a defensive collective reflex. Automatic reflexes of collective attack. Hereditary-organic reflexes of self-preservation and procreation. Collective reflexes of concentration; symbolic-verbal, personal or purposeful collective reflexes. Collective reflexes of

defense and attack. Not only the goal but the means of reaching the goal also stimulate collective reflexes. The role of propaganda in the development of collective reflexes. The maximum tension of a collective reflex. Social shifts and changes in public mood and their causes.

10 The Development of Collective Movements as Associative Reflexes

The analogy between the development of a collective associative reflex and that of an individual associative reflex. Substitution for the real stimulus by a verbal symbol or other sign. A cause that acts as an irritant which stimulates a collective associative reflex. Movements of an army, mass gymnastic exercises, dances, orchestras, choirs, theatrical performances, etc., as vivid examples of natural collective associative reflexes. Artificial collective associative reflexes. Collective mental effort during meetings as a reflex. Needs as irritants of collective associative reflexes of an hereditary-organic nature and as a consequence of a habit that evolved through numerous repetitions. Economic relations as an expression of public needs interpreted as associative reflexes. An external influence in the form of a confrontation serves as an irritant of collective reflexes. A social crisis as a form of collective inhibition. Resolution of a crisis. Various forms of the collective reflex depending on the composition and nature of a gathering. The means as a goal in itself. Stimulating conditions for collective reflexes. The role of sthenic response. Mechanization of social phenomena and the importance of habit. A collective process develops similarly to an individual one as regards the gradual nature of its ascent and descent. The importance of imitation as a stimulus for communication. The role of predisposition in action coordination owing to exercise.

11 Collective Hereditary-Organic Reflexes

Collective hereditary-organic reflexes and their varieties. An unsatisfied need as a stimulant of aggressive collective reflexes. The significance of collective experience. Socialization of individual experience. Collective gravitation toward self-preservation. The role of the collective pull toward self-preservation in forming cooperatives, in the institution of property, criminal law, succession and other economic foundations. The role of sexual attraction as the basis of the family in the development of nationalism. Malthus's findings. Dependence of marriages and births on the prices of vital products.

Public protection of the family principle as a means of ensuring population growth. The role of sexual attraction in forging closer social ties among the members of a clan and a tribe as well as in raising the moral standards in the social environment. The meaning of the social "instinct" and its origins. Collective manifestation of the gravitation toward self-preservation. Instances of looting in times of famine and consumer goods shortages. Mob lynchings. Liberation of prisoners during the first days of the revolution. Manifestations of collective sexual desire. *Ogawchnye* societies. The Asroy society. The Saratov "defenseless." Collective manifestations of the "social instinct."

12 Collective Mood and Collective Mimico-Somatic Reflexes*

The objective side of the mood or tone of a collective. Various forms of collective mood. Animated mood and the sthenic reaction. Inflammatory speeches. Mood as a determinant of the collective attitude towards individual speeches and acts. Instances of intolerance of those speeches that are at variance with the prevalent mood. Dependence of collective mood on the composition of the gathering. The rise of collective mood and its gradual fall. Its influence on elections. Changes in the collective mood in regard to events and persons depending on times and circumstances. Examples. A shift in mood. Examples. Forms of expression of the collective mood. Dissemination of the collective mood through contagion. Enthusiasm and its external causes. Statistics as a means of mood registration. Changes in the nature of public views under the influence of mood.

13 Collective Attention and Collective Observation

The advantages of collective over individual observation. Collective attention. Visual and auditory attention. Evaluation of the object of collective attention. The importance of public supervision and control. Dependence of collective illusions and hallucinations on collective attention and mood. Conditions that facilitate attention or suppress it. A specific attitude of the crowd toward various types of external influences depending on mood. Examples. The emergence of collective illusions and hallucinations.

14 Collective Creativity

A general overview of collective creative activity. Collective judgment and its characteristics. A collective decision is less biased than an individual one. Individual creativity makes use of the collective experience of the past. The best results are produced when individual creativity is discussed in a collective. Decisions taken at meetings result from two or many opposing opinions. Dependence of work productivity of a collective on its size. The result of a collective judgment in the form of decision, legislation or verdict. The press as a form of collective discussion. Various authors on crowd decision-making. Characteristics of crowd expression and activity. The crowd and the leaders. Collective creativity in the narrow sense of the word. Creativity in history is achieved through the combined efforts of the collective as well as of individual members. Linguistic collective creativity. Concepts and ideas as the principal categories of the mind are the product of society. The logic that uses the notion of identity and opposition is the result of sociality. Language as a product of collective creativity.

15 Coordinated Collective Actions

Dependence of collective work on mood. The relationship between success and people's interest in their work. To be more successful, collective work should maintain a reward system that would satisfy personal interests. The influence of different conditions on the manifestation of collective energy. The importance of active participation in collective defense. The planned and coordinated nature of collective actions and their significance. The role of mood in various manifestations of social life. Its reflection in creative works, myths and beliefs. The development of religious sentiments and mysticism among the masses as related to depressive moods.

Part 2

Editor 's Foreword—1999

Foreword

1 The Law of Conservation of Energy

All human activity is subjected to the law of energy conservation. The social process is determined by the collective energy of individual persons. When any kind of collective effort is taking place it can be described by a parallelogram of forces with a resultant force. The processes of inhibition and

acceleration. Human nervous energy. The efficiency of human work. The "spiritual" element, which is essentially energy, cannot be irretrievably lost. Nor can there be any increase in human energy without corresponding external influences. Application of the law of energy conservation to the collective. Neither collective action nor collective creativity can vanish without a trace.

2 The Law of the Proportional Relationship between Speed of Motion and Motive Power

The importance of this law in mechanics. In human associative activity, too, excitation is proportional to the force of its source until the function is exhausted. Confirmation of this law in social life. Explication and examples. Stimulating or revitalizing conditions for social movements. A frequent revival of any social reaction in general is related to its acceleration. The amalgamation of stimulating factors. The United States' involvement in the Great European War as an example of the above.

3 The Law of Gravitational Attraction

The law of gravity is valid not only for the mechanics of heavenly bodies and inanimate nature but also for the organic and social world. Taxis, tropism, and other phenomena in the biological world as manifestations of the law of gravity. Application of the law of gravity to the composite personality.

4 The Law of Repulsion

Social life is subjected not only to the law of gravity but also to the law of repulsion in the form of dissociation, rivalry, competition, and confrontation. Every composite personality displays a particular kind of equilibrium between the two forces. Rivalry and social selection. Social improvement cannot occur without competition and rivalry.

5 The Law of Equal and Opposite Reaction

The emergence of opposition in the course of the development of any social movement. Examples and explication. Growth of opposition and amalgamation of parties for the purpose of counteraction. Coalitions of states opposing the rise of one particular country. The "subbotniks." Collective

inhibition. Every social movement provokes a reaction the intensity of which is proportional to the degree of deviation by the new movement from the established social order. Competition as an inhibiting factor. Other inhibiting circumstances in a collective. Artificial inhibition of collective activity by distracting the popular movement in a different direction or by direct counteraction. Internal inhibition. Suppression of mimico-somatic responses during overexhaustion. Gradual overcoming of internal obstacles as a symptom of inner opposition. The same source of excitation may serve as an inhibitor in others. Strikes as one type of counteraction. The 1905 general political strike in Russia.

6 The Law of Similarity

The law of similarity can be traced throughout the entire organic world. The law of similarity in the social environment. Similarity in social organization, forms of worship, mythology and process of civilization in different nations. Similarity in the development of revolutionary movements in various countries. The causes of integration of associative activity in human society. Direct influence, imitation, suggestion, and persuasion as means and ways of establishing likeness and similarity between people. In other cases we must admit that identical causes have identical effects.

7 The Law of Periodicity or Rhythm

Manifestations of the law of rhythm in the inorganic and the organic worlds. The cyclic nature of movements in social environment. Every state organism and every civilization go through periods of growth, prosperity, and decline. Evolution in the world of collective formations occurs in waves. Any social movement starts on an ascending line, reaches its apogee, and wanes to the point of extinction after which another social movement of the same undulating pattern begins. The law of rhythm in the sequence of historical events. The same law in political movements. Periods of revival as manifestations of the periodicity of social phenomena. All aspects of the associative activity of a collective are subject to the law of rhythm. Periodic changes in public moods. Changes in artistic trends. The sequence of revolutionary and reactionary periods. Social revival, stagnation, and fluctuation in intellectual life. Cycles in the development of beliefs. Cycles in

the sphere of economy. The movement of any social process follows an undulating pattern. Examples and explication.

8 The Law of Inertia

Manifestations of the law of inertia in social life in the form of routine, conservatism, and sluggishness that stand in the way of any radical reform. Every nascent social movement has a potential for further development. Temporary or permanent stoppage of a social movement depends on particular inhibiting circumstances. Traditions, rituals, and other vestiges of the past as manifestations of the law of inertia. Mass opposition to any innovation is based on inertia. The influence of authority transcends its actual life span. The momentum of any social movement continues beyond its natural stopping point. A government without sufficient public backing remains in power even after it has lost all popularity and support of the population. Once a movement starts it never comes to an abrupt end but disappears gradually. Social habits and their stability. Inertia preserves national, historical, and class characteristics in the population. A collective is more conservative than individual persons. Once a revolutionary wave gains momentum it will go as far as it possibly can if it does not encounter active opposition. The same is true for new religious teachings, fashions, etc. Belief in authority. Appeal of folk tales and the remote past. The domination of the law of inertia in the sphere of economy. Social heredity as a manifestation of the law of inertia.

9 The Law of Continual Motion and Change

There is nothing permanent and unchangeable in the world. The application of this law in the organic and the inorganic worlds. Its functioning in the supraorganic world. Society as a perpetually changing interaction between two or several individuals. Not a single social movement ever remains unchanged. Society exists in continuous motion. Economic life, language, literature, science, and art constantly create new forms. The circulation of energy from the free state to the bound one and into space. The circulation theory in social life. The spiral pattern of the process of change. The law of perpetual motion and change results in the irreversible nature of ever changing forms.

10 The Law of Energy Dissipation or Entropy

The law of entropy in the inorganic world. Its application to the organic world. Application of the above law to the supraorganic world when performing physical work. Dissipation of energy spent on the development of material culture in the form of discoveries and inventions by means of absorption and imitation. Energy dissipation concomitant with the rise of a collective to the corresponding level with the inevitable expenditure of a certain amount of energy. Application of the same law to the economic sphere.

11 The Law of Relativity

Application of the law of relativity in social life. Every social movement is important only as related to the given social conditions. The same social movement can inhibit progress under some conditions and set it in motion under others. The power and the significance of any influence on a collective depends on the state of this collective at every given moment in time. Society's attitude towards certain events changes with time and circumstances. Manifestation of the same principle when an existing social phenomenon gives way to the new one.

12 The Law of Evolution

The law of evolution is the law of form creation. Irreversibility of phenomena in the process of evolution. The law of evolution can be traced in all social phenomena. Leaps in the social world in the form of revolutions. Evolution proceeds from the general to the particular. The course of the evolutionary process is determined by several factors of geographic, climatic, and socioeconomic nature. The role of inequality of collective units and individuals in the process of evolution. Application of the law of evolution to individual social phenomena. Additional information from the field of ethnography and archaeology on the evolution of nation states. The main evolutionary impulse in natural and social selection. Evolution and perfection. Social progress, or perfection. Social regress, involution, and dissolution of the collective.

13 The Law of Differentiation

The law of differentiation in the inorganic and organic worlds. The law of differentiation in the social world. The primal society as a non-differentiated society. Differentiation in government rule. Organization as an inevitable result of differentiation. Every social movement is subjected to the same process of differentiation. Inhibiting conditions are also subject to the law of differentiation. The law of differentiation in language development. Differentiation in science. Differentiation of social needs.

14 The Law of Reproduction

The law of reproduction in the inorganic and organic worlds. The factor of the "spiritual" heredity in the life of society. The legacy of language, traditions, legends, general concepts, etc. Such results of individual creativity as science, literature, and art are reflected in the functioning and state of society and affect the posterity of a collective. The importance of upbringing and heritage. Everything in social life has roots in the past. Every advance in social development results from the previous collective experience. The so-called "genius" of the people. Hereditary acquisition of individual characteristics of a collective.

15 The Law of Selective Integration

The law of selective integration in the inorganic and the organic world. In every collective the process of selective integration is taking place alongside differentiation. To achieve cohesion of its component parts every collective is organized according to the law of selective integration. Non-differentiated and split collectives become integrated. The state as an integrating organization. Alliances of nations as a manifestation of the law of selective integration. The role of selective integration in organizing collective activity; creativity as a result of synthesis. Language development and synthesis. Morality as a synthesis of conceptions about human behavior. Assimilation and amalgamation of nations. Factors that lead to synthesis and integration. The role of imitation. Irritants provoking the same reaction as integrating factors. Selective integration in collective activity. Coordination of social forces in the form of coalition and alliance.

16 The Law of Historical Succession

No social phenomenon can emerge until the necessary groundwork has been done. Examples and explications. The better prepared the organized masses are the easier the victory of a revolution. A system of government is the result of the previous preparation of the masses for a particular form of government.

17 The Law of Economy

The law of economy is the cornerstone of adaptation. Application of the law of economy in social environment. Existence of collectives is due to the law of economy. Symbolism as a manifestation of the law of economy. Establishment of general concepts and categories. Sign symbolism. Development of science and knowledge and the economy principle. The whole of social life is permeated with the principle of economy. It governs the development of traditional culture and sciences. Numerous social compromises can be explained only by the economy principle. Population distribution reflects the tendency to favor the easiest path. Military actions are based on the same law. The utilitarian principle in life and the law of economy. The role of training and habit formation in following the economy principle.

18 The Law of Adaptation

The law of adaptation in the organic world. Spencer's formula and its insufficiency. The law of adaptation as applied to a collective. Development of collective reflexes occurs according to the environment and living conditions. Every existential manifestation of a collective is an act of adaptation.

19 The Law of Selection

Regarding living conditions it is only the utility principle that justifies a particular type of adaptation. Sociality and its origins. The principle of conflict and mutual assistance as a factor of evolution. Natural selection. Work co-operatives. Development of collectivism as a result of imitation and emergence of associative reflexes. Social selection.

20 The Law of Interaction

Interaction in the inorganic and the organic worlds. The importance of the principle of interaction in collective life. The principle of interaction regarding a street crowd. Cooperation between the collective and individual minds. Responsibility for collective delusions rests with the individual mind in charge of the given assembly. The difference between the workings of the individual mind and of the collective one. Interaction of classes and between collectives.

21 The Law of Compensation or Substitution

The principle of compensation in the inorganic and the organic worlds. The principle of compensation in collectives. This principle as an explanation of social contradictions. The process of communication by signs and symbols is based on the principle of compensation. Human speech and gestures as manifestations of the principle of substitution. Development of oral language. Written speech. Other forms of symbolism. The role of signs in social life. Social heredity is related to the development of sign communication. Social symbolism. The ritual aspect of religion and social rites as symbolism. Symbolism in art. The cohesive function of symbols. Fetishism and social symbols. Symbolism familiarizes people with complex relations using comparatively simple signs. Individual heroic deeds as symbols. Personified abstractions as manifestations of symbolism.

22 The Law of Dependent Relations

The law of dependent relations in the form of cause-effect. Examples of this law in collective life: famine and its consequences. Sexual dissatisfaction and its effects in a collective. Intellectual hunger. Determination and succession of historical events. Their regularity. The statistical method. The method of genetic sociology. Historical succession implies succession of popular movements. New social movements seem to be further developments of previous movements. Analysis of this postulation using the example of the Great War and the revolutionary movements that followed it and the Russian Revolution in particular. The interrelation between the nature and the behavior of a collective on the one hand and the conditions of its environment on the other. Behavior of collectives and the physical influences produced by the vital activity of the sun.

23 The Law of Individuality

The law of individuality is a universal law. The whole of the inorganic world evolves according to individual patterns. In the organic world, too, the evolutionary process is based on the individual, and the social world also evolves on the principle of individuality. Any individuality in general can be interpreted in terms of evolutionary laws in closed systems which can be reduced to their self-regulation. Social individualities and their character traits. National individuality helps to understand national history. The principle of autarchy, or self-sufficiency, defined by Aristotle.

24 Conclusion

Index

A Note on Structure, Translation, and Transliteration

The editor and translators have faithfully preserved the original structure of the book. Thus, for example, we have retained Bekhterev's original Table of Contents, which recorded own detailed overviews of each chapter.

Every effort has been made to restore the spelling of all proper names (i.e., English, French, German, or Italian) cited by Bekhterev in their Russian transliteration. In a very few cases, when we were unable to verify problematic names in any sources available to us, we have had to rely on our linguistic expertise.

A brilliant and prolific scholar, Bekhterev was, unfortunately, not meticulous in referencing his extensively quoted material. We have made these references as accurate as possible, but have not furnished lengthy transliteration of titles in journals, etc., having instead chosen to present here only their English translations.

Except for some proper names with long-established English transliteration, such as Nevsky and Kerensky, we have adopted Shaw's System II (Shaw, 1979. p.4) in every instance, assuming that readers of *Collective Reflexology* can be fairly designated "Scholars in the natural and social sciences who are concerned with Russian Studies."

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From Energy to Collectivity: A Commentary on the Development of Bekhterev's Theoretical Views

Joan Valsiner

The publication of Vladimir Bekhterev's classic work *Collective Reflexology* in English translation is a second important milestone for North American psychology in its possibility of access to the thought of that seminal personality of the turn of the twentieth century. Bekhterev's work was certainly known in North America in the first decades of the twentieth century (see Valsiner, 1988, p. 52), but later fell into oblivion together with the extensive proliferation of America's own varieties of behaviorism. Although Bekhterev's *magnum opus* (*General Principles of Human Reflexology*) was published in English translation in 1932 (Bekhterev, 1932), the ideas it contained must have by that time fit poorly into the American social discourse about psychology as the behaviorists had agreed to view it. It is clear from reading Bekhterev that his ideas, even if close to the behaviorist credo in many ways, were actually far more sophisticated than mere reduction of all psychological complexity to associative reflexes might imply (see Bekhterev's own comparison of his ideas with American behaviorism—Bekhterev, 1932, p. 173).

The present publication of *Collective Reflexology* constitutes an effort to overcome a certain historical myopia, which has taken hold worldwide, of contemporary empiricistic psychology. In the case of contemporary psychology, the focus on collecting increasingly vast masses of data and

analyzing those by methods accepted by social conventions leads psychologists away from the disconcerting understanding that, in the conceptual domain, advancements over the recent decades might be less than profound. Of course, the exponential information explosion may make it objectively complicated to maintain a comprehensive overview of the discipline (see Thorngate, 1990). Nevertheless, the conceptualizations of person/ society relationships in the social psychology of the 1990s have not led to many noteworthy generalizations. Instead, we continue to dispute many particularistic issues of exceedingly narrowly focused theories—and call for more or better data in the hope that by empirical accumulation we might arrive at general understanding.

From this overly empiricistic standpoint of contemporary psychology, careful attention to classic psychological works of the past can bring us to the conceptual issues that our predecessors tried to address. Bekhterev's *Collective Reflexology* is clearly one of those classics that may help us to overcome some of our present problems. However, we will essentially have to transcend the culture boundary between ourselves and Bekhterev's era. We need to jump not only from the English-speaking world of today to that of the Russia of 1900, but also from our socialized view on the focus of theoretical search as a narrowly data-driven enterprise to that of vast generalizations that satisfy our quest for understanding of the social and personal world in general.

Reading Bekhterev may turn out to be an intellectual nightmare for some present-day readers. His prolific writing is filled with redundancies, polemic assertions of the truthfulness of his ideas, criticisms of his opponents and other contemporaries, and vast sweeping generalizations. All this requires intellectual stamina from the reader, paired with a readiness to play with ideas (rather than merely refusing to understand a particular nuance of Bekhterev's expression, or being bored by his insistence upon a polemic point *contra* some long-forgotten thinker). Our collective culture in a society filled with computers, nuclear electric stations, aggressive 30-second advertisements, and cults of TV stars, provides us with a remarkably poor basis for understanding a "Renaissance humanist" (see Kozulin, 1984, p. 51) who simultaneously was an administrator of numerous institutes, an opponent both to his famous contemporary, Ivan Pavlov (see Joravsky, 1989, [chapters 5 and 6](#)), and to the totalitarian tsarist regime. Finally—in the 1920s—Bekhterev became a proponent and (at least intellectually) a victim of a "new

society." All this (and more) was characteristic of Bekhterev—and this makes his role in psychology rather complicated to comprehend. The present commentary is aimed at providing an (admittedly limited) angle upon the development of Bekhterev's ideas of the relations between personality and society.

Formative Years of Bekhterev's Ideas

Bekhterev's life was filled with activities in many parallel areas (see Gerver, 1932; Kozulin, 1984, [chapter 2](#); Nikiforov, 1986; Protopopov, 1928; Valsiner, 1988). Born in 1857, he graduated from the St. Petersburg Military-Medical Academy in 1878. He received his medical doctorate at the same Academy in 1881. After a couple of years as a *Privatdozent* in St. Petersburg and a trip abroad (in 1884) to work with Flechsig in Leipzig (where he also became acquainted with Wundt's psychology on its home ground) and Charcot in Paris, Bekhterev was appointed to the Chair of Psychiatry at Kazan University. His stay in Kazan lasted eight years—in 1894 he became the Chair of Nervous and Mental Diseases at the Military-Medical Academy. From that time, until his death in 1927, Bekhterev's work took place primarily in St. Petersburg—the historic capital of Russia that later had the fate of being re-named at first Petrograd, then Leningrad, and which has by now been restored to bear the very non-Russian name it had carried since its beginning.

Bekhterev's activities in St. Petersburg were extensive in both the scientific/clinical and administrative domains (see Nikiforov, 1986). The administrative event that is of greatest relevance for the present commentary is his organization of the Psychoneurological Institute (a semiprivate institution of research and higher education). The idea for such an institute for psychological and neurological investigations led to organizational activities by Bekhterev and some of his close co-workers (among them were A.P. Nechaev and A.F. Lazurskii) as early as 1903. The founding of the Institute was approved on July 9, 1907, by the Russian Minister of Public Instruction, after which the faculty of the Institute became established. The lectures at the Institute were planned to start in spring, 1908 (see Psikhonevrologicheskii Institut, 1908).

The Psycho-Neurological Institute immediately became a highly progressive academic institution in Russia, drawing on the cooperation of many major scientists in different fields as its faculty. A number of these faculty members are referred to by Bekhterev on the pages of *Collective Reflexology*. Thus, the zoopsychologist Vladimir A. Vagner (or Wagner—as his name has been usually transliterated in German publications) was listed as a faculty member from November 1, 1907. He later even served as its director. The personality psychologist whose method of "naturalistic experiment" was essential for Bekhterev—Alexander F. Lazurskii—was among the core organizers of the Institute. The sociologist Evgenii V. de Roberti was listed as starting on the faculty of the Institute also on November, 1, 1907 (see Gerver, 1912). Another sociologist, Maksim M. Kovalevskii (also mentioned in *Collective Reflexology*) joined the faculty in 1910. The Institute also drew to its faculty a number of major Russian philosophers (Nikolai Losskii and Semyon Frank—both from October 1907 onwards), linguists (Lev Shcherba and L.A. Boudoin-de-Courteney—both from 1910), and lawyers (Mikhail Reisner—later an important figure in the Socialist Academy and Moscow Institute of Psychology—see Van der Veer and Valsiner, 1991).

The faculty of the Psycho-Neurological Institute was the basis for Bekhterev's ambitious goal of building an institution that would succeed in the interdisciplinary study of human beings. It is that ambitious goal that can be detected in the text of *Collective Reflexology*. I want to claim that the intellectual roots of *Collective Reflexology* are embedded in the intellectual environs of the Psycho-Neurological Institute from its beginning to the time of publication of the book (see also Bekhterev & Lange, 1924; Bekhterev & Lange, 1976). In terms of its wider context, *Collective Reflexology* is based on ideas that were widespread in the European traditions of the biological and social sciences of the latter half of the nineteenth and first decades of the twentieth centuries.

Bekhterev and Mind/Body Dualism

Perhaps the most general dilemma that nineteenth-century science and philosophy left for us is the issue of mind/body dualism. There are basically two possible strategies for overcoming that dualism. First, it can be

eliminated by denying one of the two parts of the whole (e.g., reducing the mind to bodily processes, or ascribing the latter to the invention or creation by the mind). The second strategy entails some conceptual construction of the unity of these opposing parts. The notion of such unity is to be elaborated, and a number of different forms of such elaboration are possible.

Bekhterev was actually facing two dilemmas at the same time: aside from the mind/body opposition, he had to address the conceptual contrast between living organisms and physical objects. That contrast was a dilemma that emerged from the biology/classical physics comparisons at the time, and that has posed formidable conceptual problems in our present-day psychology. Bekhterev's solution to these two dilemmas was to insist upon one (the clear separation of living organisms and their basic principles of existence from those of physical systems— Bekhterev, 1904, p. 178) and solve the other by way of a unifying idea—that of *energy*. Of course, the notion of energy itself was borrowed from the nineteenth-century physical sciences, and it fitted with living systems in ways that could seem rather contemporary to present-day readers.

The "Hidden Energy," its Accumulation, and Constant Transformation

Bekhterev's emphasis on energy was based on the neurophysiologist's worldview—that of observing electrical activity in the different structural parts of the nervous system. The "nervous energy" (the energetics of biochemical processes in the cells in the nervous system) that the active organism carries with itself as "hidden" was viewed as unified across its different external manifestations—those of mind (psyche) and body (Bekhterev, 1904, p. 35). In this way, Bekhterev avoided the dualism between the intrapsychological world and the immediate world of organism-environment interaction. At the level of *phenomena*, Bekhterev accepted the difference (parallel existence) of the mind and the body. However, that parallelism was explained through both those phenomena being the result of the same cause—"hidden energy" (Bekhterev, 1904, p. 39). The nerve processes and the psyche do not differ qualitatively from each other, but constitute a different form of expression of the same hidden energy. Likewise, unconscious and conscious phenomena are merely different

expressions of that energy. This energy was not construed as being hidden in a static state in some internal reservoir of the organism, but instead as being constantly changed from one form to another in the course of the organism's active relationships with the environment. The environment is the source of energy, which becomes transformed into its hidden form within the organism (building up an "energy reserve"—the basis for all subjective psychological phenomena), and through movement of the organism the hidden energy becomes again transformed into other forms of energy (e.g., by way of mechanical work). This general-biological view constitutes an open-systemic perspective on the nature of the living systems.

Bekhterev's use of the notion of energy as the unifying concept of all biological phenomena can also be seen as the theoretical foundation for his endeavors to study human beings in ways that we presently would consider "interdisciplinary." Indeed, he expected psychology as a separate discipline to vanish and be replaced by "general psychobiology" (Bekhterev, 1904, p. 44) or "reflexology," as he ended up calling that discipline later (in *Collective Reflexology*; see also Bekhterev, 1932, p. 81, on his account of the history of the use of that term). In fact, Bekhterev's discussion of nerve conductivity anticipated the 1990s fad of demonstrating the effectiveness of neural networks in the learning process (e.g., Bekhterev, 1904, p. 169). Indeed, much of present-day cognitive neuroscience seems very busy teaching computer systems' rather special, particular skills in accordance with notions that are very similar to Bekhterev's associative reflex ("neural networks" in our contemporary cognitive neuroscience).

From Evolutionary Ideas to a Theory of Activity

Bekhterev went along with the progressive Russian evolutionary biologists of the late nineteenth century who refused to appropriate Darwin's ideas without modifications (see Valsiner, 1988, pp. 3038). Two major modifications were in order. First, the notion of inter-individual *competition* was supplemented by the primacy of the idea of *cooperation* (see Kline, 1955, for overview; for an original version, see Kropotkin, 1908). Secondly, the behaving organism was viewed *not merely as a target* of the natural selection process, but an *active agent* in the process of organism-environment adaptation.

Bekhterev's emphasis on the relevance of the organism's active role in interaction with the environment was expressed by him in opposition to the ideas of Herbert Spencer (e.g., Bekhterev, 1904, p. 97), distinct from Darwin's mechanistic elaborations of the idea of natural selection. In contrast, Bekhterev emphasized the relevance of activity on behalf of the adapting organism, tracing the roots of that notion to the philosophy of zoology of J.B. Lamarck. It is therefore not surprising that, in the different versions of Russian/Soviet activity theories that have been discussed in English-language psychology literature, the "Lamarckian" and "Bekhterevian" traces of ideas surface in different forms. Bekhterev synthesized his view of the processes involved in evolution with his energy-notion:

It is obvious that the agentive role of natural selection starts only from that time when certain changes that are useful to the organism under given conditions have already emerged. In this the role of natural selection is rather more negative than positive. While it does not produce any positive novelty in the organism, it merely kills the non-adapting organisms, which opens a wide field for the experiences of more adapted ones. Natural selection is in no way responsible for the adaptation processes. Its role in the evolution of organisms is rather supplementary and not relevant. It is obvious that the relevant role (that is— the role of adaptor of the given organization to the surrounding environment) in the evolution of organisms is played by the internal forces of the organism. In other terms, [that role is played] by that very same hidden energy that is characteristic of any organism, and which is at the foundation of the exchange and alimentation of tissues and which also serves as the cause for the psyche in the wide sense. (Bekhterev, 1904, p. 100)

Organisms are thus active and constructive participants in their own adaptation to the environment, and, through that, also in the future of their species. Bekhterev's emphasis on the relevance of "internal forces" allowed him to concentrate on the active role of the organism in any setting based on the organism's history of past relationships with the environment. The phenomenon of the psyche is thus viewed as a means to the end of adaptation (Bekhterev, 1904, p. 198), and its emergence in phylogeny is explained by the notion of accumulation of energy by the organism. Hence his later emphasis on the role of personality and its socially mediated "education" in psychological development (see Bekhterev, 1912, 1914, 1916). This focus on intrapsychic energy had likewise given a relevant head-start for the work of

Bekhterev's colleague at the Psycho-Neurological Institute, Alexander Lazurskii (1906, 1916), and for Mikhail Basov (1931; see also Valsiner, 1988, ch. 5) who, as a former student of Lazurskii, later became one of the major originators of Activity Theory in the Soviet Union.

The acceptance of both intrapersonal and extrapersonal (conduct, social relations) spheres of energy accumulation and transformation is crucial for understanding Bekhterev's theoretical credo. By explaining the subjective domain of the human psyche through considering it a form of transformed energy, Bekhterev overcame the dualism between mind and body through positing a general unifying basis for both. It is another matter that he did not elaborate (beyond occasional comments) how the different unique forms of transformed energy become established, indeed forcing that explanation to the domain of formation of associative reflexes and their complexes. His polemic argument with the introspection-oriented psychology of his time surely led him to overlook the variety of specific forms of psychological phenomena and their mutual interdependence within the whole. In part, that neglect may be due to Bekhterev's style in writing (and, probably, thinking). The reader of *Collective Reflexology* has many opportunities to experience that herself/ himself: Bekhterev's sudden leaps from detailed experimental or everyday descriptions to sweeping statements about the inevitable truth of his point of view (or falsehood of that of his opponents) can leave an inquisitive reader flabbergasted.

Bekhterev's Interest in Hypnosis, and the Foundations of Social Life

Psychology of the twentieth century has had very few empirical phenomena of persisting theoretical relevance at its foundation. One of those few is the phenomenon of hypnosis, and especially that of posthypnotic suggestion. It is well known that these phenomena gave rise to the psychoanalytic belief system and its methodology. The issue of the effects of hypnosis (and its use in therapy) was a major topic that puzzled psychiatrists and psychologists in the last decades of the nineteenth century. Bekhterev, as a psychiatrist and a practitioner with practical hypnotic skills, was on target in trying to understand the phenomena of social suggestion that are involved in hypnosis as well as in the social life in general.

Social suggestion in that context was viewed by Bekhterev using an analogy with infectious diseases (*contagium vivum*) as a special kind of "psychological infection" (*contagium psychicum*). Analogously to infection-carrying biological microorganisms, that psychological infection is carried by words, gestures, and other phenomena in social interaction—hence human beings are constantly available to social suggestions (Bekhterev, 1903, p. 5). Bekhterev's emphasis on the organism-environment interaction as the framework in which the hidden reserve of energy is being accumulated fits with the notion of psychological infection: through social suggestion, each personality accumulates the particular suggested material (energy) in a way that later starts to look (from the perspective of others) as if it were the person's own willful act.

However, Bekhterev's view on the mechanisms of social suggestion (in both everyday life and under hypnosis) transcended the traditional opposition of the "self *versus* others suggestion" dilemma. Bekhterev's notion of the active person led him to emphasize that, in any case of social suggestion, the socially suggested material interacts with the person's already existing core personality. The latter, of course, was seen as a result of previous encounters with the world by the given person. Bekhterev viewed the implicit (we could call it clandestine) process of everyday social suggestion as the normative context for development of personality:

In essence, involuntary suggestion and intersuggestion (*yzaimovnushenie*), as general phenomena, act everywhere in our everyday life. Without noticing it ourselves, we appropriate to some extent the feelings, superstitions, convictions, tendencies, ideas and even personality characteristics from the persons around us with whom we interact most frequently. That kind of inoculation of psychological states takes place mutually between people who live together; in other words, every personality to some or another degree inoculates others with the peculiarities of his own psychological nature, and, in return, takes from them one or another kind of psychological trait. (Bekhterev, 1903, p. 41)

The clinical phenomena of *folie a deux* that have been well known to psychiatrists constitute a good example of these processes of mutual psychological contagion. However, even if the main route of mutual social suggestion was understood by Bekhterev to take place with minimal resistance from the person, he did recognize the active role of the personality in the process of interpersonal psychological suggestion. From his own

experience with posthypnotic suggestion, Bekhterev could draw a number of instances where the suggestion becomes transformed on the basis of internal contradiction between the preexisting personality and the particular suggestion:

The resistance to suggestion, if it occurs, is based on a critique, on a discovery of an internal contradiction between that suggested and the convictions of the person... From here it becomes obvious that suggestion in certain cases does not exclude even criticism [of the suggested material by the person], while it still does not cease to be suggestion. This can be observed during hypnosis of mild levels when the personality, or "I" that is not fully removed, still relates critically to all of the surroundings, including the suggestion.

I suggested to a person in hypnosis that when he wakes up he must take a postcard from the table. When he woke up, he almost immediately looked around on the table and his gaze became fixated on a certain spot. "Do you see something?" I asked. "I see a postcard." I said good-bye to him and prepared to leave, but he still keeps staring at the table. "Don't you need to do anything?" I ask. "I would like to take that card, but I do not need it!" answers the man, and leaves—not having fulfilled the suggestion and obviously fighting with it. (Bekhterev, 1903, p. 14)

The social world of Russia, Europe, and North America provided Bekhterev with a multitude of real-life phenomena that could be explained by social suggestions that have become fixated into persons' strong beliefs. Cases of religious sects of extreme kinds, recovery from hysterically based somatic symptoms under personal religious revelations, and so on, were all present in the Russian society of Bekhterev's time, as in any other society at any time. The strong beliefs that social groups develop among their members—by way of escalation of the belief through mutual social suggestion—constitute the basis of social hysterias that have been described widely in world history. For Bekhterev, the energy-based worldview that recognized the development of intrapersonal psychological phenomena on the basis of social (self- and other) suggestions constituted the foundations on which the ideas expressed in *Collective Reflexology* were built two decades later.

Bekhterev's Methodological Imperatives: The "Objective" Study of Developing Phenomena

All through his long academic career, Bekhterev was actively promoting objectivity of investigation, largely arguing against the acceptance of introspectively available psychological phenomena at the level of their immediate symptoms. Strong anti-introspectionist statements can be found in Bekhterev's work *Collective Reflexology*. However, in parallel with these statements one can find recurrent disclaimers that Bekhterev over many decades (e.g., Bekhterev, 1888; Bekhterev, 1903; and in the present effort) did *not* want to eliminate the phenomena of subjective nature from the focus of scientists' attention. Instead, the issue at stake was methodological—how to gain access to the *development* of subjective-psychological phenomena, rather than start from their established form (which could be accessed only by introspection). This methodological problem was relevant for other Russian psychologists of this century, despite their accentuated differences from Bekhterev's reflexology (e.g., Vygotsky, Kornilov, Basov—see Valsiner, 1988; Valsiner & Van der Veer, 1991; Van der Veer & Valsiner, 1991). In most of our Western retrospections on Russian psychology we have failed to detect the consistent emphasis on developmental methodology in the thinking of various authors whose work happens to trigger our interest.

Bekhterev's notion of objective methodology deserves careful analysis. It has become almost a legend of what was only recently called "Soviet Psychology" to consider Bekhterev an active fighter for objectivistic methodology in the sciences that study human beings. What often has been overlooked is his recognition of the complexity of human psychological phenomena, and his acceptance of a highly heterogeneous set of methods for its study. However, the main explanatory principle—the associative reflex—and its philosophical underpinnings (the worldview of associationism) remain prominent in all of Bekhterev's vast-ranging claims, and in the empirical work of his co-workers.

Nevertheless, it would be a gross simplification to consider Bekhterev a leader of the worldview of the physiological reduction-ism that is easily accepted by many empiricistic psychologists. Any reader of *Collective Reflexology* will get a glimpse of his gargantuan effort to paint the big picture of human sociality by way of an (ill-defined, but popular) energy-concept on the one hand, and different complex formations of associative reflexes on the other. It is perhaps exactly that wide scope of knowledge and readiness to approach fundamental issues of human psychology that keeps fascinating us. *Collective Reflexology* in all its complexity, persuasive efforts, and striving

for generality can thus lead us to more fundamental questions in our contemporary social psychology and beyond.

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Part 1

Editor 's Foreword—1992

Work on the first translation of Part 1 of *Collective Reflexology* has been a rewarding sojourn during my fifteen-year voyage in and around Russian and Soviet social psychology. Preparation for the journey really began in the late 1960s, when I was trying to reconcile my own mainstream-American version of what Berry (1979, p.23) would later call a social-psychological "blueprint" with the discrepant social reality based on my new home, Canada. In 1971, I spent a sabbatical leave with Henri Tajfel's unit at the University of Bristol. There, I was witness to the competition between European versions of the field and became apprised even more vividly of the cultural relativity of what constitutes social psychology.

Articulate spokespersons for the multiple and oft-conflicting American and West European variations on the main themes of social psychology were subsequently brought into confrontation in a NATO Conference at Carleton University (see Strickland, Aboud, Gergen, Jahoda & Tajfel, 1976; Strickland, Aboud, Gergen, Tajfel & Jahoda, 1976), one presented before a group of Canadian social psychologist-discussants, mostly recent Ph.Ds and graduate students; many of the papers subsequently appeared in a book (Strickland, Aboud, & Gergen, 1976).

While editing the contributions to this book, I was, one day, asked by a harried official of the Canada Council if I would supervise a Russian post-doctoral student for a year. I agreed, and this young social psychologist, Vladimir P. Trusov, of Leningrad University, and I considered at length the possibility of including Russian/Soviet ideas in the broadening focus of my work on contrasting social psychologies. After a year, we organized another Canada Council/ Carleton University conference. It had the same title as the first, "Paradigms and Priorities for Social Psychology," but it was constituted

differently. Whereas the former had brought major American and West European views of social psychology into an arena with young and/or new Canadians, the Strickland-Trusov meetings in May 1977, introduced a number of by-then-established Canadian social psychologists to an equal number of quite senior Soviet counterparts, this time with several "participant observers" from the USA and Western Europe. Inevitably, another book followed (Strickland, 1979), and this, in its turn, was succeeded by edited collections focused first on Russian and Soviet social psychological theory (Strickland, 1984) and second on empirical research (Strickland, Trusov, and Lockwood, 1986). A summary of my other efforts in this particular "context of social psychology" has been published recently (Strickland, 1991). I ended that article with a post-glasnost quotation by a young psychologist at the USSR Academy of Sciences about how ignorant Russian scholars have been forced to remain over the decades by their political circumstances:

We know neither the names, the achievements, nor the real mistakes of our predecessors. This applies not only to pedology, but also, for example, to the work of the first Soviet psychoanalysts, to the social psychology of the 1920s, to such varied figures as Bakhtin, Bekhterev and Berdyaev. (Etkind, 1990; quoted in Strickland, 1991, p. 589)

It was rewarding to see Etkind's statement in print at long last. Over the duration of my efforts to more fully comprehend Soviet social psychology, my most difficult problems were in establishing the source of certain ideas that seemed on their own merits to hold so much promise. Veiled spoken references were often made by Soviet scholars to the work of V.M. Bekhterev, but written citations were rare and nonspecific. Bekhterev's picture might hang prominently in a lecture hall, but students and young teachers, conversant with his many contributions in physiological psychology, had not read (or did not admit that they had read) his social psychological writings. At the conversational level, there were dark rumors to be heard about Bekhterev's fate (see Moroz, 1989), but of *Collective Reflexology* (Bekhterev, 1921), a book often mentioned and then left unexamined in North American histories of psychology, no details were forthcoming. The research of the late Boris Lomov and his associates at the former USSR Academy of Sciences on such topics as "joint memory," "joint psychophysics," "joint visual search" (Lomov, 1978, 1979; Lomov & Kol'tsova, 1984), taken in conjunction with the pervasive importance of a major "principle of analysis" (see Strickland, 1987) in official Soviet

psychology, "communication" (*obshchenie*), rarely failed to instigate excitement when discussed with Western audiences. Indeed, such investigations seem to have anticipated those of a number of enthusiasts in the West (Middleton & Edwards, 1990; Wegner, 1986; Zajonc & Adelman, 1987). Yet of Russian/Soviet antecedents, there was no clear record. This led to a request to the Social Sciences and Humanities Research Council of Canada for funds to permit translation of Part 1 of Bekhterev's *Collective Reflexology*.

During preparation of this translation, it became apparent that Bekhterev's work is much more than an early extension of a physiologist's reflexological views to the person/group relationship, although even as such it proves to be a remarkable statement of insights about the nature of groups and collectives, plus prescriptions of certain research methodologies for their scientific study. An historian of science may construe the book as another product of one of the greatest scientific minds of the century (Bekhterev authored 600 or more research articles and a half-dozen books and founded several journals). A psychological theorist might see Bekhterev's efforts as the first complete "systems" view of the social organism. His view of "reflex" was not restricted to the world of glandular secretions or muscle twitches—it was an energetic view of the entire organism (whether made up of one or of multiple units). In this volume, the translation is preceded by a commentary by one of the major authorities on Russian psychology, Jaan Valsiner of the University of North Carolina, who expands knowledgeably on this aspect, among others, of Bekhterev's thought.

Collective Reflexology has proved equally exciting as an early statement of political psychology, wherein the author outlined the views which probably helped cost him his life (see Moroz, 1991) on the interrelations between the person and society and the strengths and limitations of each of these social units. Reading his psychological assessments of events in early Soviet parliaments, about military morale problems, about food riots, anti-Semitic violence, etc., in the context of current news from Russia gives this historical document a new life. Finally, since the book was published in 1921 without ever a second edition, this translation testifies to what the world of social psychology, along with other spheres of psychological and social science, had lost through the totalitarian machinations of Stalin and his successors. In addition to the "progressive" ideas first voiced and then silenced, references to hitherto unknown social research and researchers will help investigators

complete the slowly emerging history of Russian social science in the years after the Revolution.

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In Place of a Foreword

I demonstrated in my *General Principles of Human Reflexology* (Petrograd, Rikker, 1918) that manifestations of the individual personality are at present investigated with a strictly objective research method. The latter should definitely be applied to the manifestations of collective, or composite, personality, for to speak of "public consciousness," "public soul," or—which is the same—of "collective soul" means no more than to use a figurative expression. The truth is that while a human individual can accomplish self-analysis by means of self-observation, the same can never be done with respect to a collective, or composite, individual.

However, sociologists as well as representatives of other humanities still operate with subjective and even metaphysical concepts. They keep talking about collective emotions, conceptions and perceptions, about the collective soul, collective consciousness, collective will, etc., whereas it is clear to everyone that in reality this is merely a generalization of the subjective state of one's own self, which should never be extrapolated to a crowd or any society in general.

Even the neopositivist sociologists generously sprinkle their treatises with subjective terminology borrowed from psychology. Let us take, for instance, E. de Roberti's interpretation of collective experience, a phenomenon undoubtedly real and objective:

It is "sociality"—that constant source of the "*spirit*" or "supra-organic phenomena" in nature—that neopositivists see as lasting, indissoluble and versatile interaction, which is necessarily established in any. permanent, rather than random, "community" of cere-brally endowed living beings between such characteristic psycho-physical and already *conscious* phenomena and processes as *sensations*, *perceptions*, *conceptions*, particular

images and particular *judgements*, as well as *emotions*, elementary *feelings* and *volitional* impulses. It is this interaction that makes up the entire inner content of collective, or communal, experience that relates, verifies, supplements, unifies and objectifies the discordant and always highly subjective data of bio-individual experience (but not of the personal one, which represents the highest state and the ripest fruit of the communal experience).¹

This quotation gives a clear idea of the peculiar manner in which neopositivism coexists with individual subjectivism.

All this subjectivism borrowed from psychology is in essence a remnant of the anthropomorphism that is being radically eliminated from the biological sciences. It should also be expelled from sociology, history and other humanities. Collective reflexology represents an attempt to found one of the most important branches of sociology, which is also often referred to as public or social psychology, on a strictly objective basis of experience and observation without venturing into the realm of subjectivism.

As we know, sociology has until now been based on two scientific disciplines: biology and psychology. Biology provided this science with a stable objective basis, while psychology as a subjective science weakened its position. From our point of view, sociology should not contain any psychological doctrines of a subjective nature. To become a strictly objective discipline, sociology should be based mainly on two sciences: biology and the reflexology that I am currently developing. The latter should replace psychology whenever we study the individual human personality and the individual members of a collective in particular.²

We shall not touch upon the question of whether this new discipline of reflexology is an abstract or a concrete science. The issue is not in the theoretical interpretation of this science but in the value of its method, which excludes subjectivism in the study of human personality. That is why reflexology rather than psychology should underlie the study of the social world.

Let us now discuss the issue of reflexology's relationship with biology and sociology. There can be no doubt that the human personality is a product of socialization, since without society human beings would not be human. In this connection one can cite certain observations that have the validity of an experiment, regarding some abandoned individuals deprived of any upbringing (e.g., Hasper Hans who lived in a barn until he was fourteen),

who were unable to talk or walk. It is also known that shipwreck survivors stranded for many years on a desert island show clear signs of mental decline. The same is observed among individuals subjected to long-term solitary confinement. We also know that village children who have had few contacts with outsiders seem backward in comparison with city children. And it is common knowledge that those who grow up in contact with a greater variety of people are more advanced than those who spend their lives shunning society.

It is obvious from the above that socialization is extremely important for the formation of the human personality: it is the principal factor of its development. In a word, personality (as such) is in its highest manifestations the product of society, which is, in turn, determined by its component members, i.e., personalities.

As we know, the neopositivist school reduces to their social origins even such categories as reason, color, space, species, number, causes, substances, etc., which places it in between empiricism and classical apriority:

In reality they (i.e., categories) represent the most general relations between objects. While surpassing all other concepts in their comprehensiveness, they satisfy all aspects of our intellectual life. Therefore, if during the same historical period people did not share homogeneous notions of time, space, cause, number, etc., any accord between individual minds would be impossible and consequently, any kind of communal life would become unfeasible. For this reason society cannot abolish these categories in favor of arbitrary personal opinions without abolishing itself. In order to be able to exist, it needs not only moral consensus but also a certain minimum of logical unanimity, the limits of which are not to be crossed arbitrarily.³

All of this is unquestionably true, but the real issue is whether these categories are also biological facts, as has previously been believed, or purely social ones, as the neopositivists claim, or—which is our contention—whether we are dealing with facts of biosocial nature. The question thus posed is in reality far more complicated than it may initially appear. In any case, it requires a comprehensive discussion and cannot be resolved in so many words. That is why we prefer to keep it out of this discourse altogether, particularly since it is outside the limits of our present task. We can only point out that individual experience must have definitely played a role in the evolution of the above categories. Apparently, collective experience acquires

a special significance here, leading through socialization to the equalization and softening of individual differences in the subsumed phenomena.

As for the evolution of the human personality, some believe that it created society. There can be no doubt that a better-developed human personality produces better social structures and that personality does, indeed, influence society. But if we ask whether society is the result of personality development, the answer can be only negative.

The first proof of this is the phenomenon of imitation. It is common knowledge that personality development is to a great extent a product of imitation, which plays an enormous role in upbringing as well as in all aspects of social life in general. One can unequivocally state that without imitation there would be no personality as a social form, inasmuch as the main source of imitation is communication with one's own kind, among whom mutual induction and suggestion are developed through cooperation.

Another proof is language, which owes its origins to imitation on the one hand and to the instinctual need for communication on the other. Language is, therefore, an undisputable product of socialization, since without preliminary contact between individuals the emergence of human language is inconceivable. On the other hand, we know that human thought evolved mostly due to the word, and in such close and immediate association with it that abstract thinking would seem impossible without the participation of inner speech. And there is, indeed, evidence that savage prehistoric man was to a significant extent ignorant of both thought and words, in place of which there were only gestures, shrieks, exclamations and sound imitations.

Writing is another proof that the human personality evolves under the influence of society. It allows the results of individual experience to be widely circulated in the collective as well as to be recorded for the future and preserved for posterity. It will be no exaggeration to state that writing transformed man into an historical being.

The transition from pictorial (ideographic) to phonetic writing also marked an enormous advance in the evolution of human personality, hardly less significant than the transformation of gestures and exclamations into articulate speech consisting of words. Indisputably, it was phonetic writing in conjunction with oral speech that gave rise to the civilization of developed nations, since it became possible for each generation of humanity to pass on the totality of its achievement to all the generations to come.

Is, however, the evolution of the human personality exclusively due to society? Heredity and what we relegate to the manifestations of instincts undoubtedly represent a biological phenomenon, but is it possible to conceive of personality development unaffected by heredity and instincts? The initial stage of personality evolution may have been determined by the role of the hand, which provided the primal language of gestures (from which human speech developed), just as the later stage was determined by verbal language. However, the evolution of the hand was the result of biological causes rather than the social conditions. The hand developed mostly due to the fact that, through natural selection or constant exercise, human beings stood up on their feet and transformed themselves from four-legged, or rather four-handed, creatures into two-legged ones.

Also, articulate speech as such could evolve only when the animal snout became the facial part of the head, which may have resulted partly from changes in nutrition and partly from the more sophisticated preparation of food, which are again biological rather than social factors. As for the family—that primal model of social relations—is it not a result of sexual attraction? Even the social aspect of human life may be a consequence of natural selection in the sense of the better survival of socially oriented beings, and natural selection is again a biological rather than a social factor. We also know from reflexology that higher, or associative, reflexes develop on the basis of common reflexes which are purely biological phenomena. Furthermore, what we define as "inborn inclinations" are determined by heredity, which is yet another purely biological factor. And finally, extraordinary giftedness, or genius, is to a significant degree produced by fortunate hereditary circumstances; education and erudition can only provide this inborn gift with the means and motivation to apply natural abilities. One can state with confidence that, no matter how substantial the social preparation for a certain discovery or invention might be, it is beyond the shadow of doubt that a poorly endowed person is not suited for making great discoveries, just as a retarded child is incapable of complex mathematical computations.

This leads us to the conclusion that the early evolution of personality was affected both by biological and social factors, while at the later stages it was influenced predominantly, though not exclusively, by social factors, which is why personality should be recognized as a phenomenon of biosocial origins.

This important thesis has been established only in recent times. It was proposed as early as 1860 by Courneau (*A Treatise on the Interrelations between Seminal Ideas in Science and History*) and the German sociologist Lazarus (*The Life of the Soul*); somewhat later this view found its way into English scientific literature in Lewis's *Problems of Life and Spirit*, 1874, and *The Physiological Foundations of the Spirit*, 1877. But it is our compatriot Professor of the Psycho-neurological Institute in Petrograd, the well-known sociologist Evgenii de Roberti, who has undoubtedly been the great champion of this proposition. He called it the "biosocial hypothesis" and used it as a foundation for his sociological research.

According to this hypothesis, or theory, the human mind is a product of the close union between the organic nature studied by biology and the supraorganic nature, or social environment, which is the subject matter of sociology. For that reason, de Roberti regarded psychology as a concrete science (when this theory was elaborated reflexology had not yet come into existence), as *biosociology* based both on biology and sociology. Therefore, it cannot be an abstract science, nor can it be recognized as a major science in the scientific hierarchy. However, we should not go as far as do some neopositivists, who claim that human individuality, i.e., personality, has no other root but the social one. Nor can we agree with Worms (*The Philosophy of Social Sciences*, 1903-1907), who does not see a satisfactory resolution of this problem and relegates its solution to the future. According to him, "No one has proved these issues insoluble and no one can claim that advances in knowledge will not shed a bright light on them with time" (vol. III, p. 95).

From our point of view, associative reflex theory also permits us to establish that personality is not after all the exclusive product of social interaction. In order to protect oneself from an attack by a wild animal it was enough to form an associative reflex between the animal's roar and appearance on the one hand, and the possibility of its attack on the other. The only escape was based on individual experience—the reflex of defense. Neither of those owe their existence to social interaction, although interaction does disseminate, through imitation, useful individual experience. In order to satisfy his hunger with shellfish or bananas, a savage did not have to possess previous collective experience: all that was needed was an associative reflex—born of the frequent search for food necessitated by hunger—between the outward appearance of the shell or the fruit and the satisfaction of hunger. In this and similar cases we deal with individual experience that subsequently

becomes universal and by itself cultivates personality—the more so since the latter is manifest among uncivilized peoples far more strongly than is generally supposed. Besides the conditions that influence personality formation through individual experience from the very first days of its existence, there also emerge conditions for cultivating the human personality via games, cooperation, etc., through collective experience in which imitation plays a special role as a reflex of exclusively social variety.

One should also bear in mind that a collective can act and express itself as a whole, as a *composite personality*, which will be discussed below and which is, in fact, the main subject of the present work.

As we know, there have already been attempts at creating so-called folk psychology as well as social, or collective, psychology (Steinthal, Lazarus, Wundt, McDougall, Kopel'man, and others), but these attempts, being based on the subjective interpretation of facts, have not met with success.

As has been mentioned earlier, the aim of the present work is to exclude subjectivism altogether from the issues of collective actions and reactions, as it has been excluded from my concept of reflexology as the objective science of the human personality.

While not permitting subjective interpretation of issues concerning the study of the evolution of social or collective phenomena, and understanding all such phenomena as collective or social reflexes, we recognize the necessity to apply to their study the same strictly objective method that we use when studying the individual personality. For this reason, the title chosen for this work is "Collective Reflexology" rather than the commonly used social, or collective, psychology.

It is hardly necessary to elaborate here on the fact that tackling the issues related to the evolution of social phenomena, or social, i.e., collective, reflexes, demanded from the author a new approach to the subject and its new interpretation, since what has until now been called social psychology could not yield much useful material in this respect because of its subjective method. One should also mention that although psychologists and subjectivist sociologists (since sociology still remains largely a subjective science, which should not be the case) could stretch their subjective method to philosophize about the crowd as an entity with a "single soul," it was far more difficult to apply the subjective method to all the other forms of the collective. That is why the crowd became the favorite object of study among psychologists and subjectivist sociologists. We shall expose in due course the one-sidedness

apparent in the works of a number of authors when they describe the crowd from a subjective point of view. Here we would like to point out a more serious mistake made by psychologists and subjectivist sociologists in their studies of the masses and their movements: some of them try to explain the latter by a single common principle.

Thus, Tarde, for instance, sees this common principle in imitation, whereas W. McDougall finds it in the instincts of human nature.

The obvious deficiency of these views proceeds from the lack of strict objectivity in research that tolerates preconceived notions.

Tarde—that talented sociologist—proceeded from a preconceived notion of the significance of imitation in social life and exaggerated its role in his "laws of imitation," paying hardly any attention to individual initiative and social creativity. When he does mention them, he, like other authors, considerably limits their importance and plays down their role in social life. At the same time, when he inevitably comes across some phenomena of social life that have nothing to do with imitation, he is forced by virtue of the preconceived nature of his theory, to incorporate them into his theory by using an ingenious term, "counter-imitation."

On the other hand, Professor W. McDougall (*The Main Problems of Social Psychology*), while emphasizing the nature of instincts in individual life, transposes the entire concept of sociopsychological phenomena: "If it is true," he says, "that human nature everywhere has always had the same natural foundation, this point of view will serve as the much-needed basis for human societies and institutions."

He then discusses the issue of instincts, unresolved by psychologists, some of whom pay it too little attention, while others, for example, Schneider and William James, maintain that humans have the same number of instincts as animals and attribute to them a corresponding role in determining human behavior. The latter view is supported by McDougall without sufficient grounds. At the same time, he admits that instinct is "of hereditary and inborn psychophysical predisposition, which enables its owner to perceive certain objects, to pay attention to them, to experience particular sensory excitement when perceiving such an object and to perform the corresponding special actions or at least feel the urge to perform them." It is on these instincts that the author builds his social psychology, by analyzing in sequence the role of the instincts of procreation, parenting, belligerence, of the herd instinct, and the instincts through which religious conceptions influence social life, the

acquisitive instinct and of the creative instinct; as for the other factors that regulate interpersonal relations in social life, the author discusses imitation, play, and habit.

It is clear that, in developing his social psychology, W. McDougall was inevitably forced to stretch his concept of instincts out of all proportion. Suffice it to say that in addition to the above instincts, the author distinguishes, among the primary inclinations of the human soul that influence social life, the instincts of escape, repulsion, curiosity, self-effacement, self-confidence and a number of other instincts manifested as fear, submissiveness, etc. It is obvious that the recognition of all instincts is, in this case, derived from a preconceived social psychological theory of the "instinctual foundation" of all social phenomena. In a word, this is a case of the artificial construction of a particular system, since the author begins his study with a preconceived idea born of the subjective method instead of objectively analyzing phenomena in question and constructing his system on the basis of the facts thus revealed.

The application of a strictly objective method to the study of social phenomena avoids such strained interpretations that lead to wrong conclusions. Therefore, the same strictly objective viewpoint should also be applied to various manifestations of the collective, or composite, personality. Only this point of view can extricate this important realm of knowledge from its infant state.

As we shall see later, a strictly objective examination of the subject leads inevitably to the conclusion that manifestations of the composite personality revealed through the analysis of social life are ruled by the same law that is revealed through a strictly objective reflexological study of manifestations of the individual personality. The very forms of this law turn out to be the same for both the individual personality and the composite one; this should certainly be recognized as an advance in knowledge, since the theory of opposition between the personality and the crowd accepted by many authors who write about the crowd is greatly exaggerated.

As will be shown below, one encounters among the manifestations of social life essentially the same reflexes, in the form of universal, eternal movements following the same path and evolution, as one finds in individual activity. This is understandable if we bear in mind that a collective is nothing other than a group of individuals interconnected through their various

interests and representing a single entity in the form of a collective, or composite, personality.

That is why the laws that govern the manifestation of collective activity are the same as those that rule the expression of individual activity. It cannot be otherwise, for a collective that represents a composite personality acts as a united group of individuals.

It follows from the above that a reflexology of the individual personality should shed light on collective reflexology just as the latter should shed light on the former, since both sciences are closely interrelated.

In conclusion, I would like to mention that the present work was begun as long ago as 1910-11 when I mapped out its basic concepts. The first chapters appeared partially in 1911 in my work *The Subject Matter and the Goals of Social Psychology as an Objective Science*. That is why the text of this work with the appropriate corrections has been incorporated into the corresponding parts of this book.

V. Bekhterev

Notes

¹ E. de Roberti, *New Ideas in Sociology* (in Russian), 2nd Collection, p. 6.

² V. Bekhterev, *General Principles of Human Reflexology* (in Russian), [chapter 2](#).

³ E. Durkheim, "Sociology and Theory of Cognition" (in Russian), in *New Ideas in Sociology*, 2nd Collection, 1914, p. 40.

1

Introduction

Social events are molded by human hands, so whatever the other determining influences are, the human factor as the agent of social events should in no instance be ignored.

Here, however, the agent is not a single person but the whole society, a crowd, a gathering or a collective—which is not the same. Consequently, individual psychology is not suitable for understanding social movements and the evolution of social events, since an individual personality cannot personify a whole society or nation. These are the grounds for the attempts to create a special so-called "social psychology."

As early as the 1860s, Steinthal and Lazarus made such an attempt to create "folk psychology." They proceeded from the premise that the national spirit is distinct from the individual soul. In their view, the existence of folk art provided evidence, as it were, in favor of the existence of a unique, suprapersonal consciousness, or a composite national spirit. Therefore, the differences between the cultures of various peoples could be attributed to the peculiarities of their national, or folk, spirit.

But this attempt failed to strike a sympathetic chord, since one can speak of supra-individual consciousness or national spirit only as a figurative comparison and not as a reality.

The fact is that if we understand consciousness as the inner content of the self—which is universally accepted—it is then impossible to divide human consciousness into an individual and a public, or national, consciousness when the latter is understood as something special, as a single supra-individual consciousness, rather than a composite consciousness of the

individual persons who make up the population of a nation and its representatives. It is, therefore, natural that the attempt to build a social psychology on such a foundation could not succeed and that it was met with opposition from various quarters.

Further development of the same concept, but in a different way, occurred in W. Wundt's work on "folk psychology." The object of his research included language, myth, and tradition as products of the collective activity of the national mind, and his purpose was to study the psychological laws they reflected.

It is obvious that here we no longer deal with suprapersonal, or folk, consciousness but with the study of collective creativity, and in this direction W. Wundt undoubtedly made a big step forward, even though he failed to produce a social, or public, psychology in the full sense of the term.

According to W. Wundt,¹ the common products of creative activity are determined by the fact that "creativity of one person can be acknowledged by another as an adequate expression of the latter's concepts and feelings, which is why large numbers of different persons can be creators of the same concept to an unequal degree." The matter is thereby reduced to the following: in social organizations there are individual mental² processes which, inasmuch as they correspond to similar processes in other individuals, are the collective products of the mental activity of a certain number of people. But in such a case there can be no social psychology, since under these circumstances it will have no new goals except those in the province of individual psychology.

The Romanic nations have made a very considerable contribution to the study of the characteristic manifestations of the crowd; one may only mention the works of such authors as Tarde, LeBon, Sighele, and others.

In Russia, crowd psychology was studied by Mikhailovskii and subsequently by a host of other authors. The crowd, however, represents only one type of social form, and the most elementary one at that, with all the features of the herd. Thus, the crowd is only one of the objects of study pursued by a science focused on the social manifestations of human activity.

Issues of a social nature were also addressed in *Suggestion and its Role in Social Life*,³ a work by the present author first published in 1898. It discussed the significance of suggestion as an important factor in the manifestation of social processes. Since then and until very recently there have been no attempts by Russian scientists to comprehend so-called "psychic" manifestations of the masses in their totality except a few pamphlets by

various authors. Abroad, a number of works were published during that period by such authors as LeBon, Tarde, Baldwin, De la Grasserie, Ross, Wundt, and others. Their common shortcoming is the subjective method used in their approach to the issue.

Somewhat later, A. Kopel'man published a work⁴ in which he attempted to establish a general point of view on this subject. His analysis led to the total exclusion of folk spirit or supra-individual consciousness. According to the author, "social spiritual products are based exclusively on individual mental processes"; therefore the root of social phenomena is to be sought "nowhere else except in individual consciousness—in an exclusively individual one—which has no social aspects."

The author admits that, as creations of individual persons, social products share the nature of the individual products they contain, but he does not think it possible to use this fact as a proof of "the unity of consciousnesses of the producers, the people."

He allows for social unity or—in civilized nations—numerous individual unities under the general term "collective," construed by the author as "any group united through the ongoing process of establishing mental unity" (p. 37). It is this kind of collective that Kopel'man recognizes as the subject matter of collective psychology. Therefore, the latter, according to the author, examines not a particular mental process but its relationship with processes that occur in other members of the collective; it tries to define the essence of this relationship, the interaction between these individual processes as well as the regularity they manifest.

This point of view is undoubtedly more correct than the one postulated by other writers. Unfortunately, the author did not go beyond the general statement of the problem and was unable to renounce the subjective standpoint in this field.

Besides, the author proceeds to discuss yet again "the unity of individual consciousnesses" and the "soul of the collective" in addition to the individual soul (p. 42), the "fusion of minds" that "form a single collective soul," etc.

We believe that in societies and gatherings one can talk about the interaction or, to a certain degree, the standardization of the products of individual associative (neuro-psychological) activity, but not about their unity in the form of a "single collective soul," which is not very different from the views of Steinthal and Lazarus opposed by Kopel'man himself.

The following quotations seem to provide the most complete representation of Kopel'man's ideas (pp. 4 and 43): "We must conceive of an individual psyche as converging to a greater or lesser degree with the psyche of other individuals....The content of other consciousnesses is added to the individual mental content, to the material which is related to, and developed in, the individual brain."

Thus occurs to a greater or lesser extent the fusion of minds, all of which form a single collective soul. It is in this single soul, in this amalgamated though initially diverse mental content, that a number of processes occur which result in the erosion of diversity and its replacement by unity: it is here that united collective creativity takes place.

In contrast to the above author, we believe that it is more correct to say that in social groups we have gatherings of individuals, their interaction and equalization which lead to an amalgamation of their creative output, rather than a "single collective psyche" or a "single collective soul," which does not exist in reality. In a word, the apparatus of association—the brain and its *associative activity*⁵—of individual members of groups and gatherings generates, as a result of interpersonal contact, the products of social creativity without relinquishing its individuality or merging with others into a single mind or a "single soul."

Furthermore, we are decisively against any subjectivism in the assessment of social activity; this subjectivism is inevitable when "unity of consciousness" is believed to be the basis of a collective.

Although consciousness has its place in individual psychology, we are dealing here with studies conducted on oneself through self-observation and, unfortunately, transferred by analogy to other persons without sufficient justification; in our opinion, we cannot apply the method of self-observation based on the analogy with one's own consciousness to a multitude of individuals in social groups while accepting a hypothetical "unity of souls" or "unity of consciousnesses" and simultaneously presupposing the unity of their content, i.e., perceptions, conceptions, etc.

We believe that in all cases that involve the manifestation of mass correlative activity we should completely renounce the subjective point of view, since only in the figurative sense can we speak of a national soul, national feeling, or conception, while what we actually mean is the feelings or conceptions of many individuals, especially those from the ruling class of

the population, and not a monolithic feeling or conception shared by the majority.

It follows from the above that in "collective reflexology"—as we shall call this new discipline—we can speak of the manifestations of associative activity of a whole group of people and their external responses under certain conditions, but not of the subjective side of their psyches, which will remain outside our field of study. In other words, collective reflexology should be an exclusively objective science in the strict sense of the word and not a subjective one.

Any social activity, irrespective of the circumstances that generated it, has an element of the highest order, or, to be more precise, a reflexological element.

The role this element plays in combat and war is common knowledge. The experts are absolutely positive that success in combat depends on the "spirit" of the troops rather than on their physical might.

According to Meillard, it is not the supremacy of weapons of destruction that wins a war. There is a force as old as the world and yet eternally young, a force more terrible than cannons and guns, which is capable of all kinds of surprises, for it can instantly produce the most unexpected means for actions both extremely divers and perfectly suited to the situation. It is moral force. This force saturates, excites and inspires the masses, making them capable of enormous sacrifices for the sake of victory. Moral force tips the scales of victory in its favor. It was Napoleon who said that in a war the moral element is three times more important than the physical one, and he was right.

This view is shared by the military historian S. Gershel'man.⁶ According to him, the Napoleonic formula regarding the role of the moral element in a war has not changed at all; on the contrary, this element has acquired even greater significance.

The same opinion is held by Korf,⁷ Golovin⁸ and other authors. A detailed discussion of this issue can be found in Dr. Shumkov's work.⁹

We would like to remark here that the role of combatant enthusiasm, which often prevails over might, has been recognized since time immemorial. In this connection, we would like to quote the German authority Bernhardt, an advocate of a large army for Germany, who, according to Sheredam, said:

One should never forget that moral and intellectual factors always prevail over others and within some limits—rather unstable, incidentally—they are manifest far more strongly than the factor of sheer numbers. However, theory

and practice have demonstrated that in the face of overwhelming numerical superiority even a genius is helpless, since a mass—if it is sufficiently large and acts according to its nature—can totally wipe out any superiority of intellect and morals.

We should not overlook the fact that in the latter case the moral factor is also at work inspiring confidence in victory in one party while dispiriting the other by paralysing its activity.

The importance of the same moral element in work becomes apparent from the simple fact that the results of labor vary depending on whether it was conducted "under duress" or of one's own free will, as well as on the system of payment—daily wage, bonus distribution or piece-rate. Moreover, we all know how important are one's mood and interest in work.

One of the reasons for decreased productivity is lack of confidence in the success of an undertaking. In work more than in any other sphere we should take into consideration the reflexological elements related to the work collective.

The phenomenon of fashion is largely based on novelty and suggestion on the one hand, and on the satisfaction of vanity of certain groups of the population on the other.

In economic life, the law of supply and demand is not the only dominating force. The factor of confidence also plays a part in all instances, especially during times of economic upheaval, when confidence then acquires even greater importance than economic factors. Let us assume that a company has sustained losses in capital and property. If it retains its credibility, it will not be declared bankrupt, but will be provided with all possible cooperation and assistance towards finding its way out of the difficult situation and recovering from the shock it has suffered. Nor can commerce survive without this element of trust. Temptation and suggestion surely play a role in the traffic of commodities. And finally, the whole of the economy and the financial sphere cannot do without the above element. In the final analysis, the entire monetary system is based on what we call confidence. As soon as confidence in the stability of a financial system has been undermined, the value of its paper money plummets.

On the other hand, this is what we find in Tugan-Baranovskii's writings on money circulation.¹⁰ When there is a significant amount of money in circulation due to greater purchasing power and higher demand for commodities, it leads to price increases in general and, in particular those

goods in high demand. As a result of the over-accumulation of money in the banks, interest rates fall, which, in turn, injects new life into free enterprise and causes prices to rise, unevenly at first, although with time prices should regain their equilibrium.

There is, however, another factor in price increases caused by the augmentation of a country's money supply; this factor acts much faster and leads to the proportional rise of all commodity prices throughout the whole market.

This factor is of a totally psychological nature, in contrast to the first two objective factors. The point is that a commodity price results from conscious individual valuation, and the resulting monetary price represents two types of valuation: that of the acquired commodity and that of the money paid for it as the means of purchase. Under normal conditions of the financial market, money is not an object of independent valuation. Every purchaser estimates the relative importance for his needs of the commodities he can obtain in exchange for money, but the question of how much this money is worth does not trouble the purchaser; usually the value of money is subject least of all to fluctuation, which is why it serves as a measure of value of all commodities without being an object of measurement itself.

Similarly, when we want to describe the length of an object, we say that "it is two feet long" without stopping to think how long the foot itself is.

There are periods of economic life when money—usually a very stable measure of value of all commodities—loses its stability and starts changing in value.

Public consciousness, accustomed to proceed from the unchanging value of money as a universal measure of value of all commodities, finds it hard to renounce its traditional view.

Facts, however, are capable of destroying the most deeply rooted prejudices and of undermining the most persistent biases.

When prices of certain commodities begin to rise, public opinion normally attributes this growth to causes rooted in market conditions rather than in the realm of money.

But if the number of goods with rising prices keeps growing and the price increase becomes of a lasting and stable nature, extending to the entire market, public opinion capitulates before the facts, loses its former confidence in the unchanging value of money and slowly begins to measure the value of money itself.

"Is money falling in value?" the average person asks, and when the answer is positive he changes his own prices accordingly. If, in his opinion, the value of money has fallen to half of what it used to be, he will double the prices of all his commodities.

In this case, the universal and proportional rise of all prices is due to public consciousness rather than to the play of the unconscious natural forces of the conflict between demand and supply.

It should be noted that the laws of reflexology are manifest in economic life in the same way as in all other spheres of social activity. Thus, demand as well as supply often obey the law of imitation and both—due to the law of inertia—occur on a wider scale than circumstances call for. On the other hand, supply and demand are subject to the law of differentiation as well as to other laws, since initially, under the influence of need, demand acquires a more general form and only later is it established as a demand for certain objects. And since any action, everything else being equal, follows the line of least resistance, the public need in any country is initially satisfied by means of exchange for money, earned by habitual labor, and only with time is this need, as a goal, met by the mining and the manufacturing industries.

Münsterberg says that

political desires had to take into account the fact that the whole of economic life in the final analysis depends on economic man and that this economic person possesses not only a body but a soul as well. The work of his muscles is regulated by the volitional impulses of his consciousness, while the wealth produced and distributed by the economy must affect human consciousness through the senses. The pricing of manufactured goods or any other economic element—no matter which—are psychological factors demanding special attention. (*Psychology and Economic Life*, p. 22)

Whenever a buyer meets a seller or an employer meets a worker—in short, whenever a particular economic process takes place in the surrounding society, one person confronts another not as an object that requires interpretation but as a subject that should be understood rather than interpreted. It holds true for our everyday contact with other people, for every conversation and every direct interpersonal relationship. (Ibid. p. 24)

We know that needs generated by habit, i.e., through repeated excitation of the same combinations of reflexes, are conditioned by established taste, fashion and imitation; does this not eventually boil down to associative reflexes? And then, if this is how needs are determined, it is obvious that

supply should adapt to these needs. Thus, for instance, an American merchant shipbuilder told de Tocqueville that it made more sense for him to build flimsy boats, since clients' tastes in this matter were subject to change. Similarly, when tradition has established a particular type of dress for a small community occupying a relatively small area, it is obvious that industrial production of this kind of clothing would prove unfeasible. Here again tradition—that stable collective reflex—regulates economic life.

Generally, when prices are established for the first time, the dominating tradition plays an important role since it determines the need. On the other hand, according to all economists, supply and demand are the major factors in setting the prices. But as we know, both demand and supply can and should be regarded as collective reflexes.

Incidentally, one should point out that economics as a science has until now paid little attention to the role of the factors that underlie the circumstances of correlative activity. Thus, as we have already mentioned, market prices are not always determined by supply and demand or, at least, not exclusively by them, for much depends on so-called public "taste" and "fashion" based on the conditions of associative activity, and lastly on suggestion and the seller's ability to present his wares. Watch an expert merchant: how he praises his goods and thrusts them on the customer. He does it mostly through suggestion and persuasion, by convincing the customer that his merchandise is of high quality. It all depends on the personal characteristics of the merchant, i.e., on his skill and shrewdness as well as on the level of his associative activity in general. Therein lies the explanation of why a commodity of the same quality can fetch different prices when sold by different merchants.

Even advertising cannot do without the help of this element and, as we know, experiments carried out by the American psychologist Munsterberg on the influence of certain forms of advertising on the public have provided some interesting results, which would be inexpedient to discuss here. The reflexological method of strictly objective study could yield considerable results here, if this line of research were seen as worth pursuing.

It goes without saying that the role of the moral element in stocks and currency speculation is well known and there is no need to elaborate on the subject. It is enough to recall notorious stock exchange panics which are typical of the periods of market fluctuations.

Another example from the economy and the financial sphere is lottery bonds. Their attraction—as well as that of all lotteries in general—is common knowledge, although everyone recognizes that as an investment they are not profitable.

Everybody knows how the revolutionary process in Russia affected the country's finances. Having restored the rights of the oppressed classes, the revolution had to substantiate them accordingly in material terms, i.e., in terms of their wages. However, the actualization of the rights of the working population to live better was not accompanied by an increase in productivity, which, on the contrary, sharply declined during the initial stages of the revolution, leading to an extensive disruption of the country's finances. How could the state cover the unprecedented growth of expenses due to the extraordinary rises in wages? Obviously only by printing more paper money which, in turn, undermined its credibility and brought about its rapid and relentless depreciation.

Naturally, the moral element also plays an enormous role in the realm of punishment and reward. The practice of capital punishment and severe repression that was accepted as a necessity in former times when society was less organized, crime prevention less feasible and mores rather crude, has been gradually disappearing from the criminal codes of various countries.

In wartime conditions, however, the introduction of capital punishment is recognized to be unavoidable as a preventive measure against desertion and other serious crimes among the troops engaged in front-line operations.

There is hardly any need to discuss here the moral significance of rewards.

It is clear from all the above how important the moral element is in social life.

It cannot be otherwise. The life of society is nothing more than the life of a gathering or a collective of individual persons. And while individual associative activity is the principal, though not single determinant of individual behavior, it is evident that this element must also play a leading role in the life of society or a collective.

Unfortunately, the study of society from the viewpoint of collective reflexology has not yet commenced. Although there have been some very impressive treatises on social psychology, they deal predominantly with various aspects of social life and, being permeated with subjectivism, do not cover the whole discipline. That is why we have every right to claim that despite a number of works on various social psychological issues, collective

reflexology as an objective science in our understanding of the term does not yet exist.

The question is, though, how this situation can be explained, particularly in the context of the enormous and detailed research in the field of individual psychology. It would seem that if certain advances have been made in individual psychology they should merely be transferred to the composite personality and this should result, it would seem, in the development of a corresponding social science that would follow in the footsteps of individual psychology. In reality, however, we observe something quite different, because the psychological method, as the purely subjective method of self-observation still used by individual psychology, is totally unsuitable for studying a composite personality.

Although we do speak of public consciousness, public emotion, public thought and public will, self-observation, which underlies these concepts related to the individual personality, is not at all applicable to the composite personality or a collective.

Regarding the individual personality, too, I have proved in my *General Principles of Human Reflexology* the necessity of studying associative activity, understood as the sum total of the higher reflexes, in a strictly objective manner. One should admit that, if the objective study of individual associative activity provides the foundation for a special scientific discipline which I call reflexology, it is even more essential for the study of the higher functions of the composite personality or the collective, where self-observation is totally inapplicable; thus, there can be no subjective social psychology reminiscent of subjective individual psychology.

Evidently, the exclusively objective method which we used as the basis of reflexology can and should be applied to the study of the higher functions of the composite personality. Therefore, to treat the manifestations of the composite personality, or collective, in a strictly objective way we can speak only of its correlative activity, while its manifestations in a collective, via analogy with the manifestations of the individual personality, can be designated by higher, or associative, reflexes. For this reason, we shall name the science which is the object of our present investigation *collective, or social, reflexology*.

We shall see later that by following this strictly objective path we can arrive at many of the conclusions that have already been reached in the field

of individual reflexology and which make this discipline a well-proportioned scientific structure.

While reflexology studies the manifestations of individual correlative activity, collective reflexology studies the same activity of the collective. It reveals, therefore, how this activity is manifested in the case of social affiliation between several (two, three or more) individuals and to what extent the characteristics of this correlative activity depend on the nature of the social bond established between individual persons.

It is thus apparent that collective reflexology must also—and most importantly—elucidate the mechanism of interpersonal relations in the collective.

Relations between individual members of a collective, whatever their outcome, first have to be established. This process is always based on the principle of associative reflex formation. Every individual act that satisfies one's own (as well as another person's) interests produces a reciprocal reflex of the same kind or, at least, a reflex with analogous results, which stimulates a corresponding reaction and, when repeated, leads to the establishment of relations based on reciprocity. In contrast, every individual act adversely affecting one's neighbor provokes a reflex of defensive nature on his part and results in the establishment of interrelations based on confrontation and revenge, which are subsequently even incorporated in the written law: "an eye for an eye, a tooth for a tooth." But, of course, prior to all the written laws, an unwritten law operates—the law of tradition instituted by life itself or by collective experience. It is this existential experience which, in one case, leads to mutually favorable relations between the cohabitants, while, in another, to mutually unfavorable or hostile relations. In the long run, these relationships become stable due to the repetition of the acts themselves, which become part of a habit and, later, of a tradition handed down from generation to generation.

Since the establishment of favorable interrelations finds more congenial circumstances within one's family and among relatives, it is only natural that, already in the prehistoric period, tribal groups were formed on the basis of cooperation between their members, whereas the establishment of hostile interrelations occurred naturally between different tribal groups. Hence the continuous warfare between various tribal groups attacking one another under conditions of prehistoric culture.

This is what happened with the establishment of collectives during the initial prehistoric stage of human existence. Mutual action coordination by individual members of a collective becomes even more reinforced due to imitation and reflex contagion, on the one hand, and to the necessity of collective defense and attack to secure the existence of the collective itself, on the other.

It goes without saying that the need for collective defense and collective attack, representing nothing other than collective defensive and offensive reflexes, excites common interests to such a degree that it becomes the connecting link between the members of a collective; but the same conditions, by provoking hostility between various groups and collectives, lead to the estrangement of these collectives.

Besides collective reflexes of a defensive and an offensive nature, there are other forms of collective reflexes, such as collective observation, collective instincts, collective emotions or mimico-so-matic reflexes, collective concentration, etc., which make up the object of investigation of collective reflexology.

Abundant material for the latter is provided by those periods in national history which, accompanied by tumultuous social movements, lead to a coup d'état. During a revolution, a country experiences in a month what, under other circumstances, spans centuries of historical experience. In such historical periods, all the manifestations of associative activity of a society appear to be more distinct and prominent, supplying thereby an abundant source of material for collective reflexology.

For this reason, much of the data used in this book as the basis for my postulations have been drawn from the Russian Revolution, which I, together with all Russian citizens, had to live through both in 1905 and from 1917 onward.

In conclusion, let us point out that social phenomena, being highly complex, cannot be explained by only one regularity or one dependence, as the well-known sociologist Tarde believed when postulating the law of imitation. We deal here with undoubtedly more complex dependences that can be revealed only by pursuing analysis of social phenomena in various directions.

Since the analysis of a physical phenomenon such as a whirlwind shows that the movement of soil and air particles is dominated by a whole variety of dependences that characterize a certain phenom-enological regularity, such as

the law of action and equal reaction, the law of attraction and repulsion, the law of inertia, the law of rhythm, etc., it is obvious that such complex phenomena as social movements must also be a result of a large number of dependences that should be elucidated through appropriate analysis.

The opportunity to reveal the regular development of individual expression and actions presented itself when we started investigating human behavior not from the standpoint of subjective causality, but from a strictly objective point of view, recognizing the need to regard all actions of the human personality as manifestations of higher, or associative, reflexes.¹¹ The latter are based on energy, and whether one calls it neurological or neuropsychological energy does not really matter, for it is similar in its nature to other kinds of energy. It is thus evident that there must also be a certain developmental regularity in the manifestations of popular masses who represent a composite personality.

In this respect, observation and experience lead us to infer that the basic laws of correlative activity of the composite personality are the same as those operating throughout the whole of the animate and inanimate world. Here we disclose through analysis the same universal cosmic laws, such as the law of energy conservation, of gravity, of repulsion, the law of action and equal reaction, of similarity, rhythm, entropy, evolution, differentiation, integration or synthesis, the law of adaptation, of selection, inertia, etc.

We shall see that the world is ruled by the same basic laws common to all phenomena, both inorganic and organic as well as supraorganic, or social, which will be discussed in more detail below.

It cannot be otherwise. If the world of animate nature is the result of the transformation and complication of inanimate nature through evolution—and the natural sciences give us no grounds whatsoever to doubt it—then why should there be one set of laws for the inorganic world, another for the organic world, and a third one for the supraorganic world?

If this, indeed, were the case, we would have to accept the existence of two or even three worlds that have nothing in common with one another, or to admit that some miracle divided our planet Earth and the universe into two or three incommensurable parts, and this would be an utter absurdity.

Notes

1. W. Wundt, *Völkerpsychologie*, part 2, p. 593.

2. Translator's note: we henceforth use the word "mental" where the Russian *psikhicheskii* might strictly be translated as "psychical," because of the infrequent present use of the latter in English.
3. The 3rd edition was published in 1908.
4. *What should be the function of collective psychology? Introduction to Works on Collective Psychology* (in Russian), Odessa, 1908.
5. The objective term I coined in reflexology instead of the subjective concept "mental activity."
6. S. Gershel'man, *The use of the moral element by Suvorov* (in Russian), 1900.
7. P. A. Korf, *On cultivating will power in military leaders* (in Russian), 1906.
8. K. N. Golovin, *Study of combat* (in Russian), St. Petersburg, 1907.
9. Dr. G. E. Shumko, *Soldiers' mentality in combat* (in Russian), Issue 1, St. Petersburg.
0. His book, *Paper money and metal*, and his article, "Should we be concerned about the increased amount of paper money?"
1. See V. Bekhterev, *General Principles of Human Reflexology*, Petrograd.

2

Definition of Collective Reflexology

The above interpretation of collective reflexology cannot in any sense be identified with Wundt's folk psychology. Collective reflexology dissociates itself from the latter because its subject matter includes not only myths, traditions and language but all manifestations of collective correlative activity in general—and not only of such large tribal social groups as separate nations but also of smaller social groups within a nation—as well as manifestations of associative activity of groups formed by nations for the purpose of carrying out joint international activity.

There is no doubt that myths, traditions and language reflect to a significant degree national characteristics and therefore provide certain material for the characterology of a nation. But valuable as this material may be, it can never cover the whole subject. The reason is that other manifestations of a nation's correlative activity such as science, art, religion, literature, etc., being original products of this nation's activity, can also contribute to national characterology. Although Wundt attributes these manifestations of creative activity to the products of individual creativity in a given nation and sets them in opposition, as it were, to such impersonal (in his view) products of social creativity as myths, traditions and language, there is no doubt that these products of national creative activity—inasmuch as they reflect manifestations of the collective activity of the representatives of a given nation, their views and opinions—cannot be excluded from national characterology.

In any case, although the study of myths, traditions and language is particularly important for a nation's characterology, it is far from being the

sole or even the principal source of research in collective activity. Besides, the aim of collective reflexology is not so much the investigation of tribal peculiarities of associative activity developed under the influence of certain historical events, as well as the climatic and geographic conditions of a given nation, but rather the definition of societal moods, creativity and actions. Therefore, in collective reflexology, as we understand it, findings obtained through the study of myths, language and traditions can occupy only a certain place but not an exclusive or even a leading position.

First of all, myths and traditions, hailing from the days of yore and being relics of the remote past, cannot characterize modern society. By the same token, language, developed over the centuries, does not by itself provide all the material for the reflexological investigation of contemporary social groups. In this case, the content and the outward form of the declarations made by a particular national assembly or representative body are just as important for reflexological description of this nation as are, the various aspects of its lifestyle and manifestations of popular movements.

Collective reflexology—as suggested by its very name—naturally impinges upon the general reflexology of the individual personality but differs from the latter in that it studies associative manifestations of social groups rather than of individual persons.

In this sense, there is a certain correlation between individual reflexology and collective reflexology. The former tries to discover the unique features of the individual personality, to identify the differences in the individual make-up of various persons and define the reflexological basis of these differences. The aim of collective reflexology is to study mass, or collective, manifestations of correlative activity and find out how its social results are achieved through interpersonal relations within social groups and through the smoothing out of individual differences.

There can be no doubt that manifestations of associative activity in social groups possess certain characteristics, derived from the fact that this activity is manifested not in individuals but in groups and gatherings welded together through specific interests, and therefore acting in cooperation like a single composite entity.

One can state with confidence that members of a gathering frequently display distinctive aspects of their associative activity which are not commonly revealed in their individual life.

For instance, when in a crowd or a gathering, a person is often transformed to such a degree that he no longer resembles his own self, that is, the kind of person he is when left to his own devices.

It also happens that a quiet and even somewhat apathetic person gets so aroused by the speeches made in a crowd or a gathering that he is soon infected by the general mood and falls into a fervor the like of which he would never express under conditions of his life as an individual.

Due to a certain elevation in mood that finds favorable conditions in a crowd, such persons by their passionate speeches sometimes sway a mass of other people towards extreme decisions, which are, however, incompatible with their quiet mode of action and demeanor in their individual life, where they are usually cautious and careful.

It is clear from all of the above why collective reflexology, which studies specific manifestations of mass associative activity, should be separated from general reflexology with its focus on individual associative activity.

Collective reflexology impinges upon sociology, for it investigates the foundations of joint activity in social groups. In this regard, we must say that collective reflexology functions as a link connecting two sciences—reflexology and sociology.

According to S. Sighele (77ie *Criminal Crowd*), collective psychology has a distinct field of research and its development follows paths never pursued by sociology: it takes over the territory vacated by sociology and legally comes into power where the reign of sociology has ended. The more transitory, incidental and unorganized the association of entities is, the further it is removed from Spencerian laws and drawn into the sphere of research pursued by collective psychology.

Therefore, the crowd, the jury, assemblies, the theatre audience, being more or less accidental gatherings, should be removed "from the jurisdiction of sociological laws and ruled by the laws of collective psychology."

That is why the crowd, as the most heterogeneous gathering, is the favorite topic of authors dedicated to the study of the laws of collective psychology, including Sighele himself.

Approximately the same definition of collective psychology is offered by De la Grasserie.¹ According to him, collective psychology should be concerned with more or less accidental and unorganized aggregates, which display neither differentiation, nor definite coordination, nor permanent relationships. Organized aggregates are the object of sociological study.

Rossi's² views, too, are close to this definition, as he assigns accidental aggregates like the crowd to collective psychology and the national or racial soul to social psychology (which is the same as folk psychology or characterology), while the initially unconscious automatic interaction that gradually becomes conscious is assigned to sociology.

These definitions, however, commit a sin primarily against sociology, which does not let the degree of organization impose any restrictions on its study of human interaction. They commit no lesser sin against collective psychology, or reflexology (to use our terminology), by relegating it to an obscure corner of human existence associated with accidental and unorganized gatherings.

If collective reflexology, as we understand it, represents the body of knowledge concerned with the manifestations of activity in integrated social groups or collectives from the objective point of view, what is then left for sociology? There is no doubt that these two scientific disciplines have not only numerous points of contact but are also closely interrelated. While one of them studies the emergence, development, and establishment of collective manifestations or behavior of social groups, the other—sociology—investigates the structure and the life of the composite personality itself, as well as interaction and correlation between social forces, primarily in connection with various conditions: racial, economic, political, geographic, climatic, and others. Obviously these factors affect collective manifestations of the composite personality, and their dominating influence on the structure and the life of the composite personality itself as well as on the balance of social forces is beyond any doubt. It is all this that makes up the province of sociology as the science of the structure and life of society under given conditions, just as the subject matter of the historical sciences is the study of the life of society in time, i.e., during certain periods of the evolution of civilization and culture.

From our viewpoint, sociology investigates the institution of social life itself, as well as the causes and effects of social phenomena and interpersonal relations, without attempting to penetrate the very mechanism of social developments, which belongs to the realm of collective reflexology. Sociology studies social institutions, facts, and phenomena, as well as their interrelations, applying wherever necessary a psychological, i.e., reflexological, interpretation. However, the actual investigation of the manifestations of mass associative activity is not one of its immediate goals.

It is merely an element of the interpretation of certain social phenomena, which incidentally points to the special significance of collective reflexology for sociology.

One of the essential tasks of sociology is undoubtedly the study of cooperation and class struggle. The latter gradually leads the impoverished classes to gain rights and, subsequently, power while weakening the social role played by the previously dominating classes. Whereas cooperation and agreement normally result in making the cooperating members stronger, confrontation is associated with the mutual weakening of both sides of the conflict, so that in some cases power is seized by a third element—such as an aggressive outsider. This latter takes advantage of the weakening of a given social organism and easily acquires power over it in his own interests. This is what happened to some republican city states of the Appenine Peninsula (e.g., Florence) and more recently to Poland. All these factors provide material for sociology, just as the struggle for survival provides material for biology. But the elucidation of the very process of struggle, its causes and development in certain directions, is the subject matter of collective reflexology. The emergence of social reflexes and their stimuli as well as their direction and expression are the indisputable subject matter of social, or collective, reflexology, while the results of this struggle and their manifestations in the life of society belong to the province of sociology. A country's socioeconomic conditions that lead to class struggle are also part of sociological research. It is clear that both scientific disciplines merge together to a considerable degree, but the objective of one of them is to study the formation of collective reflexes, the causes, and the very process of development of these reflexes, whereas the task of the other is to take into account the consequences and circumstances of the balance of power between social forces, their confrontations, struggles and reconciliations.

It follows from the above that collective reflexology should be separated from sociology, since the latter studies social life as it evolves under the influence of climatic, geographic, racial, legal, and economic conditions, as well as the interplay of interests pursued by smaller collectives (e.g., the struggle between parties). Although collective reflexology provides rich material for sociology and, consequently, the two scientific disciplines have numerous points of contact, there are no sufficient grounds to expect the convergence of the two branches of knowledge as some sociologists do, particularly those of neopositivist persuasion.

On the other hand, if we accept the popular view that sociology studies aspects of human interaction, then collective reflexology as a scientific discipline studies attitudes of human masses joined in collectives not only to other human collectives or individual persons but also to animals and the environment in general. Besides, there is another distinction to be made between sociology and collective reflexology. Since the main task of collective reflexology is to study the mechanism of collective reflex formation and behavior, we are apparently dealing with the activity of social groups possessing a certain degree of internal cohesion, since a collective cannot exist without unity. Sociology, however, cannot confine itself to the study of only those social phenomena where one or another kind of cohesiveness is established. On the contrary, its area of investigation encompasses the phenomena where one finds disunity, struggle, or open warfare, even when only two persons are involved in it. Sociology is engaged in the issues of social interrelations irrespective of whether the latter are established between united groups or between individuals. And those sociologists who classify as social phenomena only interrelations between like individuals united in a common action (Guildings, Baldwin, and others) are definitely wrong, while authors who believe that sociology should deal with the entire hierarchy of social phenomena, irrespective of the content of interpersonal or group relationships, are right. Where there is disunity and hostility there is, certainly, interrelationship between social groups or individual persons, but there is no collective.

While establishing connections between various social phenomena, sociology by itself cannot in many cases provide an appropriate explanation without turning to collective reflexology, since sometimes only the latter can supply a more precise definition of inner relationships between different social phenomena.

At the same time, collective reflexology does not regard the study of social facts and phenomena as its immediate goal; it studies only the reflexological mechanism of the evolution of social phenomena, their reflexological preparation and the conditions under which these phenomena develop in a number of individuals in the form of collective reflexes.

The difference between collective reflexology and sociology can best be described in the following way. It is not essential for a sociologist to know what changes take place in the activity and reactions of a collective as compared to individual reactions and activity. He can certainly be interested

in it, but it is not his immediate goal, for he naturally accepts as his task the elucidation of interrelations between social groups as well as the very fact of the existence of social groups or collectives. Thus, the study of the ways in which collective groups emerge and of the specific manifestations of collective activity as compared to those of the individual variety is the task of collective reflexology, whereas the number of collectives, their characteristics and the interrelations within a particular nation should be investigated by sociology.

Clearly, the method or mechanism of the emergence of collectives belongs exclusively to the sphere of collective reflexology, since one cannot study the specific manifestations of collective activity without first revealing how a collective comes into being. It is precisely this field that Tarde called "intermental psychology" and proposed it as the basis of sociology. "I mean," says Tarde,

not conventional psychology, which one can treat with the contempt it deserves [*ski*], but what I call in the subsequent discussion "reciprocal psychology" (Interpsychologie)...Interpsychology or interlogic are essentially elementary sociology, which is the only discipline capable of interpreting complex sociology or sociology in the narrow sense of the term."³

It is clear from the above that formation of collectives as the subject matter of "intermental psychology," or, to be more precise, "collective reflexology," simultaneously serves as the basis of sociology-

What we have said so far allows us to outline the objectives of collective reflexology. They include on the one hand the study of the mechanism governing the formation of a collective, and, on the other, the study of manifestations of collective reflexes, which in their totality make up collective activity, as compared to individual reflexes and individual activity.

In a word, sociology deals with social facts and phenomena and their interaction; it interprets them from the reflexological or any other point of view. Collective reflexology deals with the mechanism of integration or socialization of individual reflexological phenomena; it explains how this process generates collective reflexes and prepares the ground for a particular social phenomenon; it also investigates manifestations of social group behavior.

One should bear in mind that social phenomena that are the subject matter of sociology cannot be reduced to reflexologically explicable phenomena only, since as we have already mentioned, the evolution of social phenomena

is affected by a considerable number of factors (economic, legal, geographic, climatic, etc.), whereas collective reflexology informs us only about the reflexological mechanism of social phenomena.

We would also like to mention here that there are opinions according to which collective psychology and social psychology, or in our terminology collective and social reflexology, are not identical notions. Thus, Sighele (*The Criminal Crowd*) and De la Grasserie (*De la psycho-sociologie, Revue de sociologie*, 1912, No. 3, p. 163 and ff.) understand the former as a science that investigates phenomena of human interaction when its elements are "heterogeneous," "random," and form a "weak conscious" relationship, whereas social psychology studies organized aggregates.

This view is shared by Rossi (*Psichologia collectiva*, Milano, 1900. *Sociologia e psichologia collectiva*). We do not attach any particular importance to this fine point, for we believe that a collective is a collective both in the case of the crowd and in the case of an organized association, such as a scientific or a commercial society, a cooperative, a nation, a state, etc. For some authors, Le Bon,⁴ for instance, social psychology simply means racial psychology. From our point of view it would be more appropriate to call this subject of investigation "collective characterology" or social group characterology, since the subject matter of this branch of knowledge is the same as that covered by individual characterology but is concerned with collectives rather than individuals.

We shall note in conclusion that, depending on their school of thought, some authors incorporate the sphere of collective reflexology into sociology, while others recognize sociology as the apex of the above fields of knowledge. Thus, Rossi defines the subject matter of sociology as "initially unconscious-automatic social interaction, which becomes more and more conscious," whereas Sorokin (*The System of Sociology*, part 1, p. 20) regards the same subject as only a part of sociology or allows for the possibility of merging the two branches of knowledge into one. A considerable number of other authors also hold diverging views on this issue. Their position obviously depends on whether their interpretation of sociology as a science concerned with the structure of society and the interaction between its members is more exclusive or more inclusive:

Whatever the case might be, sociology cannot do without collective reflexology. Attempts to found sociology directly on biology, while bypassing psychology—i.e., to interpret society as a kind of organism

possessing a structure similar to that of an individual organism or to regard it as a collective in which everything is reduced to the biological laws of the species—were the results of that epoch in the history of natural sciences when biology enjoyed the dominant position and set the tone for all the sciences with the focus on human beings. Only psychology particularly in its intermental aspect, which represents a transition from psychology to sociology, if not the first part of the latter, can provide the true foundations of sociology. The social sphere belongs to the realm of mental phenomena, even though they serve as the basis for certain objective forms of social structure, which give us the right to view society not only as a simple collective, culturally united in a particular way, but also as a supra-individual entity possessing a certain structure.

The above quotation is taken from the recently published book by N. Kareev, *General Laws of Sociology* (Petrograd, 1919, pp. 1112). In a similar manner, N. Kareev also dismisses the foundation of sociology exclusively on the tenets of economic materialism based on the economic concept of relations of production with the "psychic" element eliminated. "Even if this theory were true," says the author,

relations of production, which are essentially nothing other than certain interrelations between individual persons in the sphere of material relations, become possible only due to the ability of individual members of a social collective to engage in mental interaction. It is rather questionable (to put it mildly) whether one can reduce the entire social structure and spiritual culture of society to the status of mere superstructures over the purely economic (material) basis.

As we know, at the end of his *System of Logic*, J. S. Mill had already expressed a view of sociology as applied psychology. The psychology in question was not social or collective—it was a psychology concerned with the inner world of individual persons. Nowadays, however, when sociology is—or at least should be—a strictly objective science, it is hard to see a relationship between it and individual psychology. Collective reflexology, which investigates in a strictly objective manner the formation of particular social groups and collectives, is quite a different matter.

Although the biological view of society as an organism, supported by Comte and Spencer and subsequently by René Worms, Novikov, and others, has by now outlived itself, and although it becomes increasingly clear that interpretation of the data supplied by linguistics, mythology, and political

economy is possible only through psychology, nevertheless, social, or collective, psychology is not sociology in the narrow sense of the word, even though it does explain many social phenomena. As a result, the psychological trend supported by Tarde, Mikhailovskii, Simmel, Delovt, and others came into being.

Obviously, just as there exist such schools of thought as the historical-philosophical, founded by Condorcet and developed by Comte with his social statics and dynamics; the historical-economic one founded by Marx and Engels; the evolutionist trend started by Spencer and developed by a number of authors (von Lilienfeld, Espinas, Letourneau, Vormeni, Sherle, and others); the statistical-sociological one founded by Quetelet; the social-biological direction of Benjamin Kidd and the ethnogeographic one of Montesqueue and Bockle, so there may, and does indeed, exist a psychological trend in sociology. But to say that the whole of social life can be reduced to the psychological, i.e., reflexological, interrelations between social groups means to negate all economic, marital, ethnographic and geographic correlations between individual persons and groups, when it is precisely these correlations which primarily and above all underlie the class struggle and cooperation between individual persons and parties.

E. de Roberti, however, proposes the biosocial principle as the basis of sociology as a science of human social behavior. When approached from this direction, sociology undoubtedly absorbs to a significant extent social, or collective, reflexology, but the biosocial principle cannot eliminate either economic or ethnogeographic aspects of human interactions and consequently is applicable only to some social phenomena, without penetrating them all.

Thus, such phenomena as communication as the basis of social life, the relationship between personality and society, as well as imitation and repetition, which underlie the foundation of social life, are all psychological factors, whereas everything related to the "self-sufficiency" of social groups is not of a psychological—let alone an exclusively psychological—nature.

On the other hand, a social contact established through cooperation and division of labor represents a phenomenon of an economic rather than a psychological nature. No matter how we assess the importance of these phenomena or, in other words, irrespective of whether we attach the primary significance to division of labor, according to Comte or Durkheim, or to cooperation, according to Spencer, or recognize that both are equally

important—which is a better approximation to reality—one thing is certain: neither of the two can be explained exclusively in terms of communication as a psychological factor. In any case, although stratification of society into groups and classes is related in some degree to the purely psychological characteristics of their members, it is a result of the division of labor, which is a phenomenon of an economic nature. And, finally, social strata, which play such a major role in social life, are not purely social-psychological groups, but they often share common economic interests.

"It is absolutely useless to argue," says K.M. Takhtarev, which aspect of social life—whether economic, sexual or mental—should be regarded as primary. It is equally fruitless to try to establish which of the basic human needs or forms of communication were the first to appear. At the earliest stages of human evolution accessible to the modern scientific inquiry, we find man as a social being, capable of feeling, thinking and acting; involved from the very start in the economic, sexual and mental life in its moral, intellectual and sensual aspects, all of which should be regarded equally fundamental as are the corresponding human needs.

Saying subsequently that equal primacy of different social phenomena does not imply their equivalence, the author remarks,

Economic life has thus far almost always and everywhere prevailed over mental and even sexual life. One can observe this not only among the peoples at the earliest stages of evolution but also among the more advanced nations, not only among the uneducated and most numerous classes of the population, but also among the rich and the educated in whom the craving for the material comforts of life replaces the motivation to provide the most basic means of subsistence for the have-nots. Even among the rich, sexual life is much more readily sacrificed to material gains than vice versa, and the same is true for intellectual life. The energy to pursue it becomes available only after all other needs deemed more essential have been satisfied.

Satisfaction of all needs in general, and therefore of the economic as well as the sexual one, is mediated through associative activity. In both cases the primary impulse is provided by organic needs, which underlie the corresponding instincts, or the hereditary-organic reflexes. As a result, the maturation and the satisfaction of these needs prevail over the satisfaction of the demands of purely correlative activity.

Thus we cannot but agree that none of the existing sociological schools of thought embraces the whole field of sociology as a science of social life, its

phenomena and laws. None of the sociological schools contains the whole truth, i.e., the truth in all its totality, but on the other hand, none of them—no matter how one-sided—is completely devoid of truth. Each sociological school contains a certain degree of truth and in this respect it complements the whole.⁵

Proceeding from the above, we cannot identify sociology with social, or collective, reflexology; we admit that the latter is one of the cornerstones of sociology, but not the whole of it.

There are quite a few points of contact between collective reflexology and history, for history—if we ignore the issue of influence exercised by individual persons on social events—is for the most part a history of collective human deeds. And if the study of history were limited to gathering only this kind of material, it would become a history of collective reflexes and would, or should, consequently represent social, or collective, reflexology in a historical light.

And, indeed, if we observe the history of nations not from the usual point of view in terms of the relationship between persons and events, but in terms of the movements occurring within a national collective, we shall encounter sporadic waves of revival or stimulation of certain collective reflexes generated by particular external influences and conditions. They spread throughout the collective—which often transcends the limits of a nation or a state—and are accompanied by a number of concomitant movements and reactions, which may be in harmony with the first ones, yet intersect with other movements and reactions which are in conflict with the first and inhibit their proliferation. These inhibiting movements and reactions, in turn, spread throughout the collective like rays and generate new movements and reactions. Although the latter are in conflict with the first ones, they strike sympathetic chords in other spheres and spark off new movements and reactions, for which the first ones become the inhibiting factors. In other words, in the historical perspective of social events we encounter mutually conflicting collective reflexes. While generating congruent collective reflexes they also exert a mutually inhibiting influence. Thus, the onward march of history is determined by which of the two reflexes achieves final supremacy and the most extensive proliferation.

What then is collective reflexology from a historical perspective? The difference between it and proper history is the same as between social reflexology and sociology, for history must take into consideration specific

characteristics of climate, geography, race, economy and other influential factors determining the historical existence of a given nation, all of which transcends the goals of collective reflexology.

But there is also a prehistoric period of human activity studied by archeology through objects and relics preserved in the ground for many centuries. Archeology does not limit its field to the prehistoric period exclusively, but also gathers material related to the early epochs of human history, supplementing thereby chronicles and other sources.

The specific nature of archeological research never focuses on the actions and the reactions of ancient people but on the results of those actions and reactions. Since the results enable us to ascertain the nature of activity, living conditions, economic life, etc., archeology makes it possible to recreate the collective reflexology of both the ancient and the prehistoric world. When archeology recreates the activity of individual persons and whole societies, it provides objective material for assessing individual or collective reflexology in prehistoric or ancient times.

One should bear in mind that archeology is concerned with various aspects of the ancient human collective, which yet again transcends the limits of prehistoric collective reflexology, for nowadays we can speak not only of the archeology of language, religions, mythologies and art but also of industrial, political, and legal archeology. What is of particular value here is that archeology enables us to trace the evolution of collective human reactions. Comparative linguistics, which describes and interprets the origins and destiny of each root, is nothing other than an elucidation of the evolution of language— this most essential collective symbolic reaction. By the same token, a comparative mythology concerned with the origins, evolution and destiny of each myth is nothing other than an elucidation of the collective reaction of prehistoric and ancient people to the manifestation of the mysterious forces of nature. The same refers to the issues of behavior, politics, exchange, law, and creative activity construed with the help of archeology. In short, the key to the evolution of collective reflexes in the human world of prehistory is to be found in archeological investigations, just as the key to the biological and anatomical evolution of human organs is to be found in paleontology.

In this respect, it is of interest to compare the evolution of children's language as it gradually transforms into adult language, the initial development of children's drawings, which lay the foundation of adult

creativity, and the first interpersonal relations between children, which later become the legal relations of the adult world, to what archeology reveals about the life and activity of prehistoric man. These comparisons convince us that individual evolution of human reflexes repeats, as it were, the evolution of reflexes in prehistoric man.

Notes

1. Delagrasserie, "Delapsychosociologie," 1912, No. 3.
2. P. Rossi, *Psichologia collectiva*, Milano, 1900, p. 213. See also *Sociologia e psychologia collectiva*.
3. G. Tarde, "Psychology and Sociology" (in Russian), in *New Ideas in Sociology*, Collection No. 2, St. Petersburg, 1914, p. 70.
4. *Psychological laws of the evolution of nations* (in Russian), 1906. See "The racial soul."
5. Takhtarev, *Sociology and its concise history* (in Russian), p. 13.

3

The Subject Matter and Method of Collective Reflexology

It is obvious from the above that the subject matter of collective reflexology is the study of the emergence, evolution, and activity of gatherings which, due to the interpersonal communication between individual members, manifest their *communal* associative activity as a single entity, due to the interpersonal communication between individual members.

It is absolutely immaterial whether our object of observation is a crowd gathered by chance in a street, attracted by some incident, or a public meeting focused on an event that has disturbed society, or a public or even a governmental assembly for a particular purpose-everywhere we shall encounter manifestations of public mood, communal intellectual effort and collective actions of many individuals interrelated through particular interests. Therefore, all these manifestations should be investigated by collective reflexology.

Thus, the subject matter of collective reflexology should be the manifestations of associative activity of social groups in general, irrespective of their nature and goals. Just as the subject of general reflexology is the study of individual associative activity in light of the past and present influences that have conditioned it,¹ so the subject of collective reflexology should be the products of the activity of certain social groups as entities united through the cementing material of a political, legal, religious, professional, academic or family nature in their relationship with the external and the internal, the current and the past conditions they reflect.

Clearly, collective reflexology should deal with the same aspects of associative activity as general reflexology does, but it investigates the activity of many individuals interrelated through common circumstances. Under given conditions of social activity they display manifestations of collective, or composite, associative activity. Therefore, collective reflexology should deal with the manifestations of public mood and mimico-somatic reactions in general, as well as communal observation, concentration, creativity, legislation, and action.

Irrespective of the external causes that bring social groups together, collective manifestations become possible when a certain relationship of greater or lesser coordination of various aspects of associative activity is established between individual group members. The characteristic features of these manifestations, as well as the conditions and laws of their development, should be the object of special investigation by collective reflexology.

Thus, the subject matter of collective reflexology should include all manifestations of associative activity in various social groups and gatherings, from ordinary crowds, rallies and meetings, to scientific societies and collegiate institutions of a social and administrative nature in the form of so-called Soviets, committees, etc.

The size of a given group or its permanent or temporary nature are of no consequence for collective reflexology, but it can and should take into consideration the circumstances that unite this group as well as its past, since both affect the nature and the characteristics of collective associative activity.

One should also bear in mind that collective activity develops along the same lines and according to the same laws as individual activity. Therefore, in the activity of every collective and every individual we can distinguish hereditary-organic manifestations, greater or lesser excitability, collective mood, collective ability to observe, collective concentration, collective creativity and judgment, which, as one of its manifestations, leads to a certain result in the form of a decision or a decree and, finally, to the execution of this decision or decree in the form of a particular action.

When examining social phenomena in the process of their gestation, development, and manifestation from a strictly objective point of view, collective reflexology studies the emergence, evolution, and manifestation of social activity or social movements as a kind of social reflex. Therefore, collective reflexology should not use such subjective terminology as

collective consciousness, collective feelings, collective attention, collective will, etc., which must be replaced by objective terminology.

A collective of people whose associative activity is investigated by collective reflexology is always based on some unifying element such as common mood, joint observation, collective concentration and discussion, joint decision-making, and shared goals and activity.

A crowd can serve as an example of a collective united by a common mood.

Theatres, concerts, entertainments in general and all kinds of performances are examples of collectives united through joint observation and concentration.

Soviets, conferences, committees, societies, congresses, etc., are examples of collectives necessitated by the joint discussion of certain issues leading to a particular decision.

Party meetings, professional and economic societies, cooperative organizations, etc. provide a platform for discussing the goals pursued by particular classes or certain sections of the public.

Collectives in the form of gymnastic and sport societies provide for the physical development of the masses while workers' cooperatives serve for organized production process.

Collectives in the form of army units and various patriotic organizations serve for the purpose of organized defense and attack.

Bank collectives and gambling houses are examples of collectives bound together by the goal of profit-making.

And, finally, scientific and other organizations are examples of collectives based on the unity of specific achievements.

Thus, collectives always possess certain areas of a common goal. This unity of interests can be either completely natural, resulting from daily existence or occupation (farming, fishing, commerce, etc.) or it can be formally organized and governed by certain regulations (unions, cooperatives, etc). Finally, there can be collectives—such as Soviets, committees, societies, commissions, etc.—established for the purpose of accomplishing various legislatively determined tasks.

Common interests and goals bind individual members of a collective together, and the more interests they share the more cohesive a collective is. Clearly, it is only the unity of interests and goals that stimulates a collective to undertake united actions and gives meaning to its very existence. In this

respect, a collective should be regarded as the best means of achieving a particular goal. What is unfeasible for an individual person becomes feasible for a concerted collective effort, and it is this circumstance that justifies the necessity of forming a collective.

If the collective had no advantage over individual activity whatsoever because of its lack of a positive contribution to the improvement of the human condition, it would probably not exist. Every social phenomenon must justify its existence, and the collective does so by accomplishing the tasks that are unfeasible for individual persons if they act alone.

Irrespective of the importance of the collective, there may be different reasons for its emergence and different conditions for its unity and development.

Collectives can be organized or unorganized.

Collectives bound predominantly by mood or focus of observation have no need of a specific internal organization as they have no predetermined goal. Nor are there any predetermined aspirations or plans of action, making internal organization unnecessary, especially since such collectives are short-lived (a crowd, a theatre audience, etc.). They need only external stimulation for the purpose of maintaining external order, so that the given collective does not complicate the life of other collectives.

All other collectives that possess well-defined goals should certainly have internal organization, the purpose of which is to maintain the regularity and order of collective work as well as to establish relations between individual members. To ensure coordination of joint actions, their organization into collectives is carried out by an executive committee, a presidium, a directorate, commanding officers, etc.

Generally, action coordination, planning of collective aspirations, and goal achievement are accomplished by delegating power to one person or a small collective—an executive—formed for that purpose. Internal order is also achieved in this manner, even through coercive and repressive measures when necessary.

These small collective institutions, endowed with power and the appropriate means of coercion, represent the apparatus that participates in the development and operation of collective reflexes regarding not only the internal order of a given collective but also the achievement of its particular goals.

Due to the special importance of internal organization for the existence and functioning of a collective, the formation of every organized social unit or collective starts first with internal organization, or inner order, and a certain distribution of interrelations between individual members and their activities. If a collective proves to be viable, it gradually attracts a growing number of members, thus expanding in size and achieving greater internal unity. This is usually accompanied by an internal process leading to considerable specialization of the individual units of a given collective, i.e., to its differentiation.

With time, separate entities develop in the collective itself as a manifestation of the same differentiation that divides the expanded collective into smaller ones.

Every newly developed collective discovers its relative position among other collectives, elaborates its own plan or program of task implementation and correlates its actions with the conditions of the surrounding reality.

Collectives may differ not only in their membership (colleagues in a certain profession, etc.) and in the nature of their relations, but also in their goals. Thus, we can distinguish a crowd, a theatre audience, a street gathering, a party or class meeting, legislative bodies in the form of Soviets, committees, conferences, cooperatives, congresses, associations, constituent assemblies, etc.

Some of these collectives represent chance gatherings of individuals, as in a crowd, while others emerge due to a particular need, such as the desire to be entertained or to satisfy economic needs, etc.

Finally, there are collectives that in essence are not assemblages but nevertheless form important groups of people bound by certain common interests. Here collective activity is carried out via representation or the press. Such collectives are the nation, the tribe, the state, an association of nations, the whole of humanity.

The fact is that direct interaction for the purpose of collective activity has a certain limit. The more elementary the goals of the collective, the greater size it can achieve. Thus, a crowd can number several hundred thousand people. A theatre audience can consist of several thousand spectators, whereas collectives engaged in the exchange of opinion may consist of several hundred individuals, seldom more than that. Collectives involved in accomplishing some specific tasks—preparing legislation, for instance—number no more than a few dozen, whereas collectives pursuing a still

narrower goal such as, say, editing a text, can have no more than a few individuals.

Therefore, collectives whose membership exceeds the above numbers must operate through their elected representatives.

Consequently, the means of communication between the members of a collective depends on its size and nature, and cannot be the same for all collectives. In some cases, there is direct interpersonal contact, while in others communication occurs only via representation or by the written or printed word.

But whichever route communication takes, a collective invariably implies a certain unity. Therefore, those collectives where general assemblies do not exist—e.g., nation, tribe, humanity—can be perceived as collectives only inasmuch as they constitute a unity based on the elected or appointed representation of individual assemblages and the printed word.

It goes without saying that the work productivity of a collective is not at all related to its size. In this respect, one should bear in mind that in those collectives from which more is expected than a mere expression of the common mood—for instance, collective opinion and judgment—the larger the collective itself the more slowly it works, but in this case the work of the collective is more extensive, provided its membership consists of suitable individuals. However, more narrow collective goals, such as the elaboration of some details, are attained faster and better in a smaller assemblage.

A considerable number of authors, including Ostwald,² Solvay,³ Voronov,⁴ Haret, and Partuendo y Barcelo, have tried to apply energy theory to social phenomena, regarding the latter as physical/ chemical phenomena. "And to what effect?" P. Sorokin queries. We have gained a few formulas of the type "consciousness is the flow of the energy process," "war, crime and punishment are related to a 'loss of energy'," "buying and selling constitute the reaction of exchange" (Ostwald), "cooperation means addition of forces, social strife is subtraction of forces, while social organization represents a balance of forces," "degeneration is the disintegration of forces" and "law is a correlate of forces," and so on and so forth. All this may well be true. But the question is how much our understanding of social phenomena and interpersonal interaction gains from the above analogies—in any one instance not more than from the analogies of the unfortunate "organic school" in sociology. All these formulas and analogies are truisms at best and inaccurate comparisons at worst. One has to come to the same conclusion regarding the

attempts of creating a "social mechanics," which reduce the whole of human behavior and interpersonal relations to the laws of general mechanics and physical phenomena. Such attempts were made by Haret and A.P. y Barcelo.⁵

This approach to social phenomena means that human beings are studied in the same manner as any other physical bodies. But people are people, i.e., they must be regarded as energy accumulators resulting from their past individual experience and hereditary influences. Consequently, their responses to external excitations cannot be assessed in the same way as are reactions of physical bodies, for in this case a response often results not from the effect of a given excitation but is to a far greater degree a consequence of past excitations, while the immediate one merely provides an opportunity for the manifestation of what is essentially the result of past influences. The same is true for collectives.

Therefore, the question is not whether social phenomena are reduced to physical/chemical substances or not, but whether their behavior displays any peculiarities and how they respond to external stimulation—in other words, whether these responses show the influence of established traditions and their own structural characteristics.

It is thus clear that one can and should approach the study of social phenomena using the methods of natural sciences with the purpose of discovering the characteristics of more complicated social phenomena as compared to more elementary physical or physical/chemical ones.⁶

Consequently, if social sciences—particularly, the one concerned with collective activity—move forward using the methods of natural sciences, there should be no subjectivism in their contents or their name.

However, such a newly established science as sociology has not yet settled the argument between the objectivists and the subjectiv-ists.

In Durkheim, we constantly read about collective consciousness, collective concepts, feelings, etc.

Comparatively recently, Boodin,⁷ proceeding from a theory of inter-cerebral phenomena, went back to the social soul. "Instead of proceeding from the postulate of the isolated spirit as psychology did in the past, trying to explain interpersonal cognition by analogy, we should proceed from the intersubjective spiritual continuum as an elementary fact." Other sociologists are trying to approach the study of social phenomena using a more objective method, but the majority have not succeeded in this. These divergences are well represented in Sorokin's latest work, *Sistema Sotsiologii* (*The System of*

Sociology), vol. 1, p. 44-76. The author offers a positive solution in favor of objectivism but, unfortunately, he makes some unwarranted (in our opinion) concessions to subjectivism.

At this point, it is essential to find out whether such concessions are indeed necessary, and since this methodological issue is also related to the volume of knowledge we are interested in, we should resolve it once and for all. Let us follow this controversy through in connection with the tendencies found in Sorokin's work.

The author defines sociology as a science "of human behavior in the process of interaction and the results of this behavior," having stated earlier in the text that human interaction "takes place whenever individual behavior—whether conscious or not—is a function of other people's behavior" (p. 50). The author then analyzes the methods of investigation and poses the question of whether interaction should be studied through external acts only or in relation to the corresponding psychological experiences. He first of all draws our attention to the vagueness of the very concept of the "mental" and its synonyms (consciousness, spiritual phenomena, etc.) as well as to the inaccessibility of the psyche to observation.

When turning to psychology for the definition of the mental, the author finds there a peculiar circular argumentation, while the concept itself remains undefined. According to Wundt, the common definition of psychology as a science "of the state of consciousness" is circular, for if one subsequently inquires what is meant by this consciousness whose state psychology should be studying, the answer would be "consciousness represents the totality of the state we are aware of."⁸ According to Gefding, it is impossible to define consciousness and its elements.⁹ We find the same, or approximately the same, views in other authors. Consciousness is frequently defined as a spiritual or a mental process, while the term "spiritual" is replaced by the term "conscious."¹⁰ Psychologists display no less conceptual confusion regarding the notions of "mind," "reason," "thought," etc., and are probably even more confused about the issue of which phenomena can or cannot be described as mental. Thus, such authors as Haeckel, Le Dantec, Perti and other exponents of monism talk about "cellular consciousness" and "atomic soul," while Haekkel and De la Grasserie go as far as the psyche of atoms, molecules, and even the "psychology of minerals." This school can be described as "pan-psychism." Others (Wundt, Romaines, Letourneau, Espinas, Jennings) attribute consciousness to organic nature only, still others

recognize the existence of consciousness only in those organisms that have a nervous system, while some authors recognize it only in those higher animals who have gray matter. Some contemporary physiologists (Loeb, to a degree Bohn and others) throw overboard the concept of consciousness altogether and obscure it with the unfortunate (from my point of view) term "associative memory" (Loeb).

As works by Durkheim, Kost, Wachswiler, and others show, there has recently appeared in sociology a marked tendency towards dissociation from psychology and subjectivism in general. However, even in these works one comes across psychological, i.e., subjective, terms and concepts at every turn; therefore we do not find here true objectivism either. In my *Reflexology*, I came to the conclusion that it is necessary to study "associative activity" in humans and animals as it is manifested in outward actions and reactions in connection with external influences and impacts both past and present. I attach particular importance to the latter fact, i.e., to the dependence of the external expression of an individual on past influences. I discuss it extensively both in *General Principles of Human Reflexology* and in my earlier work *Objective Psychology*. To overlook this circumstance would lead to some confusion, as will become apparent from what follows below.

I do not dismiss the study of the mental world or the processes in consciousness, but relegate subjective psychology to the area of self-study, through self-observation and experiments conducted with a detailed self-analysis of one's subjective states, which can certainly produce some generalizations regarding the subjective world in general and even philosophical speculations on the subject of pan-psychism. Scientific thought, however, should adhere to a strictly limiting rule when dealing with these expansions of subjectivism, and instead of the so-called psyche or consciousness in others, it should study the various correlations between the organism and the environment by means of strictly objective observation and experiment.

Recently, the realm of social problems has been entered by physiologists who have eliminated "subjective" terminology from their investigations of brain functioning and have spread this tendency to the study of interrelations not only between animals but also between humans.

This viewpoint, which excludes all psychology from the social sciences, is also shared by Bentley.¹¹

In this controversy about objectivism, the physiologists, unfortunately, proposed their own purely physiological point of view and extended it to a number of social phenomena, which can on no account be a part of physiology as the science of bodily functions. This false move met with strong opposition from those engaged in pure sociology. I am referring to the paper by Dr. Zelenyi (of the Pavlov school), according to whom the scientific investigation of social phenomena should be limited to the objective study of their physiological instead of their psychological aspect.¹² And, indeed, one should not forget that sociology is not social physiology and it is not without reason that the organic theory of social phenomena has long been abandoned and finds no support among present-day sociologists. It is, therefore, quite natural that one of the prominent American sociologists, Ellwood, rebuffed these physiological excursions into the realm of sociology, excursions that weaken rather than strengthen the position of objectivism in sociology.¹³ Thus, when criticizing the Pavlovian school's misplaced attempt to introduce physiological sociology, Ellwood remarks in passing that this method

may be good for studying the behavior of a rat or a rat society or at best the behavior of a human group that lived half-a-million years ago in the prehistoric world. Civilized man, however, lives in the world of ideas. For him the world of real objects is to a significant degree replaced by the world of ideas, symbols and values. These ideas, symbols and values have gradually evolved and accumulated in the course of the whole human history. This history itself can be perceived as a continuously evolving tradition or social spirit, which cannot be understood outside of its context. Human culture as such is essentially of a mental nature and this culture produced the society we know.

Even if we disagree with the author's subjective interpretation of sociology, we must admit that W. Wagner repudiated even more emphatically I. Pavlov's proposal to abolish zoopsychology (or, more precisely, comparative psychology) altogether by replacing it with a true physiology of the brain. I. Pavlov makes the following comment about one of his experiments:

A zoopsychologist is someone who wants to penetrate into a dog's soul and says that a dog should pay attention and remember that, as soon as it feels its skin irritated in a particular spot, an acid solution is being administered; therefore, when only its skin is irritated, it imagines, as it were, the administered acid and responds correspondingly by salivating, etc.¹⁴

Regarding these views of academician I. Pavlov, Professor W. Wagner in turn comments,

This is how Professor Pavlov recounts the words of representative scientists, outstanding scientists and thinkers who have been the centre of attention throughout the ages. After attributing to zoopsychologists the above absurdities, Professor Pavlov commences to judge them, etc. In order to disgrace the zoopsychologist once and for all, Professor Pavlov sets him a task, the unsatisfactory resolution of which would "render their point of view totally unscientific and useless for a profitable research." This "examination" task consists of the following: having determined by the usual route the coordination between salivary gland functioning and three skin areas, Professor Pavlov found out that when the gland stops responding to the irritation of one skin area, it resumes functioning with the same intensity when the second area is irritated, etc. The answer to this question had been given by zoopsychologists long before Professor Pavlov started his work on the physiology of the canine gland: zoopsychologists draw a borderline between the phenomena (actions) of bodily functions and the phenomena (actions) of animal behavior. Physiology deals with the former, zoopsychology with the latter. The facts presented by Professor Pavlov pertain to animal functions, which means, naturally, that they are outside the province of zoopsychology and that Professor Pavlov has not the slightest understanding of modern, objective zoopsychology.¹⁵

From my point of view, every trace of subjectivism should be eliminated from zoopsychology and comparative (as well as any other) psychology, which should be replaced by the objective sciences of zoo- and comparative reflexology. Physiology will naturally remain with its special goals and methods, as will also remain zoo- and comparative reflexology with their own goals and methods.¹⁶ Nor is the investigation of issues related to the functioning of a particular gland in the form of conditioned and unconditioned reflexes (or "associative" and "simple," in our terminology) capable of embracing such vast areas of knowledge as, for instance, comparative reflexology or even brain physiology itself, for in the latter it is not solely the method of conditioned salivary reflexes that is important but even more so our method of associative movement reflexes.¹⁷

As for sociology, Sorokin, who has followed the whole controversy between the "objectivists" and the "subjectivists," has remained undecided despite his obvious gravitation towards objectivism. "The data obtained by

the "objectivists" have so far been too inconclusive and insignificant," he says,

for the sociologist to ignore the subjective-psychological aspect of human activity altogether. Hence, there is only one route to follow: to minimize the use of psychological terms; to turn only in extreme cases to the analysis of subjective mental experiences in order to explain certain phenomena and to introduce optimal precision into the study of such phenomena.¹⁸

Let us examine the way in which the author applies his half-way policy to sociology.

First of all, he talks about "conscious" emotions and acts in terms of one's individual experience, and admits that "their frequent and principal forms are purposeful, volitional, and conscious acts under the influence of ideas."

While recognizing, in agreement with Wagner, three forms of mental phenomena—reflexes, instincts, and reason—the author, unlike Wagner, attributes only reason to mental phenomena rather than both reason and instincts, for, according to Wagner, instincts are similar to reflexes in their general nature while appearing more complicated than the latter.

Another difference of opinion between Wagner and Sorokin is in the latter's refusal to identify reason and consciousness with "goal setting." The fact is that there are acts (not only habitual ones, as Wagner believes) which, although rational, i.e., conscious, are not motivated. A child, for instance, consciously repeats his father's statement that "one should not tell lies," or declares that "to torture a cat is wicked because daddy said so," and that is all there is to it.

The author then poses the question whether emotional experiences should be totally ignored; being inaccessible to us directly, they cannot be generally understood, since any attempt at their interpretation through their external manifestations (words, movements, facial expression, gestures, etc.) inevitably leads to errors. His answer to this question is negative. According to the author, if it were otherwise, people would not be able to understand one another or read another person's mind:

However, we continually make diagnoses of the type: "X is out of sorts," "Y is feeling abit low," "L is angry," "A is delighted," "D is contemplating a nasty deed," etc., and our diagnoses prove to be correct. We continually discuss our emotional experiences with other people and in the majority of cases we do understand one another. (I.e. p. 65)

The problem, however, is not whether we understand others or not but to what extent we understand them. On the basis of the words and facial expression of another person, we reproduce not his emotional states but our own analogous emotions, if we ever experienced them. It goes without saying that if one has never gone through similar inner experiences one cannot reproduce them from another person's words. Someone deaf from birth is unable to reproduce sounds in his imagination no matter how persuasively they might be described. A child who has never tasted bananas is incapable of recalling their taste no matter how much he might hear about it. But what one has personally experienced can be reproduced by the listener on hearing the corresponding description. This process, however, is based on one's own past experience rather than on the current experience of the speaker. We can thereby recognize another person's emotional state, albeit only to the degree in which the analogous states we reproduce coincide in their external manifestations with the emotional states of the other person. When we say "X is out of sorts," "Y has been feeling low," "L is angry," "A is delighted," etc., it is our own indisposition, sadness, anger, or delight that we reproduce from the corresponding external signs. And, since the states I experience are similar in their overall pattern to those of other people, I am capable of recognizing them in another individual; this is as far as it goes. As to any attempts at more detailed diagnoses such as "It is the problem of consciousness that B is contemplating" or "C is craving sweets" or "D is definitely contemplating a nasty deed and not anything else," even if they are possible at all, result from pure guesswork when these states are not manifested, i.e., not expressed verbally. Words, however, will enable us to reproduce our own, rather than another person's, contemplation of the problem of consciousness, our own craving for sweets, our own nastiness, etc. And what about simulation? I may be contemplating a nasty deed yet not show any signs of scheming against another person; if so, is it right to use in research the methods of subjective psychology (the methods of self-observation and analogy, the combined methods of external and internal observation according to Petrazhitskii, the methods of "empathy," "intuition," and "instinctive understanding of expressive movements" as supplementary and auxiliary methods to those of the objective study of outward acts)? I do not think it is right. And it is not right precisely because the investigation of mental experiences does not add anything to the understanding and elucidation of outward acts, contrary to what P. Sorokin thinks.

In the above case the respected author gives the following example to make his point:

A postman delivers two telegrams to A. He opens the first one and reads "Arriving today, B." Mr. A smiles, rushes off to call the servant and gives orders to have his carriage ready by five o'clock. He then opens the second telegram and reads: "Your wife was run over by a train. Come. Station master X." A dies from a heart attack. Here we have two examples of interaction. The sources of stimulation are similar (telegrams). The individual affected is the same A. But the effects are totally different: in the first case a lively response to stimulation, in the second—death. The question is whether we can understand these two phenomena completely if we analyze them from a strictly objective point of view. Will these facts be sufficiently clarified and explicated if we exclude in advance from our field of vision all manifestations of subjective states such as "love, attachment, the horror and the magnitude of loss, profound grief, loneliness, etc? The answer seems to be self-evident (l.c.p. 71).

One should not forget, however, that the words in the telegrams are not significant in themselves as visual signs or sounds. They consist of symbols which, although only slightly different, refer to completely opposite facts. This alone explains the opposite effects in the two cases. Nor should we forget that an effect depends not only on stimulation as a cause of action but also on a number of preceding or past influences related to the current source of stimulation and the response it generates. But when we take all this into consideration, we have to interpret the second effect as the result of the symbolic signs of the telegram, which tells A not only about the misfortune with his wife but also about the disintegration—due to her death—of the whole chain of circumstances attending their common existence, which created a certain way of life that became part of the man himself, about the irrevocable loss of someone dear with whom he shared the good and the bad times, to whom he is attached and owes so much in his life, etc. We should also bear in mind that someone who dies from a heart attack on receiving a telegram with the news of the fatality with his wife must have suffered from a bad heart condition.

Thus, the lethal effect of the telegraphed news of the wife's death becomes quite clear and understandable from the standpoint of objective analysis. One wonders, what can subjective analysis possibly add to this? It would tell us about such hypothetical states as the recollection and experience at the

moment of receiving the telegram of the feelings of love, horror and magnitude of loss, profound grief, loneliness, etc. However, all this "subjectivity" that we would like to attribute to the man who died of heart failure on reading the second telegram may prove to be pure nonsense; a man suffering from a heart condition could, in fact, die without reproducing anything at all in his consciousness, but simply from the shock imparted by the telegram with the news of his wife's fatal accident. The same is true for another of the author's examples when "one and the same action of outstretching one's arm with a pointing finger can mean indicating a direction or giving the command 'to go there' or ordering someone 'to get out of here'." To understand this gesture we do not have to take into account the inaccessible mental states of the person in question but need only to perceive his tone of voice, the corresponding facial expression, and his relations with the other person as they have evolved in the course of time. Let us take two other examples offered by Sorokin. Everybody knows that the phrase "What a pig you are" usually has a derogatory meaning, but sometimes it stands for a somewhat affectionate form of address; similarly the expression, "Damn it," can indicate frustration, anger, or thrill. In both of these cases, however, we should take into consideration not the inaccessible subjective states of the person who pronounces the utterance—as Sorokin suggests—but only the tone, the environment, the nature and the habitual manners of the person in question as well as his past and present attitude to a given person or event.

Subjectivists, as we know, regard "goal" exclusively from the subjective point of view and believe that it can be set only consciously by this or that subject.¹⁹ A biologist, however, understands "goal" as something objective and occasionally even designed by nature as, for instance, a goal carried out by instincts as automatic acts. In acts of a personal nature, we understand "goal" as the aspirations that the human personality or the human collective manifest in their acts and behavior based on past individual experience as a reproduction of these actions in the form of associative reflexes combined in a particular way. Thus, even in this case there is no reason to turn to the subjective method which gives rise to views that often have nothing in common with scientific thought.

Therefore, there are no grounds whatsoever for supplementing the study of interpersonal relations using the objective method, which records the outward forms of human behavior as affected by various external influences, with the

analysis of subjective or mental states that we are unable to identify accurately and that cannot add anything at all to the objective facts.

Thus, the method of collective reflexology, as well as of sociology, should be exclusively objective. When studying the characteristics of the associative activity of gatherings or groups of individuals, collective reflexology cannot assume the viewpoint of an imaginary composite "self," or consciousness, of a crowd that would recount everything this crowd or gathering has experienced: this would be a mere reflection of the subjective viewpoint of an individual "self." Collective reflexology, however, can make use of the testimonies of those individuals who were present in the crowd or the gathering, since they have an objective nature, interpret external events and characterize the developments in social life. In any case, these testimonies should be analyzed in a strictly objective, rather than subjective, way.

How little this truth has been absorbed by the appropriate scientific literature is apparent from Tarde's writings. He reduces all social phenomena based on "mental" human activity to two principal forces: desire and faith, or confidence. Desire defines the aspiration while faith determines the attainability of the desired. According to Tarde, it is by reducing social phenomena to these simple forces that the issue of social facts is resolved.

On the other hand, while recognizing invention and imitation as principal elementary social processes, Tarde queries

What is being invented and what is being imitated? That which is invented and that which is imitated is nothing other than idea or desire, opinion or intention. They manifest a certain degree of *belief and* desire, which make up the true soul of any word in a language, of any prayer in religion, any undertaking in a country, any work in industry and any activity in art. Belief and desire are therefore the substance and the force, the two psychological entities revealed by analysis to be at the base of all *sensuous* phenomena combined with them. And when invention and imitation master them through usage and organization, they become the two principal social entities.²⁰

As far as its method is concerned, collective reflexology should follow in the footsteps of individual reflexology. Therefore, it should observe and record the outward aspect of mass behavior as well as determine its relationship to the preceding external influences which provided the cause for mass activity in any given case. The nature and membership of a gathering should be taken into account, since the gathering itself represents a public individuality, as it were, and just as individual reflexology is important in

regard to the manifestations of individual correlative activity, the nature and the composition of a gathering exercise an enormous influence on the nature and the direction of its activity.

Thus, collective reflexology ought to elucidate not only the mood or the "emotional" aspect, or to be more precise, the mimico-so-matic response, of a gathering and its specific sensitivity or impressionability, but also its decisions and collective actions. It should examine all these mass manifestations in relation to the nature of the gathering and its past as well as to the diverse external influences that provided the main cause and the conditions for the activity of a given gathering in each case.

Only this method can show in the right light the characteristic manifestations of correlative activity of human masses united by particular conditions.

Statistical method should undoubtedly be recognized as one of the important research methods of collective reflexology. It makes it possible to record the actions of human masses, but it should be applied with competence. Statistics should be used for the tasks that correspond to a particular goal, and should be collected in those areas where they can yield appropriate results. Thus, if you wish to determine public temperament and mood, you should gather data about the attendance at theatres, restaurants, gambling houses, churches and about the number of suicides. If you are interested in the orientation of public opinion, take the statistics of public libraries and find out which books are in the greatest demand, collect data regarding the popularity of various periodicals and journals, approach publishers for the information on various genres of publications, etc. If you want to find out the direction of public activity, you should gather data on the number of young people entering higher educational institutions, on the membership numbers of various professional associations and on the proliferation of certain undertakings. If you want to know the direction of the political views in a given society, you must count the membership numbers of different political parties, the number of participants in various political and public meetings, the number of votes for members of particular parties during elections, etc. If you are interested in public behavior from the point of view of its morality, you should collect data about the number and nature of crimes committed within a certain period of time, etc. If you want to know the needs and the tastes of society, you should gather information from

various companies and stores about the quantity and nature of merchandise sold during a certain period of time.

It is clear from the above that statistics is of use for collective reflexology only when it deals with the tasks of the latter. It is not enough, however, to obtain the data: it is also necessary to discover the cause of the changes occurring in public life, which can be achieved through correlation with numerous other data gathered with the same statistical method. If we avoid a subjective interpretation of the results obtained, we shall possess strictly objective statistical material that will also enable us to correlate human actions with various external causes. The realm of statistics includes all kinds of actions and events that make up our existence: birth, marriage, death (natural and violent, i.e., suicide); crime, various diseases; alcohol and drug consumption; postal and other kinds of communications; development of transport (by land, by rail, by air and underwater), agriculture, industrial labor, various crafts and occupations, commerce and industry; dissemination of books, development of religious and political teachings, establishment of protective organizations in the form of insurance companies, etc. Statistics determines the number of homogeneous acts committed in a collective, recording thereby how a particular collective activity or state progresses—what deviations or interruptions occur in their development—or decline. At the same time, statistics also records biological (sex and age), individual (origins and temperament), anthropological (race and character), planetary/magnetic and meteorological (nature of locality, climate, season, etc.) influences.

Besides the analysis of the curves representing the activity or state of a collective, statistics enables us to correlate them with other states and phenomena also represented by curves, establishing thereby the mutual interrelations between them, which can also be verified by other methods. Let us take an example from medicine. If we ascertain the number of patients suffering from progressive paralysis in regional hospitals we shall have figures representing the incidence of the disease in this region. In addition, we may take figures representing alcohol consumption by the population. And then it turns out that the curves of progressive paralysis and those of alcohol consumption almost coincide (Dr. Vinogradov. Dissertation, St. Petersburg). This gives every reason to conclude that there exists a direct cause-and-effect dependence between alcohol consumption and progressive paralysis. This, however, does not exclude other elements, such as the

influence of syphilis on progressive paralysis, which is proved by other means. Yet another valuable feature of the statistical method is that it enables us to determine not only the state of a collective in terms of its composition, its members' health, growth, birthrate, ability to work, etc., but also their political and religious views, moral qualities, economic status, level of literacy, etc, as well as various collective needs, their growth and decline and, finally, their dependence on various circumstances.

Thus, by means of the statistical method we ascertain the state of the population concerning different types of education, particular forms of upbringing, the need for books of a certain nature, the need for self-preservation from natural disasters (fire, etc.) by insurance, the need for communication, transport, particular clothes, foods and other consumer goods, the need for stimulants, narcotics, etc. Incidentally, the whole of commercial and industrial statistics indicates a number of needs generated by the daily existence and way of life pursued by a country's inhabitants. There is no doubt that statistics is one of the most important methods of collective reflexology. Not an independent science, its importance lies in making the conclusions of collective reflexology more precise.

The extent to which statistics is a method that enables us to deal with the issues of collective reflexology is apparent from the fact that any human need represents a habitual associative reflex, while a social need represents a collective associative reflex. The evolution of this collective associative reflex, just like that of the individual associative reflex, follows a certain pattern that can be described as follows: initially, the given need goes through a period of slow growth followed by a relatively rapid increase that reaches a certain level. The growth then slows down again and proceeds along a horizontal line until it comes to a temporary halt to be followed by a more or less slow decline of the given need as it is being supplanted by other needs.

One should point out, however, that the above development of a need may be disrupted by various attending circumstances and concurrent events, as a result of which the distribution of any given need may show oscillations and disturbances.

Thus, what happens after a social need has developed depends on numerous conditions, such as competition and new inventions, which help to supplant the need or, conversely, on all kinds of favorable circumstances.

It is apparent from the above that by using statistics it is also easy to follow the action of the inhibiting influences on a given social need or social

movement as well as the substitution of this action by the opposite, so to speak, favorable influences under different social conditions.

One example of both inhibiting and stimulating conditions (depending on the force) in a collective is the competition due to the initiative of those promoting a new invention or making a new discovery. It is these instances of competition that disturb, as we have seen, the regular pattern of the distributions representing various needs of a collective. To give an example, let us take crime statistics, which record the negative need to live at the expense of, and by inflicting harm on, other people in human society. Due to the abnormal conditions of life in contemporary society, this specific need displayed by some of its members is on the rise all over the world. In the meantime, England has been trying to cope with child vagrancy by "inventing" reformatories (which have not yet sufficiently spread to other countries) and as a result the curve of the crime rate in England has started to go downward. Another example is the funds allocated by the Russian government for education. Their increase from one year to another had been extremely slow until a sharp rise in 1906, when they began to grow rather slowly at first and then faster. One can easily guess that this increase in education funding in Russia is related to the establishment of legislative institutions.

Since the October Revolution there has been a tremendous new rise in government spending on education (of all kinds and levels) due to the new social system. Now let us take France's expenses on public works. In the period of 1843-1849, these figures increased because of the implementation of a new invention in the form of railway construction.

Tarde says that "it is only imitation and not invention that is subject to laws in the proper sense of the word" (Tarde, *The Laws of Imitation*, p. 147). The question is, however, whether imitation can exist without invention. Is invention not a kind of imitation transplanted into another area? In any case, inventions, too, are subject to laws.

One should also bear in mind that statistics establishes primarily quantitative relations, whereas every social phenomenon has, in fact, not only a quantitative aspect but a qualitative one as well. Thus, the number of school children may remain more or less the same, but the quality of education may improve or decline depending on the competence of the teachers. Similarly, the number of government supporters can remain unchanged while their

attachment to this form of government may progressively decrease and vice versa.

These kinds of circumstances should never be overlooked when analyzing statistical data. It certainly does not mean that such data cannot be used numerically, but for this purpose another kind of material seems to be necessary: that which would clarify precisely this aspect of the matter.

Statistical indications of certain correlations between social phenomena are extremely valuable, since by understanding these correlations between various social phenomena we can prevent the latter. Let us take medical statistics. With its help it is not difficult to determine under which conditions a particular epidemic disease develops and under which conditions it abates, as well as how successful preventive measures are: e.g., to what extent the rabies vaccination protects society against this disease, etc. In the same way, we find out from statistics the degree to which marriage is related to lower mortality rate; it tells us about the increasing number of suicides, crimes, and personality aberrations in the urban environment as compared to the rural one; to what extent the state's concern with homeless children limits criminal behavior or how alcohol prohibition affects crime rate, suicides, and personality aberrations. These and other numerous statistical findings leave no doubt that statistics not only helps to elucidate various correlations but also indicates how to change these correlations for the benefit of mankind.

In the final analysis, statistics is essentially mass observation which provides material for collective reflexology. The latter, however, also needs experiment for the verification of observation. The simplest form of collective experiment can be carried out by means of so-called questionnaires or inventories distributed in a considerable number of copies. The disadvantage of these questionnaires is in that the number of responses is much lower than the number of the questionnaires sent. Nevertheless, it is possible with their help to gather sufficient material for the appropriate conclusions. Besides, the formulation of the questions depends exclusively on the author of the questionnaire that may be used for verification purposes. The success of a questionnaire certainly depends greatly on the nature of questions, their formulation and even on those who collect the data, which facts should never be overlooked.

Another kind of collective experiment is the method of direct experimentation with a whole collective, which can be achieved more successfully in schools and universities. The disadvantage of this method,

however, is that the experiment can be conducted only with relatively limited collectives where individual members function under unequal spatial conditions, which leads to the experimenter's inability to affect every member to the same degree. Besides, the fact that the subjects are aware of being experimented on creates an artificial situation impinging on the members of the collective. These circumstances are avoided by the natural experiment devised by our school. The essence of this method, used by Dr. Lazurskii et al. in schools and earlier in my work with infants (see Dr. Brazhas's paper in *Vestnik psikhologii*), can be reduced to the following. The subjects are given a lesson during which they perform as during any other lesson, unaware of being involved in an experiment. The lesson, however, is conducted in such a manner as to reveal individual characteristics of children's associative activity. With the same degree of success one can use appropriately selected games for preschoolers and occupations for adults.

In both cases, however, assessment of the results in terms of their meaning in the context of the collective can be carried out only by comparing them with the results of the same experiment conducted with individual members of the same collective. Finally, the games and the occupations can be organized in such a way as to involve the whole collective as a gathering of individuals.

Until now, experimentation for the purposes of collective reflexology has been almost nonexistent, although it must play the same role here as in other branches of knowledge. The experiments with collectives conducted thus far were done not for the purpose of investigating the role of the collective in a particular process or of revealing personality changes in the collective, but for other reasons, such as verification of witness accounts (Binet, Stern, Claparede, and others).

Our so-called natural experiment used in schools identifies in the mass of people, i.e., in the collective, certain members by their individual characteristics, which is essentially an analysis of the collective in terms of its composition.

For collective reflexology, however, a different kind of issue is of greater significance: how personality manifestations change in a collective or, to be more precise, what happens to personality when it takes part in the activity of the collective. In this field, observation alone yields numerous data, but experiment can undoubtedly produce valuable results here, as our experiments with kindergarten children have shown. Another experimental

task is to find out the difference between the response to a certain event by the collective personality and by the individual one under the same or similar circumstances. The application of collective experiment in this direction is the subject of future studies.

We can find only rather general information on the findings of experimental research or, rather, observation conducted to ascertain the role of collective associative activity. This is what Karl Marx stated in the chapter on combined labor:

The sum total of the mechanical forces of individual workers differs from the mechanical force generated by many hands simultaneously participating in the same operation. The action developed during this combined labour is either impossible to achieve by single-handed efforts of individual workers, or it would take them a much longer time to perform it, or they could only perform a minuscule part of it. We are dealing here not only with an increase of an individual production force through cooperation but also with the creation of a specific production force of the mass.

On the other hand, we have the dynamometric observations by Féré worthy of our attention. First of all, this author proved that whether or not stimulation reaches consciousness it is always manifest in the growth of a dynamic equivalent or, in other words, in an increase of muscular force when measured by a dynamometer. His experiments with individual persons and with a crowd convinced the author that simply being present in a crowd sharply increases one's dynamic force. The same can be observed in walking.

Clearly, a crowd creates conditions that not only stimulate mood but also boost energy levels to a degree sufficient to invigorate motion impulses.

I have not found in the literature any other mention of experimental studies of the collective for the above purposes. It is therefore clear that experimental collective reflexology has yet to be established.

The question is, where and how should the collective experiment be conducted? One must not recommend the use of a common laboratory set-up in the sphere of collective reflexology. It is inexpedient to invite to a laboratory a considerable number of people to conduct an experiment with them as a collective, since we shall be dealing with several individuals rather than a collective or a society united by certain goals and common conditions. There is still no better place for conducting experiments on collective reflexology than a school, where we have a society of peers at work.

Another setting suitable for observing collective personality is children's playgrounds, where we actually have a proper crowd. Although it consists of children, it is, nevertheless, a crowd in the true sense of the word, and with it one can conduct a proper collective experiment. The results of such an experiment can be compared with the results of studies carried out on an individual basis with the same children (or with other children with similar characteristics) under different conditions. Gymnastic and sport societies provide a suitable setting for investigating the issues associated with collective physical exercises, muscular effort, etc.

Such settings also make it easy to assess individual level fluctuations depending on one's personal commitment as a member of a collective: e.g., during games or prize-winning exercises. Choral and orchestral societies can be used for some specialized research.

There are obviously numerous other settings for experimental studies of the composite personality, such as public lectures and other gatherings, industrial enterprises, etc.

It is clear from the above that experimentation in collective reflexology should proceed from a comparison with the data obtained from experiments with individual persons. There is no other way, since the only measure to be used for recording the influence of society on the activity of its members is the activity of a separate personality in its habitual individual work? It is only proceeding from this individual activity that we can discover how this activity changes when it is performed in a collective.

Among the questions in need of experimental resolution, collective reflexology should pose the following one: how do individual forces add together during joint work in the same direction and during their simultaneous application in the production process? Will the end result correspond to the sum of individual forces applied by each worker or will it be lesser or greater than the sum total? All these questions should be studied in detail. For this purpose a device based on the principle of traction with a dynamometer can be built. Application of force—both individual and collective—should be designed in such a way that traction produced by a number of people would most certainly occur in one and the same direction. For this purpose, the principle of A. Mosso's device can be used with appropriate modifications.

No less important are issues related to the inhibition of collective work: inner inhibition in the form of fatigue or tiredness, as well as external

inhibition, which depends on external inhibiting circumstances leading to the suppression of muscular efforts. These questions, too, can be resolved experimentally in the process of collective labor performed under certain circumstances on the appropriate equipment. It is easy to see that these experiments must be extremely important for the solution of problems related to the scientific organization of labor.

What role is played by collective work itself in this respect, in terms of inhibiting or stimulating the energy of the workers, will be demonstrated by all the experiments that compare collective and individual work. Some special experiments, however, can be done to reveal the external conditions of inhibition and de-inhibition, which should disclose the role of various external influences in the work of a collective. Here one can expect an unequal impact of different external influences on the work of a collective, depending on the nature of the influence and the nature of the work. For instance, one type of music facilitates physical work while another type affects it adversely. The same music, however, which facilitates physical work as an invigorating stimulus, as it were, may produce a negative effect on mental collective activity by disrupting mental concentration.

Another important issue in collective reflexology which can be experimentally resolved concerns the influence of induction and suggestion. For this purpose, for instance, the subjects can be shown a few lines whose length is progressively increased up to a certain limit. A group of subjects must record in their notebooks how many times the experiment with the lengthening lines has been done and on which line it has been interrupted. Another procedure may consist of the verbal suggestion that the lines shown are gradually lengthening, while, in fact, they are not. The number of subjects who recorded the lengthening of the lines with a "+" will illustrate the effect of the applied suggestion. In both cases, a comparison should be made between group and individual suggestions. To trace the role of mutual suggestion in a crowd, the same experiment could be modified as follows. A group of subjects observes the lines and anyone who notices the interruption in line lengthening says so, while the experimenter records the number of people who have come to the same conclusion.

Similar experiments can be done on determining shades of color. One can also conduct an experiment on the suggestion of collective illusions and hallucinations. In one of the simple experiments of this kind the experimenter

suggests that there is the smell of something burning or some other smell in the room where the subjects are.

Such statements can be made quite unexpectedly during mental exercise, for instance, when pupils are involved in the solution of some problem, or it can be done after a rest period. In this case, we shall see a difference in the results depending on lesser or greater fatigue. It is also very important to determine the role of attention on the success of suggestion. We should also attach particular significance to the question of imitation in a crowd. For this purpose, a number of people could be tested for the average speed of their muscular movement by asking them, for instance, to draw lines on paper. The subjects are then seated in such a way that everyone can observe the hand movements of everyone else.

If, after a preliminary individual examination, we conduct this experiment first with two subjects and then with several we shall be able to determine whether imitation is related to a lesser or greater number of participants.

This type of experiment can be done with attention disruption by external stimulation and without such disruption, thereby revealing the role of the latter. Then an external light or sound stimulus applied with a certain frequency can be used in these experiments to find out how this frequency affects the curves of individual subjects working alone or together with others.

Changes in the intensity and the frequency of external stimulation may also indicate their effect on imitation.

In the future, I believe, it would be extremely valuable for the goals of collective reflexology to determine the place of competition, which plays such an important role in any collective in general. One can hardly doubt that competition must increase work intensity, but the question is how it affects the work quality.

For the above purpose there should be a special interest in work performance, which can be provided by a mark for excellence, a prize, etc. The object of work may vary considerably. For example, they could count instances of a certain letter in a text, or count numbers, reproduce words or fill in the blanks in a text, solve a problem, measure their strength with a dynamometer, play a game, etc.

Naturally, these experiments, too, should be conducted both with a collective and with individual members using the same set of instructions.

It would also be extremely interesting to study collective creativity in the form of exchange of opinion required for the discussion of a specific task. For experimental purposes, this task should be of the kind that could be discussed and performed by a collective. This collective effort with the most complete record of individual opinions and proposals should be compared to the individual effort of the same type so as to reveal the characteristics of collective and individual work.

In the latter case, the experiment should demonstrate how the collective mind, as compared to the individual mind, functions in general from the quantitative as well as the qualitative point of view. From our everyday experience of participating in various meetings we know that life demands collective decisions, but we also know that a considerable number of authors claim that creativity of crowds and collectives is always inferior to individual creativity and that even the gifted cannot fully express themselves in a crowd. This should be verified in experiment.

There are certainly other possible ways of organizing collective activity in experimental settings. All of them should provide valuable material for a more detailed interpretation of various questions belonging to the province of collective reflexology.

As well, various methods can be applied specifically to the characterology of social groups. Here we should focus primarily on group attitudes toward certain facts and phenomena of vital or social significance as well as on professional skills, etc.

Notes

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1. V. Bekhterev, *Objective psychology* (in Russian), in *Vestnikpsikhologii*, 1908, and in a separate edition. See also my *General Principles of Human Reflexology* Petrograd, Rikker, 1918.
 2. Ostwald, *Naturphilosophie. Die energetischen Grundlagen d. Kulturw*, Leipzig, 1909.
 3. Solvay, *Notes sur de formules d'introduction a l'énergétique physio et psychosociologique*. Institut Solvay, 1906.
 4. Voronov, *General Principles of Sociology* (in Russian).
 5. A.P. y Barcelo, *Mécanique sociale*, 1910.
 6. The energy point of view is quite acceptable in this case but not in the previous sense.
 7. Boodin, "The existence of social minds," *American Journal of Sociology*, Vol. XIX.
 8. W. Wundt, *Introduction to Psychology* (in Russian), Moscow, 1912.
 9. Gefding, *Essays on Psychology* (in Russian), St. Petersburg, 1918.
 0. See "Consciousness" and "Mind" in *The Dictionary of Philosophy and Psychology*.
 1. Bentley, *Process of Government*, 1908.

2. Rassen—u, Gesellschafts—Biologie, 1912, IX. There has been a recent attempt by one of the Pavlovians to go deeper into this issue in the pamphlet *18 Comments on the "Theory of New Biology"* (The draft proposal for establishing the Revolutionary-Scientific Council of the Republic and for introducing the system of physiological passports). Published by the Revolutionary Committee of the North Caucasus, the very title of the pamphlet speaks for itself.
3. Ellwood, "Objectivism in Sociology," *The American Journal of Sociology*, Vol. XXII.
4. "Physiology and psychology when studying the higher nervous activity" (in Russian), in *Psikhiatricheskaia gazeta*, 1917, No. 6.
5. W. Wagner, *Vestipsikhologii*, 1917, Vol. 3.
6. See my paper, "On the Goals of Zoopsychology " delivered at the Conference on Brain Studies in 1920, in *Voprosy izucheniia i vospitaniia lichnosti*.
7. V Bekhterev, "The importance of the movement sphere, etc.," (in Russian), in *Ruskiivrach*, 1909,35-36.
8. P. Sorokin, *System of Sociology*, vol. 1, p. 63.
9. I cannot agree with Sorokin's comment on this matter on p. 11,vol. \,ofhis*System ofSociology*.
0. Tarde, *The Laws of Imitation* (in Russian), pp. 149-150. Italics are mine— V. Bekhterev.

4

The Interrelation of Personality and Society

The development of the personality as social formation is rooted primarily in those special circumstances that evolve at the time of its birth due to a distinct heredity and, consequently, to differences in temperament as well as in the evolution and nature of individual inclinations. This determines unequal abilities for skill acquisition, which in turn—combined with the external circumstances surrounding the personality—results in its particular status in society. Different personalities are born into different environments, which generate great variety in the inner resources of each personality that consist of acquiring skills or so-called associative reflexes and that maintain the various conditions of its existence.

As we have already mentioned, the neopositivists admit that society is not a product of personality but quite the opposite—personality is a product of society. In this respect, E. Durkheim was particularly instrumental in establishing this view. According to him, personality without society would never have matured to the level of use of categories and general concepts and, consequently, would have been unable to reach the state of the reasoning personality.¹ Dragichesko—another representative of the same school—reduces the whole of the individual psyche to social life. For him, even the evolution of consciousness is unimaginable without society. While recognizing two processes as the actual conditions for the emergence of consciousness—integration and differentiation, or combination and relativity—and asking what the source of these two laws is, he concludes that the

physical environment, having remained unchanged in the course of history, could not have been the source of changeable contrasts, novelty, and complexity, and, consequently, could not have been the source of the emergence of consciousness, since "the relative monotony of the cosmic environment could not have served as the stimulating condition—or the stimulus—that caused the emergence of consciousness." As opposed to this, the social environment with its expansion of social groups leads to specialization and the division of labor:

The simple fact of numerous small social circles and groups merging together into a single society infinitely complicates internal social life: it generates a great variety of work, behavior, mentality and religious beliefs; all of these become accumulated, complicated and undergo continuous fluctuations and changes that represent a complete contrast to the immobility and relative uniformity of the cosmic environment.

Thus, the evolution of the social environment necessarily conditions and causes personality differentiation, the mobility and instability of interpersonal relations, the original nature of activity, etc., and most importantly, an animated confrontation and irrepressible competition arise between group members, which in turn awakens and strengthens consciousness.²

In another place in the same work, the author formulates his idea in the following manner: "consciousness emerges wherever the life phenomena subordinated to the influence of the physical-chemical environment are affected by social conditions" (p. 114). "Consciousness," says the author in conclusion, "is a social phenomenon based on the experimentally proved reality of society. One can conceive it as a social phenomenon embodied in an organism and appropriately manifested in psychological language." This extreme view, however, encountered certain objections in literature elaborated particularly well by G. Palant in his book.³ He emphasizes the role of physiology, heredity and race, and mentions animals with a social organization whose intelligence is not superior to that of animals leading a solitary life. Bees and ants, for example, developed a collective instinct, but have no individual intelligence, display no initiative and are incapable of any innovations. Social differentiation is the fruit of invention, while invention

itself is "primarily a biological and psychological fact, and only then a social one." Nor is there any parallelism between social evolution and the degree of intellectual progress. Contrary to Dragichesko's belief, the role of heredity is not regressing, and, although due to intermixture, race is losing its significance regarding mental ability, the yellow, the black, and the white races do not exclude mental differences.

All of the above leads the author to the conclusion that, although the role of society in personality development is enormous, the individual factor underlying the biological nature of personality cannot be totally erased by the influence of society, and therefore personality represents—and will always represent—a biosocial phenomenon. Thus, the author arrives at the conclusion established by the studies and penetrating analysis of our compatriot, Professor E.V. de Roberti of the Petrograd Psycho-Neurological Institute, who founded the biosocial theory accepted nowadays by the majority of prominent sociologists. According to this theory, human intelligence, or reason, results from a combination of the organic properties of matter that constitute the subject of biology with the supraorganic properties studied by sociology. It can thus be considered a fact that the personality is a biosocial phenomenon, its biological element consisting of temperament, instincts, inclinations and abilities.

One should bear in mind that the preparation of the personality for social activity and its adaptation to the social environment start at a very early age, together with the development and regulation of its actions. These regulators are, on the one hand, counteractions, prohibitions, and condemnations that limit freedom of movement and act as inhibiting factors, and, on the other hand, praise, encouragement and reward that stimulate and revitalize activity. Direct observation can prove that both begin to affect the child from the first days of his life through the appropriate treatment dispensed by his natural mother or nurse. In time, the child is familiarized through experience with the general social rules in his environment and he learns in the same way, as well as through imitation, what is permitted and what is not, what is condoned and what is not. Finally, familiarization with the laws that define certain repressive measures for violating the minimal requirements of individual behavior in a particular environment completes the regulation of his actions.

Various restrictive measures established by law represent, on the one hand, an act of defense on the part of the collective, while, on the other, an inhibitor of individual aspirations that transgress social rules. For their part, all

measures of encouragement stimulate individual persons to work for the common weal. Step by step, individual behavior is thus oriented towards collective interests, and a person gradually develops into a social personality. In this case, the child repeats, as it were, the history of society, since the development of the latter followed the path of a gradual subordination of individual interests to the common interests of a collective.

At the highest rung of evolution of a human society with complex organization, individual behavior is no longer regulated by measures of actual or possible repression, but by the mere judgment of one's inner circle and particularly by the judgment of public opinion. In this case, words of condemnation—even of a hypothetical one—are enough to inhibit an action, whereas words of encouragement act as an impulse that stimulates one's activity. Apparently, at this stage of personality development, social skills have already evolved and in certain cases need only the guiding influence of approval or condemnation from one's intimate friends and family, from public opinion or from one's own principles.

A well-developed social personality does not even need either praise or disapproval from his family and friends or from society. It acts according to an already developed habit, for the associative reflexes acquired through upbringing and life experience become habitual and determine human behavior under different circumstances. An army volunteer who is risking his life is not at all motivated by approval or disapproval. His actions are unavoidable, for his external circumstances are such that he cannot act differently under given conditions. A person embarking on a political career, which also involves certain risks, follows a path prepared by all his past life in a particular social milieu. One way or another, social cooperation achieves conformity of individual actions to its rules whenever it is a matter of reaching collective goals. This, in turn, involves the individual personality in public or collective life to such a degree that the person cannot but act in accordance with his habits and traditions developed around the interests of the collective as an entity.

Children in a collective are known to be far more cruel than adults, and this is because they have not been socialized to a degree that would enable them to restrain their individual desires and coordinate their actions with common interests. Because of this, even in their collective behavior children manifest inadequate restraint and are predisposed more than adults towards a harsher condemnation of individual members. In this respect, women's collectives

possibly come somewhere between children's and men's collectives—at least at present when women have not yet fully entered public life on an equal footing with men.

One should add that different personalities exist under different conditions in terms of both their contact with other people in their milieu and their upbringing, which is of enormous consequence for the future development of every personality. Therefore, personal individuality is in a significant measure determined by society, its interests, traditions, and other social rules. In other words, a personality is a product of the society that itself represents millions of such personalities in a series of generations replacing one another. It is the totality of those circumstances, which are different for each personality, that make the latter characterologically different as a social phenomenon from any other personality.

It goes without saying that professions and occupations naturally reflected in the general development of associative activity of each personality are tremendously important in this respect. Here one should take into particular consideration division of labor—sometimes highly sophisticated—which cannot but exercise a powerful influence on the development of individual personalities. In exactly the same way, different levels of material well-being must be somehow reflected in the makeup of individual personalities.

It is therefore clear that surrounding conditions exert an essential influence on the development of the personality by reinforcing and maintaining its various characteristics through the idiosyncratic features of various professions or different living conditions.

In the same manner, the external effects of geographic and climatic condition are necessarily reflected in personality development. When assessing the latter, one should also take into consideration the state of physical health, past ailments, and other organic conditions that affect the nature and agility of movement.

Although each personality appears to be distinct from all others, being characterized by its own combination of habits, functioning as an individual and expressing itself according to its worldview, education, social status, etc., in point of fact the activity of each personality is in considerable measure social rather than individual.⁴ Thus, first of all, the language used by a personality belongs to the whole nation and not to the individual. At the same time, individual behavior, too, is subordinate to the laws of society. Every person is to a certain degree a slave of the tradition and form developed by a

society, as well as of its dominant prejudices and superstitions. Not only does one use the language common to all, but one also wears the socially sanctioned style of dress; follows the fashion; espouses a degree of nationalism; abides by the generally accepted rules and regulations; shares a common culture; has common legal and ethical concepts, a more or less common worldview, common ideals; and evaluates both past and present events in approximately the same way.

All of this is determined by the fact that, due to imitation, suggestion, and persuasion, a personality absorbs, during its upbringing, prevalent habits and views and becomes a typical representative of its own milieu.

Similarly, a person shares common beliefs, and repeats the statements about the surrounding world and the creation of the Universe that were made not by him but by others and were accepted by the whole society. He lives by ideals shared with others, and, finally, his aspirations are also common to many.

In a word, instead of pursuing unique self-expression, the personality—through most of its actions and statements—represents society rather than its own self. It is evident from the above that a personality is more of an imitator than an individual creator. In other words, it is in considerable measure a social product rather than a unique form.

Thus, a personality incorporated in society loses its own identity, becomes part of the society itself, relinquishes in this sense a considerable portion of its independent existence and becomes a practitioner of social customs. It is clear from the above that society rules over personality, for it is in charge of the complex correlations in which individual personalities act merely as executive organs. In other words, the joint associative activity of society is the resultant force, as it were, of individual activities pursued by its members, and has its own laws of development based on the interaction between its members.

A collective depersonalizes individuals by ensuring the prevalence of quantity, whereas a personality represents qualitative characteristics by revealing individual creative energy. Therefore, individual behavior in isolation cannot be the same as in a crowd. G. Le Bon says in his *Psychology of Socialism*, p. 54:

It seems, however—and this point is a most important one—that individual behavior is completely opposite to one's behavior in a crowd.

An individual personality is preoccupied with its private interests only. A person in a crowd is guided by common interests.

In a crowd, all national characteristics come to the surface, and when a crowd consists of particular social groups, it is all the peculiarities of these groups that come to the surface. According to Tarde (I.e., p. 132), "national life will always require the individual to sacrifice his most cherished habits and personal gains in the name of the public interest."

Although persons of different nationalities speak different languages, they abide by universal social customs, have the same, or approximately the same, ethical and aesthetic concepts, express the same views of an abstract nature, proclaim more or less the same universal ideals, etc.

In contemporary society, we encounter different social strata, class divisions, professional groups, and various circles that seem to divide mankind into separate parts. These subdivisions, however, are not deep enough to destroy the common element of the multiform external manifestations of associative activity practiced by various persons.

Incidentally, one should mention that these subdivisions in their role of minor societies have exactly the same power over their members, who for this reason always act in conformity with the customs, views, and regulations universally accepted in the given association.

So in this case, too, a personality proves to be a social being in the true sense of the word: it reiterates generally accepted—rather than its own unique—views, common customs and in some cases performs common actions.

In this case, all individual acts seem to be predetermined. A person in society is unconditionally dominated by social requirements. His dress has been defined beforehand, his treatment of others is dominated by certain conventions, and his manner of action is bound by social form.

In a word, a person in a public place is an automaton, as it were, who abides by the general rules and follows the generally accepted customs. One can afford originality of self-expression in actions and speech only to a certain extent—mostly in content and very insignificantly in form, for both the nature of actions and the content of speech, let alone form, are to a considerable degree determined by the membership of a gathering.

Under social pressure and under the overwhelming influence of social statutes, a person involved in social activity has hardly any opportunity to freely express his vital needs: to move spontaneously, to sit naturally, to relax

when necessary, to breathe freely, etc., since all of this is strictly regulated by social customs, rules, etiquette, certain pre-established forms, and so on. As soon as a person makes an attempt to transgress the rules established by society, the latter grips him in a social vice, as it were, with all the force it can muster, and brings him back into the fold. In case of sharper deviations, society crushes him almost to the point of annihilation, in the social sense of the word, as if taking revenge for the daring transgression of the norms established by society.

When speaking about the role of society and its power over individuals, one should not forget about the yoke of public opinion, against which even the strongest personalities cannot fight. This is how Le Bon describes the ancient world in this respect:

The Greeks who, according to ignorant rhetoricians enjoyed tremendous freedom, were, in fact, under the yoke of public opinion and tradition. Every citizen was enmeshed in a net of absolutely inviolable beliefs: no one would ever dream of challenging the generally accepted ideas and would abide by them without protest. The Greek world possessed neither religious freedom, nor freedom in private life, no freedom of any kind.⁵

One should bear in mind that social manifestations of personality are also passed on from generation to generation. In this case, the law of evolution is certainly at work, with general, or social, manifestations of a particular society being affected by continuous changes occurring with a greater or lesser speed. Nevertheless, an individual personality is invariably dominated by a given society's social manifestations, even though they constantly change due to generational evolution.

We know that fashions, views, and theories sometimes change extraordinarily fast, thus creating an impression of highly variable phenomena. However, an individual personality follows all these changes, submitting thereby to the constantly changing manifestations of social life.

It is apparent from the above that the very organization of society rests on the principle of its domination of the personality. The customs and the laws of society are categorical and demand an almost unconditional obedience. As society continues to develop they become ever more differentiated, constricting thereby individual expression in all spheres, limiting private aspirations and destroying individuality. At the same time, with the evolution

of society its laws become both more complicated and more imperative regarding common needs of a collective and, therefore, constrain the personality even more.

In the final analysis, the development of social life, too, is expressed through the establishment of a certain relationship between personality and society. Interestingly, both the self-determination of a personality regulated by the development of civic consciousness and the socialization of its various manifestations appear to be highly mobile processes, whose resultant force is continuously shifting from one direction to the other. Personal freedom and social necessity; individualism and socialism—these are the two sides of the social process that follow the path of social evolution. Nevertheless, personality and society are not opposites, since the most perfect society can consist only of the most perfect personalities. It is therefore in the interests of society to encourage the development of, rather than to suppress, personality as long as this development is not detrimental to society. Thus, society ought to promote all-round personality development, while protecting itself from disintegration by reinforcing solidarity between its members. This should be the only goal of the organized society we call the state. But this type of correlation between society and personality belongs to the future, because until now society has been suppressing personality with its traditions, customs, rituals, etiquette and other conventions and rules.

The fact that society has always suppressed the personality does not mean that this is the most appropriate type of correlation between personality and society. Continuous improvement and ordered freedom, the highest development of creative activity and the socially accomplished personality, equal rights for all, and the abolition of oppression and exploitation of one person by another constitute goals of future society that have not yet been achieved on this earth. In this case, the goal of producing the perfect personality coincides with the actualization of the most perfect society. Here personality represents the highest social value, and the suppression of its valuable individual aspects by the social environment should be recognized as a totally antisocial phenomenon.

However, one should not lose sight of the fact that personality will always remain the product of social life. In contemporary society, first the family, then the school or the factory, as well as the whole social milieu, creates the personality. The latter thus becomes a direct product of society that, in turn, developed on the basis of the given biological organization of its members.

At the same time, everything that constrains personality and represents social standards is the product of individual social activity, but in this case we mean the lower levels of individual activity, such habitual actions bordering on automatic behavior.

Just as the actions of an individual become automatic and routine due to their frequent practice and habit, so the above-mentioned social rules, too, become firmly established, generally accepted, and habitual due to their repetition and custom continued in every generation.

Regarding conditions of social relations, one should say that customs and "social standards," in fact, constrain personality only in its more or less automatic activity. They do not touch its individual manifestations and those of its aspects that make the personality a unique entity with its own views and aspirations, provided the latter do not lead to the violation of other people's rights.

This individual aspect of personality must possess a special significance in social life as well, for it is this aspect that mainly determines the progress of society.

It is only due to its individual abilities that a personality can rise above the masses and carry out the productive activity that ensures the onward march of the intellectual evolution of mankind.

One may say that unique personality development is the basis of social progress, and the best way to secure it is to grant total freedom of individual views and aspirations as long as the latter do not interfere with the interests of other individuals and especially of society as a whole.

It is therefore unjustified to hold the rather popular view of an interaction between the personality and society, according to which the two are seen as balance pans oscillating in opposite directions, so that the interests of society benefit from the suppression of the personality, and vice versa.

On the contrary, one should recognize that, while properly looking after public interests, a good social organization will also grant as much freedom as possible to the unique development of personality, for it is only this development of individual originality that guarantees the progress of nations.

The proof of this is provided by all great personalities whose significance for history is indisputable. The role of great people is in their synthesis of aspirations existing in a given race. According to G. Le Bon "their discoveries always represent the result of a long series of previous

discoveries. They erect edifices from the stones slowly hewn by their ancestors."⁶

While partially reflecting their own nation, great people undoubtedly disrupt to a degree the regular course of the evolutionary process, although they are unable to alter or halt its movement. However, even their mode of expression is conditioned by external causes whose absence would render the action itself impossible. According to Le Bon,

the truly great people in politics are those who foresee the germinating needs and developments prepared by the past, and point at the path to follow. Like the great inventors, they too synthesize the results of the long preceding work.

One cannot lead a nation without embodying its dreams. For the Hebrew slaves lashed by Egyptian whips, Moses personified their thirst for freedom harbored for years in their souls. Buddha and Jesus were able to understand the infinite sorrows of their time and to express in religion the need for charity and compassion, which in the epochs of universal suffering were beginning to make their appearance in the world. Through religious unity, Muhammad achieved the political unity of a people divided into thousands of hostile tribes. The artilleryman Napoleon embodied the ideals of great glory, vanity and revolutionary propaganda that were then the main characteristics of the nation that followed him for fifteen years all over Europe in pursuit of the most reckless adventures.⁷

A leader is a leader only when his aspirations merge with those of his army, in the same way a ruler is a ruler only when he expresses the aspirations of his nation and identifies with it. "L'état, c'est moi," was the well-known formula of an eminent French king.

On the other hand, the nation also identifies with its ruler. It is proud of his glory and lives by his happiness. Taine is right when he describes a feudal fief as a kind of fatherland

identified in the vassals' minds with their feudal lord and his family: they take pride in him, tell one another about his war exploits, greet him with loud cheers when he rides along the street with his retinue, and enjoy through empathy the splendor that surrounds him.⁸

The fact that good social organization is not hostile to personality is proved by the existence in a collective of certain persons who stand out due to their outstanding individual characteristics. Far from being suppressed by the collective, these individual characteristics, as well as the persons in question, become the object of worship.

Deification of popular idols sometimes reaches extraordinary proportions; one need only mention Robespierre, who was a veritable god of democracy during the French Revolution, or Marat, who was compared to Christ in funeral speeches. "Heart of Jesus and heart of Marat, you are equally worthy of our veneration," were the actual words of one of the speakers. Another continued, "Marat, like Jesus, passionately loved the people; like Jesus, he hated the aristocracy, the priests, the rich and the dishonest; like Jesus he led an austere and simple life."⁹

And finally, another example of deification is the great Bonaparte whom all of France blindly venerated both during his lifetime and after his death. The kind of charisma he emanated is conveyed by one of his contemporaries who describes a ceremony during which the division generals, including Augereau, were first presented to Bonaparte on his appointment as the Commander of the Italian army. One may say that being an old soldier, uncouth, heroic and proud of both his stature and courage, Augereau had already been prejudiced against the "upstart" sent from Paris, and had been incensed by his appointment. They were brought into the room. Bonaparte made them wait. He finally appeared in full military dress, and having put on his hat he explained to the generals his intentions, gave out his orders, and dismissed them.

Augereau did not say a word, and only when they were out in the street did he suddenly explode and start swearing in his usual fashion, agreeing with Massena that the little general inspired fear in him, and wondering why on earth he would feel so utterly insignificant under the gaze of His Excellency.¹⁰

It goes without saying that Napoleon's charisma continued to grow with each new success he scored, and finally equaled that of a divinity. The bravest of people lost face in his presence and trembled like children when they had to approach him.

Another proof of this charisma is his unprecedented escape from the island of Elba, when within a few weeks and without bloodshed he overthrows all the powers that be and becomes yet again the absolute ruler of the whole

country. The charm of his personality follows him into his grave, for his life and glory have become a legend. In general, the charisma of great military leaders like Napoleon, Suvorov, and others survives their death and comes close to the charisma of a divinity. Here is another example: General Vandamme, who himself had fought in many battles, was saying to a friend of his on the way to the Tuileries Palace in 1815, when Napoleon's star was no longer as bright as it used to be:

My dear, the charm this man exudes affects me so strongly that I cannot explain it even to myself; its degree is such that I, who fear neither God nor Devil, tremble like a child in front of him, and he can make me go through the eye of a needle and send me into a fire.

The effect of this charm on the masses is shown in Count Leo Tolstoy's masterful description of the crossing of the Vistula by a squadron of Polish soldiers before Napoleon's eyes. "'Long live the Emperor!' the Poles were cheering ecstatically, disrupting the front and crushing one another in order to see him...They were ordered to find a ford and to cross to the other side by wading." The commander of the Uhlans asked permission to swim across the Vistula rather than look for a ford, and having obtained this permission, the old mus-tached officer immediately threw himself into the river with a happy face, shining eyes and a cheer on his lips:

Hundreds of Uhlans followed him. It was cold and terrifying in the middle of the river, in the rapids. Although they were trying to hold onto one another, the Uhlans were falling off their horses. The horses were drowning, and so were the people; the rest were trying to swim in the saddle or holding onto a horse's mane. They were trying to swim, and were proud of swimming and drowning in this river before the eyes of the man who was sitting on a log without even looking at what they were doing. After crossing the river they cheered again, looking rapturously at the place where Napoleon was standing.

Our own military leader, Suvorov, possessed the same kind of charisma. We know that when his army was crossing the Alps the soldiers became so exhausted that they could no longer move. When informed about this, Suvorov ordered a grave to be dug for himself in the mountains. The effect of

this action on the army was such that the exhaustion was immediately forgotten. The campaign continued and ended in a total victory.

Tarde is quite right when he says that

When the crowd admires its leader, or when the army adores its general, it admires itself and appropriates its leader's high opinion of himself, which, in the guise of pride in his origins or genius, shines forth from the face of Louis XIV or Cromwell, Alexander the Great or Scipio, or even some popular leader.

It goes without saying that the charisma of a military leader creates fertile soil for suggestion, and the latter often works wonders and saves the situation. The following is an example of Napoleon's magic effect on an Austrian officer who had the force of a whole detachment behind him. This episode was related by Dragomarov (I am quoting A.S. Rezanov):

Napoleon with his retinue and a small convoy came upon a four-thousand-man column of Austrians. One of their officers approached Napoleon and demanded his surrender. "Do you know to whom you are talking? I am Commander-in-Chief. The whole army is behind me! How dare you! Inform the commander of your column that I demand your immediate surrender! If you fail to lay down your arms in five minutes I shall give orders to shoot every one of you!" And they laid down their arms and surrendered.

The explanation of charisma lies both in one's personal qualities and certainly in success. In the above example, however, personal qualities were apparently more important, since even in the absence of success—when one's charismatic power usually runs out—Napoleon's charisma survived, and so did that of another legendary hero—Charles XII. I have deliberately cited examples of those historical figures, whose names retained their charismatic appeal after their death, even though they were eventually defeated and the cause they championed was not truly their people's cause, as were, for example, the struggle of the Polish people for independence associated with the memory of the great Kosciuszko, or the struggle for the liberation and unity of the Italian nation associated with the memory of another remarkable personality—Garibaldi. In the case of the last two examples, it is quite natural that as long as the idea or the guiding principle championed by the people

survives, the hero who linked his name to this principle by personifying it, as it were, will retain his charismatic power in the people's memory and will be surrounded by the halo of glory bordering on deification.

All of the above leads us to the discussion of the art of controlling a composite personality. This art should apparently take into consideration the nature of the latter. Therefore, one set of methods may be appropriate for a crowd, another for the organized masses, yet another for a theatre audience, and finally some special methods of control may prove suitable for the collectives whose tasks involve decision making.

Let us first examine the ways and means of guiding or controlling a crowd. The power of the crowd lies, above all, in its unity and homogeneity, characterized by a common ideology or a common goal. Dissent in this regard is highly destructive. Therefore, the art of controlling a crowd consists of maintaining the common ideology at all costs and of removing any possibility of dissent by all available means. For this purpose one may use various methods, which basically come down to isolating the crowd from extraneous influences and increasing the homogeneity of the collective through every means that supports and justifies the aspirations of the crowd.

Some authors (Tarde, Le Bon, Sighele, and others) agree that the crowd is hopelessly obtuse. This is, however, a gross exaggeration. Giddings is right when he claims that "the social mind does not differ too much from the individual mind in its manifestations." It is beyond any doubt, however, that the crowd is incapable of reasoning, and that reasoning has hardly any meaning and value for the crowd. The latter is cemented into a single whole mostly by mood; therefore, one should rely on fiery words rather than persuasion in order to excite the crowd. Once this is achieved, all one has to do is to rule, dispense orders and serve as an example, since the latter acts like suggestion, and all famous military leaders use it.

In general, one should bear in mind that the more cohesive and especially the more homogeneous the mass is, the easier psychic conduction becomes, the faster social contagion spreads, and the more likely is an explosion of enthusiasm under the influence of a fanatical sermon. Every individual absorbed by the crowd experiences a weakening of the inhibiting influences and an intensification of the associative reflexes of an imitative nature.

Due to suggestion, an individual in the crowd loses to a significant degree his critical thinking in the general atmosphere of weakened and stunted moral principles, heightened sensitivity, and astounding suggestibility.

It goes without saying that the crowd is not always capable of assessing correctly even its own heroes, and often surrounds with slavish adoration those who in reality are the evil geniuses of their nation and its destiny.

Another means of controlling the crowd is its organization. Only spontaneous crowds have no organization of their own, which is why they are capable of impulsive action. They are incapable, however, of practicing self-restraint in any more or less complex collective action. Only organized masses of people, like the army, are capable of it. On the one hand, the organization of the crowd should consist of its correct differentiation into separate parts, each performing its own functions, and on the other hand, it should unite all these parts through their joint action and the principle of solidarity, unity and mutual support.

Finally, the pursuit of particular goals in any struggle should be comprehensible to the crowd and ought to inspire it. A.S. Rezanov says in *The Army and the Crowd* that "the perfection of the external organization of a regular army will not result in the desired advantages without an inspiring moral force, for such an organism is a decaying wreck reminiscent of a dying lion." The examples of this are the defeat of the Italians in Abyssinia and the long unsuccessful operations of the British in the Transvaal. However, organization and inspiration are not enough. There should also be faith in one's leader, one's cause, and one's power. This faith should be total and unshakable. It constitutes what one calls the morale of the army, without which even the best organization is powerless. As soon as faith in the leader and the cause is shaken, the crowd begins rapidly to disintegrate. This is what happened to Napoleon's army after arriving in Moscow. The same was observed in the Russian army during the World War, when the shortage of armaments, the Tsar's inept military leadership, as well as rumors about the treason of the incompetent, rude, and ignorant war minister, caused the troops to lose faith in the possibility of their victory, which led inevitably to the demoralization of the army, completed by revolutionary propaganda.

Faith in victory is undoubtedly of paramount importance to the crowd, so particular foresight and caution should be exercised in order to avoid straining this faith excessively, as was the case during the Russo-Japanese war. According to G. Martynov,

for some reason the Russian army in Manchuria regarded deception as a means of raising the spirit of the troops. For example, they would always

announce before battles that there would be no retreat, and then the very same people would give the order to retreat; they would announce our victories, which later proved to have been defeats; they would spread rumors about the deaths of Kouroki and Oyama (the former managed to die three times), about the incredible losses and the epidemic diseases in the Japanese army, about the sinking of the enemy fleet, and about famine and uprisings in Japan.

Obviously, deception will invariably be disclosed one way or another; it cannot fail to undermine the moral authority of the commanding staff and to result in a general disaffection followed by loss of energy. Clearly, faith in victory should be sustained by all possible means, and good timing is particularly important here. " 'A division is coming to our rescue!' one of Sevastopol's heroes shouted, and the defenders of the hill repulsed with renewed energy the enemy's mad attacks, while their actual reinforcement consisted of the pitiful remnants of a few detachments."¹¹

At the same time, one should not overlook the extraordinary gullibility of the crowd and its predilection for exaggeration. That is why it is just as easy for a popular leader to earn unmerited adoration as to completely lose the people's trust.

Returning to the subject of collectives, one must further add that collectives generally manifest less initiative than individual persons. The latter, however, can better implement their intentions through a collective.

As we have already mentioned, alongside unity of interests, every collective also presupposes a certain self-limitation of individual members in the interests of the collective—in other words, an inhibition of individual aspirations that go against the common interests. On the other hand, a collective supports all individual aspirations that benefit the common weal.

In the course of history, the evolution of collective life originally led to the suppression of personality by the family, by the clan and, subsequently, by the state. However, the more recent period of human history has, until the Great War, been gradually eroding those institutions which put excessive constraints on an individual, and this erosion resulted in the liberation of the personality from its fetters. With the progress of history the personality gradually frees itself from the chains of the family, the tribe, and the state, and embarks on the road to self-determination.

Undoubtedly, this is directly related to personality development and progress. The truth is that human personality represents not only something individual, but also something recurrent. Besides, a higher personality development implies a more valuable contribution to a human collective. That is why, when individual development was rather limited, the value of the personality in the collective was reduced mostly to increasing the collective in terms of its numbers, whereas the personality's characteristics were not duly appreciated by the collective because of their insignificant value in general.

At the most, an average individual can make his presence known during voting by raising or not raising his hand, or by leaving through this or that door. In any case, the minority must defer to the majority, and it is only in highly exceptional cases that the minority protests publicly. Quite often this protest is of purely theoretical significance— in other words, it is not expressed in any action. The minority merely preserves its political or social physiognomy, and falls away, as it were, from the majority, which implements its policies unimpeded. Instances of resistance and confrontation are certainly possible during particular periods, but only when the minority can find support among some actual forces that would prove sufficient to counteract the majority.

With the gradual development of the personality as a unit that contributes its initiative and unique intellectual abilities to the activity of the whole collective, its qualities become particularly important in the light of collective interests.

For this reason, the role of the modern state amounts to facilitating personality development, satisfying individual interests, and encouraging the fusion and cohesiveness of individual members into the powerful collective of the state.

However, parties themselves, as well as other collectives, including the state, also strive for self-determination. This kind of self-determination aims at establishing normal relationships between different collectives, and basically should not violate the principles of individual self-determination: e.g., personal immunity; sanctity of the home; freedom of speech, assembly, and associations; freedom of conscience; national equality and free elections of governing bodies.

Yelinek, as we know, recognized that the state wields absolute power over the personality, and determines its own rights in regard to the personality.

However, Benjamin Constant and our M.M. Kovalevskii (*Common Constitutional Law IT*) refute this theory. "I believe," says Kovalevskii,

that the presently dominant theory, supported by most German publicists and claiming that any constitution is no more than a voluntary self-restriction of state authority, cannot serve as a bulwark against the danger to freedom represented by a doctrine that allows for unlimited state sovereignty (p. 102).

In another place in the same work the author says,

From our point of view there is no need whatsoever to accept the state's absolute power, which would enable it to revoke the freedom of personal self-determination in the interests of some state necessity. No such necessity exists or can ever exist—on the contrary, human solidarity demands that we preserve and advance the freedom of personal self-determination.

And later on: "The state should have only those rights, which, instead of inhibiting, serve to protect and guard personal self-determination as an indispensable condition of social solidarity" (pp. 103106).

It is thus apparent that governmental oppression in contemporary states would be paralyzed by the decentralization of administration, by total democracy, accountability of all civil servants to courts, and by inalienable individual rights regarding the freedom of assembly, associations and conscience, national equality, and the right to elect one's government. Therefore, under normal conditions, a modern state collective presupposes a governing body properly constructed on the principle of representation, national self-determination, and local self-government elected by voting.

The quality of any collective always depends on its constituent units. When discussing this question, Sighele says that "analogy between the properties of the units and the aggregate is possible only when its parts are somewhat similar," when its components are homogeneous. A cosmopolitan action does not reflect the national goals of a particular people, and neither does a group of individuals from this people: a hatter and a scholar in a jury cannot express their opinion as correctly as would a gathering of experts. The mere similarity of constituent elements, however, is not enough to establish an analogy between the properties of the elements and those of the aggregate

they form. One also needs a connection between the components of the aggregate. That is why spontaneous and unorganized gatherings, such as a jury, a theatre audience and a crowd "cannot reproduce in their manifestations the qualities of their component units," for there is no organic connection between them as there is, for example, between the members of a family or a social group. Sighele's conclusion that the properties of the aggregate are determined by the properties of its component units is true for aggregates that consist of homogeneous and organically interrelated units. It is not accurate, however, for partly homogeneous and weakly organized units, and it is wrong for the aggregate consisting of heterogeneous and nonorganized components.

When a collective is homogeneous, its activity largely depends on whether its members are educated or uneducated, whether they possess certain technological knowledge and experience, whether they practice any profession, or whether they belong to any social class. All of this undoubtedly shows in one way or another in the activity of the collective, makes a distinct imprint on its manifestations, and affects in a particular manner its position on various issues.

As for the manifestation of collective activity, it depends mostly on the cohesiveness and organization of the collective. The weaker its cohesiveness and organization are, the less chance there is of achieving the goals that brought this collective into existence. Thus, a crowd may become awesome due to its cohesiveness. If you create division in a crowd regarding its object of interest, it turns into a powerless rabble.

Any army is held together solely by its cohesiveness and organization. If you divide the troops and disorganize them by relaxing the discipline based on blind obedience, you will transform an army into a rabble incapable of decent behavior, let alone of offense and defense.

Collectives also differ in the character of their actions depending on their unifying principle. Thus, a crowd is united by its mood, which is the sole motive power of all its actions, and whose influence makes these actions seem automatic.

While carrying out collective activity, gatherings engaged in intellectual pursuits almost invariably reveal considerable caution and foresight in their actions.

Yet even at this type of gathering some conditions can cause the mood to rise to a certain level. Then the decisions and the actions of the gathering will

show the effect of this particular mood.

Irrespective of the complexity of various collectives, their activity is defined by the resultant force of all their constituent units. The same is true for complex collectives consisting of a number of smaller collectives which send their representatives to the larger one. The only difference is that the resultant force of behavior is shaped first in the smaller collectives and then in the complex ones.

Collective reflexology should examine collective activity from the viewpoint of both the present moment and the past events. Just as in the development of an individual personality, its past circumstances and experience are reflected in its character and its aspirations, so a social collective cannot be studied without a historical perspective.

Thus, for example, one cannot examine the collective activity of the European nations drawn into the Great War without turning to the past, to the old struggle for national sovereignty, which essentially affected the whole history of the nineteenth century. The Spanish war for independence, the Holy Alliance of 1815, the restoration of Greece, the unification of Italy and of Germany, the federal structure of Austro-Hungary, the Russo-Turkish war, the Russo-Japanese war and the later Balkan war basically represent different stages of the narrowly nationalist aspirations of various nations. At the same time, the revolutions that occurred in different countries, as well as the shift of social forces in favor of the rule by the people indicate the gains of socialism. The Great War was perceived as the struggle of right against might, but it, too, concealed the nationalist aspirations of both Germany and the Allies.

The Great War divided the nations, and everywhere socialist tendencies gave way to nationalist ones. The idea of the International seemed to have died during the initial stage of the war. But the more obvious its protracted nature became, the stronger the idea of internationalism grew, and appeals for unity were issued by all the Socialist parties of the world. This was first manifested in a particularly acute form in the Russian Revolution, which, on the one hand, ended the struggle for national independence by advancing its slogan of national self-determination, while on the other hand, it made a decisive move towards internationalism.

Notes

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1. See his "Représentation individuelle et représentation collective," *Revue métaphysique et de Morale* 1898, and *Les formes élémentaires de la vie religieuse*, Paris, Alcan 1913.
 2. Dragichesko, "The Relationship between Psychology and Sociology," *New Ideas in Sociology*, Issue 2, 1914.
 3. *Us antinomies entre l'individu et la société*, Paris, Alcan, 1914.
 4. V. Bekhterev, *Personality and the Social Conditions of its Development and Health* (in Russian). A separate edition. St PB.
 5. Le Bon, *Us lois psychologiques de l'évolution des peuples*, Alcan, 1884, pp. 131-132.
 6. G. Le Bon, *ibid.*, p. 144.
 7. G. Le Bon, *ibid.*, p. 148.
 8. H. Taine, *Us Origines de la France Contemporaine* (in Russian), St. Petersburg, 1880, p. 19.
 9. Dr. Cabanis, *Marat inconnu*.
 0. G. Le Bon, *Psychologie des foules*, p. 252.
 1. A.S. Rezanov, *Le*, p. 82.

5

The Collective as a Composite Personality

Guided by the argument that the property of an entity depends on the composition of its components, Spencer postulated in his *Introduction to Sociology* that the most important goals of human societies correspond to the principal characteristics of the individual person, society itself being, according to Spencer, nothing other than an organism.

According to A. Comte (*Système de politique positive*. Paris), human society represents an "ever-existing personality," and in this sense there is a certain similarity between the views of Comte and Spencer. Subsequently, a number of other authors, including most recent ones, have supported the comparison between society and the organism. This analogy, however, is of a purely external nature, and by now it has been essentially abandoned: no one will agree that sociology should be reduced to biology.

Yet even after discarding the view of society as an organism, it is not as simple as it initially appears to be to define what a society, or a collective, or—as I would suggest calling it—a composite personality, is.

According to some authors, we are dealing with a group of individuals who provide certain services for one another and are in general mutually useful. This introduces the principle of utilitarianism into the basis of definition. Tarde, however, although admitting the clarity of this definition, regarded it erroneous and preferred a juridical interpretation of society, wherein individual members enjoy certain mutual rights established by custom, law, or rules of propriety. Finally, according to Tarde, one could develop a purely

political or a purely religious concept of social communion, since all those who are united by the same political plan and form a single political body, as well as those who profess the same religion, would have a legitimate right to be considered members of the same society. In this case, according to Tarde, the similarity of political goals and of religious belief is due to imitation. Therefore, he defines a social group as an "assembly of creatures who are prepared to imitate one another, or who are similar without imitating one another, since their common features are ancient copies of the same original."

Many sociologists see the most typical characteristic of society as the interaction of its members. "The long-term, continuous, many-sided and necessary interaction established in any permanent, rather than accidental, aggregation of living beings" is, according to De Roberti,¹ characteristic of "community." This opinion is shared by Simmel, Novikov, Sorokin, and in part by Gumplovicz (group interaction), Giddings, Draghicesco, Bouglé, Fouillée, Grasserie, Ward and others.²

It is not enough, however, to talk about interaction: one should define what kind of interaction is meant. Most sociologists mean the "psychological" nature of interaction in a social phenomenon. But "psychological" interaction, too, can be different. For example, it can be determined by mutual benefit, by a common goal, by a common interest, by common origins, by common links on the basis of imitation (Tarde) or by mutual strife and persecution.

In this respect we encounter a considerable divergence of opinion amongst authors, but we shall not discuss these views in detail. My point of view is that we should recognize as a society any association of interacting individuals brought together by any common interests whatsoever—political, economic, religious, business or any other. Tarde's view that society presupposes similarity based on imitation strikes me as being both too general and too one-sided: although imitation can indeed create a society, it is also true that everybody imitates everybody else, including those who do not form a society in the present sense of the word. On the other hand, society is based not only on imitation but also on the division of labor which, too, creates society, for the division of labor presupposes society just as imitation does, and essentially society cannot exist without the division of labor, just as it cannot exist without imitation.

A chance conglomeration of numerous individuals in a given period of time in a particular place cannot be described as a composite personality, a society or a collective. This kind of gathering is a crowd without any unifying

principle, and since each individual is here on his own and is alienated from the rest, it is clear that in this case one cannot speak of a collective at all. It is a different matter when the same mass of people gathers somewhere, not by chance, but being attracted by a particular goal or mood. Here we have a mass of people united by a single stimulus that leads to the emergence of a collective, or a composite personality. One should not forget, however, that under appropriate conditions any chance gathering can easily be transformed into a composite personality, or a collective. It is enough for someone to address a chance gathering and, by generating in it a single attitude toward a particular public matter or fact, to suffuse it with a certain mood or aspiration, in order to transform a chance gathering into a composite personality, or a collective, capable of expressing itself as a whole.

This is how a crowd comes into being. Numerous authors both in this country and in the West—particularly Tarde, Le Bon, Sighele, Mikhailovskii and many others—made the crowd the object of their investigation.

A simple conglomeration of individuals does not form a society, nor does interpersonal interaction necessarily ensure unity. On the contrary, interaction can lead to hostility, division and even to the complete rupture of all bonds, as happens in time of war, for example. Thus, collective unity can be achieved by such forms of interaction that bring individuals together, rather than divide them.

When collective unity evolves together with the emergence of society, the question arises whether we are dealing with something real or with a mere name. A number of authors (without taking into account the old representatives of the organic school)—Durkheim, Gumplovicz, de Roberti, Espinas, Fouillée, Gierke and many others—give a rather positive answer to this question. Durkheim says that "there is a society above the individual. It is not an imaginary and nominal entity, but a system of actual forces" (*The Method of Sociology*).

Tarde, on the contrary, does not believe in the reality of society. "I confess," says this author,

I find it very difficult to understand how it can possibly happen that after dispensing with individuals we shall still be left with society. If we dispense with professors, I cannot clearly imagine what will be left of a university, except its name that has no meaning if nobody knows it in the

context of all the traditions associated with it. Are we not going back to the realism of the Middle Ages? (*Social Logic*, St. Petersburg, 1901, p. 2)

Some other authors, for example, Duprat, closely approximate this view. Among the Russian authors, it is supported to a certain degree by Lavrov, Mikhailovskii, and Kareyev. This Russian school, while adopting the subjective point of view, put forward the importance of personality as the highest ethical value, and recognized it, according to Mikhailovskii, as true social individuality. P. Sorokin analyzes this question in detail in his book *System of Sociology* (vol. 1, p. 243 ff.). First of all he disproves realism by saying that "a specific object cannot exist *outside and independent of its elements*." On the other hand, expressions like "the soul of society," "the soul of the nation," "the national spirit," etc., are in constant use. But does this mean that there is a "public consciousness" or a "soul of society" that exist *independently* of individual consciousness or outside individual souls? The answer is provided by the method of subtraction. Subtract from this collective "soul of society" the "souls" of all the individuals that make up the society, subtract from the "public consciousness" the consciousness of all its past and present members—and there will be nothing left.

Giddings, too, speaks out against the "self-consciousness" of society, which attributes to the latter a process of thinking that differs from individual mentality.³

On the other hand, P. Sorokin rebels against Tarde and refuses to accept his postulation that there is nothing in society that would not be present in individuals. The author perceives "society as the sum total of interacting individuals," and claims that this interaction is precisely the new condition that transforms a sum of individuals into collective unity. In this case, there appear "a number of attributes, phenomena and processes, which are not—and can never be— present in a sum of isolated individuals." Society, however, exists not "outside" and "independently" of individuals, but only as a system of interacting units. We have already mentioned that interaction is not unity and that hostile interaction results in division rather than union. We can describe interaction as the essence of the social process, but not as a prerequisite for generating the unity on which society is based. It is true that every society is beset by strife, but as long as this strife does not cause one side to break up with the other, unity can still be maintained in some respects. Consequently, one can conceive of a society where two parties confront each

other within the same collective, but one cannot conceive of a unity wherein one side is completely cut off from the other, their mode of interaction being a fight to the death.

Some authors (Tarde, Le Bon, Sighele, Mikhailovskii, Rezanov and others) tend to extend the concept of the "crowd," or the "psychological crowd," to all types of human gatherings held together one way or another.

The description of various collectives by one common name, "crowd," as practiced by the above authors, is unfortunate, as it contradicts the established terms and related concepts. Thus, for example, A.S. Rezanov, the author of the article, "The Army and the Crowd," says, "the psychological crowd may be represented by a herd, a meeting, an army regiment, a theatre, a congress, a parliament, etc."

I would like to mention here that Tarde's division of the crowd into the male, the female, the young and the old, as well as—according to its orientation towards the surrounding reality—into the expecting, the curious, the attentive, the enchanted, the manifesting, the loving, the hating, the idle, the sad, etc., does not hold any water either, for, excluding gender and age, the same collective can successively undergo transformations into all of the above types of crowd within a few hours and without losing any basic principles of its organization.

As a result of describing any collective by this general term "crowd," we end up in such a muddle that it is extremely difficult to get out of it at all. Thus, for example, the same Tarde accords a special place to parliamentary assemblies, for, in his opinion, they are "complex and contradictory crowds—two-headed crowds, so to speak, just as there are two-headed freaks of nature." On the other hand, the same author believes that academies "are collectives of individuals who mutually influence one another, rather than corporations based on cooperation." Furthermore, a theatre audience is, in Tarde's view, "not a permanent crowd, but only half a crowd due to the sedentary position of its individual members."

In general, despite all his achievements, Tarde is rather confused when it comes to definitions, and it would be a mistake to follow him in this regard.

We are not going to discuss other authors' classifications of collectives. I shall only say that I basically disagree on this matter with Tarde, Le Bon, Sighele, and other authors. Besides, one can find critiques of their classifications in the works of more recent authors (see, for example, A.S. Rezanov's "The Army and the Crowd"). I think it more correct to describe

individual collectives by using terms that would indicate the distinct nature of various collectives.

In this regard, we can describe a crowd proper as a gathering of people brought together by a temporary cause.

This crowd may not be sufficiently organized if there is no established order for the purpose of coordinating its actions. The crowd may become organized when there evolves a more or less strict subordination to one person as its leader, or even to several individuals who form a presidium, as in the case of political gatherings, meetings, etc. The leaders of a street crowd remain in charge only temporarily, for a change of leadership may occur even while a crowd is engaged in one and the same action. Occasionally, a crowd might commence something under the influence of one leader and conclude it under the influence of another. The truth is that every leader gets ahead in a crowd by using its unity under a particular slogan.

Clearly, the very acceptance of a leader indicates the active role of the crowd that asserts, as it were, its choice of leader.

In other cases a gathering of people takes place with a particular purpose; this purpose, however, predetermines the passive role of the gathering itself that may listen to, or watch, something, while expressing its attitude to what it is shown or forced to hear by signs of approval or disapproval at the most. We describe this type of gathering that has no internal organization, but is organized in terms of its purpose as the public that one sees at the theatre, concerts, lectures, literary soirées, temples of prayer, etc.

A third category also includes gatherings with definite purpose, but these gatherings, instead of displaying a passive attitude toward the purpose that has brought them together, reveal their attitude to the subject matter in the form of a conclusion or a resolution, a sentence or a decree; this enables individual members of such gatherings to declare their own opinions, for which purpose a certain order of discussion through elected representatives is established. We call such gatherings assemblies. They include parliaments, church councils, congresses, various societies, institutions, councils, committees, commissions, circles, courts, etc. Every assembly possesses some distinct characteristics of external and internal nature. Further, we should mention collectives that are engaged in a coordinated activity: e.g., small enterprises, cooperatives, factories, monasteries, financial institutions, etc.

We can thus distinguish three forms of a composite personality that differ in their inner content: an unorganized collective (a street crowd) and two types of organized collectives represented by an audience on the one hand, and by a workers' cooperative on the other.

However, there are also some special forms of collectives that do not fit into the above framework. One of them is based on the instruction of the young by their elders in kindergartens, educational establishments, workshops, etc., and can be described by the general name, a school.

Another type of collective evolves as a result of training people to act as a whole collective. It includes troops, gymnastic societies, dance groups, orchestras, choirs, public games, organized crime, etc. Such gatherings can be described as organized hordes.

Yet another form of collective is an association of people who are related through blood or spiritual kinship, even though they may be separated by distance. This category includes family, clan, tribe, nation, estate, caste, sect, party, class, profession, readership, church, etc. Such collectives can be described as tribal alliances in the first case, and as corporate ones in the second.

Finally, there are collectives unified only through more or less common views, even in the absence of close contacts. Such a collective may be described as a world or a milieu. Here we are dealing with collectives that possess their own more or less commonly shared opinions and even traditions, and that therefore represent a real, albeit physically elusive, collective responsible for so-called public opinion. Among such collectives one can mention the literary world, the military world, the medical world, the pedagogical world, and the universal human world that unites all of the above. Finally, large and complex collectives unified through the highest, common, historically established power structure with certain prerogatives and organization are called states.

We should therefore distinguish the following types of collectives: the crowd, the audience, assemblies, workers' cooperatives, the school, organized hordes, tribal and corporate alliances, the world and the state. A more detailed description of all these collectives would at this point be superfluous, since what has already been said is sufficient for understanding their differences.

The essence of any collective is its unity; as we have seen from the above, this may not depend on the degree of this unity, which may vary in character.

Thus, unity may find its expression in common mood, common observation, common discussion and common resolution, common views and common action. Collective organization is one of the important unifying factors. I disagree with such authors as Espinas who interpret organization as subordination. There is no doubt that subordination implies organization, but conceptually the two are not identical. Subordination is an orientation concerning power, whereas organization is the establishment of interconnection and order.

Due to this interconnection and order, every collective is a composite organism, as it were, with its own history, for under certain conditions it is conceived, grows, develops and then, after a more or less long or short life, it disintegrates and dies, often transforming into other forms of collectives. A collective as a composite organism has its own mood, its own powers of observation and perception, its views, judgment, and actions, which by their nature are often at odds with the same manifestations of individual personalities that make up the collective.

In conclusion, we should say that the collective as a composite organism exists and functions at the expense of the individual organism, since the personality is always, to a greater or lesser degree, suppressed by a crowd, an assembly, a corporation and by any public or state organization; the more united the collective is, the less room it leaves for individual expression. Thus, the suppression of a personality is at its highest in disciplined troops, and seems to be at its lowest in scientific assemblies, since the issues of science are not subject to collective decision-making by vote.

Like a living organism, society represents a dynamic equilibrium rather than something static; it continuously moves—whether slowly or rapidly—in one direction or another. And Tarde has some reason to claim that "societies live as long as they progress."⁴

At the same time, society generates such social phenomena as language, abstract concepts, religion, traditions, codes, etc.

However, the notion of society as an organism is nothing other than a metaphor, for, as we have mentioned before, most authors no longer subscribe to this theory.

According to Tarde,

the metaphor of the organic society—discussed in great detail in the second volume of Herbert Spencer's writings, accepted only provisionally

and rejected subsequently— was taken up again, analyzed and apparently corroborated by René Worms in his doctoral thesis, *Organism and Society*. By taking this proposition to its extremes, and to the *summum* of its clarity and precision, as well as by reporting faithfully every criticism that was levelled at it, he—like Novikov in his highly interesting book *Social Consciousness and Will*—helps to rid science of it.⁵

In general, any society, being a composite personality, possesses its own individuality that depends on the intellectual level of its members, their occupation and career status, on the particular "cement" that unites individual members into an association, and, finally, it depends on tribal characteristics.

It is apparent from the above that, depending on their common unifying goal, various public and national assemblies generate different composite personalities.

Irrespective of their form and membership, assemblies that constitute the object of investigation for collective reflexology are invariably engaged in a joint activity, without which they are a mere gathering of individuals who have nothing in common.

Thus, a composite personality can be represented by any crowd whatsoever, by a public assembly, a council meeting, a theatre audience, etc.—in other words, by any kind of gathering united by a certain mood, a particular goal or action, irrespective of whether they are determined by a chance happening, or whether the goal or the action are prescribed by law or established by tradition or by other means.

A vivid example of a composite personality is a street crowd brought together by chance.

Any crowd at all represents a gathering of people who had previously nothing in common with one another, and who have come together on a chance occasion that provokes the same response in many persons. Due to the unity of its members, a crowd always represents a single whole, a composite personality in which individuals seem to retire to the background. This cohesiveness is not achieved immediately, but nearly always with extraordinary speed.

At political meetings, besides the external causes that provoke a particular response, the very subject of discussion functions as a unifying agent. Therefore, although the content of speeches made at such meetings may vary, the subject of discussion is of interest to all. This generates the unifying

slogans supported by many, and often maintains the common mood. Those who do not join the rest at these gatherings, and are strongly opposed to the latter, usually remain passive or leave the meeting, and consequently do not interfere with the general course of events at such gatherings.

In public, scientific, and professional assemblies, the joint intellectual activity is determined by a particular goal: e.g., by the need to analyze a situation, to search for truth or to regulate governmental procedures. Public and political associations represent assemblies whose goal is predetermined and specified in a written code that imposes certain demands on the persons who are part of this alliance or association. The goal of commercial assemblies is the success of the undertakings that brought the assembly into existence, etc.

The limits of "composite personality" may certainly be expanded to incorporate entire nations and countries whose population is united by common interests. We can therefore speak of Moscow society, Caucasian society, Polish society, Russian society or the Russian public. In such large social units, cohesion is achieved predominantly through the press and via interaction between separate groups; this does not, however, affect the heart of the matter, since here, as everywhere, we deal with larger or smaller social groups cemented by certain common interests or by a particular goal.

A nation represents an even more extended social group.

One might imagine that a nation presents such a motley image that it is hard even to speak about its unity. There is no doubt, however, that at any given moment in time a nation lives by common interests, experiences a more or less common mood, feels the same emotions, is stimulated by more or less similar social ideals, and lives by the same social aspirations, while generating one common culture. In any given nation, this unity of spiritual life evolves historically due to a number of economic, political, and ethnographic conditions, and is determined by cultural unity, by the unity of socioeconomic interests, and by sharing the same historical destiny in the course of communal existence.

Finally, the most extended group is represented by the whole of civilized humanity, whose interests are common to the entire population of the globe, and who follow the universal human guiding principles.

One may undoubtedly ask the question whether such large groups as populations of countries or states, or even the whole of civilized mankind, can be the subject matter of collective reflexology. Here we deal with

associations that, although united by the same interests, are deprived of any means of mutual communication except the press.

Obviously, the conditions that exist in ordinary gatherings cannot exist here, and, consequently, not everything that is related to the associations and gatherings dominated by verbal discourse can be applied here. Nevertheless, there is no doubt that these large associations represent groups that display a certain public mood revealed in the press as well as in individual assemblies. Likewise, through the press and individual assemblies, these large groups discuss and criticize various events, and pass the resolutions that represent composite decisions of numerous groups or individual societies of a given country or state. Thus, in this case, too, we are dealing with the subject matter of collective reflexology, although this subject matter differs somewhat from that of ordinary gatherings.

If the state is an association of people bound by certain rules, then one can apply to it the same laws as one applies to smaller gatherings. This may easily be demonstrated in smaller states with homogeneous populations untouched by tribal hostilities. When the represents a conglomerate of nations, it is the tribal groups', which live by the same interests, that function as collectives proper; however, when it comes to the solution of general issues that are relevant to the whole heterogeneous population of such states, we are dealing with the subject matter of collective reflexology.

Among other things, a characteristic feature of large groups is the slow pace of all the processes of associative activity, for, in general, huge masses of people are more conservative; besides, the means of unifying their activity are more complex (the press, separate assemblies, etc.), although, with sophisticated forms of self-government and with freedom of assembly, the cohesiveness of such masses grows faster than it does under different conditions.

Nevertheless, here, too, we encounter a mass rise in the mood of the active elements of society; their oppression; public indignation, outrage and protests; the collective mentality of entire nationalities as it is reflected in the daily press and literature, as well as in the solutions to public issues; and, finally, we encounter active manifestations expressed in declarations or demands of individual groups that may or may not be supported by the representatives of the whole nation.

Thus, all the elements of collective correlative activity are present and should, therefore, serve as the material for collective reflexology.

We all know, however, that there are forms of social activity that involve auditory and visual stimulation, which provoke a vigorous response in the listeners and viewers who are not meant to take an active part in these gatherings. The latter include public readings, concerts, theatres, shows, etc.

Here we deal with a collective impression and with the emergence of a certain mood in a human mass, just as in an individual life a book may generate a number of impressions and produce a particular mood in the person who is reading or being read to. A question naturally arises whether these phenomena belong to the subject matter of collective reflexology.

The above gatherings involve an audience united by a spectacle, and although communication between the members is here only indirect, it is easy to see that the cohesiveness of a theatre audience watching a play or listening to a concert is established due to the universal impact of the spectacle or the concert, and due to the stimulation of the same responses. Therefore, even though listeners and viewers are, in these cases, passive participants of such gatherings, they are actually united through one common interest, display similar reactions, are stimulated by more or less identical influences, and, finally, they collectively declare the same attitude to what is taking place on stage by the signs of approval or disapproval. In view of the above, there can be no doubt that these gatherings, too, provide material for collective reflexology.

A public reading with debates represents a gathering that not only perceives but also reasons, and, consequently, displays not only its common mood but also its collective intellect.

The gatherings that make certain decisions based on what has been read and discussed manifest not only collective concentration and collective mood but also collective intellect in the form of collective resolutions.

As we know, every individual in a composite personality becomes more or less inconspicuous in a mass, unless he represents the opinions of a particular group of people or is the leader of the whole gathering.

However, the society, or the collective, while always acting as a composite whole, does not eliminate the role of each constituent unit, since the latter affects the resultant force that determines the movement of the collective. The stronger the social impact of a personality is, the greater is its influence on the direction of the resultant force. This may continue until the personality and the collective come into conflict, which may lead either to mutual concessions and a reconciliation or to a rupture. Incidentally, this is how

public figures are replaced in the course of a revolution, when the collective moves spontaneously in a particular direction that no longer satisfies the aspirations of the leaders; as a result of this, the latter gradually lose ground, and are forced to step aside. However, this phenomenon can be observed not only during revolutions but also in ordinary life; in such cases the conflict is generated not only by the collective itself but also by certain circumstances associated with the personality of the leader. This point of view reconciles various discordant opinions on this matter found in literature.

It is clear from the above that, in order to influence a crowd or a collective, one should deal first with its leaders, and it is only with their consent and cooperation that one can influence the crowd in the desired direction.

As we have already mentioned, every collective possesses its own individuality reflected in the distinct nature of its aspirations that, incidentally, change in the course of time.

According to Tarde,

Some nations imbued with moral and religious aspirations are essentially peace-loving, and take up arms very unwillingly, whereas others display a true military fanaticism in the interests of their religious or moral proselytizing, or, to put it better, one and the same society, while remaining faithful to its aspirations in a particular direction, may in turn be engaged in war or in production, in militarism or in industry. Egypt, which was traditionally mystical and industrial, became from time to time—as during the Ramses dynasty—militant and expansionist, while losing none of its mysticism. Presently peaceful Islam used to be profoundly militant. Athenians as a predominantly aesthetically minded nation had for a long time been fond of war from an artistic point of view, so to speak; later on, while still remaining artists, they, unfortunately, became too fond of peace at all costs. There has never been a more trading and, at present, a more peace-loving nation than Holland; in the XVII century it was the most bellicose of nations.⁶

If we now pose the question of what exactly leads to the development and establishment of a collective, we should first mention the social instinct that causes even animals of the same species to lead a herd-like existence.

People of more or less similar views naturally come together, and, in the same manner, people with similar individualities and temperaments

experience mutual attraction. It goes without saying that belonging to the same profession generates a sufficient number of common interests that bring people closer together.

Another factor that stimulates establishment of collectives is the mutual need of individual persons—irrespective of their dissimilarities—for one another, as well as Darwin's natural selection that conditions the need to join forces for the purposes of attack and defense, since only those individuals benefit who prefer to join a group for the above purposes, whereas the rest are doomed to perish. As the third stimulus, I have postulated social selection, which consists of public support for those members who, even though they may be physically weak, conform to collective interests. This promotes the development and cultivation of social tendencies in individual person and maintains the social cohesiveness of a collective, which is passed on to the next generation via social heredity.⁷

We would like to point out that every society survives for as long as it is justified by the existing conditions. In this respect one should bear in mind Aristotle's law of the self-sufficiency of any political association.⁸

Regarding its associative activity, a collective cannot be identified with a simple gathering of individual persons; it always possesses an integrated physiognomy that represents the collective as a whole. A street crowd demonstrates this best of all.

Le Bon says in *The Psychology of Socialism* (pp. 51-52):

The psychology of a crowd is not at all the same as the psychology of the individual elements that make it up... The soul of the crowd generates aspirations as ideas that are born in the upper strata of the nation, undergo a number of arbitrary evolutions, eventually take root in the soul of the crowd and lead there a secret, vegetal and powerful life.

Then and only then do they become conventional opinions and almost unshakable convictions, the major factors of religion, revolutions, reforms and civilization. A crowd is the inherited soil in which all our metaphysical, religious, political and social perceptions of the world are rooted, and we should know it very well.

A crowd as a collective is characterized by "monoideism," by considerable suggestibility, by an extraordinary excitability that often turns into cruelty, by gullibility, extreme recklessness, reflex actions, by lack of steadfastness, by

striking changeability, relatively small differentiation and by herd-like submissiveness to its leader resulting from its hypertrophied suggestibility and predilections for imitation:

Among the characteristic attributes of a crowd we come across such features as impulsiveness, irritability, inability to think through, an absence of reasoning and critical thinking, hypertrophied sensitivity, etc. that are typical of the lower forms of evolution such as women, children and savages (*The Psychology of Nations and Masses*, p. 173).

A crowd easily becomes an executioner, and just as easily it chooses martyrdom. In order for some faith to triumph, it lets rivers of blood flow from its own body. (Ibid., p. 175)

Its sheer numbers make a crowd confident in its own power and impunity.

According to Le Bon, a crowd's suggestibility and gullibility are due to its expectant attention and its lack of critical thinking.

In this connection, habitual distortions intrinsic to human imagination may explain the origins of legends, while contagion plays a part in their dissemination.

The following example from Le Bon's *Psychology of Nations and Masses* (p. 181) illustrates the gullibility of a crowd: some time ago,

all the newspapers were bursting with stories about two drowned little girls pulled out of the Seine. At least a dozen witnesses recognized the children in the most categorical terms. All their testimonies were so concordant that the coroner could entertain no doubts and filled out the death certificate, but just as the supposed victims were to be buried, they were found to be alive, their appearance being only slightly reminiscent of that of the drowned.

The testimony of the first witness apparently generated the same illusion in others, and then spread to quite a number of witnesses.

In another case, even the child's mother was prey to this kind of suggestibility. This is what *Eclair* reported on 25 April 1895:

A child mistakenly identified a dead body as that of his playmate. This mistake immediately caused quite a few similar mistakes including the

following remarkable episode: one woman who saw the corpse of the child exclaimed: "Oh dear, it is my child!" On closer examination she noticed a scar on his forehead and said: "Oh yes, this is my poor little son who disappeared in July. He was kidnapped and killed." The woman was a concierge from Rue Dufour. When her son-in-law appeared on the scene he declared without hesitation: "This is little Philibert." Several inhabitants of this street also recognized Philibert in the dead child, and even his school teacher, on seeing the medal, recognized his former pupil in the dead boy. Well, what do you know! The neighbors, the son-in-law, the school teacher, as well as the mother—they were all mistaken. Six weeks later the identity of the child was definitively established: he was from Bordeaux, was murdered there and brought to Paris in a stage coach.

The excessive suggestibility of a crowd is clear from the following example:

In one national theatre where a serious drama was regularly performed "an actor who played the role of a traitor was in constant danger and had to be protected, for the spectators, infuriated by his fictional crimes, were ready to tear him apart "⁹

In his *Psychology of Nations and Masses*, G. Le Bon examines in succession the following aspects of a crowd: (1) impulsiveness, changeability, and irritability; (2) pliability, suggestibility, and gullibility; (3) exaggeration and emotional bias; (4) intolerance, authoritarian and conservative ideas, reasoning, imagination, the influence of race, traditions, education, and upbringing. There is no need, however, to continue quoting this extremely subtle observer lest the reader falls under the spell of his ideas, which are not always completely acceptable.

Numerous authors mention cruelty as a characteristic attribute of the crowd, perceived by them as a dangerous many-headed beast. Sighele, for example, says in no uncertain terms that

heroism, valor and kindness can be characteristic of an individual person; but they can never, or hardly ever, appear as the distinguishing features of a large gathering. ...A crowd is a substratum in which the germ of evil spreads very easily, while the germ of good nearly always dies for lack of the necessary conditions for survival.

The same view is held by Tarde, according to whom, the soul of the crowd is "murder, robbery or arson."

In his well-known work *Origines de la France contemporaine* (vol. 2, *Anarchy*) Taine says (with respect to the crowd during the French Revolution) that a crowd's mode of action is violence, which is quite in keeping with its nature: "everything that offers resistance to a crowd is destroyed by it."

A crowd that gathers by chance in response to an appeal or an alarm is an awesome and destructive force. An individual who joins a crowd is suddenly transformed into a barbarian or worse—into a prehistoric beast, a grimacing ape, bloodthirsty and lustful.

Sluchevskii, too, is close to this view when he speaks of "high" excitability and cruelty displayed in action" as the general attributes of a crowd.¹⁰

A thirst for revenge makes a crowd particularly violent if the resistance it meets results in the loss of lives on the part of the crowd. When it gains the upper hand, no one can tell where the limits of its blind and unrestrained fury will be.

One can cite as an example the destruction of the cadets (who were supporting the Provisional Government) on 28 October 1917 in Petrograd during the second revolution. The newspaper descriptions of this event were so depressing that we do not dare to reproduce them here.

Without denying the facts on which the above interpretation of the crowd is based, we must point out its one-sidedness, for the attitude of a crowd is determined by its mood. In some cases, a crowd acts as a cruel beast, while in others it is capable of worshipping and defending its idol to the last, and is ready to die for his sake.

In this respect, G. Le Bon is closer to the truth than other authors, when he says that "there exists not only a criminal crowd but also a virtuous crowd, a heroic crowd and many others." When discussing further the destructive power of a crowd, he comments that only a crowd is capable of manifesting "the greatest selflessness and the greatest devotion."

And, indeed, there are countless examples of such devotion, including the self-sacrifice of many when their leader is in mortal danger. All revolutions have been fought by crowds that at times had to face certain death in order to defend the rights of the people.

In Le Bon's words (I.e., p. 52), psychologists always saw in a crowd only "a wild beast never sated with plunder and blood." However, if we look into this question more carefully, we shall soon discover that the worst instances of crowd behavior were often motivated by very noble and selfless ideas, and that in the final analysis a crowd is a victim rather than an executioner.

A book entitled *The Virtuous Crowd* would be incomparably more appropriate than a book entitled *The Criminal Crowd*. According to the author, the main underlying distinction between an individual and a crowd is that the former is always motivated by personal interests, whereas a crowd obeys shared, rather than unstable and selfish, interests:

Heroism and self-abnegation motivate crowds far more frequently than they do an individual person; beneath collective cruelty one most often finds a certain conviction, a concept of justice, a need for moral satisfaction, a total disregard for personal interests, and self-sacrifice for the sake of common interests—in a word, the exact opposite of egotism.

However, a crowd as such has no proclivity towards either crime or virtue: it can generate both, depending on the cause that brought it together and on the direction into which its aspirations are channeled by its leaders; a hungry crowd is as egotistic as any individual who demands a piece of bread, while a political crowd crushes everything in the path of the social principles of the new order it has adopted, and eulogizes its heroes. That is why G. Le Bon is wrong when he says (I.e., p. 53) that "Egotism—so deeply rooted in an individual personality—is totally alien to a crowd, precisely because of its inability to either contemplate or rationalize."

A crowd is not devoid of the self-preservation instinct, and, consequently, neither is it devoid of egotistic aspirations. One thing is true: when we deal with a criminal crowd, the motives of the perpetrated crime often lie in the correctly or falsely interpreted common weal, whereas individual crime is fairly seldom socially motivated: here egoistic aspirations are far more common as the source of motivation.

In this respect, a crowd seems to have a leveling effect on individual persons who, being unable to lead, have no opportunity for self-expression save, perhaps, by casting their ballots.

Each individual person may be rational, but in a crowd he is often deprived of the opportunity to protest; at worst he gives in to the influence of the

crowd, participates in its actions and even in crime.

It goes without saying that the noble aspects of the individual personality in a crowd, including its leaders, cannot find favorable conditions for their expression, and, consequently, a crowd functions on a far lower level than many of its members.

Besides, one should bear in mind that an individual person in a crowd invariably conforms in a greater or lesser degree to the collective view inculcated in the crowd by the demagogical methods of its leaders.

I happened to talk to some army officers who survived the Russian Revolution and who knew their detachments well. All of them claimed that when they would talk to an individual soldier it was easy to convince him to adopt a sensible view of things, but as soon as this soldier found himself in a crowd at a political meeting, he would accept the collective slogan together with the resolution that had been drafted in advance, and would express his agreement by shouting "Right! Right!"

In this case, the collective suppresses a personality not only through the power of suggestion applied by its leaders but also because the collective is stronger and more influential than any individual personality. An individual would never dare to accomplish what a collective would. In folk reflexology this attitude is expressed in the well-known saying, "two in distress make sorrow less."

Irrespective of the above, any collective whatsoever suppresses with its sheer mass all individual self-expression.

If the mood of the majority or its attitude to a particular fact has become apparent, few people would be brave enough to go against the stream. Even meetings that are specifically convened to discuss a certain subject, and where discussion is the purpose of the meeting, an individual may find it difficult to speak out against the opinion of the majority after this opinion has been established. It is even more difficult to do this in a street crowd, where there is no discussion, but only a leader who holds the crowd in almost blind obedience, particularly if he has previously won its favor.

The more cohesive and homogeneous a collective is, the greater its moral power and its activity are, and vice versa. The truth of the former is exemplified by communist parties that have always been known for their strict discipline. The opposite case is represented by the Socialist-Revolutionary Party that played an extremely important role in the February coup d'état in Russia. However, when after this coup a mass of new members

(who essentially had little in common with socialism) joined the party, the latter—despite the enormous number of members—could no longer play an appropriate role either in the Democratic Assembly or in the Preliminary Parliament. In the end, a perfectly natural split occurred in this party, whereby one group assumed a more rightist position, while the other—the more leftist one.

Another example is the Moscow Assembly that proved to be fruitless precisely because, being convened at the most critical moment of the Russian Revolution, it had to face two public movements: the census group and the democratic group. It was counteracted by the Democratic Assembly in Petrograd which adopted a far more radical point of view, but again failed to succeed due to the heterogeneity of its membership. As we know, this Assembly ended in an indeterminate vote on the distribution of power: whether to include the census supporters in the coalition or not. It was clear that neither Assembly possessed the moral power, and that the Provisional Government could not follow their legislations.

The role of the Preliminary Parliament, from which the Bolshevik faction immediately split off, proved to be equally fruitless at the time of the Russian Revolution. Obviously, there are moments when a collective is unable to come to any decision, when party proposals are rejected one after another, and no common decision can be agreed upon—in other words, no single general formula can be worked out. Such a situation may develop predominantly during periods of difficult social crises and public dissent.

This is what happened in the Soviet of the Republic, where the debates on state defense generated so much discord between the parties that none of the proposed resolutions gathered a majority of votes in the assembly.

In this case, assemblies themselves undergo a crisis due to the fragmentation of votes and become ineffective. They are either sabotaged by one particular party that wants to cause their downfall, or they make suicidal decisions as was the case with the *Convention Nationale* during the Great French Revolution. To illustrate this we shall quote from the well-known historian H. Taine, who thus describes the crisis of the Convention:

They approve and prescribe what they themselves find reprehensible—not only absurdities and madness but also crime and the murder of the innocent. Unanimously and to the sound of thundering applause, the left joins forces with the right, and sends to the scaffold Danton, its natural

leader, the great organizer and standard-bearer of the Revolution. Unanimously and, again, to the clamor of applause, the right joins forces with the left to vote for the worst decrees of the revolutionary government. Unanimously and to the accompaniment of enthusiastic cheering and open sympathy for Collot d'Herbois, Couthon, and Robespierre, the Convention, through arbitrary and multiple re-elections, maintains a homicidal government which is hated by some for its murders and by others for trying to annihilate them. The Plain and the Mountain, the majority and the minority, ended up by agreeing to cooperate in their own suicide.¹¹

It goes without saying that greater homogeneity enables an assembly to arrive at more clear-cut decisions and to exercise a greater moral power, which affects both the majority group and its opponents: the former find in the assembly the most vivid reflection of their own aspirations, while the latter must respect its decision as the most clear-cut and definitive one that dots the i's and crosses the t's.

During the most critical periods of human history, half-way policies fail to gain any moral power, while neutrality of behavior breeds discontent and condemnation—and occasionally even resentment—not only in one particular group but in all; as a rule, half-way policies prove fatal for their practitioners.

One can cite as examples the role of the Girondins during the Great French Revolution, the role of the Constitutional Democrats in the first Russian Revolution, and the role of *VIKZheV*¹² in the second, October, Revolution. Although the *VIKZheVs* neutrality was to the advantage of the Bolsheviks, neither the Bolsheviks nor the more moderate elements were happy with its role.

In the final analysis, the actions of a collective are determined by its composition, level of development, and dominant mood.

As for the composition of a collective, this concept must include national characteristics interpreted not so much as the natural peculiarities in the organization of its members, but rather as the conditions of social heredity inherited from their ancestors via social traditions, views, customs and, finally, the moral dimensions of the environment in which the collective is formed. But, above everything else, the actions of a collective are determined by its dominant mood, which is conditioned by the given circumstances.

Finally, the last determining factor of a collective action is a cause that acts as a challenge.

Thus, the behavior of a collective expressed through its actions, gestures, and declarations depends on the totality of the above conditions, and in this respect it should be recognized as natural as individual behavior.

Notes

1. E. de Roberti, *The New Formulation of Issues in Sociology* (in Russian), Sytin Publishing House.
2. See P. Sorokin, "The Boundaries and Subject of Sociology" (in Russian), in *New Ideas in Sociology*, Issue 1.
3. Giddings, *The Elements*, p. 119.
4. Tarde, *La logique sociale* (in Russian), p. 171.
5. Ibid., p. 153.
6. G. Tarde, *La logique sociale* (in Russian), p. 246.
7. V. Bekhterev, "Social selection and its biological significance," *Vestnikpsikhologii*, 1912. "The Importance of Hormonism and Social Selection in the evolution of organisms," *Priroda*, October 1916 (in Russian).
8. The essence of this law, known as antarchy, is as follows: every political association is viable only as long as it satisfies the needs of its members for whose sake it has been created (more about this elsewhere).
9. G. LeBon, *The Psychology of Nations and Masses*, p. 202.
0. Sluchevskii, "The crowd and its psychology" (in Russian), *Knizhki nedeli*, 1893, Nos. 4 and 5.
1. H. Taine, *Us Origines de la France Contemporaine*, vol. 7, pp. 284-285.
2. The abbreviation of the All-Russian Executive Committee of the Railway Employees' Union (ed.).

6

On Collectivity

Natural sciences have not yet offered an adequate interpretation of sociality—the tendency to form collectives—in the animal world. The best-known theory in this respect is that of natural selection, according to which individuals who live in societies have certain advantages over those who do not, and therefore the former survive, while the latter die out.

However, this theory is insufficiently comprehensive, for it applies only to those animals who lead a social life on a permanent basis—following an inborn instinct, as it were—e.g., bees, ants, termites, etc. It cannot explain the existence of temporary social groups, although we do know that social groups in the animal kingdom are often formed on a seasonal basis and as a result of particular living conditions—e.g., for the purposes of attack and defense. When these conditions undergo changes, social groups, too, cease existing until other circumstances arise and force new alliances into being.

In my view, the above theory should be supplemented with the theory of social selection.¹ The latter secures the survival of a newly formed community by supporting its most socially oriented members—even those who are physically weak—by providing for them appropriate living conditions and better protection, which to a large degree ensures their survival in defiance of natural selection.

We should point out that some authors (e.g., McDougall and others), who find the concept of natural selection insufficient, put forward a theory of social instinct, which interprets the latter as the motivating factor in adopting a communal existence. Yet, if we consider the above-mentioned association

between communal living in the animal world on the one hand and the seasonal variations and certain living conditions on the other, it will become clear that instinct alone cannot explain sociality, nor can it explain some symbiotic phenomena.

Generally speaking, the concept of instinct has undoubtedly been abused, for it is always brought in when there is no other suitable explanation. In any case, if a social instinct does exist, it should be decoded. In the meantime, associative reflex theory can offer quite a satisfactory explanation of communal living.

We know that various external conditions force individuals to congregate in certain locations—e.g., in places that offer protection from cold and inclement weather, or plentiful nourishment—which, naturally, promote close contact and interaction. This closeness and interaction, repeated many times and accompanied by a positive mimico-somatic (emotional) reaction, soon becomes a need reinforced by the appropriate response during encounters with other individuals under different circumstances. This generates mutual attraction among creatures of the same species. Even different species who live together definitely become attracted to one another due to the above condition. Observe a dog that meets its owner by displaying signs of affection. Its animation and visible affection reproduce a mimico-somatic reaction that occurred previously. This reaction, which undoubtedly quickens vital processes, is certainly beneficial for the animal, which is why it seeks contact with its owner, maintaining and enhancing thereby its attachment to him.

It is therefore clear that sociality, or the tendency to form associations, is a result of individual experience; it is based on the emergence of positive mimico-somatic reflexes invariably associated with an approach reaction resulting in closer contact.

The same holds true for human society:

It has long been known that similar individuals become attracted to one another. We feel close to those who share our language, worldview and ideals, and who look like ourselves—in short, to those who are similar to us in external appearance and inner qualities. This similarity acts as a force that attracts socially similar individuals to one another. Children seek the company of other children like themselves. Adults seek the company of other adults like themselves; the attraction between two

Europeans is far greater than between a European and an Asian or an African. People of the same social status understand one another better than individuals who belong to different social classes, groups, estates, professions, etc.²

In general, this thesis is advocated by a number of authors. As we have already mentioned, McDougall explains it as a manifestation of the herd instinct. According to him,

the herd instinct of an animal is satisfied only in the presence of similar creatures, and the greater this similarity the greater the satisfaction. One observes exactly the same phenomenon among humans: similar elements come together in a society precisely because we do not feel equally attracted to all social aggregates, and our herd impulse is best satisfied in the company of the people who are most similar to ourselves. Uncultured nations usually live in tribal communities, with tribes closely related through blood ties inhabiting adjacent territories....The same tendency was at work in the United States, where immigrants from various European countries formed their own large communities within certain territorial limits. It is manifested in our big cities in the segregation of people belonging to similar race, occupation and social standing.³

The same phenomenon is mentioned by Durkheim (*The Division of Social Labour* (in Russian), p. 83 ff.), Waxweiler (*Esquisse*, pp. 16-57), Pareto (*Trattato di sociologia generate*, Vol. 1, pp. 11151133), Giddings (*Principles of Sociology* [in Russian], pp. 19-20), and others.

Another unifying factor is represented by individual differences ("opposites attract"), but only of the kind when the qualities and idiosyncrasies of one individual complement those of another individual. In this case, it is mutual need that acts as a unifying impulse. What one individual possesses can benefit another person, and, conversely, what the latter has to offer may prove useful to the former. Here the connecting element is the principle of division of labor, which makes itself felt not only in public affairs but also, as we know, in industry. Individual insufficiency in a particular sphere provides an impulse, as it were, for an individual to seek compensation through another person.

It is easy to see that in both cases the decisive motive for establishing contact is its social benefit. In the first case, this benefit results from higher productivity achieved through the combined efforts of similar forces; in the second case, the same benefit is achieved via mutual compensation of insufficiency in two or more individuals on the basis of the division of labor.

The benefit is the end result of coming together, the primary impulse being the favorable mimico-somatic reaction in mutual encounters, which has been discussed above.

As we know, P. Kropotkin (*Mutual Assistance as a Factor of Evolution*) regarded mutual assistance as the basis of associations. P. Sorokin, however, believes that mutual assistance is only one of the unifying factors. When elaborating on his objections, he says that animals, as well as humans, come in contact not only for mutual assistance but also

for mutual or unilateral aggression and for causing harm. A lion looks for prey and makes contact with it; parasites interact with the organism they live on, and this interaction may last for long periods of time. People, too, communicate and interact with one another not only for the sake of mutual assistance but also for the purpose of mutual annihilation and oppression (wars, fights, the interaction between the capitalist and the worker, between the slave-owner and the slave, between the oppressors and the oppressed, etc.)⁴

All of this is true with respect to contact and interaction, but unity and association are completely out of the question when it comes to fighting and wars. For one should emphasize that interaction and unity are not all identical concepts.

"Ideological antagonism," says P. Sorokin (I.e., p. 311),

frequently leads to more acute forms of antagonistic interaction, which bring individuals together in a lasting antagonistic collective unity. The simplest example of this kind of phenomenon is an argument between two opponents that ends up in a fight, or a squabble between members of different parties in a parliament. Other examples of extensive and lasting collective unities generated and maintained by the antagonism of ideas and convictions include the antagonistic collective unity (interaction) of the Nikonians and the Schismatics in Russia, their long struggle accompanied

by the imprisonment and exile of the Schismatics. Another and even more extensive and prolonged interaction caused and maintained by antagonistic ideas and beliefs was the collective unity of Catholics and heretics persecuted by the former (the interaction between the "Holy Inquisition" and the Waldensians, Albigensians, Cathars and others as manifested in wars, persecutions, burnings of heretics, etc.). Religious disagreements and wars locked opponents into a lasting collective unity (e.g., the struggle between Catholicism and Protestantism). (I.e., pp. 311-312)

P. Sorokin elaborates this point in a footnote to this text: "I emphasize that from my non-normative point of view a group of interacting Schismatics and Nikonians represents a collective unity, a real sum total and an association."

It is clear from the above that, on the one hand, Sorokin substitutes the term interaction for the term unity, implying thereby that these concepts are of the same order in conventional use. On the other hand, he interprets collective unity as an actual sum total of individual members, as an antagonistic association—consequently, as something that has nothing in common with what we mean by the "spiritual" unity of a crowd or any other gathering.

It goes without saying that individual behavior in this kind of antagonistic interaction is governed by causality, and one cannot but agree with Sorokin that these interactions should become an object of study for sociology as well as for social reflexology.

We should, nevertheless, distinguish between collective unity, on the one hand, and antagonism between two or more interacting individuals, on the other, for the nature and consequences of these relationships are radically different: it is mutual assistance in one case, and struggle and mutual annihilation in the other. The statement that "mutual attraction can be caused not only by altruistic motives but also by hostile ones" is undoubtedly true. This does not imply, however, that both harmony and antagonism fit into the concept of "unity." By removing this confusion we shall free ourselves from the ambiguity it naturally engenders in various discussions of collective manifestations.

Further, there is no doubt that antagonistic and extremely hostile groups may form a cohesive unity, but this is achieved by subordinating one group to another by force, as well as by enforcing punitive measures for separatism and rewards for good performance of appropriate undertakings. In general,

unity is often established on the basis of collective subordination that gradually becomes habitual and no longer incites protest in the subordinated party as it did initially. In other instances, a prolonged struggle exhausts the energy of the opponents, who make peace for this reason and subsequently coexist in solidarity. Here we are dealing with natural internal inhibition, whereas in the first case one group is subjected to external inhibition. Nevertheless, there exists in both cases a social relationship between the two groups, which in time may lead to a more or less real, or internal, unity generated either by a seeming fictive unity or by an enforced unity.

Thus, unity can be of two kinds—internal unity, or unity in the true sense of the word, and external unity imposed by force or through necessity.

The first may develop from common interests, social similarities, and the division of labor, when the groups that are part of a given unity do not appear self-sufficient; the second kind emerges from the confrontation between the strong and the weak and from the oppression of the latter.

A crowd presupposes a direct association of a predominantly emotional—or, objectively speaking, a mimico-somatic nature. Therefore, the more homogeneous individuals are, in terms of their associative activity, the sooner this amalgamation occurs and the faster a crowd is moved into action. An assembly expected to solve a complicated problem and united by this common task, on the contrary, often benefits from the presence of heterogeneous individuals, since the objective here is not so much the speed of problem solving as the quality of the solution itself, which is usually of a multifaceted nature. Therefore, in an assembly, where one finds representatives of different schools and movements as well as divergence of opinion, debates proceed in a more animated fashion compared to a society where everyone more or less agrees with everyone else.

In any case, every collective presupposes the existence of common interests in one or another respect, which is precisely what maintains the unity among the members of a collective.⁵ Some collectives—as, for example, a crowd—are united by a common mood generated by a certain event; other collectives, such as a theatre audience, are united by a performance that arouses particular emotions; a third type, such as trade unions, are united through a common occupation; a fourth type—e.g., financial and other cooperatives, syndicates, trusts, etc.—are bound together by various economic interests; a fifth type of collectives, such as religious communities, are united through religious goals; a sixth type of collectives

are united by shared political views (e.g., political parties) or by a common danger (e.g., military collectives). In short, a collective as such is inconceivable without some kind of group unity.

So what conditions promote collective unity?

Discourse in the broad sense of the term—i.e., the spoken and printed, or written, word—is undoubtedly the most important and the most convenient means of communication for exchange of opinions in human society.

Unquestionably, the spoken word that enables us to engage in personal exchange of opinion, accompanied by such powerful stimulants of emotions as gestures and expressive movements, is the most important means of exchange and communication between people. Its influence, however, is severely restricted by certain spatial conditions. Nevertheless, in some cases, the spoken word may function as a connecting link, not only in interpersonal communication within a given large group, but also in contacts between various social groups, for the spoken word can establish communication even between spatially separated groups of people through intermediaries, envoys, or delegates.

This method of mediated verbal communication, commonly used in olden days and occasionally practiced today (e.g., when sending representatives from one group to another), is significantly inferior in practical terms to the written or printed word. For although the latter lacks such important concomitant elements as gestures and expressive movements, it can evoke them in the reader, and its main advantage is the ability to travel across vast distances.

However, discourse is not the only mediator in social-group cohesion. Expressive movements and gestures undoubtedly play a significant role in this respect. By acting directly on a person and by stimulative imitation, the language of gestures may sometimes have greater communicative power than discourse: one can observe this in an excited crowd.

It is also beyond any doubt that there are forms of social-group cohesion in which action rather than words functions as a cohesive agent and produces an affective state, as, for example, in an audience watching a theatrical performance. An analogous phenomenon occurs in places of prayer and occasionally in a crowd, where action often stimulates imitation.

The cohesive factor in professional groups is to a certain degree represented by the uniform activity pursued by the majority of the group.

Complex coordinated forms of movement, such as gymnastics performed by many individuals or choir singing that produces a coordinated vocal effect, are in themselves important means of social cohesion. An equally important cohesive function is performed by those complex forms of movement in which the division of various roles (as in the theatre) or of labor (as in industry) are in pursuit of a particular goal: to convey basic meaning in the first case and to achieve a joint result in the second.

Evidently, there are thousands of other forms of movements, or stimuli, that help to establish interaction between two and more individuals. When this interaction proves to be useful, it naturally brings the interacting individuals together. If it happens to be harmful for one side or both sides, it leads to separation or repulsion. This kind of interaction can be maintained only by some form of coercion. Consequently, when the latter is abandoned, the union naturally disintegrates. Examples from international relations are Russia and Finland, Russia and Poland, England and Ireland, etc.

Among the stimuli that promote interaction and often cohesion there are objects that may possess an emblematic and, occasionally, symbolic significance. These include various dress codes and insignia that indicate one's belonging to a particular level of bureaucracy, to a particular circle of people, etc. Other objects of material culture such as monuments, palaces, temples, even household utensils and furnishings—in short, all objects of material culture and everything produced by human creativity—function as mediators of interaction and unity. They

are the true intermediaries between people: for some they provide an opportunity to record the results of their labor and actions, for others—whom they affect as stimuli—they make it possible to perceive these results and respond to them. From this point of view, human mediators who transmit stimulation from one person to another may be regarded as being of the same order as material mediators. Their difference from the latter lies in their ability to move. One can therefore describe them as dynamic-material conductors.⁶

There is hardly any need to add that the mediating role of people is far more significant than that of mere "dynamic-material conductors." Letting alone the unsuitable term "conductors" applied to a stimulus, one must say that human language as an interpersonal mediating agent is far more valuable

than man's role as a moving object. When describing people as points of contact in a chain of conductors, P. Sorokin gives some vivid examples of the mediating role of people in the sphere of interaction (I.e., pp. 142 ff.). Stimulation generated by a particular individual is first transmitted by one conductor, then it is picked up by another conductor, then by the third and so on, until it reaches the addressee.

One should not overlook the fact that in these cases the very effect of the stimulus depends to a significant degree on the personality that functions as a conductor—on his past experience, views, and predilections, as a result of which each conductor perceives and interprets material stimuli in his "own" unique way. This is how the actual fact of stimulation may become distorted beyond all recognition. Such distortions explain the genealogy of rumors and legends that sometimes acquire a truly fantastic and even monstrous character.

Among all commodities, the most active stimulant of interpersonal cohesion and interaction is undoubtedly money as an established measure of value attributed to various objects. It may replace them when necessary or restore them through purchase. P. Sorokin is right, however, when he emphasizes the special significance of relics belonging to the family, the church, or the state:

a family relic, such as a lock of Mother's hair, is a sacred object: by passing it from generation to generation they pass on a stimulus that will speak of the family glory and honor, will stir guilt in the soul of an undeserving descendant and pride in the soul of a deserving one; in short, this lock of hair is a true conductor that links the past with the present, and the present with the future. The same *mutato mutandis* may be applied to all other material conductors.^{7, 8}

Interpersonal communication and interaction can be immediate and mediated. When people communicate without any mediator, by using their organs of movement on the one hand and their organs of perception on the other, one can speak of immediate interaction and communication. All other cases involve mediated interaction and communication established through letters, telegrams, sign language, colors, or other objects. Acoustic equipment, such as the telephone, musical instruments, etc., is just as important in mediating interaction and communication. Tactile mediators

include the alphabet for the blind, various forms of touching, kisses and other caresses, sexual intercourse, beatings and other kinds of mechanical impact, thermal and electrical stimulation, etc. Olfactory mediators include various aromatic substances, while gustatory ones—foodstuffs and other edible substances.

It is noteworthy that mediators can establish contacts not only between individuals separated by vast distances but also between different epochs. Parchments, monuments of antiquity and archeo-logical finds provide our link with the ancients and even with prehistoric people! In the same way monuments of art, architecture, and engineering can mediate interaction and communication between people of different nations and epochs.

Mediation expands the limits of interaction and cohesion to an incredible degree: "I am interacting with a friend who lives in America," P. Sorokin writes in *The System of Sociology*:

Today I received a letter from him; this stimulus made me perform a number of actions: I wrote a reply, went to buy a book he needed and visited a post office. Besides, his letter made me terribly happy. In short, my friend who lives in America has affected in a particular way my emotions and actions. As we know, there is an infinite number of such events happening every hour. Scientists, journalists, newspapermen, industrialists, heads of states, etc. separated by enormous distances interact and communicate with one another. We know from our everyday life that people can participate in the process of interaction irrespective of the time that separates them. A caretaker paid me a visit yesterday and, since I was out at the time, left a message with my servant requesting me to provide proof of my residence permit. Today I have spent almost all day attending to this business. Hence, the caretaker determined my behavior and emotions, even though I did not see him and came home several hours after his visit.

Furthermore, there may be interaction between the living and the dead. Although Spencer and Marx are no longer alive, they continue to influence many of our contemporaries, and this influence is not likely to cease in the near future. Through their works that have reached us in the form of books, they stimulate a number of emotional responses and actions in many persons. The will of a deceased Mr. X generates a number of

emotional responses and actions in many individuals—his heirs, executors, notaries, judges, faculty members of the university to which he bequeathed his library, charges of the asylum to which he left some money, etc.

It is clear from the above that *in fact neither space nor time are obstacles to human interaction** due to the involvement of the various mediators P. Sorokin calls "conductors" (a rather unfortunate term from our point of view). A letter, a book, a telegram, a will, an ancient statue as a relic of antiquity, various symbolic signs, etc.— all these are mediators of interaction rather than its conductors, the latter referring to environment in general and two certain types of energy in particular: e.g., that of sound, electricity, etc.

Three main organs of perception—touch, vision, and hearing—play an important role in unifying popular masses into a single social group.

According to Guyau, "touch is the most primitive as well as the surest means to achieve communication, harmony and socialization between two nervous systems, two consciences and two lives."⁹

Tarde, too, recognizes the importance of mutual contact or friction ("collision," to use Sidis's term) in a crowd, which has an electrifying effect. The unifying significance of touch is corroborated by a state of mutual love whose manifestation inevitably demands mutual contact. The latter has a stimulating effect on those preparing for a dangerous attack, it encourages defense in moments of uncertainty, and provides comfort when misfortune or calamity strikes. On the other hand, mutual contact facilitates coordination of movement, as in walking, running, holding hands, dancing and acrobatic exercises. It is clear from the above that the denser a crowd, the easier its unity evolves, which is manifested in uniform movements and shouts. While providing relative freedom of movement, a standing position is more conducive to closer physical contact and promotes a greater cohesiveness among the masses.

Vision is another unifying factor. According to Guyau,

the perception of movement or sentiment in others evokes an echo within ourselves. Here the principle of imitation is particularly apparent. The affective facial expression of one individual generates the same affective state in another person. In some cases a gesture by the leader may produce a magical effect.

Finally, everybody is familiar with the contagious influence of movements. How many dancers can practice self-restraint at the sight of someone in the crowd performing an exciting dance?

We all know that the rhythmical marching of troops carries in its wake gawkers and street urchins who march along with the same speed and rhythm of movement. And, finally, the movements of a concert-master, as we know, coordinate the performance of a whole orchestra or choir.

In the animal world, too, movement provides guidance for the whole herd, as certain movements of the leader are imitated by other members of the herd. It is also common knowledge that members of the same family may pass on to each other characteristic speech patterns, gestures, handwriting, habits, etc.

Finally, movement attracts and focuses one's attention. Observe a young kitten concentrating on and watching every movement of a folded piece of paper. He becomes completely indifferent to the same piece of paper as soon as it stops moving.

According to Morgan, newborn birdlings "instinctively grab a worm not because it is a worm but because it is a small moving object," for, to quote the same author, "if one moves food before newborn birdlings, they are quicker to notice it."

One can deduce from the above that the movements, gestures, facial expressions and actions of a crowd's leaders and heroes in some cases stimulate concentration and thereby increase collective impressionability, while in other cases they provoke blind imitation.

The power of contagion in a crowd is well known. An individual who attracts the wrath of a crowd might be spared if the person standing next to him would not strike him. This serves as a signal for the rest to fall upon him, and the lynch mob has its victim.

Hearing plays an even greater role in uniting popular masses. First of all, it is the ability to hear that makes uniformity of movement possible. One need mention only the role of music in marches, dances, and gymnastic exercises involving large numbers of people. Singing performs the same function in workmen's teams.

The unifying role of music is also evident in gatherings involved in mere listening to music without actively participating in the performance. When a musician plays a composition by Musorgskii or Skriabin, the physical nature of this process can be reduced to the following phenomena: the performer's fingers, by moving the bow against the strings of the violin, cause their

vibration, which through the airwaves reaches the organ of hearing, where it generates vibrations in the form of a neural wave that arrives at the auricular regions on the cerebral cortex. The latter sends waves back to the periphery and to the internal organs, which stimulates a certain mimico-somatic response and a number of other mental reactions. It is precisely these neural waves generated in the listeners by the same mediating agent that unite a mass of individuals listening to a musician. They receive the same charge of physical energy, and although its effects may vary from one person to the next due to individual differences, the fact that their mimico-somatic responses have a common denominator promotes general unity and interaction among the listeners, as well as between the audience and the composers who died a long time ago.

Human voice is the most significant means of stimulating concentration. A whisper in a complete silence is enough to rivet everybody's eyes on its source.

Even in a noisy crowd a scream or an exclamation will attract general attention and increase the impressionability of the crowd.

On the other hand, voice stimulates imitation by having the same effect as contagion. We all know how difficult it is to resist joining a sing-along. When a crowd's emotions are on the rise, shouts and slogans produce a contagious effect and are repeated by all or most of those present.

The contagious influence of the voice on a crowd and its unifying power appear to be particularly important. In the leader's voice, intonation alone plays a significant role in terms of its impact on a crowd.

As a unifying factor, however, discourse is far more influential. I have discussed its suggestive function at length in my book, *Suggestion and Its Role in Social Life* (St. Petersburg, 3rd edition).¹⁰⁻¹¹ We know that in human society it is primarily through discourse that gatherings are organized. On the other hand, the so-called inflammatory speeches delivered at these gatherings by the leaders produce not only a stimulating effect on the crowd but a unifying one as well, for it is the general mood that functions as a unifying factor. The role of discourse in this aspect will be discussed later.

In addition, the process of organizing the masses, which leads to their unity and to the proper coordination of their activity, occurs through the mediation of the spoken or written word.

All of the above makes it clear why communal existence, work, and activity, as well as collectively experienced sorrows and joys, promote by

their very nature the unity of a collective. The enormous importance of schools, institutions, communal residences, etc., in this regard is universally recognized.

One should emphasize that collective experiences—especially those of adversity—have a particularly powerful unifying effect on a collective.

Generally speaking, shared misfortune unites people better than shared prosperity.

It is evident from the above that a true collective, i.e., a united whole, is such a collective where the influence of some members on others results in their unity in one respect or another.

Notes

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1. See V. Bekhterev, "Social selection and its biological significance" (in Russian), *Vestniknaniia*, 1912. Also by the same author "Hormonism and social selection" (in Russian), *Priroda*, 1914.
 2. P. Sorokin, *The System of Sociology* (in Russian), part 1, p. 777.
 3. McDougall, *The Main Issues in Social Psychology* (in Russian), Moscow, 1916, pp. 220 ff. [In English—*Introduction to Social Psychology*, London, Methuen, 1908.]
 4. P. Sorokin, *System of Sociology* (in Russian), vol. 1, p. 283.
 5. According to Tocqueville, "no society can prosper or even merely exist without common beliefs, for without common ideas there is no common action. People can exist without common action, but a social organism cannot."
 6. P. Sorokin, *System of Sociology* (in Russian), vol. 1, p. 139.
 7. P. Sorokin, *The System of Sociology*, vol. 1, pp. 140-141.
 8. *The System of Sociology*, vol. 1, pp. 116-117.
 9. Guyau, *L'art du point de vue sociologique*, Paris, Alcan, 1923, p. 3.
 0. It is available in German: *Suggestion und seine Rolle in Sozialleben*, Wiessbaden.
 1. Editor's note: It is now available in English: *Suggestion and Its Role in Social Life* (L.H. Strickland, ed.), New Brunswick, NJ: Transaction Publishers.

7

Mutual Suggestion, Mutual Imitation, and Mutual Induction as Unifying Factors

Among the factors of a unifying nature, we should examine in greater detail mutual suggestion, mutual imitation, and mutual induction. As far as mutual suggestion is concerned, I have discussed it in a separate work.¹ Therefore, I am not going to dwell on this subject now, but shall return to it later in this book.

Imitation has been extensively discussed by such authors as Mikhailovskii, Tarde, Rossi, Le Bon, Gardner, Sighele, Sidis, myself, Vigouroux, Juquelier and others.

The general conclusion common to all of these studies is that the mere gathering of individuals into a crowd is conducive to mutual contagion, which is precisely what leads to collective action.

According to Binet, who conducted some experimental studies with children, "the very fact of bringing children together in one group makes them more susceptible to suggestion."²

According to Vigouroux and Juquelier, "the mere perception of yawning induces the repetition of this act in others. Yawning is the most contagious reflex...We happened to witness a passenger in an omnibus start a veritable epidemic of yawning."³

Our experiments on so-called mental suggestion⁴ prove that, besides contagion, there also exists direct induction.

In this case we are dealing with those forms of impact within a crowd where individual persons act as mutual resonators, as it were.

As we know, every movement that does not provoke defensive counteraction stimulates imitation, which in itself unites the masses. There is something spontaneously contagious in movements that are more or less familiar and that we find easy to perform. Such, for example, are the movements of yawning, laughter, dancing, etc.

Tarde thoroughly examines the phenomenon of imitation in his works. He interprets the term not only as individual imitation in the form of so-called "mental" contagion, but expands it to include social classes and nations. The heart of the matter lies in the unifying significance of interrelationships established in a collective, for the latter cannot be an integral whole unless its individual members accept the general characteristics of the collective, while, on the other hand, individual achievements are recognized as beneficial to all and become collective achievements.

This is precisely the source of the "leveling" effect produced by any more or less organized collective, whether it represents a professional society, a nation, or a state.

We are dealing here with the extension of individual achievements to the whole collective through their appropriation and assimilation by other members of the same collective. This appropriation and assimilation occurs through simple induction, imitation, suggestion, and persuasion.

Thus, the above factors play an enormous role in the social environment, for it is due to them that interpersonal relations and collective unity can be established, or, in other words, it is due to them that a human collective, or a composite personality, can come into existence. Out of these factors, the earliest to develop (with the exception of induction) is the ability to imitate, which is also characteristic of a large class of animals, as we know. One can observe it in newborn infants: it probably helps them to establish their first social contacts. The child's first imitative sound, his first smile in response to the mother's smile or his imitation of a movement once observed indicate the beginnings of sociality in the first weeks and months of his life.

In general, imitation should be understood in a broader sense than the one we are used to. Imitation may depend on reproduction, in which case it can be viewed from a historical perspective.

Cicero's manuscript on the republic is discovered 2,000 years after it was written, it is reprinted, it provides inspiration; this posthumous imitation could not have occurred if the molecules of the parchment had stopped existing and vibrating, and if human reproduction from Cicero to our days had been disrupted.

Reproduction is a free oscillation whose waves form individual worlds. Imitation goes even further: its action is transmitted not only over very large distances but also over enormous intervals of time. It establishes a relationship pregnant with consequences between the inventor and the imitator, separated from each other by thousands of years—between Licurgus and a member of Convent Nationale, between a Roman artist who painted a fresco on the walls of Pompeii and the modern painters inspired by it. Imitation is long-distance reproduction.⁵

It is clear from the above that imitation can become a cause of epidemics and uprisings when a revolutionary movement spreads from city to city, or it may cause a reproduction of historical events. Was it not imitation that revived the ancient Olympic games, javelin throwing and marathon running, rhythmic gymnastics and the ancient prosodic forms such as the hexameter?

Tarde, who interprets the whole of social life in terms of imitation, thus describes its significance:

The laws formulated by Malthus and Darwin regarding the geometric progression of individual reproduction within a given species are the true laws of radiating reproduction of living individuals. In the same manner—due to imitation—a local dialect used by a few families is gradually transformed into a national language. In prehistoric societies, the art of chiseling stones, taming dogs, making arrows and bows, and, somewhat later, of baking bread, producing bronze and iron, etc.—must have spread through contagious imitation. Each arrow, each piece of bread, each bronze hook and each chiseled stone represented both a copy and a model. This is how dissemination of diverse useful information propagates nowadays, too, the only difference being that the increased population density and the overall progress that has occurred in the course of time accelerate this dissemination in a most amazing manner, just as the velocity of sound is directly proportional to the density of the medium.⁶

The special significance of imitation as the outstanding factor of social life, so brilliantly analyzed by Tarde in his book, is beyond any doubt, but one should not forget that society consists not only of imitators but also of creators—hence the existence of competing individuals. Imitation, therefore, does not always achieve its goal because of interpersonal competition. Thus, the life of the social organism consists of imitation and competition, or rivalry, which interact in a particular way that leads to a dynamic equilibrium.

We have mentioned before that models for imitation are provided by persons in a position of power. Thus, children imitate parents, pupils imitate teachers, the younger imitate the older, etc. The same holds true for a collective environment, where fashion always originates with the upper classes and then trickles down to the lower classes. The reverse tendency is certainly not out of the question, but it invariably occurs on a lesser scale and only in certain periods, as well as under special circumstances. Such is, for example, the preaching of the simple life that gains popularity from time to time with its emphasis on simple dress, plain food, hard work, etc.

Usually, the "plebeians" imitate the "patricians," rather than vice versa; the village imitates the city, and country dwellers imitate city dwellers more readily than the latter imitate the former; the vanquished imitate the victors, and those who love imitate their loved ones; the aristocracy imitates the monarch or the president and the courtiers, rather than vice versa, etc. A language of high literary merits and rich phraseology—particularly if it happens to be the language of a victorious nation—spreads through imitation among the uncultured nation to a greater extent than the language of a less cultured nation may spread among the highly civilized nations.

According to Tarde,

all similarities of social origin observed in the social world represent a direct or indirect consequence of imitation in its diverse manifestations: imitation-tradition, imitation-fashion, imitation-attraction or obedience, imitation-education or upbringing, blind imitation or conscious imitation, etc.⁷

Naturally, Tarde recognizes language as the "great transmitter" of all these varieties of imitation.

Language itself is imitation. The origins of the European languages in Sanskrit indicate boundless imitation, where each deviation or transformation

represents the discovery or invention of an unknown author, which, in turn, stimulates imitation:

The extraordinary proliferation of all kinds of fashion—in dress, food, housing, needs, ideas, institutions and arts—results in the transformation of the entire European population into the kind of people who remind one of a typeset edition printed in millions of copies.⁸

One should not forget, however, that imitation in a collective encounters certain inhibiting circumstances that represent obstacles arising from the life experience of individual members of the collective. Indeed, if the Russians are imitating the Germans, the English, and the French, it is clear that their imitation is confined to those phenomena that do not contradict their life-style and is in their interests and of benefit to them at this particular point in time. On the other hand, everything that goes against their life-style and interests encounters natural inhibitions and is rejected. The same holds true for the Japanese who imitate the West-European countries and Russia. Thus, imitation is to a certain degree restricted by the collective inhibitions determined by life experience and life-style, as well as by resistance to everything that disagrees with national tastes, customs, and established traditions.

As we know, imitation is inhibited by individual experience; therefore, imitation is always more prominent in children than in adults. The same phenomenon is evident in collectives. Younger collectives follow the example of older ones; younger nations imitate older nations to a greater extent than vice versa. Thus, Russians in many ways imitate the culturally older Western countries that borrow from the Russians on a far lesser scale, if at all.

As a nation that entered the arena of public life after the Russians, the Japanese imitate to a significant degree both the West-Europeans and the Russians, whereas the latter two imitate the Japanese hardly at all.

One should bear in mind that imitation is an important aspect of progress, for any forward movement of society, as well as of individuals, is based on the imitation of improvement wherever it occurs. The achieved improvement assimilated by a collective or an individual leads to further improvement, which is transmitted through imitation to a third collective or individual and results in further improvement.

In addition, imitation reduces and facilitates the imitator's effort. When someone achieves improvement through one's personal experience, those who imitate him do not have to go through the same experience: they assimilate the result, while omitting all preliminary work. Besides, imitation occurs at all stages of development and improvement, so that an undertaking begun by one person can be passed on to another individual while still in its early formative stage. This leaves room for the original contributions to the project, which yet again may prove fruitful.

In social life, the law of imitation is manifested in the fact that no invention ever remains in the exclusive possession of one individual, but sooner or later becomes the common property of a collective. This is why alongside national differentiation there occurs—albeit gradually—a leveling process among all nations as a token of their unity, for no nation created its culture single-handedly, but borrowed some of its elements from other nations. This borrowing through direct or indirect imitation is inevitable for any collective and any nation, however unique its creative potential might be. One may only say that a nation imitated by others most often is more original in its creativity, compared to the nations that are not as readily imitated. On the other hand, one may say that a nation capable of borrowing the best from other nations has advantages over other nations. Naturally, everything original affects others one way or another, stimulating their desire to borrow and imitate. In fact, dissemination of myths, fairytales, legends, certain forms of art, scientific discoveries and inventions is based on borrowing and imitation. Myth borrowing that results in a wide dissemination of myths is particularly remarkable.

Similarly, national art forms are essentially borrowed from other, older, forms of art, and are, in turn, borrowed by other nations.

Science is also subject to the law of social leveling. However impressive the development of science in some countries may be, its achievements inevitably tend to spread wherever conditions are favorable. Quite often we are dealing with simple borrowing, but scientific creativity as such does not remain isolated either and tends to spread—again through borrowing—which brings about a certain degree of social equalization.

The same is true for technology and industry.

If one nation achieves a certain level of technological and industrial progress, the benefits of this progress necessarily spread among other nations,

equalizing them to a greater or lesser degree in the utilization of the achieved improvements.

It goes without saying that this tendency towards leveling may unfold faster or slower, depending on the conditions of the social environment and other circumstances, but no collective remains in isolation, since all nations, societies, and circles are in contact with one another. Although local conditions, a greater or lesser geographic isolation, as well as routine—typical of all societies as a manifestation of the law of inertia—offer a certain resistance to this process of leveling, the latter occurs wherever it encounters favorable conditions.

Thus, in ancient history the Romans appropriated the highest manifestations of culture from the Greeks, while the barbarians they conquered borrowed culture from the Romans and subsequently became so successful in the martial arts that they were able to break the resistance of the Roman army.

In modern times, Japan as a naval power quickly adopted the benefits of European culture and within a short time became one of the civilized countries of the East.

Any invention capable of improving human existence and of providing mankind with such material benefits, such as agriculture, the domestication of animals. The invention of gun-powder, book-printing, the discovery of steam power, the knowledge of electricity, etc., as well as all the achievements of the human genius in the spheres of literature, art, and science, after making their appearance in a particular place, are quickly appropriated by the masses and become universally accepted.

As a result, every civilization represents a product of appropriations by the human genius in various spheres, adopted by a given nation that, in turn, makes its own creative contribution to civilization.

It is worth mentioning that borrowing yields permanent results only when it proceeds from the general to the particular, for one nation borrows from another first the established principles and only then the institutions that embody these principles. Thus, Renaissance in Western Europe was preceded by the preliminary study of Greek literature; familiarity with the works of the English thinkers paved the way for the adoption of English institutions by other countries; and the popularity of the French political doctrines prepared the ground for the revolutions in other countries.

On the other hand, if minds are not sufficiently prepared for particular principles, then the institutions introduced simultaneously with these principles or shortly thereafter will prove unstable. This is why parliamentarism failed in Turkey before the Great War. It also explains the instability of the parliamentary system in South American republics. For the same reason, borrowings in every nation are usually disseminated first among the higher or the educated classes, gradually spreading to the lower ones, since the former are quicker to adopt abstract principles and new ideas developed in other countries and to transplant them into their native soil. Besides, due to special circumstances, it is easier for the higher classes of society to become familiar with the innovations introduced in other countries.

Let us take fashion. It is first adopted by the higher classes and then gradually spreads to the lower ones, after which the former discard it in order to borrow and adopt a new fashion. Here one also observes the influence of the dominant class that enjoys a certain prestige and authority among the populace and stimulates imitation among the other classes.

In a large collective, nationalism is a manifestation of its unity, and everything that sustains national characteristics and traditions in the life of a given nation also sustains its unity. Even the components of class—these guardians of caste differences—support the nationalism that manifests the unity of a given nation, although by their very nature they tend to isolate themselves from other collective groups that belong to the same nation.

When discussing imitation, our main concern is not the fact that it is determined by sociality or that social dynamism is an imitational dynamism, but why it is so. What is the reason for two creatures or two groups of creatures to begin imitating each other?

According to Bordier, every human being is predisposed to imitation, but this ability reaches its apogee in human gatherings; to prove this point it suffices to mention public assemblies where an applause or a whistle is enough to stimulate the audience in one direction or another.⁹

While recognizing imitation as an inborn ability, Sighele notes that not only does it double but it increases a hundredfold in a crowd, where everyone's imagination is easily stimulated and where the unity of time and space in a remarkable way—with the speed of a lightning—facilitates the exchange of impressions and emotions.¹⁰

Solly, too, speaks of a predisposition to imitation.¹¹ The essence of this predisposition, however, remains unclear.

Tarde, as well as a number of other authors, reduces social contagion to the phenomena of hypnotic suggestion.

When discussing the influence of one personality on another, which underlies social life and determines imitation—a fact already known to Espinas and even Cabanis—Tarde writes extensively on hypnotic influence, trying to convince the reader with the help of several examples and comparisons that this influence does not essentially differ from a hypnotic one and is similar to somnambulant state induced by hypnosis. "Imagine a somnambulist," writes the author on page 86,

who imitates his medium to such a degree that he himself becomes a medium for a third person who, in turn, imitates the somnambulist, etc. Is not this what social life is all about? Such a cascade of consecutive, mutually linked magnetizations is the general rule, while the mutual magnetization I have just discussed is only an exception. Usually, a charismatic individual provides an impulse, which is immediately reflected by thousands of others who copy him in everything and appropriate his charisma that enables them to affect millions of other individuals further removed from the center. Only after this impact (descending from above) has been exhausted can one observe in democratic times the emergence of new actions: millions of people begin to manipulate their old mediums and make them obey...It is neither fear nor the violence of the victor but amazement and resplendence of the perceived powerful superiority that produce social somnambulism; society is imitation, and imitation is a kind of hypnotism...

Such is the author's final conclusion.

Had this been a figure of speech, we would not object to it. If we bear in mind, however, that suggestion in the waking state is a far more widespread and extensive phenomenon than hypnotic suggestion, it becomes obvious that the impact of one person on another can never be identified with a hypnotic impact. Can we agree with the statement that "a social person is a true somnambulist" and that "a social state, just like a hypnotic state, is nothing other than a sleep induced by an order—a sleep in an active state"? We regard this as overstatement typical of this author in general, and of his interpretation of the "laws of imitation" in particular. Tarde, however, is followed by Siddis who recognizes that the "self of a crowd consists of

subconscious individual "selves," which explains the heightened suggestibility of the crowd. Prior to both these authors, our Mikhailovskii—who is undoubtedly the founder of this hypothesis—discussed this subject in the same vein.

The above authors are followed by a number of others who treat this issue from a purely subjective point of view. They see the main condition for crowd suggestibility in the restriction of free movement, as well as in narrowed consciousness and monoideism, or mental preoccupation with one idea. We shall not follow the subjectivists, but shall try to discover the objective conditions of crowd suggestibility. They can be reduced to three principal conditions: first, confinement to the same position for long periods of time, which besides restricting active movement leads to physical exhaustion; second, prolonged concentration on the same subject (usually on the leader and his speech) undermines one's ability to concentrate; third, the leader's demagogical methods, accompanied by appropriate gestures and facial expressions, determine the uniformity of the mood, which, in turn, defines the direction of the active attitude of the crowd, since a rise in mood is associated with readiness for action. As we know, under these conditions a single word or even a gesture that acts as an order may prove sufficient for the crowd to accomplish a certain act. A fall in mood, on the contrary, provides a fertile soil for panic, which may sometimes erupt immediately after a scream or a declaration, no matter how absurd the latter may be.

One should not forget, however, that imitation can be observed not only in the crowd, but wherever there is human contact of any kind and where hypnosis is out of the question. Since we know that suggestion also takes place in the waking state,¹² this circumstance should undoubtedly be taken into consideration and, hence, the hypnosis theory should be regarded as one-sided at the very least. However, one should not overlook the fact that verbal suggestion cannot totally explain imitation, for in many cases there is no trace of verbal suggestion at all. Besides, instances of imitation among animals are well documented.

Espinas came early to the conclusion that the only reason wasps guard and protect the entrances to their nest and warn others in case of danger is the sight of an enraged individual, for there is a "general law in the sphere of intellectual life according to which the sight of an excited state stimulates the same state in those who witness it." This law, however, is not new. It is

mentioned by Cabanis,¹³ and Sighele claims that even Horace was familiar with it.

Here we are dealing with direct imitation, or contagion, in the true sense of the word, as described by Vigouroux and Juquelier in their work, *Le contagion mentale*.

According to Rambasson (*Phenomenes nerveux intellectuels et mentaux, leur transmission par contagion*), imitation is based on the fact that for every "psychic" phenomenon there is a corresponding movement of a cerebral nature, manifested externally through facial expression, gestures, and posture. This movement is transmitted in space to other people, stimulating them to perform the same actions. It is due to this transmission of movement across distances that laughter, yawning, sadness, and other phenomena—the simple as well as the complex—occur.¹⁴ Here, too, the issue in question is nothing other than direct contagion, but its cause remains unexplained.

From our point of view, the source of imitation in the phylogenetic process of evolution lies in the competition and confrontation of social life. When one of a community of animals finds a source of food and attacks it, all the rest assault the same source of nourishment, reproducing the offensive reflex, whereas those animals who have not developed this reflex of direct imitation lose in the battle for nourishment and survival and are destined to die, according to the law of natural selection. In this case, we are dealing with an imitative reflex of an offensive nature. In other cases, an animal that encounters its natural enemy responds to being attacked with an analogous act of offense or with an act of passive or active defense, since any animal that would not perform this act would perish, which is why defensive imitation is vitally important for any living creature that exists under conditions of rivalry and confrontation. Other defensive imitative responses in the animal kingdom are of the same origin, for when all means of attack are exhausted and the only alternative is to run for one's life, any animal who is less competent in this respect than its neighbors will inevitably perish. The truth of this observation is corroborated by the mimicry of the lower animals, who perform the imitative act in relation to inanimate objects. This, in turn, reveals the vital necessity of the imitative response in general.

Further, one should take into consideration the fact that any imitative act is facilitated by virtue of its being an act of replication and therefore easily executed, just as a copy is easier to execute than an act of creation. Since collective actions always require a certain coordination they are possible only

due to imitation. Hence, a collective experience would be altogether impossible without imitation. This makes the biological significance of the ability to imitate particularly obvious.

Is it possible to transmit directly the energy of the neural centers from one individual to another? To date, there has been no positive answer to this question, for, as far as I know, all the attempts to find scientific proof of telepathic transmission across distances—so artistically represented in Lermontov's poem, "A Dream,"—have not yet yielded any convincing results.

One should bear in mind that quite a few scientists—mathematicians, physicists, physiologists, psychologists, neuro-pathologists and psychiatrists—have tried various approaches to this issue, but all the available data have not yet provided the final solution to the problem. All the experiments in this field have so far focused mainly on guessing chosen objects or identifying their location, on performing a certain action, etc. If this ability to guess is observed at all, it usually occurs in nervous and rather impressionable individuals in a hypnoid state—e.g., in mediums who are able to get into a trance or a special hypnoid state induced by autosuggestion.

However, such direct effect of one individual on another can also be observed in the waking state in people of a nervous disposition who are capable of so-called direct perception.

Incidentally, special experiments in this area were conducted abroad by Charles Richet, Lehmann, and many others, and in this country by Khvorin, Zhuk, Kotik, and others. Not all of these experiments are of equal value, and in any case they have not yielded sufficiently convincing results regarding transmission of thought over distance. As we know, Richet's experiments have failed to convince others. Dr. Zhuk's experiments might have been used to better effect towards a positive solution of this problem. As for Dr. Kotik's experiments, even though he firmly believes in and defends transmission of thought over distance, his experiments involving Sophia Shtarker were, undoubtedly, a result of inadequate caution on his part regarding the tricks of the subject and her father, as I showed in one of my papers.¹⁵ Dr. Kotik's other experiments suggest some conclusions that could satisfy the most fiery imagination, but one can discuss them only after they are confirmed by other authors.

At this stage, we are interested in a simple fact: is direct induction of one individual by another possible, i.e., can one person influence another without

using signs or any other types of mediation? It seems to me that a question posed in this manner could be positively resolved on the basis of some specially designed experiments with dogs, as well as with humans.

I conducted my first experiments with dogs trained by V. Durov in my own apartment in Petersburg and then in Moscow—both in Durov's apartment and in his so-called "Little Corner." These experiments were conducted by me personally, with or without Durov, and in the presence of several assistants (Vorobiova, Nikonova-Bekhtereva, Shchelovanova, Fleksor, Trirodova-Kazachenko, Professor Feldberg, I. Lev, and others).

These experiments were subsequently continued by my former students Drs. Perepel and Karmanov, then by Dr. Fleksor together with Dr. Einhorn, and by Dr. Ivanov-Smolenskii together with Dr. Fleksor. They were carried out according to the plans I had devised, and were later repeated by me under different conditions.

The overall number of the experiments conducted with dogs at various times is between fifty and seventy-five. Three dogs named Lord, Nickie, and Daisy served as subjects. All these dogs had been previously trained by Durov to obey ("domestication"), so that during the experiment the dogs would remain calm and ready to perform the required task. Thus, Lord, a Saint-Bernard, barked as many times as the experimenter mentally suggested. The number, however, should not be large (no higher than nine), otherwise the dog would become confused due to the developing automatism in barking. I discovered the same ability (to bark as many times as suggested) in the bulldog, Daisy, with which I conducted some successful experiments on mental suggestion, or induction, even from another room with closed doors. A fox-terrier, Nickie, performed some rather complicated actions, mentally suggested by the experimenter: e.g., she would jump on a chair that stood near the wall, then jump from this chair onto a little table, stand on her hind legs and scratch a portrait on the wall; or she would jump on the piano stool and strike the right side of the keyboard; or run into another room and pick out one of the many paper balls scattered around the room.

These experiments have been described in detail by me and my former students (Drs. Ivanov-Smolenskii and Fleksor) and presented at the 1920 Conference at the Institute of Brain Research in a number of papers.¹⁶ I would like to mention at this stage that in the experiments with Nickie and Daisy every precaution was taken to prevent a situation in which the experimenter could—albeit unconsciously—provide clues for the animals.

The later experiments were designed so that the assignment was known only to the experimenter (a doctor) who was separated from the dog, i.e., did not touch it, but made eye contact only while mentally concentrating on the task that the dog was supposed to perform; then the experimenter would isolate himself from the dog by closing the doors behind him, while the actions of the dog were observed by an assistant who was not familiar with the assignment. The results were that mental suggestion was followed with a sufficient degree of accuracy even under these conditions.

The experiments with the bulldog Daisy were designed to test whether the dog would bark as many times as mentally suggested by the experimenter. These experiments were conducted in such a way that the suggestion, or induction, was directed from another room with closed doors, and even so the results were positive.

As for experiments with people, our first subject was an eighteen-year-old girl who had never been involved in such experiments before, but who was highly impressionable and possessed an extremely sharp visual memory (she could reproduce up to seventy-five words and twenty-five two-digit numbers), as well as motor memory (she was a fine dancer). The experimental procedures were as follows: on the table in front of her a number of most common objects were laid out—e.g., a match, a little box, a cigarette, a comb, a sea-shell, etc. The number of the objects varied from seven in some experiments to eight or twelve in others. The experimenters—myself and Durov—were sitting on two chairs next to each other, with our backs to the girl, at a distance of about 6.5 metres away from her. With the exception of the first six experiments we were separated from her by high, thick screens.

The names of all the objects on the table were written down on small pieces of paper (tickets) that were folded and put in a hat from which one of the doctors who observed our experimental procedures would pick one out without unfolding it and hand it over to me. Having unfolded the ticket, I would either conduct the experiment myself, without mentioning to anyone the name of the object, or would pass the ticket on to my neighbor who would then carry out the experiment. Thus, one of us would concentrate on the visual image of the object mentioned in the ticket, while the girl was supposed to indicate which object was being suggested. Here are some of the results: seven simple objects were laid out on the table; Durov acted as an experimenter. He conducted six consecutive experiments, all of which were

quite successful, i.e., every object was correctly guessed by the girl, and only in one case was the object guessed on the second, rather than on the first, attempt.

In another case, we selected twelve objects as common as those in the above experiment (a match, a pencil, a hook, a toothbrush, a piece of paper, an ashtray, a key, a seashell, a cigarette, a box, an electric bulb, and a button). Durov conducted six new experiments: in the first three he was separated from the subject by screens in the same way as before, while in the other three experiments he was separated from the girl by a large metal screen. During both sets of experiments I was sitting next to him. Folded tickets with the names of the objects were placed in a hat, then our assistant, A. Lev, would take one of them and pass it on to me without reading it. After glancing at the note, I would silently give it to Durov, who would concentrate on it by intensely visualizing the object and the girl's intention to pick it up. The results were as follows: the objects selected in the first three experiments were a box, a piece of paper and a toothbrush; the box and the paper were guessed at once, while the brush was picked out on second try after an initially wrong supposition. In the other three experiments, all the objects—a seashell, a toothbrush, and a key—were guessed correctly immediately.

My next five experiments were equally convincing. The experimental procedure was the same, the only difference being that after the experiment I handed the note over to a third party for control. Out of the above five experiments, two failed and the remaining three proved successful, i.e., the girl guessed correctly in one case on the first attempt and the other two on the second.

In other experiments with the same girl, eight objects were placed on the table, and the experimenter was ten feet away from her in another room, behind closed doors. Out of the four experiments conducted by Durov two were successful—with a bulb and a cigarette. The latter was not guessed at once: after the girl asked "Is it a bulb?" and received a negative reply she stated with certainty "Then it is a cigarette." Out of my two experiments one proved successful and the other did not. I must say that in these experiments we avoided repeating those attempts that initially failed. Thus, approximately half of the experiments on suggestion from another room with closed doors yielded positive results, which cannot be explained either by the theory of probability or by anything else.

Interestingly, when the girl was asked why she would pick out a particular object during the experiments, she explained that she visualized very clearly the object in question; on another occasion she said that she visualized half of all the objects and one of them particularly clearly. This was the object she felt compelled to pick out. Another interesting fact is that when the experimenter initially thought of one object and then concentrated on another, the girl often visualized the first object.

Analogous experiments conducted later with other subjects under similar circumstances also produced positive results. They will be published elsewhere.

All of the above brings us to the conclusion that experiments on so-called "mental" suggestion, or—more precisely—direct induction, are successful when conducted with excitable people, as well as with animals. This convinces us that in a crowd, besides mutual suggestion and the so-called contagion that stimulates direct imitation, there must also exist a special factor in the form of direct impact via the immediate transmission of excitation from the brain centers of one individual to the corresponding centers of another individual. At this point, a detailed discussion of the mode of this transmission would transcend the limits of our present goal; however, one can already see the paths to follow in search of the correct explanation of the above transmission in the form of direct impact.¹⁷

Thus, verbal persuasion, mutual suggestion, contagion through imitation, as well as direct induction are the factors that operate in every gathering as in a collective united by a common mood or a slogan.

In conclusion, we shall discuss several other factors that play a particularly significant role in the formation of a collective.

It is commonly known that biological factors—such as the need for nourishment—lead to socialization for the purpose of obtaining food. The same bonding role is performed by the need for self-preservation, which unites different people in various groups with the purpose of cooperation and in the interests of self-defense and attack. In the same way, sexual function based on the carnal relationship leads to a "spiritual" relationship, which usually even precedes the physical one. A marital union is followed by the family that forms a natural unity due to the material, as well as "spiritual," interdependence of family members. For children this interdependence remains primarily unilateral until they become of age. Subsequently, we are

dealing with blood or tribal ties that maintain "spiritual" kinship and form a familial or tribal collective.

One should not forget that the habit of communicating with others, fostered by life itself, becomes an organic need, which from time to time draws us into society for the mutual exchange of reflexes. "Any idea—no matter how primitive it might be— moves a person to communicate it to a fellow human being," says P. Sorokin:

The feelings of joy, grief, sorrow and fear demand a response from others. Joy must be shared, grief demands consolation, fear and indecision require sympathy and approval, while hatred calls for revenge. A desire to accomplish something forces an individual to approach other people for the purpose of involving them in the effort towards achieving the goal in question. All these emotions tend to "spill over" the bounds of an individual soul, making this interpersonal exchange indispensable, and the more intense these emotions are, the stronger attraction to fellow human beings they generate. These needs unite people by invisible but effective ties and attract them to one another—just like the invisible yet effective force of a magnet attracts iron.¹⁸

One should bear in mind that all members of a collective interact with one another either directly or indirectly. This means that individual A acts on an individual B as a stimulating agent, while B in turn acts on A as a stimulus. Naturally, this interaction may prove— and does often prove—unequal, since one individual may have a greater influence over another individual than the latter over the former.

Interaction is also established between the leader and the crowd, the actors and the public, between the organizer of a gathering and the gathering itself, or the chairman of a meeting and its participants. Here, too, the interaction can be direct and indirect.

Due to mediators, a collective association may evolve irrespective of spatial and temporal conditions: a society may be dispersed all over the world and still preserve a certain unity of its members. Similarly, a "spiritual" unity maintained through traditions and other mediators, such as writings and oral communication, may exist in a collective of individuals living in different periods of time. Such are religious communities, states, and other collectives.

In general, favorable circumstances for promoting interaction and unity are represented not only by the spatial proximity that brings people together mechanically, as it were (i.e., stimulates mutual exchanges), but also by the existence of mediators that include all the objects that may be of interest to many individuals; historical monuments, literary and scientific works, technological structures, various inventions, methods of teaching and upbringing, various events and, finally, the common interests aroused by one or another phenomenon—these are the principal mediators of human unity.

Notes

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1. V. Bekhterev, *Suggestion and Its Role in Social Life*, (in Russian), 3d edition, St. Petersburg, K. Rikker.
 2. P. Binet, *La suggestibilité*, Paris, 1900, p. 209 ff. 6. Ibid., p. 16.
 3. A. Vogouroix et P. Juquelier, *La contagion mentale*. Paris, 1905, p. 27.
 4. See the papers presented at the 1919-20 Conference sponsored by the Institute of Brain Research. *Issues in the Study and Education of the Personality*. Issue II, 1920.
 5. Tarde. *The Laws of Imitation*, pp. 34-35.
 6. Ibid., p. 16.
 7. Ibid., p. 13.
 8. Tarde, Op. cit., p. 15.
 9. D. Bordier, *La vie des sociétés*, Paris, 1888.
 0. Sighele, l.c., p. 31.
 1. See *Union médicale*, 1869, vol. 8, p. 369.
 2. See V. Bekhterev, *Suggestion and Its Role in Social Life*, 3d edition. St. Petersburg, K. Rikker.
 3. *Oeuvres complètes*, Paris, 1824, vol. 3; see introduction, p. 14.
 4. Sighele, l.c., p. 50.
 5. V. Bekhterev, "Concerning Experiments on the So-Called 'Clairvoyance' on Stage" (in Russian), *Russkii vrach*, 1911.
 6. See "Issues in the Study and Education of the Personality," Issue 2, 1920.
 7. I mean the latest research of academician E. Lazarev on the ionic theory of excitation (*Recherche sur la théorie ionique de l'excitation*), as well as the electric currents generated by the functioning cortex (see Dr. Trivus's dissertation and Dr. Larionov's work—both from my laboratory). The waves of electric energy produced by the excitation of the cortex and the conductors are inevitably accompanied by the Herz rays, which, as we know, are transmitted in various directions.
 8. P. Sorokin, *System of Sociology* (in Russian), vol. 1, pp. 310-311. See also Ross, *Soc. control*, p. 14.

8

Language as a Unifying Factor

The deeper we go into the study of the social world, the more we realize that large collectives only appear to be homogeneous, and that any collective, once it has been formed, breaks into groups; this fact had been overlooked by the first sociologists (Compte and others). Ross is right when he says,

The moment the sociologist undertakes to "explain phenomena," he is staggered by the variety of reactions, the dissimilarity of responses to like stimuli, exhibited in a given group. Here are contrasts of devout with undevout, of backward-lookers (traditionalists) with forward-lookers...of risk-lovers with risk-shunners.¹

A population may look gray from a distance, but from close up it is seen to be made up of multicolored particles which, when grouped like with like, give rise to all manner of contrasts and effects.²

Ross is wrong, however, when he explains various social groupings in terms of "innate individual differences." These innate differences are certainly meaningful (e.g., the significance of gender, race, etc.), but in this case a far greater significance belongs to the environment that produces persons with certain views, beliefs, and tendencies often absorbed with their mother's milk. Since, as we know, language itself is not an innate phenomenon, views, and tendencies of a social nature are even less likely to be so.

It goes without saying that collective unity is established more easily in homogeneous than heterogeneous groups, but everything depends here on the unifying stimulus. When the unifying stimulus acquires a particular significance for everyone, then most dissimilar individuals—even those in opposition to one another—exhibit similar reactions. This tends to happen in times of national catastrophes, such as wars, natural disasters, etc. Thus, the Great World War united all parliamentary parties within the warring countries, as well as bringing together such nations, dissimilar in their tribal and cultural characteristics, as Bulgaria, Turkey, Austria, and Germany on one side and England, France, Italy and Russia on the other.

One of the most important conditions that lead to and maintain collective unity is undoubtedly the common character of symbolic reflexes in the form of one common language, which, in itself, is a result of collective unity. We should dwell on this subject in greater detail due to its special importance.

The unifying significance of language is so apparent that our discussion of this matter can be quite brief. Even the language of facial expression plays an enormous role in this respect, serving as an important unifying factor in the whole of the animal kingdom. In the human world, too, facial expression is not devoid of unifying significance for, as we know, it often functions as a most important unifying factor, (e.g., in a crowd that produces the same exclamations, and same gestures and other expressive movements). Facial expression and gestures are also an important auxiliary means for a public speaker who is trying to unite his listeners by a common goal or bring them together in one aspiration. An incomparably more significant role in this respect is played by sound language, or discourse, that acts over distance as far as sound can travel and, consequently, can play a unifying role even when people have no opportunity to see one another.

P. Sorokin writes the following about language:

People who speak the same language are members of the least costly and most popular telephone network in the world. They can easily connect with one another at any moment and without any assistance from a telephone operator. Those who do not know the language are not part of this network; they have no way of connecting with people who speak other languages.³

If one also takes into consideration written discourse that permits people to communicate by mail and telegraph, it becomes clear that discourse is capable of manifesting its unifying significance irrespective of any spatial circumstances.

Compared to the primitive language of gestures, sound language has the advantage of being a means of communication that is not as closely tied to spatial circumstances as the language of gestures, which is generally a rather primitive mode of communication.

In the course of social progress, language departs further from being a primitive reaction that replaces the language of gestures and becomes a more versatile form of communication in which gestures and intonation are only supplementary; written language, being symbolic and devoid of gesture and intonation—these most important tools of self-expression that provided the primary base for the development of oral language—is the next consequence of sociality:

The role of language as a symbolic human reflex is enormous.

Our words are similar to an electric current that arises between people, enables them to influence one another and to condition their interpersonal relations. Their effect can sometimes be astounding. "A word can kill." In some religions, as for example in Brahmanism, the word has become the magic force that rules the world and even the will of the gods. Significantly, the apostle says in the Gospel: "In the beginning was the Word, and the Word was with God, and the Word was God."⁴

In any collective, language is involved in generating its unity. After many arguments and conflicts, a collective reaches a state of relative consensus; this achievement due mostly to the dominant role of language. A social person submits naturally to the power of persuasion exercised by his fellow human beings, which results in a collective consensus.

Persuasion is the usual mode of influencing those who participate in collective assemblies. In essence, any discussion of a particular issue by an assembly is aimed at persuading the assembly to accept as correct the points made by the speakers, which in the end enables the assembly to arrive at a certain decision.

The press uses persuasion to influence a collective in a similar fashion.

In any case, some of the most effective means of uniting a collective through persuasion are the press and public speaking. Oral speech has a stronger effect than the printed word, for it affects the listeners with the emotional impact of the speaker, arousing in them an appropriate reaction that is further intensified by the common mood of the assembly. The impact of the press on the members of a collective can never be as strong, but its sphere of influence is considerably larger, for in one way or another it affects nearly the whole literate human collective that speaks the same language and, through translation, the educated portion of all humanity.

How persuasion can affect a mass of people in confusion was exemplified by Kerensky who, after assuming the post of War Minister at a critical moment in recent Russian history, at a time when our troops were fraternizing with the Germans while we had to show our allies whether we were with the Germans or against them, succeeded by the sheer power of persuasion in rousing the troops from their passive state and leading them in the attack of 18 July 1917, which was highly successful. It was, however, the last spark of energy displayed by the demoralized collective of our huge army.

Verbal suggestion, which is also realized through language, plays an enormous role in uniting the masses. Naturally, we are speaking about suggestion in the waking state, or what we call "personal influence."

"As we know, the personal influence of one individual upon another is an elementary social phenomenon that, except for its scale, does not differ from the influence of the hypnotist on the hypnotized."⁵

Since I discussed this influence in more detail in my work, *Suggestion and Its Role in Social Life*, there is no need to dwell too much on this phenomenon here.

We shall only point out the role of suggestion in collective illusions and hallucinations, common mood, collective actions, etc.

Examples of collective illusions and hallucinations based on mutual suggestion were cited in my above-mentioned book. Suggestion may produce mirages similar to the illusions and hallucinations of the organs of perception in higher levels of associative activity. This diminishes somewhat the unifying role of suggestion in collectives in comparison with persuasion, despite its advantage with respect to speed of action.

Everyone knows that one emotional speech delivered at a gathering is enough to charge the whole assembly with the same emotion, when the very

words and gestures of the speaker are, as it were, directly absorbed by the audience.

We also know that to scream "Fire!" in a crowded room is enough to start a panic. A lynch mob often begins in the same way, with someone thoughtlessly inciting the crowd. In another case, a single call "Get him!" acts like a wave of an invisible hand, sending the crowd after its victim.

The epidemic scale of some social phenomena is based to a significant degree on suggestion.

Suggestion has always played an enormous role in social life. G. Le Bon's opinion is significant in this respect:

If we were guided in the process of adopting our political, social or religious views by at least a small dose of the lucid and consistent reasoning applied by the humblest shopkeeper to his business affairs, then in the sphere of political and religious issues we would not depend—as we do now—on fashion, environment, and emotions, and would not be swept along by current events and opinions.⁶

In society, all propaganda is in general aimed at suggestion rather than persuasion. The following example of organized propaganda that placed a special four-minute limit on its agents' public speaking has been borrowed from the press:

Americans stationed in Petrograd have received quite definite information concerning the United States' intention to continue the war in close alliance with England, France, and Italy on condition of the strict coordination of plans of military operations and genuinely mutual control over all measures and undertakings related to the issues of war and peace.

It is impossible to conceal that there exists a certain "defeatist" movement in some American circles. Specifically, some groups believe that it is not in the interests of the United States to assist England in defeating Germany.

These "defeatists" are devising all kinds of strategies to end the war in such a way that America alone—and not Western Europe—would be the victor.

President Wilson has turned his attention to this kind of defeatist propaganda and has set up counter-propaganda units in the United States.

The most curious aspect of this matter is the four-minute limit imposed on each of these speakers.

The nickname for these 15,000 propagandists is "four-minute people." They do not frequent large meetings that, as we know, attract people with definite party views.

Every day they visit theatres, cinemas, railway stations, etc., where they address their "four-minute" speeches to a chance audience.

They also take streetcars, trains and steamboats where they can address passengers with their short speeches.

To quickly produce the desired mood in their audience, speakers from the "four-minute" units use a method of comparison we find rather interesting.

"Some people say that America's victory over Germany is desirable, while England's victory over Germany is not," begins the speaker, "yet what sensible man could wish that the thieves who broke into his home would steal all of his wife's belongings on condition they would not touch his own things? Or who would wish that his shop would make good sales and order the cashier to daily throw the contents of his till into the fire?"

This kind of propaganda was quite successful with the Americans.

Naturally, the role of language in a collective is not reduced to its unifying significance, for due to language a given collective develops its own worldview. We owe to language the evolution of religion and morals, of science and arts and even technology, for in the final analysis both religion and science, literature (including fiction) and technology are the products of the evolution and richness of language.

Obviously, none of the above could exist without language, and even culture, in the modern sense of the word, would have been impossible.

In the cultural life of all nations, language functions as a tool that is used to satisfy various needs arising during the evolution of the composite personality, including the success of its literature, that reflects the totality of

its intellectual acquisitions. It is due to language that the evolution of knowledge in human society has become possible.

The wizard—the primeval scientist who, at the dawn of civilization, was trying to disclose nature's secrets and to subject its mysterious forces to his own will—uses primarily language in his craft. Whether his purpose is to cope with a disaster, to punish a thief, or to find a lost object, he is dealing with verbalization or some other symbolization related to the event or its participants, and here one finds primitive reasoning by analogy.

When, chanting a spell, he throws wood shavings in the fire, it symbolizes the death and scorching of the enemy or thief. When the wizard pierces the heart of the enemy's effigy, it symbolizes the suffering and impending death of the person thus represented. These methods also depend on the power of suggestion, for the intended victim is bound to hear of the magician's spell.

With time, the wizard is replaced by the medicine man, who appears at the stage of social evolution when faith in the power of identification and symbolization is on the decline, while faith in mysterious forces of nature, which are revealed to those who devoted themselves to their study, is still alive.

The medicine man is followed by primitive natural philosophers, such as astrologists and alchemists, who are striving to understand the essence of things and to achieve universal scientific values through the discovery of the philosopher's stone.

It is only after these natural philosophers that the true scientist arrives with his analytical methods of investigation.

However, the scientist, too, when recognizing the mysterious forces of nature, created by dialectics and speculative thinking, initially pays homage to the precocious natural philosophy, in order to explain phenomena he does not understand.

It is very slowly that science begins to discard these useless fetishes and to advance the value and role of positive knowledge based on the strict logic that results from the observation of surrounding nature. With the development of logical thinking, the role of language is therefore invaluable in the evolution of knowledge.

We also owe to language the evolution of public opinion.

With the emergence of family hearth, clan, and tribe, there began the establishment of corresponding positions of a social nature. Before these alliances came into being, human actions were guided solely by an egoism

that at most permitted mercy for one's enemy; now, for the first time ever, human actions began to be regulated in the interests of an alliance. This regulation is everywhere guided by so-called public opinion, which defines good and evil from the viewpoint of the public interest, rather than from the viewpoint of individual aspirations alone. In addition, special court procedures have been developed in the course of time, and punishment for antisocial behavior, as well as rewards for civic-minded actions, have been instituted. The court system, however, did not decrease the power and importance of public opinion, which plays a role in the most primitive societies as well as in later—and more developed—social organizations.

Public opinion invariably turns out to be a more perceptive judge of human behavior: it qualifies even those moral actions that fall outside the court's province and passes its own unequivocal verdict on them. This public opinion is related to the phenomenon we call popularity, fame, or celebrity, which stimulates many people to become engaged in actions of a social nature and social interest.

When a Spartan or an Athenian hero sacrificed his life so easily in order to win fame in a small country, he was convinced that the whole world with all its invisible forces was watching him, that the great spectators of the whole world were applauding him, and that the echo of this applause would resound forever in the infinity of time and space.⁷

In all these cases, the role of language is indubitable, for only language makes the existence of public opinion possible and functions as its means of expression. It is only due to language that one can become popular, well known, or famous.

According to Le Bon, one

has to take this into consideration even in politics, where the language of a defeated nation facilitates the introduction of new ideas, laws, religion, beliefs literature and institutions that are being imposed on it, and which it could probably assimilate without the psychological resistance produced by its dialect.

The Low Bretonne dialect probably did more to prevent Bretagne's assimilation by France than what Christianity did to achieve it; similarly,

the Arabic language represents no lesser obstacle to the French domination of Algiers than Osmanism itself.

Therefore, all state governments prefer one state language for their citizens, so that commands to soldiers can be given in one language—otherwise the state would no longer be united and its army would fall apart. Moreover, without language there can be neither sociality nor political associations and collectives in the true sense of the word.

Tarde is wrong, however, when he claims in his *Social Logic* (p. 210) that a child enters social life only when he begins to speak.

Even before acquiring verbal language, the child uses facial expression and gestures to make contact with the environment.

In exactly the same way, animals, being deprived of language in the human sense, form communities and achieve concord by using their facial expressions. Thus, language in the form of articulate speech is only an improved tool for reaching unity in a human collective.

According to Drummond, the study of the content of language will take us back not only to what people used to produce with their own hands, but also to what they used to tell one another while working, and to what they used to think while talking:

The word is the abacus for the brain, the palpable manifestation of a mental state, an inherited estate of the cultural wealth of the race. An ancient word, like an ancient coin, tells us about a bygone mentality, while by its image and inscription it reveals the mental life and aspirations of those who minted it.

We had the opportunity to discuss the origins of primal language elsewhere;⁸ but let us imagine that a language has already evolved— what next? It develops further, through differentiation and expansion of its parts, roots, prefixes, flexions, etc., and through the formation of compound and composite words by combining different roots. Finally, the language branches out into different dialects or even several languages, which depend, on the one hand, on the geographical and economic isolation of the population and, on the other, on various local influences, such as the influence of neighbors speaking a different language, of immigration, of assimilation of nationalities, etc.

Irrespective of the above, linguistic evolution is a legitimate phenomenon; this evolution, however, does not have to follow the same pattern even under the same external conditions. One population may live through a period of political upheavals, another may be a successful trading nation, while a third one is quietly engaged in agriculture. Such conditions must necessarily affect the language of these nations. These and other circumstances explain how, after the fall of the Roman Empire, the Latin language gave birth to several Romance languages and numerous dialects, just as Sanskrit in the past provided the basis for many modern languages. Linguistic differentiation, however, occurs parallel to linguistic assimilation that, in turn, leads to an amalgamation of languages, which separates them even more from the basic para-language.

So many languages have already disappeared from the face of the earth. If the language of a cultured nation—such as Greek or Latin—has been preserved to this day, even though no longer in use, this is due only to the surviving literature written in these languages. Conversely, why did a great many languages sink into oblivion? Simply because they could not compete with other languages that gained currency in their country, whether due to the conquest and rule of victors who spoke a different language, or due to cultural amalgamation with another nation. The fact is, however, that a language does not vanish without a trace. The disappearing language introduces into the language that absorbs it some local words, expressions, accents, etc., which make it possible to speak of linguistic fusion and amalgamation, rather than the destruction of one language by another. And, finally, amalgamation of languages is manifest in the universal aspiration to use a common language in international relations, for which purpose artificial languages have been created. Thus, after the unsuccessful Volapük, we have recently seen increasing use of Esperanto, so that one can foresee the time in the distant future when all the nations of the globe will join together, using a language common to all humanity.

Since language is, generally speaking, the most powerful and most indispensable means of interpersonal communication, it is no exaggeration to say that such social changes as the assimilation of individuals and social groups, which occurred in a given territory due to the replacement of horse-drawn vehicles by locomotives, are nothing compared to similar transformations caused by the victory of a major dialect over a minor jargon, or of a language over dialects.

Linguistic similarity is a sine qua non of all other social similarities and, consequently, of all the great and glorious forms of human activity that presuppose existence of these similarities, which provide the framework for their work.

According to Tarde (*The Laws of Imitation*, p. 121),

It is due to language that a very strict order is established in the chaos of contradictory visions and hallucinations that assault the ears of the primitive man.

Language represents a logical arrangement given to the social human being, just as time and space are given to the individual.

There is no need to analyze in detail the erroneous nature of this statement, for language evolves together with society, rather than society evolving due to language, while space and time are not given to each individual person, but are the result of the collective experience of the social human being (see below).

Tarde's interpretation of the origins of language is also rather peculiar. In his words, languages are born out of religions, although they do subsequently facilitate the development of the latter:

I am quite willing to believe that words represent the logical consequence of the fluorescence of myths transmitted from generation to generation. And, indeed, to speak means essentially to divinely personify and animate every object whose qualities or actions acquire their own independent existence and magic power (p. 118). I can interpret the primitive word only as a peculiar acoustic fetish that emerged by itself, under the influence of the deification of nature as well as the most extraordinary human actions, whose number began to increase as religious enthrallment, or awe, began to be related to new objects and people, losing its connection to nature (I.e., p. 120).

Thus, in a language, every root is nothing other than an idol still used and preserved, while language itself is nothing other than an ancient remnant of prehistoric fetishism and naive, gradually extinguished religions; it is in a sense the ashes of the ancient sacred fires.

I am quoting these statements for the sole purpose of illustrating the conceptual chaos that dominates this subject in general.

Our own understanding of linguistic evolution is quite different: language originally evolved as a result of collective experience, as a supplement to facial expression in the form of sound imitation and sound reflexes, conditioned by external and internal stimulation: e.g., being suddenly hurt or annoyed produced such expletives as "Ouch," "Oh," "Ah," etc., that were subsequently, according to the law of associative reflexes, transformed into interjections from which language then developed *via* doubling, differentiation and combination, prefixes, flexions., etc. The original sounds that provided the basis for future human language came from the animal world, and now, due to Professor Garner's research in this field, we have acquired some more or less definite ideas on this subject. In any case, it is clear that primates undoubtedly possess a number of acoustic signs for identifying objects that are a part of their simple everyday existence.

Generic definitions develop subsequently from specific ones. For example, the word *devu* that used to refer to a fir-tree, a conifer, became a generic term, since conifers happened to be the most common trees. On the other hand, in Iceland, where there are no trees at all, it is the word *eik* (oak) that is used as the generic name for tree. God was originally identified with the sky, the sun or the sunshine (Sanskrit *deva*) or lightening, and it was only gradually that man arrived at the concept of God as the creator of the Universe, by discarding the physical attributes of the object defined by the symbol, so that God came to signify a superhuman being.

How words lose their original meaning is exemplified by the Russian *Spasibo* (meaning "Thank you," but literally "God save you") and the French *Adieu*, that are nowadays completely devoid of all religious significance. In general, all religious terms originate in the everyday world, which is exactly the opposite of what Tarde states in the above quotations.

Notes

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1. E.A. Ross (1905), *Foundations of Sociology*, New York, Macmillan, p. 291.
 2. Ibid., p. 293.
 3. P. Sorokin, *The System of Sociology*, vol. 1, p. 227.
 4. Ibid., *The System of Sociology*, vol. I, p. 127.
 5. See Tarde, *Social Logic*, p. 93.

6. *The Psychology of Socialism*, p. 44.
7. Tarde, *SocialLogic*, p. 303.
8. See V. Bekhterev, *Objective Psychology* (in Russian), St. Petersburg, 1912. Also *General Principles of Human Reflexology* (in Russian), 1918, by the same author.

9

On Collective Reflexes in General

Since there is no doubt that some collective reflexes belong to the order of hereditary organic reflexes, while others to the order of higher associative reflexes acquired through life experience (we shall exemplify this below), then a question arises whether a collective can manifest reflexes of a simpler nature, ones which we call "ordinary" in general human reflexology and which, as we know, occur in a machine-like fashion. It seems that under certain conditions a collective, too, can manifest ordinary reflexes. The latter, in my opinion, include such collective reflexes as panic in a crowd and unmotivated assault—i.e., collective reflexes of a defensive and offensive nature that develop through contagion.

Panic is an unconscious defensive reflex of flight displayed by a whole collective; it is unrelated to external circumstances and often occurs without a legitimate cause or for an insignificant reason. Such panics are also known to occur among animals—e.g., so-called stampedes in herds of horses, pigs, etc. The Gospels mention a herd of pigs possessed by an evil spirit and driven into the sea, where they perished.

More recently herds of sheep were observed to throw themselves off a precipice and die under the influence of panic.

We all know how easily an unexpected event can spark off an extremely dangerous panic among a crowd of people. It is enough to shout "Fire!" in a dense crowd that is listening to an exciting speech, and a fatal panic that will claim many lives will break out.

There have been cases when some ill-intentioned individuals used this device to enrich themselves at the expense of those who were panic-stricken.

At this point there is no need to elaborate on the development of panic, as the reader can learn more about it from my book *Suggestion and Its Role in Social Life*, where I give a detailed description of panics partially based on my own observations. A characteristic feature of all these panic attacks—besides their standard and purely automatic manifestation in the form of uncontrollable flight—is the unconscious nature of the flight reflex itself, for after the panic no one can usually explain why they ran and how it all happened.

As for the automatic reflex of collective attack developed through contagion, this phenomenon, too, is known to occur both in the animal world and among the humans. Thus, when we encounter a pack of dogs, it is enough to accidentally provoke one dog for all the rest to join in the attack. There is no need to elaborate on the danger of such attacks. Likewise, if someone in an angry crowd assaults another person for the most trivial reason, the whole crowd will attack the same unfortunate individual without even knowing what it is all about.

Hereditary-organic reflexes and individual reflexes of the same type have an identical basis, the only difference being in the collective nature of their manifestations. Thus, the looting of food stores and markets during a famine or unrestrained sex orgies exemplify the hereditary-organic reflexes of self-preservation in the first case and those of sexual nature in the second.

As for collective reflexes of a higher order, they, too, may be observed under certain conditions. Thus, collective reflexes of concentration are easy to observe in any concert hall, theatre, or auditorium.

Symbolic collective reflexes—verbal ones in particular—can be observed in the gestures and exclamations of a crowd, in choral singing, etc.

The emergence of individual or collective purposive reflexes that, like other associative reflexes, evolve on the basis of experience needs further elaboration. An individual reflex evolves in the following manner: stimulus A evokes in individual B a particular reaction of concentration, attack or defense, or in some cases it may suppress a reaction, depending on whether A affects the given individual beneficially or adversely. A beneficial effect is accompanied by a positive mimico-somatic reaction on the part of the individual, while an adverse effect by a negative one. Both, however, depend not only on the stimulus alone, but also on the state of the individual. A repeated action of the same stimulus produces the habit of reacting to this stimulus in a certain manner, while the habit itself creates a specific need in

the form of internal stimulation caused by the absence of the mimico-somatic reaction that followed the external stimulation. This results in the internal stimulation emerging independently, i.e., without any external stimulation, whose presence always increases this need to the maximum.

When the external stimulus produces an adverse effect, the process of internal stimulation occurs along the same lines, albeit with the opposite result—i.e., with a defensive reaction or with the suppression of all reaction. In other words, one develops an inner need for defensive action or avoids the given stimulation altogether.

The same result is achieved by mediated stimulation. Thus, a certain stimulus may affect a given individual not immediately, but following the impact of a preceding or accompanying stimulus.

Thus, a food object may be hidden inside an inedible shell or under the leaves of a certain plant or in particular places in the soil, etc. On the other hand, a harmful stimulus concealed behind some external obstacles may begin to produce its adverse effect after they are removed. In these cases, the acting stimulus is accompanied by another, and the impact of the latter is preceded by that of the former.

However, in accordance with the law of associative reflexes, the accompanying stimulus begins to act like the primary stimulus after the repeated action of the latter. For this reason, the stimuli that do not produce either beneficial or adverse effects directly on an organism provoke a specific reaction that attracts or repulses the individual, depending on what kind of primary stimuli accompanied these originally neutral stimuli.

What is true for individual persons is also true for a social group as a collective entity, which is why the group response to external stimulation is in certain cases exactly the same as the individual one, the only difference being that here the individual reflexes are combined due to mutual communication and suggestion.

Thus, for example, one can interpret the development of an uprising in the following manner: through one's own individual experience every person develops a sense of protest as an associative reflex of defense that evolves from the ordinary defensive reflex under unfavorable physical stimulation. When enough external reasons for protest are accumulated by many individuals, manifestations of this protest in a collective are combined, and due to interaction and contagion a certain collective unity evolves, which, in turn, enhances the manifestation of this protest. When this protesting unity

emerges in a collective, the most insignificant cause provides a sufficient stimulus for the collective protest to reach the level of social unrest that keeps growing to the point of collective explosion in the form of an uprising.

The development of the collective reflex of attack in the form of pogroms, wars, etc., proceeds in an analogous fashion.

According to the above law of associative reflexes, not only the goal itself but also the means of achieving this goal—i.e., the preceding stimulus—must be sufficient for triggering the reflex. Thus, food stimulates not only every individual but a collective as well. Since it is the acquisition of money that ensures the acquisition of necessary household objects and food products, it is clear that, due to experience and according to the law of associative reflexes, money, too, becomes a stimulus that triggers the same reaction as the main goal, i.e., the necessary objects and food products, as a result of which the money itself becomes an immediate goal. All processes of exchange in collectives, banks, and other organizations are based on this process of development of associative reflexes in which not only the goal itself provides stimulation as a result of experience, but, as well, all the means of achieving this goal.

One should not overlook the fact that any social movement to a significant degree owes its existence to special propaganda that often consolidates, as well as stimulates, collective action. This propaganda is usually carried out by agitators who have no scruples about using every means at their disposal to electrify a crowd.

One of the classic examples of this kind of agitation is Anthony's speech after the death of Julius Caesar in Shakespeare's play.

In any case, agitators know from experience when and where to address a crowd with their inflammatory speeches and to achieve the desired effect by the appropriate means.

Agitators themselves—these temporary leaders of the crowd—are often far from courageous, but those who sometimes skillfully hide behind the crowd. When arrested, they usually show no signs of repentance and at times behave in an extraordinarily cowardly manner.

Thus, according to newspaper reports, an agitator who during the Kronshtadt uprising of July 4 and 5 did not waver to agitate for shooting defenseless women and children in the streets behaved in the most disgraceful, cowardly fashion when arrested:

He was trembling with fear like a leaf, terrified of being executed, and begged the military authorities tearfully not to let the Kronshtadt sailors and soldiers escort him to Petrograd.

"They are angry with me and will shoot me," this agitator and crowd leader kept babbling.

A social movement as such is a result of the combined activities, struggle, and cooperation of individual persons, groups, and associations or unions. Like any social process, it must produce either a borrowing based on imitation or a creative act. In the latter case, there appears in the collective life of the society a relatively new and original phenomenon, even though it may prove similar to one that was once experienced in the past.

Every social movement usually develops until it exhausts all of its potential power generated by the corresponding stimulation in a given environment, or until it encounters the corresponding counteraction in another social movement or in some external force; in other words, a collective reflex—as any associative reflex—is terminated by internal or external inhibition. While developing in the form of a collective reflex, a social movement at a certain point reaches its climax, after which the wave of the movement begins to ebb and finally disappears, just like an associative reflex of an individual person. In such cases one can speak about a social shift at the initial stage of the social movement, about a turning point in the mood of the popular masses at the peak of the social movement, and about its decline when the movement begins to weaken and is then replaced by another social movement.

Obviously, the shift itself, as well as the turning point, is conditioned by some specific causes that force the masses to change direction.

Thus, during the Great Russian Revolution, when at the innumerable political meetings in the streets and in public buildings one heard the appeals to destroy the bourgeoisie, to rob the banks, few of the authorities and of the bourgeois population realized the danger of this kind of propaganda. They believed that an ideological sermon of murder and undisguised robbery had as much legal right to exist as any other, and should not therefore encounter any legal counteraction in the name of the proclaimed freedom of speech.

Similarly, newspapers and leaflets that were disseminated among the troops and called for the disobedience to their superiors were not

counteracted, in the name of the same freedom of the printed word.

Then masses of armed workers appeared in the streets of Petrograd, the shooting began, and we ended up with the July uprising that demanded military intervention. Those in power woke up to reality, and, at the same time as the uprising was suppressed, a radical shift occurred among the large segments of society that turned against Bolshevism.

As a result, a considerable part of society and the authorities moved to the right, and the freedoms of assembly and speech were curtailed. Administrative measures against individuals were introduced. Some newspapers were prohibited, and the Bolshevik movement was temporarily suppressed.

For at least some time, the Bolshevik leaders did not dare to make speeches at the meetings where they were usually ridiculed; for a period of time, the unfolding social movement was suppressed or temporarily inhibited.

All of this notwithstanding, the propaganda among workers and soldiers was doing its bit, and on the 25th of October of the same year the Bolshevik *coup* took place, the Provisional Government was arrested, its head Kerensky fled, the cadets and the women's battalions that offered resistance were disarmed and—in the case of the cadets—shot. Thus, the wave of the Bolshevik movement reached its apogee.

The first months after October 25 were a period when the Bolshevik movement was still on the rise and, finally, its wave reached its highest point; this wave, however, can never stay at its peak and must gradually ebb. It is already on the wane, for the Bolshevism of today is in many ways different from what it used to be in the first months after October 25.

The same processes occur in all other collective reflexes. Some barely noticeable symptoms signal the appearance of a new social movement that grows, develops, reaches its highest point, then gradually weakens and enters the quiet flow of social life.

10

The Development of Collective Movements as Associative Reflexes

We know from reflexology that an associative reflex evolves on the basis of an ordinary reflex and is essentially its reproduction. The same is true of collective reflexes. Let us take a crowd pacified by a police detachment and fleeing in panic when shooting begins— this is an ordinary collective reflex. When a collective is subjected to a violent assault and one can hear swearing, groans, and heartrending cries from its midst, this is also an ordinary reflex. But when one hears the same swearing, groans, and cries from a collective at the time when it is perturbed by an emotional account of the atrocities inflicted on those it holds dear, then we have an associative reflex. When a crowd disperses under the influence of a simple threat, this, too, is an associative collective reflex.

It is clear from the above that there exists a permanent analogy between the development of a collective associative reflex and that of an individual associative reflex, for it is evident that in both cases the associative reflex— whether individual or collective—evolves from an ordinary reflex as its reproduction under the impact of the stimulus that has accompanied the principal reflexo-genic stimulation.

One should point out that the stimulus that evokes an ordinary reflex can be replaced by its symbol—a word or another sign—and the end result will be an associative reflex of the same type. For example, a person who has never experienced the impact of firearms himself and has never even seen it in action, will, nevertheless, turn away from it in fright exclusively under the

influence of stories about its lethal effect. In this case the sight of the firearms alone or even its mention is enough to stimulate an associative reflex of defense. The fact is that the symbol is inevitably associated with its meaning as a weapon whose threat one can avoid only by removing oneself from its sphere of action.

The same is true of a collective. In the above example of crowd pacification one does not even have to attack the crowd. Everybody knows about the lethal power of firearms or side-arms, if not from one's own experience, then from the verbally communicated experience of others. Consequently, in the course of one's life, everyone develops a reflex of defense at the sight of a weapon and when being threatened by it, which is enough to induce a crowd threatened by weapons to disperse.

If this does not happen, even a single blank-fire shot is enough to put the crowd to flight.

In this case, too, we are dealing with an associative collective reflex that follows a symbolic stimulus in the form of a simple threat, thus reproducing the reflex caused by real stimulation under different circumstances. It is obvious from the above that for a collective associative reflex to develop it is not at all necessary to have the entire collective subjected in advance to the kind of external stimulation that would lead to the development of an ordinary reflex. The fact is that an associative reflex can occur without such collective experience, due exclusively to the influence of secondhand accounts of the consequences of the above impact, as well as due to the previous experience of some individual members of the given collective. In the latter case, the experience of certain individuals is disseminated throughout the entire collective through discourse and becomes its common property. This is clear from the fact that the quality of a collective and its relationship to the surrounding world depend on its individual members who can be subjected to appropriate training. It is therefore natural that the experience of certain individuals has some effect on the activity of the entire collective.

Every social movement has a cause that acts as a stimulus. The Swiss uprising developed on prepared ground after the well-known incident with the humiliating demand to shoot at the apple on the son's head. The Russian Revolution of 1905 was started by the 9th of January shootings at the mass demonstrations. The revolution of 1917 started with the dissolution of the State Duma instead of the setting up of the responsible ministry demanded by

public opinion. The Spartacists' uprising in Berlin on 5 January 1919 started with the dismissal of the Berlin prefect Eighorn, who had been elected by the workers and who decided not to abandon his post as given to him by the workers.

It is clear that every social movement develops as a mass associative reflex under the influence of a particular stimulus that acts as the final impetus to a previously prepared situation.

To establish an analogy between the manifestations of social or collective activity and the associative reflexes of an individual personality, one has to analyze a particular social movement.

In this respect there is no better example than an army where genuine collective associative reflexes are developed through training and discipline reminiscent of those used to train performing animals. On hearing a single word "Form!" a mass of people assume formation. The phrase "Stand easy!" frees the troops from the constraints of the previous command. At the command "Dress to the right!" one and all turn to the right; at the command "Dress to the left!" everybody turns to the left. At the command "Ready!" all rifles take aim; at the word "Fire!" everybody fires simultaneously. Here the words of the command act as an associative stimulus, while the main stimulus is the mandatory nature of army drill.

If we take mass gymnastic exercises, we shall come across the same learned associative collective movement when an instructor's command causes an entire mass of people to perform simultaneously, one and all, the same movements. The command in this case, as well as in others, can with the proper training be replaced by any sign—by a wave of a hand or a scarf, by a flag, a shot, by the sound of a horn or a drum, etc.—which results in the performance of the same movements. Finally, after appropriate training, the complex gymnastic exercises, consisting of a particular sequence of movements, can be performed to music.

Those who ever attended the so-called Sokolskii rallies had a chance to see for themselves how an enormous mass of people, numbering tens of thousands, performed complicated gymnastic exercises to the sounds of music. All of them would simultaneously raise their arms to a particular sound of music, lower them to another sound, kneel with the third sound, then rise with the same tempo, turn sideways, etc.

The same occurs in dancing. Here again music is an associative stimulus, while the need to move is the principal one, just as in the above example.

Here, too, a mass of people moves to the rhythm of music that acts as a familiar associative stimulus, while the words of the instructor provide a particular signal.

Another example is a large orchestra. All members of the orchestra have before them a musical score from which they have already learned the piece to be performed. A sign from the conductor focuses everybody's attention on him, and then, to the movements of his baton, the musicians begin the prepared and previously coordinated performance on a number of instruments, some of which are supposed to start at one moment, others at another and so on. Finally, the last wave of the conductor's baton and everybody stops playing instantaneously. Is this not a conditioned associative collective reflex?

The same phenomenon occurs in choral singing, as well as in theatrical performance, where the function of a conductor's baton is to a certain extent assumed by the prompter.

Further, the actions of firemen performing a certain task or the coordinated actions of workers engaged in a collective effort provide still more examples of the artificially conditioned collective associative reflexes that are perfectly similar to the associative reflexes of an individual personality.

It goes without saying that in everyday life the same kinds of collective associative reflexes are conditioned quite naturally.

Thus, to avoid misfortune a crowd is immediately put to flight by a scream like "Fire" or "Run for your life"; on hearing "Get this thief it beats him up. The first sounds of gunfire send a crowd into a chaotic flight and so on.

All of these are examples of natural associative collective reflexes cultivated or conditioned by life itself.

When a collective is subjected to a particular stimulus leading to a certain reaction, any concomitant circumstance becomes capable of producing the same reaction.

Let us presume that one nation is engaged in a war against another nation. It is common knowledge that defensive and offensive actions by the troops are determined by enemy actions. The enemy tries to conceal these actions. If, however, its offensives were frequently accompanied by a particular signal, it is natural that the defenders will be alarmed by this signal, even though it may not signify the enemy's attack. If the mounted police often fired at a crowd, the latter will immediately disperse at the mere sight of a

few mounted patrols, even unarmed ones, or on hearing the clatter of the horses' hooves.

Even artificially cultivated collective associative reflexes adapt to various natural circumstances. Thus, the sounds of a march induce a crowd to start walking to the beat of the music, and the sounds of a song accompanying collective work cause workers to move in rhythm with the song.

When a theatre audience applauds an actor and displays other signs of approval, it reproduces ordinary gesticulatory reflexes and facial expressions, doing this as a collective. When a crowd laughs together with an actor on stage, it is infected by his laughter, reproducing the same mimic reflex. When listening to an orchestra, it manifests collective concentration just like any individual person.

When a crowd flees from a firing squad or attacks someone, perpetrating a mob lynching, it performs a collective defensive or offensive associative reflex.

When it goes on a rampage, looting bakeries or butchers' shops in time of famine, it follows the instinct of self-preservation. When a choir sings or a crowd screams, we are dealing with a collective symbolic or verbal reflex, etc.

Clearly, all of the above cases provide valid reasons for comparing the activity of a collective to that of an individual.

The situation is much more complicated when we deal not with a crowd that performs a particular movement or action but with a mass of individuals in an assembly, when after prolonged discussions these persons arrive at a certain decision. There seems to be little analogy with individual human activity. But let us take a case when a person is given a certain task that he is supposed to accomplish in one way or another. In this case, the main stimulus is the goal of providing a certain solution as a result of necessity or need; the performance of the task is realized through complex, previously cultivated forms of judgment, considerations *pro* and *contra*, evaluation of all available information and various events from one's own life experience.

It is not difficult to perceive in the performance of this task a certain analogy with collective work in assemblies.

In assemblies, we deal essentially with the same kind of activity that is pursued by an individual person, but here the work is accomplished jointly by an entire collective, as a result of which the resources of individual experience are replaced by the experience of the entire collective. Here, the

artificially cultivated need for solving a problem and the performance of duty, or a sense of responsibility that guides collective and individual actions in such cases, may and should be perceived as the major stimuli that accompany all collective intellectual activity performed in an assembly and leading to particular solutions.

Thus, in all the diverse forms of the manifestation of collective activity one can observe a complete analogy with individual associative activity; and just as the latter consists of individual associative reflexes, one encounters in a collective associative reflexes of a collective nature, i.e., performed jointly by the entire collective.

Every social movement always follows a path that delineates the behavior of the entire collective as an entity.

Every major collective, however, consists of a number of minor collectives that, in turn, fall into smaller collectives, etc., so that every collective, irrespective of its size, follows a distinctive line of behavior that represents the resultant force of all the aspirations pursued by the individual members of a given collective. Thus, the line of behavior of any large collective represents a resultant force of the resultant forces of the smaller collectives that make it up.

Since the smallest collective consists of two individual persons whose associative activities differ in character and content, and since all collectives, in general, have a heterogeneous membership and therefore display different lines of behavior, it is obvious that every collective is either supported by others when their interests coincide or counteracted and inhibited in its actions when there is discord and disagreement. Perfect accord between all collectives, just like perfect accord among individual members within every collective, is not possible.

In general, one should bear in mind that every collective, like any organism, lives, i.e., moves, develops and evolves, and its movement is determined either by its needs or by external stimuli. The former constitute an inalienable part of any living collective, for only a dying and disintegrating collective would lose its needs. A need, however, arises either in the form of hereditary-organic reflexes determined by biological conditions, or from a well-established habit conditioned by the numerous recurrences of particular associative reflexes.

In the first case, the requirements of a collective arise from its past experience of satisfying the needs that are inherent in the very essence of the

organization itself, just as in an individual organism some needs in the form of the so-called instincts are determined by the nature of the organization. In collectives these include conjugal needs, economic needs, needs for movement, and so forth. In the second case, the needs arise from existential experience, e.g., the need for education, the need for play, entertainment, and so on. The latter result from existential experience under conditions that lead to the consumption or utilization by a collective of a particular material, and from the concomitant sthenic reaction characterized by corresponding mimico-somatic and offensive reflexes aimed at a possible prolongation of the sthenic reaction.¹ When the latter is frequently reproduced, it becomes habitual to such a degree that its insufficient reproduction turns into a stimulus, generating the need for utilizing the same material. Excessive gratification, on the contrary, induces a sthenic reaction accompanied by defensive or offensive reflexes that put an end to the further gratification of the need.

In collectives one encounters various needs born of existential experience. These needs lead to the same consequences and are satisfied correspondingly.

Thus, the need to realize manufactured commodities brings about measures that encourage export; an overflow of imported goods on the market leads to the introduction of protectionist duties; an increased immigration of foreigners results in counteractive measures against it, etc.

Needs as a source of motivation for activity and offensive reflexes are the most important stimuli of progress, for they stimulate inventiveness where every success is accompanied by a positive sthenic reaction that provides motivation for further discoveries and inventions; it is the need for innovations that sets in motion the human genius. The latter satisfies this need when favorable conditions and appropriate prerequisites become available. That is why one can state with certainty that genius makes its appearance in the life of a nation only when its time comes.

Even those phenomena that we perceive as being of an economic nature such as exchange in the form of buying and selling, services, usage, property, securities, inheritance, rent, prescriptive right, etc., are, in fact, nothing other than the result of established social needs entrenched in law; consequently, they, too, are primarily associative reflexes in the economic sphere—a sphere that satisfies the established needs of a collective. The market itself, with its exchange and monetary units, is on the one hand a result of demand, or need, and on the other of supply as a need to dispose of the products of one's labor

when they are in abundance in order to exchange them for others that are in short supply; a need, as we have seen, is nothing other than an associative reflex. It is therefore clear that the whole economic aspect of social life develops along the lines of associative reflexes, while society, as a living organism, is in continuous motion, as a result of which the correlation between various social groups within it is constantly in a state of dynamic equilibrium.

A collective need can be stimulated by any kind of external influence, due to which one collective, in the presence of competition, may enter into a confrontation with another collective and exercise the right of might. Thus, the once disfranchised Jewry, when liberated by the February Revolution from its former social slavery, found itself on the crest of the revolutionary wave in the persons of many prominent revolutionary leaders; yet, despite the declared freedoms and equality of nationalities, the very first months of the revolution were followed by pogroms, even though the country was enjoying all kinds of freedoms and was in the process of preparation for the first elections based on the universal, equal, direct, and secret ballot. Another example: the horrific despotism of the tsarist regime resulted in the most tumultuous revolution in the world since the Great French Revolution and brought the proletariat, previously engaged in a hopeless struggle against capitalism, to the forefront.

A third example: the advancement of the proletariat during the *coup* of October 25 led to Kornilov's counterattack and stimulated in various strata of society—among its bourgeois elements in particular—an attraction towards monarchism, which forced the ruling classes to take severe repressive measures. The same collision of social aspirations is apparent in the reaction of the so-called democratic elements of the country to the Bolshevism that was trying to deepen the revolution. This is what we read on this account in one of the newspapers of 19 October 1917:

The more furious and sweeping Bolshevism grows, the more tightly those elements of the country who think in terms of state interests close their ranks. Slowly but surely the mood of the broad masses of the population is undergoing an evolution: the democratic Soviets of Workers' and Peasants' Deputies are moving to the left, while the rest of the country is moving to the right.

This process began a long time ago, almost from the first ecstatic days of democratic creativity unleashed by the revolution. The early demarcation between the state and the class mentality became clearly apparent during the April days when the left-wing democracy had its first experience of overthrowing a "bourgeois" government. That was the beginning of society's move to the right.

This process of reaction-formation, resulting from a collision of interests, proceeds more or less evenly only when it does not encounter any new attendant circumstances in its way. For example, in that same movement to the right among the democratic elements of the country, which was associated with the growth of Bolshevism, the role of such new attendant circumstance that slowed down this movement to the right was played by the Kornilov affair. In the same paper we read the following regarding the above-mentioned evolution in the popular movement:

The Kornilov days broke the thread of this evolution. The democrats forsook the "bourgeoisie" yet again, having made a rash evaluation of the Kornilov affair. The "counterrevolutionary" nature of the bourgeoisie has again become a bugbear even for the responsible and state-minded democrats who were only recently agreeing to concessions for the sake of the unity of all national forces in the name of the country's defense and freedom. The chasm between the classes has again deepened and widened, the two sides are now further apart than before the August days, and both of them are further away from the common goal. The ones who reaped the fruits of this false storm were the Bolsheviks.

In the democratic country, the Bolsheviks started seizing power, and the state-minded democratic elements began to lose power among the masses corrupted and deceived by demagoguery.

When two mutually incompatible processes of social evolution reach their climax, this leads to a social crisis that results in a mutual inhibition followed by the turn of the popular movement in one or another direction.

In the Russian Revolution this kind of crisis occurred in the Democratic Assembly that ended in a completely discordant and fruitless vote.

Like any crisis, a social crisis must certainly lend itself to a resolution, and this resolution may occur rapidly through an open confrontation in which a

more powerful popular movement suppresses the weaker one by force. In other cases, a crisis is resolved slowly, and the gradually ebbing tide of the popular movement outlives itself little by little.

Whether the resolution of a crisis proceeds stormily, "critically," or slowly and gently, depends on the nature—whether it is more active or less so—of the popular movement, as well as on some other conditions (e.g., the speed of the movement's expansion, etc.).

Every social collective may at a particular moment in time undergo its own crisis.

Thus, in every party, a difference of opinion between two camps or two movements may lead to the kind of crisis that will be followed by the victory of one trend or the other.

Incidentally, the debates on defense in the Soviet of the Republic ended in a crisis and a rejection of the proposed formulae for the resolution.

As for the form of the collective reflex, it may differ, depending on the nature of the gathering or the collective. Thus, in parliamentary assemblies, protests are mostly limited to vehement rebuttal or, at the most, to an eccentric action, altercation, exit from the assembly during somebody's speech, etc.—greater excesses are rare; in a crowd, we deal with more extreme manifestations (e.g., mob lynching) that often lead to fatal consequences.

Associative reflexes² can also explain the striking phenomenon whereby a means becomes an end in itself, substituting the latter, as it were. As we know, the means precedes and even accompanies the end; yet, although initially the end as a stimulus provokes the reflex, with time the means, as a phenomenon preceding and accompanying the end, begins to act as a stimulus that provokes the associative reflex. Thus, in times of great revolution, the sermon of liberty and equality as a means of achieving the common weal becomes, in the long run, an end in itself, initially leading to the destruction of the public welfare.

As we have mentioned above, the instinct of self-preservation stimulates a desire to achieve optimal well being under all possible living conditions. Consequently, everything related to obtaining appropriate nourishment and a better position in life results from stimulation of a special need associated with this instinct. Since money and profit are, in general, a means of satisfying these needs, the acquisition of money is associated with the latter as a stimulus that invariably precedes it. As a result, money itself, as a means

of satisfying one's needs, is capable of arousing the same reactions that are stimulated by the satisfaction of the corresponding needs. Thus, here, too, the means as such becomes the end.

In those individuals whose needs stimulated by the instinct of self-preservation are not inhibited by other circumstances, they are naturally maintained and enhanced by various living conditions, which engenders the attraction to profit and acquisition manifested through the appropriate reflexes.

The passion for acquisition, however, is in certain cases best satisfied through joint efforts, which naturally gives rise to all kinds of economic organizations that exist in human society and satisfy the instinct of self-preservation in collective life.

One can prove that the oldest habits survive longer in a collective than those acquired later on, which provides yet another analogy with individual reflexes. Established ritual, etiquette, and custom are examples of old social habits that survive throughout centuries without the slightest changes, whereas fashion, failing to become a habit, is replaced in time by a new fashion, which, too, will have but a transitory existence.

We would also like to note that as the relationship of an individual to a collective evolves, there unfolds a process of social identification that reaches such a degree that not only do individual aspirations become subordinated to public interests, but the interests of the collective are adopted by the individual as his own; in other words, they become identified with his personal, or individual, interests. These are the origins of the line of individual aspirations called patriotism. Such collective interests as, for example, the prosperity of the state, constitute an object of identification with one's own interests to the extent that everything favoring this prosperity stimulates in every individual a positive sthenic reaction and vice versa.

As we know, a sthenic reaction in individual life accompanies not only the satisfaction of a physical need, but also the activity of the relating apparatus, or the so-called mental work, unless it leads to exhaustion. Here, too, what we call creative mental activity and even the very acquisition of knowledge as such is usually accompanied by a sthenic reaction, which phenomenon is reflected in the language itself in such words denoting attraction to knowledge as *liuboznatelnost* (literally, love of knowledge) meaning "curiosity" and *liubopytstvo* (literally, love of inquiry) meaning "inquisitiveness." Thus, a sthenic reaction that accompanies mental activity

until it becomes tiring appears to act as a reliable stimulus towards the intellectual growth and self-improvement that occur according to evolutionary laws. This is also the basis for the role of initiative in associative human activity.

The same processes take place in a collective where inquisitiveness and curiosity are enhanced through example and imitation. A crowd of busybodies gathered in the street around a gawker is a good example. It is enough for someone in the street to start gazing at the sky, and many of the passers-by would do the same. Let someone in a big city stop on a bridge and start peering into the water of the river flowing underneath—within a short time he will, surely, be surrounded by a curious crowd.

Collective curiosity is behind every new social movement; no matter how vast the blind faith in the authorities, fascination with the glorious past and conformity to the traditions of forebears might be, a collective is, due to curiosity, gradually penetrated by a new current, by one or another innovation, that through imitation gradually spreads among a considerable number of people and then among the entire collective. Thus, what one may call a social shift begins to occur at a faster or slower pace, depending on the circumstances.

Here one has to note that the longer social stagnation based on imitation of the past lasts, the greater the thirst for innovation becomes; this is why epochs of stagnation in society are usually followed by epochs of rejuvenation and upheavals, the latter occurring the more vigorously the longer has stagnation lasted and the more atavistic has been the old order.

Habit and mechanization associated with acceleration of various processes operate in every social phenomenon, just as in any individual action. Thus, many industrial processes, such as the manufacture of fabrics, clocks and watches, various commodities, agricultural machinery, etc., occur at a much higher speed than in the past. Besides, every recurrent social phenomenon becomes established and turns into a habitual associative reflex, as it were.

For example, a foreign language used instead of one's mother-tongue becomes habitual with time, replaces the mother-tongue, and, due to habit, even its alien structure is assimilated as the national one, as it were. We know, for instance, how rapidly the characteristics of the Greek language with its periods, meters, and other attributes became the flesh and blood of the Latin language and formed the solid heritage of Roman culture. The same is true of other social phenomena. Thus, an introduction of any new social

reform— whether educational, judicial, or administrative—initially proceeds with great difficulties and must overcome various hurdles and frictions; with time, however, these same reforms, as it were rendered automatic, become part of the tradition and are adopted as permanent cultural heritage.

In exactly the same way the generally accepted gift-giving in the East that initially used to be voluntary, turned first into a custom and then into a mandatory favor, virtually becoming a kind of obligation. Charitable donations that used to be collected for victims of fire were with time replaced by insurance companies. Religious rites were initially performed by the clergy free of charge, and the parishioners used to reward them with voluntary donations; with time, however, these voluntary donations turned into a duty and became compulsory. Special honors bestowed on the powers that be gave rise to flattery and politeness, the former becoming obligatory in the East, while the latter became compulsory in the European high society. The old practice of prostrating oneself before the ruler assumed with time the form of bowing and taking one's hat off when meeting any person one wishes to pay attention to, primarily one's acquaintances. According to Tarde (*Laws of Imitation*, p. 359), "the widespread need to hear flattery, receive visits and bows in imitation of high officials and gentry served as a hidden cause of the gradual and universal spread of politeness in France and other countries."

Due to the same circumstances, the long-term exposure of the popular masses to theatrical performances in ancient Rome generated a need for them, just like the need for bread. It is therefore understandable why the popular masses in Rome began to demand both, shouting out "*Panem et circenses*."

Every innovation or discovery is generated by a social need, i.e., an associative collective stimulus that, in turn, arose from the appropriate recurrent circumstances. Let us take the following example: the human need for communication is based on the development of culture and leads to the improved means of communication. Introduction of these means results in the intensification of exchange, which with time becomes a new social need, etc. If we look in retrospect at the whole process of the cultural evolution of mankind with its inventions and needs, we shall reach back to the primal foundations of communal living, where individual persons, due to imitation, experience the need to form a collective, while imitation itself also becomes a need in virtue of its inevitable recurrence under conditions of communal existence.

Imitation, however, inevitable as it is in social life, implies the loss of novelty. At the same time it involves a reduction of the imitated process. Hence, the "necessity of new inventions and fresh sources of imitation, which could resuscitate the depleting social energy" (Tarde, *Laws of Imitation*, p. 216). Thus, art gradually degenerates, religious beliefs lose their power and intensity, discourse becomes shorter and idioms fade. "Honorary titles (*seigneur* reduced to *seur*), greetings (feudal genuflection replaced by a slight nod of the head), compliments and manners are increasingly reduced, weakened and simplified as they become more widespread." The same was proved earlier by Spencer.

On the other hand, no social movement ever starts abruptly, but develops gradually: it first goes through a latent stage when, due to social conservatism or to inertia, it slowly—sometimes over a period of many years—forces its way through and enters the public arena. Before art began to degenerate it had to flower. Every religious doctrine goes through an initial phase of development and only gradually gains a wide popularity. Before losing its meaning, every title had to assume a particular form required by social need. Before fading away, every term and every expression had to come into existence and occupy their place in the public vocabulary, affecting the minds with their novelty and characteristic flavor; every manner of greeting and parting is a vestige of social relations that used to have a far more significant and real meaning.

In a word, a collective reflex in this respect evolves according to the model of individual associative reflexes. Initially, after a latent period, it grows more or less rapidly. While growing, it gains ground, differentiates through imitation during its increasing dissemination, and then, having become habitual, it loses its novelty, weakens and fades, i.e., loses its role as a social process. It revives, however, with each new stimulus. Thus, in a collective reflex too, one can distinguish its own developmental stages completely analogous to those one observes in an individual associative reflex. A latent period, a more or less rapid growth at the beginning that leads to a climax, followed by a gradual weakening and final decline (retaining the ability, however, to reappear under favorable and stimulating conditions)—these are the stages of a social movement as a collective reflex.

We know that a sthenic reaction associated with a particular stimulation is subject to the general law of associative reflexes. Initially, therefore, this reaction accompanies the satisfaction of an organism's immediate needs, but

in the course of one's life experience it begins to accompany those means that are not immediately connected with the satisfaction of existential needs. Thus, we have already mentioned that, for instance, money by itself—as pieces of paper—would be unable to provoke a sthenic reaction, but as a means of possible acquisition of vitally indispensable objects it stimulates a sthenic reaction in anyone in need of the latter. On the other hand, achieving a goal while, for example, obtaining food supplies through one's own or other people's experience stimulates a sthenic reaction resulting from the satisfaction of one's existential needs; with time, however, the process of achieving any goal whatsoever becomes a means of stimulating a sthenic reaction, even though the goal itself under different circumstances would not directly lead to the satisfaction of existential needs.

In this respect, what is true for individual persons has the same significance for a collective. Such, for example, is the development of a collective sthenic reaction that accompanies the satisfaction of patriotic goals by the state collective when securing the means that make a future satisfaction of patriotic goals possible. A victory that guarantees a favorable outcome of a country's military operations, or an alliance that inspires hope for satisfying a country's existential needs in a more or less remote future already stimulate a collective sthenic reaction.

As we have seen, one should include imitation among collective associative reflexes. It is a reflex that ensures attainment of collective successes achieved by individual persons. If at the lower levels of the animal kingdom imitation in the form of mimicry serves primarily for the preservation of the species, at the highest evolutionary levels of the animal kingdom imitation serves social goals and provides the basis of sociality. In the final analysis, the whole of culture as we know it is based on imitation that permeates the life of every creature whatsoever from its earliest days to its particular level of development, as well as social life starting from the primitive peoples, the infant nations, and ending with the nations of the highest culture.

According to Féré's experiments and some other findings, imitation is not only conducive to acquiring reflexes but it also facilitates their execution. As we know, imitation in collective work has a stimulating effect and facilitates the work effort itself.

We have already mentioned elsewhere that seeing a dancing person makes one start dancing himself, that a laughing person causes another person laugh,

that someone who yawns makes his neighbor yawn and so on. It is thus evident that imitation, being a lighter reaction by its very nature, on repetition easily turns into a need that motivates an organism towards communicating with other individuals.

That is why imitation is, on the one hand, a stimulus for social life and, on the other, an inevitable consequence of contacts between two and more individuals leading a social existence.

One may positively claim that sociality emerges not only due to its benefits for the species but also due to the immediate, i.e., biological, individual benefits of imitation itself, which acquires particular significance as a means of the borrowing by one individual every new success and invention from another.

If, as we begin to see, repeated imitation easily turns into a need, then contact with one's own kind as a prerequisite for imitation also becomes a need, and this resolves the issue of the whole social problem. We realize why contact with one's own kind emerges in the organic world. It is generated by the urge to imitate, which turns into a need whose satisfaction is inevitably accompanied by a sthenic reaction. This is the main prerequisite for a social instinct.

Any game in general is based on imitation and associated sthenic reaction.

Observe how excited two puppies are by their encounter, as a result of which they immediately start playing with, and imitating, each other. What stimulated the previously dormant energy in both of them? It was the opportunity to indulge in contact and imitation as a stimulating agent accompanied by a sthenic reaction that leads to play and is in itself inevitably associated with imitation. And, indeed, if we look carefully at any play, we shall see that it consists of mutual imitation—particularly by the less experienced of the more experienced—which is the social meaning of play as a method of natural learning for an organism in the process of development.

It becomes clear from the above that, as an inevitable result of imitation, one of the vitally useful conditions for an organism is communication; this is a natural stimulating agent needed by the organism under all circumstances when it is in its prime and not suffering from fatigue. This need appears to be as organic as the need for light, warmth, nourishment, etc.

It is worthy of attention that repetitiveness of collective action causes the masses to adopt a particular disposition regarding coordination of individual actions, and this disposition is to a significant degree determined by

imitation. Everyone knows how much training is needed to achieve the properly coordinated actions of army detachments, how many rehearsals are needed for actors to achieve coordination in their actions on stage, how many sessions of singing practice precede a public performance of a choir, or how many rehearsals are held by an orchestra preparing to show its program on stage. And yet, this coordination is invariably achieved as a result of practice.

It is therefore clear that experience and imitation have an enormous significance for a composite personality, not only with respect to reproducing an action performed on a single occasion, but also with respect to the collective coordination of actions performed by a mass of people in a particular order and sequence.

Notes

¹ On the sthenic reaction, see V. Bekhterev, *Objective Reflexology* (in Russian), 2nd edition.

² See V. Bekhterev, *General Principles of Human Reflexology* (in Russian), Petrograd, 1918.

11

Collective Hereditary-Organic Reflexes

As we have mentioned before, one may describe all manifestations of collective activity as collective reflexes or collective reactions.

Manifestations of these collective reflexes, however, differ from one case to the next. To achieve a better understanding of various manifestations of collective reflexes, we are going to examine in sequential order collective hereditary-organic reflexes (instincts), collective mood and collective mimico-somatic reflexes, collective concentration, collective observation, collective judgment, and collective actions.

Let us begin with collective hereditary-organic reflexes.

As in individual life we distinguish hereditary-organic reflexes, so in social life one has to be aware of the manifestations of the same type of collective reflexes. Their immediate goal is self-preservation—particularly nourishment, satisfaction of sexual needs, attraction to communal living, etc.¹

Satisfaction of natural needs with respect to food, drink, protection from inclement weather and cold is usually accompanied by the corresponding positive, or sthenic, mimico-somatic reaction on the part of the vegetative functions, manifested among other things by increased cardiac and respiratory activity, as well as by movements and actions of an offensive nature that lead, or are supposed to lead, to the achievement of the most complete satisfaction of the corresponding needs (the process of eating is associated with obtaining food products, processing them appropriately and

consuming them with the purpose of satiation; protection from inclement weather is associated with using clothes, setting up a dwelling, etc.).

Let us assume that a certain country or region is suffering from famine, as was the case during the revolution in Russia—particularly in its capitals, Moscow and Petersburg. As a result of the social conditions of our existence, the problems of food supply move into the foreground. They fill the pages of newspapers and magazines, and wherever you go the main subject of conversation is our daily bread. Whether one is visiting a friend or runs into someone in the street, the conversation will sooner or later turn to food rations, food products, and eating. Naturally, while standing in long lines in front of the bakeries, people not only talk about bread but also argue about the supposed perpetrators of famine. The same is true for individual and group actions: here one deals with the major influence of the unsatisfied need for nourishment that resulted in innumerable cases of theft and robbery. In a word, the whole society, formerly preoccupied with the most diverse interests, is now transformed: its most important interests concern nourishment and food. The rest seems to recede into the background before the interests of the stomach, which dominate everything else.

In all cases, the actual stimulus of an offensive reaction is a need that arises with particular force when it is not satisfied, whereas total satisfaction—oversatiation in particular—is accompanied by a defensive reflex.

The nature of the reaction itself is determined by the effect of external influences on the collective. Beneficial external influences stimulate a sthenic reaction and lead to offensive actions aimed at intensifying and prolonging these influences, whereas adverse external influences are accompanied by an asthenic reaction that stimulates a desire to be free of them or, in other words, creates an impulse towards movements and actions of a defensive nature.

Since natural needs or attractions underlie the very makeup of individual mentality and are caused by organic conditions that are inherited rather than acquired like other needs, we classify them as hereditary-organic reflexes leading to actions that ensure satisfaction of natural needs.

Due to the fact that satisfaction of various hereditary-organic reflexes, such as defense from unfavorable conditions, is in many cases easier to achieve through collective rather than individual efforts, this circumstance often serves as an impulse for joint action. Every socially raised individual who is a member of a community strengthens the latter by virtue of his or her membership, since every community provides protection and defense of its

individual members. It is the socialization of this hereditary-organic reflex that led to the emergence of the norms of communal living and to the establishment of power as an organizing principle that provides internal and external protection of the community itself.

Hereditary-organic reflexes are subjected to the appropriate regulation by the same concomitant sthenic reaction mentioned above. When hereditary-organic reflexes—whether in animals or in humans—lead to results that satisfy the needs of the collective, the latter, as we know, develops a sthenic reaction that maintains the reflex and its further development until its completion; on the other hand, when the reflex is incompatible with the needs of the collective at the given moment, no sthenic reaction is stimulated and the reflex itself is inhibited. In the end, the goal achieved by this reflex becomes, due to the previous experience, a more or less permanent stimulus and as such stimulates, along with the hereditary-organic reflexes, personal ones as well, which, in this case, assist the hereditary-organic reflexes and develop in the same direction. In this way, due to past experience, a habit is eventually formed, which contributes to the achievement of the goal even before the impulse that develops under the influence of organic stimuli reaches the appropriate tension. Furthermore, for the purpose of achieving a goal a collective experience can develop a means to which any individual—irrespective of the goal to be achieved in accordance with the reflex—would be totally indifferent; with the development of the collective experience, these means, due to an associative reflex, begin to act as the stimulating agents of a sthenic reaction and, consequently, of the corresponding actions. Such are the above-mentioned money and other signs and documents (as, for example, treaties etc.), which evolved as a result of the collective experience of exchanging commodities and which led to the satisfaction of needs for food, acquisition of clothes and other goods that enhance one's prosperity.

One should also bear in mind that a collective organism can draw on the support of the individual experience of its members, which under certain conditions is socialized through imitation. Let us presume that a given locality is suffering from famine. It stimulates the population to engage in active behavior, which under the influence of acute hunger and malnutrition may assume the form of pogroms. In such cases the perpetration of pogroms and pogrom behavior as such are initiated by the individual members of the crowd. When a crowd stimulated by malnutrition behaves aggressively, it is enough to point out the culprit of the people's tribulations for the crowd to

perpetrate its mob law, without bothering to find out whether he is, indeed, guilty. Under the same conditions, a crowd that has approached a baker's shop stops its actions when coming against an obstacle deemed unsurpassable. But then someone brings a crowbar or an axe and strikes the first blow to the door, and the crowd joins in breaking the door in order to get inside the shop and loot it. In this case, individual initiative and actions meet with collective support and involve other elements of the composite personality in joint participation through imitation, which results in socialization. Therefore, individual experience and individual initiative are of colossal importance in collective actions. It is due to them that a given person usually becomes the leader of the crowd that follows him blindly.

Thus, it is not necessary for the whole mass of people who form the crowd to possess any previous experience of achieving a particular goal. It is enough for at least one individual in the whole crowd to possess such experience, and the crowd is capable of a collective action under his leadership, as long as we are dealing with means and ways accessible to everyone. This certainly does not mean that a composite personality needs no experience whatsoever or that an action performed only once leaves no traces. On the contrary, in a mass of people, experience is always valuable as it generates a collective habit. A successful robbery encourages a band of robbers to engage in the same kind of robbery henceforth. A successful pogrom encourages a crowd to perpetrate pogroms in the future. Thus, collective habits come into existence and encourage future collective actions in the form of reproductive reflexes of a collective nature.

A collective inclination towards self-preservation also underlies the basis of party tactics and state politics. It guides the actions of the crowd when it flees from shooting soldiers. The same inclination directs the actions of organized masses—e.g., troops—when they are threatened with one or another kind of danger. And finally, the same inclination guides the actions of state authorities in defense of their international rights and their position in the country. This is why any power, under the influence of its inclination to self-preservation, defends its own existence, even though this may go against the interests of the popular masses. The best, though not perfect, remedy against this phenomenon is re-elections of the authorities after a certain period of time, as soon as the majority of the elected representatives forms an opposition regarding some important issue of government. One should not forget, however, that frequent change of power has its disadvantages in other

respects, while during the periods of such popular crises as wars, for example, this is totally undesirable, for it is difficult to set in motion the mechanism of power that would possess sufficient experience and authority at a time when the country is in danger and the house is, so to speak, on fire.

The same inclination to self-preservation in a collective organism leads to preparatory work on defense in the form of constructing fortresses, manufacturing the means of defense and attack, etc., to the exploitation of natural resources such as fuel, and finally to the exchange of commodities and the institute of trade and commerce.

In all of these cases, cooperation facilitates the satisfaction of the collective inclination towards self-preservation, which naturally results in this inclination leading to a particular type of cooperation and reinforcing the communal principle. The same inclination, however, lies at the root of all wars and plundering raids that divide national collectives into hostile camps opposing one another. Inclination towards self-preservation also explains the emergence of religious doctrines, for they reveal human need for protection against the unknown and mysterious forces of nature and the surrounding enemies.

One should note, incidentally, that the inclination to self-preservation also manifests itself in an antisocial manner in all those cases when the whole collective or the majority of its members face mortal danger. This will be discussed below.

Attraction to nourishment in social life is at the basis of cooperation in hunting and fishing; it led to the domestication of animals, to poultry farming, the exploitation of plant and animal food products, the development of farming and cultivation of cereals, vegetables, fruit trees, and fodder crops, as well as to the industrial production of food products. Inasmuch as the procurement of food products and the exploitation of the environment for this purpose require cooperation, this attraction maintains and stimulates sociality.

In this respect a significant role is played by the cooperative protection of the products of labor associated with the procurement of food.

It is common knowledge that farming became possible only after reliable protection against predators had become feasible.

It goes without saying that an unsatisfied attraction to nourishment is a frequent cause of social movements against predators or even against authorities that are incapable of organizing proper food supplies. This is why

a shortage of food products is the principal organizer of the so-called revolts of the hungry.

The collective inclination towards self-preservation led among other things to the establishment of the right to own land for the purpose of procuring food products, to building houses for protection against inclement weather, etc. Both tasks usually require collective efforts achieved through the interactions of an economic nature, which underlie the living conditions of a collective. The same interactions led to the establishment of civil law.

In the course of time, due to the abuse of the right of property by certain individuals and the exploitation of others for one's own benefit as its consequence, this right evolves in the direction of socialist tendencies. However, in this case, too, the heart of the matter in terms of providing for the needs of every member of a given collective remains unchanged, the only difference being a more equitable distribution of commodities that provide for the living conditions of the populace.

Out of the hereditary-organic reflexes of self-protection against attack and enemies there emerged in collectives the institution of criminal law wherein the protection of individual life in society is entrusted to a specially elected organ in the form of a judicial tribunal and executive power.

As the longevity of a family collective is not measured by the lives of its individual members but includes posterity and the transition from the older generations to the younger ones, it is natural that in capitalist countries the right of property inheritance by descendants comes into existence.

This right of inheritance presupposes the continuity of the personality as a unit of the collective through one's descendants or close relations, and inheritance therefore assumes the form of the property of one's descendants or closest relatives. With the changed views on the relationships between individual members of a collective, it is also possible to change the right of property inheritance in a socialist direction, i.e., to make inherited property the common possession of the collective, recognizing it as state or public property, which, in this case, is the same.

The manifestations of the same hereditary-organic reflexes of self-preservation also generate in collectives other economic institutions, such as, for example, the processes of exchange and commercial relations, the invention of banknotes as the compensation for commodities, etc., as well as the proliferation of stealing.

As for another hereditary-organic reflex characterized by sexual attraction, in civilized societies it provides the foundation of the family —the small cell of peaceful cohabitation, which is at the basis of family life, family traditions, and institutions that underlie family law. It was the family that originally introduced cultivation of cereal crops and exploitation of domestic animals. The family was also responsible for the emergence of clans and tribes, which led to the appearance and defense of national principles that give rise to tribal and national strife and wars.

The significance of sexual attraction responsible for reproduction has become particularly clear since Malthus published his work. It was he who pointed out that if this attraction did not encounter any inhibition in the social environment, the population growth was particularly fast; in civilized countries, however, reproduction is much lower than its maximum potential, which implies that these countries have certain correctives with respect to the given attraction. In the final analysis, they are reduced to its inhibition by placing natural limits on it or by taking artificial measures against reproduction. In this respect, a relatively slow increase in food products is of particular importance. Because of this, the population of most European countries was growing extremely slowly over many centuries, with the exception of the latest historical period, starting from the nineteenth century, when the necessary means of subsistence began to increase with the invention of the machines that are so important in every industry. Economic reasons among other factors undoubtedly exercise a particular influence on the state of this instinct. Thus, statistics positively established that the number of marriages and births in different countries is directly dependent on the prices of vital products, as well as on the conditions in agriculture and commerce. As we know, due to the same conditions, the population of modern France stopped growing except for immigration.

The limitation of sexual attraction in civilized countries is manifested through the establishment of the "two-children system," maintained by means of miscarriages and "incorrect" sexual relations, voluntary self-restraint, infanticide, etc., which in the long run results in a quantitative decrease in population growth and in the declining role of the family in social life. Although infanticide assumes enormous proportions among the uncivilized peoples, it is not even regarded as a crime.

The parental "instinct," which is closely connected to sexual attraction, is another pillar of the family community. The family exists among all higher

animals whose males and females take care of their offspring on the basis of cooperative parenting.

When turning our attention to modern humanity, one should bear in mind that ethnography does not know of a single tribe where the family does not exist in one form or another. Only among the Negro tribes is the joint parenting of the offspring restricted to the period during which the child is breast-fed and unable to walk. Even here, however, we are dealing with the family, albeit in a more limited form.

In the civilized countries of the European and Asian continent, the stability of the family is recognized as a prerequisite of the healthy condition of the society and the state.

The family as a consequence of the sexual and the parental "instincts" is the progenitor of motherhood that ensures the upbringing of young children, and since both are essential in the existence of modern society, the latter supports and protects the family principle in every possible way through the appropriate laws.

Due to the particular importance of the family with respect to the replenishment of the population, any state collectively supports the institution of the family, as well as the institution of family law associated with it.

For this reason both the institute of marriage and the parental duties are protected in all civilized countries by civic and religious sanction, through a special dispensation that became part of tradition, religious ritual, and state laws. This protection and support of marriage is the stronger the more conservative are the state structure and society composition.

One may presume that those societies that did not maintain and improve this protection in due measure were doomed to degeneration and extinction. The weakening of this social protection in the form of the parental "instinct" played a considerable role in the disintegration of the highly civilized societies of ancient Greece and Rome.

Incidentally, enormous support of the family structure is provided by the veneration of ancestors, which leads to the stability of the nation. Ancestor worship greatly contributed to the stability of ancient Rome, as nowadays the same ancestor worship contributes to the stability and power of Japan and China.

It goes without saying that the sexual attraction responsible for reproduction has as its secondary consequence the development of closer social bonds among the members of a clan or a tribe, as well as the

improvement of moral principles in the social milieu, for maternal upbringing establishes the norms of social structure, eliminating rough treatment of subordinates. Regardless of economic and political reasons, the abolition of slavery and serfdom, too, occurs to a significant degree due to the gentler conditions of upbringing itself. Further, the emergence of charity in society and humanitarian regulation of the most cruel manifestations of state egoism such as war, for example, is also to a significant extent a product of the parental "instinct" and of the gentler conditions of upbringing associated with it. Even the demands of the popular masses meet with greater responsiveness on the part of society's ruling classes due to the same parental "instinct" and to a gentler upbringing.

However, one should bear in mind that society did not evolve out of the family, as some presume, and that sociality in the animal kingdom emerged far earlier than the family did. Society emerged primarily from natural cooperation established for the purposes of obtaining food by hunting, of protection from enemies and the pursuit of predators. These joint efforts are based on the inclination towards imitation and sociality, which brings together individuals who possess the means of communication and cohesion in the form of appropriate facial expression and sound language. Little by little, interpersonal communication becomes a need, due to the mutual exchange of individual experience and the ability to imitate, which is also useful as a skill. This is mainly what lies at the basis of social life maintained through natural and social selection.^{2, 3}

Biological data leave no doubt that social attraction is characteristic of all animals and even plants that exist in communities. In the human world, however, communities assume the form of political societies.

Admittedly, not all biologists recognize the "social instinct" as such, but the real issue concerns the facts, rather than the words. We know, however, that chickens, hatched artificially away from their family, keep together from the very start and follow one another without any preliminary training.

The "social instinct" provided the foundation of the human language—this important means of sociality, which unites people to a greater degree than even religious or economic conditions.

If individual hereditary-organic reflexes caused by the imperative internal stimuli (hunger, sexual needs, etc.) in some cases overwhelm all other desires, it is quite obvious that collective hereditary-organic reflexes can be manifested even more prominently due to the fact that in a crowd, in addition

to all the other conditions, there is an element of contagion, while at the same time there are no sufficient conditions for restraining the desires, and as a result once these reflexes manifest themselves there is no restraining or inhibiting them. There are more than enough examples of this both from the sphere of inclination to self-preservation—food in particular— and from the sphere of sexual attraction.

Inclination towards self-preservation manifests itself particularly strongly when a mass of people face the issues of life and death as, for example, during major disasters that affect great numbers of people.

Alongside what one can describe as heroic actions of relatively few individuals who sacrifice themselves for the sake of saving others, the majority in these cases manifest the inclination towards self-preservation in all its crudeness and nakedness when people destroy others in order to save their own lives. This is what happens during big fires that break out in crowded public buildings. This is what happens during shipwrecks. This is what happens at the time of any other mass disaster. Those who have read the description of a fire in one of the summer theatres in Paris, or who have read the details of the destruction of the ocean liner *Titanic*, will easily recall the striking and at the same time repulsive manifestations of the reflex of self-preservation in a collective environment under the above conditions.

One observes a different type of manifestations of collective inclinations towards self-preservation in cases of food shortages. Here we usually deal with riots and pogroms that are often accompanied with extraordinary violence. The following account of a riot at the Apraksin Market in Petrograd during the revolution illustrates how this kind of unrest arises from the shortages of vital products:

Barrels with merchandise that have just arrived from the railway station were being unloaded from a cart into the Apraksin Lane fabric store owned by Mr. Yagodkin. Some passers-by noticed that the merchandise was packed in barrels rather than canvas sacks, as it is usually done.

Some ten busy-bodies gathered around and approached the carter, asking him what was in the barrels. One of the shop-assistants overheard the questions and shouted irritably:

"It's none of your business. This is our merchandise and that's that."

At this moment one of the barrels unexpectedly slipped out of the carter's hands, fell on the ground and broke into pieces. Sweet tea-biscuits spilled out of the barrel.

Within a minute the shop was surrounded by a crowd of common people swearing and shouting:

"You damned marauders! Hiding food! Kill them!"

The crowd was growing by the minute. Its mood was extremely volatile.

A few people started breaking open other barrels from which candies, crackers and various metal objects—horse-shoes, cramp-irons, files—spilled out.

Edged on by some individuals calling for reprisal, the crowd stormed into the shop. The frightened employees ran away. The owner of the shop, Mr. Yagodkin, escaped from mob law by hiding in the Commissariat.

One person who suffered grievously, however, was an innocent Jew who happened to be in the crowd by chance.

Ignorant of what was going on, he asked someone:

"Why has a crowd gathered? Did they catch a thief?"

Instead of an answer a hooligan yelled "Kill the Yids!" and struck him on the head, after which the blows fell thick and fast on the unfortunate man.

He was barely alive when some militiamen managed to save him from being lynched. Due to the extremely militant mood of the crowd, troops were called in to assist the militia. The soldiers of the Pavlovskiy Semionovskii and the 1st Machine-Gun regiments arrived, as well as a military detachment.

When they heard the crowd shouting "The marauders are hiding the food, we should search all the warehouses," some soldiers supported this demand.

Sokolovskii's warehouse and Ginzburg's shop were immediately cordoned off. The crowd and the soldiers carried out the search and found large amounts of sugar, biscuits, sweets, as well as rolls of various fabrics.

All the discovered goods were loaded on carts and sent to the Commissariat.

The next day the unrest recurred on an even larger scale.

The owner of a fabric store, Mr. Anter, was in the process of dispatching several rolls of merchandise to the station to be transported to Pskov, Yuriev and Revel.

At this moment a soldier approached him and with the words "Aha, you are going to send off some foodstuffs, aren't you?" hit him in the face with his fist. A crowd gathered immediately and rushed off, shouting threats, into several shops in search of the allegedly hidden food.

Rolls of merchandise were thrown onto the ground, boxes were broken open spilling various goods—galoshes, shoes, etc. All in all the search was thus carried out in 15 warehouses and shops—Ivanov's, Yeliseev's, Ponizovskii's, Tiukov's and others. The rest avoided the search by closing down their shops in a hurry.

They say that in Tiukov's store they discovered such incompatible goods as boots and almonds. Under the crowd's pressure, the militia removed from Yeliseev's store five cartloads of textiles.

All day long carts with the requisitioned merchandise and food were coming out of the Apraksin Market.

The crowd rushed to every shop from which the owners or shoppers were carrying out merchandise and demanded that the wholesalers should sell by retail only. The crowd and the soldiers prohibited the sending of merchandise to Finland and to other cities, threatening to loot those shops whose owners would not comply.

Many frightened merchants ran to the Commissariat, threw in the keys from their shops and declared: "Do what you like, we are not going to work under these conditions."

Another case we are going to mention here concerns mob rioting in a meat line:

Due to the food shortages, a confrontation between the crowd on the one hand and militia and the army patrol on the other took place at the corner of Degtiarnaia and Rozhdestvenskaia in front of a city meat shop.

On that day the shop was selling the usual meat rations.

Some time later the meat sales were stopped, and the shop-assistants declared that everything was sold out.

The crowd did not believe them and began to push into the shop. Shouts were heard: "They are lying. They are deceiving us. Beat up the crooks."

A militiaman tried to calm the crowd down and explained that the meat was, indeed, sold out. The crowd did not take any notice and began looting the shop.

Another militiaman and a military patrol from the Pavlovsky regiment were summoned for assistance. All attempts to prevent mob violence were in vain.

The enraged crowd attacked one of the militiamen, Mr. Grinberg, who was trying to stop the looting of the shop, and beat him up mercilessly.

Seeing the futility of verbal appeals, the military patrol started firing in the air. The gun-shots to a certain extent brought the rioting crowd under control. The military detachments that arrived soon after dispersed the mass of people.

The militiaman Grinberg, beaten up and bleeding, had to be sent to hospital for medical help.

And here are some telegraph communications about the revolts of the

hungry in the German city of Stettin; brief as they are, these new items paint a vivid picture of the bloody consequences of this kind of riot:

Stockholm. 13th (26th) June. A Swede, who arrived from Goteborg and who had been in Stettin during the latest unrest, conveyed to the employees of the Danish newspapers the following details:

The riots started on Monday, 5th of June. They were provoked by the rumours that food products were being transported out of the city, even though the city was suffering from food shortages. The crowd destroyed and looted several shops, breaking the windows. Police were acting with extreme cruelty. One young woman sustained sabre wounds. Seven people were killed near the Ganse bridge. Machine-guns were positioned in the centre of the city, and the soldiers behaved as if they were on a battlefield. Machine-gun fire continued through the night. Ambulances were constantly driving all over the city, transporting the wounded to hospitals. The next day there were soldiers with machine-guns posted on many street corners. The city appeared to be dead. Only in front of the shop run by the commission on food supplies a crowd gathered, even though the sales were not to start until the next day. On Wednesday the factories resumed their work.

To further illustrate the manifestations of crowd animosity caused by the shortages of vital products one can cite recent instances of unrest that took place at the Sennaya Square in Petrograd:

In the morning, a crowd of women standing in line for bread in front of the city shop at 6 Zabalkanskii Lane noticed that boxes with soap were being carried out of G. E. Dolgov's fish shop. The women asked the drayman where he was supposed to take this soap to. The drayman said he was taking it to the Baltic Railway Station.

The women began to protest and called in a militiaman on duty. The owner of the shop produced a permit for transporting the soap out of Petrograd; this failed to convince the women, however, and they insisted on dispatching the cartload of soap to the city general store at the corner of Sadovaia Street and Tairov Lane, where the administration was ordered by the crowd to sell it for 52 kopecks a pound.

At the same time two carts drove up to A. I. Znesin's meat shop at 42 Sadovaia Street to be loaded with leather for making gloves. Passers-by were extremely surprised to see this kind of merchandise appear out of a meat shop.

The Militia was informed, and the Deputy Commissar of the 3rd Spassky district, Mr. Sharikov, and six militiamen arrived. When Sharikov began to defend the owner of the meat shop and to explain to the crowd that Znesin possessed all the documents permitting him to send the leather to the Novgorod province, the crowd became even more agitated.

"This leather belongs to the commissars themselves—they have been hiding it here," someone in the crowd said, and they assaulted the Deputy Commissar Sharikov. He was knocked down and beaten mercilessly all over. When the Deputy Commissar lost consciousness, they bound him with strips of leather and hung a poster on his chest with the inscription "This Commissar is a marauder." A passer-by wanted to intercede on behalf of the victim, but the crowd attacked him as well; trying to escape he ran into the district judge's chambers, but the crowd stormed into the chambers, dragged the defender of the militia out into the street and beat him up so severely that he had to be sent to the Obukhov hospital.

The crowd stopped a cab that was riding by and put the beaten-up Deputy Commissar on it under the escort of three soldiers.

A cart was loaded with several boxes to make a platform on which they put the beaten-up shop-assistant, also covered with pieces of leather and a poster on his chest saying "A marauder merchant."

The cab and the cart accompanied by a part of the crowd moved along Sadovaia Street towards Nevsky Prospekt, while the other part of the crowd went to the Commissariat to demand explanations.

The appearance of two thousand people caused a panic in Gorstkina Street where the offices of the 3rd Spasskii District Commissariat are situated. Having realized that the crowd was in a militant mood, Commissar Ozol decided to summon a military detachment or a fire brigade to protect the Commissariat, but neither arrived. Some of the crowd broke into the

Commissariat, shouting threats against the Commissar and militia in general. Nearly all the militiamen had recently been disarmed, but fortunately two militiamen happened to have revolvers, and the Commissar ordered them to fire in the air. These two shots produced such a magic effect on the crowd that it started leaving the commissariat in a hurry. It was only at 2 p.m., when life in Gorstkina Street had already returned to normal, that the military detachment arrived.

In other cases, the inclination towards self-preservation assumes the form of collective actions aimed at protecting property and national wealth from any infringement and at defending religious beliefs. These include mob law applied to thieves and robbers, lynching in America, national conflicts and wars.

In general, any society—just like an individual personality—is interested in its own welfare under the influence of its inherent inclination towards self-preservation. Everything that sustains the welfare of society, stimulating in it a sthenic reaction, generates the desire to possess, whereas everything that acts adversely, stimulating an asthenic reaction, generates resistance and counteraction. The longing for freedom and for defending one's rights also arises from the inclination towards self-preservation, since damage to freedom suppresses the vital activity of the composite personality.

People generally do not put up with the heavy fetters imposed, for instance, by the capitalist state on the working masses, and underneath these chains of economic slavery there always smolders a desire to break them and escape from bondage. When the state regime and the oppression of the capital reach a degree that the national collective can no longer tolerate, a *coup d'état* called revolution ensues.

A state regime can gradually increase repressive measures to a considerable degree until they reach a certain limit followed by an explosion of popular discontent manifested in a riot or an uprising. The explosion of wrath and revenge occurs sooner or later, depending on the living conditions of the national collective itself, on whether the nation is disciplined, civilized, industrious, rich or poor, etc., but it takes place only under certain conditions as an inevitable effect of particular causes.

The collective attraction to self-preservation and freedom is also manifested in the fact that when the masses win their rights through a revolutionary upheaval, they first of all try to liberate their jailed brothers in

prisons and penitentiaries. This recurs during all revolutions, as a rule and a law. We observed the same during the last Russian Revolution, and this assault of the liberated people on jails occurred not only in the main centers of the Russian Revolution—Moscow and Petrograd—but in all Russian cities. The very seats of judgment, the so-called temples of Themis, were subjected to destruction and arson, as we saw in the case of the Petrograd Regional Court burned down by the crowd.

Some reprehensible manifestations of collective sexual desire have been observed in certain sectarian gatherings, special get-togethers in the form of Athenian evenings and even in the organized youth societies (*ogarochnye* and others),⁴ which were active here during the period of the first revolution. One comes across the same phenomena, however, even in uncivilized countries. This is, for instance, what we read in Ch. Letourneau:

On the Society Islands, as well as many other archipelagoes there even existed a famous religious fraternity—the Asroy (*Areoi* in French —E. L.) association whose purpose was the licentious satisfaction of sexual needs. It was a religious society. Their patron was the god Oro. All members of this Polynesian Freemasonry were regarded as superior beings irrespective of their gender. They were protected by special divine patronage, and after their death their shadows were taken to a special paradise. The society recognized total equality of its members; naturally, most of them belonged to the aristocracy, but commoners, too, were accepted. The only difference was that for the nobles the formalities were made simpler. This fact is extremely curious, for in these tribes the nobles regarded themselves as being made out of a far superior material than other mortals. The requirements for being admitted to the fraternity were very strict. First, in a fit of religious ecstasy, they had to prove that they possessed the divine spirit; then they were accepted as novices. Months or even years later a novice was subjected to another trial and had to take a great vow to kill any children he may have from then on. He thus entered the seventh and last rank of Asroy hierarchy; he was initiated into the sacred rites, songs, dances and special mimic movements. Ascending all the steps of this ladder was an extremely slow process; one had to go through new trials and to reveal his abilities in oratory, singing or poetry. Finally, each new stage of initiation was marked by a special tattoo.

The aim of the society was stimulation and unbridled satisfaction of erotic passions. All female members of the society belonged to all male ones, and couples came together for no more than two or three days. The life led by the Asroys was an eternal festivity. It consisted of endless feasts, songs and competitions. They even took trips together to other islands and were met everywhere with great respect. Women would perform for the members of the society an extremely sensual dance called *Timoradia*, which is accompanied by appropriate singing and is very popular in Polynesia. The sexual desires stimulated by it were satisfied immediately and publicly, but children were not admitted and infanticide was enforced as a strict duty. To be able to save her newborn, a woman from the Asroy society had to find for the child a foster father among its members, but then both of them were expelled from the society.

To belong to the Asroy fraternity was considered a great honour. One Tahitian brought to England by Cook declared that he regarded himself equal to the King of England because he was an Asroy.

One should note, however, that although we are dealing with the country where chastity is totally unknown, this society established an erotic religious cult similar to our sects of *khlysty* and others.

Sexual desire in a crowd is usually manifested in a crude and ugly form that at times exceeds all likelihood. When the Austrian troops occupied Lorraine during the Great French Revolution, the Croats burned down the monastery of Saint Nicholas, and many nuns fled to Chalons. Most of them were raped by the enemy soldiers. They called it suffering martyrdom. According to the nuns, some of them suffered this kind of martyrdom two, three and even four times, while one nun suffered eight. This is what we read in the article "The Defenceless" published in a Saratov newspaper:

To the noise of the events—both momentous and disturbing—some nightmarish phenomena occur in our city.

I am talking about the unfortunate residents of Petin Street.

What is going on there turns one's heart to ice and paralyses one's mind.

Here are some quotations from Dr. I.V. Vyazemsky's report recently presented in the Saratov Society of Physicians: "According to the commander of the N division of the NN reserve artillery brigade, since early in the morning long lines of soldiers wait their turn in front of the houses of assignation in Petin Street: the number of soldiers per woman reaches 40; there were cases when girls ran out of the houses crying "Save me, I can't take it any more."

Having discussed this statement and having come to the conclusion that in the centre of the city, before everybody's eyes, women are being tortured, the physicians' committee arrived at the unequivocal decision to have the houses of assignation closed.

A. Kalmanovich adds to this description:

The houses of assignation have not yet been closed, and some truly horrible things continue to go on there: three days ago the drunken crowd was rioting, breaking the windows, wrecking the furniture and beating the women, while the "tortures," mentioned in the decision of the physicians' committee, in some cases assumed unimaginable proportions when the number of unrestrained men per woman reached one hundred.

One can no longer procrastinate with closing down the houses of assignation. Protective measures have no effect whatsoever. Although Petin Street has been closed off from both ends by chains, men get there through one of the houses in the adjoining street: they climb up on the roof and from there they climb down the drain-pipes.

One could also cite other examples of striking sexual licentiousness particularly in a crowd, where people are not embarrassed to perform in front of everybody the indecencies that individual persons out of propriety conceal from the public eye.

It goes without saying that a crowd does not know—and has no desire to know—any restraint, and often leaves nothing but corpses in the wake of its violence, as it was the case in Kalushcha during the war after the start of the revolution. We often observe it in other cases, when a crowd commits violence upon unfortunate young women.

At this point there is no need to dwell on the manifestations of the "social instinct" in collectives, but there is no doubt that it, too, is expressed in an imperative manner. This is confirmed by a more than cruel attitude of a collective to telltales, informers, spies, stool-pigeons, strike-breakers and, in general, all those who one way or another violate the rules of collective life.

It goes without saying that under conditions of combat life the "social instinct" is expressed through some exceedingly striking manifestations, and its violation often leads to tragic and bloody events; law, as we know, imposes capital punishment even for attempted treason in the army.

Notes

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- 1 The available literature offers a wide variety of opinion on the number of "instincts," or biological needs, in man. Thus, James, as well as Ellwood and McDougall, distinguish a great number of instincts, while Darwin recognizes only a limited number of instincts in man. According to Wagner, there are only three of them: nourishment, reproduction, and self-preservation. We regard nourishment as a particular kind of the instinct of self-preservation. Attraction to community, however, should also be included in the category of instincts.
 - 2 V. Bekhterev, "Social Selection and Its Biological Significance" (in Russian), *Vestnik znaniya*, 1912. "Individual and Social Factors of Progress and Sociality as a Prerequisite of Progress" (in Russian), *Vestnik psikhologii*, 1913. "The Significance of Hormonism and Social Selection in the Evolution of Organisms" (in Russian), *Priroda*, 1916.
 - 3 See V. Bekhterev, *Suggestion and Its Role in Social Life* (in Russian), 3rd edition, K. Rikker.
 - 4 Ch. Letourneau. *Evolution de la morale*, 2nd edition, p. 142.

12

Collective Mood and Collective Mimico-Somatic Reflexes*

When dealing with an individual person, one can evaluate his mood from a purely subjective point of view, as is done by a psychologist, by analogy with his own self, guided by the evidence provided by the person who experiences the given mood. Reflexology, however, does not need this subjective evaluation, and collective reflexology has an even lesser need for this kind of subjective evaluation of such states, for even if it does discuss collective mood, it assesses this mood, or tonus, exclusively from a strictly objective standpoint, taking into consideration facial expressions of the members of the collective, their gestures, the nature of their exclamations, intonation, signs of approval or disapproval, a particular attitude to a certain fact, etc.—in a word, everything we describe by the term mimico-somatic reflexes.

As in individual reflexology, the mood of a collective may be lively in one case, apathetic and depressed in another, or friendly, unfriendly, angry, patriotic, religious, etc.

It is common knowledge that a friendly mood is characterized by a collective's benevolent attitude toward certain persons and phenomena, all of this occurring in predominantly calm tones and without any particularly intense manifestations. Occasionally, however, it is characterized by an outburst of a friendly disposition and touch in the form of embraces, kisses, etc.

An unfriendly mood of a collective is characterized by an inimical attitude expressed in a milder degree through cautious actions, avoidance of contact,

etc., while in a more pronounced degree this mood is characterized by the facial expression of hatred, by threats and the desire to engage in verbal abuse and fighting.

A collective patriotic mood, typical of most people living in an organized society, is characterized by the manifestations of attachment to one's collective and by the desire to support and defend it when necessary through all possible means. Patriotic excitement represents the apogee of patriotism, as it were, characterized by appropriate exclamations, gestures, and threats directed at the enemies, etc.

The religious mood of a collective can be observed in any temple, where it is characterized by a calm, silent concentration on the words of the prayer and on the external manifestations of worshipping the deity. A higher degree of religious mood is characterized, as well, by the appropriate facial expression, by emotional declarations of profound repentance of one's sins, by wringing one's hands, beating one's chest, and similar manifestations of self-torture (in some sectarian rituals).

During certain periods, a collective mood, or tonus, can reach an extraordinary degree of intensity and change into an acute mimico-somatic reflex of a particular kind in the form of anger, for example, or depression and fright.

One should not forget that an animated mood and the corresponding mimico-somatic reflexes are invariably accompanied by a sthenic collective reaction characterized by a certain readiness for offensive actions, whereas a depressed mood and the corresponding mimico-somatic reflexes are accompanied by an asthenic reaction leading to defensive actions and religiosity, or to total confusion in its more severe manifestations.

Thus, mood prepares the ground for particular collective decisions and actions, just as in individual reflexology.

An animated mood together with a sthenic reaction always raises the energy of a collective, whereas a depressed mood leads to the depletion of its energy.

Therefore, at the beginning of any meeting, in order to stimulate a mass of people for action, inflammatory or stimulating speeches are made to electrify the audience, elevate its mood and encourage thereby the adoption of a particular decision that will lead to the appropriate action.

To provide a vivid example of an inflammatory speech let us dwell for a moment on a description of the concert-meeting "The Loan of Freedom" that

took place in Pavlovsk during the period of the February Russian Revolution.

The Committee of the Employees of State and Private Credit Institutions that organized 'three days of the Loan of Freedom' moved the centre of its energetic activity to the Pavlovsk railway station last night...

"Buy the bonds of Freedom," one hears everywhere. "Fulfil your duty to the Motherland!" "Exchange your rubles for bonds." "The Motherland expects everyone to fulfil his duty."

The beautiful summer evening contributed to the success of the concert-meeting. The public turned out in huge numbers, as if it were the opening of a concert season.

At this meeting it was the speech by the sailor Fedor Batkin, who was greeted by a storm of applause, that attracted the greatest interest.

"We are now living through the most tragic, the most horrific and at the same time the most glorious period of our history," begins the speaker,

Yesterday we were slaves, today we are free citizens. Yesterday we were people without the right to think, today we are a nation that won the right to demand.

What can be happier and greater than this? But at the same time what can be more terrible for a noble and great nation that has always responded generously to the needs of her neighbors than to become pathetic and ostracised, and perhaps even to lose the right to be called a state?

"Much has been said about the causes of this decline," continues the speaker,

but where do we seek salvation—this is the burning question. We must get at the root of the danger that threatens us. So let us ask ourselves where this danger is? It is not in a handful of traitors in khaki greatcoats deserting from the front and shamefully abandoning their positions to the attacking enemy; nor is it in the worm that has been eating away at the core of the state organism, for a grassroots explosion of self-renewal and

self-purification can easily get rid of it. The danger is in the indifference and passivity of Russian society, which is enveloping its large segments with an ever-growing power and persistence.

"Shame," energetically exclaims the speaker,

shame to a great country, which is unable to achieve unity even at the most crucial turning point in its history. Shame to all those who are to blame for the present destruction. Shame to all those who do not bother to think about the fate of the motherland and who have to be reminded of the example set by the tiny, heroic Belgian nation of six million who showed how to die for the honour of one's country.

Do the Russian people know what the honour of the Motherland is? No, they do not, but if there are some who know this, they—those few—will be able to speak out about it.

"We are 'resolutionaries' rather than revolutionaries," says the speaker.

Are we indeed incapable of doing and fulfilling even a part of what we need—desperately need—to fulfil?

Now we are facing this part, this small portion of what is urgently needed to be done, which is the obligation to give our state a loan of money for saving the motherland. If Russian citizens fail to comprehend the sacred duty towards the fatherland, which compels one to make sacrifices for its welfare without thinking, let those whose conscience is paralysed with the deadly and heavy sleep of lethargy understand that they should do the same from sheer egoism.

If they do not wish to be the architects of the temple of freedom, let them realize that they are building with their own hands a grim prison for themselves and their children.

"Our motherland is in danger," concludes the speaker, "your mother is dying. Is there indeed a son who would refuse her the means of recovery!"

After Batkin's speech, drowned in prolonged applause, they opened a subscription for the bonds, which continued throughout the whole evening and produced results that exceeded the organizers' expectations.

Mood also determines the attitude of a gathering towards the speeches and actions of individual persons or toward certain phenomena of social life in general. That is why, when making his speech, an adept speaker who wants to sway his audience always takes into account its general mood. When he disagrees with the general orientation of the gathering, he should express it rather cautiously and gradually.

Everyone knows how difficult it is to go against the general mood of a gathering and how relatively easy it is to speak in tune with the general mood.

To prove this point, I shall take an example from the history of our revolution. At the night session of the Congress of the Party for National Freedom, one of the delegates who participated in the debates on P. N. Miliukov's speech protested against the inclusion of party members in the Government and attacked A. F. Kerensky when the latter's star was still shining brightly:

This man is represented here as a victim of the Soviet of Workers' and Peasants' Deputies. And we—the Party for National Freedom—are supposed to save this victim at any cost. So we have come here and are trying to save him, and this is what Nekrasov is magnanimously praising us for. But is A. F. Kerensky, indeed, a victim? Was he protesting against the destruction perpetrated by the Soviets?

Then the speaker declares unexpectedly:

"The author of Decree No. 1 was N. D. Sokolov, but we are persistently told that A. F. Kerensky himself participated in drafting it."

"Rubbish," shouts a voice from the audience, and a pandemonium breaks out. "Shut up! Down with them! Nonsense..."

One can hear feeble applause here and there.

The speaker tries to continue his speech but a storm breaks out in the audience once again.

A particular mood in a gathering may cause some characteristic incidents. Here is one example. During the days of the Russian "Bonds of Freedom" campaign, when these bonds were being sold in the atmosphere of general support for this campaign, a big scandal broke out at the Luna Park theatre when the actor Koshevskii appealed to the public to buy the "Bonds of Freedom." As soon as the actor Koshevskii started talking about the goals of the "Bonds of Freedom" campaign, someone shouted from a box:

"Go to hell, liar."

Pandemonium broke out. The whole crowd, one and all, rushed to the box, intending to lynch the insolent fellow. Order, however, was soon regained, and the culprit escorted by several people from the audience was sent to the Commissariat.

Singing is occasionally used for raising the spirits, although just as often singing itself serves as an expression of the mood. During the same "Bonds of Freedom" campaign, the sounds of the *Marseillaise* resounded from the open-air stage of the Summer Bouffe Theatre during the first intermission.

"*Allons enfants de la patrie...*" Kachenovskii was singing.

The singer's powerful bass attracted all the public who was asked to buy the "Bonds of Freedom."

The public, one and all, rushed to the tables and bought nearly all the available bonds.

In other cases, even external circumstances enhance the mood. Thus, during the "Bonds of Freedom" days, a decorated automobile with improvised bank booths arrived at the Zoological Gardens.

The arrival of the automobile created a furor, and the subscription was highly successful.

In fact, not a single social movement can be launched without the appropriate mood.

On the other hand, the nature of the mood determines not only the corresponding energy of the action but also its direction.

One can observe in every gathering a particular mood that may change even during the course of the gathering, depending on the particular circumstances and on the speeches of the participants, i.e., it can either grow in intensity or become weaker and depressed. It is invariably characterized by the prevalent facial expression, gestures, exclamations, signs of approval regarding particular expressions and theses, and finally, by the prevalent

content of the speeches and by the relative intolerance of those speeches that are incompatible with the common mood.

Here is one of the numerous examples. At the time of general disarray during the first period of the Russian Revolution, an officers' council held a tense meeting:

The first speaker was the chairman, general Amosov. He painted a horrific picture of disarray at the battle-fields and on the home front.

Other speakers, most of whom arrived from the front-line, confirmed the dangerous situation in the army.

Officer Malevanov declared that no one seemed to be in charge and used a highly inappropriate expression regarding our Government.

It caused an uproar in the audience.

People were shouting "Get out of here, down with him, this is disrespect of authority, this is intolerable!"

The chairman ordered the speaker from the floor.

Collective mood also depends to a significant extent on the composition of the gathering itself, on its greater or lesser homogeneity, on the subject of discussion and its greater or lesser relevance to the interests of the participants, on the given external circumstances, i.e., social environment, and even on the greater or lesser importance of the decision to be made by a gathering. Finally, the mood of a gathering reflects the disagreements existing in the gathering itself, for they exacerbate the passions of individual speakers, due to which this passionate attitude spreads across the whole gathering by means of contagion. Therefore, the fewer disagreements there are in a gathering the more tranquil the course of its activity. And on the contrary, the more heterogeneous a gathering is the more tumultuous its proceedings are.

Occasionally, there may be not one but two or more issues whose discussion raises the mood of a gathering to the highest degree.

This happens particularly in those cases when a gathering whose participants represent several different parties is preoccupied with a complex

subject. Thus, at the former State Assembly in Moscow, the first and highest rise in mood appeared on the second day of the session, during the confrontation between two opposing views on the ways of revitalizing the state held by the socialist parties on the one hand and by the Party for National Freedom on the other. A second rise in mood, albeit of a lesser intensity, occurred on the third day of the session during which the issue of improving the fighting efficiency of the Russian army was discussed.

The mood of any gathering is seldom determined from the start, unless it has been prepared in advance by certain events. Usually, it is only after several speakers voice their opinions that the nature of the collective mood begins to become apparent and reaches a certain level when, just before the final decision is made, the most intense confrontation of opinions takes place. Therefore, the highest rise of the mood in a gathering usually coincides with the moment of the sharpest discord. At this moment, one may always fear certain excesses on the part of different members of the assembly.

After the mood of a gathering has reached its apogee, it tends to drop more or less quickly, and the gathering calms down.

The greater the size of a collective the more difficult it is to create a common mood. It is natural, for example, that, because of the heterogeneity of the individuals who make up a particular state collective, it is impossible for the whole country to share the same common mood; in the final analysis the latter represents the mood of individual circles, parties, associations, classes and nationalities that populate a given country, and depending on the degree to which certain events or undertakings correspond or fail to correspond to the goals and aspirations of individual collectives, they generate a depressed mood in some quarters and high spirits in others. Thus, the term "common mood in a country" should actually be interpreted as the prevalent mood or the mood in the majority of individual collectives in the country; this mood may differ—and very radically—from one collective to the next, yet one can still perceive in the total mass of the collectives a certain tendency towards a particular direction exhibited by the majority of them.

Incidentally, a common mood is invariably reflected in election results, although here much depends on the activity of one or another party. In any case, the mood manifested at pre-election meetings and gatherings finds its expression in the voting-papers dropped into the ballot boxes. Thus, the outcome of the elections determines numerically, as it were, the nature and the direction of public support for certain parties and persons; the final choice

seems to represent the resultant force of the preferences and aspirations of a given society.

It stands to reason that the mood of a collective is, generally speaking, unstable—just as is the mood of an individual person. Regardless of how high the mood may rise as a result of a particular event, it inevitably goes down with time. The degree to which public mood determines the attitude of a collective to a particular event is exemplified by the cases when one and the same speech provokes different responses depending on the nature of public mood. Thus, there was a time when Kerensky's belligerent speeches were enthusiastically received by the population; but could they possibly provoke a similar response during the subsequent period of the Russian Revolution when a well-defined decidedly negative attitude to the external war prevailed in the army? Certainly not.

When laying down the main principles on which peace should be established, President Wilson made the following statement:

We believe so sincerely in the inevitable triumph of these ideas that I am convinced I am expressing the mood and the will of the American people when I declare that the United States are prepared to participate in any possible coalition of nations aimed at achieving these ideals and at protecting them against any violence.

This militant speech received the clamorous approval of the audience. This was in May 1916, when the Great War had already lasted for nearly two years, by which time the underwater war was threatening to violate seriously the interests of the United States, and the public opinion of the American nation learned much during this long world war. If, however, President Wilson had made this speech before the Great War broke out, it would have undoubtedly provoked a storm of protests. It goes without saying that the mood does not appear to be the same in various social circles of the country, as a result of which, for example, at the time when Kerensky was making his speeches that were received with stormy approval at the front before the offensive on the 18th of July 1918, one could hear all over Petrograd speeches against the war that received no less approval by the popular masses at meetings.

The story of Louis the Beloved illustrates the changes in the mood of the popular masses: "The name *Bien-Aimé* (the beloved)," reports Carlyle,

quoting Hénault,

was given to Louis XV on the following occasion. When in 1744 this monarch was rushing from one end of his kingdom to another, having interrupted his conquest of Flanders in order to render assistance to Alsace, he was overtaken in Metz by a malady that threatened to cut his life short. When this news reached Paris, the city stricken with horror resembled a city stormed by enemy troops. Supplications and moans resounded in churches, the prayers of the clergy and the people were constantly interrupted by sobs, so that this loving and tender interest naturally produced the name *Bien-Aimé*—an honorary title that surpassed all the others acquired by this great sovereign.

This is what was written in memory of the year 1744.

Another 30 years have gone by and the "great sovereign" is again confined to bed by another illness, but what a change in the circumstances: Paris is stoically calm; church services are not disrupted by sobbing because no services are held except those ordered by the state for a fixed fee per hour, and this kind of service is not usually interrupted by anything. The pastor of his people whose heart is heavy with grief has been transported from the small Trianon to his Château de Versailles and put to bed. His flock knows about it but remains indifferent. At the most, a mention of the royal illness appears among other news once in a while in the endless stream of French conversation (that flows without interruption day in, day out and comes to a halt only during the brief hours of the night). One cannot help betting on this subject after all.. ..Moreover, some people even dare to speak loudly in the streets; the May sun is shining over the green field and over the city's bell-towers, the May sun is setting, and the people are going about their own useful or useless business as if no Louis were in mortal danger.^{1, 2}

Such was the change that occurred over the period of thirty years in the Frenchmen's attitude toward their king, but during the periods of revolutionary movements, when popular masses experience in weeks and months what they normally experience in years, this change in the mood and in the attitude to their heroes does not take as long and occurs much faster. An edifying example in this respect is provided by Kerensky, the hero of the

February Revolution in Russia. Destiny pushed this man—a former barrister's assistant and subsequently the leader of the labor party in the State Duma—into the front ranks of the revolutionary leaders immediately after the overthrow of the monarchy. First he was the Minister of Justice, then Chairman of the Council of Ministers and the Minister of War and Navy, and then Head of the Provisional Government with unlimited powers as Chairman of the Council of Ministers and the Supreme Commander-in-Chief.

Having joined the first revolutionary ministry as the Minister of Justice from the Social-Revolutionary party, Kerensky immediately assumed in the new ministry the status of a person who became the focus of attention and hope for all the forces of revolutionary democracy. Subsequently his popularity started growing by the hour, and his every step was discussed in public. His handshake with a watchman and his condescending attitude toward the highest bureaucrats in the ministry when he assumed his post made him universally popular with the revolutionary masses. His remarkable oratorical talent rapidly reinforced his popularity, for during the revolution the masses needed speeches, more speeches and still more speeches.

At the same time, being the Deputy Chairman of the Soldiers' and Workers' Soviet, his status in that milieu was such that any leader would envy his popularity. That is why, when the first Chairman of the Council of Ministers, Mr. Lvov, resigned, his post was given by right—unwritten but universally accepted—to none other than Kerensky.

At the critical moment for the Government, when army discipline among the troops began to disintegrate after the Declaration of Soldier's Rights was published by the then Minister of War, Mr. Guchkov, who resigned from his post with the hopeless statement that only a miracle could save Russia, all eyes turned on Kerensky. He was perceived as the only person who could lead Russia out of the dead-end and save the country. Then all the ministers agreed that the War Minister's portfolio, as the most important one at the time, should be given to Kerensky. His acceptance of this post while retaining the title of the Chairman of the Council of Ministers made Kerensky not only the ruler of the ministry, where he was worshipped to such a degree that at any governmental crisis he was given the portfolios of all the ministers, but also the ruler of all Russia. His presence among the troops and his propaganda about revolutionary discipline sparked hope among the Russian democratic forces that the revolutionary army, like the army during the French Revolution, would enthusiastically assume the offensive and show

the world what heroic deeds the Russian revolutionary people could perform. And, indeed, the Russian army that at the time was more involved in political meetings than in fighting, that refused to obey officers and even dismissed them arbitrarily, seemed to have come to its senses for a while and on June 18 mounted a successful offensive on the South-Western front, advancing towards Lemberg in Galicia. The enemy troops began to sustain one defeat after another. The day of June 18 turned out to be the triumph of none other than Kerensky. Even though he did not participate in planning the military operations, he won the glory, and the troops presented him with the Cross of Saint George. At this point, his glory reached its zenith. His name began to have a magical effect on the crowd. Thus, during my speech at a concert-meeting in one of our large theatres, the mere mention of Kerensky's name in connection with the presentation to him of the Cross of Saint George caused the whole audience, as if it were charged with an electric spark, to jump to their feet and start a loud and long ovation in his honor.

It goes without saying that ministers were mere pawns in his hands. Some of them fawned on him in the full sense of the word, others barely managed to remain within the limits of decent obedience. He manipulated everybody as he pleased and fired the most deserving and prominent generals with a wave of his hand. Thus he dismissed one after another such prominent military commanders of our army as Alekseev, Brusilov, Kornilov, and Yudenich. General Ruzskii was unfairly discharged even earlier, when the post of War Minister was occupied by Guchkov and when the Council of Ministers was headed by that same Kerensky. Finally, during the initial period of the revolution, all the papers were full of news about Kerensky. His every word was devoured and printed. The papers did not stint their praise and glorification of Kerensky, who was sometimes described as the "pearl of the revolution." But only a few months later his undecisive and maneuvering policies brought about his downfall. His lucky star already began to set after the shameful retreat of our troops in Galicia. Everybody realized that speeches alone, even those of Kerensky, were unable to stop the disintegration at the front and that the role of the "Persuader-in-Chief" of the army was of no help either. His perpetual tendency to adopt a line of conduct between the Soviet of Workers' and Soldiers' deputies on the one hand and the so-called democratic strata on the other repulsed all bourgeois elements, while the Soviets' gravitation towards bolshevism resulted in the left wing of the democratic parties breaking up with him.

It was the Kornilov "revolt," however, in which, according to the bourgeois elements, he played the role of an *agent provocateur* that sharply exacerbated the attitude of the bourgeoisie towards him, while the introduction of capital punishment and his constant hesitations and indecisiveness undermined the trust he had previously enjoyed among the democratic elements.

It goes without saying that the Bolsheviks did not trust him from the very start, and, after their first insurrection of 3-5 July, he became their actual enemy.

As a result of all of the above, at the most critical moment in its history—during the 25 October coup—the Provisional Government of which he was the head failed to find any defenders for itself in the entire, numerous Petrograd garrison, except for a few hundred cadets and a detachment of the women's shock battalion. These forces were, naturally, defeated, the Government was arrested, and the cadets were massacred in a bloodbath. After that no one spoke with regret about Kerensky, who managed to escape. He was, however, remembered, as the leader of the counterrevolutionary troops he was bringing to Petrograd. But the headquarters focused their attention on the troops and the outcome of future developments, rather than on Kerensky.

Here yet again Kerensky demonstrated his indecisiveness, and his military adventure ended up with the capitulation of the troops and his escape. Now he was called a criminal by the very same people who had worshipped him before. Soon they stopped talking about him altogether. No one cared any more where he was and what he was doing. He was simply forgotten, and forgotten forever.³ Thus, over the period of several months, the mood of the popular masses had been changing with developments of historical events.

Similarly, the Great World War at the very beginning awakened patriotic sentiments in every country; but after several years of warfare, the mood gradually deteriorated even among those who had the opportunity to boost it with victories. In the final analysis, Hindenburg, who declared that victory would be granted to the side with the strongest nerves, was proved right.

However, irrespective of the natural decline or internal inhibition of collective mood, the latter may also change in the opposite direction under the influence of events. This kind of change in the mood is called its turning point.

During a turning point, the mood of the masses completely changes its direction due to the current circumstances. Thus, in the course of the Russian

Revolution, the Bolsheviks' insurrection in July was followed by a turning point in the mood of Russian society's democratic circles, expressed in their highly negative attitude towards the movements on the extreme left, which used to attract many sympathizers. That did not last long, however, for after the Kornilov affair on the 27th of August the public mood swung in the direction of bolshevism yet again.

During the year 1905, people were clearly sympathetic to the revolutionary movement, but following the subsequent excesses of a revolutionary nature, and the so-called expropriations, the mood began to change rapidly: its final turning point in the opposite direction occurred after the onset of the "bomb-throwing" movement.

A vivid example of this kind of change in the mood of the popular masses is furnished by the replacement of the militant aspirations of the Russian army by its "fraternizing" with the former enemies during the initial period of the Russian Revolution.

Here is another example dealing with the change in the mood of the rebellious troops after the suppression of the Bolshevik insurrection of July 3-5 in Petrograd. General Polovtsev, who had suppressed the insurrection and had been working hard during those days, was relating in his hoarse, tired voice how he was being thanked by the very commanders he had suspended. The way they formulated their gratitude was also quite original: "For demonstrating your authority!" Some may claim that only army-drilled people, conditioned by fear and subordination, can feel gratitude for being punished, offended, and suspended from their duties. But gratitude was expressed even by the soldiers and sailors who had been rioting the day before.

They say that when K—v was visiting the Commander-in-Chief, soldiers of the Preobrazhenskii division were enraged and demanded that K—v be handed over to them; Polovtsev had to convince the soldiers that K—v would not be allowed to leave without their permission; as soon as comrade Gots persuaded the soldiers to agree, K—v was spirited off down the backstairs.

Subsequently, after they cooled off and recovered from their rage, the soldiers were thanking those who were around for having prevented them from committing an act of insanity.

Here is another example of a rapid change in the mood of a collective. According to the newspapers, some time after shooting their officers, the soldiers of the ship *Sebastopol* started holding funeral services for them.

A turning point in the public mood completely changes the attitude towards former idols and institutions that used to represent sources of national pride. We saw this in the cases of Louis the Beloved and Kerensky, but let us take another example: General Kornilov, the hero of the successful offensive of 18 July 1917 and subsequently Supreme Commander-in-Chief of the Russian army, was surrounded by an extraordinary aura of glory and esteem. But when he led the so-called "revolt" against the Provisionary Government, everybody or nearly everybody turned away from him, and his former admirers became his most vehement detractors.

Another example: the former Chairman of the State Duma, Mr. Rodzianko, was surrounded with no lesser aura of esteem and respect not only during the period of existence of the Duma itself but also during the first days of the revolution in which he participated to a certain extent. But what followed afterwards? When the revolution gave power to the proletariat, he was showered with all kinds of accusations, and resolutions of protest were adopted all over the place. These kinds of examples abounded during the Great French Revolution, as well as during all revolutions in general, and even in all social movements of any significance.

Another example: during the Maillard tribunal, when aristocrats were massacred under the pretense of the purely external legality of the tribunal, very few escaped from the guillotine. But then an aristocrat, Jourgniac de Saint-Meard presented the judges with evidence of his civic virtues, mentioning that his soldiers loved him. He was pardoned, and as soon as the verdict was pronounced, the crowd broke out into the jubilant: "Long live the nation!" and the prisoner was enthusiastically accompanied from the courtroom by the same crowd that was ready to gloat over every death sentence.

This is how the volatile mood of the crowd was described by Marie-Antoinette's foster brother who happened to be among the fortunate few who had been spared the lot of the majority to be guillotined:

As soon as the guards raised their hats on the tips of their sabres and cried out "Long live the nation," the crowd broke into furious applause; having noticed that I was wearing white silk stockings, some women rudely stopped the two soldiers who were holding me by the arms and said to them: "Look out, you are making this gentleman walk in the gutter." They were right: the gutter was full of blood. I was particularly struck by this

kind of attention on the part of those shrews, since they had just been applauding violently the massacre of my fellow prisoners.⁴

Under normal circumstances, the faster and the higher the mood wave rises, the faster and the lower it then drops, and vice versa, the slower it rises, the slower it goes down. One should not forget, however, that the very rise in mood can be caused by certain external stimuli outside the given society, in which case the above rule cannot operate. Thus, for example, a declaration of war by another country can immediately stimulate a tremendous upsurge of the patriotic mood that will subside only gradually, as the developments unfold. In the same manner, an unexpected military disaster can lead to an explosion of patriotism in a given country, as occurred, for example, in October 1917 in Italy where the population was fatigued by the war and the disintegration of the army—similar to ours—had already begun. However, the military catastrophe that resulted from the German offensive turned socialists into so-called social-patriots, while the general patriotic mood manifested in the slogan "Save the Fatherland" became dominant. Besides, foreign troops quickly came to Italy's assistance, while her own wavering detachments were moved behind the lines, which saved the situation.

How quickly society's mood can change is also illustrated by the bourgeois sabotage after the October coup in Russia in 1917. Initially, bourgeoisie represented by the Union of Unions declared a total boycott of the Bolsheviks and rejected any cooperation with them, assuming an implacable position in this regard. A few months later, however, the same Union of Unions was forced to announce its repudiation of sabotage. The same was done by the representatives of higher educational establishments, and then the mood of the majority became oriented towards the possibility of cooperation, which gradually began to develop. Some accepted it because of the necessity to earn a living, others considered sabotage to have been a major tactical error, and some others believed that the country was in need of work and that politics should step aside, etc., but the truth of the matter is that sabotage came to be disapproved by nearly everybody.

The Civil War that broke out simultaneously with the Russian Revolution also provides numerous examples of changes in the mood of the popular masses. While the Reds were in power, the popular masses, depressed by the state of general disarray, showed signs of discontent, greeting the White troops with great pomp; but within a few months or even weeks the White

regime would cause general discontent, and people would be awaiting the Reds as their saviors. This kind of change in the mood during the Civil War would repeatedly occur in both directions.

Another illustration of changes in the mood of the masses, particularly during the periods of tumultuous political coups, is provided by Germany: after the impressive victories achieved by that country during the Great War, when the population was under maximum military strain, it proved impossible as the revolution unfolded to draft into compulsory military service any number of recruits. Thus, in the 25 February newspaper dispatch from Berlin we read that during the discussion of state defense a Social Democrat, Mr. Schefflin, made the following statement:

It is impossible at the present moment to build a military force by drafting recruits; the state that has until very recently been one of the greatest military powers in the world is now forced to recruit volunteers through newspaper ads. It is necessary that the army of national defence respect the Constitution, obey the Government and protect the laws of the land. It is also necessary that the volunteers obey their superiors and adhere to strict discipline.

As we know, various moods are manifested through corresponding mimico-somatic reflexes. Thus, a more pronounced manifestation of depression induced by certain circumstances is expressed in individual life through sighing, moaning, weeping, sobbing, lamenting, wailing, and complaining, as well as through other phenomena.

Through a gradual evolution, a depressed mood is transformed into a wailing song reminiscent of moaning, sighing, and weeping.

On the other hand, a lively mood is characterized by chuckles, laughter, exclamations of delight, excited speech, as well as sounds and statements of joy, and then through a gradual evolution it is transformed into a lively song of a cheerful nature.

In both cases, to enhance its effect, the song is accompanied by the appropriate sounds of music, with the music itself functioning as a manifestation of one or another mood and, through contagion, as the stimulus of the corresponding mood in others.

As we know, mood is manifested not only through voice but also through body movements: a depressed tonus suppresses movements, whereas a lively

tonus stimulates them, provoking a whole range of various body movements in the form of a discharge of nervous energy, as it were. Therefore, a lively tonus at the peak of its evolution leads to a dance.

Naturally, all collective moods are accompanied by this variety of phenomena. Thus, a depressed state during a funeral is characterized by mournful singing, funeral marches, silent standing in assemblies to honor the memory of those, who died, etc.; on the other hand, a collective celebration in cases of a friendly meeting, a victory, etc., is accompanied by collective feasting, joyous and triumphant music, cheerful marches, lively speeches, greetings, solemn ceremonies, balls, etc.

Collective moods tend to spread and in certain cases can even hold sway over a number of collectives.

Symbolism supplements the evolution of collective moods, enhancing them to an even greater degree. For example, black flags on days of national mourning, national flags on solemn occasions of a patriotic nature, appropriate speeches, toasts, etc., undoubtedly have an enhancing effect on the general mood of a collective, acting at the same time as further manifestations of mimico-somatic reflexes.

When a cheerful collective mood is accompanied by the desire to accomplish appropriate actions and reaches its climax, we speak about the enthusiasm of the popular masses.

This enthusiasm, however, is manifested under the influence of more or less strong external causes. It is usually encountered together with the general rise in mood conditioned by a certain cause.

Thus, just as in individual life, where an insult or some other cause can provoke irritation, so in collectives, under the influence of certain events that hurt the honor of a collective and its independence, a hostile mood, an explosion of enthusiasm and militant energy directed against the perpetrators of these events may develop. There are many examples of this phenomenon, particularly during wars. Incidentally, the Americans provided a vivid example of this during the present war when German submarines started sinking American ships near the Atlantic coastline. The fact itself was certainly inevitable in the course of the war, but at the same time the news about this fact, which indicated the arrival of war in America, immediately resulted in a great rush of army volunteers and stimulated the increase of patriotism and the rise in combat spirit among the Americans, which is evident from a considerable number of newspaper items. We shall quote from

those American sources that deserve our attention. The *New York Times* makes the following comment: "Germans could not have chosen a better way of enhancing military ambitions in the hearts of all Americans and of driving patriotic fervour to fever temperatures." When discussing the increase in recruitment, the *New York Herald* says among other things:

Soon after special issues of the evening papers hit the streets people thronged to the recruiting stations and demanded to be drafted into the army. As the dusk approached, ever-increasing numbers of would-be recruits flowed into the offices and inundated them with their applications for joining one or another branch of Uncle Sam's armed forces.

At the naval recruiting station by 3 p.m. about 500 young people were trying to join the navy, and more of them were arriving every minute. One of the officers at the station said: "The office was inundated by the crowd, and it was impossible to attend to every recruit immediately on his arrival, even though our numerous staff are working very hard."

The same story was repeated at all the recruiting stations. Masses of people thronged there in the hope of joining the army, and their patriotism and bellicose spirit had been reinforced by the news of the Huns' pirate raids. Ever since the United States entered the World War there has been no comparable drive to join the army and participate in the conflict.

Women displayed an equally strong patriotic spirit manifested in their flocking to the Red Cross in great numbers for the purpose of joining the staff of the American military and naval hospitals. The *Evening Mail* writes the following on this account:

At six in the morning those who wished to join the navy arrived at the naval recruiting station in such huge numbers that the queue stretched from the third floor where the recruiting station was situated down to the street and reached the next block. At all the other recruiting stations people had been waiting since early morning for the doors to open.

One should add that, according to the dispatches from the coastline cities, the actions of the submarines did not produce any panic. The lights are not

extinguished anywhere except in two cities, which in compliance with their mayors' orders are submerged in darkness in the evenings.

All of the above indicates that the mood of a collective can be measured numerically. And indeed, if we consider a precise measure of collective mood, there is hardly a more correct method than counting the votes during an election. Just like the counting of votes in cases of controversial issues gives preference to one or another decision, so during elections the counting of votes always determines as accurately as possible the political mood of the country. However, voting can never be complete, as there is always a number of those who fail to vote. If the latter cannot be explained by certain reasons that complicate the process of voting, it should be attributed to a greater or lesser degree of an indifferent attitude towards political issues and in particular to the population's attitude towards the issue that is supposed to be resolved by these elections. It is thus clear that correctly gathered statistics is, generally speaking, the best and most accurate indicator of the country's collective mood and the changes it undergoes in the course of time. It is noteworthy that, just as in individual life, together with the change in the mood of a group or a nation the whole nature of public views is changed accordingly.

I can cite an example from French life when a change in public mood was followed by the change in the general outlook of the whole nation:

A quarter of a century ago, not long before the Great War, reminders of her past military glory were considered a crime against France. In those days, Jacob, the well-known professor and philosopher, who was teaching ethics in *école normale* for girls in Sèvres, said the following: "If our motherland is destined to play a brilliant role in the world, it will be through science, arts, fine crafts, good taste, magnanimity of comportment—in a word, due to our hospitable nature; I do not see anything in the future except peaceful means." The supreme representatives of intellectualism in France in the latter half of the last century were Renan and Taine who preached a true cult of reason and science.

Reaction against this trend was developing imperceptibly and gradually. Many were encouraged by the prospect of support from the allies. Achievements in technology, the automobile industry, and especially

aviation provided new stimuli for reawakening the French bellicose temperament. The ground had been prepared when the crude excesses of the Eastern neighbours provoked loud protests. The expedition of a naval ship to Agadir was the last straw, and the whole of France started talking about war.

In the intellectual sphere this manifestation of nationalism found a very radical expression. According to the well-known critic Emile Faguet, "reaction against Kant, Taine and Renan was manifested with far greater force than one could have expected. When they are mentioned, it is only with utmost contempt....The general trend is a strongly pronounced anti-intellectualism."⁵

Under such conditions it is not surprising that everything which is conducive to expanding one's horizons and promotes peacefulness is nowadays regarded by many Frenchmen as humiliating and indecent....One English author, troubled by the bellicose mood of young Frenchmen, expressed herself in the following manner:

Those of my friends whom I used to perceive as champions of peace, enemies of militarism and restrictive nationalism, and as admirers of Goethe, Nietzsche and Wagner, have undergone a striking transformation.

They only whisper their former words of peace and progress, while their every word, every emphasis and every glance makes one aware of their desire for war.

Similar to the simultaneous changes in the public mood and in the outlook of French intellectuals, we, too, experienced the same transformation of our worldview between the 1860s and the late 1870s when the Russo-Turkish war broke out. Statistics concerning the book market and the books borrowed from the libraries would yield some particularly edifying findings in this respect.

Notes

- 1 In subjective psychology, the term "mood" is used to describe feelings or emotional disposition, but the word "mood" also represents an objective term, and we use it to describe a positive, negative or neutral tonus of the higher nervous centers.
- 2 Thomas Carlyle, *The French Revolution*, Russian translation, p. 5.
- 3 Although on 22 November 1917 he himself reminded people of his existence by publishing in the papers a letter under his signature, the letter itself was more like a cry of despair. Needless to say, it failed to make the slightest impression on society. His former friends could only regret that the letter ever appeared in the press.
- 4 Cabanis and Nass, *La névrose révolutionnaire*, Paris, 1906, pp. 26-28.
- 5 I. Mechnikov, *40years, etc.*, p. 30 (in Russian).

13

Collective Attention and Collective Observation

With the emergence of a collective, its attention—irrespective of whether it is directed towards defense, attack, or towards anything that excites its curiosity—is increased in direct proportion to the number of its members.

Naturally, a collective always has an advantage over an individual with respect to observation and ability to notice, which is corroborated by numerous examples, particularly from animal life.

To illustrate this point it is suffice to cite one example from H. Drummond:¹

Here is a herd of deer spread out, as they are fond of doing, over a quarter-mile distance; each animal in the herd participates not only in the physical strength of the rest but also in their observational abilities; its caution in case of possible danger is the caution of the whole herd; it has as many eyes, ears and organs of smell as the whole herd; its nervous system embraces the whole space occupied by the line: in a word, its environment consists not only of what it hears, sees, smells, feels and tastes, but also of what every individual member of the herd hears, sees, smells, feels and tastes.

Attention as a kind of collective reflex does not differ essentially in anything from the individual reflex of observation in the form of visual concentration or looking, auditory concentration or listening, etc., the only

difference being that here the act of looking and listening is performed by thousands of eyes focused on one and the same object and by thousands of pricked-up ears. Here one finds the same forms of muscular contractions as one observes in acts of individual looking and listening, but these forms of muscular contractions are performed simultaneously by a multitude of people. However, there is a big difference in the results of looking, i.e., in the observational ability, or the ability to notice.

Thousands of eyes see better than a pair of eyes. The minutest weakness in performance is immediately picked up by a watching collective, whereas a single spectator may not have paid any attention to it. One may say that a collective's ability to notice is as many times greater as the collective is numerically larger than a single individual. In a concert hall one can observe a listening collective in the particular poses that characterize a certain tension of the audial apparatus. And again, a listening collective displays *caeteris paribus* a sharper observation of sound effects than any individual person does, for even the smallest fault in the performance will be noticed in a collective by a certain number of people, whereas a single individual—no matter how good his musical ear might be—may fail to notice it, if only due to a temporary distraction.

Collective attention exists in any crowd listening to its speaker, in any theatre audience watching a performance or difficult acrobatic feats, in any concert hall where an audience is listening to musical pieces or to singing, in any school where students are concentrating on the words of their teacher, in any difficult collective work where people are involved in a joint project and focus their attention on their supervisor in order to coordinate their movements, on which the correct result of their work depends.

During certain periods of time, a collective concentrates on a particular event and displays a heightened sensitivity towards this event. Thus, in wartime, collective attention is directed at the news from the scene of battle. In times of famine, collective attention is focused on issues related to the delivery of food supplies. During a sensational court case, collective attention is directed at the process of criminal investigation and trial. In the case of a devastating social disaster, collective attention is directed at the various details of this event, etc.

Thus, public attention in social life is attracted by any sensational news, such as the declaration of war, a grandiose undertaking, a new discovery or

invention, even by an outstanding book—generally, by everything out of the ordinary.

The above-mentioned greater effect of collective attention and the greater result of collective observation ability are achieved in animal and human collectives by nothing other than the unifying principle of the collective, due to which a characteristic observed by one member is immediately communicated to others by means of signs or language. This is enough to make it a communal possession and an object of further joint discussion and identification, which due to the increased concentration by many on the same object leads to an even greater observational ability in the given direction.

As a result of all this, the evaluation of a particular object of concentration—whether a painting, a theatrical performance, a musical piece, etc.—turns out to be more accurate on the part of the collective than on the part of each of its individual members.

Let us take an example from life. At the beginning of the February Revolution, P. M—ov, one of the most popular leaders of that time elected by the popular will to the Provisional Government, during one of his speeches that followed the coup mentioned in passing the name of a possible future monarch of Russia. This was enough to provoke an almost universal condemnation of the speech and to cause an immediate loss of the speaker's credibility among the leftist political circles of the country, whereas an individual person exposed to the same speech may have failed to pay any attention to this place in the speech or during an exchange of opinion the speaker could have improved the impression from his speech one way or another.

It is clear from the above why collective, or public, supervision has an advantage over individual supervision.

Therein lies the importance of public control, which is, generally speaking, significantly more attentive and strict than individual control.

However, all of the above is true in the case when the watching collective has an opportunity to display its unifying principle through open criticism and exchange of opinions. In the case when the whole matter is reduced to collective observation alone and the collective can at the most express only its approval or disapproval—as is the case with a theatre audience—the collective loses a significant degree of the intensity of its power to observe, so that the end result of the latter is reduced to the impression produced by a given performance on the majority of observers. But since here, too, the

unifying element in the collective is expressed through signs of approval or disapproval, a collective observational ability obeys to a certain degree the law of imitation or contagion. Everyone knows that when a group of people begins to applaud at a certain place in a play, this applause is picked up by others and frequently by the entire audience. This phenomenon, as we know, is used in the theatres by claquees hired specifically to support a certain play.

The same condition of imitation and suggestion plays a certain role in the emergence of collective illusions and hallucinations, which will be described below.

Without going into detail on this subject, I would only like to draw your attention to the fact that every collective hallucination begins with a collective illusion or even represents an illusion based on collective concentration and oriented in a particular direction by the appropriate mood and the expectation associated with it. Thus, when a family is struck by grief as a result of the death of one of its members, it sometimes happens that other family members see the dead person under certain circumstances or conditions that give rise to a mass illusion. When troops are in a state of constant anxiety they are often struck by collective illusions and hallucinations in the form of an advancing enemy. Such illusions are particularly common at dusk or at nighttime. Here again, the emergence of collective visions is generated by one or another external cause that initially produces an illusion in an individual person, but subsequently this illusion is passed on to others through contagion and, while spreading, it is transformed into a collective illusion and sometimes a hallucination.²

As in the case of individual attention, collective attention is characterized by the fact that every small detail of the event that has attracted collective concentration provokes a collective mimico-so-matic reaction and numerous discussions in society. The event itself becomes the object of common judgment and a topic of fashionable conversation everywhere.

It goes without saying that collective attention is distinguished by the same characteristics as individual attention.

Thus, when its object is monotonous it is subject to fatigue; on the contrary, a variety of external impressions, particularly after an interval, revives it. A single event, no matter how important it might be, is unable to sustain collective attention for long. The latter is gradually blunted or inhibited, subsequently switching to another event.

In general, as in the case of individual attention, the change of events that serve as objects of concentration represents an extremely important factor in maintaining collective attention.

The latter becomes particularly receptive towards an event if, over a certain period of time, collective attention has been completely unoccupied, in other words has been idle and resting, as it were. Naturally, the intensity of concentration is determined by the social importance of a particular event, which, in turn, depends on the extent to which this event touches upon the interests of individual members of a given community or collective.

One is struck by the specific attitude of a crowd, predisposed in a particular way, towards external influences of one or another kind. It turns out that, just like the individual person, the crowd remains indifferent to all influences that fail to match its mood, and, on the contrary, it is highly sensitive towards the influences whose nature is in one way or another in tune with its general mood. In this respect, its observational ability and concentration become heightened. This is completely analogous to the mood of an individual person. When one is in a low or depressed mood, one is partially or totally unreceptive to anything that contradicts his mood or can provide some diversion and raise his spirits. Conversely, when one is in high spirits, one is unreceptive to other people's grief.

On the other hand, during the periods of political coups, disturbances, and uprisings, political tensions generate collective fears and collective suspicion. During the Great French Revolution, as well as during the Russian Revolution, the fear of a possible counterrevolution persisted in society during nearly the whole period of the revolution, which provided a breeding-ground for collective suspicion, and as a result almost any person who did not belong to the ruling parties was perceived as a counterrevolutionary and was called to account under the most trifling pretexts. In the same manner, a heightened collective suspiciousness with respect to spies develops in wartime. This becomes particularly pronounced in countries like Germany, where military espionage is practiced on an extraordinarily high level. One may say that the extent of "spy mania" depends on the extent of espionage, for the better espionage is developed in one's own country, the greater one's suspicions are that the enemy state practices espionage on an approximately similar scale, and it is therefore natural to suspect the existence of enemy espionage all around you. This is why in Germany, for example, more than in any other country during the last war, one was struck by the abundance of

posters with warnings against the danger of espionage in the streets, squares, railway stations, shops, restaurants, cafés, and other public places.

Naturally, German "spy mania" was directed first of all against those foreigners, including Russians, who had not managed to leave Germany before the declaration of war. The brunt of this spy mania was then borne by the prisoners of war. In this respect, we are more familiar with the facts that refer to Russian prisoners of war. In this case, it is undoubtedly the prisoners of war themselves who provide the most valuable testimony, which is reproduced below from N. Vavulin's words:³

Germans were capable of seeing a spy in every prisoner. Searches were a common occurrence in the camps for prisoners of war. Our physicians, officers, soldiers and civilians were subjected to searches. It was nearly always unexpectedly that guards would appear in the POW camps and hospitals, cordon off the huts and block all the entrances and exits. The camp authorities would arrive together with detectives and occasionally even with police dogs. The fuss would start right away, and German officers and sergeant majors, brandishing their weapons and red from the strain, would yell at their own soldiers as well as at ours. This swearing directed at us and the shrill orders mixed with threats unnerved us in the extreme. Physicians and officers were subjected to more humiliating search procedures than soldiers. They were usually led out of their own premises into an empty hut where they were told to strip naked. While the physicians and officers were undressing and the German officers were rummaging in their underwear, majors or superintendents would go into the living quarters of those being searched where with the enthusiasm of treasure-hunters they would dig into the dirty linen, crawl under the beds, tap the walls and open the bedside tables.

Soldiers were searched in a less elaborate manner but more frequently: a block sergeant major, as one responsible for his block, would usually order the prisoners to line up on the drill-ground with their belongings. And when the gray ragged mass lined up with their equally ragged sacks, a sergeant major would appear with interpreters and sometimes with a pack of police dogs; they were always on the look-out for something decidedly criminal, but all they would ever find were rags, occasionally notes,

diaries, books and newspapers, which were immediately confiscated and sent to the commandant's office.

"Spy mania" terrorized not only the German population but their army as well. The Germans' cruel treatment of us was quite understandable, for it was rooted in their fear that their humane treatment of the prisoners of war might give rise to the accusation of being in illegal contact with us, which might allegedly endanger the security of the German empire.

Germany guards its military secrets extremely jealously. Searches, censorship and interrogations were all auxiliary means for determining the "loyalty" of a person under suspicion. Those who expressed any disapproval of the German empire were subjected to harsh punishment. Prisoners of war who for reasons of health qualified for exchange but aroused some suspicions remained in the camps for ever.

When discussing the POW group that returned from the fortification works to the hospital of the Altdamm camp, the author notes,

They were half-dead, withered men, skeletons in the full sense of the word, their yellowed skin dry and taut. Most of them were in the last stages of tuberculosis, and some were crippled. So only because these people had been driven to exhaustion while working for the state, the German Ministry of War issued an order not to exchange them, since they might give away military secrets. And such prisoners were ruthlessly detained to their dying day.

For the purposes of preventing espionage, all the correspondence from prisoners is kept in quarantine for 10 days and longer before undergoing censorship, and most of it is destroyed altogether. For the same reasons, no outsiders have the right to visit the POW camps, and prisoners themselves are forbidden not only to live in private houses but also to go from one camp block to another.

The author does not even touch upon the special protective measures that were applied to anyone leaving German borders, and a description of those measures would add a great deal to the general picture of the German spy mania.

According to Dr. Kreslin, who personally inspected the camps of the Russian prisoners of war in Germany, the picture appears to have been even gloomier, but we shall not dwell on this subject, so as not to re-open the wounds of the recent past.

Let us note in conclusion that depending on its mood—particularly in the periods of anxiety or anticipation of a certain event—a collective, just like an individual person, develops an increased sensitivity of its organs of perception to the point of generating collective illusions and hallucinations.

To illustrate the above statement I shall limit myself to citing a case of collective hallucination that occurred, according to the description in *Razvedchik*, on the night of June 2-3 in the Andizhan garrison soon after the well-known uprising of 18 May:

The Andizhan garrison was languishing under the yoke of the ever-increasing problems of army life. The number of prisoners was growing by the hour; the convoy duties and distant reconnaissance raids tied up daily the whole of the Andizhan cavalry; there were justified fears of a possible outburst of fanaticism as the time of the execution was approaching: rumours were afoot that the locals were prepared to rush to the rescue of their holy ring-leaders and chiefs, and it seemed that the numerous crowds of daredevils could easily wipe from the face of the earth a handful of Russians, among them a temporary Commander-in-Chief with his retinue and the top officials of the military tribunal.

If those who were fully aware of the Andizhan circumstances did not view the situation through rose-coloured spectacles, just imagine the feelings of those who received information that was inflated to chimeric proportions and you will easily imagine the state of the ignorant mass of soldiers. Batmen were particularly ardent disseminators of all kinds of nonsense among the soldiers.

The night of the Andizhan alarm happened to be pitch dark and cloudy. It was preceded by rain and possibly by a thunderstorm.

The newly arrived riflemen were billeted in the barracks and camps. In those days barracks were very poorly lit inside, and the huts were lit nearly as sparsely as on the fatal night of 18 May. The rest of the space in both the city and its environs was drowned in thick darkness. The street-lights

in the city of Andizhan were not lit up that night in accordance with the glorious Russian custom that prohibited it when the moon was supposed to be shining, even though it was obscured by the clouds.

It was after midnight. Soldiers in the barracks and in the camps were sleeping side by side, holding on tightly to their rifles and touching the cartridges in their sleep. A truly anguished and depressing silence reigned supreme.

Suddenly there was a muffled noise somewhere in the distance, which made us all start, a second later feel terrified and then gasp, for the noise kept growing and growing; then it turned into the hubbub of a crowd in which one could discern occasional wild screams and then whole sheaves of screams, squeals and shrieks.

Then all of this was joined by the cries of "Hurrah" that were scattered at first, then loud and then a thundering victorious "Hurrah," and "Hurrah" kept expanding and enveloping the camp and the barracks.

Suddenly gunfire started. An eye-witness arrived at the camp when yet another soldier on duty was firing the last charge in the direction of the retreating enemy. At that moment all the soldiers were already lined up in their huts and were carefully examined and calmed down by their superiors. Everyone was frankly explaining where he was shooting and how many times, but no one in the camps or on the outposts really knew where the shooting and the alarm started.

In fact, the false alarm was caused by a wounded soldier who jumped out of bed in his delirium and started running and screaming; after that all the other wounded jumped out of beds.

The riflemen detachment closest to the infirmary was woken up, grabbed their rifles and shouting "Hurrah" dashed to the rescue of their wounded.

This stirred the whole camp! People started jumping to their feet, pulling on whatever clothes was within their reach, clamouring and yelling. They shouted the "Hurrah-that grew and reverberated and reached the farthest corners of the garrison.

Then someone fired his rifle into the darkness and the mass hallucination was born in all its glory—everybody was seeing, hearing and shooting at the enemy. For one second people became deaf to the voices of their superiors and to the signals.

Notes

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- 1 H. Drummond, *Progress and Evolution of Man*, p. 177.
 - 2 For details and numerous examples of collective illusions and hallucinations see my book, *Suggestion and Its Role in Social Life* (in Russian), St. Petersburg, 3rd edition, 1908.
 - 3 N. Vavulin, *Vestnik Znaniia* (Herald of Knowledge), issue 6—"German procedures to secure a victory."

14

Collective Creativity

The view of collective creativity has recently undergone an essential transformation. Whereas in the recent past it was believed that creativity was predominantly a product of individual persons, distinguished by the power of their intellect, and that with respect to the processes of reasoning and creativity a collective was unable to produce anything but mediocrity, nowadays, on the contrary, one has to recognize as a fact that collective creative activity plays a dominant role in mankind and that reasoning itself owes its existence to nothing other than collective activity. Incidentally, this point of view is also shared by the outstanding modern sociologist E. Durkheim. At this point, we shall not dwell on this fact but merely note that when it comes to individual reasoning, a great deal, if not everything is determined by previous individual experience, that defines the value of a particular judgment. Application of this experience to a new task is eventually reduced to supplying pieces of evidence "for" and "against" and to counterbalancing both categories for the purpose of summing up the final results of this weighing process in the form of a particular conclusion.

A similar phenomenon occurs in collective reasoning, the only difference being that here one deals with the application of experience generated not by an individual person but by the many people who make up a given collective. As a result, collective judgment usually appears to be more comprehensive than an individual one in the same measure as the experience of many is more comprehensive than the experience of one. For this reason, a critique of a particular object or its details in a collective is invariably more

comprehensive and multifaceted than an individual critique, and the judgments and decisions themselves turn out to be more prudent.

However, due to the participation of many individuals in the discussion, the effort of summing up the results in a collective is much slower than an individual effort.

Therefore, in those cases when a decision has to be made quickly and hastily, a collective effort is out of place, even though it might be better than an individual effort; and, on the contrary, when there is a need for greater prudence and caution in the absence of urgency, then a collective effort should be preferable to an individual one.

The nature of collective judgment is affected by the composition of the collective that determines the views and interests of the majority of its individual members.

It is obvious from the above that, depending on the composition of the collective, its judgments and decisions may in some cases be tendentious and even somewhat biased. As a general rule, however, collective judgments and decisions appear to be more impartial than individual judgments and decisions. In order to ensure a reasonably unbiased collective judgment, the collective itself should consist of diverse elements.

When tackling a particular problem and being guided by one's own experience and erudition, individual intellectual activity essentially utilizes collective experience, synthesizes it, and makes certain conclusions on the basis of the appropriate material. At the same time, it is distinguished by creativity, determined by its own experience, erudition, and talent. However, this activity, fruitful as it may be, always bears the stamp of individuality, and when the erudition and the breadth of one's views are inadequate, it bears the stamp of one-sidedness. For this reason, the best results are achieved in those cases when individual creative activity is discussed in a collective, where, under cross-criticism from people who work in other areas, the conclusions and decisions made by an individual person in the quiet of his study are polished, supplemented, expanded or generalized. Naturally, they are disseminated and popularized in gatherings.

In this process, however, the work itself is often stripped of its most idiosyncratic features. In order to preserve them the author should be allowed to utilize only those comments of the collective that he finds acceptable.

The content of speeches in an assembly usually reflects the personalities of individual members of the assembly, but one should bear in mind that those

who speak in assemblies unconsciously take into consideration the nature of the assembly as well as the views of other speakers; consequently, even these speeches reflect the influence of the collective, but the final collective judgment results only from mutually opposing and mutually criticizing opinions and is eventually transformed into a particular decree, decision, or resolution.

As we know, every assembly may proceed in a sluggish or lively manner, depending on the importance of the subject, on the composition of the assembly, and on the interest displayed by its participants.

If there is not total discord of opinion at the assembly and there is a possibility of finding a common meeting-ground, the decisionmaking follows the line of the resultant force of two or more opposing opinions, but the more divergent the views are, the longer it usually takes to produce each result of the discussion, i.e., a decision.

It goes without saying that the skillful guidance of debates is conducive to a greater success and a greater restraint of an assembly, as well as to its progression towards a certain goal. On the other hand, inadequate regulation of debates may rapidly lead to an incident of one kind or another, while the inability to resolve an incident in good time occasionally ruins the assembly altogether. The truth is that any incident that arises from a disagreement in an assembly brings this disagreement into sharp relief, undermining thereby the proper work of the collective, if not bringing it to a complete halt in cases of the total divergence of opinion among various parts of the assembly. Collective productivity also depends to a certain degree on the size of the given collective. Usually, the larger a collective is, the slower its work proceeds, but it provides a better guarantee of satisfaction of the interests of a greater mass of people. However, the size of a gathering should have some limits for the purposes of a successful collective discussion of a subject, and assemblies with more than 500-1,000 participants make it very difficult to guide the discussion properly.

When the subject matter becomes clear to the majority of people and when the process of collective work and exchange of opinions reveals the heart of the matter, then one may speak about the collective elucidation of the subject.

When we talk about a collective analysis of a subject, about a collective critique, a collective view of things, etc., all of these are in essence only particular stages of collective associative activity, which finally assume the form of collective judgment.

The result of the latter is a resolution or a decision, a decree or a sentence approved by the majority of the assembly. These decisions, decrees, or sentences are essentially a summary of the whole discussion and the final conclusion at which the collective arrives as a result of its work, and their formula is proposed by one or another side, or by any participant in the debates.

As to the verbal collective statements in the proper sense of the term, there is essentially only one case when a collective articulates a speech, all parts of which are well coordinated. This occurs in choral singing, but here the coordination is aimed at producing a harmony of sounds, as well as verbal symbolization.

On the other hand, collective statements may be more or less coordinated in the form of exclamations and cries in a crowd; in other cases, however, these statements are made by each person individually, even though their content may be more or less coordinated. Such are statements and demands put forward by a mass of people who have gathered for a particular reason or are united by common interests.

In other cases—and this is what happens most often—a collective expresses its opinion by using one common formula (it makes no difference whether it is written or verbal) transmitted through an elected person. This common formula—a sentence, a resolution or a decision—may be very laconic, but before it is generated it always demands some preliminary work in the form of a collective discussion of a given issue.

The press is a form of collective discussion that involves a whole nation and is therefore broader in the sense of a wider spectrum of people who are attracted to the discussion and wield a skilful pen; besides, due to the nature of the press, the discussion itself is devoid of the specific expressiveness possessed by the live word.

Naturally, even though such discussion may be particularly valuable in the sense of elucidating various aspects of the issue, its very nature makes it even more time-consuming and eliminates the possibility of arriving at a particular decision by direct vote, with the exception of the questionnaire method.

We have already mentioned that numerous authors claimed that only individual activity could be of a creative nature and that collective activity was incapable of creativity or, in any case, could not be compared to individual creativity in terms of its quality.

Thus, many authors consider a crowd to be dumb and its mental processes to be inferior to those of individual persons, for a crowd's decision is allegedly confined within the bounds of mediocrity.

Incidentally, Gabelli illustrates this idea with respect to the process of election—for example, the elections of a university's president:

They elect a person who is least threatening to their self-esteem, who does not stand out and is totally unprepossessing. They often wish to see in this position the most tolerant, amenable and agreeable person of minimal importance—in a word, the least energetic and decisive man of little influence. Thus, the person who is elected is not trusted by either his supporters or his detractors, each of whom is very well aware of the reasons behind his own vote. Occasionally there were even cases when after the elections the supporters of the chosen candidate were less inclined in his favor than his adversaries.

The above views are not exceptional. They are shared by other authors who wrote on the same subject. Max Nordau:

Bring together twenty or thirty Goethes, Kants, Helmholtzs, Shakespeares, Newtons, etc. and ask them to pass a decision or a judgement on some practical issues of the day. Their reasoning might differ radically from the judgements of an ordinary gathering (although I cannot guarantee this), but when it comes to conclusions, they will not differ in any way from the conclusions of an ordinary meeting. Why is that? Because each of the 20 or 30 participants, besides his inherent originality that makes him unique, possesses the inherited characteristics of the species; the latter make him similar not only to his fellow participant but also to any stranger he may have passed in the street. One may say that all normal people possess certain characteristics which represent a certain homogeneous value X, to which, in personalities who are above the average, another value is added which varies depending on the individuality and which should be defined in various ways—<s.g., b, c, d, etc. Once we accept this proposition, we shall have in a gathering of 20 people who are presumably all geniuses 20 X and only 1 b, 1 c, 1 d, etc. There is no doubt that 20 X will prevail over the individual b, c, d, etc. This means that the generic characteristics of the

human kind will prevail over the individual personality and that a worker's cap will cover a doctor's or a philosopher's hat.¹

E. Ferri expresses the same idea in a different way but just as vividly: "A gathering of sensible people may result in an assembly that lacks common sense, just as in chemistry a combination of two gases may produce a liquid."²

According to Sighele,

Collective psychology is full of surprises: hundreds or thousands of people are collectively capable of committing actions that not a single person out of a hundred or a thousand would ever commit individually. These surprises are nearly always of a regrettable nature. A gathering of good people hardly ever produces a fine result, occasionally a mediocre one and in some cases a very poor one.³

While looking for an answer to this question, the author comes to the conclusion that "in a crowd the good qualities of certain individuals are not added to one another but subtracted from one another." This means that the best qualities of some people are overshadowed by the lower qualities of most members of the crowd. It is thus clear that the latter should prevail over the former. Besides, the influence of mutual suggestion in a crowd should have a similar effect. Finally, the level of morality in a crowd is relatively low. The same phenomenon is observed in assemblies with respect to intellectual powers, which are, in the final analysis, inferior to the most moderate talent, even though a crowd that most often behaves like a bloodthirsty beast is in some cases capable of displaying a degree of nobility and heroism.

The unity of a crowd as a social personality is, as we know, a prerequisite for mass actions. This unity is often established independently of membership composition and under the influence of emotional states generated by a particular fact.

However, once this unity is achieved, a crowd as a social personality is no longer the average element of its individual members but possesses some totally distinctive characteristics. This is presumably caused by the fact that higher products of associative activity of various individuals are subtracted from one another, as it were, and follow the middle line, while everything

valuable and heterogeneous is effaced. Conversely, everything which is associated with the lower forms of associative activity, such as impulsiveness and selfishness, and is suppressed in individual persons, is here accumulated, as it were, and overshadows the higher manifestations of associative activity. This is why, according to G. Le Bon,

it is stupidity and not intelligence that is accumulated in a crowd. What they call "the whole world" cannot be more intelligent than Voltaire, but quite the opposite, it is Voltaire who is more intelligent than "the whole world," if the latter phrase means a crowd.⁴

On the other hand, due to their considerable numbers, the members of a crowd develop a greater confidence in their power and authority which no individual person possesses in the same measure. This is associated with the belief in the impunity and infallibility of a collective decision. Then a crowd develops an extraordinarily high susceptibility to suggestion and a predisposition towards contagion.

All of this results in a predisposition towards impulsive behavior and in the inability to assess one's actions critically.

The above leads to what one may recognize as a general phenomenon—on the one hand a crowd displays a diminished intelligence, and on the other an increase in mimico-somatic reflexes. All of this, naturally, refers to a street crowd and under no circumstances may be applicable to all assemblies, particularly those of a special nature.

According to G. Le Bon, "the activity of a crowd is always and everywhere inferior to the activity of an individual person."⁵ This statement, however, is disputable, for even a crowd that is unorganized but united by a particular mood is capable of heroic deeds and actions that embellish the pages of history up to the present. On the other hand, while listening to the reports prepared in advance by specialists, assemblies smooth over certain biases and overstatements, and the usefulness of this process is corroborated by the fact that the specialists-speakers themselves often agree with the suggested corrections and even renounce their own statements.

Much has been said in general about the greater stupidity of a collective in comparison with an individual person without a thorough analysis of all circumstances. First, one should not confuse street crowds united solely by their mood with other collectives. On the other hand, actions of an individual

person often represent the fruit of long searches and efforts based on the synthesis of collective work, whereas a crowd is for the most part forced to make decisions that cannot be postponed for any length of time. At the same time, a crowd's decision is the resultant force or, more precisely, the arithmetical mean, of the opinions of many people who make up the crowd, therefore the nature of the decision is determined by the composition of the crowd, so that when it is dominated by mediocrity its opinion will also be mediocre, but it has not at all been proved that a crowd of intelligent people will produce an opinion that will be inferior to those expressed under the same conditions by each individual separately.

Incidentally, Bentham points out the following difference between the decrees of political societies of a permanent nature and identical societies of an accidental and short-term nature: the former find it easier to arrive at decisions that are in the true interests of their members than do the latter ones.

According to G. Le Bon, even parliamentary assemblies represent a dual danger of "unproductive waste of finances and a progressive restriction of individual freedom." However, this statement, too, is in need of correction. It is indisputable that parliaments regulate finances better than a minister's individual power. Public control alone immediately cuts off from the national budget those expenditures that cannot be justified from the standpoint of public interest. Even though parliaments do not spare money for needs ignored by the crown, they also discover the sources of payment for new expenditures regarded as productive.

As to the right to personal freedom, it is a debatable question who promulgates more edicts that restrict personal freedom—a parliament or an administrator who on busy days churns out a cornucopia of mandatory decrees that are for the most part completely inane.

Nor can one accept without reservations Le Bon's statement regarding the one-sidedness of opinions voiced by a crowd, even if it is a parliamentary crowd. If in the interests of its elected representatives a party votes unanimously in compliance with party discipline, one should not forget that prior to voting in parliament the issue was thoroughly discussed at party meetings where many opposing opinions intersected and where the adopted decision reflected the median line of behavior, which happened to be the most expedient one in the given situation of party politics. This predetermined voting explains the indifferent attitude of the audience to the

speeches of lesser known speakers, whereas the speeches of the main party leaders always appear to be significant and are received with due attention.

One should note that until now a crowd has been described as a more or less independent community of people, whereas an active crowd cannot exist without its leaders; the crowd and the leaders mutually complement and influence each other. The leader is guided by the mood of the crowd and by the effect of his speech on the crowd, whereas the crowd listens to its leader and obeys him in its actions.

Any assembly whatsoever represents a combination of individual intellect and the collective one. Undoubtedly, not everybody can lead a crowd, and then again no two crowds are alike. Not every leader of a parliamentary party can lead a street crowd and vice versa.

For this reason, characterization of the crowd by various psychologists has suffered from a one-sidedness of one kind or another, since they failed to take into account, as they should have, a close interrelationship between the leader and the crowd, to which subject our Mikhailovskii devoted some interesting pages and which was subsequently discussed by such prominent Western sociologists as, for example, Tarde and some others.

As for collective creativity in the narrow sense of the word, we do not share the accepted view that one cannot compare collective work and individual work in terms of their quality. Sighele, G. Le Bon, and others claim that with respect to the quality of work a crowd can never reach even the average level of individual work. This opinion, however, should be regarded as extremely one-sided. Although it is true that initiative and inventions are more often generated by an individual rather than by a group, one should not forget that an individual genius himself is a product of society, one who personifies a synthesis of collective work and is inspired by the crowd, that the implementation of an initiative requires participation of the crowd, that inventions are produced for the crowd and are totally meaningless without it, and that an individual impulse is often completely powerless unless it receives the support and the enthusiasm of the crowd. It is clear that the individual without a crowd, or a composite personality, is nothing—just as a crowd without the individual is nothing. It is only the joint effort of an individual and a crowd, or a composite personality, that ensures progress. Under these circumstances, each side is destined to play its own part in the progress of humanity.

In order to express itself, the individual mind needs the assistance of the collective, for it creates for the collective, it invents what is useful for and needed by the collective, it gathers the material that has been prepared before it by the collective, it synthesizes the facts that have already been prepared by the collective, it carries out a particular task in the interests of the collective and follows all the instructions provided by the collective. On the other hand, the collective mind makes use of the products of the individual mind, while smoothing over all the sharp angles in its creative activity, eliminating the superfluous details and rough spots that may be of some significance to an individual but of no general importance. It generalizes the activity of the individual mind and renders it universally acceptable and meaningful. It is therefore completely wrong to claim that only an individual mind can create properly, whereas the mind of a crowd or of an assembly is incapable of creativity or at least of the same measure of creativity as an individual mind, and that the latter should in this respect be regarded superior to the collective mind. I am more inclined to the idea that better creative activity is achieved through joint work in which both the individual mind and the collective mind participate and where the former takes upon itself to prepare and process the material, while the latter criticizes, supplements, and generalizes it. Do we not know the cases when even creations of geniuses were corrected as a result of criticism and evaluation meted out by the collective that was supposed to judge it? And if this is the case, it is obvious that the collective mind will always play its role in appropriately correcting and smoothing over the shortcomings and one-sidedness of individual work.

Collective or public opinion is the result of discussion of a particular fact or event by individual members of a collective or by its various circles, and in large collectives it is individual parties and societies that are involved in this discussion disseminated by word of mouth and in an educated society by the press, which generally plays an enormous role in all collectives without exception. In cases of particularly important events in countries that enjoy political freedoms, a certain role in shaping the public opinion is played by grassroots meetings and partly by theatres, cinemas, fair-booths, etc.

In general, any task which is delegated to an assembly is in any case performed as a collective exchange of opinions that are subsequently synthesized in the form of a conclusion by the collective mind, or more often by an individual mind which, in this case, reflects the effort of the collective mind.

In general, as I have already said, one should not overlook the fact that individual creativity makes use of the accumulated riches of the collective, that every discovery is the fruit of the accumulated facts and that even a generalization of these facts, which is needed for making a new discovery, is often prepared to such an extent that the only thing left to do is to summarize everything previously accomplished; the discovery itself is in the air, as it were, and an individual mind is nothing other than a vivid manifestation of this generalization.

When posing the question of how the process of discovery occurs, Tarde says, among other things, the following:

First we notice the emergence of all kinds of inventions and mutually opposing ideas and fantasies that keep appearing and disappearing until the arrival of lucid formulae, and some useful machines make us forget everything else and from then on serve as the foundation on which further improvements are built for future development. This process therefore approximates collective thinking without a collective brain, which is possible due to the solidarity (as a consequence of imitation) of many brains of inventors and scientists who exchange information about their discoveries.⁶

However, we also know some forms of true collective creativity. They include language, common law, beliefs, myths and, to some extent, folk epics. Even artistic creativity can be pursued by a collective, the evidence of which one can see on the walls of some old Russian churches that were painted by groups of artists, such as "Rublev and fellow-painters"; some works of literature, too, were created collectively. For example, a well-known Russian literary work, *Kuzma Prutkov*, belongs here. It is thus indisputable that to deny the role of the collective in creative activity means to go from one extreme to the other.

In the final analysis, creativity in human history is achieved by the joint efforts of a collective and of individual persons. The lot of the former is for the most part to accumulate experience and facts and to bring together the results of individual experience, in the interests of the whole entity, whereas the lot of the latter is to synthesize these facts from a personal point of view. It is clear that a creative act is not possible without the first process, nor is it possible to complete a creative act without the second process. But both

together, through mutual enrichment, lead to creativity, progress, and improvement.

Here we shall discuss only the most important collective invention—the human language. P. Collet says that the emergence of language necessarily presupposes an already existing community and is therefore explained as an exclusively social fact:

Modern linguistics.. deduces the formation of language from gestures, cries and exclamations through which the savage expressed and communicated to others his sensations and sentiments; then also from the consonances (onomatopoeia) with which he tried to imitate cries of animals and sounds of nature, which is still done by the most backward savages and by our own children in the first months of their lives.. Renan and Max Muller deduce the roots of the words from these imitative consonances - the primitive monosyllabic sounds that formed the first words by joining together in various combinations. Thus, language must have been a spontaneous and unconscious product of the collective, rather than the individual, mind.

At the same time, thought could not have emerged, let alone develop, without this indispensable tool.

The origin of writing was also of an exclusively social nature. Initially ideographic and subsequently phonetic writing signifies a transition from animal existence to intellectual life which, together with language, brought about the most important factor of civilization—the very condition that made the evolution of tradition and upbringing possible and that enabled each generation to make use of the progress achieved by the previous generations.⁷

In his paper, "Sociology and Cognitive Theory" (*New Ideas in Sociology*, issue 2), E. Durkheim derives the origins of logical judgment and, consequently, of thought itself from social life. He begins by pointing out that the content of logical thought consists of general notions (concepts) and poses the question, "What enables society to participate in concept formation?"

He claims that as long as one perceives a concept as a general idea this question remains insoluble, for an individual is capable of comparing his own

perceptions and images and of distilling something common from them. The author defines concepts in a different manner. Sense perceptions are fluid: they may appear and disappear while undergoing changes each time; these transformations may result not only from a change in the perceived object but also from a change in the perception itself; whereas a general notion remains "outside time and evolution," being isolated from these fluctuations, as it were. A general idea is a mode of thinking, "which is fixed and crystallized at every point in time." The system of ideas is already provided with one's mother tongue, for each word signifies a concept, while the language itself is more or less fixed and if it undergoes any changes at all their pace is slow.

Any logical notion, even if it is "not universal, it is in any case capable of becoming such." An idea is not my exclusive possession—it can be generalized, which is impossible to achieve with "perceptions," as "any intellectual communication between people consists precisely of exchanging concepts." Thus, by its very nature, a concept is not something individual, but is by origin a result of the collectivity, just as language itself, which consists of concepts, is a result of that same collectivity. In the final analysis, ideas are nothing other than objects of collective experience; they are "collective representations" in a form that has been established by language, and they are shared by the entire group.

Thus, logical judgment becomes essentially possible when an individual acquires in addition to his individual ideas the concepts that are shared by many intellects and are of an impersonal and stable nature. In the end, impersonal thought evolved from collective thought.

Furthermore, the most recent neopositivist school of sociology recognizes that not only concepts, or ideas, but also the principal intellectual categories, are nothing other than products of society, which is perceived not as an "imagined and nominal entity but as a system of actual forces." Such intellectual categories as the concepts of time, space, genus, number, cause, substance, personality, etc., which reflect the most common attributes of objects, are in the end a product of society. According to E. Durkheim, one encounters the most fundamental of these categories when analyzing religious beliefs. "Religion is an essentially social phenomenon." On the other hand, one cannot represent time "without taking into consideration the devices we use to divide, measure and express it through particular signs." Time is impossible to understand unless "one is able to distinguish its various heterogeneous elements." And so, according to Durkheim, a personal, i.e.,

individual, experience is inadequate for "generating the concept or category of time, as time is an abstract and impersonal frame that encloses not only our own individual existence but also that of the whole humanity." But if time is not only my time, but the time that belongs to all people as something objectively conceived, it is apparent that it can only be a result of collective experience. In reality, the order of phenomena distribution in time is generated by social life. Days, weeks, months, years, etc., correspond to the periodicity of rituals, festivals, and public ceremonies. According to the author, perceptions and representations of the sequence of events are a result of individual experience, whereas the category of time shared by the whole group is a national creation. The same holds true for space and causality. With respect to space, one should first of all renounce Kant's view of it as something pure and absolutely homogeneous that does not serve any purpose. As a result of individual experience, spatial representation consists of the primary distribution of findings provided by this experience, so that some of them are located to the right and others to the left, some are up and others are down, still others are situated in the north, south, west, east and so on—similar to the distribution of time according to dates. It is the identical division of space that proves the social origins of these differences. There also exist more direct indications pointing to the social origins of spatial distributions. Some Australian and American societies are of a circular shape and their space is divided in the same way as the tribe's camping ground. Thus, the nation of Zuni (Pueblo) consists of seven groups and their space consists of seven parts of the world closely connected with the corresponding group. And since each tribal group has its own symbolic color, each part of the world has the same color. It is thus clear that spatial organization is directly dependent on social organization, and even the distinction between right and left is associated with religious beliefs.

The concepts of genus, force, personality, and action are also of the same social origin.

Even logic with its concepts of identity and contradiction is apparently of social origin, for we know that in mythology a part may equal the whole, or a certain object may exist before the beginning of time—in other words, we encounter irreconcilable logical contradictions and "are constantly faced with the problem of existence that possesses the most contradictory attributes, such as unity and multiplicity, matter and spirit, ability to be divided to infinity without losing integrity and so on." Therefore, the principle of

identity is far from eternal and depends at least in part on historical—and consequently on social—facts. Clearly, the claim that categories are not derived from experience but precede it and make it possible, as well as Spencer's view of categories as originating in individual experience are equally unacceptable. If empiricism had some foundation, categories—these most universal concepts—would have to part with their common characteristics that determine their application to everything real and at the same time establish their independence of each individual experience, whereas perceptions and images are both "individual and subjective":

These are the two types of knowledge that represent two poles of intellect, as it were. To deduce reason from experience under these conditions means to make it disappear, for to arrive at such a conclusion means to reduce the universality and necessity that characterize reason to mere appearances and illusions that may be convenient in practice yet lack any real foundation; it also means to recognize the objective reality of logical life whose orderliness and organization are precisely the main functions of categories.

On the other hand, to view categories as a product of a particular type of mind perceived as an "emanation," as it were, of a supreme or divine intellect means to adopt a hypothesis that cannot be scientifically corroborated. Besides, the categories of human thought have never been set in a single unchangeable form. They have been continuously created, destroyed and re-created; they have been changing depending on time and place. Divine intellect, however, is endowed with the opposite quality.⁸

In a word, "knowledge consists of two types of mutually irreducible elements." Empirical knowledge is generated under the direct influence of an object and, consequently, here we are dealing exclusively with individual states, whereas categories are a product of collective experience.

We believe that this issue appears much more obscure when the object of investigation is discussed from a subjective point of view, for in this case one has to deal with collective notions, impersonal thought, etc. The issue is significantly simplified when one discusses it from the objective point of view that perceives language as a tool of logical thought. And indeed, we can definitely say that judgment is impossible without language, for in a

subjective world notions in the form of judgments do not and cannot exist. There can only exist an associative interaction of notions based on contiguity, similitude, belonging, etc., whereas a connection that we call a judgment becomes possible only in the presence of verbal and other symbolic combinations. However, words, gestures, and written signs represent, as we know, symbols that are a product of collective experience—hence, judgment, too, is nothing other than a similar product of collective experience. It is thus clear that logic evolved together with language, and when language is at a primitive stage of its development, logical thinking cannot achieve its full potential; this is corroborated by primeval mythology.

When pursuing logical thinking, a person goes beyond his sensual experience, enters the realm of abstraction and acquires an insight into the true nature of the outside world, but he is able to do this only with the help of a symbol as a product of collectivity and as the first step towards abstraction characterized by impersonality. It is only through the mediation of collective experience that a given symbol acquires a certain stability and becomes an impersonal sign for denoting particular objects, as well as the first generalization of all analogous objects of many individual experiences. Obviously, it is only through this kind of generalization that people reached the level of intellectual interaction, which is therefore a product of collectivity.

It is beyond any doubt that without the collective experience that resulted in the symbolic identification of various phenomena of the external and the inner world, man would have remained in the position of an inferior being, whose inner world would have been limited to images and sense experiences, which would have precluded his communication with others, as well as have prevented him from creating generalizations in the form of concepts. It was only through language that man acquired the ability to form concepts that belonged to all and whose meaning was verified by all, and at the same time every individual acquired the ability to engage in further juxtapositions of concepts and new generalizations and conclusions, which was destined to result in the emergence of logical discourse. It is only under conditions of collectivity that the universality of certain symbols-concepts is generated, for each individual mind uses its own personal experience to verify to a certain extent the given symbols-concepts. At the same time, a given symbol is correlated in a collective with all the other symbols through a mutual exchange of these symbols, so that symbols in general and concepts in

particular acquire a certain significance, impersonality, and stability in the furnace of joint collective work, for these symbols become a creation of the whole society, rather than of a single individual. It is clear from the above that our logic, too, is a product of collectivity, and as the latter was evolving, so was the logic whose evolution is still going on; this does not mean, however, that the individual plays no part in this process.

As we know, categories have been and are recognized as absolutely meaningful and invariable forms of any judgment. This may indicate that from the very beginning of intellectual exchange between those who existed and worked in a human collective there was total and unconditional unanimity regarding categories, and this unanimity became part of the character, so to say, of every individual who grew up and was brought up in a collective. Durkheim is right in recognizing that not only are categories of social origin but that they also signify various aspects of social existence:

Thus, the category of clan was initially inseparable from the notion of the human soul; the category of time is based on the rhythm of communal life; the category of space was modelled on the space occupied by the group; collective force served as the prototype for the notion of active force—an essential element of the category of causality.

We believe, however, that categories, as well as concepts, are products of both individual and collective activities. It is precisely here that one observes complete accord between the results of personal experience and those of collective experience, which developed and consolidated them as the immutable forms of human intellectual exchange. An individual person exists in space, and the very structure of his organs (semicircular canals, etc.) enables him to orient himself in it. He exists in time, being endowed with the faculty of perceiving a sequence of stimulations (acoustic ones and others), and is capable of measuring the sequence of external phenomena with the help of his breathing rate, the number of completed movements, such as steps, etc. He acts, being the cause of changes in the world, using his extremities as if they were tools like levers; he observes similar objects reproduced with a particular ease when encountering other similar objects, and through juxtaposition he establishes kinship between them; he perceives a group of objects that are interconnected and that form an integrated whole, etc. However, even though all these products of individual experience are

based on the organic foundation of the perceiving apparatus, they would have remained underdeveloped and discordant in their various parts without collective experience. An individual whose perceiving organs possess the organic structure that enables them to form categories would not have been able to create them as universally meaningful phenomena without a mutual exchange in this respect with other individuals and without the collective experience that alone is capable of establishing categories as an indisputable basis of any judgment, which, in turn, results from the mutual exchange of personal experience between individual persons.

Categories reflect the collective mind, but the latter became part of them on the basis of individual intellectual activity. It is essentially the individual mind that records the succession of the seasons, of day and night, respiratory movements, etc. But only the collective mind could make this succession universally meaningful by establishing further necessary subdivisions of this succession in the form of particular periods. In the same manner, through experience, the individual mind determines spatial dimensions with the help of its movements for which purpose it uses the dimensions of its body or movements, such as, for example, the thickness of one's fingers, *vershok* (a Russian equivalent of an inch), a quarter, a foot, *arshin* (one step), etc., as well as the body movements to one or the other side, forward, backward, etc. However, these definitions, too, acquire universal significance only through collective experience, which correlates all the necessary results of individual experience, supplements them with the necessary details, and creates thereby universal categories, distilling them, as it were, from the sphere of individual experience because of their universal significance. Being a product of individual-collective experience, general categories nevertheless preserve their theoretical value, for they cannot but reflect the very nature of things.

We already know that abstract thinking is possible only with language, and since language is a collective creation it is clear that categories, like all general notions, could not have been generated without collective experience. This, however, does not at all exclude the possibility that with the emergence of language one or several individuals may have initially created symbols for some general notions in response to a practical need, after which these symbols became universally popular through their generalization. One can provide some proof of this, which amounts to experimental data, from the sphere of inventiveness of the gesticulatory language of the deaf-mute, which can be verified by any observer.

We are dealing with a group of deaf-mute children aged ten to twelve. In the evening of 27 June, I came into their dormitory together with Dr. Sh. The children were in their beds though not yet asleep. They were gesticulating with one another in a very lively manner, as all deaf-mute children of this age do. Incidentally, every adult is identified in their language by a gesture that indicates a particular characteristic. For example, a lady with full cheeks is described by an index finger pointing to a cheek; a moustached gentleman is indicated by a movement of both hands from under the nose, as if smoothing out one's moustache; a doctor who wears glasses is described by forming a circle with the fingers of the right hand and placing it in front of the right eye, etc. I was warned that it would be extremely interesting to see what gesture the children would invent specially for me, as I was a new person they were seeing for the first time in their life. So what did they do? Within a few seconds one child used his right hand to make a gesture showing my beard descending down my chest and indicating thereby the characteristic external attribute of my face, which was sufficient for the gesture to be adopted by the children for referring to my personality among themselves.

Thus, here an appropriate gesture made by one child and rapidly adopted by other children is generalized through collective exchange and is transformed into a universally accepted gesture that has become a communal possession understood by one and all. Do we not perceive the same in the oral language of the hearing population? Someone attaches an appropriate term to a new object and this term is accepted by others, entering general usage. Thus, not so long ago such words as "hooligan" and others entered general usage to be joined somewhat later, due to their popularization by the press, by such new scientific terms as "X-rays" introduced by Roentgen and "reflexology" suggested by me as the name for a new discipline I was developing that focused on the objective study of human personality.

It is evident from the above that one cannot exclude creative participation of an individual person. There is even more reason to recognize this with respect to the formation of categories that depend in certain measure on the organic conditions of functioning of each individual's organs of perception.

All of the above, however, does not diminish the creative role of society, and E. Durkheim is a thousand times right when he says that

it is precisely the society that possesses a creative power that no other creature possesses. In reality all creation, except the mystical operation

that eludes intellect and science, is a product of synthesis, and if the syntheses of separate representations occurring in the depth of each individual consciousness can create something new, then how much more effective are the vast syntheses of a multitude of individual consciousnesses that are societies. Society is the most powerful focus of physical and moral forces that exists in the world.⁹

Notes

¹ Max Nordau, chapter VII, quoted from Sighele.

² E. Ferri, *Nouveaux Horizons*, 2nd edition, p. 484.

³ I.e., p. 54.

⁴ I.e., p. 168.

⁵ I.e., p. 322.

⁶ Tarde, *Laws of Imitation*, p. 153.

⁷ P. Collet, *Elements de sociologie*, Paris, Riviere, 1908. Quoted from the Russian publication *New Ideas in Sociology*, Issue 2, pp. 84-85.

⁸ E. Durkheim, *Sociology and Cognitive Theory*. Quoted from the Russian publication *New Ideas in Sociology*, 1914, Issue 2, p. 33. See also Windelband, *Zeitschrift f Volkerpsychologie* VIII, p. 166 ff.; Steinthal, *ibid.*, p. 178 ff.

⁹ E. Durkheim, *Sociology and Cognitive Theory*. Quoted from the Russian publication *New Ideas in Sociology*, 1914, Issue 2, pp. 65-66.

Coordinated Collective Actions

Much has already been said about a crowd's actions when we were discussing the characteristics of the crowd and hereditary organic reflexes. In those cases, however, we described mostly the actions of a street crowd, which is a chance gathering of people united by the same mood. It is therefore natural that this kind of crowd will always be impulsive and in some cases extremely cruel, while in others, noble, depending on its leaders and on the cause of the gathering.

At this point, we would like to draw your attention to another aspect of collective actions when, due to the coordinated efforts of many individuals, they often achieve the results that are unfeasible for individual persons. In this case, the joint result of collective actions depends on the greater or lesser coordination of actions and movements of the members who make up a given collective.

Examples of such coordinated collective actions are an orchestra, a theatre, joint collective work of drafting resolutions, dances, acrobatic and gymnastic exercises, movements of troops, actions of a crowd under someone's leadership, etc.

Owing to collective activity, modern civilized society generates creative work that assists the appropriately directed collective efforts in producing some marvelous creations of architecture, that before our very eyes works wonders of modern technology in the form of various structures of public and private nature, and that attracts collective labor to mines, factories, plants and to any undertaking that is beyond the powers of an individual person.

It has been known for a long time that there is a direct relationship between any kind of work and mood. This has been proved with precise figures in the case of individual work.

This is no less true with respect to collective work. Here, too, general mood affects the intensity of labor in a most dramatic manner.

There are grounds to believe that musical sounds, too, can influence collective work in terms of its increase or decrease, as is the case with individual work. This underlies the significance of songs and music for performing certain forms of collective work.

When discussing the relationship between the successfulness of the work itself and its stimuli, one might compare the productivity of collective effort undertaken in the interests of a common cause and the productivity of the same work effort accomplished in one's own interests. Experience shows that comparison should favor the latter, and the reason for this is that the satisfaction of one's own interests provides a stronger stimulus for mimico-somatic (emotive) reflexes and raises the energy level higher than the stimulation provided by the synthetic complex of common interests if the person is not sufficiently pervaded by sociality. Work for the sake of the common interest often weakens the desire to do more than one's neighbor and thus weakens individual work effort. Uniform evaluation of work and uniform distribution of the products of work among all frequently eliminates competition, an important stimulus for work in general. It goes without saying that establishing universal control over distribution not only diverts masses of people from productive labor but also, by requiring a special kind of guardianship over the personality, oppresses it and simultaneously inhibits its energy.

At the same time, improvement of production processes is inhibited by the establishment of mandatory social conditions of work in a certain form. Elimination of individual ways of production and procurement weakens private initiative and personal interest in improving one's work. On the other hand, restriction of needs to a universal standard leads to the suppression of impulses to activity and inventiveness. Finally, the inability to use the products of one's labor at one's own discretion suppresses the impulse to economize and accumulate the products of labor in the interests of the future. It is thus apparent that, while recognizing the particularly important value for society of work on collective principles, one has to bear in mind that for the sake of greater success in improving the speed of production and its quality,

as well as in utilizing the products more economically, individuals should not be deprived of encouragement measures such as bonuses and other forms of satisfaction of one's personal interests.

The actions of a collective, however, are expressed not only through physical labor in the true sense of the word.

Thus, the creation of spiritual and cultural institutions should also be regarded as collective actions that pursue a particular goal.

One should not forget that stability of energy level in a collective depends on the circumstances of its manifestation. Thus, in a struggle, a passive defense is always exhausting, as it does not generate energy, provide training or skill, nor does it temper the fighting elements in the struggle; the enemy's unexpected blows produce the oppressive effect of anxiety and perpetual danger, leading to a fruitless expenditure of energy consumed by the anticipation of this danger. The situation is quite different in the case of so-called active defense or attack, in which fighters become experienced and where maximum energy is generated with each success, which invariably stimulates a sthenic reaction.

One should also bear in mind that it is not only collective undertakings of an active nature that can be successful but that in certain cases passive measures may prove no less effective than active ones. Let us take a strike as an example of a collective work stoppage. In a certain sense it is often more effective than any other active measures.

The planned nature of collective work presupposes a degree of organized behavior in the collective itself, which is achieved with the help of an executive bureau or authority elected by the collective.

It is apparent that the planned nature of collective activity is directly related to the degree of organized behavior of the collective itself, while this latter organization is, in turn, a direct relationship with what is represented by the executive bureau or authority and with the degree of ability of these organs to exercise their rights and achieve their goals.

It goes without saying that coordination of movements not only determines the productivity of collective actions but also provides them with power that is beyond the reach of individual persons. If you remove this coordination you will be left with the chaotic movements that hold no hope of overcoming considerable obstacles.

Thus, every army is based on the coordination of its movements learned throughout the period of preliminary discipline training.

One may say that the energy of a collective is directly proportional to the coordination among its individual members established through organized behavior, and inversely proportional to their disunity.

On the other hand, collectives that lack the appropriate organization more often display a destructive, rather than creative, force, as exemplified by the unrestrained actions of a crowd. This is so mostly due to the particular importance acquired by the demagogical element represented by the leaders.

Social aspirations and social undertakings, as well as collective work, are directly related to the mood of the community, while the nature of the actions of the given community depends on the moral qualities and the level of advancement of its popular masses. Thus, an elevated mood is associated with the revival of social activities that may occasionally assume an expansive and even aggressive nature; whereas a low mood is associated with its passivity, lack of vitality, and a limited scope of action.

Literature—this mirror of public moods—also provides a vivid manifestation of this rule. In some cases, when the mood is low, one is struck by the public whining and inaction; under other circumstances the aspirations of the national heroes and of society itself acquire a wide scope.

With respect to the Russian reality one could observe an animated mood during the 1860s, while the period of reaction during the 1870-1880s was associated with a depressed mood, when public whining and *Weltschmerz* were in vogue, particularly among the young, which forced the healthy forces of society to fight against it.

Just as the mood of individual persons is reflected in all spheres of their activity and is expressed through its suppression and through reactions of a negative or defensive nature, so the mood of a collective is reflected in its activity. Thus a low public mood is reflected in visual art and its colors, in literature, in collective activity, in the sluggishness of external reaction, and in the negative direction of foreign policies. In a word, everything acquires a coloring that corresponds to the mood.

On the other hand, we know that victory, by raising the mood in troops, increases their combat energy, while a defeat, on the contrary, paralyses it. That is why it is so much more difficult to enter a battle with troops that have once been defeated than to do this with victorious troops.

It seems, however, that nothing reflects the nature of the popular mood better than a nation's creative activity and, incidentally, its mythology and beliefs.

During the periods of suppression of national spirit, religion and mysticism have always acquired a particular vitality among the popular masses. This was the case in pre-revolutionary France, and also here in Russia, when the first open revolutionary act on the 9th of January was even led by the priest Gapon, with a cross in his hands and gonfalons and icons carried in the front rows of the demonstration.

It is quite natural that when all political movements are suppressed, the masses reveal a proclivity towards mysticism, which provides a fertile soil for religious fervor. Thus, during the eighteenth century in France, this was manifested in Swedenborg's theosophy, Saint-Madard phenomena, and Cagliostro's mystical charlatanism.

In Russia, this was manifested in the activities of various sects, discoveries of a great number of new relics of newly proclaimed saints, and in recent times, in the rapid dissemination of occultism and "Rasputinism," which was nourished by religious mysticism as well as by the unsatisfied sexual needs of aristocratic ladies.

Revolution does not immediately sweep mysticism away, and besides, during a revolutionary period a certain segment of society that has been depressed by the events searches for the solution to its problems in religious beliefs. In addition, as reasoning ability declines during the periods of popular movements, the impressionability of the crowd, on the contrary, increases, which nourishes the tendency towards mysticism expressed in the heightened attraction to everything supernatural and miraculous.

At the very height of the French Revolution, the mysticism of the popular masses did not disappear but was merely transformed. This is what we read on this subject in the book by Cabanis and L. Nass (*The Revolutionary Neurosis*, p. 328):

Under the influence of the popular festivals that replaced church ceremonies there was a considerable spread of mysticism among the masses. On memorial days, when the virtues of a particular hero were publicly glorified or the triumph of an abstract principle was celebrated, a kind of prophetic exaltation transformed these people, who aspired to the new ideals with all their heart and soul. Only newly-born religions are capable of generating such lofty impulses.

We must admit that, in the process of transition to the ceremonies of the cult of reason and then of the supreme being, the terrorized society was at the time obsessed with a true religious delirium that was very close to madness and that kept growing with the successes of Robespierre's dictatorship.

In Russia, during the Revolution we observed huge crowds of worshippers in the temples and even greater crowds flooding the streets during religious processions—all of this happening at the time when some of the zealous revolutionaries were preaching condemnation of religion in the crudest manner from the pulpit and were publicly denying the existence of the divinity and mocking popular beliefs.

Part 2

Editor 's Foreword—1999

It has been five years since the publication of the first English translation of Part 1 of V.M. Bekhterev's *Collective Reflexology* (Strickland, 1994), an interval during which the frustration involved in a job half-completed was counterbalanced by satisfaction from seeing that this great Russian scientist's views were at last attracting positive interest. This is not to mention the sheer pleasure gained from exploring his ideas in other contexts (e.g., Strickland, 1997), and consequently heightened respect for the achievements of his far-ranging intellect. The review by Duane Rumbaugh (1996) of the first translation of Part 1 was, as far as I am aware, the first informed treatment of Bekhterev's social psychological contribution in English beyond those of Jaan Valsiner (Valsiner, 1994; Van der Veer, R. & Valsiner, J., 1991) and ourselves, and it produced an immediate reduction in our feelings of isolation. We have little doubt that Valsiner's and Rumbaugh's commentaries made it easier for Tzvetanka Dobрева-Martinova and me to gain support for and to publish during this interval a translation of *Suggestion and Its Role in Social Life* (Bekhterev, 1908/1998), also with Transaction Publishers.

During this half decade, colleagues and I have been discussing Bekhterev's contributions before a variety of sympathetic audiences (Dobрева-Martinova and Strickland, 1995; 1996; Strickland, 1997; Strickland and Dobрева-Martinova, 1996). These efforts also left time to appreciate the enormous gaps in our picture of this great scientist. Perhaps especially, we attempted to become more comfortable with the scope of his central concept, "reflex," than we had been on the basis of those social-psychological (or, in his terms, "biosociological") writings on which we had concentrated in Part 1.

An early firsthand introduction to and description in English of Bekhterev's scientific extension of the reflex concept was provided by the latter's younger

colleague, Prof. A. Gerver (1933), who, toward the end of his commentary on *General Principles of Human Reflexology*, stated that Bekhterev's

scientific activity is displayed with peculiar vividness in his creation of a new and autonomous science which he calls "reflexology." According to Bekhterev's definition, reflexology is a branch of science which studies the human personality from the strictly objective biosocial standpoint. Reflexology, as Bekhterev teaches, investigates all the manifestations of the "so-called psychic activity," or, in other words, "the spiritual sphere," from the objective standpoint and confines itself to the external peculiarities of the activity of man: his facial expressions, his gestures, voice and speech, as a coherent integration of signs, in correlation with the exciting external influences—physical, biological, and, above all, social—but also with the internal, regardless of whether either of these two types of influence is referable to the present or the past. From the standpoint of reflexology, man is not only a living organism but a "bio-social being" acting in dependence not only on the natural but also on the social environment. The divisions of the earlier psychology dealing with intellect, feeling, memory, will and attention Bekhterev replaces in reflexology by divisions into higher and personal reflexes, mimico-somatic reflexes, reflexes of concentration, of reproduction, etc. (Gerver, 1933, pp. 11-12)

The reader may startle at the presumption involved in broadening the notion of "reflex" with its traditional, restricted, typically Pavlovian construal, to facial expressions and gestures, as mentioned above, to the realm of mob action and inter-army conflict, as discussed in Part 1 of *Collective Reflexology*, and to major social movements, as evidenced in Part 2. Hence, it seems useful to review the notion here, extending the introductory commentary of Jaan Valsiner to Part 1. Some of the following paragraphs are saturated with quotes from other of Bekhterev's works, particularly from the rarely noted *General Principles of Human Reflexology* (Bekhterev, 1933), to indicate both the extent and insistent nature of Bekhterev's conviction about the scope of his reflex formulation.

The concept "reflex," referring to a "reflection" and transformation of energy, was seen by Bekhterev as applicable to every variety and form of life

—from one-celled organisms reacting to bright lights to complex human groups under conditions of stress and panic, and to societal upheaval.

In his *General Principles of Human Reflexology* (1933), Bekhterev painstakingly traced the route by which the reflex notion could be extended:

As we have reason to suppose that the existence of all living beings depends on the irritability which characterizes their protoplasm, an irritability which is an energetic manifestation at the basis of which processes of ionization lie, we must keep in mind that the nervous current which develops in the nervous system of the higher animals and man is only a derived phenomenon which is a qualified kind of irritability.

But if the nervous current is a qualified kind of irritability of cellular protoplasm, the question arises: What is the source of that energy which is manifested in the form of nervous current as a special kind of cell irritability?

It is obvious that the primary source of this energy lies outside the organism, and is to be found in food and those stimuli which exert their influence on the organism through its internal surfaces, the cutaneous and mucous coats, the movements of the limbs and the whole body in space (muscular stimuli, semicircular canals) and through special receptive organs (eye, ear, nose, and mouth).

As early as 1896,¹ advanced the hypothesis that our receptive organs are merely transformers of external energies, so that these energies produce the nervous current when they impinge on the receptive organs. The nervous current must be regarded as a manifestation of one kind of the universal energy. (Bekhterev, 1933, p. 174)

If one were to accept Bekhterev's view of the generality of the reflex, i.e., of an event sequence involving "energy in—energy transformed—energy reflected" in the lives of both lower and higher organisms, then one could also, by extension of the laws applicable to energy, accept his argument that the behavior of individuals and groups becomes explainable in the same terms that applied to external energy and its transfer (e.g., in mechanics) in the universe-at-large. Thus, with this concession, one could reasonably explain the activities of persons and groups in terms of the law of

conservation of energy, the law of gravity, of equal and opposite reactions, of inertia, of relativity, etc. This explanatory effort reaches its zenith in *Collective Reflexology* (with groups and collectives in Part 1, with larger social units in Part 2), but, at the level of the person, it is in the rarely cited (but here frequently quoted) *General Principles of Human Reflexology* (Bekhterev, 1933) that the derivation is at its most detailed in English.

Inhibited Reflexes and the Experience of "Mind"

If the social events of general interest to psychologists and sociologists are acknowledged to be in some manner the products of thinking minds, one may wonder how Bekhterev proposed to get from reflected energy to mental phenomena? This question follows from the common but erroneous assumption that such a materialistic scheme as reflexology could offer no place to the "spiritual" event. Bekhterev seems to have evaded the mind-body problem by asserting that it simply could not pose a challenge for an energetic/reflexological analysis. Psychic or conscious experience was but a qualitatively different variety of reflection. The private conscious event was itself generated by the *inhibition* of a behavioral reflex, but one property which made it amenable to objective scientific analysis was that it was publicly *accountable*: one could observe and tell another, or otherwise reveal to him/her, what "was on one's mind." As a consequence of the communicability of accounts, scientists could analyze them, not necessarily assuming their validity, just as they could any other data. The scientist could also tap unconscious psychic activity through other associative reflex routes. Bekhterev had first outlined a model of mind with special reference to issues of the conscious and the unconscious (for him, the "personal sphere" and the "general sphere") in *Suggestion and Its Role in Social Life* (1908/1998). These concepts were modified as he developed his reflexological doctrine, and in his *General Principles of Human Reflexology*, after a discussion of creativity during sleep, Bekhterev (1933) said:

In the subjective world of our thought processes, we are, as you know, by no means conscious of everything, but rather of only the final result of our thinking, while the process itself by which we come to the given conclusion usually takes place to a great extent in the sphere of the unconscious or unaccountable processes.

Taking into consideration certain processes of the unconscious, i.e., direct knowledge in the form of intuition, and also unconscious "intuitive" creativeness, we must, on the contrary, come to the conclusion that the conscious is not at all an indispensable factor in establishing of the appropriate correlations between the individual and the environment.

On our part, we do not think of denying "psychic" reality, but we are justified in maintaining that it is not something isolated from the brain processes. We say that the psychical process is inseparable from brain process and, therefore, when studying brain processes in their higher manifestations, we study along with them the course and development of psychical processes, (p. 44)

It is significant, as noted above, that one of Bekhterev's criteria for the occurrence of a conscious mental event was that it be communicable (in his terms above, "accountable"). This turns out to be a social psychological standard for what we nowadays consider to be the subject matter of "personality." Bekhterev (1933) addressed the issue as follows:

As we have already stated, we regard the physical and the psychical as a unitary phenomenon in one process of nervous current, as energy passing through the higher ranges of the nervous system. From experience, we know that this phenomenon as such can reveal itself in man in its complete form—with the participation of the psychic ingredient—only by means of active concentration, which is usually closely associated with the stimuli which issue from the somatic sphere. If we call one reflex conscious, and other subconscious or unconscious, this does not mean that the latter are not accompanied by the element of the psychic, or by consciousness, during their activity, but only when they are not reproduced as are other activities, and, consequently, are not "accountable." To the category of the accountable belong all reflexes which occur when active concentration participates, (p. 104)

Prior to attempting to compare his "objective" formulations about the psyche with leading "subjective" formulations, Bekhterev characteristically reiterated his basic position, perhaps so often because he felt he had to remind his audience that it was neither possible nor necessary to define a link between a material event and a parallel, immaterial "soul":

Everything said above leads to the conclusion that psychic or subjective processes are, as we have already mentioned, a result of tension of energy or of nervous current in certain areas of the cerebral cortex, while there is simultaneous inhibition of other areas. In cases in which the conditions of nervous conductivity are facilitated and the nervous current moves unimpeded, the subjective process or consciousness is weakened and even disappears. We have already said that, for instance, every action performed for the first time is mostly conscious, while every habitual action is unconscious. But even an habitual action may become conscious, if it is performed slowly and with concentration on each component movement.

Concentration, as a process of excitation of a definite center, is invariably bound up with impediment or inhibition of other centers; hence it happens that everything that produces concentration is accompanied by conscious perception; everything that is not accompanied by concentration goes unperceived and, consequently, occurs without participation of consciousness. (Bekhterev, 1933, p. 401)

To again reinforce his view that the mental event has ties to something ultimately observable objectively, Bekhterev (1933) stated:

Finally, what are we to call ideas and thoughts, by which we understand the subjective process which has attained a certain degree of completeness and development? Is this process distinct from other manifestations of association-reflex activity? As a matter of fact every idea is accompanied either by weakly expressed external movements, in the form of words or actions, or by internal movements and secretions, (p. 402)

With this line of thought, Bekhterev certainly anticipated subsequent conjectures that thinking was a variety of internal speech, but his preoccupation with objectivity was doubtless directed at those contemporaries who construed thought as a purely "spiritual" phenomenon, and he felt (probably correctly, in tsarist Russia for one reason and in communist Russia for another) that the materialistic/ energetic argument needed insistent repetition at every opportunity:

It is clear from this that thought is also a higher association reflex, but a reflex inhibited in its external manifestation, as we have already said. Consequently, there is but one process, in which the inner phenomena, in the form of thought, are merely inhibited reflexes in which the subjective process is more marked in comparison with the external objective process. (Bekhterev, 1933, p. 403)

The talk of "concentration" (today we would frequently substitute the term, "attention") and "ideas" may sound as if Bekhterev's accountable psychic phenomena involved only cognitions in relation to one another and to behavioral events. However, as Gerver (1933) indicates, he took a similar line with respect to feelings: were "feelings" primary or secondary phenomena? He said,

for us, who hold the energetic view, which we make the basis of reflexology, this question does not arise, for, as a result, of the manifestation of the nervous current, we have simultaneously, in the given case, both external or objective phenomena manifested in the sympathico-somatic sphere and inner or subjective phenomena, the origin of which may be, in one case, a result of a higher association reflex, in another, a result of lower reflexes arising in the somatic sphere....Let us take the sphere of feeling....when we weep over the coffin of a dear person, the primary impulse arises under the influence of an external stimulus and, by establishing an association reflex, excites a reaction in the form a general physical state which leads to tears. In this case, the beginning was a higher or association reflex, which evoked the same cardio-vascular reaction as appears in a physical state conditioned by strong, external stimulations, for instance, intense destructive processes. (Bekhterev, 1933, p. 405)

Thus, feelings as "mimico-somatic" reflexes were central for Bekhterev as of course they were for his better-known European contemporaries, the psychoanalysts, who were also transforming the view of thinking persons.

Bekhterev and the Psychoanalytic Movement

Most of one's thoughts, these inhibited behavioral reflexes, might reasonably be expected to be shared with the rest of one's culture; however,

some would become problematic for the individual, might not "fit" with the rest of the person's world. Bekhterev's psychiatric work on interpretation of these troublesome psychic reflexes inevitably brought him into contact with contemporaneous formulations concerning the human personality, in particular those of Freud, Jung, and Adler, and he compared his own approach with theirs in surprising detail (see Bekhterev, 1933, Ch. 51-53). With respect to Freud, he said:

We cannot help seeing the correlation of Reflexology with Freud's doctrine, known as psycho-analysis, first of all in so-called catharsis (purification by confession) which is equivalent to discharge of a "strangled" affect or of an inhibited mimico-somatic drive. Is this not the discharge of a reflex which, when inhibited, oppresses the personality, shackles and diseases it, while, when there is a discharge of the reflex (catharsis), naturally the pathological condition disappears? Is not the weeping out of a sorrow a discharge of an impeded reflex? (Bekhterev, 1933, p. 413)

Psychic trauma as a cause of neurosis is referred by Freud chiefly to the sexual sphere. This, however, cannot be accepted without very extensive reservations and has met with much opposition in scientific literature. (Bekhterev, 1933, p. 414)

Freud sets out from the supposition that associations are grouped into complexes, and, consequently, neither the individual's words nor his gestures are casual when his thoughts are concentrating on anything, but refer to the complex to which concentration is directed. (Bekhterev, 1933, pp. 414-415)

Of the "reflexes" which were the subject matter of early investigations of Breuer and Freud, Bekhterev (1933) said:

These reflexes are abnormal partly because they are a pathological response to external impressions and partly because they are not a direct response to an impression received, but are a response only through the mediation of pathological memories. These latter, isolating themselves from the patient's own observation, or his ego, give hysterical symptoms

their apparent arbitrariness and this, by the way, distinguishes hysteria from traumatic neuroses.

Dissociation of the ego is by no means unusual. On the contrary, even in the normal state, this splitting may be observed to a certain extent. We may look without seeing, hear without being accountable to ourselves in respect of what we hear. Needless to say, the splitting is much deeper and more considerable in pathological states. In hypnosis and suggestion, we have obviously a similar, only deepened, process of dissociation of the subject's ego or personality. (Bekhterev, 1933, p. 414)

To clarify his disagreement with Freud, Bekhterev turned to the issue of sexual energy, and he expressed qualified support for Adler:

It must be noted here that Freud's method, since it is exclusively subjective, has led its initiator to the gravely erroneous conclusion that sexual trauma is the basis of the common neurosis. Such error frequently results when investigations are given a subjective basis. The Great War produced an enormous number of neurotics, and here the traumata were of a different kind, and rarely sexual. Thereupon, Freud's disciple, Adler, advanced the doctrine, not of sexual trauma, propounded by his master as the cause of the common neuroses, but of social conflict, in which the neurotic manifests an active attitude toward the illness ("flight into illness"), which for him is something desired as an inevitable escape from the given situation. Adler's theory also, though considerably wider in fundamental principles than Freud's, is based to a considerable extent on subjective analysis, and therein lies its weakness. (Bekhterev, 1933, p. 415)

Bekhterev seemed most favorably disposed to Jung's approach:

Let us point out that Jung's investigations (mentioned above) in respect of the association method, which takes the form of stimulus-words, have afforded an important objective indication of what has been experienced and done, and this indication may be used in revealing both the causes of neuroses and of criminal acts, fields in which it is already beginning to be applied. For this purpose, a special list may be compiled in which words that relate to what has been experienced are inserted among indifferent

words, and it transpires that, when a word is presented to the subject in order to stimulate free association with other words, a more or less marked delay results in respect of the reaction, when words bearing reference to grievous experiences are presented. (Bekhterev, 1933, p.415)

After reminding us that interpretation of such delays using subjective states, like "guilt," is "exceedingly relative," Bekhterev adds that

the objective understanding of phenomena from the reflexological standpoint throws a light, hitherto unavailable, on Freud's theory, with its processes of repression (inhibition), conversion (substitution), etc., and at the same time, through the scheme of abnormal reflexes, supplements those processes, which were regarded as purely subjective states. (Bekhterev, 1933, p. 415)

In closing his general comparison with psychoanalytic theories, Bekhterev (1933) said,

It is true that psycho-analysis as presented by Breuer, and more so as presented by Freud, who developed it into a system, is, as we have said, teeming with a subjectivism often too involved and difficult to be explained even from the viewpoint of contemporary subjective psychology. Nevertheless, it is clear from all that we have said above that the psycho-analytic method of treatment is easily understood by means of the reflexological doctrine of association-reflex activity.

It is unnecessary to enter here into the details of this problem. We merely state that, from the viewpoint of reflexology, a strangulated affect is just an inhibited, that is, externally unexpressed, mimico-somatic reflex, which is revived by the psycho-analytic method, that is verbal stimuli, is ultimately released from inhibition, and is liberated, so that there results a cure, which presupposes the liberation of other reflexes also which have been suppressed in inhibition of the fundamental reflex. (Bekhterev, 1933, pp. 415-416)

Dreams and the Reflexological Approach

It will not come as a surprise that Bekhterev went even further with his reflexological explanation of mental phenomena to the point of examining dreams, which he judged to be so vivid precisely *because* the person was not overtly acting at the time of the dream's occurrence:

There is no doubt that dreams also, the detailed psychology of which has been elucidated by psycho-analysis, may be harmonized with the data of reflexology. Freud himself, when outlining the scheme of dream development, remarks that the psychic apparatus is constructed after the model of a reflex. It is true that he intends this comparison to be taken figuratively, but, clearly, in discussing complex psychic phenomena, even he cannot dispense with the scheme of the reflex. (Bekhterev, 1933, p. 417)

But Bekhterev disagreed with what he took as Freud's position on the source of the dream:

According to Freud's view, in dreams there is a process which consists in a regression of the nervous current to the mechanism of perception. The phenomenon of regression he regards as nothing new, for thought in general often regresses along the path of memory, but usually does not transgress the limits separating it from perception, for memory never presents such vivid pictures as does perception. In dreams, on the contrary, regression extends much farther, almost as far as hallucinations. (Bekhterev, 1933, p. 417)

Bekhterev's alternative to the hypothesized regression of mental processes to the perceptual level was an ingenious and provocative formulation:

Dreams are, like thoughts, reflexes inhibited in their external manifestation, but, in contradistinction to thoughts, dreams are distinguished by special vividness, because the suppressing influence of the active part of the personality—that part which controls internal stimulations—is removed. (Bekhterev, 1933, p. 418)

But psychologists are mistaken in thinking that we have, in dreams, a process regressive in respect of the sequence of the thinking process, for they maintain that here there is a regression to processes of perception.

We have referred above to the fact that processes freely discharged in motor reactions in the waking state occur without intense subjective images, while all inhibition of motor acts produces extreme tension and intensity of the subjective state. Therefore, there is no necessity to have recourse to the hypothesis of a regressive movement of the current in order to explain dreams, but we have only to recognize that, during sleep, there are conditions inhibiting external discharge in the form of movement, and this, under the influence of inflowing stimulations, enormously raises the intensity of subjective states. (Bekhterev, 1933, p. 418)

And, indeed, in sleep we have obviously inhibition of all motor reactions, for, notwithstanding even vivid pictures of movement, conversation, etc., all the sleeper's motor organs remain motionless. We shall not here enter into an explanation of this phenomenon, which is obviously associated with suppression of the active part of the personality, this part being centered in the frontal areas. But a fact is a fact, and this fact explains the existence of dreams, for according to the thesis propounded by us, inhibition of an external reaction, when this inhibition is accompanied by simultaneous excitation of a certain area, always raises somewhat the intensity of the inner or subjective experiences. (Bekhterev, 1933, p. 419)

Bekhterev (1933) extended his analyses to other of Freud's basic concepts (e.g., "censorship" by the super-ego being construed as inhibitory associative reflexes that stand in opposition to a given complex; phantasy; poetic creativity, etc.). Likewise, he developed a reflexological analysis of affective phenomena such as moods, motives, tendencies, etc. But it really is not necessary to "dilate further on this here," as Bekhterev himself would often say. It is enough for present purposes to acknowledge the nearly super-human feat that he executed in developing such a "grand theory." One can only speculate about what the immediate future of his efforts might have been, perhaps in correspondence with Freud and other contemporaries, had he not left the scene so suddenly.

From Individual to Group Reflexological Processes

If Bekhterev's group psychology, i.e., collective reflexology, had been called "biosociology" from the start, perhaps its general reception would have

been a bit more sympathetic, but even then one might have anticipated some reluctance to accept the following paragraphs from Part 1—"In Place of a Foreword"—(p. 27, present revised translation):

As we shall see later, a strictly objective examination of the subject leads inevitably to the conclusion that manifestations of the composite personality revealed through the analysis of social life are ruled by the same law that is revealed through a strictly objective reflexological study of manifestations of the individual personality. The very forms of this law turn out to be the same for both the individual personality and the composite one; this should certainly be recognized as an advance in knowledge, since the theory of the opposition between the personality and the crowd accepted by many authors who wrote about the crowd is to a considerable degree exaggerated.

As will be shown below, one encounters among the manifestations of social life essentially the same reflexes, in the form of universal, eternal movements following the same path and evolution, as one finds in individual activity. This should be quite clear if we take into consideration the fact that a collective is nothing other than a group of individuals interconnected through their various interests and representing a single entity in the form of a collective, or composite, personality.

In his Introduction to Part 1, Bekhterev restated his position:

While reflexology studies the manifestations of individual associative activity, collective reflexology studies the same activity of the collective. It reveals, therefore, how this activity is manifested in the case of social affiliation between several (two, three, or more) individuals and to what extent the characteristics of this associative activity depend on the nature of the social bond established between individual persons, (p. 40, present translation)

At the close of his Introduction to Part 1, he rephrases his argument once again:

The opportunity to reveal the regular development of individual expression and actions presented itself when we started investigating

human behaviour not from the standpoint of subjective causality but from a strictly objective point of view, recognizing the need to regard all acts of human personality as manifestations of higher, or associative, reflexes. The latter are based on energy, and whether one calls it neurological or neuropsychological does not really matter, for in any case it is energy similar in its nature to other kinds of energy. It is thus evident that there must also be a certain developmental regularity in the manifestations of popular masses who represent a composite personality.

In this respect, observation and experience lead us to infer that the basic laws of associative activity of the composite personality are the same as those operating throughout the whole of animate and inanimate nature. Here we disclose through analysis the same universal cosmic laws such as the law of energy conservation, of gravity, of repulsion, the law of equal and opposite reactions, of similarity, rhythm, entropy, evolution, differentiation, integration or synthesis, the law of adaptation, of selection, inertia, etc.

We shall see that the world is ruled by the same basic laws which are common to all phenomena in general, both inorganic and organic as well as supra-organic, or social, which will be discussed in more detail below.

It cannot be otherwise. If the world of animate nature is the result of the transformation and complication of inanimate nature through evolution—the natural sciences give us no grounds whatsoever to doubt it—then why should there be one set of laws for the inorganic world, another for the organic and a third for the supraorganic world? (pp. 42, present translation)

The Fate of Bekhterev's Social-Reflexological Theory

It would be satisfying to be able to complete this Foreword with a comment on the reception which the social science community gave to this "grand theory," but in introductions to Part 1 and to *Suggestion and Its Role in Social Life* we have discussed how Bekhterev's efforts in the area of social psychology remained hidden for at least a half century. In the ordinary course of events for our field, Bekhterev's reflexological doctrine would most

certainly have received more attention—in social psychology, at the very least, by Gordon Allport, (who did acknowledge one Bekhterev paper published in German and one book (*Suggestion and Its Role in Social Life*, in Russian) in his discussion of "simple and sovereign theories" in the influential chapter, "The Historical Background of Modern Social Psychology" (Allport, 1954). In this widely cited source, Allport compared with one another the positions of all those (Western) theorists with whom Bekhterev compared himself (e.g., Spencer, Tarde, Binet, McDougall, Le Bon, Sighele, etc.). As one possible consequence of more detailed consideration by Allport, Bekhterev's heroic efforts would have been at least acknowledged in subsequent literature concerning theories of social behavior. Alas, this has happened almost not at all.

On the other hand, it is also possible that, by now, whatever the attraction such a general theory of reflexology might have held in the 1920s, or even the 1950s, it would have faded with the evolution of criteria to which a "good scientific theory" should hold. One is reminded of a provocative paper by Warren Thorngate (1976), wherein he pointed out in his discussion of "postulates of impotence" that, while psychologists have been long taught to value most highly a theory that can boast simplicity of formulation, accuracy of prediction, and generality of application, it is the unfortunate case that it is impossible to generate such a theory: the theory that is general and simple will not be able to make accurate predictions; the theory that is simple and accurate will have limited generality; the theory that is general and accurate will not be a simple one. Thorngate cites a similar contention by Levins (1968), applicable to biological systems, to the effect that "it is not possible to maximize simultaneously generality, realism, and precision" (p. 7). The reflexological theory of Bekhterev, which began in biological systems and was elaborated to explain social systems, would doubtless have proved to be as vulnerable in the one system as in the other. But it is not timely to criticize reflexology's vulnerabilities in terms of philosophy-of-science formulations developed a half century after the theory was first presented. It is perhaps enough to admire the genius and bravery of the person who would have attempted such a venture in his own time, and to regret that his ideas did not receive the attention they deserved.

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Foreword

Once we have analyzed the various facets of associative activity, it becomes essential to investigate the laws that apply to the activity of the collective, since no science should be satisfied with only a description of its phenomena. To the contrary, it must explore the dependence of these postulates on external factors of some form or other. This aspect of scientific discipline is especially important, since in discovering the fixed regularities between two phenomena and the law-determined reliance of some phenomena on others, we gain the potential to predict the course of phenomena or to foresee the future. In these two facts lies the true value of scientific knowledge.

If a mathematician predicts the divisibility or indivisibility of a number, then he does so employing a previously ascertained dependence. If an astronomer forecasts solar or lunar eclipses, he can do so only thanks to the discovery that the movement of planets depends on the laws of gravity, established by Newton.

If a chemist can foretell the atomic mass of an existent, but not necessarily located, body and, to a point, its characteristics, then he also does this only because of the discovery of law-determined dependence, which is, in turn, explained by Mendeleyev's periodic system. If a mechanic builds a new car, having planned in advance how it will work, he achieves this through the knowledge of a certain dependence of phenomena on the correlation between one lever and another. In the same manner, a physicist can calculate the number of joules needed to warm a body of a certain magnitude, by employing a certain dependence. We have the same ability, to some extent, in biological investigations, although the extent of this ability does not come close to that of astronomy, chemistry, mechanics, and physics.

Nevertheless, Mendel's laws provide us with the opportunity to anticipate the characteristics of offspring, and Darwin's teachings help clarify the growth and evolution of organisms and hypothesize on the nature of species whose extinct forms are discovered in geological strata.

Thus, a paleontologist can, using the same concept of dependence, reproduce from one or more parts of a skeleton an extinct species. The humanitarian sciences should obviously be pursuing the same end; indeed, some of them have already established a certain relationship between phenomena which is sufficient to postulate the course of phenomena in the future.

They were able to establish this dependence thanks to a strictly objective method, such as the one which is employed in political economics but which is nonexistent in sociology. The sociological law of historical illusionism shows, among other things, how the subjective understanding of phenomena differs from objective reality. For example, it is proven that the participants in the French Revolution felt that they were fighting for freedom, equality, and fraternity when in truth the battle was fought for the release of the bourgeois from the domination of feudalism; in other words, the conflict forwarded the interests of capitalism. The same can be observed in Christianity. Did it stand for the teachings of Christ, as its proponents stated, or for the scholastics, for the hierarchic organization of the Church, and for the rule of Christian materialism, as is revealed today? There is a large discrepancy between the ideas promoted by the proponents of Christianity and the objective facts of its historic development, which even the leaders of the movement failed to notice. It is only necessary to remember the Inquisition in order to prove this point. Another example of illusionism is the recent World War, at the root of which lay the economic rivalry between England and Germany. Unquestionably, each side in the conflict strove to vanquish the other in order to leave it in economic ruin, which, in the end, served to undermine the capitalist ideal and instigate a general trend towards socialism.

For the above reasons, all subjectivity must be excluded from sociology and from the study of popular mass movements in general, and only objective study can provide the opportunity to establish certain correlations and laws.

We must strive for the same thing in collective reflexology. The fact that in individual associative activity we have been able to confirm certain dependences instills the hope that in collective associative activity these dependences will also be found.

In addition, the analysis of associative phenomena displayed by the collective brings us to the conclusion that the dependences here are the same as those observed in relation to the associative activity of the individual.¹

Each of these factors, then, serves in the given manner as a confirmation and reference point for the other.

In the end, it must be acknowledged that if the reflexology of an individual is based on energetic science, which recognizes all the external characteristics and manifestations of an individual as being the result of the energy he accumulates with the processes of birth, nourishment and the influence of external factors, then collective reflexology can only become a science—with the ability to outline the laws of social life—when it takes an energetic point of view and explores the vital manifestations of each collective in relation to the circumstances which led to, and the reasons for, its creation; in relation to the characteristics of the individuals which make it up; and finally in relation to the external factors that have affected the collective in the past and the present. This will enable us to look at the results of collective activity and the factors that influence them, as well as to investigate the correlations or laws that govern the actions of the collective.

In the text that follows, we will analyze these correlations under separate categories, defined as laws. As was previously mentioned, the basic principles or laws of the supraorganic world, in its objective study, are the same as those of the organic and inorganic worlds.² This is the best proof that reflexology, as a strictly objective science that studies individual and collective associative activity, comes closer to being natural science than does any subjective discipline such as general or experimental psychology, which employs a method of self-observation that cannot be applied to the sciences of living and nonliving nature.

Notes

¹ See V. Bekhterev, *General Principles of Human Reflexology* (in Russian), Petrograd, 1918.

² On the same topic, see my investigation (1920/21): *On the basic laws of the world* (in Russian), and others in *Issues in the Study and Education of the Personality*, issues 2,3 and 4.

1

The Law of Conservation of Energy

The Law of Conservation of Energy, discovered by R. Mayer, widened in scope by the work of the English physicist and technician Joule, and finalized by Helmholtz, needs no explanation in the present day because it is widely accepted and understood. Basically, this law stipulates that the amount of energy in a closed system does not diminish or increase, and that all forms of energy in nature are, in relation to each other, equivalent. As for the Principle of Conservation of Matter, it is simply a subsection of the principle of conservation of energy, since matter is nothing more than a bundle of (restricted) energy. Besides, in the present day it is difficult to speak of the conservation of matter without making exceptions for the discovery of radioactive bodies.

In my book, *General Principles of Human Reflexology*, I prove that since our sensory organs are, in essence, transformers of external energy (which was already stated by me in 1896 in a separate work¹), any activity a person executes is subject to the law of conservation of energy.

The social process is dependent on the collective energy of individuals and is always manifested in some activity, work, and the like. This, in turn, can only be done with the consumption or expenditure of energy.

It goes without saying that very rarely does the collective perform a task as a whole; more often, the initial work is done by one or several people, while the others are onlookers, listeners, or even imitators. However, this can also be called collective work—granted, it is unequally distributed—since some expend stored energy, while others transform the byproducts (external

stimuli) into reserves of energy, or release their own energy under the influence of a stimulus that causes imitative behavior, i.e., do so very economically. Nevertheless, in one way or another, that the whole collective participates in a single activity is indisputable.

In this way, social or collective energy is made up of the combined energy reserves of all the participants in a single task. It is important to keep in mind, however, that the energy expended by these participants is not all converted into useful output or work; some of it is wasted on overcoming the inertia of the collective, on internal friction between workers, on overcoming various external obstacles, etc.

Thus, only a fraction of the originally available energy is channeled into useful work, as is the case during the execution of any mechanical work.

It is possible to apply to the performance of collective work the parallelogram of force used in mechanics, namely: the result of action is manifested in a certain direction, and is equal to all efforts made by the separate individuals participating in the given collective work.

To social energy are also applicable the processes of suppression or inhibition, and those of augmentation, which are dependent on the application of new strength or on the release of individuals' energy reserves; this will be discussed later in more detail. A social movement that encounters opposition either becomes part of another movement or is temporarily inhibited, but it does not disappear.

Already, these facts support the assumption that the law of conservation of energy can be applied to the activity of the collective as well as to the individual person. However, up until now the application to mankind of the principle of conservation of energy has been hindered by subjective outlooks on the human personality, and as a result this understanding could not be established.

The strictly objective point of view, taken by reflexology and eliminating subjective phenomena, acknowledges that every individual is basically an accumulator of energy, which has already begun to be collected at conception and continues to be accumulated with the ingestion of food and with the help of sensory organs functioning as transformers of energy. In its turn, reserved energy is used to perform kinetic and molecular work in the flexing of muscles and in glandular secretion. There can be no doubt that the law of conservation of energy can be applied to humans, as to any living organism, since otherwise it would lose its status as a universal law.

Today, we know that the energy reserves of our brain centers are held, primarily, in the granular section of neurons, since cellular grains disintegrate as a result of continuous function and fatigue. In this way, the very substrate of nervous energy is known to us.

Recently, it has even become possible to determine the percent of useful work the human organism can do in relation to the amount of energy it receives with food. Specialized research has shown that useful work executed by human muscle has an efficiency of approximately 21 percent, whereas a steam engine gives only 13 percent useful work for all the energy it uses.

Moreover, fitter individuals efficiently use up to 36 percent of the energy they accumulate. Of course, in all these situations, the expenditure of energy on the functioning of internal glands and other somatic organs (sweat production, faster blood circulation, etc.) is not taken into account. The fact that humans utilize energy more efficiently than steam engines is explained in this way: the human "engine" replenishes, to a certain extent, its deficit with the stored energy that is always in the body's centers, thanks not only to the absorption of nutrients, but also to the transformation of different forms of energy in the external environment, which affect our external sensory devices.

In my article, "The immortality of the human personality from a scientific point of view" (in Russian), *Vestnik Znaniia*, 1918, and other publications, I discuss in detail the fact that the creativity of human associative activity lives forever.

"A person's 'spiritual' self," I write in that paper,

in possessing self-contained value, can never vanish without trace. In this way, every human individual who combines the experience of his ancestors with his own does not terminate his existence with the termination of his life. On the contrary, he continues to exist in all those people whose lives he affected and, eventually, is preserved in posterity. It can be said that during his lifetime, an individual distributes his energy among those who surround him, who in turn transfer this energy to others, continuing this cycle to the farthest reaches of human communication and interaction. In a composite human personality, everyone is connected in an intricate relationship, such that no single event can occur without having widespread repercussions. One heroic deed breeds another, just as one crime begets another.

But if the action of one individual cannot be erased, then conversely there is nothing that can produce energy without the related external influences; in other words, no extra human energy arises outside certain mutual correlations. While on this subject, it is impossible not to mention the observations that L. Tolstoy made, quite correctly, in his famous work *War and Peace* (p. 272), and which I have already stated in reflexology:

The recognition of individual freedom as a force that is able to influence historical events, i.e., as one not subject to any laws, is to history what the recognition of the total freedom of planetary movement would be to astronomy.

The above admissions eliminate all possibility of the existence of laws, and thus the existence of any knowledge. If even one freely moving planet exists, then the laws of Kepler and Newton are nullified, as is our understanding of the movements of heavenly bodies. If one free human action exists, then there are no historical laws and we have no understanding of historical events.

From here, naturally, stems the conclusion that human energy, like any other, can neither be created nor destroyed and simply changes from one form of work or energy to another.

It is evident, then, that when analyzing the collective personality or the collective we must keep in mind the same law of conservation of energy. Here, too, this law should be considered of equal importance, since if it is possible to apply it to the individual, then it is equally possible to apply it to the large number of individuals that make up a collective and are connected to each other in one way or another. A crowd, in the same way as a solitary individual, is stimulated by its surroundings thanks to its sensory detectors, sight and hearing; it is likewise an accumulator of energy that is transformed into kinetic energy or work, and emitted as a reaction to external stimulants of one form or another.

Of course, we cannot hope to test this law in practice, since it is not possible to take into account the amount of stored energy in every individual, nor can we allow for the way in which this energy is combined when individuals are brought together in a crowd. However, arbitrarily designating a certain quantity of energy to both, it can be said that the energy of the

collective personality is expended depending on the amount of its stored energy, and depending on the strength of the stimuli causing the latter's release; and that the direction and power of these releases of energy, manifested as work, are subject to a certain regularity, as statistics show. This last fact leaves no doubt as to the fact that all phenomena of social life show a relation between given causes and their respective effects.

In any case, it is erroneous to assume that the individual mind controls occurrences. Only the above conditions can serve to bring forward one or another individual, and allow him to become the spokesperson for social moods and collective actions. Naturally, it follows that in social matters the role of the majority is limited by individual attributes, and, as a result, an individual representing a certain view can influence a crowd and even crush the said majority. This is, in a nutshell, the role of the individual in popular movements.

In times of social upheaval the role of the crowd becomes so much greater that its so-called instincts, or inherited/organic reflexes take the upper hand, and the intellectual control of those individuals who act as leaders of the crowd is reduced; whereas in calmer periods of social life, the role of the director of the crowd's actions is more significant.

In any case, not only do the actions of separate people not disappear without trace, but a crowd's creativity is also preserved through its transformation, over time, into various forms of energy.

Does a national collective, having created a culture, ever "spiritually" die? It can fall apart, separate, or be subjected to foreign rule. But in all these cases, the results of a nation's creativity are passed on to, and used by, other nations who mold that creativity into new forms, as was done by the peoples that inherited the "spiritual" riches of ancient Greece and Rome.

In any case, for those who maintain a strictly objective view of the human individual in accordance with the teachings of reflexology, i.e., reject the subjectivity of contemporary psychology and observe in humans only associative activity, there can be no doubt that the law of conservation of energy must be applied to the individual and his actions. In concluding, I must mention that the reflexology of humans has been studied by myself only from an *énergie*² point of view, which I have already stated in the book, *Psyche and Life*,³ and analyzed in more minute detail in the French and German editions of the same name.⁴

Notes

- 1 V. Bekhterev, *Obozrenie Psikhiatrii and Neurologisches Centmlblatt*, 1896.
- 2 See V. Bekhterev, *General Principles of Human Reflexology* (in Russian), Petrograd, 1918.
- 3 V. Bekhterev, *Psyche and Life* (in Russian), St. Petersburg. SPB.
- 4 W. Bechterew, *Psyche und Leben*, Wiesbaden; also by Bechterew, *L'Activité psychique et /a vie*, Paris.

2

The Law of the Proportional Relationship between Speed of Motion and Motive Power

In mechanics, there is a long-known law which states that any increase in the speed of an object's movement is determined by its motive power, which is proportional to and of the same direction as the movement. This law is a general one, and can be applied not only to the inanimate but also to the organic and supraorganic worlds. In the organic world, every function can be subject to acceleration, depending on certain stimuli, and this increase in speed is proportional to the strength of the stimulus and the direction of an organism's actions. However, since organisms are made up of essentially closed systems in which the extent of development and the direction of most functions is predetermined, the application of this law is limited.

In associative activity, as we know, a stimulus is proportional to the force of its agent until functions are exhausted and the system begins to break down. As for the orientation of the stimulus, it is determined by the structure of motor organs and therefore can only be in one direction. However, in this case the stimulus can function as such only if it has the same orientation as the organs, and so conforms to the law.

In the social world we find confirmations of the above law in many different situations.

Society is in continuous mobile equilibrium as a consequence of the interaction of stimulating and inhibiting conditions; social movements take

on a form that is determined by the dominance of the latter over the former and flow in the direction of the resultant force. In view of their significance, it is important that we investigate both of the aforementioned stimulating and inhibiting influences in greater detail.

When a social movement is born, it is dependent on the dominant stimulating, facilitating, or vitalizing conditions. This last condition can be either natural or artificial. Natural vitalizing conditions encompass, first of all, the forces that are in direct correspondence with the main stimulus, the one that initiated the social movement. Let us assume, for instance, that we are dealing with a patriotic movement that was instigated by wartime conditions. It is quite natural that in such a situation all instances of apparent military success would increase and strengthen national patriotic feelings in one way or another.

The development of a religious movement is strengthened by the execution of various religious rites and ceremonies. On the other hand, the growth of an antigovernmental social movement is vitalized with every event that makes the success of the movement more likely, etc. Furthermore, natural vitalizing conditions of a social movement include all those conditions that, although they are not in direct contact with the movement, are in harmony with its objectives.

Thus, movements of a moral nature are encouraged by religious circles; sports or gymnastics movements receive support from military spheres, since physical exercises help in the military preparation of the population, etc. This also explains the fact that the more extremist parties receive support in revolutionary times, while during calmer periods moderate parties are given primary consideration.

If we are presented with a military collective in the form of troops, it is evident that every increase in strength results in the more rapid fulfillment of goals, the speed of fulfillment being proportional to the force that was added. If we observe a working collective, we can see that the augmentation of strength proportionally accelerates the work being done.

However, social life manifests itself not only in manual labor. Revitalization of social activity occurs, *in esse*, with every new discovery or invention. These findings act as a kind of force, in the form of an impetus for social activity, in the sense that they uplift the mood of, and arouse interest within, the social body.

On the basis of what has been said, it becomes clear that if there exists a general trend in society towards a certain profession, then there must be a stimulus that drives people to choose that profession. We can say with certainty that the attraction to that profession would be directly proportional to the size of its stimulus.

If immigration occurs, then it is also controlled by the stimulus that encourages people to come to a given country, and the more enticing the stimulus, the greater the number of immigrants. Conversely, emigration from a given country is stimulated by unfavorable conditions in that area, and this emigration again increases as conditions worsen in relation to those in other countries.

If a collective, while spreading and growing, funnels its actions in a certain direction, then we can say without a doubt that the interest drawing it in that direction does so with a rapidity that increases in proportion to the increase in interest.

The advances of science are governed by the same law. Any discovery, like a revitalizing force, turns minds in a certain direction; the greater the discovery, the more intensive the work related to it. Something new in the arts will attract artists, and cause them to work in a style that imitates the innovation.

Finally, we can make the same observations in the economic/fiscal world. Every industry develops in the direction of, and with the strength needed to meet, the demand for its product. This is also true for trade, which is always conducted in connection with the demand on communication routes: the lesser the latter, the greater the growth of trade.

It should be added that trade is also stimulated by manufacture and by the amount of money in a country, which complicates the usual simple relationship. However, this does not affect the crux of the matter, since in this case we are dealing with conditions of growth of capital and competition, which are also subject to the law of proportional correlation between the speed of motion and motive power.

When a relatively intense conflict between parties is underway and the balance of their strengths seems insufficiently steady, it becomes very risky for one of the parties to make a sharp deviation in its own favor without first deciding whether or not it is strong enough to stand up to opposition; otherwise, the matter can come to provocative events and to the stimulation of a defensive reaction from the opposing party, and the situation may have

unfortunate consequences for the instigating party. In fact, this is exactly what happened in Russia with the decision to dissolve the fourth State Duma, after which revolutionary movements openly developed both in the Duma itself and in the nation. A similar progression of events led to the second March revolution in Germany (1920) and to the seizure of power by the military party with Lustwitz at its head.

In general, during periods of imbalanced social forces, every careless step on the part of either side of the conflict can function as a challenge, provocation, or excitant for a widespread national movement. History gives many, many examples of this. However, closest to ourselves would be the events of January 9, 1905, which gave a substantial boost to the revolutionary movement in Russia and greatly precipitated the first Russian revolution, which had already been instigated by Alexander the Third's regime and by the unfortunate Russo-Japanese war. Another example is the Lena executions of 1912. These events, which took place far off in the taiga of Eastern Siberia, should seemingly have had no effect on the masses of inner Russia. However, they served as the first of several blows that led to the birth of the proletariat in Russia. Finally, the agonizing death of the workers of Lena was reflected in a renewal of the revolutionary movement.

A wave of strikes swept across Russia, workers began to listen more attentively to revolutionary slogans and to unite in the name of a fight for their rights. Thanks to this, by the time war commenced in 1914, the workers' movement in St. Petersburg had reached the level of a barricade rebellion, and it was only the sudden war that, for a time, suspended the revolutionary movement. However, it arose with renewed strength two and a half years later in connection with our failures at the front and the stubbornness of the tsar, who turned everyone against him except his most deferent bootlickers as a result of the Rasputin scandal.

But the bourgeoisie, in siding with the revolution, did not take into account the fact that it had always served as a support for capitalism. The no longer restrainable worker's movement naturally had to take its course, and almost all of the bourgeoisie died in the fire of the revolution, while the intelligentsia affiliated with it was subjected to all kinds of persecution after its pointless decision to sabotage bolshevism.

Temporary rest or suspension of action is another revitalizing condition of social movements.

No social movement can maintain high levels of energy without a break. It generally quiets down of its own accord, due to internal inhibiting conditions that are manifested in the growing fatigue of the followers of the movement. However, rest or a temporary break in the movement give it new life and new impetus to proceed.

Artificial revitalizing conditions encompass all those that are created by the collective itself. Stimulating conditions, such as praise, rewards, and other encouraging measures taken by social and class establishments, religious communities, etc., fall under this category. On the other hand, repressions such as punishment, isolation, arrest, etc., are artificial inhibitors set up by the collective to deal with people who disrupt the normal living conditions and activity of the collective.

When a social movement is well established, usually only a very small, ordinary spark is required to vitalize it, while during the initial stages of its development achieving the same vitality would have demanded great effort.

This is why, when the conditions are right, even the most insignificant impetus can create turmoil in a given movement.

As we already stated, the contemporary Russian Revolution started only with the edict on the dissolution of the State Duma, though the conditions for a revolution had long been favorable.

The Swiss revolution—which, likewise, had been brewing for a long time—began with the ridicule of a prominent Swiss citizen. In every social gathering or crowd, when the prevailing mood reaches its peak, it suffices to shout one slogan, to drop one incautious word, and the crowd or gathering becomes active.

An example of this occurrence can be found in the depiction by L. Tolstoy, in *War and Peace*, of the lynching of a petty criminal on Count Rastopchin's porch. Naturally, the transition from inaction to action comes more easily to a crowd than to a gathering, since a crowd acts much more impulsively, more automatically, than any organized meeting.

Finally, it is important to keep in mind that the revitalization of a social movement can also occur in the form of a reaction to another, opposing, social movement or state measure.

In this way, the persecution of Christians in the first centuries of that religion's existence did not weaken the Christian doctrine in any way; to the contrary, because it gave Christians the appearance of martyrs, this ill-treatment supported their cause.

The same is true for the persecution of religious renegades (for example, the *raskol'niki* in Russia) and the harassment of revolutionaries. In general, we can say that every new social movement that wants to change the existing way of life, and yet does not represent the majority, must in the course of its establishment make sacrifices and be mistreated.

The battle with cannibalism cost many human lives during the European expeditions to and conquering of distant countries. The fight to provide international aid to those maimed by war was stimulated by the suffering of many thousands of injured people. The struggle for political freedom claimed a great many lives of so-called political prisoners. The fight for freedom of one nation or another was expiated by that nation's protests, which were often put down by force.

The battle for equality of nations was accompanied by an array of wars fought against imperialism, which brought innumerable deaths. We need only remember the Napoleonic wars, the Sebastopol war, and the Great European War to see the costs of fighting for state equality.

The effectuation of permanent peace and the disarmament of nations also requires sacrifices, and the struggle of the Internationale must bring with it many losses before this admirable and, shamefully, as yet unrealized goal can be attained.

A huge revitalizing force in the progression of social movements is rivalry, a result of the competition between two, three, or more social movements that are more or less similar in character and goals. Here competition serves as an impulse for the collective to exert its internal forces, adding to the strength of the social movement. If it then demonstrates viability, the movement will be revitalized through competition and will consequently receive a mighty stimulus to its development. So many important social movements owe their existence to competition, and so many have developed thanks to it! It would be difficult to enumerate all the examples that support this observation, since they are visible to everyone. However, we must not forget that competition is only a revitalizing factor when it is not overwhelming, since in the latter case competition has the totally opposite effect, slowing development in movements not able to stand up to rivalry. Among other things, stimulating or encouraging conditions that aid the distribution of an innovation throughout the collective include, on the one hand, all subsidiary discoveries which are congruous with the innovation, and, on the other, the improvement of living conditions in the collective.

It is important to mention that the frequent revival of any reaction is irrevocably connected to its acceleration, as is the case in mechanics. The same can be observed in social movements. In any collective, social movements that are frequently repeated and that become habitual always develop more quickly than those that did not have an equivalent in the past. The leaders of revolutions know very well that people must be made accustomed to revolutionary demonstrations if an organized insurrection is to take place. Any military leader knows that seasoned troops—that is, those who have had experience under fire—are stronger in battle than those who are new and unaccustomed to fighting.

If factors furthering a certain phenomenon are repeated one after another, then they seemingly build on each other and, all together, act with greater force in a certain direction. It would be easiest to observe this in relation to a social phenomenon. For example, let us look at an incident of great social importance, such as the entrance of the United States of America into the Great European War despite the fact that the interests of America, it would seem, were not affected by that conflict. America, which not long before the war had been promoting the idea of international arbitration, could not have entered the conflict without the influence of social and national movement. Let us try and determine how, in the end, this seemingly improbable event occurred.

We must first of all say that, at the dawn of the war, the international position of the United States boiled down to an adherence to the following principles: (1) the Monroe Doctrine, which guided the American government to take upon itself the protection of the New World from attacks by European nations and at the same time to distance itself from European transactions; (2) the defense of the concept of freedom of the seas; and (3) an overwhelming preoccupation with international arbitration, which would provide the opportunity of preventing the recurrence of war.

The eruption of the European war in 1914 came unexpectedly for America because the latter could detect no visible signs of the pending events. Also, every warring European country had many expatriates in America, and their respective arguments confused the scenario and made it difficult to decide on whose side the truth lay. From the very beginning of the war, America declared neutrality, which was accepted as natural and necessary. Then, because of the diversity of opinion in the States, the president was forced to make a speech in which he cautioned that the differences in views held by

Americans could inhibit the fulfillment of the responsibilities of that great, peace-loving nation, which consisted of giving advice to, pacifying, and negotiating between nations "not as a follower of one warring faction or another, but as a friend." This is how the president expressed himself on the twentieth of April, 1915:

We are the negotiators for all nations of the world; since our nation was made from the mixing of all nations, we are, naturally, the intermediary between their blood, their traditions, their interests, tastes, and desires. We are able to understand them all as a whole, not each individually, for we are not on the side of one nation or another but rather understand, bring together, and represent all nations. In this sense, I name the American people the negotiators.

American neutrality, then, had every reason to be complete and infallible—and it was, in the beginning. As the war wore on, and as the atrocious behavior of the Germans after their invasion of Belgium became known, American public opinion turned against Germany. The commission of the Belgian government, which arrived in America in September of 1914 with complaints about the conduct of the Germans, did not achieve all the results it intended; however, the president, in his response, in order to prevent the necessity of the United States entering the European conflict, retained the right to state his decision on the subject should the need arise. This was, of course, a purely diplomatic move, an attempt to get out of a difficult situation; the sentiments of most Americans were already on the Belgian side, and they sent money, food, and clothing to that country. This aside, the necessity of sustaining American neutrality was nationally supported, and the political actions of the president were judged unanimously as being proper.

Things began to change only when it became apparent that America was seriously threatened by the war at sea. In August of 1914, the American Minister of Foreign Affairs sent a memorandum to each of the warring countries which referred to the serious danger facing neutral countries as a result of their uncertainty in the guarantee of their rights at sea. England agreed to follow, with some exceptions, the stipulations of the London Conference, whose declaration it later rejected. The diplomatic controversy which arose over the subject between America and England revealed that the latter was not inclined to support the American view that the oceans were not

sole property of the Old World. England's stance came down to the fact that it would not agree to the reduction of its naval fleet until its neighbors agreed to the reduction of their armies.

Further into the conflict, America's primary goal became bringing about the end of the war as quickly as possible. On 27 May 1916, the president spoke to the League of Peace and outlined the principles on whose basis peace would be established. At this point, it became obvious that America was not disinterested in the conditions of the future peace; this is revealed especially in the following section of the president's speech:

We so honestly believe in the necessity of the triumph of these ideas, that I am certain I express the thoughts and wishes of the people of America when I declare the willingness of the United States to participate in every possible alliance of nations, to attain these ideals, and to safeguard nations against any abuse.

It goes without saying that this speech was greeted with enthusiastic approval by the American people, who had decided, under the influence of certain events which demonstrated to them that without participating in the war on land it would be impossible to protect America's rights at sea, that it would be best to fight for these rights. However, the point of America's sword was not directed at Great Britain (as could be expected in lieu of the above-mentioned diplomatic polemic), but at Germany, and this is why: there was no agreement between the States and Germany about arbitration, which was Germany's own fault; it was very vague on the matter, as it was in the rejection of the offers of the Hague conference. Moreover, a conflict with Germany had already occurred when that country had protested that America, in the role of a neutral nation, should not have been providing weapons to warring countries, although Germany itself had given arms to countries in the Balkans not long before. The righteousness of America's conduct in this case could not be questioned, not only because countries in question, not being prepared for war, would find themselves in an unfavorable position, but also because America, in the event of war, would not be without support. Of course, in the end, this state of affairs would allow the nation with the greatest number of arms-manufacturing plants to rule the globe.

The establishment by Germany of a self-declared zone within the boundaries of which it could sink any commercial ships had also served as a

provocation to the Americans. When the *Lusitania* was sunk, causing a large number of deaths, the indignation of the American people knew no bounds, and yet this terrible incident also blew over at no cost to Germany. However, further incidents, including derisive and negligent treatment by the German government; conspiracies and improper relations with German diplomatic agents within America; Germany's intrigues in Japan and South Africa, which were meant to turn those countries against the States; and even attempts to instigate revolt in Cuba, Haiti, and San Domingo, had distinctly provocative characteristics and greatly tested the patience of the United States.

At the ceremony of presentations in appreciation for the maneuvers of the Austro-Hungarian and German troops and navy, there were neither apologies from the German government, nor punishments for the participants; indeed, the commander of the submarine that sank the *Lusitania* received, from that crowned degenerate Wilhelm, the highly prestigious medal *Pour le Mérite*. Meanwhile, the actions of submarines against American citizens further aggravated the situation, and on the 2 February 1916, the president of the United States stated, in a speech given at Topeka, that if the rights of American citizens were contravened it would become necessary to revert to forceful action.¹ Negotiations that were taking place as a result of the sinking of the ship *Sussex* partially appeased the United States, and war was temporarily prevented.

Near the end of 1916, President Wilson planned to draw up a note in which he would request all warring countries to outline the purpose of the war. Germany discovered his intention, and, along with its confederates (Austro-Hungary, Bulgaria, and Turkey) addressed itself to the Allies with a request to begin peace negotiations. In reality, this was hardly a peaceful offer; it had the tone of a conqueror asking a favor of the conquered. Moreover, neutral states found themselves threatened by Germany, for they did not participate in the peace process and so could not prevent Germany and its allies from declaring themselves free from all responsibilities pertaining to the protection of the interests of neutral countries. This meant that German submarines would have no scruples about sinking neutral ships. The distribution of President Wilson's note addressing the purpose of the war clarified the situation. Germany gave a rather indefinite reply, but the responses of the Allied Nations were honest and to the point, and agreed with the stance taken by the American government.

Then came Wilson's famous speech to the Senate, which was addressed to both the American people and the citizens of the world, and in which he outlined the conditions of the future peace accord, as approved by the United States. The essence of the speech was as follows:

A pact cannot be stable unless it is in accordance with the principle which states that governments are directed by the people and express their desires, and that the transfer of people from one governor to another, in the manner of a material possession, is equivalent to the deprivation of civil rights.

What were the consequences of this address? The media in France and Great Britain, as a reflection of public opinion, approved of the points that had been brought up in Wilson's speech. In Germany, on the other hand, there were either very few who actually supported Wilson, or else they did not have the opportunity to express their views. Besides, on 16 January 1917, the German Minister of Foreign Affairs sent an order to the German envoy in Mexico, in which he wrote of Germany's refusal to be involved in any discussions of the sinking of the *Sussex*, and offered New Mexico and Arizona to the Mexican government if, along with Japan, it would attack the United States.

It cannot be doubted that these fresh incidents turned the Americans more and more against Germany, and set the stage for cooperation between America and the Allies. Indignation naturally grew in the States over Germany's naval piracy and cruel conduct in occupied territory. Things could perhaps have still been resolved without war, but it so happened that in January 1917, Count Bernstorff sent the American Minister of Foreign Affairs a memo with the declaration that the German government planned to increase submarine action, contrary to the wishes of America. German/American diplomatic relations were subsequently suspended. Even then, the president was reluctant to start a war, and told the Senate: "We are true friends of the German people, and wish to remain on peaceful terms with the government that speaks in their name." This statement, however, had no effect; America's demands were not met. In the beginning of March, it became evident that the German government planned to put its threat into practice, and so the president asked the Congress to approve America's becoming an armed neutrality. Five hundred of the 531 members of both houses backed this

measure, and on the twelfth of March there followed an order to arm American trade ships. But Germany did not relax its piratic politics in relation to the trade ships of all nations. As a result, on 2 April 1917, Wilson spoke to the Congress once more with the proposal to declare war with Germany, and on the sixth of the same month the lower chamber voted in favor of a decision which had earlier been accepted by the Senate:

Taking into account the fact that the German imperial government has made many openly hostile movements towards the government and people of the United States of America, the Senate and the House of Representatives of the United States of America have decided, upon meeting in a general sitting of the Congress, to formally declare war—which Germany has thrust upon America—between the German government and that of the United States; this gives the president full authority to use the naval and armed forces of the United States, as well as the other resources available to the government, in waging war against the German imperial government. The Congress of the United States guarantees, by this very act, that the collective strength of the country will be directed towards the victorious resolution of this conflict.²

It is obvious that, in order for the outlook of the peace-loving American nation to change, and for the relations between it and autocratic Germany to be determined by a collective militant reflex on the side of the American people, there must have been a systematic reinforcement of the mood in America through constant aggravation on the part of Germany.

Thus, this example clearly proves how a social movement is sustained and developed in a certain direction under the influence of the incentives and stimuli which support it, and eventually cause the destruction of all restraining factors; the law of inertia is conquered, and a peace-loving nation, excited by military feeling, sends its sons to a bloody battle fought on the distant European continent.

Notes

¹ See *How Did America Become Involved in the War?*, June 1918. (Brochure in Russian.)

² *How Did America Become Involved in the War?*, July [sic] 1918. (Brochure in Russian.)

3

The Law of Gravitational Attraction

In my work, *On the Basic Laws of the World*,¹ I prove that laws such as that of gravitational attraction are not only valid for the mechanics of heavenly bodies and inanimate nature, but also for the animal and plant kingdoms and, likewise, for the social world.

In biotic nature, we meet with the law of gravitational attraction in phenomena such as positive taxis and positive tropism. These phenomena are so well known in biology that there is no need to go into detail about them here. However, it is important to understand gravity in the biological sense as more than simply mechanical gravitational pull, since any cooperation or symbiosis is a special kind of biological gravitational attraction.

Symbiosis is also quite well known, and needs no detailed explanation. Some facts must nonetheless be touched upon. Careful, in-depth research has proved that the transformation of nitrogen into nitrates is aided by root nodule bacteria. There are also many investigations into the mutually beneficial relationships between various species of mushrooms and woody forest plants. Thus, mycelia, weaving themselves around the roots of woody plants, protect them from dehydration, cold temperatures, and parasites. Moreover, they help the root system in that they extend it further and increase the rate of its absorption of nutrients. In growing together with the thin, secondary roots and in entering their tissue, the threads of the mycelia play the role of root hairs which absorb water and associated minerals into the root system.

Bernard proved another important fact in this area: that the mycelia of mushrooms which are cohabiting with a plant are essential to the latter for the germination of its seeds. These do not germinate, even in the most favorable conditions, in the absence of the mycelia whose fungi are subsequently parasitical to the plant. Especially good examples of this can be found when observing orchidaceae. Similar phenomena are known of the lycopodia, which reproduce via spores (Traube). Aitkinson's, Metenius', and Grevilius' observations lead to the same conclusions about blackthorns, sometimes classified as ferns.

All mushrooms are essentially parasitical plants. They live and breed on the body of one plant or another because they can only survive with its help. From this naturally stems the mutual attraction of the two plants. They are connected by their living conditions in the same manner as two stars or two planets are connected in the solar system.

Not less known are the symbiotic relationships in the animal kingdom. But here there are different conditions defining the mutual gravitational pull, or attraction, between organisms. These are the cooperation and mutual help that animals give each other. Interesting details on the subject can be found in Kropotkin's work, *Mutual Aid as a Factor in Evolution*. It would be impossible to imagine the animal world without cooperation or mutual aid, for this is what determines the mutual attraction of one individual to another.

In this way, the law of gravitational attraction is also applicable to the social world.

The instinct of self-preservation leads to the development of socialization, bringing individuals closer not only mechanically, but also binding them in a collective, in which the mutual attraction between individuals is expressed. Some, such as Waksweiler, call this attraction "social affinity" (*l'affinité sociale*). However, no matter how we interpret that "social affinity," relations between people, which are inevitably formed by nature during child-rearing, lead to the development of an attraction to those similar to oneself and to a habit of reacting to their influence in a particular manner. And since the influence of those similar to ourselves is most often expressed in mutual need, then it is clear why, in any place where there are mutual interests, social units are formed very easily. Moreover, unity is brought on by a need for protection from the unfavorable influence of an aggregate of forces:

Travelers under threat of attack draw closer, band together, and fend off the attackers with combined forces; in the case of danger, many formerly unacquainted people come together to form a single group. The joining of people in permanent offensive and defensive unions, starting with the agreements of two or three individuals, and ending with the conjunction of cities, regions, and kingdoms, is a process that is caused by certain biological conditions. These not only drove people together, but also created a certain order and organization in human relations.²

In general, individuals who have similar internal and external characteristics are mutually attracted; on the other hand, it can be said that dissimilar people, who are, nonetheless, mutually complementary, attract. (This category encompasses the mutual attraction of opposites.) In the above cases, it seems that the useful outcome of a union is somehow instinctively foreseen—in the first case from the union of identical forces, and in the second from the combination of mutually complementary forces.

Thus, the collective truly owes its conception to the law of gravitational attraction, with the unification of the collective occurring, in one case, thanks to the similarity in character, opinion, or goals of its component individuals, and in the other thanks to the union of unlike, but mutually complementary, personalities. However, no matter what the main conditions influencing the education of a collective are, it cannot be doubted that the basis of the above-mentioned phenomena is the mutual attraction (pull) of individuals to one another.

The application of the law of gravitational attraction to the collective personality is characterized by the fact that in social life we encounter certain gravitational centers, which can be large or small. The latter also experience an attraction to each other, with the strength of pull depending on the size of the centers, as can be observed in the gravitational attraction of heavenly bodies. However, here the size of the center is less important than its social significance. Obviously, the attraction between these centers is dependent on the distance and ease of communication between them; here, also, is applicable the formula which states that the gravitational pull between two bodies is inversely proportional to the square of the distance between them.

What, then, defines the strength of attraction of certain centers in the social world? There can be no single answer, since even the attraction itself can be

determined by such diverse factors as the geographical, economic, political, or scientific/literary conditions in a given area.

Obviously, the geographical conditions influencing gravitational attraction are ones such as the convenience of transport systems leading to a given center, the proximity of a center to areas of greater development in extractive industry, etc. Economic conditions of attraction include the location of a place in relation to trade routes and its nearness to areas encompassing manufacturing industries. Political conditions would encompass the location of internal government and of political leaders' living quarters. Finally, scientific and literary attractions are comprised of conglomeration of scientific or literary heritage, as well as scientific discoveries and inventions.

Indeed, every discovery or invention immediately attracts individuals, including those who wish to learn about it or discuss it, and the more important the invention, the greater its power of attraction. If a scientific discovery is of an experimental nature, it attracts people with the same interests who wish to look over and make observations on the discovery, listen to lectures, etc. If the discovery is technical, plans for its exploitation are made immediately, and people interested in this new concept gather to work on it. If a new theory is devised and taught in one of the higher institutions of learning, individuals flock to that school. If there is a new trade venture taking place, which is in itself a form of discovery, it similarly amasses a number of participants and supporters. Furthermore, any factory or plant attracts a range of people who function either as organizers or as workers. Any school, as a discovery of an imitational nature, becomes an attractive force for young people. Because of this, a large city, which contains many of the above factors, acts as a strongly gravitationally attractive center.

Thanks to these conditions, centers of gravitational attraction are those places on the globe which collect the most institutions that cultivate discoveries and inventions in various areas of human activity, from the economic to the scientific and artistic. It is due to this law of gravitational attraction that world centers of civilization are created.

Depending on the creative activity of a nation, its living conditions, and its cultural level, various centers of civilization have been created through the ages. Thus, at one time, the centers of civilization were at Babylon and Nineveh, at another time at Alexandria, later in Athens, and finally Rome.

In the middle ages, many cities—but primarily Paris—could be considered such centers. However, every country has its own centers of civilization,

which attract its population.

Notes

¹ See *issues in the Study and Education of the Personality* (in Russian), issues 2 and 3, 1920-21.

² P. Sorokin, *System of Sociology* (in Russian), Part I, p. 269.

4

The Law of Repulsion

As well as being influenced by the law of gravitational attraction and cooperation, social life is subject to the law of repulsion and confrontation. Every society, from the moment of its conception, begins to differentiate or break up into various parts; one form of differentiation fosters the division of labor and therefore promotes cooperation and rapprochement, while the other form is characterized by the diverse ambitions of individuals in a society. The consequences of the last type of differentiation include mutual repulsion and dissociation, and later rivalry, competition, and confrontation. This is why no society can evade rivalry, competition, or confrontation; however, even these conditions can eventually lead to its success and improvement. In a developed, mature society the very structure of social life brings about confrontations between parties, in which each one strives to hold the dominant position.

Thus, every social enterprise breeds many individuals who aspire to be at its head and, to that end, endeavor to display their most favorable characteristics. This, in turn, spurs other candidates into action.

The same can be observed in all areas of manufacturing work, where we often meet with repulsion and dissociation, leading to rivalry and competition. This is what makes every company try to improve the quality of manufacture, or at least increase the distribution of its product, to attract the attention of consumers. Furthermore, rivalry and competition in the work force of industrialized nations bring about not only the intensification of manufacture, but also the development of a proletariat—thanks to the

increased number of would-be workers—and then the devaluing and thoughtless exploitation of these laborers.

An identical process of mutual dissociation and repulsion, followed by rivalry and competition, can be found in any line of work, including the pedagogical and scientific, since the followers of these disciplines strive to achieve a higher standing and better results than the others of their profession. Working alongside discoveries and inventions, the rivalry and competition have—in accordance with the law of repulsion—decentralizing influence on the collective personality. This influence is dependent on the necessity of propagandizing and popularizing every discovery and invention. Propaganda diverts some of the populace from the larger centers, spreading it throughout the surrounding areas. This is accompanied by the dissemination of the given invention in the interests of making it accessible to rural sections of the population, which allows for the creation of smaller, secondary centers. The location of these centers depends on certain regional conditions, patterns of communication routes, etc.

In this way, along with the law of gravitational attraction, inventions and discoveries themselves exert an attractive or repellent force, and consequently every composite personality attains a certain equilibrium between both forces. This, in turn, determines a population's distribution over the earth's surface and its grouping in one area or another:

All the aspects of society that we now consider to be accepted, proven, habitual, were once the subject of heated debate. There is not one peaceful organization which did not originate from discord. Grammar, poetry, predominant industry, the legal code the constitution, the catechism, all this, which makes up the basic foundation of society, was slowly developed through societal dialectics. Every grammar rule is in essence nothing more than the expression of a habit of speech that was able to triumph over all others. Every article of the code is an agreement, a peaceful treaty following bloody civil conflict, aggressive polemic in the press, and oratorical tempests in parliament. Every principle embodied in the constitution triumphed only after many revolutions, etc. (Tarde, *The Laws of Imitation*, pp. 173-174)

Dissociation and rivalry, based on the very narrow political interests of individuals, can be traced through all of human history. Confrontations

between families and tribes; battles fought in the initial stages of social development by the younger members of society to overturn domination by the father, or head, of the family; struggles of the individual against caste or class rule; constant confrontations over economics; the struggle of the individual against assimilation into mass society; battles for liberating the individual from slavery, then from economic and religious servitude; conflicts between parties in every caste, class, and organized society; internal strife in families, artels, communes, co-operatives, parties, parishes, indeed in any social group—these are many, but far from all, of the types of mutual repulsion, rivalry, and confrontation that inevitably develop with the mighty strength of an immutable law in the conditions of social life. Yet at the same time, the law of repulsion and dissociation, acting as a stimulus of competition and confrontation, serves as an important factor in the development and improvement of the human personality.

Thus, although imitation and borrowing, which determine the degree of similarity, that is, the basic condition of mutual attraction and conjunction between peoples, encourage nations to unite, this alliance is hindered by dissociation, competition, national prejudice, historical traditions, national, class, or party pride, etc., all of which correspond to the force of repulsion in mechanics. The basic reason for this repulsion and dissociation is the natural and social inequality between people, of which the latter can be eliminated by the spread of culture.

In the words of C. Letourneau,

man can be called a hierarchical animal. With the exception, perhaps, of the utterly primitive hordes of Tierra del Fuego, inequality exists in all human societies. Everywhere, the weak are dominated by the strong; however, in societies with a complicated structure and organization that depend on the inherited transferal of privileges, the power of the strong is replaced by the power of the rich. In this manner, the function of the fight for survival, as it exists in the rest of the organic world, is distorted, and often the weaker but more privileged individual conquers a stronger and more intelligent adversary. This situation is especially evident in societies with strict divisions into castes, but our contemporary states are, to a considerable extent, still rife with similar instances, even the more democratic ones. (C. Letourneau, *Morality*, p. 347)

From the above, it can be seen that, in the case of humans, intellect—or the higher forms of correlative activity, which lead to wealth as a result of innovation—is the best weapon to employ in the battle for survival, and has replaced strong teeth, sharp claws, and healthy muscles. However, the evolution of intellect, a result not of individual, but of social life, since only interaction and the exchange of personal experience encourages the development specifically of intellect, does not exclude the possibility of physical conflict. This is proved by the countless wars humanity has waged throughout its entire history, a practice which it, sadly, cannot seem to abandon. To humanity's shame, it has not yet produced the most important innovation, one which would forever terminate wars between nations and conflicts between individuals.

Competition and rivalry occur not only between individuals, but also between various institutions. Every government institution competes with and rivals other, analogous government institutions. Every industrial or trade firm, every undertaking, every factory is the competitor or rival of similar establishments within a given country. Likewise, institutions involved in intellectual work compete with each other and strive to prove themselves better than their rivals.

Different schools, such as academies, universities, institutes, etc., compete with and rival each other in the same way as do all scientific and pedagogical institutions, not to mention the scientists and academics themselves.

In a word, no matter which branch of work we look at, we inevitably meet with competition and rivalry between institutions and their representatives. However, in this competition or confrontation, in this seemingly antisocial act, lies the guarantee of social improvement.

The fact of the matter is that, thanks to this confrontation, there occurs a social selection that protects and provides for the development of socially valuable institutions, and precipitates the extinction of institutions that do not answer to social needs. This social selection, in contrast to Darwin's natural selection, does not depend on any mutation within organizations and is not based on adaptation to the natural conditions of existence, but rather has as its object one principle: social usefulness, which determines the fate of any social institution. This institution is preserved for posterity not as a result of the biological inheritance that is so important in Darwin's natural selection, but as a result of social inheritance, which I elaborate on in another text.¹

The above proves that the law of repulsion, brought about by competition, rivalry, and conflict, is one of the basic laws of social improvement, and if without social rapprochement and cooperation there can be no realization of social goals, then without competition, rivalry, and conflict, which are the basis of the law of repulsion, there can be no improvement in the social world.

Notes

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- 1 V. Bekhterev, "Social selection and its biological significance" (in Russian), *Vestnik Znaniia, Nord u. Sud*, 1912. V. Bekhterev, "The significance of hormonism in the evolution of organisms and of social selection" (in Russian), *Priroda*, 1916.

5

The Law of Equal and Opposite Reaction

The subject here is a law in agreement with which social movements, led by various social groups, interact in such a way that the growth of one social movement stimulates opposition from other, contrary movements. Thus, the vigorous growth of nationalism instigates the growth of socialism, and increasingly powerful authority is accompanied by the strengthening of opposition. With the development of anarchy comes a desire in other groups to uphold the governing power and establish unity.

In all situations of this kind, opposition gains momentum at the same rate as the given social movement grows.

Sociopolitical life is defined by the aspirations of various parties or circles and usually follows a zigzag path, leaning first towards one party, then towards another, depending on the party's influence. There have been moments in history when inter-party conflict has led to a social crisis.

Every new social movement meets with some form of opposition from certain social groups. But when one movement gains the upper hand, the combined forces of other parties begin to act against it. Thus, despotic rule is the lead-up to revolution, and a revolution, in turn, brings about a counterrevolution.

Any discovery or invention in general, once made, strives to be disseminated in its immediate environment; then it gradually spreads further and further, meeting with inhibition in the form of the conservatism of society and opposition from the established inventions that it encounters. Let

us say that the invention in question is the electric light. It is known that its distribution was, and still is, inhibited by the conservative public, and also by opposition from representatives and supporters of gas- and Auer-lighting. In the end, the analysis of any analogous case will show that reaction is equal to action, and opposite to it in direction.

If there is pressure exerted by the government on certain cross sections of society, there is also opposition from the social circles involved in the conflict, which grows in proportion to the extent of the reaction displayed by the governing power, and as a result the inertia of the masses plus the action of the opposition is balanced by the reaction of the government. Over and above the inertia of the masses, every revolution meets with resistance from counterrevolutionary, less radical elements; as a result, reaction must be equal to action and opposite to it in direction.

In conflict between parties, the power or dominance of one party is resisted by others, and the first will continue to expand its influence until the opposition of the other parties becomes equal to its power. Similarly, in war the action of one side will outweigh the reaction of opposing forces until this reaction becomes equal to the action.

Socialist tendencies arise in society as a reaction against the bourgeois; internationalism and pacifism develop as a reaction against war.

If, with the passage of time, patriotism is weakened in some sections of the population under the influence of internationalist views, then it is inevitably strengthened in other sections. Thus, when Russian soldiers refused to fight under the influence of Bolshevik internationalist propaganda, death battalions, officers' battalions of honor, and women's battalions arose as a reaction against Bolshevism.

In general, with any reinforcement of a social movement, another opposing movement develops as a counterweight. In this way, every time the equilibrium between separate sectors is disturbed, a movement is established in society which attempts to correct the disturbance through reaction.

Conflict between parties is also governed by this law, and whenever one party gains the upper hand, other parties rise up against it and, if necessary, form a bloc. On the same basis, the governments of constitutional countries must accede to the views of the parliamentary majority, which is formed either of one governing party or of a bloc of several parties, which are, in turn, in conflict with the opposition.

Here, all efforts of the opposition are directed at resisting the majority and gaining compensation in certain areas.

But there are times in history when the state as a whole is threatened, as for example during war; in such instances, all parties seem to merge in order to avert the given danger, and even the parties that would otherwise make up the opposition support the government.

In other words, in order to ward off impending danger, all powers in the state merge into one; there is no disagreement either between parties or between representatives of the people and even governments as to the military mobilization of the country. Moreover, the governing spheres usually reorganize themselves into a so-called ministerial coalition to attain even greater combined power whereby to oppose the external danger.

In accordance with the given law, when a single group has insufficient strength, coalitions and blocs are formed for its aid. The same law is applicable to large collectives such as nations and states. When France became powerful and militant in Napoleon's time, coalitions of other countries rose against it. When Russia gained strength in the first half of the nineteenth century, English, French, Sardinian, and Turkish coalitions opposed it.

When Germany emerged as the next great power, Russia and France united to counterbalance its alliance with Austria and Italy; when the Great War began, Russia, France, England, Italy, Japan, and, later, America consolidated to oppose Germany's four-way alliance with Austria, Bulgaria, and Turkey. All this conforms to the law of action and equal reaction.

As another example of the embodiment of this law, let us take the contemporary Russian social movement known as *subbotnichestvo*, which lasted from 1919 to 1920. The following was written by myself for the magazine *Zhelezno-Dorozhnaiia Tekhnika i Ekonomika* (Railway Technology and Economics), when the given movement was in the early stages of its development:

Subbotniki, or *subbotnichestvo*, is a social movement that cannot be passed over in silence. Indeed, is it not surprising that, at a time of such instability in Russia, when everyone is subject to already habitual undernourishment and weakness, and complains of the shortage of bread, resources, and all other necessities, masses of people go out on Saturdays—ostensibly a time of rest after six days of work—to perform hard labour,

doing so not for their own benefit, as would be expected, but for the benefit of others and of society. Until now, such occurrences, if they existed, were only singular happenings, brought about by extraordinary circumstances such as sudden catastrophes, which necessitate the input of combined social powers. This is that so-called community aid, which has survived from ancient times in "communal" Russia. But here we are dealing not with a single phenomenon, brought on by the sudden advent of some unforeseen incident, but with a phenomenon that has taken on a systematic form and spread throughout Soviet Russia. What, then, could be its cause? Undoubtedly, the movement is a reaction by the healthy sections of society to the destruction and lack of peaceful constructive labour that we are experiencing.

Subbotnichestvo first arose in the environment which felt the greatest need to demonstrate, by example, how work should be conducted in the "new" Russia; but it is spreading throughout society independently of any party or view. The centuries-long slavery and ignorance of the Russian people, as determined by the previous regime, is the reason why, when the labourers received freedom and even a position of some power, a totally unforeseen event occurred. The fact of the matter is that, upon gaining liberty and rights, those workers who had previously executed unbearably hard forced labour who had so to speak toiled beneath a yoke, began to free themselves from work altogether. This added a more serious type of ruin to the general disintegration that was inevitable after a complete overthrow of traditional social life: a lack of discipline, or "slovenliness," on the one hand, and an evasion of work, or "mischievousness," on the other. This destruction of the labour force, the only guarantee of the progress and creativity of the masses thanks to its ability to overcome any obstacle, was the most dangerous phenomenon of our society at the time, caused by past sins and present conditions. It was leading us to certain death, a fact which there seemed to be no way to avoid; for where a fountain of force and life should have sprung up to replace the old, stagnant ways, there gaped an awful abyss, ready to swallow even the greatest efforts and cloud over the sunniest hopes for a new social life.

However, social life is subject to certain laws or regularities, which are as immutable as the movement of celestial bodies, though not well known or

understood. And so, while the widespread ruin in our country was inevitable in the world of social phenomena, so was the reaction to this ruin, in the form of that amazing and unique social movement, *subbotnichestvo*. In the mechanics of inorganic nature there is, as we know, a law which states that reaction is equal to action, and also opposite to it in direction. That same law is applicable to the "social" mechanics of mankind. Thus the aforementioned lack of discipline has been replaced by the coordination of social forces, which should eventually lead to the evolution of structured social forms. Presently, healthy and productive collective work, *subbotnichestvo*, has replaced evasion of labour. The most intriguing aspect of the situation is this: notwithstanding the widespread evasion of forced or involuntary, and therefore difficult, labour at the beginning of the revolution, we are now, during the formation of a new social life, witness to the rise of freely chosen, collective, selfless, almost heroic, and in any case honourable labour; honourable not in the old, false sense of the word, but according to its true meaning. This collective labour in the form of *subbotnichestvo* is being disseminated throughout Soviet Russia, permeating all healthy elements of society. This is nothing short of initiative, the "great initiative," in Lenin's words, within the productive work of our new life. What makes this social phenomenon amazing, albeit short-lived, is that, contrary to the democratic principle that every labour must have its recompense, this movement demands no payment. Here there is only selfless work, dedication to the well-being of society. *Subbotnichestvo* is both a social impulse and an example of the best way to serve society.

It is interesting to trace the development of the given "epidemic" of labor. This is discussed in various newspaper reports, excerpts of which are presented in an article written by an observer (*Railway Technology and Economics*, August 1919):

It all began with communist railwaymen of the Moscow railway junction on the Kazan' line. At first, a resolution was made which stipulated that every Communist must work two extra hours without pay. By the tenth of May, 182 people on that line were working six hours overtime, and with incredible efficiency; they performed work in those few hours which

would otherwise have taken one week to complete. The productivity of labour reached 270%, with some groups working even on Sundays.

The Kazan' example was followed by the Moscow-Alexandrovsk railway line, where a resolution on *subbotniks* for Communists and sympathizers was made in a general meeting of Communists and affiliated causes, and already, on the 17 May, the first *subbotnik* raised the productivity of labour to three times its original number. Next came the decision to create, as of the 22 May, an additional four-hour working day for Communists; then the movement in favour of *subbotnichestvo* spread to the countryside, first in railway service shops in Saratov, then in Ruzaevo, later in railroad service shops in Tula, still later in Sormovo, Voronezh (8 and 15 June) and at the Maloyaroslavets station (13 and 15 June), etc., etc.

In this way, in the course of one month this epidemic of labour infected many Moscow organizations and spread to many provincial cities, and has presently become an almost universal phenomenon. In relation to this, we cannot overlook the importance of uplift in mood, which not only greatly increases the productivity of labour, but also stimulates people to continue that labour on Sunday. It is obvious that in every type of work there is an honourable impetus, and this case is no exception: the honourable impetus has become infectious, attracting many people and uniting in the name of labour all those who have not lost their will to live. Moreover, when reading reports of the previous week's *subbotnik*, of the increased number of those going out to work, of the participation not only of the Communist initiators, but also of non-Party people, of the greater amount of labour executed—one comes to the conclusion that the framework of a renewed industrious life is being established, and one seems to distinguish the first rays of the dawn of a promising future on the horizon.

This is how we see *subbotnichestvo* from the domestic aspect. But from the point of view of social life, the phenomenon was, as we mentioned, a reaction to destruction or ruin in the form of a possible counterbalance. Thus, *subbotnichestvo*, combined with measures taken by the government to prevent ruin, will develop until the period of decline has been overcome, at which point reaction will finally equal action.

The law of equal and opposite reaction is also supported by incident of collective inhibition, which are comparable to the effect of friction in mechanics and physics.

When speaking of instances of inhibition in social movements, it is necessary to keep in mind that they are as inevitable in social life as revitalizing conditions; inhibitors act as regulators of social movement—bringing it into line, so to speak. What would follow, for example, if a large collective was left without government, that most important inhibitor within any social collective? The answer is that absolute freedom, whose extreme is anarchy and disorganization, which, in turn, inevitably leads back to organization and limited freedom, would result. This is why mankind's aspiration towards ideal and absolute freedom is a "Blue Bird," a dream whose realization can never last longer than a brief moment, and can only occur during periods of social turmoil; this was achieved, for example, under Kerensky at the time of the Russian social revolution. As there is no absolutely free "will" of separate individuals, so there cannot be absolutely free "will" of collectives, since it is in its turn limited by pressure from neighbors, organizations, and the conditions of life in general.

All so-called acts of abstention, endurance, and silence, in other words, all forms of passive and active inertia, which are devoid of or else have insignificant external influence, are part of inhibition in individual life. The latter also encompasses restraint upon receiving insult; abstention from participation in any given undertaking, spectacle, or from the fulfillment of instincts; endurance upon confiscation of belongings; silence when provoked through taunting or personal affront to join a conversation, etc., etc. In collectives we encounter the same phenomena, though on a mass scale.

In the following, we will focus on the development of acts of inhibition in the collective as a manifestation of the law of equal and opposite reaction.

In any society, we have, on the one hand, agreement, cooperation, and assistance and, on the other, disagreement, opposition, and inhibition. The first set of conditions is one which supports and revitalizes social progress, while the second inhibits it or holds it back. Equilibrium between these two factors causes social stagnation; domination by inhibiting conditions leads to decline; and the predominance of helpful conditions leads to progress. And, since at any given moment some forms, being more complete, receive support, and less complete ones meet with opposition, progress is

characterized by the perfection of social forms and the accumulation of social abilities.

The very development of social movements does not escape acts of inhibition.

According to Tarde (*The Laws of Imitation*, p. 130),

Every belief, every need always comes to light with great difficulty, since it must first conquer old habits and established beliefs; then, having overcome these obstacles, it can only be disseminated until its unbridled success breeds a new opponent, which blocks its path and introduces an apparently insurmountable obstacle. New opponents of a given need consist primarily of habits caused directly or indirectly by the need itself. Thus, for a given belief, which is always in some measure false, adversaries are opposing ideas that had originally been encompassed by it, or that had been discovered separately with its assistance. Examples include heresy, or teachings that arose from a certain dogma but are opposite to it and strive to terminate its victorious procession across the globe, as well as scientific theories or industrial applications that sprang from previous theories, and that now disprove the latter or hinder their progress.

The great amount of effort which Christianity expended in order to carve a niche for itself in ancient countries is widely acknowledged. The new religion took precedence over older ones very slowly and eventually reached its zenith during the middle ages, after which its influence and importance in Europe gradually declined, while being replaced by scientific and philosophical ideas in the intellectual segments of the population.

As an example of an episode of conflict between religion and science, one can point out that in Galileo's time the religious viewpoint collided with the new scientific theory of the rotation of the earth; the latter met with great opposition on the part of the former, until it was able to finally and completely eliminate religious superstition.

The same can be seen in the development of revolutionary movements under monarchies. At first, the revolutionary movement encounters strong repulsion from the greater part of the population, but gradually accumulates more and more followers, until it finally triumphs. In Russia, for example, it is possible to observe a complete reversal of outlook and values in the span of

one or two generations. During the hanging of the murderers of Alexander II, the people of Petersburg threw clods of dirt at the twisted, jerking bodies of the condemned Perovskaia, Zheliabov, and others; at the time of the revolution of 1917, the names Perovskaia and Zheliabov were practically idolized, and the populace swarmed with red socialist flags onto the same square on which they were hanged.

It can be said, as a rule, that every new social movement instigates an opposing reaction, whose magnitude is proportional to the extent to which the movement is more radically different from the traditional social structure. If the new movement, notwithstanding the obstacles placed in its path, forces its way forward, then—as with the collision of physical objects—the force of opposition is equal to that of action; however, the opposing mass is usually smaller than the acting one, and so must forfeit its position.

The existing social system is in itself a kind of internal inhibiting factor, which is gradually overcome by a new movement, which acts as a social reflex; and here, too, we observe once again the law of equal and opposite reaction.

In this way, the spread of any innovation in a social environment meets with opposition on the part of any other innovation that has a similar propensity to distribution. The end result could be that two innovations are mutually balanced and both receive equal—though limited—distribution, or else that one of the innovations gains precedence over the others and is most popularized.

Tarde represents the "collision" of innovations as their interference, and compares their dissemination through imitation to radiation (rays). This interference, however, does not always cause mutual resistance to dissemination, but in certain cases leads to the merging of rays and so to their amplification and acceleration. When the interaction is simultaneous, it leads to conflict and mutual weakening or destruction; however, if the interaction is sequential it results in rhythmical progression.

In science, one theory replaces another. Thus, for example, Copernicus' system displaced that of Ptolemy, and the latter, being less successful than the former, was inhibited by it. However, all scientific facts, whether they are eventually found to be applicable to real life or not, are nonetheless kept and collected, since they make up material for evaluation—a reference point, if you will—and are not directly related to human activity. The same occurs in other areas of human activity—in literature, in the arts, and in industry. One

trend replaces another. Romanticism is pushed aside by realism, the classical art of one school is replaced by that of another, and finally all classical schools give way to modernism, while the accumulated wealth of preceding works becomes simply examples of art. In technology and industry there are also certain styles, tastes, and views that replace one another without destroying any of the antecedent, gradually accumulating discoveries.

In the evolutionary process of the above areas, the inhibition of some phenomena and the arousal or revitalization of others can likewise be seen:

The accepted philosophical system survives all the blows of critics until another, rival system arises. No matter how excellent a famous school of art is, it only exists until another comes to replace it. The Gothic style eliminated that of the Romanesque period; the Renaissance style was needed to eradicate Gothicism. Similarly, in spite of criticism, classical drama would have survived had not Romantic drama arisen (a hybrid form, as a matter of fact). An industrial product disappears from the market only after the advent of a new product that answers the same need, or after the lessening of demand in accordance with a change in fashion or tradition, of which the last circumstance could simply be the result of the popularization of a new style (and not necessarily distaste towards the old or new principle) and not only opposition to the former principle. (Tarde, *The Laws of Imitation*, pp. 167-168)

Any invention is difficult to conceive; however, once it is complete, it immediately enters into competition with the tools that perform the same tasks as the invention but do so less efficiently. It gradually replaces these tools, at the same time encountering—from them and from conservatism in the environment—reaction, equal to the action of the invention, but opposite to it in direction.

In this way, initiative, imitation, and competition determine the development and direction of any social movement. Moreover, imitation is directly dependent on the homogeneity of the environment, and is itself a contributor to uniformity, whereas competition is determined by the diversity of its environment, and initiative on the activity of its supporters. Because of this, imitation is easily spread in the living conditions of closed societies, parties, etc., while competition is strongest when one social factor or another

goes outside the bounds of a given society, party, or any more or less homogeneous environment.

The analysis of rivalry does not fall under the scope of our task, since the subject has already been dissected by a large number of eminent sociologists, the first being Marx. We will simply mention competition as a factor of a correlative nature, one which counterbalances imitation and which is no less important in the social environment than imitation. Only thanks to rivalry do social forms change, since imitation on its own could only lead to a single form, which would bring us to question the very existence of society, its life. It seems Tarde also faced this question, but did not resolve it, leaving it open to discussion. On page 33 of his work, he points out the difficulties encountered in resolving the social question:

Would it be good or bad [he writes] if someday there was established complete unanimity of human outlook, attained via the exile, or forcible conversion, of discordant minorities, and will this type of structure ever be realized? Would it be good or bad if commercial, professional, individual, political, and national rivalries were eliminated by organization of labour or by state socialism, by a worldwide federation or by European stability—the first step towards the union of the European states—and is this in store for us in the future? Would it be good or bad, if, once it had removed all control and opposition, a free and strong social power emerged, one which was completely autocratic and had the potential for great deeds, one which had absolute power in the face of a certain Caesar, party, or nation, a government which was the most humanitarian, the most sensible imaginable, and is it even advisable to establish such a system? This is a dangerous question.

The author goes on to say that mankind is heading in the direction of the greatest truth and ultimate power, but the method by which this apogee will be attained is uncertain—whether "by the development of contradiction, competition, and criticism, or by the elimination of all these and by the infinite dissemination (through imitation) of a single will, a single idea which will root itself more and more in the masses" (Tarde, *The Laws of Imitation*, pp. 33-34). Can it be doubted that neither method can either lead to the proposed result, or be realized in practice, since each of them complements the other and leads to the maximum attainable well-being?

It is clear, then, that to the list of inhibiting conditions must be added, in certain cases, competition, which does, indeed, act as a very strong inhibitor for any unstable social phenomenon. Because of this, Tarde is wrong in trying to reduce social life to imitation, since competition, as we know, is far from always an inhibiting force, and is often a stimulus, which was discussed previously.¹

An example of competition in the area of culture could be the displacement, near the beginning of our era, of the cuneiform writing system in Asia Minor by the Phoenician alphabet that had spread throughout the Mediterranean coastal regions. Undoubtedly, this displacement occurred with the military and political successes of the people settling the shores of the Mediterranean, but we know that even conquerors sometimes have an inferior culture to the vanquished, and the subject must consequently be analyzed, for the most part, as the competition or rivalry of two cultures that history has brought together in conflict.

The same can be observed with respect to language. The languages of less cultured nations are gradually supplanted by the languages of more culturally developed nations, since the latter inevitably come into close contact with the former in the role of conquerors.

Furthermore, in languages there is likewise rivalry between different linguistic forms, and, consequently, mutual inhibition, under which one form survives the competition and becomes established and popularized through imitation, while the other is inhibited and disappears completely. Of course, the result of the conflict or competition is tied to possible simplifications of the language, and to the form of its expression.

Inhibiting conditions in the collective are created by the government. In the end, all power—which is most often the result of the predominance of one class over another—is a form of coercion, and without constraint there can be no power. It is nonetheless true that in some cases government is the result of an agreement between two or more conflicting parties. However, in the latter case, it would still suppress the elements that do not partake in the agreement. To realize its plans, a power cannot do without certain instruments, which become a part of it and at the same time are executors of its will.

Whether the power is the result of agreement between parties or classes, or of the predominance of one of them, it has the opportunity to act on behalf of society as its representative. But it is enough for the main instrument of a government—its military—to forfeit its loyalty to the given power, as

happens in most revolutions, and the power is deprived of its basic support and collapses.

Every social movement, once commenced, has a tendency to continue developing in its original direction in accordance with the law of inertia. If, in reality, this does not occur, then this is caused by certain obstacles, both internal and external, that inhibit the movement. The inhibition of collective labor can depend on any new stimulus that attracts social concentration. Take, for example, a collective that is busily executing its usual, day-to-day labor. Someone new arrives, bringing with him an interesting piece of information, and the work of the collective is suspended while he speaks.

Let us next imagine a meeting whose object is the discussion of a certain issue, and the attention or concentration of the collective is focused on the speech of the orator, who is developing, step-by-step, his theme. But suddenly an alarm bell is sounded, or an unusual noise is heard outside, and the collective's concentration, formerly directed at the orator, is immediately transferred to whatever has interrupted him.

Or let us say that a crowd in the street is displaying collective anger towards an individual. However, if the object of the anger is adroit enough to amuse the crowd, to provoke its laughter, then that is often sufficient to calm the collective. In a different scenario, another person might distract the crowd with his exhortations, and redirect the anger of the collective at himself.

In another scenario, we may be dealing with an enraged crowd that under the influence of whatever provocation or rumor begins to commit excesses. The advent of armed forces is enough to disperse the crowd or assuage its rowdiness.

Or perhaps it is a hungry crowd that begins protesting against a person selling produce. It often suffices to find out that the merchant does not even have any groceries, since he has not received them, for the crowd, originally geared for a pogrom, to disperse.

Let us look at another example: a collective is beating someone. As soon as armed forces arrive on the scene, the person, who would certainly have fallen victim to the crowd's fury, is saved.

On the other hand, the primary factor inhibiting a social movement is its encounter with movements of a different character. These are, once again, external inhibitors for the social movement.

When any social movement is developing or growing, it does not usually meet with opposition from other social forces until it reaches the level of

development at which it clashes with the interests of other movements.

At this point, social circles of the "enemy" camp begin preparing themselves for a suitable reaction, which eventually confines the social movement within certain boundaries.

All conditions of social and state imposition can be considered more or less constant or, to be more specific, long-term inhibiting factors in social life.

The above include laws, customs, morals, traditions (even so-called prejudices), class establishments, generally accepted conditions of social morality or propriety, religious demands, established decorum in speech, dress, and manner, the demands of fashion, etc.

Temporarily inhibiting conditions include all those that are consequences of current events.

A social movement can be artificially inhibited through diversion or through direct opposition. Thus, when war breaks out, the opposition, in view of the patriotic mood of the nation and of the imminent danger, terminates its conflict with the government. To this end, governments have more than once resorted to war, guided by the principle that the prospective victories will distract or temporarily suppress a revolutionary movement.

The thirteenth century was notoriously replete with the persecution of the Jews—who were often accused of ritual killings—and was so because the lower classes began displaying acute discontent, pushing the clergy and the nobility to redirect, out of trepidation over the possibility of any independent step being made by the masses, the mutinous thoughts of the people at religious persecution of the Jews. Mythical fables, telling of Jews using the blood of Christian infants for ritualistic purposes, were invented to serve as a diversion of national discontent into a channel that would not be threatening to the clergy and nobility. Judaism was to be the scapegoat for the sins of the governors and ecclesiastics, and so the massacre of Jews began, with murder and robbery stemming from the most insignificant occurrence. Another way of heightening the religious fervor of the medieval citizen was the burning of witches and sorcerers, and the persecution of heretics. An analogous endeavor was embodied, though of course on a larger scale, by the Crusades.

Similarly, an inhibiting factor for national movements is any new social event that implicates the whole of society. This could be a war, an unusual discovery, or even the organization of public entertainment on a massive scale. No wonder the hungry Roman crowd, which was diverted, in the

interests of the government, by various spectacles, cried for "bread and circuses."

We will take for the illustration of our point a quotation from Tarde (*Social Logic*, p. 233), which discusses the achievement of social unanimity:

When, in one region or another, parties or classes are dying of starvation and are at each other's throats, the discovery of a coal mine, an asset for all, establishes accord and good feeling. When a country is divided by internal strife, the establishment or conquering—for a military nation, this is equivalent to establishment—of a colony generates a great amount of energy, which is both pacifying and enriching. Columbus' incredible discovery eliminated or postponed for several centuries many European wars and revolutions; how insignificant, in comparison, seem all the petty colonial disputes that it caused. It provided a vent for the spirit of disagreement, bitterness, and inhumanity which, finding satisfaction in the maltreatment of savages, no longer plagued Europe.

In the realm of inhibition must be placed the following factors: the limitations placed on the opinion of one party when it encounters the opposing view of another party, and also the opposition that is organized against the selfish plans of one representative of power or another; the constitutions which limit the functions of power; and finally the laws that create barriers to the various desires that oppose the basic requirements of communal life.

The suppression of a social movement through direct opposition by, for example, the government or the military, is also a form of external inhibition. In these cases, we are usually dealing with temporary inhibition, since when the power is weakened the movement once again develops.

Thus, we must not forget that taking drastic measures against one social movement or another can have only a short-term effect. Such measures are inevitably followed by more rational ones, and even these do not always result in permanent inhibition, since when the inhibiting conditions are taken away the original movement can be revitalized.

No less valid examples of enforced, i.e., external inhibition of the social outlook are easy to pinpoint in the Russian Civil War (after the Revolution). Areas that were previously under the control of the "Red" regime jubilantly celebrated the advent of the "Whites," even showing a certain amount of

enthusiasm. However, as time passed, the populace began to despise the new, harsher regime, and welcomed the return of the "Reds." Some regions were transferred from power to power, and underwent this change of affiliation, many times.

Aside from external inhibition, it is equally important to look at internal inhibition, which is made up of regional conservatism, of habits and customs, and of routines that hinder the development of any social novelty.

Any collective associative reflex develops on the same ground as a regular reflex and, appearing as the result of an accessory aggravation, is easily subject to internal inhibition.

Let us say that, under the influence of one condition or another, an associative collective reflex is established, satisfying certain individuals. Let this reflex be in the form of the right-wing organizations that were so powerful—and active—in Russia during the period preceding the Great War. With the advent of war and the "call from above" requiring all parties to unite, those organizations became more subdued, and when the revolution developed they disappeared completely, since they no longer had a sensible reason for existence, and no longer felt a corresponding gratification.

What happened was, in large part, internal inhibition, as opposed to external inhibition which would be, for example, the result of a prohibiting law or of limitation by administration or force.

Internal inhibitors in collectives are not rare. Thus, in history we often come across conditions that necessitate the manifestation of energy and enthusiasm, which is nonetheless rendered impossible for reasons of internal organization, class disagreement, etc. Such events include the delay of a cultural movement whose followers are torn apart by civil dissension, civil war, disparity of religious beliefs, with skepticism on one side and religious fanaticism on the other, or constant conflict between conservatives and their opposition. It is due to these same conditions that the very struggle of a nation for its historic existence is often unrealizable, since the products of its military initiatives collapse and are lost under conditions which it would be pointless or impossible to withstand. All these situations usually lead to the stifling, and limitation or weakening, of national impetus, and consequently to inactivity and paralysis.

In general, wherever there is action between opposing forces, there is suppression of social movements as well as a prevalence of social deviance.

Even a simple disagreement between individuals paralyzes both of them to a greater or lesser extent.

As a matter of fact, the so-called collective frame of mind or "mood" is also subject to internal inhibition, one sentiment being replaced by another. Let us take an example from the history of Russian society. In the beginning of the Great War, all of society followed even the most insignificant events of the war: every success at the front excited the masses and provided reasons for triumphant speeches at social meetings, whereas military failure was considered a national misfortune, and plunged the majority into depression. However, a period consisting solely of military failures gradually dulled the social reaction to these events.

Even during the first stages of the Revolution, the July 18 offensive gave powerful incentive to the development of the patriotic mood in the country—since the Revolution itself began under a patriotic flag—but that was to be the last burst of patriotism. The following military failures finally uprooted any belief in the successful continuation of the war and provided favorable conditions for internationalistic teachings. This bred an attitude of indifference towards the war, and when the Bolsheviks seized power this attitude reached such proportions that the secession (from Russia) of Poland, Lithuania, Latvia, and then of Ukraine, Finland, the Caucasus, and the Baltic area, passed over with no specific reaction from Russian society, as though it was not the dismemberment of a living social body that was taking place, but rather that of a lifeless corpse.

A phenomenon of collective reflexology deserving particular attention is the natural inhibition caused by executions and shootings, found in the mimico-somatic reflex of a crowd. The shocking effect of a murder on the nerves under peaceful circumstances, and the contrasting gradual accustomization to death which develops during war, is well known. A soldier who is accustomed to battle no longer feels afraid or conscience-stricken; he is excited in conflict and seems almost drunk, disregarding danger. A mimico-somatic reaction, in this way, is gradually dulled through constant repetition, and, in the end, the reaction—even in the face of an individual's own death—is completely inhibited. The constant expectation of death makes people indifferent to real hazards.

Here we are speaking of a more or less total suppression of a mimico-somatic response, resulting from over-stimulation. In support of the aforementioned, we find the following passage in the memoirs of Cattelain, a

member of the Paris Commune and the chief of the Criminal Investigation Department:

In the end, as people must tire of anything oft-repeated, so the soldiers no longer destroyed or annihilated captives, but opened fire on them without fury, without wild howlings, rather in perfect order and with wise economy of cartridges.

Two or three shots from a rifle, sometimes only one, directed without hesitation into the skull, and it was over. Eventually even the victims allowed themselves to be placed against the wall with calm hopelessness. There were only a few modest complaints:

"My poor boy, what is it you want to do, you didn't even give me time to kiss my children."²

Greater indifference in the face of death, on the part of both sides, is difficult to imagine.

Even simple people, seeing themselves surrounded with executions, become as accustomed to the idea of their own death as to any everyday action. Thus, in Orange, the peasants danced upon the scaffold and a fifteen-year-old boy asked the executioner: "Will you hurt me very much?"

The endurance and courage of the leaders of revolutionary movements, who usually die with heroic calmness, go without saying. It is known with what indifference Danton went to the scaffold: "You will show my head to the people, it is worthy of such an honor" were his last words. There were even jesters at the gibbet. Thus, one jokingly wished the onlookers greater success than he had attained; another, at the guillotine, taunted the spectators as well as the executioner.

It is known with what stoicism conspirators die during periods of bloody revolution. Thus, the victims of the "red shirt" conspiracy went to the scaffold, one by one, without revealing their secrets. An even better example in this respect is the Girondins:

The heroic death of the Girondins makes them deserving of immediate immortality. Every measure was taken to prevent them from drinking anything "invigorating" before the execution. This did not prevent the Girondins from singing the Marseillaise, not as a funerary psalm, but as a

victory march, as a symbol of hope for their mother-land. They sang it with enthusiasm, loudly and clearly, on the cart taking them to their doom, and going up the steps of the scaffold. Only the heavy knife of the guillotine silenced their voices forever. (Cabanis and L. Nass, p. 98)

The crowd which witnessed so many executions was no less indifferent.

Sacrifices to the guillotine were inculcated upon society more and more; they began to be habitual, became a national necessity, and the crowd, accustomed to the sight of blood, no longer felt disgust; soon it could not do without such stimulus as the spectacle provided. Mothers brought their children to the scene of the execution, fathers hoisted them to their shoulders, "nurturing in them a feeling of hate towards the tyrants and a sense of courage." (Cabanis and L. Nass, p. 101)

Analogous phenomena can be observed in a crowd on the attack.

In the beginning, the crowd still feels negatively, or at least apprehensively, towards blood, but as soon as the first blood is spilled, the fury of the crowd knows no bounds; blood ceases to arouse disgust: to the contrary, it means absolutely nothing to the crowd, and even excites it, which explains the ridiculing and maltreatment of the corpses of those murdered.

There are many more examples, which, however, need not be given since they are known to everyone from history. It is clear from the above that the accustomization to something which previously aroused a mimico-somatic reflex is in itself an act of internal inhibition.

For the same reason (the surmounting of internal obstacles) any social innovation is not immediately accepted, but enters life gradually through the re-education of society. If an innovation drastically changes the usual tenor of social life, then it always meets with more or less strong opposition, because of habits and routines that are expressions of the law of inertia. So, the greater the expected break in social tradition, the greater the counteraction on the part of society, and the greater the internal inhibition, which for the most part can be stifled only with force. The opposition, briefly mentioned previously, which followed the introduction of the steam engine, the construction of railways, the first attempt at cultivating potatoes, the enforcement of reforms, etc., can serve as an example of this form of inhibition.

National disturbances and rebellions find a favorable environment in identical encounters between established social conditions and innovations for which society is unprepared.

To avoid this, all more or less radical social innovations are most easily introduced through younger generations, which have become neither entrenched in the rut of old traditions nor accustomed to them; consequently, in these cases a certain role must be assigned to the re-education of youth.

In general, every innovation in a social environment that affects the interests of the majority must encounter opposition, whose magnitude depends on how deeply those interests are affected. With the passage of time, the given innovation becomes habitual, and is later defended as something acquired by and belonging to society, despite the fact that it was at one point forcibly thrust upon the latter. Thus, Roman law was once violently enforced among the GalloRoman people, who then recognized the defense of Roman law as their sacred obligation.

Obstacles to the popularization of one innovation or another in the collective include contrary opinions rooted in society, as well as various predispositions, and even national, tribal, or class pride based on the past life of the collective. Depending on these conditions, one collective accepts an innovation more easily than another, in which the popularization of the same innovation meets with much greater adversity. It goes without saying that the popularization of an innovation in the collective meets greater opposition if it does not originate in an authoritative environment, and, conversely, it is accelerated if the innovation comes from prominent and trusted spheres, which stimulate imitation in their surroundings. Similarly, the spread of an innovation from one collective to another is facilitated if it originates in a collective of higher status, as compared to one which spreads from a less important collective, since the former has greater influence than the latter.

In general, internal inhibition is facilitated by frequent repetition of social facts. "As a result of frequent use [says Tarde (*The Laws of Imitation*, p. 216)], a word is shortened, faded, worn, like a pebble bathed by waves; religious belief loses its intensity, art degenerates, etc." The same can be observed in every social movement which, with the passage of time, wanes, losing its novelty, and neither attracts the interest of others nor retains the attention of its own followers.

Because of this, a new stimulus is needed to revitalize the social movement. Even a break in the movement may turn out to be beneficial,

thanks to the fact that its followers have time to accumulate energy, and the movement may eventually excite public interest again.

The previously mentioned collective concentration, like its individual counterpart, is subject to fatigue and internal inhibition. In view of this, collective labor, which demands strenuous concentration, must be accompanied by certain breaks, after which it can continue with renewed energy, thanks to the resultant revitalization. In this case, the point at which inhibition begins depends on the nature of the collective labor, and on its greater or lesser monotony.

Analogously, an orator who speaks unenthusiastically and monotonously tires his audience in a very short time, whereas an orator who is able to diversify his speech in expression and content can entertain a crowd, without exhausting it, for two hours or more. So, to sustain concentration during continuous mental effort, both interruptions and changes in subject matter are employed. When these factors are taken into account, collective pursuits can continue, as is known, for five or six hours without excessive fatigue, and various other collective activities can continue for a much longer time. It is acknowledged that eight hours of physical labor, coupled with appropriate rest periods, is the optimum duration of work; in most cases, this schedule prevents undue fatigue and exhaustion under conditions of continued physical effort. However, overwork depends not only on the number of hours of labor and length of breaks, but also on the strenuousness of the labor and even on the interest in the work.³

If an inhibitor of social life is something which hinders society's natural course of development, then it should not be forgotten that in collective, and similarly in individual reflexology, the same factor can be both inhibiting or stimulating, depending on the conditions in which it is present. Thus, the use of force, such as military execution, often leads to the suppression of a national movement in the first stages of development; however, when national aggravation has reached a higher level, executions can only instigate a renewal of national displeasure and strengthen the national movement. Thus, during the historic strike of February 25, when the Cossacks refused to shoot the populace and shouted, with the people, "Down with autocracy," police were introduced into the army, and the shooting of rioters was conducted from machine guns arranged on the roofs of houses. But it was too late, and the political strike quickly turned into a general one. Every day, renewed slaughter increased national indignation and rooted the national

movement more firmly, so that it eventually gained the upper hand when a crowd of people in the streets mingled with troops, took control of the situation, and executed a coup with lightning speed.

Another example: competition, as we saw, is often an inhibitor for fledgling social movements, while in other cases it is a stimulus. It is known that competition is often a stimulating factor in the development of one social movement or another. Only movements that are insufficiently strong and lack vital capacity can, as was mentioned, be stifled and inhibited, since they are unable to withstand competition; as a matter of fact, it is on the basis of this principle that protectionism is established. In all other cases, competition serves as a revitalizing factor of social movements, allowing for increased activity of the collective, which can then advance beyond its competitors.

In all the above cases, acts of inhibition in the collective create conditions that breed opposition to any social movement, and in the end this reaction always equals action, as represented by the given movement.

In conclusion, let us note that strikes must be added to the ranks of important collective acts of inhibition, having in some cases political and in others economic significance. Here we are dealing with an act of social negativism or with a display of social inhibition that forces previously vigorous social activity to terminate, sometimes immediately. This inhibition is usually manifested under the influence of whatever external conditions are powerful enough to affect the masses, at a time when the latter cannot otherwise oppose such—usually coercive—influences. In other words, when active opposition is unrealizable, its place is taken by inhibition. A strike is an act of great social importance, not only as a defense, but also as an actual form of counteraction, often attaining positive results. One of the most significant, in magnitude and results, is the general political strike that broke out in Russia in 1905. It would be of interest for us to follow in this work the development of the strike that eventually led to the convocation of Russia's legislative institutions.

The general political strike in Russia of 1905 has deep roots, whose growth began long before the strike's eruption. The pointless Japanese war, which demanded of Russia colossal sacrifices, first aroused the discontent of the Russian people. Propaganda against the tsarist government spread across the country. The government would not capitulate, and continued to uphold a very strict regime. But the long-developing discontent with the tsar's regime increased rapidly. In order to paralyze the agitators who opposed the tsar's

rule, the latter organized its own agitation among workers; to this end, "societies" were formed among the workers, at the meetings of which orators, bribed by the government, proved to the people that it was in their interests to depend on the tsar, that they should not believe the tsar's adversaries, that the tsar honestly wished to help his subjects, that the tsar loved his subjects and wanted only the best for them, and that if there was discontent with the tsar's regime in Russia, then it was the fault of over-zealous government officials, etc. One such society was established in Moscow. Another was organized in 1904 in St. Petersburg, on the police's initiative and with the participation of the priest Georgiy Gapon; one of its goals was to distract workers from "criminal" propaganda.

Gapon, who was at first trusted by members of the government and clergy, was an excellent orator, well able to entertain his audience. When he depicted the plight of the worker, tears came to the eyes of his listeners, and they displayed childlike joy when he dictated that workers must depend wholly on the tsar. Needless to say, the tsar's name was, at that time, still associated with a kind of aureole, handed down from father to son in the manner of an inheritance. The number of members of societies and followers of Gapon and other orators grew with incredible speed. Audiences seemed to recognize, in the words of the speakers, their own hidden hopes, and the crowds of workers were overcome with a feeling of excitement.

This propaganda, abetted by the police, did have a reverse side, since to ignite hope was equivalent to letting it burst into flame, and it is quite understandable that the simple masses made their own deductions to that end. If the tsar was the "father" of the people, then he had to listen to his "children," in view of which the idea arose to appeal directly to the tsar, in order that he hear the "needs" of the people and help them in such dire circumstances. It must be noted that the people had previously appealed to the tsar through messengers. In this case, however, it was decided not to depend on messengers, but to go en masse to the palace, carrying portraits of the tsar, icons, and gonfalons.

This idea was realized on the ninth of January. The capital spent a worried night, awaiting grave occurrences, since no one could believe that this direct appeal to the tsar could pass without unfortunate consequences for the unarmed crowd. The situation unfolded in the manner that had been foreseen by those who understood the true state of affairs. The Winter Palace was surrounded with troops that impeded the advance of the workers; shooting

followed, many workers fell victim, and the ground around the palace was stained scarlet with blood. In official bulletins, the wounded were counted in the hundreds, and the dead—in tens; rumors spread of much larger numbers. Witnesses stated that children, who had climbed trees in the Admiralty park to watch the goings-on, fell like hunted sparrows when the shots rang out. The disillusionment among the workers and lower classes was total. Their last hopes, centered on the tsar and based on the belief in him instilled by the orators, had yielded bitter fruit. It became clear to everyone that appeals would not serve their intended purpose, and alternate means of influence on the tsar had to be used. Almost all of the country was in a state of upheaval, which increased to greater and greater dimensions; this led to the replacement of the governmental regime and the outline of a future constitution through the Duma. But that which could have appeased the people, more or less, in the spring of 1905, when opposition to the government could not have reached the corresponding height or tension, was now an insufficient measure after the increase in social strength following the unsuccessful publication of the Duma's legislative act. Since the rights of the people were not properly addressed, whilst the unfortunate Japanese war showed the more unpleasant aspects of the tsar's regime, the national movement developed more and more, and at last, in the circles opposing the government, there came the decision to organize an offensive in the form of a general political strike among the working classes, which was later joined by railway stations and, on the twelfth of October, by the postal/ telegraph workers. The country's heartbeat seemed to have been suspended. Trains were forced to stop when they encountered the strike. The post and telegraph terminated their activities. Factories and plants closed. Trams in the cities stopped running, and stood—or even lay—empty in the streets. Needless to say, the strike threatened the well-being of the entire country, and the government understood its sinister meaning; as a result, the act on the powers of the legislative Duma was quickly drawn up, and on October 17 the corresponding manifesto was publicized, a manifesto which gave wing to the hopes of the stifled Russian people.

At that moment, the strike ended. Trains resumed their functions, wheels turned once more at the factories and plants, trams made the rounds of city streets once again. In this way, the act of the seventeenth of October served as a lubricating oil that did away with the inhibition in the wheels of the social machine, which had ground to a halt when social needs were not met.

From the above, it is clear that the law of reaction being equal to action is as immutable in social life as in mechanics. In society, it encounters a more complicated situation and so is not manifested in such simple forms as in mechanics, but this does not change its essence and lessens neither its significance nor its importance.

Notes

- 1 Aside from imitation and rivalry, every new factor requires an initiative or stimulus, without which no discovery, invention, slogan, or appeal could exist. This is the role of personal initiative in the social process. But initiative only becomes useful when it encounters imitation by and rivalry from others. In this way, initiative, imitation, and rivalry all determine the social process or social movement.
- 2 P. Cattelain, *Mémoires inédites du chef de la Sûreté sous la Commune*, Paris, pp. 231-37.
- 3 See V. Bekhterev, "The main tasks of the reflexology of physical labor (in Russian), *Issues in the Study and Education of the Personality*, issue 1, 1920.

See also my presentation at the conference on the scientific organization of industrial labor, which was given in Moscow in February of 1921, and which can be found in the conference proceedings.

6

The Law of Similarity

There is a theorem in mechanics which states that two geometrically similar systems, whether they be dimensionally different or not, will always have geometrically similar movements, so that even after every possible displacement they will still resemble each other.

This theorem flows naturally out of the law of similarity, which can likewise be observed throughout the organic world. Thus, we know that, thanks to the relatively constant basic composition of the living matter of the protoplasm, all basic functions such as nutrition, breathing, and digestion, are similar. Everywhere in the organic world there is a commonality of organism structure, represented by cell colonies and the greater or lesser centralization of parts of the body. Chemical and mechanical processes, in turn, are common to all organisms. Even external defense mechanisms, such as the production of poison, are of a similar character in such dissimilar organisms as the bee and the snake. Finally, even the ontogenetic development of organisms proceeds according to one common plan, creating all the forms of phylogenetic development.

In social life, this law of similarity can be observed in various facets of collective activity. Thus, all social organization is defined by one and the same formula. In the collaborative form of social activity, there is always an elected presidium with a chairman at its head; an autocratic form of government has an appointed head which delegates people to execute its will, etc. In different countries, under a wide range of circumstances, we meet with more or less identical forms of worship, while the concept of God is universal. It is known that the mythology of the Old and New Worlds has

common aspects which cannot be explained by interaction between nations, since the similarity between the myths was ascertained upon the discovery of America. Quite a few common rituals and customs were also found between nations that historically had no communication among themselves.

While discussing the resemblance between native Mexican religion and the Catholic one, C Letourneau notes:

This is not the first time we come across the fact that, at all times and in all places, people belonging to different races have done similar things and come to similar and sometimes very unique conclusions, which can only be explained by their common humanity. Recall the ancient Egyptians, the African negroes and the redskins, who cruelly decided to cut off the noses of their unfaithful wives; or the lip *Botoque* of the Botocudos, Nutka-Columbians, and negroes of central Africa, etc. (C. Letourneau, *Morality*, p. 227)

Furthermore, the historical development of civilizations seems to follow a single plan. The hunting lifestyle of primitive nations was exchanged for a nomadic one after the domestication of livestock, and this nomadic period was replaced by agriculture with the advent of increased security. In terms of the socioeconomic order, the slave-owning system was replaced by the feudal one, the bourgeois-capitalist regime replaced feudalism, and finally bourgeois-capitalism was exchanged for socialism.

If we observe the revolutionary movement in Russia from 1905 to 1906, and especially from 1917-onwards, and compare it to the following revolution in Germany and to preceding insurrections such as the Great French Revolution, we inevitably note its conformity to the regularity of complex human activities and, in turn, the latter's correspondence to the law of similarity. Who can deny that the Russian Revolution of 1917 was, in large part, a repetition of the revolution of the late 1700s, notwithstanding the fact that both revolutions occurred in different historical periods, that one of them developed at a time of peace and another in the midst of a colossal world war, that both were realized by nations of different race under different climates and historical conditions.

As for the German Revolution, it was almost a duplicate of the Russian one, although it did not reach the level of development of the latter and advanced at a faster rate than the original. Like its Russian counterpart, the

German Revolution began in the army, with mutiny quickly spreading across the whole Western Front; the murder of officers, the creation of soldiers' and peasants' committees, the division of the nation based on the principle of the self-determination of peoples, and the first coalitional ministry likewise followed.

The German Revolution even had its own Kornilov in the person of Hindenburg; its counterrevolutionary opposition in some military sectors, its Bolsheviki represented by Spartacists; its leader of Bolshevism in the person of Liebknecht; its July uprising and December days in Berlin, etc., and all these events unfolded in the same order as they had in Russia, despite the fact that there were seemingly more than enough reasons for the revolutions to follow different paths. Although the German Revolution developed during the same war as the Russian, Germany was a conqueror nation that could be proud of its military successes and of the extension of its borders; a nation that led an especially disciplined life, and whose people were more highly educated than Russia's; a nation that lived under the aureole of a glamorous military history; a nation that was governed, more prudently than ourselves, by a dynasty that was not weakened by Rasputinism, but rather that enjoyed great popularity in the person of Wilhelm II; all this, it would seem, should have led the revolution in Germany along a different course. And yet the revolution in Germany, nonetheless, developed along the same lines as that in Russia; however, it could not, in view of special conditions in its country of origin, withstand the pressure of the counter-revolutionary movement and so did not attain its goal.

Obviously, this common resemblance of circumstances can only be explained by the fact that the same reasons or laws are active in determining their development. If this is so, then we cannot doubt that this similarity of phenomena must exist in social movements that develop during more peaceful periods of social life, since in periods of revolution the resemblance is simply more apparent, and stands out more obviously than at other times.

Without providing many other proofs of this law in the supraorganic world, let us now point out the congruence between the phases of the ontogenetic development of correlative activity and the various phases of ontegenetic development in general, since a child, throughout its mental development, undergoes all the changes that humanity has experienced, from savagery to cultural enlightenment.

Since the mechanism of association, or brain, has essentially the same structure in individuals of a given type, the so-called associative functions of these individuals develop along parallel lines. This is the basic explanation for the ease with which separate species of similar character, upon communication with each other, are united for joint activity when necessary. This can be observed most often in the communication of humans. What conditions, then, explain this union through associative activity in human society?

Recently, some authors¹ have hypothesized that the unconscious activity of all individuals, which represents the experience of previous generations, is the same; differences are only noticeable in their conscious aspects. This would, of course, help elucidate the common actions of the masses in certain cases. But the fact of the matter is that the inherited experience of preceding generations is manifested in simple reflexes, and not in those "unconscious" processes which refer, as do conscious ones, to associative activity that implies past individual experience. From this it is clear that, no matter how tempting the above hypothesis is, it cannot be useful in the explanation of the common associative activity of a mass of individuals.

In our opinion, the unifying factors of the associative activity of individuals in social groups are the following: direct influence or induction, imitation, then suggestion, and finally persuasion.

The least studied form of mutual influence in the collective is induction, or the direct action of one person upon another. Previous investigations in this field have given, for the most part, unconvincing results. But recently I was able to conduct an investigation, in the form of a laboratory experiment, both on animals and on humans. This investigation, which served as the subject of my talks at the conference on the study of the brain and psychic activity in 1920 (see *Issues in the Study and Education of the Personality* for 1920-onwards), leaves no doubt as to the possibility of direct induction from one individual to another.

If this is so, then it is clear that in a collective, especially one such as a crowd, there can and doubtless does exist direct mutual influence or induction, which should always be taken into account.

Imitation is essentially a natural consequence and further development of the process of reproduction, since any reproduction is self-imitation or imitation of a once-performed act; muscular stimuli that were experienced at

one point leave an impression which, upon revitalization, instigates new actions of a similar nature.

The imitation of others has to do with the reproduction of what has been heard, seen, or felt, with the impulse in this case being an external stimulus that arouses, thanks to previously acquired individual experience, the muscle movements needed to imitate. Imitation is the basic social reflex, just as reproduction is the basic reflex of individual associative activity.

Without speaking of imitation as a means of biological defense in the form of so-called mimicry, it is necessary to keep in mind that imitation is a primary form of communication among all living creatures.

This is why, in all inferior creatures, imitation alone is the basis of social life.

If caterpillars are moving as a clump in one direction, as I have had the occasion to observe, tumbling over each other with identical movements, then this is happening only thanks to tactile/muscular imitation.

If we hear, on a quiet summer evening, the chirping of grasshoppers, then this is also due to imitation. When a swarm of mosquitoes circles our heads, moving up and down in unison, then this is likewise explicable only by imitation.

In the kingdom of the bees and ants, we meet with imitation everywhere. Even when the hive is attacked by foreign bees who begin eating the young larvae, the bees that inhabit it must perform the same cannibalistic act.

If a frog's croak is heard on a marsh, its friends will invariably make their voices heard in response. The cry of a rooster is echoed by its feathered neighbors.

Imitation can also be heard in the chorus of birds. As soon as the "guard" of a group of birds flaps his wings and flies away, the whole flock follows suit.

In the world of mammals, imitation has likewise developed to an incredible extent. Everyone has witnessed how a herd obeys its leader, how the braying of a donkey or neighing of a horse is reciprocated by others of its kind, etc. A massive amount of observations have been conducted on the propensity to imitate among various animals, and the monkey's notoriety in this respect has even become proverbial.

Imitation, generally, is a phenomenon that has spread among all higher animals. It includes such actions as the stampede of a herd from danger, or the hunting down of an animal by the pack. Everyone knows that when one

of the animals of a pack comes upon prey and begins to pursue it, the rest of the pack soon follows. In herds of sheep and other animals, the leader's movements are imitated by the other members of the herd. Thus, his inspection of some object or other incites his confreres to do the same, as does his leap at a certain place. If a dog begins barking in response to a noise, its neighbors will follow suit. The cry of one goose stimulates the honking of the whole gaggle. When a flock of birds settles on a tree, or a group of wild seagulls rests, from fishing, somewhere on a sandy bank, the cry of one bird can be heard spreading throughout the flock, giving the impression of a curious conversation.

All these are examples of imitation in the life of animals, of which many more could be provided, if it were necessary to do so.

For animals, imitation seems to be an instinctive, natural act. For instance, let us say we are observing a chick that has just hatched from its egg. Being separated from its mother, it cannot orient itself among the food, or eggshells, surrounding it, and dies of starvation because it is not "taught" to eat. But how is it possible to teach it to do so? To this end, it suffices to tap your finger, imitating a mother's pecking, and the chick immediately begins pecking also, imitating the movement of your hand. The first piece of eggshell to enter its mouth is swallowed thanks to reflex action; after a few of these imitational ventures, the chick begins executing the pecking motion, which leads to ingestion, on its own.

Imitation is inevitable in the interaction of two similar individuals because it is always accompanied by a sthenic reaction; direct experience has proved that physical labor increases with the observation of synchronous movements. Féré's experiment can be considered convincing in this respect. If the subject performs work on an ergo-graph, and at the same time another person across from him performs the same movements on an ergograph, the amount of work done increases. This sthenic influence of imitation is accompanied by a positive emotional, that is mimico-somatic, reaction, leading to a more or less extended period of reproducing the same movements. This is why imitation is a universal law in the animal kingdom. This situation brings us to the conclusion that at the base of sociality lies not only social selection, but often also a biological or, more specifically, bioreflexological factor that holds a stimulating role in imitation, causing at the same time a mimico-somatic reaction of a positive nature.

In the world of humans, imitation can likewise be encountered at every step, notwithstanding the fact that the process of inhibition plays a much larger role among humans than in the rest of the animal kingdom. Look at children, and you will see how they imitate each other and their elders. The shout of one initiates the shouting of others. If one begins to be naughty, the others will follow his example.

Imitation is likewise at the basis of the development of human speech, gesture, and partially mime, since deaf people remain dumb from birth, but have well-developed mime, while blind people have very limited miming skills, while their other senses are acutely developed.

The entire upbringing of humans, as that of the majority of animals, is based on imitation, thanks to which the experience of elders is passed down to youth. Even in the social life of adults, imitation is found everywhere. If somebody yawns, especially in the evening, everyone begins yawning. It is known what vitality is brought into society by one animated and happy person. A previously bored and listless group begins, under his influence, a lively conversation. The infectiousness of laughter is also generally acknowledged. Certain people cannot see someone crying without also bursting into tears.

On the other hand, imitation through mimicry between people who live together a long time can be so strong that it leads to similar characteristics, as can be observed in a married couple.

It is likewise known that imitation causes stuttering, hysterics, and other nervous conditions.

In some diseases, the ability to imitate reaches its highest level, causing a given word or motion to be immediately repeated (as in so-called echolalia and echopraxia). Finally, imitation is manifested in all complex actions, including work; every student repeats what his teacher dictates, and only gradually does he free himself from imitation by creating something more or less original. This kind of imitation is also widely known. But even complex social phenomena are based, in some measure, on imitation. Thus, the course of the Russian Revolution was, in large part, an imitation of the Great French Revolution. Doubtlessly, the reason for this resemblance was the common conformity to laws of social phenomena by both revolutions; however, the fact that imitation played a role here shows the similarity even between the purely external forms of certain social phenomena.

We will not discuss the nature of imitation, one of the most basic reflexes in biology, for it is only vaguely understood. Let us be content with saying that the impulses registered by sight and hearing seem to ease the reproduction of what has been seen or heard, in the same way as the previous execution of a movement eases its reproduction, and that the facilitated movement is stimulated by an accompanying sthenic reaction.

A special form of unification by means of conversation is suggestion and persuasion. Suggestion is, in the true sense of the concept, indirect imitation, since simple imitation involves the reproduction of one act or another under the influence of its relative ocular, aural, or tactile impulse, whereas suggestion is the fulfillment of one act or another by means of its corresponding symbol (word). The difference, consequently, is that in the first case an external impression stimulates the coinciding act, and in the second the same act is reproduced through its corresponding symbol.

Can there be suggestion among animals? Seemingly, yes; at least, among the more highly developed of them. When a flock of wild birds is busy eating somewhere, for example in a field or marsh, it assigns to one of its members the responsibility of being "guard," and as soon as that guard extends his neck or looks up, the flock is put on the alert; when he gives a worried cry, the flock quickly flies away. In this way, his voiced warning results in a certain action by the flock. What is this, if not suggestion? At least, there is nothing in this case that would isolate this phenomenon from that which we know to be human suggestion.

In this way, it can be acknowledged that animals employ, depending on their living conditions and the usefulness of imitation in their fight for survival, mimicry and other forms of imitation, and later suggestion.

Among humans, we can observe induction, imitation, and suggestion; the latter can be accomplished not only by a cry or gesture, as in the animal world, but also, and most often, through the spoken word.

The mechanism of suggestion, in its turn, boils down to the fact that under certain conditions connected to the interruption of concentration, a word can stimulate the facilitated reproduction of a movement or condition previously acquired through experience.

As is known, science has devised a method of suggestion both under hypnosis and normal conditions. In real life, however, we are dealing with unpremeditated suggestion in the conscious state, which is imperceptibly passed from one person to another, differing from imitation, as was

mentioned, in that the latter entails the reproduction of something seen or heard, whereas the former involves the stimulation of a certain movement or condition with the help of a word, cry, or gesture.

The significance of suggestion in social life is immense, and arises everywhere in a manner similar to that of imitation, often becoming interwoven with the latter by almost imperceptible threads.

I will not set out to discuss the role of suggestion in any greater detail, since I dedicated an entire book—*Suggestion and Its Role in Social Life*—to the subject; it is to that work that I direct the reader.

At this point, we should go on to analyze another process, known as learning, based on instruction and persuasion. Here, we are dealing with an even more complicated imitational process, in which, in order to reproduce a foreign act, an individual neither blindly imitates it nor is directly verbally stimulated to perform it, but first evaluates its significance and usefulness and only then, after a positive outcome of this evaluation, begins reproducing that act. As a result, the act is recognized as having been learned, with the object of learning being not so much the form of the act, but its substance.

When learning is realized, and the learned action can be reproduced thanks to training, then the act can be executed with especial speed and ease upon a certain spoken or written command, or even a certain sign, for example a gesture playing the role of a suggestion.

It goes without saying that if learning, that form of more complicated imitation, is inferior to the latter in terms of its speed of execution, then it is infinitely superior to the same in view of the evaluation that takes place before what is to be learned is accepted.

There is no need to elaborate on the significance of this form of complicated imitation in social life. It suffices to say that it lies at the base of all civilization and culture.

Some symbols doubtlessly owe their existence to metaphorical assimilation, based on similarity and imitation. This fact is not difficult to prove. The movements of a conductor's baton are based on the spatial interpretation of tone and of the extent of the register; colored symbols are based on the assimilation of forms existing in reality. Thus, a red flag is the sign of blood and so is appropriated by revolutionary movements. It is likewise the warning of any danger.

A green flag or banner is the color of growing plant life, and so is associated with hope for future flowering and prosperity.

Black is likened to the darkness of the grave and is, therefore, the symbol of death and terror.

White, conversely, signifies glittering brightness and is appropriated by the monarchy.

Blue, corresponding to the color of the sky, is often a symbol of purity and innocence, and so is an emblem in some religious decorations and rites.

In all these cases, the symbol is a reproduction of another phenomenon that has a certain meaning in our surroundings.

The cases in which certain actions of people have an effect on a given bystander must be categorized as a special kind of influence. Examples of this include a situation in which an infant, seeing one person beating another, begins crying, or in which an adult, observing a horrific scene, loses consciousness. Here there is neither imitation nor suggestion, but nothing less than direct influence in the form of an associative reflex.

A similar example is provided by the following incident. When Mendeleyev represented the Petrograd chamber at the hundredth anniversary of Yuryev University,² he stumbled on his way to the podium. The audience rose as one in respect to his person. This was not an ovation that could be accompanied by public applause. This was silent homage to national pride and fame.

Another example is the reception of Kerensky among troops at the peak of his fame. Here is one of these scenes:

At precisely nine o'clock, the orchestra began playing the Marseillaise, the Cossacks stood at attention, and Kerensky appeared in an open automobile. A crowd of thousands of people roared "Hurrah." The faces of the welcomers betrayed excitement and love towards their revolutionary chief and hero. Kerensky rose to his full height in the vehicle and, holding his cap high, hailed the people. His sincere, spiritual face attracted attention by virtue of its expressiveness and inner charm. When Kerensky's automobile flew through the crowd, *everyone, as though hypnotized, ran after it, constantly shouting "Hurrah" and waving caps and hats* (from newspaper articles).

Influence, as understood in the aforementioned manner, is exerted by means of associative reflexes, in a way similar to the stimulation of associative reflexes with the help of symbolic signs, whether this

[stimulation] is caused directly in the form of suggestion, or in the form of logical persuasion.

This type of associative influence seems to come somewhat later in the developmental process, with the establishment of certain relations between people. Thus, if a child sees a fist-fight through the window and begins crying, then this is undoubtedly an act of induction; but the latter can appear only after imitation in the cycle of development, since it requires a certain amount of experience in relation to even such simple acts as the beating of one person by another. In exactly the same manner, when one person rushes to help another who has fallen or dropped something, he is executing an act of induction that once again requires the corresponding life experience and therefore evolves later than imitation.

As for suggestion, it arises at the point when a child has sufficiently mastered the meaning of gesture and speech, which occurs after the learning of the meanings of simple acts, and so suggestion may begin simultaneously with or, more commonly, after induction.

Finally, persuasion by deduction requires well-refined speech and completely developed logical premises, and so is manifested at an even more advanced age, when experience has accumulated the knowledge of causative correlations between separate phenomena.

In this case, it is of primary interest that imitation and suggestion, as well as induction, lead to coordinated actions of the collective or of a group of people in the collective, since this is especially significant to the social conditions of life. It must be kept in mind that a person's imitation and suggestion, and likewise induction, never remain at the same level. They are dependent on his condition at one time or another, on his greater or lesser concentration, and, partially, on the source of a given influence. Thus, a condition of passivity and distracted concentration is conducive to the development of imitation, induction, and suggestion.

On the other hand, a developed person who possesses fresh strength, and who is consequently not fatigued, can oppose imitation, induction, and suggestion. An invigorating reaction of a mimico-somatic nature always promotes the development of imitation, induction, and suggestion in a direction which is correspondent to the character of the given emotion. It is known that in a crowd, imitation, induction, and suggestion meet with especially favorable conditions, and this is because a crowd's emotive state goes up sharply under conditions of passivity or fatigue.

Also, children and less developed adults are, as is known, more easily subject to induction, imitation, and suggestion.

However, the source of influence itself is not unimportant in terms of the realizability of the influence in one form or another.

Thus, influence exerted on a student by the gesture or word of his friend will be completely different from that exerted by the gesture or word of his teacher; in exactly the same way, the influence exerted on an uneducated person by the gestures or words of someone similar to himself will not be the same as the influence exerted by those of an educated person, etc. Lastly, a person's character is not insignificant in terms of the strength of his influence.

Experience, in any case, leads us to acknowledge that some people are by nature predisposed to imitation, induction, and suggestion.

Finally, the physical and abnormal conditions that increase a person's susceptibility to imitation, induction, and suggestion should be mentioned.

These include such conditions as somnambulistic states, which encompass hypnosis or artificial sleep and some of the general neuroses, for example hysteria, psychasthenia, and others.

Countersuggestion is the condition in which a recipient opposes the influence being exerted on him by suggestion, which explains the habit people have of protecting themselves from outside influence. This opposition sometimes occurs in relation to the influence of deductive reasoning, when a person disagrees, without foundation, with everything of which he would otherwise be easily convinced. He is told, for example, that it would be beneficial for him to take a walk, and he, without sufficient logical basis for his response, says, "There is no need for me to do that," and so on. This habit of negativism in word and deed is the result of upbringing and, partially, of mood.

However, we should keep in mind that although countersuggestion and logical disagreement are not, in the end, insurmountable obstacles to suggestion and persuasion, they can in some measure limit the influence of one or the other.

As for imitation and induction, opposition has no place, or at least is not obviously apparent, in these processes. As a matter of fact, this observation largely holds true only in relation to imitation in the sense of the mimicry of actions or words, and not in the broader understanding that Tarde discussed in his sociological treatise, which included all processes of sympathy, suggestion, and imitation in general.

Under normal conditions it is not always easy to separate suggestion and persuasion, since both processes of influence are often put to use at the same time, in the same situation. We can only say that in one case we are dealing with direct induction upon another individual and, consequently, the provocation of certain conditions, actions, or statements due to the speech or gestures of another person, whereas in the other case we are dealing with the provocation of actions or statements in an individual by means of reasoning or of so-called logical assumptions, which support the necessity or usefulness of the given actions or statements at the present time. These actions, etc., can only be brought about if the person being influenced, after weighing the corresponding pros and cons, agrees with the reasoning of the influencer, i.e., acknowledges the necessity and usefulness of the suggested actions under the given circumstances and makes them the governing principles of his conduct.

In view of this, it is clear that logical influence or persuasion through reasoning requires time, since it is accompanied by criticism, while suggestion is imparted directly, and therefore rapidly; however, the former always corresponds to the interests of the individual who carries out the suggestion, whereas the latter gives the interests of the recipient no consideration.

The fact that imitation is the basis of community was already suggested by Tarde. The study of speech, the first and basic weapon of community, shows that imitation plays such a great role that without it the very development of speech would be impossible. Likewise, upbringing is based in large part on imitation. The latter also allows humans to adopt societal conventions and customs. Thanks only to imitation does mankind become a social creature in the true meaning of the word.

On the other hand, aside from imitation it is necessary to keep in mind suggestion, i.e., the propensity to directly obey verbal influence. Here, we are dealing with mutual influence, by means of the word, of one group of people on others, in a form of suggestion that can be called social suggestion. Already, when being taught at school we encounter suggestion at every step. When a person enters one group, one circle or other, he immediately receives suggestions concerning external conditions of society, customs, and traditions, and involuntarily follows them.

The process of persuasion, which leads to learning, acts on collectives in the same manner as on individuals. When the leader of a crowd explains an event to it, he is not necessarily acting solely through suggestion. A certain

portion of the influence in such cases must be attributed to persuasion, which is also an important factor in the unification of the neuropsychic activity of masses of people. In terms of the significance of persuasion as a unifying factor, the fact that at school everyone is taught, and hopefully learns, the same things is especially important, and helps to level the classes of a given society.

The unification of members of any social group is accomplished with the help of the aforementioned factors, based on induction, imitation, suggestion, and learning (as a result of persuasion). The external impulses that lead to unification are, however, predominantly class and professional conditions that pull an individual into a given circle of habits and customs. These conditions, as was already stated, are the determinants of many external characteristics of one person or another, depending on the group of individuals or social class he belongs to.

In every social group there are certain rules, customs, and traditions that are usually adopted by every new member relatively quickly, thanks to the aforementioned factors that often act in a manner imperceptible to most individuals; however, as soon as a person transfers from one group to another, for whatever reason, he realizes to what extent the conventions and customs of the former group are a part of him, and sees that it will take some time to become accustomed to the conditions of the group which he is entering.

Here, too, the law of similarity is at work. Upon the unification of people, the associative activity of separate individuals has to manifest itself in correspondence with everyone else's and consequently, when it finds itself under new conditions, must regain resemblance to fellow members of the new association.

However, the above facts do not make up the law of similarity in its entirety, since we know of cases in which there is no evidence of communication among nations whatsoever, and so there can be no question of imitation or borrowing between them; nevertheless, we can observe similarities between some of their social phenomena. Examples of this have been given already, including similar forms of worship in the Old and New Worlds, the resemblance between myths of very different nations, etc.

It is clear that in the above cases identical causes lead to identical effects, which conclusion satisfies the basic principle of the law of similarity.

Notes

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- ¹ Vierkandt, *Naturvölker u. Kulturvölker*, Lamprecht, *Moderne Geschichtswissenschaft*, et al.
 - ² Today's Tartu University in Estonia (translator's note).

7

The Law of Periodicity or Rhythm

The omnipresence of oscillation is well known in physics. In essence, no form of energy exists without oscillating motion. The movement of celestial bodies, like the movement of atoms and molecules, has its own cycles; organic nature likewise displays cyclic functions such as growth and reproduction. The vascular, respiratory, and metabolic systems, as well as sleep and wakefulness, are arranged in phases and cycles.

All processes of collective activity are subject to periodicity. Change of mood, stimulation, depression (as a result of fatigue), and other analogous phenomena characterize this periodicity in manifestations of the associative activity of separate personalities. Even the talent of prominent figures is subject to periods of development, prosperity, and decline, and consequently has its own cycle as an expression of a rhythmical form of movement. In the very activity of gifted individuals can be observed periodicity, which is supported by a detailed analysis of their life.

Similarly, a political or social environment displays cyclicity in its development and movements. First and foremost, any collective undergoes its origin, development, then stages of prosperity and decline. Even such a collective as a national organism, strengthened by a state system, has phases of development, during which it increases its power, reaches its apogee in relation to power and prosperity, after which it gradually enters a period of decline and decay. Thus, one form of state system is outdated over the centuries, after which a nation begins to live a new state life—which will also be subject to growth, prosperity, and decline.

In essence, any state that is subject to the law of rhythm undergoes a period of primary development, then of prosperity, succeeded by a period of decline, and then, in a renewed form, the state can once again display its energy and power, attain new prosperity, decline, etc. In this case, the state's military power is also subject to the same law of development, predominance, and subsequent weakening.

Even political regimes are subject to periodic fluctuation, since government reaction is usually replaced by liberal reforms, followed again by reaction and then by governmental liberalism. It can even be said that any political measure, brought to its extreme, prepares its own downfall by inciting an opposing reaction in society.

However, not only a state organism, but any civilization has its periods of development, prosperity, and decline. In this respect, I will quote G. Le Bon's work *The Psychology of Nations and Masses* (pp. 327-29), in which he characterizes the cycles that the ancient civilizations experienced.

At the dawn of these civilizations we see a handful of people, of varied descent, who have joined together thanks to chance emigration, invasion, and conquest. The partial acknowledgement of a single chief is the unifying factor between these people, who otherwise differ from each other in language, religion, and origin.. Gradually, time has its natural effect. The sameness of the environment, the continual interbreeding, the demands of social life eventually leave their mark, and the agglomeration of different individuals begins melding together to form a race, i.e., an aggregate which has common features and feelings that are fixed more and more through heredity. A crowd becomes a nation, one which nonetheless cannot immediately escape from barbarism. Usually, it will rid itself of barbarity only when, after much effort, constant conflict, and countless failures, it attains an ideal. The nature of this ideal is of small importance; it could be the cult of Rome, Athens, or Allah, it could be anything, but this ideal is enough to create unanimity of thought and sensibility between all individuals of a race. Only then, can a new civilization be born, with all its institutions, beliefs, and arts. Carried away by its dreams, a race acquires all that gives it brilliance, power, and majesty... Having completed its constructive work, time inevitably begins the process of destruction, which can be evaded by neither gods nor people. Once a civilization has attained a certain level of power and complexity, it stops

growing and is condemned to decline. Soon its hour of senility comes, drawing closer as the ideal that upheld the spirit of the race is weakened. As the ideal fades, the edifices of politics, society, and religion supported by it begin to totter.

The final surrender of the ideal causes the race to lose its spirit; it degrades into a handful of isolated individuals and reverts to what it was at the very outset of its existence, a crowd.. Civilization loses its solidarity and finds itself subject to the sway of every chance occurrence. Rabble gains power, barbarity arises. A civilization may still seem brilliant, because its external façade still stands, but in reality the building behind the façade is undermined, has nothing to uphold it, and collapses in the first storm . . .To go from barbarity to civilization, by following a dream, and then to gradually weaken and die as that dream is lost—that is the life cycle of every nation.

It would have been better to say "the life cycle of every civilization." After the fall of a civilization, new conditions of a political nature generate a new culture that carries new ideals, often originates at another point on the globe, and comes to dominate the nations of the old civilization.

A civilization that has already undergone periods of development and prosperity, and is at the stage of decline in one part of the world, is often adopted by other nations and revitalized, though on a different basis and with a different nature, under the influence of new impulses. From what was said, it is clear that just as there are encouraging or revitalizing influences in collective manifestations, there are also inhibiting influences, and these dominate consecutively, determining the character of the periodicity of social phenomena.

Even the energy of separate nations is subject to the law of rhythm. After being manifested in one direction, it weakens while trying to proceed in another direction.

It is clear that, where in the physical world we have wavelike and oscillating motion in the form of rhythm, in the world of collective entities we likewise have evolution through wavelike motion. Any society goes from stagnation to activity and, conversely, from exaltation to downfall. Observe the history of any nation, and you will be convinced of the truth of this statement. In any collective in general, a conservative government is replaced

by a more liberal one, which is, in turn, replaced by another conservative government; interest in an innovation is displaced by reversion to the past, to custom, indeed eventually any innovation itself becomes customary or ancient.

Inhibition plays a certain role in this process of transition from progress to regress and vice versa. The manifestation of a single trend eventually leads to a lack of social interest in that trend, which internally inhibits the process of its social development and forces society to take another, opposite direction. Once a direction has been taken by society, it is brought, in accordance to the law of inertia, to those extremes that stimulate opposition and protest in significant sectors of the population, and in the end create a shift in social opinion, inducing social movement to proceed in an opposite direction.

As is known, the sociologist Tarde clarified in detail the role of imitation as an important reflexological element in the life of societies. However, the objects of imitation are creative acts that are usually attributed to separate individuals, although we should not forget the existence of collective, or national, creativity. In this way, the life of society is regulated, on the one hand, by creativity or discoveries and inventions, and on the other by imitation; social movement is accompanied by uplift during a period of discovery and invention and by stagnation during the periods following the discoveries. During this time, primarily imitational social labor develops and facts requiring explanation and synthesis accumulate. At some point during this accumulation, a great mind appears that makes the corresponding generalization, and a discovery arises once again, stimulating social uplift. Social and scientific thinking moves in this wavelike manner, as do all areas of human creative activity.

Any more or less significant social movement develops along an ascending line, reaches its apogee, and then wanes to the point of extinction, after which a new social movement arises, most often born of the old one, but having a new form which, in its turn, develops on an ascending line, reaches an apogee, and wanes, which is characteristic of the periodicity of social movements. And so it is clear that social movements never follow a straight line, but always travel in waves, one replacing another. The increased energy at the beginning of any social movement is exchanged for its subsequent decline, after which the same movement can be revitalized. An uplift in social mood is usually replaced by depression. The aspirations of society in one

direction, meeting with unfavorable or unwanted phenomena, cause social change, after which society strives in a different direction.

Thus, the revolution in Russia reached extreme bias towards Bolshevism during the uprising in Petrograd in the beginning of July 1917, causing a break in the social movement and creating a counterrevolution without having achieved its goal; but the counterrevolution, in its turn, led the social movement to support Bolshevism once more.

It should be kept in mind that in revolutionary periods the waves of social mood flow much more turbulently than under normal conditions. Usually, a revolution explodes when the oppression of the people reaches its pinnacle, and so it is all the more violent if it is intended to free the people from slavery. Everything that is connected in one way or another to the old order is destroyed or abused.

Let us take, for example, relations with the church and religion. As the servant and bulwark of the monarchy, during revolutionary coups religion is subject to abuse and degradation, which increase as the revolutionary movement develops, once again as stipulated by the law of periodicity. To convey at least a fraction of the hate that was displayed during the Great French Revolution towards the objects of Catholicism, we will list here some of the more characteristic events of 1789, involving so-called iconoclasy. We will speak neither of the purely barbaric destruction of bell-towers, monumental church fences, tombs, holy vessels, nor of the utterly pointless burning of fabric and lace. However, we will pause to cast our eye on scenes that are even more vivid in this respect.

Starting in August of 1789, fires sprang up across all of France on which, to the accompaniment of the wild screams and dances of drunken rabble, thousands of priceless objects were burned: marble tables, fireplaces, mirrors, windows, stained glass, statues, carved chairs, and suchlike.

"Igniters" of castles did not even spare the ancient trees growing nearby; all were cut down and burned on the spot.

Iconoclasts attacked not only images of "former" angels, "former" Christs, and "former" saints, but also canopies, gonfalons, candlesticks, lamps, cups, vessels, bowls, and all the decorations used in customs of the "former" Catholic cult. Local workers were forced to aid the "cause"; they

were made to abandon their work in order to plunder the lands of the church and nobility.

Thus, here we are dealing with the rage of a frustrated people against the products of human labor and talent generated over many centuries.

Analogous phenomena during revolutionary periods can be observed in relation to art. It must be said that one of the more saddening circumstances of national upheaval is the vandalism of art, when valuable masterpieces are destroyed to satisfy no real need; many treasured objects have been destroyed in this manner in the bonfires of revolutions. Does this mean that leaders of revolutions direct the people against art? No, emphatically not. It is known, for example, what efforts the Convention made to revive art and in general give the masses all that had been the sole privilege of the wealthy, holding grandiose public ceremonies to this end, and yet all this time the people continued to vandalize objets d'art. During the Russian Revolution, the Bolsheviks strove to give the proletariat everything that had been the exclusive prerogative of the upper class, establishing the actual and legal rule of the proletarian classes and poorest peasants. Some of the latter's leaders gave lectures on the importance of art, created new museums from former palaces; but was it not at this time that palace treasures were subject to the most destruction and plunder? This astonishing vandalism can be explained by the underdeveloped nature of the people, their lack of proper upbringing and education. However, this state of affairs does not fully explain the pointless destruction of art which could have been spared with out detriment to the revolution. It is clear that the reason for the vandalism lies in something else: and that is the indifference, even hatred, towards all that was connected with the aristocracy expressed by the people's thoughtless pilfering. When a nation is held in the chains of slavery and these chains are suddenly broken, the nation, left to itself, gives full rein to its instincts, and here national hatred towards all reminders of the past order destroys everything, good or evil. The heavier the chains of slavery were, the more powerfully is national vandalism manifested in times of upheaval.

This is related to another situation.

In all periods of their history, nations have striven to represent their political and religious/moral beliefs as certain symbols, which are reflected in art. However, as these beliefs begin to wane or disappear, a nation turns against the symbols that it previously accepted with a fury whose strength

depends on the degree to which a nation's new political convictions oppose its former ones. This is why everything that reminds a nation of the preceding structure of its life, or represents in symbolic images its previous barbarity, falls victim to its blind hatred. And so, the French Revolution of the Convention's time was marked by a decree on the destruction of all royal statues, tombs, and burial-vaults, and during the Bolshevik period of the Russian Revolution, two months later, a decree was made on the public removal of statues that offended national sensibilities. The results of these actions are widely known and there is no need to go over them here. They can be found in the very history of such periods of social upheavals.

In a word, we also meet with the law of rhythm in consecutive events, in the sense that one social movement is replaced by another, opposing one. To a certain extent, a correlation between the power of a social movement and the following declination in another direction does exist. Thus, the greater the social uplift in one direction, the greater its consequent fall. Compliance with this law is even manifested in the greater or lesser acuteness of a given social movement. In other words, the more turbulent an uplift's development, the more speedy and abrupt is the consequent decline, and, conversely, the more gradual the uplift, the slower the following decline.

In political movements, the law of rhythm is manifested in the fact that the greater the power possessed by political movement of a given orientation, the stronger the reaction towards it. The excessive oppressiveness of the tsarist regime in Russia led, in the end, to Bolshevism, which was foreseeable even at the beginning of the revolution, since on the twentieth and twenty-first of February cries of "down with Milyukov," and "down with the Provisional Government" could be heard in the streets of Petrograd.

As a matter of fact, periods of revival, which recur from time to time in the history of a nation, are classic demonstrations of the periodicity of social phenomena. Although it might seem that certain nations and their accompanying cultural valuables have long disappeared, and that even the inheritors of their culture have become the legacy of history or even archaeology, that they have gone beyond the reach of the contemporary world, a period of revival nonetheless comes, during which they are once more the center of social attention and imitation, having been reproduced under new conditions.

It must be noted that the same pattern can be observed in other manifestations of national life. Thus, society may experience a burst of

nationalism, followed by indifference to national issues and even by the burning of nationalist literature; it may swing from a period of materialism to idealism and vice versa, sometimes pausing on the middle course. Every scientific profession undergoes the same wavelike motion in its development, which is characterized by various trends during different periods.

In general, all manifestations of the associative activity of any collective are subject to the law of rhythm. Thus, it is first of all important to establish that the mood of the collective experiences periodical shifts in one direction or another, which are determined not only by external conditions which could be favorable or unfavorable to the collective, but also by the instability of the mood itself, which cannot remain unchanged for long periods of time. Usually, one social mood is replaced by a calmer or opposing mood, which is then subject to a similar change, etc., examples of which were put forward in the first part of this work.

The mood of national masses is not an exception to the general rule. Today's hero is made the idol of the crowd tomorrow, but the day after tomorrow the same crowd is indifferent, or even hostile, towards him. So many former idols have undergone a period of national enthusiasm and fascination displayed at the simple mention of their name, and ended ignominiously with national indifference, oblivion, or even degradation and accusation. Such was the fate of Hannibal, of Julius Caesar, of Napoleon I. Even the fate of the most recent hero of the Russian Revolution, Kerensky, may in time prove to be the same. Only the impartial court of history gives every personality that plays a part in the fate of nations its place, evaluating its positive features without concealing its negative aspects.

The fluctuation of social mood can occur quickly or slowly, depending on the period of social life. In revolutionary periods, change of mood occurs with kaleidoscopic speed. Displeasure with one political regime leads to rebellion; the conditions of the resulting new regime do not usually appease the people for long, and soon the voices of dissenters rise once again, attracting more and more supporters, preparing a retro-insurrection.

In the evolutionary process, we have, aside from the alternation of opposing social moods, the transfer from one trend in the creative process to another. It is likewise widely known that, after a turbulent manifestation of social life in the form of a revolution, there is often a subsequent process of social reaction.

But even outside these periods of national indignation, every nation goes from vitality to stagnation, after which, under the influence of some stimulus, national life is revived and steps are taken towards progress; however, stagnation eventually takes over again.

Any doctrine in general, after developing, reaches dominance and finally is subject to transformation. The new doctrine that replaces it is doomed to the same fate; it must blossom, then wither away and undergo a metamorphosis.

Neither literature nor philosophy can evade a similar displacement of one trend by another: romanticism is exchanged for naturalism, after which comes mysticism, then again naturalism, etc.

Any social progress in intellectual life experiences the same fluctuations, flowing now quickly, now slowly, going from stagnation to revival and so on.

Such is the progress of literature and science. Art moves in much the same manner, with one trend replacing another, after which the new trend, having completed its cycle of development, is replaced by a different, often opposite one that has been previously experienced but has acquired new characteristics, corresponding to the times and to the progress achieved in art.

Even religion does not escape a similar fate.

Any belief undergoes a cycle of birth, prosperity, and decline, eventually clearing the way for religious quests of a different nature to arise.

Primitive Christianity had its own period of development and prosperity, attained in the third century. However, when it took the Emperor Constantine to its bosom, granting him all its property and privileges, the pure Christianity of yore signed its own death sentence. Soon, it became a state religion, gaining special significance in state life and receiving a powerful impulse to its popularization. In this new form, Christianity reached the pinnacle of its development in Europe during the Middle Ages, when it acquired control of state life, and even kings were forced to worship at Canossa. Later, however, state power triumphed over Christianity, and gradually, with the evolution of state political life, it began to take on a more and more secondary role; with the development of science, Christianity is gradually losing its influence on the minds of the people.

If we turn to economy, we meet with the same cycles. Every nation develops past a stage of fledgling industry and reaches its economic peak, after which, thanks to competition or new conditions, it loses what it had attained at the time of its prosperity. Every branch of industry, every undertaking suffers the same fate. Some invention or other stimulates a

certain industry, enabling it to reach its full development; but then a new invention appears and the enterprise falls into decline, forfeiting its place to new industry.

In this way, the manifestations of collective activity, and likewise the manifestations of individual life, are subject to the law of rhythm, which consequently has universal significance.

From the above, it is clear that the social process follows a wavelike path. After the tsar's regime, after proclamations of war until death or victory, the Russian Revolution, as a reaction against the tsarist regime and proclamations of war, brought Bolshevism about very quickly; the latter should, in the not-too-distant future, lead to an inclination to the right, which may be expressed as a dictatorship. In the way that tsarism brought itself to the extremes of despotism during the war and thus destroyed itself, so Bolshevism may eliminate itself through careless and excessive reform. Only a gradually agitated social sea can be calmed through the slow diminution of its waves, until finally it enters quiet shores. This is why individuals and parties taking the middle line in turbulent times always lose. To become accustomed to their environment, they must be as changeable as kaleidoscope figures.

"The uprising of the crowd," says G. Le Bon (*The Psychology of Socialism*, p. 133),

is a matter of minute-long fury, and under the renewed influence of its conservative inclinations the crowd soon returns to its former views and demands the reinstatement of those ideals that it destroyed in a fit of anger. Our history of the past hundred years repeats this to us on its every page. A revolution, for example, barely completes its destructive acts, and already almost all the things that it overthrew, be they religious or political institutions, are reinstated under a new name.

A good example in this respect is the Russian Revolution. As is known, mutiny in the troops began with the murder of officers, most of whom consequently had no choice but to flee; they were degraded and humiliated, even in uniform they were reduced to the level of simple soldiers, while in some places they were forced to be cooks. However, with the advance of the Germans after the Brest peace, in view of the shortage of regular officers in the Red Army, they were once more called to military activity.

In the course of social evolution, the social process, having begun with a certain phenomenon, generally becomes complicated by new ingredients and then returns to a simplicity analogous to the original phenomenon. This fact can be illustrated by a number of examples.

However, I will outline only one: that of the church. Initially, the church represented an institution free from state control in the form of a small commune; then it became closely connected to the state, as a part of its government; finally, it once again distanced itself from the nation, regaining its independence and becoming a hierarchically organized institution, acknowledged by the government and state as a free and self-contained establishment.¹

Note

¹ We do not undertake to discuss whether we are dealing with the universal dissemination of energy or its accumulation under certain conditions, e.g., with the formation of stars.

8

The Law of Inertia

It is important to acknowledge the law of inertia among the other laws affecting collective life, since it is universal for all nature. Every social trend is subject to this law, which causes any movement, once it has developed, to tend to be popularized across the whole collective, attracting more and more individuals in the manner of a snowball that grows unlimitedly as it rolls through snow.

"Society could not exist, could not make a step forward without the priceless treasures known as routine, simian imitation, and totally ovine herdability that are constantly increasing with every new generation" (Tarde, *The Laws of Imitation*, p. 76).

Even conservatism is a manifestation of the law of inertia, as is any routine in general.

As is known, a crowd is especially conservative. In the words of Le Bon, (*The Psychology of Nations and Masses*, p. 191):

The changeability of a crowd is expressed only on a superficial level; in essence, the crowd's conservative instincts are still active, and are as indestructible as any human's. The crowd feels a holy respect for tradition and an unconscious, deep terror of any innovation that has the potential to alter the conditions of its existence.

The most turbulent revolutions often result only in the re-christening of time-honored institutions.

The ideas of any society always lag behind those of its scientists and philosophers by several generations.

A vivid example of the law of inertia is the reinstatement of the Catholic Church due to popular demand after it had been abolished during the French Revolution. Here is the speech of a former member of the Convention, Fourcroix, cited by Taine:

Widespread celebration of the Sabbath, and likewise attendance of church, indicates that a mass of French people wish to return to former customs, and this is not a time to oppose the national tendency or disposition.. .The majority of people displays a need for religion, ritual, and priests. The mistaken belief of some contemporary philosophers, which I also shared, was that education, if it attained sufficient popularization throughout the nation, could destroy religious prejudices. But these prejudices serve as a source of solace for a huge number of unfortunates, and consequently priests, altars, and ritual must be left to the national masses.

Try to pass a radical reform in any environment, and you will see how great the conservatism of people is, since, in the end, the reform will not be passed due to that very conservatism. Why do great revolutions rarely achieve their objectives? Once again because the conservatism of the masses cannot be overcome.

On the other hand, a social movement that has already commenced tends to experience further development.

Even such artificially created obstacles as persecution can only detain the movement for a short time, if at all, since a social movement that is deeply rooted in the external or internal conditions of social life continues to fix itself in people's minds notwithstanding persecution, and even its sacrifices are beneficial since they give it an aura of martyrdom. If the movement is for some reason manifested too early, it can temporarily be vanquished through inhibiting influences; still, eventually it is revived under more favorable conditions.

Any social movement, once developed, continues to develop in its original direction, provided the conditions are favorable.

If this does not occur, and one social movement or other is terminated or temporarily arrested, then this depends on the internal or external inhibiting conditions of the environment in which it develops. If, on the other hand, the

direction of the social movement changes, then this is once again dependent on external phenomena.

The law of inertia leads to the fact that any action that was once performed is reproduced upon repetition of the original stimulus, leading to the development of habit and practice.

We are dealing, in the end, with collective reproductive reflexes that are manifested with greater and greater ease upon repetition. This results in social habits and tendencies that are based on the imitation of elders by youth and of older members of a given association by newer elements. Every generation more or less follows the path of the previous generation, conserving the latter's institutions and customs, and only new living conditions can lead to the innovations that mark a new generation.

It goes without saying that the law of inertia acts more powerfully in places where living conditions lead to national isolation and to the greater unity of the old and the young, which can be observed, for example, in China, which is geographically separated from other nations, as well as in the lifestyle of all nomadic peoples in general. This results in the unusual conservatism of such nations, which retain for centuries their patriarchal lifestyle and the corresponding tenor of life; this structure has the same form now as it did at the dawn of civilization.

Conversely, warlike peoples that often collide with other nations during periods of conquest are marked by progressiveness, although there are other factors at work here such as competition with neighbors in everything related to progress, without which conflict could not be successfully resolved.

The reasons for many customs and habits are lost in the mists of antiquity, and at the present time even their significance cannot be discovered; nonetheless, they are recreated from generation to generation through tradition or routine. Take, for example, fortune-telling: who can say with certainty why it takes place on Christmas Eve or at a christening, rather than at Easter?

Likewise, various contemporary establishments are encumbered with a large number of pointless ceremonies whose origin is murky or unknown, but which are nonetheless executed in corresponding situations. Finally, many useless social establishments, including various forms of etiquette, are upheld in accordance with the law of inertia that lies at the base of social conservatism.

The suspension of one social movement or another does not depend on its own loss of strength for no apparent reason, but follows either the desertion of the movement by its supporters due to the conditions of the time, or the combination of certain events, or even the temporary inhibition of the movement itself after it meets opposition in one respect or another.

This is best proved through example.

Take Christianity, for instance. Appearing as a reaction to the crude, materialistic views of paganism, it attracted more and more people and reached its acme during the middle ages, when it triumphed completely over paganism. If, with the passage of time, Christianity has ceased to occupy its former social role, then this is explained by the new structure of European social life that came with the development of industry and science. The same can be observed in other social movements.

Thanks to the above law, any social movement, once developed, tends to continue in its original direction. This forms a kind of routine in social activity. In connection with this, the social environment is generally marked by widespread conservatism. It seems that the less educated a society is, the greater its conservatism. From this stems the retention of customs, beliefs, and traditions which have long ago lost their significance.

Any innovation comes into existence only when the corresponding environment is prepared for it, and when the old structure has already lived—or outlived—its time. So many laws have remained on paper only because society was not properly prepared for them. On the other hand, the forceful introduction of something into social life always meets corresponding opposition and resistance.

It suffices to remember the potato rebellions in Russia, the opposition to smallpox vaccination and to the disinfection of communities during cholera epidemics, etc., in order to gain a clear concept of the latter, forceful, introduction.

Even the use of steel instead of goose quills stimulated universal opposition, and only gradually did it acquire popularity.

Finally, tradition and habit in social life can also be traced to the same law of inertia.

Does not prestige or authority in the eyes of the crowd serve, in its turn, as an example of the law of inertia? Everyone knows that authority, even after it is completely exhausted, still has the power to subject the masses, which follow it due to inertia, finding themselves fascinated.

On the other hand, if a shift in a certain direction occurs in society, it usually has the tendency to progress further than necessary, and the more turbulently the shift occurs, the greater the tendency outlined above.

Any social movement, once born and rooted in social life, continues to develop, if the conditions are favorable, until it exhausts all available material; unless it is opposed through whatever measures, it will develop further until it finally loses its social *raison d'être*.

In this way, a social movement always grows to a certain limit, after which it wanes, gradually dying due to lack of supporting material. During this process, a movement will usually develop to its greatest extent, if it is not inhibited.

At this point, the movement meets with opposition from groups that have joined together for a rebuttal, and undergoes its crisis. Thus, a revolution that has reached its furthest extent usually meets with repulsion from a counterrevolution.

However, thanks to the previously outlined law, a movement can go outside the limits where it should have stopped in its wavelike progression. Thus, the contemporary Russian Revolution, instead of being all-encompassing, as it was in the very beginning, was subject to a strong bias towards the rule of the proletariat and became, in essence, party-oriented, answering only to the interests of the proletarian class.

As is known, the Revolution of 1905 likewise resorted to expropriation and bomb-throwing that, in turn, stimulated a reaction in most of Russian society, which was horrified by these wild forms of revolutionary movement. After the Revolution of 1905, when the period of reaction reached its peak, the inclination of society toward the right once again attained such heights that it far from corresponded to the existing correlation of social forces.

For the above reason, intermediate groups hold predominant influence in times of social stagnation, and lose all their influence during national upheaval. The first situation is explained by the fact that the masses, if manifesting discontent, are disposed to pay attention to promises of better living conditions but do not initially follow extremist movements to the same extent as moderate ones, since the former seem Utopian and their propaganda is made difficult by social stagnation and reaction. Because of this, during periods of national transition or change the first to make history are the more moderate parties; however, when the real revolution begins and the conditions for propaganda are more favorable, extremist movements take the

upper hand thanks to the fact that they give greater satisfaction to the demands of the nation by acting decisively and consequently opening wider horizons to the future. Finally, they are once again in opposition to moderate currents and win over their followers.

This can be compared to the movement of a cart. In order to start it on its way, the efforts of the wheel-horse are most important, and the traces play only a secondary role. However, when the cart begins moving, the action of the traces give it such momentum as could not have developed with the help of the wheel-horse alone.

Moreover, it should not be forgotten that national movements are frequently joined by the most conservative elements, which previously constituted stagnant water in the social bog, and consequently strive to pull themselves out of obscurity by enlisting in the most advanced, i.e., more extreme, groups.

Meanwhile, the more moderate elements, accustomed to obeying their leaders, remain loyal to their slogans and resist the social movement. In the end, these moderate currents lose ground in the large, agitated mass, they are no longer heeded, and gradually they are abandoned by their followers who perceive the success of others. Because of this, they enter into agreements with more radical parties in order to save their reputation, are nonetheless forced out of the social scene while taunted and pursued by extremist currents, and often die. Such is the fate of all moderate trends under conditions of national upheaval. An example could be the fate of the Girondins at the time of the Great French Revolution or the downfall of the Russian Cadets during the Bolshevik coup.

It is clear that the law of inertia acts here in two ways. Thanks to it, some elements lag behind, remaining in one place; others, caught up in the flow of a movement, climb further and further, reaching the most remote slopes of social change.

The same law of inertia is responsible for the fact that an established power remains long after it loses public sympathy and support. It is likewise known that panic continues for some time after there is no longer a basis for its existence. Under the influence of a success on the battlefield, troops advance further than originally planned, and often meet with unforeseen opposition.

When a national movement is growing, it opens new horizons with every victory, as a result of which the demands of its leaders become greater and greater. The power opposing the movement, however, takes into account only

recent results and in order to protect itself tries to make as few concessions as possible, while its actions are guided by reactions to current developments rather than by forecasts of what is to come. From this stems the common phenomenon of government concessions coming too late; this circumstance always aids strongly rooted movements, since they find in belated and insufficient concessions new stimulus to attack and promote their demands. Then, when the victor has been established as a given side or party, the latter usually abuses its position and more or less completely ignores the interests of the conquered side, in this way giving reason for the growth of displeasure and opposition, or even open indignation and revolt.

Let us take an example from the Russian Revolution. According to the central organ of the Russian Socio-Democratic Labour Party, its *Labour Newspaper*, Commander-in-Chief Kornilov's so-called "mutiny" was brought on by the mistakes of the revolution. This is what we read on the subject in *Den'* (31 August 1917):

The central organ of the Russian Socio-Democratic Labour Party was brave enough to openly declare, at a time when the mutiny instigated by Gen. Kornilov was still not liquidated, that the mutiny was prepared, to some extent, by the mistakes of the revolutionary democracy. The neglect of national defence, the concessionary attitude towards anarchic groups and slogans, the policy of isolation of the proletariat from the progressive, wealthy classes, which separated these classes from the revolution with a speed that corresponded neither to their interests nor to those of the democracy itself—all this taken together laid the groundwork for an eruption of counter-revolutionary forces, in this case in the form of Kornilov's march on Petrograd.

Compromise with revolutionary adventurism and obstinacy towards the wealthy classes were equally provoking with respect to a counter-revolution, and from the unfortunate July Days onward we progressed irreversibly towards reactionary coup.

Fortunately for Russia, the latter took the form of the badly executed Kornilovsk adventure. However, it would be a grave mistake to think that the eradication of a badly executed adventure is also the elimination of a counter-revolution.

It is enough for general living conditions to worsen, and dissatisfaction with the heads of a nation immediately develops, even if the worsening is not the fault of the governing party.

Whenever general discontent arises, it first of all takes the form of a protest against the prominent figures who are assumed to be responsible for the misfortunes of the people. This is the case during all revolts caused by starvation.

This is why a kind of instinctive aspiration to create national, rather than party, government arises during terrible national ordeals, in the hope that all the elements of society will be represented and that the crisis will consequently be properly addressed.

This is why it became necessary to create coalitional ministries in such parliamentary countries as France and England at the time of the Great War.

In Russia, this was echoed in demands for a ministry of trust, and then for a ministry of responsibility.

Of course, these demands were not met, but the Tsar was punished for his disregard by the revolution. In the course of the latter, the voices and slogans of the majority of people once again requested a coalitional ministry made up of representatives of socialism and the bourgeoisie, in view of the fact that single-party government cannot be responsible at such important moments in history.

Finally, the law of inertia is found in the fact that once the unity of a collective has been attained, it is upheld solely through internal discipline, in the manner of training; however, it exists only until some stimulus is able to push the collective out of its precarious balance and thus make it fall apart or become disorganized.

Likewise dependent on the law of inertia is the impossibility of the immediate termination of a movement that has just begun. Even forceful suppression does not suddenly, completely liquidate a movement, which is most often only temporarily inhibited, and in due course it once again manifests itself with former or even greater power—provided, of course, that it is not redirected. Thus, the first Russian Revolution of 1905, extinguished by the heavy heel of reaction, flared up with even greater strength in 1917 and swept away in no time at all the power that had trod upon it in 1905.

However, inertia is also manifested in another aspect of social life:

A series of socializing acts, repeated many times for one reason or another, acquire the power of a habit that is subsequently maintained only out of inertia, independent of any other cause or condition. After a certain number of repetitions of mutual influence, collective unity that has already emerged and has existed for a certain time continues its existence through inertia, up until the moment when a given force creates a new condition that renews the connection between interacting units. The inertial force resulting from the repetition of a series of actions is well known. It suffices to say that the multiple and monotonous mechanical reaction to one stimulus or another causes that reaction to be spontaneously reproduced when the corresponding stimulus is given. In other words, repetition creates habit and habit creates routines of action, which spontaneously force us to act in agreement with these routines. In this respect, habit—in the literal sense of the word—can be called biological inertia.

There are countless examples in social life in which habit/inertia plays a leading role. Such examples include the preservation of marital unions when they have lost any real unifying basis; the sustenance of trade ties with certain individuals when thousands of trade correlations with others exist; the continued participation in one circle, party, society, organization or other after there is no longer any agreement with their views. Accustomization to executions, shootings, and similar occurrences should be attributed to social inertia: although unusual punishment may at first stimulate a sharp social reaction, with time the attitude towards it becomes indifferent, as that towards any ordinary occurrence.

In accordance with the law of inertia, when the authority of a separate individual has begun to grow in the eyes of a crowd, it continues to do so regardless of subsequent events, and sometimes remains even when the merit of the individual has long faded away and no further services have been executed by him or will be in the foreseeable future.

This is not difficult to observe in any context, but it is especially obvious in the army, where one success or another quickly exalts the originator of the triumph in the eyes of the crowd; this person's popularity grows until it reaches its peak at a certain level. *Caesare, Caesare, morituri te salutant—that cry* embodies the boundless authority of a military leader over the crowd.

The law of inertia has greatly abetted the union of societies, as well as their homogeneity. Thus, some of the smaller associations, thanks to the features of their lifestyle that were passed on in accordance with custom from generation to generation, remained immune to foreign influence and preserved their collective attributes throughout the centuries. In this way, social inheritance is one of the manifestations of the law of inertia.

In France, for example, marriages and funerals have long been known to proceed without the participation of the church, and the French workers commonly deride anything related to religion; yet according to G. Le Bon (*The Psychology of Socialism*, p. 40), marriages without participation of a representative of the church are very rare among laborers. A marriage-contract concluded in the city hall seems insufficient. The situation comes down to a certain persistence of religious beliefs in the face of many efforts to eradicate faith and religion. The same can be observed during the second Russian revolution.

Notwithstanding the decree on the separation of the church from the state and on the permission of marriages in the commissariat, there were very few who wished to take advantage of such a simplified form of matrimony.

Thanks to the same law of inertia, forms of government, for example old monarchies, outlive themselves and so collapse like houses of cards with the first onslaught of a revolutionary movement.

In terms of politics, we can point to England as an example of the conservation through inertia of the outer shell of a monarchic state structure, which, in reality, covers a republican form of government.

An analogous example is the Tsar's rule in Russia during the latter days of the house of Romanov. In the end, it was the embodiment not of the power of the tsar, but of the handful of rascals who used the organ of the tsar's power in their own interests, since in Russia true tsarist power did not exist throughout the reign of Nicholas II.

In the end, inertia is the conservative force that preserves the national and class peculiarities, as well as the historic features, of popular character.

Tarde categorizes some phenomena explained by inertia as counter-imitational, but this counter-imitation is, in the end, the product of his obsession with the alleged universality of the laws of imitation in collective life. In fact, his counter-imitation ultimately proves that collective life is subject not only to the law of imitation, since it contains phenomena which

oppose that law, which cannot be classified under it in any form, and which were classified as counter-imitation in an utterly unscientific manner.

Generally speaking, the collective is much more conservative than the separate individual. It is more difficult for the former to accept any innovation, more difficult to force it to turn its back on the old way of life. All customs, social traditions, leftovers of antiquity, and remnants of the past in general must be made part of the law of social inertia.

Look at how contemporary society has sustained the most awful ordeals in a period of war. In view of recent traumatic events, it would seem that social life should at certain points have been thrown completely off balance; and yet the public persisted in its former habits of attending the theatre, café-chantants, cabarets, and other entertainment-oriented establishments as if nothing had changed, as if everything remained as it always was. To a detached viewer, the situation gives the repulsive impression of a feast during a plague. When foreigners expressed dismay upon observing how, at the time of the unfortunate Tsushima incident, Petersburg filled the theatres and café-chantants, they were right to do so, since they were temporary visitors to Petersburg and did not enter its life, and so did not surrender to the law of inertia; however, to the common citizen, Tsushima meant nothing. The former was accustomed to visiting during the day, being at the theatre or café-chantant in the evening, and was not likely to refuse these pleasures, since tradition was a powerful force in his actions.

Russia underwent even more terrible ordeals in 1917, when the drawn-out war resulted in the complete disruption of the internal life of the country and the reduction of food to minimal hunger rations. But what of that? It would seem that a national uprising was in order. Not so: throughout the festive days of Christmas, notwithstanding the hunger bread-rations and the difficult times, preholiday commotion continued in the usual manner.

Here, for example, is a characteristic description of this pre-holiday bustle in Moscow, where the bread rations went down to V_4 pound per person with the warning that even the V_4 pound could be reduced. The Moscow correspondent for *Nash Vek* (31 December 1917) writes on this subject:

I can confirm that Moscow has fulfilled the expectations placed on it and on its oft-tested qualities; it accepted the reduction in bread rations and the prediction of their possible transition into a sweet memory with admirable calmness and even, somehow, indifference. "Why bother with the ration

when we need to worry about getting the goose?" Not so long ago, we could cry with pathos: the demands of everyday life are not dead; no, they are very much alive. These demands are vital and tenacious, they hold tightly to existence. In Moscow, perhaps, they hold tighter than anywhere within the boundaries of the Soviet republic. And these demands of everyday life have managed to inject into the current tragic events all the grandeur of festive occasions, to retain all of the latter's details and accoutrements.

From the point of view of, say, an expatriate Russian prisoner of war it would seem that the Great Hood had begun in his country, and he would not know where to turn to find the ark in which Russia could save itself:

But upon scrutinizing the situation more closely, his surprise would know no bounds, for against the background of the flood he would see almost ordinary festive designs. He would be astounded to see that into the flood have squeezed bustling streets; overflowing delicatessens and confectionery shops, out of which pour streams of people carrying parcels, packets, and boxes; gently glittering Christmas tree lights and happy evening gatherings; posters made of sheets advertising countless theatres, cabarets, café-chantants, cinemas with repertoires of day and evening performances.

The situation in Moscow was echoed in Petrograd, and in the latter even during the most grueling moments of the political situation, posters flashed by with headlines such as "Entertaining Ball," stipulating that "Ladies should not forget to invite their partners, nor gentlemen their ladies," and promising "continual dancing" with the participation of renowned artists, as well as confetti, serpentines, snowball-fighting, games of "post-office" and other ostentatious nonsense, not forgetting that the lady who received the most cards would be presented with a prize in the form of a silver surprise. And all this was taking place at a time when old Russia was governmentally and economically degenerating, when Finland had seceded, when whole areas to the west were breaking away in Germany's interest, when robberies took place everywhere, when you could not walk down the street in safety for you risked being stripped of your clothes.

Furthermore, although during a period of conflict with an established regime there is no lack of blame attributed to the governing powers, when the opposition is at long last victorious and takes the upper hand it usually manifests great severity towards the losing side. Only gradually do these relations smooth over as general calming occurs; however, if afterwards another overthrow is enacted, it is once again accompanied by greater repression than was displayed before the coup.

The same occurs in battles between nations. In the beginning, the attacker suffers significant losses from the resistance of the enemy, but when this resistance is broken, the victor always rewards himself with a savage reprisal of the vanquished. Obviously, for the first while a kind of vengeance plays a role here, determined by the necessity of giving resistance, the greater the latter, the greater the animosity that is manifested quite vigorously in the period immediately after the conqueror emerges victorious from battle. But the law of inertia is also at work in such situations, and causes the animosity that had developed during conflict to continue to grow after victory when, it would seem, there should no longer be a place for bitterness since the conditions suitable to its development have ceased to exist.

Why do wars that do not bring decisive victories end, for the most part, in mediation? This is because the national enthusiasm of one side knows no bounds and cannot terminate its aspirations and desires at a certain limit. However, the same occurs on the opposing side, since the force of inertia is so great that national masses cannot limit themselves or stop their movement at a determined point.

On the other hand, does an agitated national sea calm down all that quickly? Does not a revolutionary war, once developed, continue to its limits, only if it does not meet with opposition?

An event as important as a war does not end without the kinetic energy of the participatory nations flowing into another form. In Russia, after the Great War, this energy was transformed first of all into a revolution, and then into a civil war. Of course, with adept direction of the national organism a more expedient channeling of kinetic energy could have been attained. In any case, here we are dealing with nothing less than the suppression of kinetic energy in one direction in order to guide it along another course.

Revolutionary movements also know no bounds or limits during their development, destroying everything to which they take a dislike or which sabotages them, and overturning, without basis, things that could seemingly

have been left untouched. When one movement or another develops in art, it once again goes outside the bounds of permissibility. Examples of this can be futurism in art and poetry, cubism, and other branches of modernism.

New religious doctrines likewise take forms that break the limits of acceptability. The development of every fashion eventually reaches such a level that it becomes obviously ugly and is only then replaced by a new fashion which, in its turn, grows to incredibly strange forms, after which it is again exchanged for another. Even science, exact science where, it would seem, there should be no room for such fads, carries one new doctrine or another to its furthest limits through inertia. This explains the various extremes in scientific/practical areas of knowledge, for example in medicine, pedagogy, etc. Thus, everyone knows to what extent went the humoralists, the representatives of cellular pathology, the various investigations involving tuberculin as a remedy for consumption, Erlich's medicine for the treatment of progressive paralysis, etc.

It is clear that the formulas and concepts that govern people's minds were devised and developed slowly, gradually, through evolution and are equally slowly losing their significance. And at a time when these formulas are no longer ultimate truths for educated individuals, they continue to play the role of governing concepts for the uneducated masses.

The influence of the Greek and Roman worlds in science and art lasted for many centuries after both languages died; in essence, Europe is still in the thrall of Greek art, just as various European establishments are under the influence of the creations of the Roman mind.

When minds are taken over by a new idea, much time and effort is required to completely crowd it out with another:

Before finally dying, the idea will for a long time make up part of the legacy of inherited ideas, which are called prejudices, but which we respect nonetheless. An old idea, even when it is no more than a word, a sound, a mirage, has incredible power, and is still able to subject us to its influence. This explains the tenaciousness of our ancient inheritance of outlived ideas, opinions, and conventions which we awfully accept, although they would not be able to stand up to the smallest amount of criticism if we were to analyze them. (G. Le Bon, *The Psychology of Nations*, p. 117)

In general, old, tattered ideas hold on to their influence for a long time, taking the form of a prejudice.

Let us bow to the force of the idea, says Le Bon (*ibid.*, p. 122):

When it reaches a certain period of its development, there are neither deductions nor proofs that can conquer it. For nations to come out from under the yoke of some idea or other, either hundreds of years or violent revolutions, and sometimes both, are necessary. All that mankind has the power to do is count the chimeras that it has invented for itself, and whose victim it has subsequently become.

The law of inertia likewise explains belief in authority and in anything that relates to social convention and habit or to the will and influence of ancestors, all of which are so powerfully manifested everywhere. Finally, the role of tradition and its predominance in life falls under the same law. In the words of Tarde (*The Laws of Imitation*, p. 241):

The appeal of traditions bequeathed by ancestors prevails everywhere over aspiration to innovation. Even our contemporary society does not present an exception in this respect, so rich is it in all kinds of adoption in language, ideas, institutions, literature. Let us compare, for example, "new" words, recently borrowed by the French from the English, German, Russian languages on the one hand and the old French vocabulary on the other; or the newest ideas of evolutionism and pessimism with the already acquired traditional views; or the most recent legal reforms and the entire French code, which is in essence contemporary to Roman law, etc. In this way, imitation of a new fashion is quite naturally only a tiny stream next to the great river of tradition and custom.

Generally speaking, the element of tradition always holds a predominant role in the collective.

The appeal of antiquity is transferred into manners and even into language. Thus, in the words of Dostoyevsky (*Notes from the House of the Dead*), to please someone in Siberia you tell them that they have an elderly appearance; the person conversing with you is dubbed "elder brother"; and when addressing a twenty-year-old youth, you pay your "respects to grandfather so-and-so." These customs were apparently brought from China. But in ancient Egypt, in Rome, in all places age was highly respected, as it is even now.

On this basis, national character (the so-called national consciousness of subjective psychology) results from the acknowledgment of the value of a peoples' history, and determines the extent of a nation's aspiration to increase and improve this legacy. The conservation of the traditions of previous generations gives the opportunity to hold a certain definitiveness and inherited consistency of views which, reproducing the past, seems to pay the necessary homage to the dead, as well as to prepare the soil for future generations. This living heritage preserves the cultural valuables of a self-sufficient nation, which are, in turn, dependent on the fusion of the individual's creativity with the collective process, and the consequent alliance of people into a single unit.

It is thus quite natural that in every social organism conservative parties are the keepers of the legacy of the past and the customs of the nation. These parties, subdivided into smaller collectives, are expressions of collective inertia and are at the same time inhibitors that stand in the way of any social progress, since while progressive parties strive to attain new rights and borrow all kinds of innovations from foreigners, conservative parties reject both rights and innovations, maintaining that inherited customs and laws are a more certain guarantee of a good life.

Finally, the predominance of the law of inertia in economics is also more than obvious.

In agreement with that law, the established cost of some item of produce or other is remarkably stable, independent even from the conditions of supply and demand. In fact, more or less acute influences are necessary for the price to fluctuate. Prices easily augment, but do not so easily diminish, which is understandable because in the first place they receive a stimulus in the form of the material gain of the merchant determining the cost, and in the second place, even if such a stimulus is lacking, the law of inertia comes into effect: it is manifested on the one hand in the accustomization of the populace to the higher cost, and on the other in the merchant's habit of receiving higher pay. We also know that, as a result of the same law of inertia, a demand that has once originated does not soon disappear. This can be proved with many convincing examples. It is clear, for instance, that the force of habit will prevent bread from ever being replaced by imported rice in Europe, just as rice was not replaced by bread in Asia.

All that is related to social inheritance should be placed under the law of inertia. It is true that Tarde categorized social inheritance under the laws of

imitation, but the name does not change the essence of the matter. In any case, inheritance is a conservative force that retains social forms for a longer period than necessary.

Any collective is a living organism, existing thanks to reproduction and realizing its own reproduction in the form created by history at every given moment. The cult of ancestor-worship and respect, preserved over the centuries, determines the passage from generation to generation of social traditions, customs, and even superstitions, which ensures the continuity of the descent to posterity of the previous generations' achievements.

Let us recall, says G. Le Bon (*The Psychology of Nations and Masses*, p. 245),

how tenaciously religious superstitions persisted for many centuries, contradicting even the most simple logic. In the course of more than two thousand years the most brilliant geniuses bowed down before their power, and only in contemporary epochs has there arisen the opportunity to question their correctness.

In the words of Tarde:

A person is freed (though never completely) from under the yoke of custom, only to be shackled once again in order to curb and establish the accomplishments of the period of temporary freedom. If he is sufficiently vital and talented, then he will once again throw off his chains and make new conquests, but then he again reverts to captivity, etc. Such is the historical fate of all great civilized nations. (Tarde, *The Laws of Imitation*, p. 245)

The same must be said in relation to rituals. It is known that the pagans kept their heathen rites long after they were converted to Christianity. When the Renaissance came, the supporters of new views nonetheless preserved the habits of feudal life for some time afterwards. In general, it is known that social forms are conserved long after the very basis of law and order has been changed. When matriarchy was destroyed, the custom of fictitious maternity in relation to the father, in the form of a *kuvada*, stayed on. The bonfires on Midsummer Night on the eve of St. John's Day remind us of the worship of pagan gods. The abduction of women ceased long ago, but the memory of it is retained in certain traditions accompanying wedding ceremonies. A

religion that replaces another retains, to an extent, the ceremonies of latter. Finally, in language we also meet with the conservation of previous roots when a given word obtains new meaning.

In general, it is not difficult to find expressions that are apparently devoid of all meaning and that are passed down as such from generation to generation. Furthermore, certain features of law are retained through the generations, notwithstanding the fact that they arose under different circumstances, at another time, and are an anachronism in the given period. In art and literature, it is not a challenge to point out many peculiarities that are simply remnants of the past and have become a part of routine. Finally, all so-called ceremonials—those "great museums of history," in the words of Paul Viollet—are manifestations of the law of social inertia, which takes such an array of unique forms.

Court etiquette, religion, are these not remnants of the past carried through to our time? In a word, the law of inertia controls all of social life, and so it is very difficult to shift society from former foundations and mould it to new forms.

The inevitability of revolutions is also apparently contained in the law of inertia, which brings a society that exists within the confines of old forms, at the same time acquiring new unanswerable needs, to a dead end. The only way out of such a predicament is the violent, forcible casting-off of the old, long outlived structure of life, and its radical renewal through revolution.

9

The Law of Continual Motion and Change

One of the ancient Greek philosophers maintained that everything in the world is moving, changing, flowing. This idea can be found in the later works of other authors in a more or less definitive form. Thus, for example, in C. T. Letourneau's *Morality*, p. 5, we read:

We now know, and are surprised at how we could not previously have noticed, that everything in the world is constantly changing, everything is developing in a positive or negative direction, that the doctrine of immobility is only the dream of metaphysicists. Our forebears despised changes. We thirst for them.

And yet the principle of continual motion was for some reason not included among the basic laws of the world, perhaps because it was crowded out by the widely acknowledged law of evolution. However, the law of evolution is a law of development, and this is not one and the same as the principle of continual flow and change.

This principle states that nothing in the world is permanent or unchangeable. Every thing, in the end, is not eternal, it is created and then destroyed, as a result of which any constancy is but an illusion. In its initial forms, this principle was suggested by Heraclitus, and was later reflected in the metaphysical teachings of Hegel. Even in antiquity, Leucippus and Democritus put forward the atomic theory of matter; these atoms, and

likewise molecules, are acknowledged to be in continual motion. The bonding of these atoms creates various substances. Presently, we know that atoms are divided into elements of energy in the form of negative electrons and positive nuclei or corpuscles that can separate, leading to the decay of the substance which they make up (the so-called radioactive phenomena). The atoms themselves consist more or less of complicated systems of electrons moving around a positively charged central nucleus. Such is the situation in the inorganic world, which must be represented as energy continually transforming from a free to a stored state and vice versa, with even the stored energy, although in the form of a substance, being in a kind of eternal, albeit internal, motion.

If we turn to the organic world, we once again encounter movement—and only movement—everywhere. A plant germinates from a seed, then grows thanks to processes of nutrition and respiration, to the motion of fluids, and to chemical transformations instigated by the sun's energy; in time, the plant produces flowers, fertilization occurs, and seeds ripen; they in turn give rise to a new plant, and so forth. In a word, we can see here a continuous process or movement and changeability.

The same can be observed in the animal kingdom. Its processes of birth, nutrition, respiration, growth, reproduction, and uninterrupted transformation of external energies, are in essence processes of motion and change. The human personality is itself an entity in constant flux, which daily enriches its reserve of knowledge and skills through experience. But if what we have outlined above is correct in relation to the inorganic and organic worlds, it is no less applicable to the supraorganic or social domain. There, too, we encounter motion and change everywhere.

A society is created through the interaction of two or more individuals, then grows and develops, borrowing the experience of the preceding collectives and neighboring societies with which it interacts; the society differentiates as it grows, communicating and uniting with other societies. The society borrows and imitates, as well as creates of its own accord; changing with the passage of time, it undergoes the corresponding transformations. In a word, we are looking at perpetual movement, continual transformations, and there is neither stoppage of this motion nor rest.

Whatever phenomenon we select from social life, we observe in it, once again, only motion. Moreover, the relations of an individual with society and vice versa are not constant, but change first in one aspect, then in another.

Political forms of government in a state vary as the preceding ones cease to exist. The very form of a state's government does not remain permanent and, although relatively slowly, one form supercedes another. The political ties of a nation change depending on the international situation: alliances and agreements are replaced by discord and war.

Economic life is still more mobile. It is enough to follow the quotations in the exchange newspapers to understand how quickly the attitude of the public changes towards valuables and how ceaselessly the money market fluctuates. As for trade, it is composed entirely of cycles and motion.

If we turn to language, it is hardly necessary to prove that it is not stationary, that any language lives, continually developing, with some words falling into disuse and oblivion, while others are reincarnated. Furthermore, the definitions of words alter, as a result of which their meaning often does not correspond to their original roots. This continual change of language goes on to such an extent that, with the passage of time, new languages, whose origin can only be traced through the comparison of roots, are formed from the combination of others.

Even meanings do not remain invariable; to the contrary, in due course one and the same word or definition acquires significance or content completely different from its original interpretation. One of the numerous examples of such transformation is the meaning of the Russian word for "autocracy," which during the epoch of Ivan III referred to state independence, but has since changed its meaning to "unlimited tsarist power."

Let us next glance at literature: we know that it is likewise a living organism, since any style or work of literature goes through an early period of its existence, a period of further development and maturity, and even a period of decrepitude. In literature we also have a constant exchange of trends and even more frequent shift in mood, which reflects the eternally changing social disposition.

No less is experienced by scientific literature; in this realm, the accumulation of facts causes one hypothesis to be replaced by another, and uninterrupted laboratory work gives, each day, new basis for old theories or lays the groundwork for new scientific theories and disciplines. The same must be kept in mind in relation to the arts, where artistic creativity takes on newer and newer forms and where we once again meet with the uninterrupted exchange of schools, trends, and styles.

In this way, everything in the supraorganic world, as in the organic one, is subject to change; everything flows, and one can agree with the fact that the entire universe is, in the end, simply a process of continual change and creation. As was already stated, this doctrine is old and has its origins at the time of ancient Greece, although traces of it can be found even earlier.

The very form of universal motion was defined differently by different people. For some, including Vico, Nietzsche, Le Bon, and Blanqui, all phenomena could be reduced to eternal and constant circulation, the return of previous forms, and the reintroduction of one and the same changes; when the last stage of a given pattern is attained, everything reverts to the first stage, as a result of which the world experiences a kind of cyclical flow. Even in Ecclesiastes (ch. 1, lines 4-10), we read:

One generation passeth away, and another generation cometh: but the earth abideth for ever. The sun also ariseth, and the sun goeth down, and hasteth to his place where he arose. The wind goeth toward the south, and turneth about unto the north; it whirleth about continually, and the wind returneth again according to his circuits. All the rivers run into the sea; yet the sea is not full; unto the place from whence the rivers came, thither they return once again... The thing that hath been, it is that which shall be; and that which is done is that which shall be done: and there is no new thing under the sun. Is there any thing whereof it may be said, See, this is new? It hath been already of old time, which was before us.

If we compared this poetry with the most recent findings, for example those of G. Le Bon: "The evolution of the world is, according to recent analysis, composed of two very different phases: the phase of accumulation of energy in atoms, and the phase of the release of that energy," then we would observe, in essence, the same idea in both, the latter simply being expressed in another language in agreement with the most recent understanding of energy.

In this way, we are dealing with the circulation of energy from a free state to one tied up in the form of atoms, and with its release into space, after which, over millions of years, the energy once again is formed into atoms. According to Le Bon: "The origination of anything is, without a doubt, no more than a repetition of the past. There is nothing that could allow us to believe that something exists for the first time, in the same way that we

cannot concede that it disappears forever and never arises again" (G. Le Bon. [1909]. *The Evolution of Matter*).

In the supraorganic or social world, this theory was applied by Vico, according to whom nations in their historic development go from a stage of deism, through a stage of heroism, to a stage of humanism, after which the first stage begins again, obviously with some alterations. This doctrine, as is easy to see, assumes that there is a limited number of forms in the world that can be translated into inevitable changes.

However, it is also possible to take the diametrically opposite viewpoint, i.e., that the number of forms that the changes can take is unlimited, and the process of alteration in the world does not repeat itself and is, consequently, irreversible. In this case the process of change can be realized in only one direction, but the way in which this change takes place is unpredictable.

However, there is also an intermediate opinion that maintains that the process of change, although irreversible, remains more or less parallel to its initial stages. In this case, the process of universal change can be figuratively portrayed as a spiral; such is the view of Goethe, Tarde, Mayer, and others. It is a fact that the world is continually changing, but this variation is not in the form of a cycle, as was so beautifully stated by Ecclesiastes, since, in reality, even the circular movement of planets is only an apparent phenomenon. If planets orbit around the sun and their axes, then they, in essence, do not repeat one and the same movement, for the sun itself moves in a certain direction, and the ellipsis of the planets causes continual deviations.

There are many facts supporting the belief that the cyclical development of phenomena is nonexistent; without speaking for the inorganic world, it can be said with certainty that the organic and social worlds contain many examples of the above. In the organic world, we can point to the birth of plants and animals, their development, growth, and reproduction, the product of which has, however, a different individuality. In the social world, state structure initially existed on the basis of democracy in its primitive forms. In due course, an elected head holding limited power came into being, followed by the advent of an unlimited monarch; then came limited monarchy once again, and finally a period of democracy returned, but in the form of parliamentary republics or even councils. It is clear that in the above case a cycle has occurred, in which the initial form has been altered but nonetheless reinstated. This is demonstrative of the aforementioned spiral motion, in which ceaseless progression creates forms that cannot be retrieved in their

original configuration, but return in a slightly different form. If we turn to the development of religion, we see, initially, the worship of animals or people, e.g., of one's ancestors (totemism), then the worship of heroes, after which came the worship of gods, then once more, with the weakening of religious views, the worship of people in the form of society or mankind (as in socialism or internationalism).

If we look at the history of marriage, we observe that at first marriage was a more or less free institution; then family ties were firmly cemented by religion, and the breach of familial fidelity was severely punished; later marriage was liberated from religious shackles and was subject, for the most part, simply to state marital requirements; then gradually marriage once more became a relatively free institution, with marriage or divorce needing only mutual agreement or officially registered agreement of one of the parties.

Societies or tribes themselves go from conditions of discord and conflict to alliances, eventually merging into one large state organism which once again divides into factions.

Literature is at first national, in the form of national epic; then, distancing itself from the people's interests, it becomes a caste institution, fulfilling the demands of the more intellectual classes, and finally re-emerges as the literature of the masses and serves the interests of the entire nation rather than those of a separate class.

Even in language such mutability can be found. Without going into philological details, let us note that, for example, in communicating with another person we used primarily only *ty* (Russian informal "you"); later that was expressed, depending on the situation, as *ty* or *vy* (Russian formal "you"), while in the languages of more developed nations, such as in French, the word *tu* is no longer employed in interaction with others—*vous* is always preferable.

Folk wisdom undergoes comparable transformations. In the beginning, information acquired through experience by primitive nations is replaced by religious views that, with the passage of time, almost crowd out the facts that stem from experience and observation and that lie at the basis of science; however, in due course, science takes the upper hand and experience and observation reclaim their rights in the life of human societies.

There is no need to multiply these examples. They are sufficiently convincing for us to determine that the world is continually changing, that everything in the world is flowing, that the array of newer and newer forms to

which the phenomena of the world are molded are irreversible, but at the same time there is a constant restoration of conditions and forms that are similar to those that have previously existed; yet these conditions and forms inevitably contain something new that has not existed, is not repeated, and cannot be reverted to in the future. From the above it is clear that the law of continual motion and change is essentially contained in the concept of the irreversibility of eternal change.

10

The Law of Energy Dissipation or Entropy

As is known, the so-called second principle of hydrodynamics, according to which it is accepted that energy strives towards dissipation or towards the transition from uneven concentration to even distribution, has been established as an immutable law of physics since 1824, the time of Sadi Carnot. This law was later presented in a more precise fashion by Klausius in Germany and by Thomson in England.

Such transition of energy from uneven to even distribution always occurs directly, while a transition in the opposite direction requires preparatory work, i.e., the expenditure of other forms of energy. In other words, as expressed by Professor Khvol'son, transformation possesses "a kind of direction."

For example, the performance of work releases a corresponding amount of heat; however, when heat is transformed back into work only a fraction of its energy goes to the execution of useful work while the rest, to an extent depending on the situation, is wasted. Thus, when a rock falls to the ground it loses kinetic energy, at the same time developing heat which cannot be transformed into kinetic energy again.

This basic law of physics, which can likewise be proved true in chemistry—for example, in irreversible reactions—is unquestionably applicable to organic nature, since all organisms are, essentially, carriers of stored energy that strives to be disseminated, naturally shifting into one form of mechanical work or another and being transformed into heat and other more simple forms

of energy during the processes of decay. The support of life itself, i.e., the nutrition of an organism, cannot occur without a certain amount of work. Finally, the adaptability of organisms is based on the same law, since adaptability presupposes the lessening of mechanical and other influences on an organism through the transformation of mechanical energy, e.g., an impact, into other less harmful energy forms (heat, electricity, etc.).

But this law of entropy or dissemination of energy also has indisputable significance in relation to the supraorganic world. In essence, every individual who makes up part of a collective is the carrier of the stored energy of his colleagues, which he disseminates directly in the surrounding environment in the form of motion or actual work. Otherwise, his activity can be in the form of signs or signals that stimulate work in others; however, the renewal of the energy spent here always entails work, in that sources for the restoration of energy must be sought out and acquired.

If the human organism, while performing physical labor, can get up to 33 percent useful work from the energy reserves it acquires primarily from food, then in the execution of collective work there is likewise a known percentage of useful work that cannot be employed for the restoration of spent energy in an amount equal to the latter's expenditure.

Furthermore, we know that the development of culture is realized through the creative work that is done on discoveries and inventions. These discoveries and inventions—in other words, the energy spent on them—are disseminated directly among the masses through learning and imitation, without the people having to expend the same amount of energy as was used to make the discovery. The progression of any collective in general, be it a class, party, or nation, to a certain level requires first of all the expenditure of one amount of energy or another on the part of the individuals making up the collective. This progress, however, is inevitably accompanied by the dissemination of the accumulated energy through the direct transferal of the results of the work to other collectives. The achievement of whatever results by one collective or another will again require the expenditure of new energy by the individuals making up the collective, which will be disseminated in the aforementioned manner, and so on.

Let us now look at all instances of conflict between nations. Before these encounters between rival parties occur, widespread preparations are conducted in the form of armament, of the accumulation of equipment and food, of the establishment of communication routes, etc., requiring a huge

amount of collective energy. During the conflict, this reserved energy is used for activity or work, but the latter does not completely replenish the expended reserves, on whose recovery a corresponding amount of energy must be spent.

The same can be observed in economics. The manufacture of one product or another requires a certain amount of energy; the end product, however, does not provide the means for the recovery of the energy expended on its manufacture. To this end, the product must be turned into a commodity and sold, in other words energy must once again be spent on it in the form of its exchange, through sale, for money.

It is clear, in this way, that collective human energy is turned into work that cannot be transformed, of its own accord, back into human energy in an amount equal to the work conducted; this is an expression of the law of dissemination of energy or entropy in the supraorganic world.¹

Note

¹ We do not undertake to discuss whether we are dealing with the universal dissemination of energy or its accumulation under certain conditions, such as the formation of heavenly bodies.

11

The Law of Relativity

We will not touch on the principle of relativity in the world in general as delineated in Einstein's famous paper,¹ but will examine only its manifestations in collective activity.

The law or principle of relativity in social life is a principle thanks to which any social movement, like any individual action, has significance not on its own, taken separately, but in correlation with the given social conditions.

Thus, in tsarist days, the demand for confidence in the Russian government was acknowledged as an extremely radical form of social movement, whereas during the revolution, only a few months later, even the demands for a republican Russian state were not considered radical. This was because many social circles already supported the concept of a democratic federal republic, and some went still further in this respect, reaching out to accept Bolshevik socialism.

Similarly, the program of the Party for National Liberty was, for the majority of the Russian population, too radical in tsarist times, and more moderate elements turned from it, while the party itself was forced into the underground by the governing power; yet, during the revolution, even the transformed Constitutional Democratic Party, which leaned quite sharply to the left, was considered extremely right-wing and was, in its turn, pushed into the underground by the Bolsheviks. A comparable contradiction can be seen in the fact that during a period of great triumphs, small victories are all but ignored, whereas when victories are lacking even insignificant military

successes are treated as events of great importance to the state. In a time of social tranquility, even an insignificant event becomes the main topic for discussion throughout society and is subjected to social criticism from all sides, while during large-scale social shifts and crises, important occurrences are given little or no attention. In the same way, at a calm point in social life even an ordinary, so to speak mundane, murder creates a sensation. In periods of national turmoil, when blood flows freely as a result of civil war, there is no mention of separate murders in the public or the press. Even the mysterious, nightmarish execution of seven innocent students by unknown people in Petrograd in the beginning of March 1918, failed to create a furore in society. The lynchings and murders of blameless people or of petty robbers that decorated the daily newspapers during the Russian Revolution need not even be mentioned. They passed by practically unnoticed.

Furthermore, the law of relativity is manifested in the fact that the strength and nature of any social movement cannot be assigned a single, constant assessment, since under certain conditions and at a certain time it could be acknowledged as progressive, and therefore as abetting social development, while at another time it may be considered retrograde or as hindering healthy social development.

One and the same social movement or action, depending on the social situation, can act, in one case, as an inhibitor of progress, and in another as its stimulus. Thus, inflexible rule is a destructive beginning when it hinders healthy social currents and, conversely, is essential and beneficial when it is used for the control of anarchy.

A patriotic mood is advantageous and essential for a healthy state in a period of war, when national unity is indispensable, whereas that same patriotic mood, expressed in excess, can be quite harmful in peacetime, limiting the cultural aspirations of the separate nations that make up the state, and consequently inhibiting the development of their unique cultures.

In the end, it can be said that there is not one social inhibitor that, under different conditions, and even at one and the same time, is not stimulating for some sectors of society, and limiting for others.

Let us look at more or less constant social inhibitors, such as the demands of laws, religion, social traditions, morals, decency, etc.

They are all, in essence, inhibitors in some cases and stimuli or incentives in others. Thus, the demands of the law limit freedom of action in relation to so-called crimes, but also assist social life in sheltering the population from

theft and violation. Similarly, the demands of religion forbid immoral actions, since they contradict the service of God, and in doing so aid their suppression in life, while at the same time inciting in others the will to execute highly moral deeds.

Social traditions limit the freedom of the individual, but at the same time exert a stimulating influence in relation to everything that abets the establishment of social beginnings.

In this way, there is not one social phenomenon that could be acknowledged as always having absolute meaning; to the contrary, every social phenomenon must be analyzed as having relative significance.

Finally, one and the same organization, depending on the conditions surrounding it, reacts to a given event differently at different times.

Thus, the Petrograd Council of soldier and worker deputies was opposed to the issue of Liberty Bonds at the outset of the Russian Revolution, since those bonds supported the war; yet at the end of June, during the Liberty Bond promotion campaign, that same Council appealed to the populace to purchase bonds.

The law of relativity is generally applicable to all collective movements.

When starvation affects a country, the instinctive reflexes of a crowd are manifested in numerous thefts and outrages that do not meet with sufficient counteraction on the part of the governing power or of society, whereas at another time the same phenomena meet with their natural opposition not only from the government, but also from all sections of society, and so their occurrence is rare.

When a crowd is happy, it is difficult to force it to cry and, conversely, it is not easy to enliven a crowd that is depressed.

When a mass of people is reared in accordance with certain views, it is extremely difficult or almost impossible to imprint upon it opposing views, and at the same time equally easy to imprint views that correspond with those acquired through upbringing.

If collective concentration is directed at a certain quarter, then it is once again extremely difficult to distract it with other events and, conversely, concentration is easily transferred to objects related to the subject at hand.

The same can be observed in collective actions.

The tactics of parties in their mutual conflict and in their conflict with the government change in accordance with the conditions of the day, since at one time a certain mode of conduct may be suitable, while at another the same

conduct may not only be unsuitable, but can even be harmful to the interests of the given party, and the latter is forced to employ other tactics. On the other hand, repressive measures, applied in a country in the form of inhibitors, can be unsuitable in some instances and acceptable in others.

It must be noted that the power and significance of any influence on the collective depends on the condition of the collective at any given moment, since sometimes even powerful influences can have a weak reaction and vice versa. Thus, at the beginning of a war any event connected to it is greeted by society with a burst of patriotism, whereas when the war has been overly long, the same sort of event no longer stimulates an equal reaction in social circles. This proves that, when evaluating social movements, various social conditions, which in one way or another are reflected in the force of a social movement, must be taken into account. Of course, this does not at all lessen the important and all-encompassing significance of the law of relativity.

Statistics show that more than one thousand steam-powered and sailing vessels sink yearly at sea. The number of victims must be huge, but this does not amaze anybody, since each disappearance occurs separately, on diverse occasions. Yet if we are dealing with the supposed sinking of one large passenger liner, a tempest of anxiety sweeps through society.

The same is manifested in relation to the significance of other undertakings.

Thus, when troops did not obey the command of their leader during the tsarist regime in Russia, their disbandment was considered a drastic measure, whereas under the conditions of the revolution, with the weakening of military tradition, the measure was acknowledged to be insufficiently strict and was replaced by reorganization, reeducation, and mass repression.

Another example: at the time of the Great Russian Revolution, the death sentence was revoked, but, after a short time, when excesses and traitorous acts became widespread in the army, it was reinstated by the same individual—Minister Kerensky—who had originally revoked it.

The very attitude of society to the death sentence changes in relation to its living conditions. In peaceful times, an execution deeply shocks the social conscience and often serves as a subject of accusation of the government on the part of the majority of the population, while at a time of war even mass shootings do not stimulate a reaction in society. In general, the attitude towards the execution of convicts is far from identical in cases when the occurrence of an execution is rare and in ones when executions are conducted

on a massive scale. Earlier, we spoke of the fact that, at the time of the constant use of the guillotine during the French Revolution, there were those who, going onto the scaffold, taunted their executioners or made completely uncalled-for jokes about them. This is reminiscent of the attitude of those victims that end up on the dinner table in countries where cannibalism is dominant.

Thus, it is known that the South American Guarajos, upon catching their human prey, strangle each captive successively. "The victims, who were as bloodthirsty as their captors," we read in C. Letourneau's *Sexual and other Morality*, p. 85, "died in the manner of redskins, singing a death-song in which they taunted their executioners, reminding the latter of the many friends and even relatives of theirs that had been eaten by the former."

The principle of relativity is evident in many other manifestations of social life, and especially in the replacement of one social phenomenon by another.

Thus, the heavier the yoke from which a nation frees itself, the greater the repression of the former oppressor by revolutionaries and victorious dissidents. However, that repression is also in direct correspondence with the extent to which the nation is freed from guardianship. This explains much of what we observe in the development of the French, Russian, and any other revolution.

According to G. Le Bon (*The Psychology of Socialism*, p. 74),

The end result [of national government] is first anarchy, then the despotic rule of one of those tyrants that a crowd always turns to when driven by a slave-like instinct of self-preservation.

Such was quite often the case in the history of democratic disorder. After Sulla, Marius and the civil wars came Caesar, Tiberius, and Nero; after the Convention came Bonaparte; after the year of '48 came Napoleon III, and all these despots were adored by the masses. How could they have remained in power, if the nation had not felt sympathy for them?

In the words of the same author, "Socialism may triumph; but it will not survive for long, and the regime following it will be one of oppression that for a long time will suppress or reverse the civilization of the country in which it is based." It is possible to disagree with the author over the substance of his views on socialism, but his knowledge of social conditions cannot be

refuted. The essence of the issue lies in the fact that an extreme shift of social life in one direction is usually followed by a shift in the opposing direction.

In this way, the more powerful and damaging is the destruction of former foundations of society in revolutionary periods, the more furiously is a reaction manifested—such is the law of history. Similarly, the more street robberies develop, the more frequently and violently society's self-defense is manifested in the form of lynch-ings and brutal punishments.

A broken and degraded army's morale always falls, and this is expressed in various thefts, drunkenness, coercion, and depravity. This is the typical reaction to despair, similar to the way in which a person drinks and abandons morality when subject to despondency or grief.

Even external symbols of a military nature are subject in the above case to disregard and destruction. Weapons are discarded, sold, or plundered and signs of excellence are removed; outfits are destroyed or thoughtlessly wasted, etc. The Russian Revolution is a perfect example of this situation, and the decrees on the removal of shoulder-boards and decorations were only a reflection of the general views among soldiers on the superfluity of both, while the decrees against the plunder of government property were publicized when, unfortunately, hardly any military equipment remained.

During the Russian Revolution, in order to protect against a counterrevolution, two battering rams were deployed—the declaration of soldiers' rights and the land question. The first was directed at the commanding officers of the former army and was reflected in discipline; the second, directed against landowners, or the bourgeoisie of the countryside, was meant to stimulate an uncontrollable pull towards the land. However, both these conditions were fatal not only for the army, but also for the revolutionary government itself in the question of democratic peace with Germany. It took incredible effort for the revolutionary government to reinstate in the army its former principles of command.

We have not listed a large number of those phenomena that are manifestations of the principle of relativity; but the given examples are enough to allow us to assess the significance of this universal principle in social life, in which it is as vividly expressed as in other manifestations of organic and inorganic nature.

Note

¹ *Annalen d. Physik*, 17, 891, 1905.

12

The Law of Evolution

Continual change on its own is not yet evolution, since change is the result of continual motion, whereas evolution is the development of one form from another and, consequently, the gradual or rapid transition from one form to another. It is true that change, although it does not entail the creation of forms, is usually accompanied by variation in situation and form. From this it becomes clear that the law of evolution complements the law of change. We, nonetheless, associate evolution not only with change, to which any movement in general is related, but also with the transformation/ conception of new internal or external forms, which represents a certain level of accumulation or deficiency of a substance, as well as its displacement. In a word, evolution, representing the development of new forms, is essentially a creative act, and the law of evolution is in essence a law of creation.

The significance of the discovery of this universal law cannot be overrated. According to Haeckel, the formation of our contemporary theory of evolution is superior to all other achievements of the human spirit. Already foreshadowed one hundred years ago by Goethe, it was satisfactorily expressed by Lamarck only in the beginning of the nineteenth century, and acquired its final form thanks to Darwin in 1859. Darwin's theory of natural selection filled the blank left in Lamarck's teachings in relation to the interaction of heredity and adaptation. We now know for certain that the organic world developed slowly, step by step, according to the same "eternal, unchanging laws" that Lyell, in 1830, proved applicable to the inorganic body of the earth; we know that the countless species of animals and plants that inhabited our planet in the course of millions of years are simply branches of

one family tree; and we know that mankind is the newest, most perfect shoot of the family tree of vertebrates.¹

The first investigations that firmly established the law of evolution (those of Lamarck, Spencer, Darwin, et al.) led to the conclusion that the creation of new forms occurs slowly, as a result of which a formula was established that stated that nature does not advance by leaps and bounds; however, further investigations showed that the creation of new forms hardly ever proceeds gradually, and even makes large leaps (de Vries), in which some saw the revocation of the law of evolution. In essence, this instance is only an extension of the understanding of evolution as a development of new forms or as a creative process, only travelling in more rapid shifts or jumps and not in the small and barely noticeable changes that were earlier acknowledged to be instrumental to the law of evolution.

The law of evolution is a universal law, scientifically proven in all areas of physico-biological knowledge, so it is quite natural that it should also be applicable to social life and, consequently, to the development of collective reactions:

It must be noted that in the process of evolution we meet with changes in phenomena that must be considered irreversible. The progression from fetishism to idolatry is as irreversible as the transfer from animal gods to human gods, from zoomorphism to anthropomorphism, and even from creature-gods to gods of domestic animals. (Tarde, 1. c, pp. 205-6)

Examples can also be given of irreversibility in linguistics, such as the transformation of hard endings of verbs into soft ones in a large number of Romance languages that have their roots in ancient Latin. To give further examples would be unnecessary.

The law of evolution can be traced in all social phenomena, beginning with language, which, coming from one common root, changes into newer and newer forms through gradual differentiation and integration. Every custom has its own history in which it is likewise not difficult to find manifestations of the law of evolution. It is still easier to trace evolution in myths—be they about Cerberus or Hercules. In both cases we can follow the myth's primary development and all the further changes to which it is subjected in connection with the views of various nations. Similarly, does not the development of religious doctrines, beginning with Vedism, through to Brahmanism and

Zoroastrianism, then from Moses to Christ and Mohammed, and finally to the various branches and sects of Christianity and Islam, give the opportunity to establish in full the role of the law of evolution?

Furthermore, evolution can be traced in any social process, even in the mood of the masses, which, originating as an almost invisible spark, gradually gains strength and reaches its apogee. The creativity of nations is also subject to a corresponding evolution; after all, does not artistic, literary, and scientific creativity evolve, in its turn, as does everything that is the result of the creativity of the human mind? Is not Theocritus reflected in Virgil, Menander in Terence, Plato in Cicero, Byron in Lermontov, etc.?

Since the law of evolution is common to all social movements, it is understandable that every social movement, wherever it is manifested, undergoes the same stages of development. However, when one collective or another reproduces a social movement that has already been experienced by a different collective and which spread to the former collective thanks to imitation, then the various stages of the movement are not recreated in their entirety, but are experienced in a more or less condensed version, in a manner similar to the way in which, in the biological world, an embryo goes through diverse stages of development. This gives young nations, which must go through the stages of development of social movements that have already been undergone by established nations, the opportunity to live through them in a shorter time, with the many interim stages of social movements passing, in these cases, in an edited form. This is why our Russian Revolution, which was in many ways a copy of the French one, went through the various stages of its development much faster than did the latter. The same can be observed in relation to all other social movements that have developed on different areas of the globe. Thanks to this, nations that have attained the level of development at which cultural achievements become realizable through initiative and imitation can experience a number of historical steps at a speed much greater than that which was attained by nations whose cultural success was achieved primarily through their own efforts. Russia of the seventeenth and nineteenth centuries, and Japan, which until recently was an isolated, largely uncivilized country and now enjoys a high level of cultural development, can serve as such examples.

It is customary to think that social evolution does not make sudden leaps, although it would seem that this is contradicted by, for example, the

acceptance of new religion by a nation. But here it only appears as though a rapid displacement takes place:

Everyone knows that all great religions—Brahminism, Buddhism, Christianity, Islam—stimulated mass conversion among whole races, which apparently accepted them immediately; but upon entering more deeply into the analysis of these conversions it can immediately be seen that, though a nation may have changed its religion, it did not in actual fact discard it; in reality, the accepted beliefs underwent the alterations necessary for them to be made a part of the race's former beliefs, and were only a continuation of the latter. This explains the changes experienced by a given religion upon its adoption by a nation. Chinese Buddhism, Islam on the Deccan peninsula, etc. can serve as examples of the above. (Le Bon, *The Psychology of Nations*, pp. 79-80)

However, G. Le Bon's opinion cannot be accepted unconditionally. As there are leaps in the biological world, in the form of the so-called de Vries mutations, so there are revolutions in the social world, which are quite lawful phenomena in the course of historical process.

It should also be kept in mind that evolution always goes from the general to the specific, as for example with the emergence of the proto-language preceding the development of separate forms and, above all, mythology and sciences. The development of science also went from more general to more specific fields, and so it is in all cases.

In general, the evolutionary process in human society is determined by many factors of a geographic, climatic, or socioeconomic nature and, as a matter of fact, assigns a very small historic role to separate individuals, be they geniuses or no. Everything comes down to the evolutionary process of the society itself, i.e., of national masses. Separate personalities, however, can nonetheless play a huge role in the latter if they manage to discern astutely the pulse of national life and the direction of historical process, thereby applying their energy towards the corresponding development of the national masses. It is possible, of course, that a powerful personality, employing its high position, can hinder to some extent the proper evolution of the social process. However, in such cases the individual must be analyzed not on his own, but as a personality that has better comprehended the legacy of past generations and is supported by one collective or another, embodying

its hopes and aspirations, as a result of which the role of that individual is determined by the social significance and power of the collective he represents. In this way, the role of a ruler, military or national leader, demagogue, or representative of a certain class or party is not so dependent on the qualities of the individual himself, which are obviously not without significance, as on the importance of the mass of people that is associated with that individual.

In the manifestation of the law of evolution, the inequality of collective units and individuals has enormous significance. If nature had given everyone the same characteristics and abilities, then obviously there could be no imitation—that mighty factor in social life—since there would be nothing to imitate. There would likewise be no aspiration to perfection, the other important factor in evolution, no conflict or rivalry, those still more important impulses of social movements. In exactly the same way, it would be almost impossible to realize social division of labor, specialization, or differentiation of social strengths, which greatly assist the success of a civilization. In a word, in place of progress we would have the more or less complete stagnation of social life. The proper organization of social life assumes the equality of all people in relation to social rights, and only this principle guarantees fair treatment of individuals. But nature does not treat all people equally, and the living conditions of separate individuals, as of separate nations, are not the same. This lies at the basis of individual national peculiarities, which have enormous significance in the evolution of human society.

Shall we apply the law of evolution to various social phenomena? Let us begin with the evolution of power. State power or government arose from the concept of the family, which is in itself a kind of self-sufficient state. These states, or families, reproducing and merging into tribes, created tribal states which, in turn, grew and became nations. States also arose through the establishment of collectives whose members pursued a shared goal, for example, plunder. Runaways from surrounding villages would join the horde or band and, in this way, the pillagers would eventually grow in number and subject the villages to their rule, which would become lawful upon an agreement between the attackers and their victims. Such was the origin of Rome in all its glory, and, subsequently, of states that owe their existence to Mongol invasions.

First to be established in a developing state is a communal structure; the latter evolves into a monarchy, which does not arise before the development of private land-ownership, under which system free landowners can be stripped of their fields by their more powerful neighbors:

The ability to acquire large land areas and the so-called process of feudalisation and serfdom transforms formerly free landowners into the servants of their stronger rivals, and removes any remnants of a free, communal system based on the concept of a popular assembly (in Russian, *veche*). Inks stead, a council arises, made up of the best and noblest men, who govern their estates and unite under the most powerful landowner—the prince, who is acknowledged by all of them as their sovereign. (Takhtarev, *Society and State* [in Russian], pp. 12-13)

It is clear that with the acquisition of material possessions comes division into classes, and along with that comes the beginning of class conflict. It must be noted that the path of political forms was already charted by Aristotle, who, based on an analysis of the 157 nations known at that time, drew up the sociological law of creation and displacement of various political forms of state existence and rule in connection with the correlation between classes and the conditions of class conflict.

Generally speaking, continual transformation in collectives through evolution usually leads to the complete restructuring of their organizations, either towards the centralization of power or towards its decentralization. Thus, for example, the Catholic and Orthodox church went from being an evangelical, democratic commune to a hierarchic organization ruled by an episcopal aristocracy, and then to a monarchy controlled by a patriarch and a Roman high priest limited by synodal edicts. Once the Pope proclaimed himself infallible, Catholicism became monarchical absolutism.

In exactly the same way ancient Rus', with its independent princes, evolved into a limited monarchy, then into an absolute monarchy.

However, that absolute monarchy has once again been exchanged for democracy, first in the form of a limited monarchy, then in the form of a republican structure with decentralized local rule and a federational state organization.

This exchange of one condition for another in the organization of society is determined by the fact that, according to the law of inertia, centralized power

gradually reaches an absolutism that becomes unbearable for the nation and instigates a reaction in the form of fierce opposition, leading to the deposition of the absolute monarch; conversely, democratic power can also bring a country, in accordance with the law of inertia, to impossible conditions, sometimes causing civil war and stimulating opposition in society, disposing the latter towards monarchy again.

Archaeology and ethnography complement still more our knowledge of the evolution of states:

We can now establish to some extent the embryology of all our social institutions. The most distant past of our societies has unfolded before our eyes. It is possible, for example, to study the creation of societies *de visu*. These are at first the primordial and anarchic hordes of the early days of the Quarternary period, such as can be observed in the present day on Tierra del Fuego. Then these hordes join into classes and tribes, the latter divide into castes, then become subject to leaders and, finally, create despotic monarchies, and so on. (Letoumeau, *Morality*, p. 25)

In general, investigations into the law of evolution of social conditions and the development of civilization find extremely valuable material in ethnographic research. Letourneau, in speaking of the evolution of the human race, remarks (*Morality*, p. 23):

History, even with the aid of legends, reveals to us only one moment out of the evolution of the human race. With the help of ethnography, we can hope to go to its very sources, reconstruct all the missing links, and tie the beginning to the end. The steam engine is in this way connected with flint, the steamship with primordial boats, the palace with cave-dwellings, polysemantic languages (i.e., those in which one word has various meanings) with simple ones, differential calculations with the primordial numerations of an aboriginal who futilely attempts to count his own fingers, the great Aryan religions with the animism of an African Negro who bestows upon his external surroundings a conscious life similar to his own. Raphael becomes a distant descendent of the primitive artists of the Department Lozère, France.

In its turn, the ownership of land arose through the population of certain areas by members of the tribe that had annexed them in some way or other. In

due course, the land was divided, becoming the property of classes, families, and finally separate individuals. The way in which language developed leads us to assume that in the distant past language could have arisen from onomatopoeic signs, associative reflexes, and ordinary sound reflexes: oh, ow, ah, hm, etc.² The associative reflex of speech that arose in this way gradually differentiated through additions, inflections, reduplication, and integration by means of figurative expressions or words that denoted general concepts. Thanks to imitation, language would then have spread among families of the same race, class, or tribe or in a certain collective, be it a commune or a horde. In this way, a race that spoke a single language, which would, however, have its own dialects thanks to the division of the populace into separate collectives, would gradually have been created.

It can likewise not be doubted that during its development and perfection, language progressed from gestures to words of concrete meaning, and from concrete words to abstract meanings. The language of our contemporary savages or so-called primal nations is still rich in gestures. It is furthermore established that there are no abstract and comparatively few collective nouns in primitive languages, since words and gestures express concrete objects. Because of this, primitive languages are rich in substantive nouns, verbs, and prepositions (L. Brühl, *Les fonctions mentales*, p. 175-onwards). The language of gestures serves to represent concrete phenomena, such as action, placement, etc., and is not composed of separate gestures, but is made up of complicated complexes of them which characterize a certain action. These combinations are a kind of mobile description or imitation. For example, making a motion as if raising liquid with cupped hands represents water, etc.

The further evolution of language is relatively well traced thanks to the successes of comparative linguistics. But we will not go into detail on the subject. We shall limit ourselves to pointing out that a basic language can change in pronunciation due even to its social surroundings. Thus, for example,

various Latin words, going from a Roman environment to a Spanish or Gallic one, underwent regular and characteristic changes, with each letter being replaced by a certain other letter, so a particular consonant in German or English is equivalent to another consonant in Sanskrit or Greek; this essentially means that a root language, going from a primitive Aryan environment into a Germanic, Greek, or Hindu one changed its

consonants in the manner outlined above, acquiring an aspirate sound instead of a hard sound, and vice versa. (Tarde, *Social Logic*, pp. 22-23)

The evolution of views on the causality of phenomena is what we call the evolution of human thought. As is known, Comte differentiated the three stages of thought: theological, metaphysical, and positive.

But Stuart Glennie maintains that the basic law of intellectual development is the law of transition from a quantitatively undefined to a quantitatively defined conception of the interaction of objects; in other words, that evolution moves from extremely incomplete views on the causality of phenomena towards the establishment of more precise views.

In place of the three Comte stages, the author divides the development of human views on causality into the following steps: magical, supernatural, and scientific.

The first stage encompasses that period in which mankind, knowing nothing about the reasons for natural phenomena and finding activity in itself through self-observation, naively assumes that this activity influences its surroundings. This is a period of national infancy, throughout which mankind believes that in order to cause rain it is enough to water the earth to the accompaniment of certain incantations; that to wreak vengeance on an enemy it is necessary to pierce his effigy; that to cure a person of an ailment it suffices to figuratively remove a certain object from his body, etc. Here, we are dealing with the association of one's actions with a certain magical force, which supposedly influences external phenomena and live objects.

In the next stage, mankind acknowledges the reason for the phenomena of the surrounding world to be a certain spiritual, invisible force. This step arises with the development of an understanding of the spirit as a special creature that actually exists.

At this stage, mankind populates nature with invisible creatures that have characteristics similar to his own, and thinks that the phenomena of nature are a result of their intervention.

Here, we are looking at animism in the form of the integration of one's subjectively understood activity with natural phenomena, and consequently at the integration of one's own spiritual conditions, evaluated from the point of view of a spiritual force.

The third stage acknowledges strictly mechanical causes for all the phenomena of nature, with the exception of their basic origin in the form of

the Creator of the universe, who controls the world in accordance with established laws; the supernatural has no place at all in natural phenomena.

We will say, in addition, that a further and final stage of development of human thought must entail an exclusion of supernatural forces not only from natural phenomena, but also from the concept of the beginning of the universe, which would conclude the evolution in our perception of the surrounding world.

Let us likewise note that we cannot imagine a case in which one of the above stages is consecutively exchanged for another. Usually, in one and the same period it is possible to meet with manifestations of the first and second, or only second, steps in one sector of society, while in other more intellectual sectors even the third stage can be observed. It is hardly necessary to say that the latter stage is the result of the application of experience in life and in the laboratory in order to prove the complete regularity of all natural phenomena.

As for the evolution of scientific knowledge, it goes, as does the evolution of language, from the general and undefined to the specific and more defined. The progress of knowledge, uncovering deeper and deeper correlations between phenomena, also finds many inherent discrepancies that were previously not noticed. Thanks to the fact that important integrations of the accumulated facts are achieved by geniuses, it is natural that the evolution of knowledge advances by leaps and bounds. It is thanks to this powerful synthetic ability of the human mind that mankind achieved its greatest discoveries, beginning with Galileo's teachings, Newton's laws, and so on. Vast collective work of the human intellect was realized in a similar manner, and expressed in encyclopedias of scientific knowledge, in codes of laws, in social morals, in artistic creativity, etc.

In art, as in knowledge, evolution proceeds by startlingly rapid progress, thanks to new ideas and new understandings that seem to refresh faded ideas and outlived knowledge. Thus, the Romanesque style was transmuted into the Gothic one thanks to new influences stemming from the Crusades. Later, Gothic art underwent a transformation into Renaissance art under the influence of renewed interest in all aspects of Greco-Roman life.

From the above it should be seen that there is nothing that would suggest or justify the parallel development of science and art. Each branch of correlative activity develops successfully at a different point in time. Let us also note that ideals, as symbols of the beautiful, change along with race and

epoch, and so each race creates its own art, just as every era is associated with a distinct art form.

What, then, is the root impulse of the evolutionary process? In biology, the process of evolution is determined by the process of fighting for survival, which is based on the needs of an organism on the one hand and the inequality of different organisms' resources for attack and defense on the other. But in our opinion—and likewise in Kropotkin's—cooperation, based on the profitable achievement of a goal, i.e., the satisfaction of needs, also plays a role in the evolutionary process and creates conditions that improve an organism's chances for survival via better offense and defense. The same can be observed in society, at the foundations of which lie rivalry and conflict, which are based on the same principles of collective needs and unequal distribution of resources for offense and defense. Cooperation also has significance here, since it makes the achievement of goals, and therefore the satisfaction of needs, more likely and creates better conditions for defense and aggression in conflict.

Can there really be aspiration toward activity without the stimulus of needs? Obviously not. But to what conditions are rivalry and conflict subject? Apparently, only the inequality of strength and resources. What determines cooperation? The aspiration to strengthen one's resources for rivalry and conflict.

From the above it is clear that social equality is possible only as an ideal, since with its advent there would no longer be an impulse to strive for new social achievements. This is why, if socialism has equality as its ideal, then it is only equality of social rights; it cannot aspire to level the abilities of separate individuals. For, if this ever occurred, it would lead to the complete termination of any social progress.

The evolutionary process is often confused with progress or even improvement, which is incorrect if only because evolution is development without distinction, and can go forward, towards progress, or backwards, towards regression.

However, the understanding of progress requires its own definition. It must be noted here that the question of what is perfection and what is progress cannot be resolved without a certain preface, i.e., a clarification of the aspect from which we should approach the question itself, since that which we call improvement from one point of view can be considered regression from another. Let us imagine that we have a machine which is designed to execute

a certain task, for example a flying apparatus. Upon becoming acquainted with its component parts, we would perhaps find it more complicated in comparison to other existing systems, but the degree of perfection of the apparatus itself, independent of its complexity, would be determined by its greater or lesser suitability to its application, i.e., to aviation. And so, in all cases involving tools or machines, the degree of perfection is determined by how well a tool or machine is suited to the purpose for which it was designed.

The concept of progress is still more general. It deals with forward advancement, but not necessarily with perfection, since progress can occur in the direction of backward development. Because of this, although perfection is inextricably tied to progress, progress itself can be accompanied not by qualitative improvement, but, conversely, by degeneration. Nonetheless, the word progress is often wrongly used to signify perfection.

Going on to the discussion of the perfection of living nature, it is necessary to keep in mind that here we must also have two separate fields of study, depending on whether we look at the subject from a strictly biological point of view, in terms of the perfection of one individual or another's features, or from a social point of view, i.e., encompassing life as a whole. With respect to the former, we have no basis on which to recognize one organ, one part of the organism as being superior, since an individual is whole and indivisible. Due to this, the opinion that greater development of the nervous system and its central organs partially characterizes greater perfection, which some biologists hold, has no significance here.

An entire organism is a self-sufficient machine, the essence of which comes down, on the one hand, to the acclimatization of the organism to its surroundings, and on the other to the exploitation of nature's external forces through the establishment of multiform interaction of the organism with its environment. From the above it is clear that the role of the nervous system, and partially of the central one, in the establishment of interactions is huge, yet not to the exclusion of everything else, since some parts of the body are tools through which the nervous system fulfills its role, and therefore the quality of these tools should be taken into account. In the end, the degree of an organism's perfection should be evaluated in terms of the extent to which the organism has established interactions with its surroundings, and not in terms of the size of its brain.

Furthermore, the exploitation of a given environment, and likewise acclimatization to it, is ensured by the differentiation of organs and the

coordination of their functions, so we could say that a more advanced organism would be the one that comprises the greatest differentiation of organs or the greatest coordination of their functions; however, the final result of the external manifestation of both these processes is the establishment of multiform interaction, so the latter is a more reliable measure of an organism's perfection.

Some think that, in analogy with the artificial machine, acclimatization to an environment and the exploitation of surrounding conditions are executed in order to fulfill the goal of self-preservation and so are a manifestation of perfection; but this would entail holding a teleological point of view, whereas nature itself knows no goals. This is why we retain a strictly objective definition. If we were to uphold a teleological definition, which is wrong in its very essence, then we would have to acknowledge that an organism that is better equipped to preserve its own life is also more complete, but, in such a case, we would discover that single-cell organisms, in not knowing death, are more advanced than ourselves. Ephemera would be less perfect than lower species, and elephants, parrots, crows, and others would, in having longer life spans, be more perfect than humans.

In this way, the only correct definition and understanding of perfection is as self-establishment through adaptation to one's environment and through exploitation of that environment via diverse interactions with it.

The same can be applied to species. It can be said with certainty that every biological species is the product of its environment and its era. This is why we cannot speak of the greater or lesser perfection of a species in terms of its greater or lesser ability for self-preservation. In a certain era, with the alteration of surrounding living conditions, one species could become better adapted to the achievement of the above goal than another, and vice versa. But all species can be compared to one another with respect to the diversity of their interactions with their surroundings, and so, depending on the extent of the above, their greater or lesser perfection can be determined. Since the nervous system is the apparatus that undertakes more complex interactions with the environment, it is clear why its superior development is considered a sign of perfection; however, not only the nervous system, which some species do not even possess, is important here.

If we turn to social progress in the sense of perfection, then here, too, a teleological explanation is often given. Thus, according to some people, progress is simply a change in society that brings it closer to its ideal, and

that ideal must appear better than the contemporary situation, even if in practice it turns out to be worse; in other words, we are dealing with an evolutionary process that goes from worse to better, no matter what the understanding of "better" is. Simply speaking, the determination of what is "best" is based on other theoretical considerations, while the ideal itself is acknowledged to be a previously established norm. There is no need to say that this definition of progress does not stand up to criticism.

Another definition of social progress corresponds to our understanding of social evolution. However, in this delineation it is assumed that the most recent form is the best one, which is essentially wrong, since evolution is not necessarily synonymous with change for the better. Evolution is change connected to the creation of new forms, but these new forms can in reality be worse than the old ones, which is not a rare occurrence since society can not only progress, but also regress. Thus, for example, one reform or another is a manifestation of the evolutionary process, but it can turn out to be a regressive rather than progressive factor and must lead to the corresponding consequences.

Every existing theory of social progress can be classified, in essence, as one of the above views. Thus, the first group would include all authors who have proposed principles as ideal norms of the social conditions of existence.

The history of science is full of many such norms that were thought to be ideals. These include:

the principle of the Golden Mean (Aristotle), the principle of happiness or benefit (Hedonism, Eudemonism, Utilitarianism), the principle of love (Christianity, Petrazhitskii), the principle of the imperative (Kant), the principle of life (Nietzsche, Guyau), the principle of community (Wundt), the principle of unity (Nathorp), the principle of solidarity (M. Kovalevskii, L. Bourgeois, and almost all contemporary French sociologists), the principle of growth of knowledge (Comte, Bockle, de Roberti), the principle of general well-being [success] (Hoeffding), the principle of conservation of strength, etc.³

Of course, this enumeration does not nearly encompass all existing principles, and far from exhausts the list of names that accompany one principle or another. Utilitarianism ("the greatest good for the greatest

number"), for example, has many more disciples, including Mill, Bentham, Spencer, Tarde, N. K. Mikhailovskii, P. L. Lavrov, and others.

Another trend, analyzing progress as evolution, likewise has a number of followers, including Condorcet, Comte, Spencer, Ostwald, Novikov, and others.

The obvious incompleteness of such concepts of progress led to the complementation of that understanding by E. V. de Roberti. According to that author, the process of change that occurs in a biological environment is evolution, and the process of change within a social environment is progress. However, since teleology is the basic precondition of social existence, E. V. de Roberti states that "progress is a type of evolution that has, for us, become a goal" (*Qu'est ce que le progrès?*, pp. 34-35).

But evolution does apply to the supraorganic world, though it may not satisfy the requirements of progress. It cannot be limited to a biological setting, since in the supraorganic world it is obvious that evolution can and does take place either towards improvement or towards degeneration. If we include in the understanding of supraorganic evolution the concept of teleology, then this view becomes very similar to one of the aforementioned principles. In this way, the preceding theory of progress does not have any advantage over the others.

We maintain that here, as in other cases, teleological and subjective points of view must be excluded. And so, as in the biological world, the social process must be characterized by the establishment of diverse interactions with the environment in terms of adaptation to it and the exploitation of it through greater social differentiation (division of labor) and greater coordination of the separate components that make up the society. The reverse would characterize social regression. However, the general understanding of progress can be separated into more specific parts. Thus, the form of relations between the individual and society, which in the present day best corresponds to the development of individual and social creative activity, that leads to the subversion of surrounding nature to the aspirations of mankind as a social individual and to the aspirations of the collective itself, should be considered a specific form of progress. This subversion of nature to the aspirations of the human collective is accompanied by growth in the productivity of the given society's labor, which leads to the broadening of its vital interactions.

Social regress, meanwhile, is characterized by the suspension of the progressive movement of social life and of its creativity, is determined by certain internal or external inhibiting conditions, and may even result in the disintegration of the collective itself.

The disintegration or destruction of the collective leads to its complete ruin, which consists of its dispersion, or more precisely of its transition from a structured and harmonious state to an unstructured and discordant one, its transition from social life to socially unstructured life, just as the death of an organism is a transition from the collective existence of its cells to disintegration into comparatively structureless molecules and atoms.

In collective reflexology, as in the rest of the world, the general rule is that creation in the evolutionary process always goes more or less slowly and sequentially, whereas destruction and disintegration nearly always proceed relatively quickly. Because of this, social, strictly organized collectives, whose creation has taken centuries, can be destroyed in a relatively short time, as is demonstrated by the history of the Roman Empire as well as by that of many other states, and even by the history of tsarist Russia.

Notes

¹ Haeckel, *Monism as the Link Between Religion and Science*, p. 45.

² See V. Bekhterev, *Objective Psychology* (in Russian), 3.

³ P. Sorokin, "What is progress?" (in Russian), *Novyie idei v sotsiologii*, issue 3, p. 146.

13

The Law of Differentiation

At the present time, the law of differentiation is an established truth in the organic and inorganic worlds, so we will not delve into its role in those areas. We will simply provide a quotation from a speech by the well-known naturalist Haeckel, which will show us how the process of the gradual differentiation of the world is now perceived:

In the very beginning, there is nothing in the endless space of the universe except mobile, elastic ether^{1, 2, 3} and countless, homogeneous particles dispersed throughout it in the manner of dust in air; these are *primary atoms*; it is possible that the latter are in essence only "points of condensation" of a vacillating substance, the remnants of which form ether. By combining in certain ratios into groups, these primary atoms, or atoms of mass, form the elements identified by contemporary chemistry. In agreement with the popular Kant-Laplace hypothesis, from such a vacillating "primordial nebula" arise rotating celestial bodies. Among the many thousands of bodies in the universe is our sun; it gathers planets around itself that have broken away from its mass thanks to centrifugal force. Our little *earth* is only one of the planets in the solar system; all individual life on that sphere will be the result of the action of sunlight. When the white-hot planet Earth cools down to a certain degree, water droplets that are prerequisite for organic life settle on the surface of the earth's hardened crust. *Carbon* atoms begin their organic activity and join with other elements into plasma that is able to expand. A small clump of

this plasma escapes the constraints of cohesion and individual growth and splits into two identical parts.

This first *monera* initiates organic life, along with its corresponding function of succession. In the monera's homogeneous plasma, soft internal mass coagulates into a solid central nucleus; this differentiation of the *nucleus* and the protoplasm creates the first organic cell. For a long time, our planet is populated solely by such protists or single-cell organisms. From the union of these protists result the *lower histons, multicellular plants and animals*}

We provide this quotation only to illustrate how, in accordance with the law of differentiation, the diversity of living things is thought to originate; we do not necessarily consider all of the author's opinions as correct. It is known that many physicists now reject the hypothesis concerning ether; furthermore, the author does not touch on the construction, and consequently on the creation, of atoms from electrons, on spinning nebulae, etc. The essence does not lie in whether or not everything in the above scheme is true to reality, but in the outline of that scheme, in how the evolutionary process is thought of as the differentiation of primitive forms, gradually leading to greater and greater diversity and, consequently, to the complication of the external world.

We know, moreover, that in the organic world the law of differentiation can be traced throughout the development of every separate individual, beginning with the moment of the fertilization of the egg.

However, in the social life of every collective, as in that of mankind, we must also keep in mind the law of differentiation or of the isolation of individuals according to certain characteristics, and the law of selective integration, which leads to the grouping of individuals in relation to their peculiarities and aspirations.

The historic development of mankind proceeds in accordance with these two basic laws.

Ribot (*The Evolution of General Ideas*, p. 31) states, while discussing the development of a child's mind:

The only suitable formula is the following: the mind goes from being undefined to being defined. Even if we mean undefinitiveness in the sense of generality, then we can maintain that it is neither the specific that arises

first of all, nor the general in the exact sense of the word, but the undefined.

Something similar can be found in relation to society.

Primitive society is undifferentiated or, more exactly, very slightly differentiated. It is held together only by joint existence, common interests, and mutual interaction, established thanks to experience. This is the society of the most primitive savages, who know neither state life, nor division into castes. In the civilized world, such a society can be represented by a group of children being raised in a common environment. Only with age is differentiation and organization originated in this society, which simultaneously brings with it the integration and coordination of the differentiated sections of society.

The state as a political organization of society can vary in accordance with the stage of development and condition of a given society, taking forms ranging from republican to absolute/limited monarchical. In both cases, there is a number of further subdivisions, including *veche* democracies, feudal states, elected and hereditary monarchies. Even a single state can differentiate into self-governing areas upon decentralization. This process of differentiation is based on the fact that the satisfaction of physical needs is better realized when some parts are isolated from others, as is any process of division of labor.

In this way, the differentiation of the state as a political and social organization reflects the level of development and condition of the given society, and so can create self-governing communities under a *veche-type* democracy and then a feudal state with an elected or hereditary suzerain at its head. More complex state forms such as monarchies and democratic or bourgeois republics are the products of further differentiation in the state organization of society.

Any viable collective in general, once formed, immediately differentiates. Already, in the very beginning, a presidium or bureau is formed in the collective, as well as an executive, a treasury, an organization of special sections, commissions, etc. In larger collectives, for example in states, differentiation is much more complicated.

Here we are dealing with the isolation of organs of government, i.e., departments, and with the division of the remaining mass of the population according to descent (differences in nationality), labor (professions and

unions), property (classes), education (intellectual vs. nonintellectual), views, etc. Further differentiation leads to the formation of separate organizations, co-operatives, commissions, organs of self-government, a central government, various ministries, etc., etc. Moreover, differentiation in the collective occurs in relation to vocation and even to the very tools of production.

According to Waltman (*Darwin's Theory and Socialism*, p. 242), the development of mankind's social life is subject to the biological laws of differentiation, acclimatization, selection, etc., as is the development of organic life. The main contrast, however, lies in the fact that humans' organic forces are complemented by technical tools, i.e., forces separate from the organism that interfere with the law of organic selection of individuals, organic succession and perfection.

The law of selection of organs and organisms can also be seen in the realm of technical tools, in that the tools—thanks to humans, of course—differentiate, conform to new goals and produce newer and better forms. In the history of tools there is likewise a constancy of form.

Lafargue is, in this sense, correct in saying that perfection and the fight for survival are enacted not among people, but among artificial organs.

The law of differentiation in social life encompasses the division of labor and, in general, the gradual separation of a primarily simple and homogeneous collective into various parts in the form of separate circles, parties, etc., and likewise in any manifestation of individualism in society:

In the lower races, says Le Bon (*The Psychology of Nations*, p. 34), all individuals, even when they belong to different sexes, are at almost the same psychic level. Being quite similar to one another, they enjoy that equality of which contemporary socialists dream. In the higher races, the inequality of individuals and of the sexes is, conversely, taken for granted.

In this way, whereas savages have few dissimilarities in relation to personality, qualities of the mind, and correlative functions, historic nations in general are quite disparate. Take a Russian, an Englishman, a Spaniard, is it not possible to distinguish one from the other by observing their manner, their expression, their actions? Does not history show deep-seated differences between these nations in the entire makeup of their social personality?

One of the main results of civilization, says the author quoted above (I.e., p. 35),

is, on the one hand, the differentiation of races through the constantly increasing mental labour experienced by nations that have reached a high level of culture, and, on the other, the greater and greater differentiation of the diverse layers that make up every civilized nation.

It is clear that organization is an inescapable result of the differentiation of any community. Organization can take various paths, proceeding, for example, in the following manner:

(1) acts of interaction, (2) their repetition, (3) custom, (4) juridical norm, (5) juridical institution, as the totality of norms of behaviour in a certain sphere of relations (family institution, institutions of ownership, power, inheritance, court, etc.), (6) a social organization as the totality of the juridical institutions of the given population sector. (P. Sorokin, I.e. Parti, p. 340)

In another case we could be dealing with a shorter path, consisting simply of an agreement among members of the collective, of the formulation of its statute, and of its design and implementation.

Any movement is at first more or less chaotic and general in nature; only upon development does it gradually acquire a more and more defined shape, in other words a more specialized form, attaining a definite physiognomy.

In this way, any social movement goes from being slightly defined to, over time, being more defined.

This process of the internal organization of a social movement, when it goes from being general and chaotic to being more specific and defined, is what we call differentiation.

Let us take any crowd that is gathering as a consequence of a given dissatisfaction. In the beginning, almost nobody knows why he is in the crowd and what the result of the gathering will be. When the first small groups are formed, a mutual exchange of opinions takes place within them. The basic reasons for unrest become more and more clear upon the above exchange of opinions in separate parts of the crowd and later within the entire body, and gradually this general dissatisfaction takes on a concrete form, following which definite accusations are directed at certain individuals.

In this way, the animosity that was at first very vague and undefined differentiates into a concrete dissatisfaction aroused by the actions of certain individuals and by the order that they established. It is obvious that this differentiation of a social movement, in the given case of crowd movement, goes from undefined and chaotic enmity to concrete dissatisfaction with certain personalities, and can even lead to violence.

However, all that was presented in the above example can only take place if the crowd's mood is firmly established against certain individuals, which may in fact not occur. Let us take any meeting at which some sort of question is being discussed. At the outset, no one can predict the course of discussion or what conclusion the meeting will reach, since at first everything is chaotic, uncertain. But now the discussions begin. At first, the question is gone over from various points of view, and still it cannot be said what path the discussion will take; however, in due course the essence of the question is more or less clarified. The views of various parties seem to become more focused, guaranteeing the differentiation of diverse opinions, which is nowhere more easily observable than in the light of collective criticism. Along with this, the resultant majority of opinions is established, and in this way the point of view of the majority is determined. Finally, a resolution is suggested in accordance with the opinion of the majority, which is accepted by the gathering after a vote.

This ends the collective labor that was at first chaotic and undefined, but which through gradual differentiation attained greater clarity and became a single, concise formula.

Let us take, furthermore, the development of a large social movement, such as an uprising. First of all, the matter commences with the silent dissatisfaction of the populace, which from time to time is manifested in the statements of separate parties, the actions of separate groups, and then of whole bands, but in the beginning nothing is specifically determined and the path the uprising will take is unknown.

Then, an organization arises in the dissatisfied nation that attracts like-minded rebels, and the uprising gradually becomes more ordered, changing into a national movement with slogans and a list of demands.

In this way, the national movement that began as hollow, almost totally undefined dissatisfaction gradually achieves complete organization, in other words differentiates into an armed uprising that has a certain program of action and all the necessary resources at its disposal.

Let us take military operations as another example. First to be developed are plans of offensive action, but not one officer is misled as to the fact that in practice the plan is almost never followed to the letter.

The prepared plan is thought of more as a goal towards which it is necessary to strive than anything else, but in the end the achievement of that goal will depend on the various conditions that are encountered. Even when the maneuvers themselves are developed, usually no one can predict how their execution will proceed, since much of what will happen is unexpected, uncertain, and only with the passage of time does the situation and, consequently, the mode of achievement of goals become more or less defined.

It must furthermore be kept in mind that when a collective encounters external influences of one kind or another for the first time, its reaction is, in the beginning, undefined and chaotic and only later does it take on a certain form.

In the above cases, the law of differentiation, common to all social movements in general, is once again manifested.

Let us say that a government, for purposes of self-defense, undertakes certain measures in order to crush one social movement or another. These measures incur widespread indignation in opposing circles. As a result of this, a reaction begins in the latter which is at first expressed in an undefined form, serving only to sharpen the antagonistic mood, and then, upon discussion of the state of affairs in various circles and party gatherings, takes on a defined form of opposition, having at its disposal a corresponding program of action and the resources necessary for the achievement of its goal.

The very government endeavors against social movements, which serve as inhibitors for the latter, are at first little defined and chaotic, and only gradually acquire a definitive form.

In the same way, any social opposition to one movement or another starts out insufficiently defined and only takes on a more or less concrete form with time.

Thus, every social movement, and even the conditions inhibiting it, undergoes from beginning to end the various stages of differentiation, living through a preparatory, and most often undefined, stage and then through the primary stages of a relatively well-established social movement.

This movement, however, is at first far too general and only gradually acquires a more and more precise character, obtaining, in the end, a more or

less concrete goal.

The same law finds support even in collective concentration, since the object of attention is primarily one event or another as a whole, while later collective concentration is directed at various details of the event. Furthermore, if we are dealing with a collective mimico-somatic reaction, then at first it is undefined in the extreme, manifested as more or less general dissatisfaction, but then it takes on a certain form, as a result of which the mimico-somatic reaction acquires a concrete, specific nature, which is evident from the preceding examples.

All other manifestations of the collective are likewise subject to the law of differentiation. As a matter of fact, this law can quite clearly be traced in the development of language.

According to A. Schleicher:

The point of departure in the development of all languages consisted of simple sounds that served to illustrate views, ideas, or concepts, and could function in any context, i.e., as any grammatical form, although no specific phonetic expression existed to represent these functions. In this way, at this primitive stage of language development there were no especial phonetic differences between verbs, nouns, conjugations, declensions, etc. Let us attempt to represent this in at least one example. In place of words that in contemporary German are pronounced *tat*, *getan*, *Töter*, *tätig*, there existed, at the time of the birth of the primary indogermanic language, the word *dha*, since that word (meaning to suppose or do, and pronounced *dha* in Sanskrit, *da* in ancient Bactrian, *de* in Greek, *de* in Lithuanian and Slavonic, *da* in Goth, *ta* in High German) is the root of all those stated above. At a later stage in the development of the Indo-Germanic language, roots were employed twice to express certain relations, joining one word or root to another, but each of the elements remained independent (at that time, roots were used as words in their own right). *Dha*, *da*, *orta*, for example, was used to indicate the first person. Upon further development of the language, when elements fused into a single whole and roots could be adapted, *dhadhami* was formed from the above three words (*dhadhami* in Sanskrit, *dadhami* in ancient Bactrian, *tdm+tuom* to form *tetomi* in Old High German, *tue* in New High German). The original *dha* contained various undistinguished and undeveloped grammatical forms, including verbs and nouns, along with all the above

derivatives. The same relationship can be observed even now in languages that have remained at the lowest stage of development. (Ravenarius, *Philosophy as a World View in Agreement with the Principle of the Least Measure of Force*, 1912, p. 70)

The differentiation of words proceeds in another direction. L. Geiger gives historical examples of such differentiation, in which one and the same word, sometimes with a few variations, serves to indicate different objects, for example *Der See* and *die See*, *Belt* and *Belt*?

It is hardly necessary to prove that the development of language proceeds via differentiation and integration. It suffices to remember that all Indo-European languages are descended from a single source, Sanskrit.

The very process of differentiation of languages depends on the migratory patterns of nations and on the creation of new words that, little by little, add complexity to a primitive language which is at the same time filled out with borrowed words. It is also known for certain that climate is not without influence on language.

Thus, for example, in a warm climate there are many words resplendent in open vowels, whereas in cold countries aspirated vowels predominate. Similarly, racial peculiarities are reflected in the character of a language in the pronunciation of words, which proves the relation of race to language development.

If we were to analyze any scientific movement, then we would be convinced that it develops through differentiation and subsequent selective integration or synthesis. In science, new facts are constantly accumulated and the material, sometimes colossal amounts of it, begins to differentiate; however, at the same time synthesis is taking place, with facts being integrated through categorization into theories. The same goes for literature and the arts. Here, we first have new works that stand out from the rest thanks to their peculiar qualities, while from the aggregate of these peculiarities grows a new trend, a certain governing system.

Even the acquisition of skills and knowledge is subject to the law of differentiation. What does a person do in order to learn how to play an instrument? He must first study separate notes, i.e., be able to differentiate between sounds when playing them on the given instrument, and can only then integrate them, making the transition to melodies. What does a person do in becoming literate? In the beginning, he studies the alphabet, in other

words differentiates grammar, and then learns to put together separate letters, syllables, and whole words.

In fact, the same general law of differentiation can even be applied to demand, since any need arises from experience and, consequently, develops only after a discovery or invention provides its source. Thus, the need for narcotics such as tobacco is the result of the discovery of their anaesthetic qualities and of a mode in which these could be exploited. Of course, even prior to this discovery there had been a demand for the stimulation of nervous activity in general, which could have been satisfied in another manner, for example with alcohol; but this demand could not have taken the form of a need specifically for tobacco before the latter's discovery. It is likewise correct to assume that the necessity to cover certain parts of the body arose not before clothing, but as a consequence of it, since the requirement for it came about as a result of climatic influence.

If we now glance at the development of demand within a society, we must take into account the fact that the development of any social need also undergoes differentiation. Let us take, for example, the aforementioned need for sources of stimulation. In the beginning, it was an undefined and more or less general social requirement, which arose with the growing complexity of civilization, and it found no satisfaction except, perhaps, in the consumption of certain fruits; however, mankind very soon discovered how to make wine from grapes, and from then on the demand for stimulating substances became more defined in terms of the need for wine. With time, new inventions came in the form of various types of wines and of the preparation of alcohols from grain, and the same demand developed into a need for a certain quantity of wine or vodka, thus making the original, uncertain demand more defined and differentiated, divided into a number of needs. With the discovery of the effect of opium, morphine, cocaine, and other narcotic substances, that same demand for stimulation differentiated still farther.

Thus, all of the above information leaves no doubt that any social manifestation must develop via differentiation. This principle is as universal for social movements as it is for associative reflexes in individual life.

Notes

¹ Provided this substance even exists, we might add.

[2](#) Haeckel, *Lectures on Natural Science and Philosophy*, p. 46. (Russ. trans.)

[3](#) L. Geiger, *Ursprung d. Sprache*, Stuttgart. S. 58.

14

The Law of Reproduction

The law of reproduction is applicable not only to the organic world. Already, in the law of rhythm or periodicity, we have encountered the fact that once-existent processes are eventually reproduced, even if in a somewhat modified form. It is impossible not to be reminded of Ecclesiastes' beautiful prose in this respect. On the other hand, in the inorganic world we have one body dividing to create another that retains the definitive characteristics of its source. An example: a planetary world whose progenitor is the sun. Another example—the creation of crystals.

In the organic world, the law of reproduction is manifested in the production of offspring, or in biological heredity. It can also be seen in the basic characteristic of cellular tissue, i.e., the way the latter responds to stimuli. Once the initial response has taken place, any later response to a similar stimulus is greatly facilitated. The response is often reproduced even following a very insignificant external stimulus.

As for correlative activity in the form of higher or associative reflexes, reproductive processes permeate the very foundation of its manifestations, characterized by the reproduction of corresponding reactions, including the motile, cardio-vascular and secretory, on which there is no need to tarry.

Let us now go on to the evaluation of the law of reproduction in the supraorganic world; we shall be brief, in view of the clarity of the theme itself.

There is considerable evidence indicating that "spiritual" heredity, alongside biological and individual heredity, plays a prominent role in the life of society. Spirituality is inherited by society through upbringing and

ancestral values, and is transferred to one social organization or another from the past in the form of established forms of social life.

The above is related first of all to collective "spiritual" wealth, in particular language, customs, legends, general concepts, etc., that are passed down to descendants from preceding generations, and likewise all that is known as established tradition and the structure of social life in general.

Often it is impossible to know when and under what conditions a custom was formed, or to recall its original purpose, but it nonetheless retains its force and influence, even if the conditions that caused its creation no longer exist. Examples include the burning of bonfires on Midsummer Night, which dates from pagan times; fortune-telling at yuletide; Christmas trees, etc.

No less tenacious are the traditions of intellectual society, which withstand any influence of fashion. These include the wearing of singular clothing such as a suit or top-hat for official occasions, and all the so-called worldly customs that indisputably govern intellectual circles, notwithstanding their widely acknowledged absurdity.

It is necessary, furthermore, to keep in mind that all the products of the creative activity of separate personalities are reflected in the "spiritual" structure of social organizations and become the legacy of society. These include such results of human creativity as science, literature, and the arts.

Without a doubt, all these products of the creative activity of separate individuals, as we already stated, reflect social moods and views, and in this respect they serve as indicators of the creative activity of a nation. But at the same time science, literature and the arts are factors whose existence and inheritance must be reflected in the social life of nations, since the results of the past mental work of every social group are not without influence on its activity.

It is likewise necessary to bear in mind that the activity of societies reflects their spiritual property, which consists of the acquisitions made by humanity with every step it makes in its progress in the realm of industry, and which is directly related to aspects of social life such as the perfection of methods of communication between people across long distances, the improvement of transportation, etc.

The above clearly demonstrates what significance social heredity holds in the collective life of nations and social organizations.

Without a doubt, this social heredity is made possible thanks to the transfer, via the printed or spoken word, of the products of human creativity

from one generation to another. A special role in the above relation is played by the upbringing of youth; however, simple successive transfer of information from person to person, which occurs within every social organization and serves as a foundation on which to strengthen social traditions, is of no lesser significance.

Because of this, nothing in social life is done without former influence, and nothing disappears without trace.¹ Even that which seems to originate solely through personal initiative is found, upon closer scrutiny, to arise from social demand and is doubtless related to many other factors of social life. In a word, there is nothing in social life which is utterly new and does not have its roots in past social life. Even genius, that powerful initiator in human society, is the product of preceding sociobiological conditions, and realizes its creative activity thanks to the accumulation of material suitable for its synthesizing work. In general, any social development is accompanied by the accumulation of collective experience as a result of the corresponding expenditure of collective energy.

Conversely, backward development or social decay is the result of the insufficiency or impossibility of collective experience, following the inhibition of social energies. This corresponds to Spencer's biological law which states that any development is accompanied by the accumulation of a substance, and that decay goes hand in hand with the reverse.

In this way, any step forward in the development of society is the result of past collective experience. It is clear that the law of social heredity encompasses the establishment of dependent relations between past and present social events. The latter fact consists of the certainty that if one event or another occurs, it will be followed, provided no impediments arise, by a certain consequence which represents a new social form.

The most convincing example of social heredity is custom. "There is no government," says Tarde, (*The Laws of Imitation*, p. 312),

more mercenary and despotic, no legislative power more strictly obeyed or more severe, than custom. By this word I denote the countless old and new habits that regulate private life not abstractly or from on high, as it is controlled by law, but directly and in all areas, encompassing all artificial needs, all sympathies, antipathies, peculiarities of manner and morals that are attributes at a given time of one person or another.

Custom, in a way similar to government and legislation, is tied to religion; it is an off-shoot of ceremony.

At one time, predispositions thought to be contained in large social groups were often defined as the genius of the people. The predispositions that were attributed to nations included grammatical forms, certain forms of government, religious beliefs, etc. It was accepted that certain social phenomena, and not others, were peculiar to the character of one nation or another. Thus, for example, China and Japan were considered countries in which European institutions were not natural, while polytheism was believed specific to Greek thought.

Yet, although it is indisputable that one nation or another has its own temperament and a certain level of efficiency that depends on climatic, economic, and other conditions, it is likewise indisputable that all else which characterizes a nation depends on its social life and structure, established over the ages. Because of this, the introduction of new forms into social life via enlightenment of the masses can radically change a nation's predisposition or character, as may be observed in Japan.

Every social individual has his own physiognomy and personal qualities that are handed down from generation to generation in the form of dominant signs. That is the genius of the people, the genius of their language, the genius of religion. That is the essence of Party spirit, that is what defines the character of a certain area, its customs and habits. In a word, wherever there is a certain collective there is likewise collective expression of the former's individual characteristics. Thus, all of the above, with the exception of temperament and mental development, are not inborn biological peculiarities, but rather ones influenced by the living conditions of a given community and so anchored in it that they have come to be considered symptoms of its character, and are studied with especial consistency.

Note

¹ For specifics, see my work, *The Immortality of the Human Personality, as a Scientific Question* (in Russian), Sep. pub.

15

The Law of Selective Integration

Once again, we meet here with a universal law—the law of selective integration. The latter can easily be observed in inorganic nature wherever systems of bodies are formed, wherever we deal with the mobile equilibrium of forces, and wherever complex chemical bonds are created. But the organic world is likewise subject to the law of integration. Is not an organism a synthesis or integration of cells in the form of tissues, a chemical integration through the media of blood and bodily fluids, and, finally, an integration of energy achieved by the nervous system? Is not the above synthesis of various functions performed in the interests of the whole organism, of its vital activity and self-preservation?

We will not linger on the above manifestations, since it is not our object to discuss the organic world, but in passing let us note that the same law is applicable to society. We have already seen that, alongside differentiation and collective activity, we meet with the law of integration or synthesis in the collective. The development of collective associative reflexes does not proceed without selective integration.

In the associative reflexes of separate individuals, the above law is manifested in the fact that if we create, along with an excitation that stimulates a reflex, a secondary stimulus, then the reaction to the given excitant will always also be accompanied by a reaction to the secondary stimulus. The latter's occurrence is due to the nature of the phenomenon itself, and so is impossible to avoid in most cases. In life experience, this dependence is in one way or another taken into account, with the consequences of a certain action later becoming the causes of action in

similar instances. For example: the hunger a newborn infant feels is sated by its mother's or wet-nurse's breast. In this first life experience, we are dealing with lack of nourishment or hunger as an organic irritant, and with the satisfaction of the given need by breast milk. Until life experience suggests another way in which to alleviate hunger, the infant will satisfy its demand for food by suckling. As a result, the elimination of hunger as a stimulant or irritant is so closely associated with the consequent well being of the organism that a certain need for the latter is established, and nourishment becomes the necessary goal for satisfying the given need.

In this way, a goal is formed as a result of a certain experience, and it controls conduct in some situations, i.e., when similar conditions arise.

However, not every goal is established through individual experience, and in such cases it is the result of the collective experience of others, an inevitable consequence of the law of associative reflexes. Thus, life experience states that all people die sooner or later. A person himself, in this instance, does not and cannot have experience in relation to death, but the collective life experience of the death of others acquires a governing significance, and a goal is established— on the one hand, to postpone one's own death, and on other to be prepared for its advent.

Another example: experience shows that people who come in contact with a highly infectious disease also contract it; this leads to prudence and precaution when encountering those stricken by the infection. A given person may not have personal experience in this respect, but he nonetheless integrates the encounters of others into his own knowledge, and in taking the corresponding measures creates a personal goal to evade infection.

In this way, we are dealing with the creation of a defensive reflex that develops through the experience of people in general, and is engraved on a given individual not through personal experience, but through the natural cultivation of a collective reflex and the application of foreign experience to himself, thanks to synthesis or integration.

In general, in any collective the process of selective integration goes on, alongside differentiation, in a manner comparable to the way a heterogeneous crowd is quickly homogenized, drawn into a unit around a single, common goal. Similarly, the individual sections of any collective band together thanks to the establishment of organization. In every organized crowd, agreement or harmony inevitably arise among its component parts and a common goal is established, which is evidence of the beginning of integration.

In agreement with the law of selective integration, organization is realized in any collective in order to consolidate its parts for common activity.

Thanks to the above law, any long-lasting collective unity, depending on the duration of its existence, goes from being indefinite to defined, from irregular to regular, from structureless to structured, in short, from disorganized to organized (P. Sorokin, *The System of Sociology*, Volume 1, p. 33).

In this way, the initially fractionated and differentiated sections of society are subject, in due course, to amalgamation, under the influence of common language, custom, religion, and other institutions or causes that play the role of stimuli. Thus, the ancient Greek states were united thanks to the Olympic games and the Delphic oracle. Nations of the Roman empire were allied not only by force, but also by a shared language and a common religion that was able to accommodate countless new gods, and likewise by the dissemination of a single culture. Christianity and a common culture unite the civilized nations of our time.

State integration, with its accompanying political organization, always goes hand in hand with the differentiation of the population into class and tribe structure. Nations, in borrowing cultural innovations from one another, are already becoming united in some measure. Even wars, which, it would appear, must separate nations, since they are born of rifts between them, in reality lead to unions and alliances, and do so thanks not only to the conquering of one nation by another, but also to the appropriations that a less civilized nation gains from a more culturally developed one, especially if the latter is victorious. This integration is facilitated by state organization that similarly joins culturally and religiously dissimilar nations into one powerful unit.

Furthermore, economic and cultural propinquity can develop notwithstanding the political separation of nations. The same occurs in relation to smaller collectives, such as castes, which unite on the grounds of common cultural and economic interests and, through imitation of and borrowing from one another, gradually merge, eliminating class and caste divisions.

In general, differentiation in a society's political organization is always accompanied by integration. In the words of Takhtarev:

Even the more complex forms of political society and state are created simply, through integration. Thus, tribes merge into peoples, and peoples take on the form of alliances of self-governing nations.

Feudal principalities, upon losing self-sufficiency and consequently becoming dependent on more powerful estates, fall under the control of their rulers and are molded into a more complicated feudal state. When urbanized republics of an autonomous nature can no longer be independent in terms of the ensurance of their external safety, they must inevitably unite in order to fight external enemies. By forming alliances and federations, they lay the groundwork for more involved republican forms. Basic state forms, becoming similarly dependent on one another, give rise to more complex forms of the nation-state. Finally, even nation-states and nation/class societies, under the influence of international communication and the loss of self-sufficiency, form unions and become international states which, in their turn, strive to unite and be transformed into an all-encompassing world republic. (Takhtarev, *Society and State* [in Russian], p. 10)

The organization of power always has a certain connection to the interrelations of the social classes and groups that make up a nation, while the manifestation of power as a force, in accordance with H. Jephson (*The Platform: Part II*, 1901, r.p. St. Petersburg.) and Arthur Bentley (*The Process of Government*), depends on pressure from one large social group or another. In the end, power inevitably falls into the hands of the ruling social group, becoming a tool used in the achievement of its goals.

In this way, it is obvious that integration goes hand in hand with differentiation in the political life of nations. The process of integration goes still farther than described above, so that when a state can no longer satisfy the requirements of its existence, they are supplemented through the necessary agreements, alliances, and mergers.

The process of selective integration likewise accompanies the realization of any collective activity. Collective actions are at first insufficiently organized and undefined; with time, they take on the form of better coordinated actions, geared towards the achievement of a certain goal.

Thus, if troops are learning maneuvers, their movements at the outset are uncoordinated, and only gradually do they acquire the requisite harmony, and

consequently become defined or precise.

The same process of transition from discord and chaos to harmony can be found in orchestra practice, in choral singing, and in other similar collective movements. Here, too, we are dealing with integration.

Examples of integration can also be found in industrial alliances, in various financial unions, in co-operatives, etc.

It is hardly necessary to prove the importance of integration or synthesis in scientific thought, as in any creative act, since it is obvious that the accumulation of facts is followed by their union into a single system, their integration into one principle or under one law. Any creativity in general, be it scientific or artistic, is in large part synthesis and integration. According to Tarde (*Social Laws*, p. 106), it is absolutely indisputable that the sciences were composed at first solely of small, separate discoveries that had no common ties; then these discoveries were amalgamated into groups (where every separate grouping was a discovery in its own right) and theories that, in their turn, were united into broader theories, supported or corrected by a number of other discoveries. Finally, all this was connected by bridges of hypotheses.

The life of every society is characterized by creativity, which in a collective is not manifested identically in all individuals, but more or less unequally, being present to a greater extent in some than in others. Creativity is the work which, in the form of impulses, instigates any social process. Creativity is expressed through an invention or discovery that sooner or later leads to change in social correlations. Thus, the discovery of fire totally altered human existence by allowing for the use of artificially prepared food, the employment of malleable metals instead of stone, the occupation of heated enclosures in much colder climates, the illumination of darkness, etc. The discovery of explosives gave the opportunity to use firearms, to more easily alter the earth's surface, to better exploit its underground resources; the invention of steam power speeded up routes of communication and changed the organization of human work, replacing manual labor with factory and plant industry; the discovery of electricity made it possible to contact a person over long distances in a written or spoken form (via telegraph or telephone), as well as making a large number of improvements in various areas of technology.

In the above cases, we are dealing with inventions and discoveries that have to do with improvements in the sphere of technology which originated

new eras in human history. Discoveries in the area of higher manifestations of human correlative activity are no less important. Such discoveries were made by motivators of science and art, by founders of religion, who laid new paths to greater knowledge and moral perfection. This, in turn, led to the establishment of better living conditions for mankind, as a result of the ability to predict events, the creation of perfected tools that facilitate human work, the abolition of slave labor, the termination of feudalism and serfdom, etc.

It goes without saying that alongside these important discoveries and inventions we meet minor ones at every step, whose magnitude is correspondingly less; however, their impact in terms of the character of their influence on social activity is significant. Take, for example, an invention like the safety pin. Did it not economize human activity by eliminating the necessity of sewing those areas of material that could be pinned?

What, then, contains the force and significance of creation? Where does that novelty, which is found in every discovery and invention, come from? It cannot be doubted that creativity is the result of the synthesis or combination by a powerful mind of existent phenomena and lesser discoveries; this opinion can be supported by the analysis of any important invention or discovery.

It can be said with certainty that any discovery or invention is based on the synthesis of natural phenomena. Separate facts are discovered by many, but their combination is not evident to all; it is that combination which creates something new, heretofore unknown. Let us take Newton's discovery, for example. Who did not know that apples fall from the tree to the ground; yet that obvious fact, in combination with others no less commonly known to astronomers, led to a discovery which surprised the world.

Thanks to such integration, scientific prediction becomes possible, as is proved by many widely acknowledged cases.

The significance of large numbers, as it is revealed in statistics and in the theory of probability, also has at its basis the law of integration.

Not only science itself, but also various views, assumptions, erroneous theories, fantastic suppositions and outlooks, beginning with primitive humans and ending with civilized races, in a word, all animistic, fetishistic, totemic, and other inane views and theories, as well as religious conceptions and fantastic explanations of natural phenomena given by uneducated people, are products of synthesis —fallacious, it is true, and originating in incorrect beliefs, but synthesis nonetheless as an integrating principle.

The same should be said of the arts. Thus, for example, the original art of the Hindus owes its creation to the merging of ancient Egyptian and Assyrian art. A similar trend can be traced in the development of European art, and in that of other nations.

The same process of integration, as we saw, lies at the base of the development of categories and general notions.

Finally, the development of language itself is affected both by differentiation and integration. How, for example, were the present-day Romance languages derived? Obviously, through the commingling of Latin with local tongues; and in that process of combination, synthesis or selective integration played a role.

Even the spread of a language does not occur without the participation of the law of integration, since a new language appropriated by one people or another is altered as it merges with the local dialect.

Thus, as a consequence of two centuries of Roman occupation, the Gauls adopted Latin, but that language was gradually adapted to their needs and their linguistic forms, resulting in the creation of French.

In the end, any social phenomenon arises due to integration. Thus, countless poor, minor languages of separate peoples unite via integration into one great and powerful language that satisfies everyone.

Even social morality is gradually established thanks to the evaluation of individual conduct under various conditions by individuals and groups, as a result of which public opinion evolves in the form of social morality.

The assimilation and union of nations is also relegated to the ranks of integration, when we are dealing with the internal affinity of one to another. According to G. Le Bon (*The Psychology of Nations and Masses*, p. 15), "When a nation has achieved this unity, an instinctive agreement of all its members on all important questions is established, and serious differences of opinion can no longer arise within it."

Finally, how many architectural styles are products of the process of integration? A true connoisseur of art can speak of this better than the author of this book. Even civilizations owe their development, in the end, to integration, since there are no pure civilizations made up of a single people, only mixed ones in which a thin coating of one civilization covers another, more ancient culture.

If we turn to the question of what factors lead to social synthesis, it is necessary to keep in mind that in organized societies, the goals of collective

union or synthesis are forwarded by meetings, conferences, discussions, etc. The same purpose is served by all factories with their distribution of labor, corporations, banks, government institutions, scientific societies and other collectives in which representatives of different specialties work towards a common goal.

One of the most important forces leading to the manifestation of integration, agreement, and alliance by separate individuals is imitation and appropriation. Notwithstanding the differentiating role of various factors, in the form of geographic, economic, and other influences that distinguish nations from one another and divide societies, imitation, based on adoption, establishes a similarity between the differentiated sections of nations and societies, which leads to their union and integration.

The difference between the civilized nations of Europe and, for example, the Albanians is so great that it is difficult to even contemplate the possibility of their alliance, whereas between Switzerland and other West European nations there are already many forms of cultural attachment, thanks to a long history of mutual imitation of a cultural nature. In exactly the same manner, the lowest class of a society, thanks to imitation in the area of education, gradually merges with the higher classes in cultural and economic respects, which leads to greater integration and union of those groups. Imitation in language, i.e., the borrowing of a language by one nation from another, likewise serves as an important stimulus to integration, leading to the merging of nations. In general, imitation plays an integratory and unifying role everywhere. Social beliefs and aspirations, thanks to mutual communicability and imitation, grow in force and significance even in relation to the number of members of a collective.

The discovery of one or another compromising situation in the life of any public figure is sufficient to arouse society's mistrust of his every action.

Selective integration always arises when a certain influence on society is treated with a similar attitude in different social groups.

Thus, opposition to one act of the government or another can develop in both right and left circles. When a calamity such as war or rebellion occurs, the various sectors of the population strive to show their patriotism. One and the same goal can excite different social collectives. For example, the exploration of tropical countries on the African continent and of the North and South polar regions spurred the activity of many scientific collectives and merchant classes of different nations, which joined forces on that basis.

Those cases in which different movements can originate from and be upheld by a single source fall under the same category of phenomena. Thus, the Russian Revolution of 1917 led at first to the union of all opposing elements in the country, due to the fact that the preceding tsarist rule had been disliked by all socialist and liberal-bourgeois sectors of the population, as by the rural populace. Finally, a great social misfortune usually motivates aid-giving movements in diverse social circles.

All these are examples of the law of selective integration, in accordance with which various social movements are inspired by one and the same stimulus.

Selective integration can be observed in relation to any collective activity. Take, for example, a meeting at which a question of great importance is being discussed. Notwithstanding the disparity of views of different political parties, some of those parties can agree on a single decision.

In the case of the development of collective disorder on the basis of a food shortage, we often see, through integration due to collective associative reflex, looting not only of shops containing provisions, but also of shops which display goods of a totally different nature.

In general, collective arousal, as well as collective concentration, that has been generated by a certain object is later directed at all objects which have a certain relation to the first.

Finally, conflict with power sometimes integrates even the most extreme social movements. There is no need to search for examples abroad; we can simply point out the joint activity in Russia of extreme left-wing parties, of the notorious Union of the Russian People, and of former police chiefs and gendarmes. Here is what can be read on the subject in the newspapers:

A few minutes before the arrival of the special train bearing the minister, in the gilt hall where tsars had rested on voyages from Tsarskoe Selo to Petersburg and back, commander-in-chief of the military district Gen. Polovtsov indignantly handed me a card, found in the palace seized by the maximalists, painted in two colours—black and red.

Caricatured in black were long-nosed, grey-haired, skull-capped Jews, and in red— a sea of blood, pouring into a vase held by a Jewess from the throat of a fourteen year old Hungarian girl who had been "slaughtered" in Tissa-Eslar.

"Well, you see," said the general curtly.

"I see," I shrugged. It was as in a good detective story: all the threads were tangled. Extreme left, extreme right—everything boils down to a ritual pogrom.

In this episode there is perhaps something exaggerated, even accidental, but there is no doubt that many former monarchists manifested a certain inclination to work with the extreme left, to the exclusion of all other revolutionary parties.

In all the above cases we are dealing with selective rather than simple integration, for it occurs between social movements of different natures which have a certain relationship to a single phenomenon that plays the role of a common stimulus. However, true merger is also possible when two or more related organizations, pursuing similar goals, unite into one whole, while preserving their individual physiognomy. In this instance, the individual response of each organization is lost, and from the moment of merging it is the new, united organization that displays a reaction to external events.

On the other hand, it is known that for the purpose of mutual support, two or more social movements, striving to coordinate forces, can form a bloc.

In this way, we are looking here at cases in which a few different social currents, in order to better attain their objectives, integrate with one another without losing their social physiognomy. Often, this integration goes no further than the definite goal, the achievement of which is acknowledged to be the only activity necessitating a combined effort, while other activities entering into the scope of the united organizations follow diverging directions. Thus, for example, political parties unite into blocs to combine action, pursue certain political goals, form trusts, etc. There is no need to look far for examples of the preceding in the political history of nations.

K. Marx, discussing the civil war in France, points out that the downfall of the Paris Commune was facilitated by the traitorous agreement between former revolutionaries and socialists headed by Louis Blanc, who chose to support the forces of order at the Versailles Assembly. Here is what the historian of the Paris Commune, Lissagaray, writes on the subject:

The votes of the rural deputies and Thiers' past too obviously contradicted his republican declamations; the former heroes of the front likewise instilled little trust. Thiers sensed this and turned to the purest of the pure, the revolutionary aristocracy that had returned from exile. Its prestige was not damaged in the eyes of provincial democrats. Thiers pursued aristocrats in couloirs, told them that the fate of the republic was in their hands, flattered their senile vanity and so skillfully handled them that they became a submissive tool in his control, and he could telegraph that they applauded the speeches of March 21 that anathematized Paris.

Louis Blanc, who had innocently grumbled against the foundations of French society all his life, now bared his teeth at the members of the Commune, who, in his words, sentenced him to death. With whom, he asked the delegates, with whom can one negotiate in Paris? The nobles who are already disputing each others' power are fanatics, idiots or swindlers, not to mention those involved in Napoleonic and Prussian intrigues.

These or similar forms of the selective integration of different parties, for the purpose of conflict with others, are also encountered in the contemporary conditions of a revolutionary movement. Examples are there for everyone to see.

In exactly the same way, scientific societies that pursue the same end often work together in order to attain the best possible results. Thus, for example, national and international associations of scientific institutions represent such integration of scientific collectives. The same can be observed in relation to social organizations of an economic or professional nature.

Their coming together in unions for the combined protection of their interests is yet another example of selective integration.

The first step in the gathering of a crowd is the arrival of separate groups of people who are attracted by the same situation, after which these separate groups along with newly arriving individuals aggregate into a single dense crowd.

In its primary stages, a rebellion is likewise composed of a few bands that later unite, forming one common organization. The latter then directs the actions of its component parts toward a definite goal.

When common effort, such as collective labor, is executed by inexperienced individuals, there is at first insufficient coordination of work; often, one person's work inhibits another's. Little by little, however, the work of many people reaches complete coordination and so allies the participants in the common labor.

Orchestral or theatrical performances can serve as suitable examples in this respect. Every musician or actor must begin by rehearsing his role separately, which could be seen as supporting the law of differentiation, but later these separate roles will be complemented in a single body, which corresponds to the law of selective integration. Finally, choral singing is another example of the same phenomenon.

In all these cases we are dealing with the same phenomenon as it is observed in the reflexology of individual persons.¹

Thus, one and the same goal, as a basic stimulus, can motivate a large number of individuals, who consequently draw closer together and eventually unite, realizing the principle of social synthesis or selective integration.

Note

¹ See V. Bekhterev, *General Principles of Human Reflexology* (in Russian), p. 19 and ff.

16

The Law of Historical Succession

While observing the phenomena of social life, we must come to the important thesis that no social movement can arise before the necessary prerequisites are in place. This law is immutable to such an extent that, if a genius predicts one discovery or another several centuries before its realization, that discovery will, nonetheless, not come into being until the preconditions are established that make it a natural and indispensable link in the common chain of historical events. Even firsthand observation of a phenomenon is sometimes unable to shift scientific thought from an established position before the groundwork is laid for the phenomenon's acceptance. A classic example of this is the reaction of the Paris Academy of Sciences to the first human record of a meteorite reaching the Earth's surface:

A "heavenly rock" fell near the town of Jouillac in 1790, and the mayor of the town, astounded by the occurrence, decided to send a notification of the fact along with the signatures of 300 witnesses to the Academy of Sciences, and what happened? The speaker, having read this notification to the members of the Academy, expressed the following opinion: "How unfortunate it is that an entire municipality supports popular superstitions, whose very existence is regrettable, in a formal report. What can I say of this? Only that any philosophically educated person, having read this certification of an obviously false fact, of a physically impossible phenomenon, can himself make all the necessary conclusions. '¹

It is obvious that the unpreparedness of minds for a new, apparently unbelievable phenomenon led to the lamentable rejection of a fact witnessed by 300 people—a fact which, given the absence of prejudice, it would be easy to verify *in situ*, and thus dispel all doubt; but even that most simple and natural method of confirmation would not be resorted to in the given case, due to the inacceptability of the fact itself to unprepared minds.

A no less amazing example is the stance the same Academy took on the impossibility of human flight in view of the fact that it contradicted human nature, while the Medical Academy of Paris found nothing extraordinary, outside the phenomena of imagination, in Mesmer's experiments with hypnosis. In both cases, it was necessary for mankind to live another few decades before these previously inconceivable facts and discoveries were acknowledged.

Because of this, there is no doubt that the development of science is a gradual process of preparing social thought. Even Mendeleev stated this: "In attempting to know the infinite, science itself has no end, and, being universal, inevitably acquires a national character, even more or less personal nuances" (*The Foundations of Chemistry IV*, 1905-1906). The author, comparing science and art, goes on to say:

In essence, science and art pursue the same goals; alongside the arousal of a sense of pleasure, science and art are occupied with the study and description of the real world, expanding our mental scope and developing our knowledge of nature and life. And so to the psychological similarity between the two fields is added a biological resemblance, since both have an independent significance in practical life, in the relationship of a person with his surroundings, and lead to the employment of the products of human creativity.

Academician P. I. Valden notes on the subject:

But the scientist and inventor, artist and poet, is a living part of nature; his will is a part of the peoples' will, his imagination—a part of the fantasy of the masses, his creative force—part of the hidden spiritual force of the people. This is why the roots of an individual's creativity are embedded in the imagination and will of the masses. This is why the analysis of

scientific creativity must touch on collective, national imagination. (*Priroda*, Sept.-Oct. 1917, p. 90)

From the aforementioned it is clear that scientific and artistic creativity is based on the preexisting spiritual "forces" of a nation.

On the other hand, it can be said with certainty that no invention can be realized before the corresponding factual knowledge is obtained through previous investigation. Thus, the invention of fire could not be realized before man made use of tools and could prepare material for the maintenance of fire. The cart could not be invented before the wheel and the domestication of large animals such as the horse or ox. The steam engine appeared only after better acquaintance with the function of the piston and lever and the action of steam, etc. In these cases the role of the genius is to unite the disparate components in such a way that they form a tool which performs a certain task:

One discovery, writes Tarde,

always has in it the potential for other discoveries, but it is uncertain whether the latter will be realized, and if they are, when and in what order. The discovery of the surveying compass contained in itself the discovery of America and Oceania, in the sense that the practically impossible detection of that continent or archipelago without a magnetic indicator became very narrowly probable after its discovery, more or less probable after 50 years, slightly more probable after two or three centuries and highly probable or even certain after 1000 years.

In any case, it is clear that though of these two discoveries the first had to precede the second, Florida's existence could have been known before or after that of Brazil, and New Caledonia's before or after New Holland's.

The discovery of writing obviously had to occur prior to the invention of the printing press; the discovery of Arabic numerals was essential to the success of our mathematicians. Before all else, the discovery of articulate speech is the sine qua non condition for all other discoveries. (*Social Logic*, p. 124)

The more prepared the national organized masses are, the more easily and painlessly any upheaval proceeds. Thus, the February Revolution in Russia

was hailed in that country as a deliverance from the terrible oppression and traitorousness of war, and caused an immense emotional uplift in the populace, to the extent that people on the streets embraced and congratulated each other on the advent of the "rebirth." In a word, there were no protestors, since the remaining handful of monarchists hid from the common exultation. Troops voluntarily shifted to the side of the new government, exchanging their former imperial pennants for red flags and marching to revolutionary songs.

The October Revolution, organized by the Bolsheviks, who staged a new coup thanks to their seizure of power, presents a different picture altogether. Its slogans—immediate peace, immediate division of land, and the rule of the proletariat—attracted the troops and laboring masses, but the remainder of the working democracy and bourgeoisie, not to mention the intellectual elements, saw in this coup unfavorable consequences for itself and so entered the ranks of enemies of the new revolution, supporting the already almost nonexistent (thanks to arrest and displacement) Provisional Government; as a result, the masses were instantly organized into committees dedicated to saving the motherland and the revolution, the city Duma became the center of new organizations that opposed the latest power, and employees of ministries and other government institutions announced a boycott and would no longer serve under the new government. Inevitably, a civil war broke out, ending with the victory of the Bolshevik forces at Petrograd, followed by the execution of the cadets, the dreadful events in Moscow and widespread pillaging in other towns, not to mention countless arrests.

Notwithstanding its military success and the gradual elimination of all bourgeois publications, including socialist newspapers, the new government could not fully master the situation, which led to a drawn-out civil war, and all because the population of the country was not and could not be prepared for Communism.

The development of the cultural life of nations is subject to the same law of historical succession. The stone age was replaced by the bronze age, in turn supplanted by the iron age, the steam age, and the age of electricity, but this succession was determined by corresponding discoveries and inventions, before which none of these periods of human cultural development could have been realized.

In exactly the same manner, the hunter-gatherer way of life does not become a nomadic or agricultural existence before the safety of the

population from aggressive neighbors is guaranteed, while the industrial structure of a country is dependent on an excess of people unable to support themselves by working the land, as on a few other conditions, of course.

The political structure of the state itself is the result of the previous preparation of the masses for a certain form of government.

A social phenomenon of such huge importance as the establishment of a new political system, in the form of, say, a socialist republic, is not fully realizable without the corresponding prerequisites, as is discussed by R. P. Pinkevich in his article (see *Vecherniaia Zvezda*, 25). These prerequisites are that the proletariat be literate and have a sufficiently wide scope of knowledge, a certain moral fortitude, that a significant majority of it be consciously socialist, i.e., know what socialism is and how it should be approached, and, finally, that it understands the harm to the common interests of work and the whole caused by the satisfaction of private interests of the working class, etc. In other words, the proletariat should be sufficiently intelligent to be able to achieve a given minimum level of mental and moral development, if not become the brains of the country, and be educated in a corresponding manner; if this is not the case, any socialist restructuring of the state ultimately ends in failure, as is portrayed by G. Le Bon in his book, *The Psychology of Socialism*.

And so, when there is a lack of historical prerequisites, the law of historical succession leads to disastrous consequences.

Note

¹ Dr. M. Kemmerich, *Kulturcuriosa*, Munchen. Quoted from K. Krall's *Thinking Animals*, p. 233.

The Law of Economy

We owe the principle of least expenditure of resources in nonliving nature to Zellner (*Ueber d. Natur. d. Cometen*, 1872, Leipzig). The law of adaptation, to be discussed later, inevitably leads to the principle of economy in the organic world. There is no need, however, to discuss this aspect of the principle in detail in this work, since we are concerned primarily with the supraorganic world.

The philosophy of Avenarius (*Philosophy as a World View According to the Principle of the Least Effort*, 1913, St. Petersburg) maintains, as is known, that the "spiritual" sphere achieves its goals through the least possible expenditure of strength; however, it does not achieve this immediately, and does so through practice and habit.

This principle can be considered as also holding true for correlative activity. The creation of concepts, the search for regularity of phenomena, numeration, and mathematical calculations all correspond to the principle of least expenditure of strength. Whether we analyze correlative activity as a result of nervous or other energy, it is clear that the principle of economy must be acknowledged as immutable in relation to any exercise, and consequently any repetition, that is accompanied by the economy of energy.

We will not go into detail on the application of this principle to the correlative activity of separate individuals. Let us simply note that the principle of economy lies at the basis of any adaptation, since the process of adaptation essentially involves the economy of expended strength as a consequence of the more precise definition of the work being done.

Obviously, the law of economy must also have significance in a social environment. Here, the process of adaptation strives to achieve the greatest satisfaction of needs with the least expenditure of effort and resources. All socioeconomic life is based on the principle of the development of greatest productivity with the least possible expenditure of effort and resources.

Moreover, we owe even the inception of collectives themselves to the principle of economy, since the assimilation of people into societies can only be justified by that principle. Where separate individuals cannot execute a given task, a collective is needed to unite those individuals and allow them to achieve the necessary result through joint effort. This is why any type of collective is, in essence, a realization of the principle of economy of strength, as is any manifestation of social life. Avenarius is right in saying:

The influence of the principle of least effort in practical life is not only obviously manifested in any aspiration to freedom, to division of labour, to the union of administrative, commercial, legal, state, and social life, but is likewise apparent in the institutions of trade, political economy, legislation, government, politics, etc. (I.e., p. 95)

Any symbolism can also be explained by the principle of economy, for the former's purpose is to replace complex phenomena with eye-catching or, in any case, expressive and easily understood signs. Language itself, a product of social creativity, is a realization of the principle of economy. With the help of words, a number of objects and the relationship between them can be easily defined, and at the same time the whole process of interaction between us and surrounding individuals is facilitated, as are all processes of correlative activity in general.

The establishment of common concepts and classifications, as a result of social living conditions, also satisfies the law of economy of strength.

The principle of economy is manifested even in the creation of words that are abbreviated to an astonishing extent. The Psycho-Neurological Institute is often contracted to "Psycho-Neuro"; the Institute for the Study of the Brain and Psychic Activity is called the "Institute of the Brain": for example, "Let's go to the Institute of the Brain," or even just "the Brain." Instead of The Soviet of Worker and Peasant Deputies, people say "the Sovdep" or "Smol'ny." All complicated names are contracted in agreement with the law of economy: "Ponto," "Mono," "Santrud," "Glavsanupr," "Nkps," etc.

Similarly, commonly known and frequently used terms are often abbreviated in writing, for example "etc.," "e.g.," "i.e." Telegrams necessitate still further abbreviation of words and names, while stenography replaces words with certain signs. The use of numeration and mathematical calculations in the interests of economy was already noted.

Furthermore, the symbolism of signs is no less vividly expressed in, for example, cockades, shoulder-boards, clothing, various conventional designations and badges, even in the attributes of callings and professions (the wearing of corresponding regalia, e.g., the official's cockade, the judge's chain). In some cases, symbolism comes down to the reproduction of one characteristic quality or function of an object, which replaces the object itself. This is manifested in onomatopoeic words (whistle, buzz, cluck, bang, etc.) All these are diverse forms of economization in correlative activity.

In essence, the development of all science or knowledge, and its practical achievements, pursues the same goal of the realization of one aim or another with the least expenditure of effort.

The principle of economy is realized in the process of scientific creativity with every generalization, every establishment of one dependence or another, and every foundation of a law that encompasses a number of dependences and phenomena.

Here is what we read, for example, in A. A. Chuprov's book (*Essays on the Theory of Statistics* [in Russian], 1909, pp. 9-11) on the subject of the law of gravitation:

The discovery of this law signals an upheaval in the theory of gravitation. Why? Even before Newton it was acknowledged that the sun attracted the planets; the inverse proportionality of attraction to the square of the distance between bodies was likewise given consideration. Newton simply made the transition from the analysis of forces acting between bodies of finite proportions to the study of forces that attract infinitely small parts. This transition is attended by such economy of mental energy that it rightly deserves the place given it in the history of the study of gravity. If memory had to retain every single fact established in this area, then with the growth of knowledge the complexity of the problem would go outside the limits of the capacity of human intelligence.

Meanwhile, with Newton's discovery of this law "we immediately get out of our difficulty: the variegated mass of separate facts instantly becomes easily surveyable. The abundant accumulation of observations is summarized in a compact formula."

Even art is subject to the same principle of economy, discarding, in its aspiration to stimulate uplifted emotions, all that is spurious and cumbersome in order to achieve the same result with the least expenditure of effort.

Any social movement develops in the direction of least resistance as long as this does not contradict the basic object of that movement, which also satisfies the principle of economy.

The achievement of greatest results with the least expenditure of resources is the guiding principle of any conflict in general, including military activity and the triumph over obstacles.

In the end, all social life is permeated by the principle of economy. The ideal of individual and social life is the achievement of greatest results with least effort. It can be said that the satisfaction of the most demands with the least possible expenditure of resources is the basic rule behind all socioeconomic politics. Any collective activity tends to take the path of least resistance, which also conforms to the principle of economy of effort. Thus, any new institution gains the potential of future development only when it is needed to fill a void in society and when its realization does not require excessive resources; in other words, when it conforms to the principle of satisfaction of the greatest number of needs with the least expenditure of effort and resources. An established institution always develops, in relation to its goals, in the direction in which it encounters the least resistance.

The development of national culture once again is subject to the law of economy of effort, since it is determined by the existence of available material on the one hand, and by the corresponding preparation of a work force on the other. The same can be observed in relation to the development of mining and refining industry and to the development of various professions and arts.

The development and teaching of sciences is no exception in this respect, since it is similarly directly dependent on favorable conditions and on the preparation of the corresponding personnel, which creates the possibility of economizing resources and efforts.

It is obvious that when an undertaking comes in contact with one obstacle or another, it cuts through the impediment where the latter is weakest.

Naturally, this corresponds to the law of economy of effort. For this reason, the path along which national dissatisfaction develops is always in the direction of least resistance.

The same principle of economy explains the compromises reached when, for example, an established custom collides with a sensible innovation.

In this case the expediency of the innovation conquers routine; the essence of the latter becomes a new doctrine, while the form remains the same. In this way, only the form of the previous structure remains. Thus, Christianity, while spreading among the European nations, left the organization of the state untouched, only altering social life.

All vestiges of the past in general are partially explained from this point of view.

In many cases a new religion, new political form, or new law limits itself to the introduction of a new norm, leaving the essence of its former equivalent applied to new conditions.

The settling of a nation once again follows the path along which it meets with the fewest obstacles and where people find the greatest number of conveniences and corresponding resources. Emigration usually takes place in the direction of least resistance, least competition, and greatest number of fundamental comforts. If we are dealing with the movement of a rioting crowd, it always occurs in the direction of least resistance and in a direction favorable in terms of the achievement of the given goal.

The same can be seen in relation to a military collective during battle. It commonly moves in the direction of least enemy resistance, i.e., proceeds where the opposition is weakest. A mission may require that the point of greatest resistance of the opposing side be crushed, but this is likewise done while bearing in mind the eventual achievement of the best results with the least expenditure of resources.

The governing principle of usefulness (utilitarianism), no matter what its objective, also agrees with the law of economy, since a useful thing often not only satisfies one demand or another, but also does so with the least expenditure of effort. And the actions of any collective in general correspond to the principle of utility only if, satisfying one need or another of the collective, they are executed with the least use of energy and resources.

It is necessary to point out the role of training and habit in the realization of this law, since in the life of social institutions and in the collective activity of any society, former experience and acquired habit play a significant role in

terms of the least expenditure of energy resulting in the greatest achievable results. The fact is that in collective and individual activity, training leads to facilitated activity in a certain field. In other words, the multiple reproduction of a certain collective action creates habits that, in leading to the facilitated activity of the collective in a certain direction, realize this law in practice under the conditions of collective activity.

From all that has been outlined above, it is clear that in the life and activity of a collective the law of economy has the significance of a basic principle, and it can be said that there is no manifestation in the life of a collective in which this principle does not have its place. In any case, this is one of those principles that must be taken into account when discussing all correlations in the life and activity of the collective.

18

The Law of Adaptation

The law of adaptation is applicable not only to organic nature, but should be extended to encompass all of the inorganic world as well. Is the evolution of the inorganic world not based on the mutual adaptation of interacting bodies and energies? Is not the movement of heavenly bodies, in its turn, mutual adaptation of a kind? On the other hand, when a rock is smoothed by the ocean, when a river or stream chooses to flow through the most pliable soil, is this not once again adaptation in inorganic nature?

But we are interested primarily in the organic and supraorganic worlds.

As for the former, the formula of life, given by Spencer and consisting of "continual maintenance of the adaptation of internal interactions to external ones" (*The Foundations of Biology: Part I.* 8 I and ff., par. 30), cannot be acknowledged as satisfactory. We are not dealing solely with the adaptation of internal interactions to external ones, but likewise with the use of external interactions for the sustenance of internal ones. If the first, Spencerian part of the formula reveals the passive role of a living creature, then the second, added by ourselves, exposes its active role. The higher the species of animal, the greater the extent to which the second part of the new formula, i.e., the use of external interactions in order to support internal ones, is manifested.

In the plant world, we seem to encounter primarily the first part of this formula; however, there is some exploitation of external interactions, if we take into account a flower's rotation with the movement of the sun, the turning of leaves in order to best make use of solar energy, the guiding of the root system in order to most efficiently absorb the nutrients in the soil, etc. If the adaptation of internal interactions to external ones depends on the

changes in internal processes brought on by a change in external conditions, e.g., the increased production of heat and so less significant loss of it in cold external temperatures, the protection of essential organs from weight loss upon scarcity of nourishment, etc., then the use of external interactions is expressed in external movement, tropism, and taxis, and in the development of free movement for the procurement of food, of protection from bad weather, of shelter from the elements in a burrow or cave, and, among the higher species, in the construction of burrows or dens in preparation for winter, or of nests in the mating season, etc.¹

Finally, in the human world all material culture is distinguished by the exploitation of various natural forces, for the purpose of sustaining the internal interactions of the organism at a corresponding level.

One of the vital apparati for the execution of this adaptation of internal interactions to external ones and the use of these in the interests of the organism is the nervous system and its receptors, or transformers. The latter select stimuli from the external world that are significant to the organism and, transforming them into electrical impulses, connect the influence exerted by those stimuli to the corresponding motor and secretory apparati (muscles, vessels, and glands) thanks to the associative system in the skull and spinal cord. It is especially important that some receptors, such as the retina, the Corti spiral organ, the Schneider membrane and the heat receptors on the skin, give the opportunity to sense external excitants from a distance and thus allow the organism, with the help of experience, to prepare for the corresponding reaction. Thanks to the higher centers of the nervous system, diverse interactions between the organism and its environment are established, depending on individual experience and in connection with the changing conditions of the individual and the surrounding environment.

Thanks to this act of adaptation, a number of higher or associative reflexes is created, through which the complex and multifaceted adaptation of every individual to the conditions surrounding him, and likewise his exploitation of the latter in order to survive, is established.

Moreover, both adaptations are improved thanks to the fact that the nervous system separates excitants into their component parts. Acquiring the ability to react, depending on experience, to one part or another, the nervous system can at the same time determine the relationship in space and time between interdependent excitants, which makes it possible in many cases to prevent subsequent irritants.

If for the first, i.e., passive section of the aforementioned formula, the analysis of external irritants was sufficient to work out the adaptations to them of internal interactions, then it is not in the least sufficient for the second, active section of the life formula, where synthesis is of especial significance. This is why the areas of gray matter in the nervous system, by means of which associative reflexes are carried out, cannot be studied only as analyzers, as they have recently been branded by physiologists of the I. Pavlov school. They are, in my opinion, not only analyzers, but also synthesizing apparati or associators, which is proved by direct experimentation.²

The fact of the matter is that in living conditions all forms of complex adaptation of an organism to its environment and its exploitation of the latter require not only analysis, but also the selective integration of external influences, as well as the coordination of corresponding responses. It cannot be doubted that, among higher animals and humans, both the one and the other are realized with the participation of the cortex as a superior organ which establishes all the complex correlations of an organism with its environment, and alters them in correspondence with changes in the latter.

Not only does a given external influence cause the corresponding reaction to it, but the latter is also able to reproduce, to a certain extent, a corresponding influence. In laboratory experiments, this is demonstrated when people who are especially impressionable under regular conditions of arousal of associative reflexes are subjected to electrical stimulation of the skin applied simultaneously with a sound or light stimulus. After the establishment of an associative reflex, the examinee quickly withdraws his hand or foot whenever the sound or light signal is given because, in his words, he experiences an electrical shock that in reality does not exist. In other, more common cases, although there is no such hallucination, the test subject reproduces the possibility of an electrical shock to which the trained associative reflex conforms.

On the other hand, a reflex itself is always associated in a certain manner with a given stimulus, since a harsh and unusual stimulus incites a defensive reflex, whereas a moderate stimulus having a favorable effect on the organism arouses an offensive reflex, adapted to the stimulus in a way that employs the latter to the utmost. This continuous relation established between the reflex and its external influence, becoming more and more entrenched, leads to the mechanization of actions whenever the reflex is repeated multiple

times. Thanks to this mechanization, reflexive action is executed more easily, speedily, and without excess movement. Concentration, as a preparatory reflex on the part of receptive organs, is likewise associated with a certain external influence, and that is sufficient to arouse concentration upon the reproduction of a certain influence, or to reproduce a given external influence as a result of concentration.

These conditions are at the basis of the facilitated reproduction of a previously experienced reaction, which has exceptional significance not only in view of the fact that facilitated reproduction satisfies the law of economy, but also in that, through multiple repetition of one reflex act or another under the corresponding influence, certain habits are created and new "supraorganic" demands of the individual arise, guiding his conduct.

That which relates to the individual is also applicable to the collective personality. Here, adaptation follows the same path as described above, and is manifested not only in the establishment and maintenance of continual adaptation of the collective's internal interactions to its external ones, but also in the corresponding use of external interactions in order to preserve the collective. The collective, like the individual, needs to satisfy its vital demands while it develops and grows. The adaptation here is realized by means of associative reflexes of a collective nature, forming under various external influences, both of nature and of other collectives or individuals.

In this way, the collective fights for its existence, adapting to the conditions of locality and climate, to surrounding living and nonliving nature, and to those collectives with which it interacts, at the same time employing all three components of its environment in its interests. The development of collective associative reflexes occurs in correspondence with the collective's living conditions and surroundings; these reflexes include concentration, imitation, symbolism, and offensive and defensive acts that satisfy demands of an inherited/organic nature as well as those acquired through experience of repeated external influences of one kind or another, in the form of customary collective actions.

There is no need to examine such adaptational activity of the collective in detail, since its nature is clear from what has already been stated. Let us simply underline the fact that any existential manifestation of the collective in terms of one reflex or another is, in essence, an act of adaptation to surrounding conditions or of their use in the interest of self-preservation.

Notes

- [1](#) See V. Bekhterev, *Psyche and Life* (in Russian), St. Petersburg.
- [2](#) See V. Bekhterev, *General Principles of Human Reflexology* (in Russian), 1918.

19

The Law of Selection

The collective, as any phenomenon, must have its basis in the common laws of nature; otherwise, its existence would have no justification in the surrounding reality.

It cannot be doubted that, in application to living conditions, only the principle of usefulness can justify one phenomenon or another, and in this respect the followers of Darwin's doctrine study sociality in the animal kingdom as a condition conducive to success in the struggle for existence. Thanks to it, animals that lead a social form of life gain precedence in the struggle for existence over others and, by virtue of natural selection, increase the probability of the survival of their species.

So, from the point of view of Darwinism, we owe the development of associations to the struggle for existence, i.e., to external reasons, since the struggle of communities with external foes and with nature is more successful than that of individuals. But external conditions alone are insufficient for the formation of communities, since in the animal world they are determined not only by struggle, but also by the conditions resulting from the organization itself. Not to mention the fact that the struggle itself arises as a consequence of an organism's living conditions, in connection with the need for nourishment. There must be other reasons for the formation of groups or societies.

When individuals interact, they exert an influence on each other, thanks to which any feeling—for example, that of satisfaction—is augmented if achieved communally, since the effect of one's own satisfaction, complemented by the impression of the same satisfaction in others, reaches

the highest possible level thanks to the action of direct induction or infection. People eat with greater appetite when seated at a table with others. In exactly the same way, dancers enjoy dancing in a group much more than individually.

The aspiration to interact with others, in its turn, develops with the increase in energy that accompanies greater satisfaction of vital needs, whereas a depressed condition is attended by a desire for solitude and the evasion of society. The communication between individuals, leading to imitation which, in turn, facilitates the acquisition of social habits, is also of great significance. This is why the reason for the creation of collectives is by no means confined to their usefulness in the struggle for existence, but is connected to the conditions outlined above. In a word, struggle alone is not the incentive for the formation of a collective, but also individuals' needs for society, which can originate on a basis having nothing in common with struggle.

The law of associative reflexes also has significance in this instance. The latter law, as is known, states that any secondary stimulus that accompanies the principal one becomes, upon frequent repetition, an independent stimulus of the same reflex. Let us say that in the struggle with nature, in hunting, and in other undertakings primitive man met with more success when in a group. This led to the creation of communal living conditions in accordance not only with the law of natural selection, as regards the survival of the fittest, but also with the law of associative reflexes. By virtue of that law, the better results achieved as a community, arousing the corresponding sthenic reaction in separate individuals, were accompanied for each person by secondary stimulants following association with other people. The latter was in itself a favorable stimulant and so attracted individuals.

But since communal living conditions must have their own laws, society establishes corresponding conditions of sociality, connected to the inhibition of the aspirations of the individual and the limitation of his rights for the benefit of society, along with his acquisition of special comforts and advantages (e.g., better protection). From this it is clear that the conditions of social life are indirect results of the law of associative reflexes.

It should be kept in mind that the unquestionable prevalence of the principle of the struggle for existence has lately been more and more disputed, especially since Kropotkin published his well-known work, *Mutual Aid as a Factor in Evolution*, which contains abundant factual material on the given subject. We can acknowledge with Kropotkin that the principle of

solidarity and mutual assistance is as widespread in organic nature as the struggle for existence. This explains the development of sociality and the formation of collectives better than does the struggle for existence principle.

Collective attacks, although they guarantee success, also necessitate the division of the spoils into as many parts as there are members of the collective; consequently, the majority of predators prefer to hunt alone.

Collective defense is an altogether different matter. Here there is no need to share something throughout the collective. Of course, it is quite obvious that the collective is not assaulted by solitary individuals, but, conversely, any attacker corresponds the strength of his forces to the potential of resistance on the part of the attacked, which is why collective defense usually meets its match in collective offense.

Thus, it is obvious that the existence of collectives is not sufficiently justified either by conditions of offense or by those of defense.

No, it is mutual assistance under general living conditions—not only those of defense or attack—which accounts for the existence of collectives, since mutual assistance is unrealizable without the collective, and vice versa. In any case, mutual aid as a factor of evolution is definitely manifested in the collective, at the same time justifying the existence of the collective itself.

In separate works, I tarry on the subject of solidarity and mutual assistance as natural phenomena that proceed alongside the struggle for existence,¹ and as phenomena that guarantee improvement and progress.² In those compositions, I prove that in social life we meet with a particular law of nature called "social selection" which, to counterbalance natural selection, acts not in terms of the survival of the strong and physically capable, but rather of the weak and less adapted to the conditions of life, yet more sociable or, in general, answering more to the interests of society.³

Thus, the support of the elderly, of the sick or handicapped, of helpless infants—all this is executed in the collective as something proper and unavoidable, and yet it clearly contradicts the law of struggle for existence in the sense of a struggle against everyone, solely for self-preservation.

It is surprising that the acknowledgement of the universality of the latter law for a long time forced biologists to study cooperation and mutual assistance, which are expressed in the human collective as well as everywhere in nature, as principles subject to the law of struggle for existence, and consequently having no apparent justification in the conditions of organic nature. On the basis of what was outlined above, we cannot agree

with the majority of authors who analyze the formation of the collective as an inevitable result of the law of struggle for existence.

Grant-Allen (*Die Neue Zeit*, 29, 1893-94, p. 71) accuses the defenders of unlimited natural selection of discussing it not in a broad biological sense, but in a very narrow one. They focus only on the individual, not on a group or species. Yet biology should adopt the view that a race enters the struggle for existence as a separate unit.

In the words of G. Le Bon, association replaces powerless personal egoism with mighty collective egoism that benefits everyone. With the exception of groups held together by the ties of religion, blood, or politics—ties whose influence weakens from day-to-day—solidarity of interests can unite people quite firmly:

At the present time, the word solidarity is used far more frequently than the old terms liberty, equality and fraternity, and soon it will supplant them altogether.

Yet it in no way symbolizes those terms. Since the end object of any great coalition of interests is struggle with other interests, it is obvious that solidarity is but a partial form of the universal battle of organisms and classes. Solidarity, in its contemporary understanding, inevitably excludes liberty and confines our former dreams of fraternity in the very narrow frame of association. (Le Bon, *The Psychology of Socialism*)

Conflict is without a doubt one of the important stimuli in the establishment of sociality and of the collective; however, if it were the sole stimulus, the creation of collectives with aesthetic or moral goals would be incomprehensible.

How, for example, are we to understand the reason for the formation of orchestral, choral, literate, or artistic collectives? It is obvious that life is controlled not only by struggle, but also by other guiding motives, the significance of which, for some reason, has been obscured without sufficient basis.

According to N. Ferri, Marx's doctrine is a further development of the theory of evolution. It should be noted, though, that if this can be acknowledged at all, then certainly not from a historical point of view, since Marx preceded Darwin and came from a different background. However this

may be, the doctrine of pure socialism, which presupposes not the equality of people, which does not exist, but the equality of rights to a human existence, should be taken into account alongside the struggle between classes. The division of labor is unavoidable owing to the disparity between people's natural or acquired talents. However, the situation should never arise in which some are idle, while others are overworked like pack animals.

Science acknowledges the vitality of mutual assistance based on solidarity of interests. "As for solidarity," writes G. Le Bon, that ardent antagonist of socialism, "no observation can deny its existence, because such solidarity can be seen in relation to all creatures. Some animal societies, especially lower (i.e., weaker) ones, only exist thanks to the narrow solidarity which makes possible their protection from enemies." But solidarity, in our opinion, should not be analyzed only as a means of protection from enemies. Solidarity is a factor of evolution of the same significance as struggle, so its roots are deeply embedded in living nature, and if in certain cases it depends on struggle between individuals, then struggle is in the same measure determined by individuals' solidarity. Most importantly, solidarity is a factor conducive to and supportive of sociality.

Of course, the law of struggle for existence, along with solidarity and social selection, does have a place in the human world, and only the interaction of these laws determines human progress. What would happen to humanity if competition and rivalry did not exist in its midst? Without rivalry between individuals, tribes, and classes, in a word, "without universal struggle, man would never have departed from his primitive barbarity. He would even now be living in caves, guarding his meager spoils from wild animals, and would never have known the benefits of civilization" (G. Le Bon, *The Psychology of Socialism*, p. 77).

According to the above author, the doctrine of equality and fraternity received a significant setback in the form of

the doctrine of evolution, which proved that a continuous struggle, always resulting in the death of the weakest, is enacted in nature—a law that is doubtlessly bloody, but that is the source of all progress, and without which humanity would never have come out of its primitive barbarity, while civilization would never have existed (Le Bon, p. 61).

But the struggle for survival, whose motto is "Death to the weak and unfit," leads in the human world, as in that of many animals, to the development of conflict between societies in which solidarity must govern; consequently, the latter must be acknowledged an important factor in evolution in general. V. Bittner recently expressed this final deduction from Darwinism:

If in the first stages of its development the law of struggle for existence entailed "death to the weak," then human struggle, leading to the development of social instincts, forms the basis for protection not only of the emotionally connected, but also of the distant and weak.

It is necessary to keep in mind that it is not so much the struggle for existence that plays a role in the development of the organic world as the survival of the adapted, while adaptation is the correspondence of the structure and function of an organism to its environment, and is realized not only through struggle.

Generally speaking, adaptation is not backed by a certain principle. It does not matter to nature by what means this correspondence is achieved—whether through an organism's progress or through its regress, as is the case in the parasite world; moreover, correspondence can be achieved, as was mentioned, via the adaptation of the environment to an organism. The latter condition, almost nonexistent among the lower organisms, is manifested to greater and greater extents with the progress of organization, and reaches its highest level in the human world.⁴ Thanks to this, man is the only higher animal that has populated the entire earth.

As a matter of course, the mighty source of the adaptation of nature to one's own interests is creativity, which in turn is based, in large part, on the conditions of social life.

According to A. Parshin,⁵ the adaptation of physical surroundings to the interests of the human organism proceeds in two directions: (1) "the creation of clothes, tools, optical instruments, modes of transport, etc., in order to sustain the human organism, and (2) the re-arrangement of the physical environment in the form of houses, roads, cultivation, etc."

All this is, in the end, material culture that can be attained only by means of cooperation. However, sociality arose neither from these conditions, as A. Parshin maintains, since it can be found in the animal world, nor from the

Darwinist struggle for existence. It has a self-contained basis in the principle of best satisfaction of demands as a community and in the principle of protection of any member of that community so long as this is in the interests of the group. The support and protection of children, of the sick and of the elderly could not be accounted for from the point of view of the law of struggle for existence or from that of natural selection.

Only gradually, according to A. Parshin, did "familial-cooperative groups" replace individually egotistical hunters. The family was a cell, and for the adaptation of nature to its needs animals were domesticated and the first attempts to cultivate grain were made.

Later, cooperation resulted in the creation of clans and tribes, which represented the first groupings of humans and which were formed under the influence of ancestral ties. With the growth of the family into a tribe, ancestral ties became insufficient, and necessitated the intermediation of a religious element that arose solely from an instinct of self-preservation. Religion led to the conception of a strict tribal god, who protected the family and punished parasitism and rapaciousness within the tribe.

The further development of peaceful cooperation was the consequence of the development of state, national, and religious principles.

However, it should be noted that the family, clan, or tribe did not lead to the development of extensive cooperation in labor; the latter was the product of the expedient combination of energies of individuals who pursued the satisfaction of the same demands. This is why sociality is manifested in the animal kingdom long before family. The family and sociality are factors of the evolution of social cooperation in the human world: the first of tribal and national, and the second of political and partly of religious cooperation. Under contemporary conditions, humankind lives in nation-state or simply state forms, in which parasitism and rapaciousness are restrained by legal norms. States are the cooperations that, in protecting themselves, unfortunately manifest animal instincts in the form of rapaciousness and parasitism towards other nations and states, which inevitably leads to wars.

As for religion, having appeared, as many maintain, out of an instinct of self-preservation, or having been, according to others, the fruit of social living conditions, it assisted the family, clan, and tribe, and then also served the state. However, religious doctrines from the time of ancient Greece began to reject tribal and state origins, which is proved in the Iliad's description of Zeus' chagrin at Achilles' persecution of Hector; ever since the emergence of

Christianity, religion has promoted the peaceful coexistence of humankind in its entirety, condemning rapaciousness and parasitism as evils common to all of humanity.

Religion, however, being the support and stimulus of the uneducated masses, will not be the decisive factor in the union of humanity; it is more likely that the development of governmental systems as well as the elimination of economic slavery and of the so-called capitalist structure will play a significant role. Humanity would undergo a union of states, with state isolation being eliminated and states becoming autonomous areas federatively connected into a single whole under conditions of mutual assistance and cooperation.

The development of so-called moral principles that go alongside the development of collaboration in the human world is likewise of assistance in the establishment of unity, but the development solely of moral principles, without the establishment of expedient conditions of communal living, is not enough to bring mankind closer to unity.

There exists an opinion that the development of collective and altruistic motives arises on the basis of human aspiration to personal gain. This view is upheld by N. Shavrov:⁶

Man is egotistic, as are all creatures.

Altruistic motives arose only as a consequence and on the grounds of human aspiration to personal gain....

The struggle with nature and unfavourable external conditions, the procurement of food, the hunt, self-defence from attacks by animals, etc. were significantly more successful when executed in a group. This left a lasting impression on the human psyche, and originated certain ideas that were partly conscious, and partly remained in the subconscious. Being repeated a countless number of times over thousands of years, this left such deep impressions in the neuro-psychic area that it originated notions which seemed original and peculiar to humankind from the very beginning.

The author precedes his views with facts derived from the law of associative reflexes, in agreement with which any accessory irritant that accompanies the main stimulus, after a number of reproductions, is able to incite the same reflex as that stimulus.

While applying that law to collectivism, the author notes:

Mankind's first motives were egotistic; collectivism was an accessory phenomenon in the life of primitive humans, a means of achieving objectives such as the satisfaction of elementary demands. Later, it became an independent stimulus.

Altruistic concepts, which also became independent motives for humans, came as a result of collectivism. This was a natural occurrence, since without the former all sociality and collective forms of struggle would fall apart like a building made of sand. Accessory phenomena and resources for the achievement of goals gained the significance of independent stimuli, (l.c.p. 18)

In essence, this definition of collectivism corresponds in part to Darwin's doctrine, since both explain collectivism by the principle of struggle for existence, placing egotism, as an embodiment of the struggle against everyone, at its foundation. However, the principle of struggle for existence leads to survival by means of natural selection. How can collectivism, which arose from egotism, play a role in evolution?

Obviously, we cannot do without inheritance. In order to save his theory, the author allows for such inheritance: "Undoubtedly, the higher psychic reflexes leave quite stable impressions on the organization of the brain's gray matter, and can be transferred by one method or another to posterity" (p. 40). It is understood that the basis on which these psychoreflexes can develop is inherited, for example in the form of connections between psychic centers of molecular balance, etc.

However, no matter how attractive the hypothesis of physiological inheritance may seem, there are as yet no facts that would allow us to discuss with certainty the heredity of acquired characteristics and influences, excepting those which act upon the embryonic plasma; in any case, experiments with animals, especially lower ones, are not convincing in this respect.

If we do not acknowledge the hereditary transmission of collectivism, then it must be a phenomenon in the existence of every species that depends on the circumstances and conditions of the organism's environment, as do all associative reflexes.

From my point of view, both external and personal factors are significant for the development of collectivism; in other words, it is contributed to by the law of associative reflexes, by imitation—which is also an associative reflex—and by the law of natural selection. The transmission of the inclination to collectivism must occur through the upbringing of each successive generation, which is supported by the fact that the conception and nurture of offspring is impossible in the life of higher animals without a family collective.

Moreover, the collective itself maintains conditions that make it a self-sufficient institution, since it is upheld not only by the success of collective labor, which leads to mutual encouragement, competition, and to the division of labor through specialization, but also by the factor that I call "social selection."

The latter ensures common mutual support within the collective—support not of the most adapted to the conditions of the surrounding environment, but of the most adapted to the collective, for though they may be the least strong in the physical sense, they are strongest in the social respect.

Notes

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- 1 V. Bekhterev, "Social selection and its biological significance" (in Russian), *Vestn. Znaniia, NordundSud*, 1912.
 - 2 V. Bekhterev, "Individual and sociological factors in the development of organisms and sociality as the condition of progress" (in Russian), *Vestn. Psikholog*, 1913.
 - 3 See V. Bekhterev, "The significance of hormonism in biology and social selection" (in Russian), *Priroda*, 1916.
 - 4 V. Bekhterev, *Psyche and Life* (in Russian), St. Petersburg.
 - 5 A. Parshin, *Scientific Religion* (in Russian).
 - 6 Dr. N. Shavrov, *Thoughts on Man and the State: A Fighting Army* (in Russian), 1917, p. 15.

20

The Law of Interaction

Among the other laws which govern the existence and activity of the collective, it is necessary to analyze the law of interaction. This law can be expressed in the following formula: if two systems have a certain relation to each other, the influence they exert on each other is directly proportionate to their force. This law can first of all find support in the movement of celestial bodies. The interaction of physical forces is, as is known, the subject of study in mechanics; the phenomena of molecular and chemical interactions are studied in physics and chemistry. But the same law belongs in the organic world, and includes the correlative activity of separate individuals.

The organic world consists entirely of interactions. Presently a new scientific discipline has even been isolated under the name of "ecology,"¹ which is occupied with the study of interactions between the organism and its environment and, in part, of interactions between plants and animals. In the latter instances this science essentially represents phyto- and zoosociology,² and is crowned by homo-sociology.³

From the above it is clear that the law of interaction has especial significance in human collective existence, since the creation of the collective itself, requiring the union of individuals, presupposes their interaction. This interaction occurs not only between individuals of the same collective, but also between every individual and society as a whole, and to an even greater extent between society and its directors, or between a crowd and its leader. On the other hand, all government is based on interaction between the ruled and the rulers.

All relations, says P. Sorokin (*The System of Sociology: Book 1*, p. 81), from manufacturing and economic to aesthetic, religious, and scientific ones, can fall under the category of interactive relations. Simply speaking, the interaction of two or more individuals is the fundamental concept of social phenomena; it can serve as a model of the latter. In studying these models, we can discover the structure of all social phenomena. Having decomposed interaction into its component parts, we will have decomposed the most complex social phenomena.

Lasalle was correct in saying that a constitution is worthy and acceptable when it is a proper expression of the relationship between forces acting in a state at any given moment, since any social existence presupposes the conflict of classes and parties and consequently the interaction between them, and, in essence, every collective interacts with all the other collectives it encounters during its existence, and even with the entire surrounding world. In order to clarify the principle of interaction, let us take the most primitive collective—the street crowd—and see how this principle is manifested.

There have been a number of investigations into the crowd, but they, generally speaking, contain much one-sidedness and exaggeration. Some see in the crowd only the tendency to commit crimes, not noticing that it can be heroic in the fullest sense of the word, and forgetting that history owes many of the most noble human impulses to the crowd. Others do not sufficiently distinguish between the characteristic peculiarities of a street crowd and those of a crowd presiding in courts in the form of a jury, in parliaments, in committees, etc. That which is correct in relation to a street crowd is not applicable to an organized meeting, and vice versa. Thus, for example, among the characteristic qualities of the crowd, authors name impulsiveness; irritability; lack of reasoning, analytical, or critical ability; excessive emotionality; extraordinary suggestibility, etc. G. Le Bon, as a matter of fact, compares these qualities to those observed among such lower specimens of evolution as women, children, and savages.

However, can it be said that the form of reasoning of councils, committees, or parliaments is comparable to that of women, savages, or children? If this were the case, could it be conceivable that where the greatest impartiality, prudence of judgment and circumspection in decision-making is required, humankind would allow the collective mind to act? How could we resolve with this view the fact that the foremost nations of Europe leave the most important questions of their existence, often deciding the fate of nations, to

be resolved by the collective mind in the form of parliaments and international tribunals?

It is obvious that this is not the way things stand, and that which relates to the disorganized crowd cannot be carried over to the organized meeting, where different outlooks and deductions, cultivated by various thinkers and accepted by individual minds from a certain viewpoint, are presented and evaluated.

We can acknowledge the fact that the decisions of meetings cannot lay claim to genius, although that too depends, in essence, on the significance of the leaders' minds and on their ability to defend their opinion; but at least the decisions of meetings are relatively rarely one-sided, which is more an attribute of the individual mind.

Similarly, we cannot agree with those authors who maintain that reason cannot play a role with respect to crowds, and may even have a negative connotation to them. In the words of the same G. Le Bon, "it is impossible to influence a crowd with reasoning because only a crude association of ideas is accessible to it. This is why orators who can make an impression on a crowd always appeal to its emotions, not its reason. The laws of logic have no effect upon it" (*The Psychology of Nations and Masses*, p. 243). In relation to the disorganized crowd this opinion may be considered just, but only in the case when the crowd is already in the grip of a certain mood, which occurs quite often. But if the crowd is calm, such as one that has collected not under the influence of dissatisfaction or indignation, but simply to hear an orator, then how can it be said that the laws of logic have no effect upon it? The orator who would take Le Bon's advice and, throwing logical argument to the winds, put emphasis on impact in an attempt to arouse the crowd's emotional side would gain little respect. In any case, a calm crowd listens to arguments which must be clear, simple, and convincing, and better still accompanied by corresponding examples or images; emotional overtones are important only when the crowd must be motivated to act.

As for the organized crowd, such as jurors, I have attended famous trials and have several times encountered the fact that the lawyer who presents irrefutable arguments has a greater influence on the jury than the one who emphasizes sensation and attempts to arouse an emotional reaction in his audience. At one point G. Le Bon (I.e., p. 293) writes:

Jurors succumb as easily as any crowd to the influence of compassion, and very little to the influence of reasoning... .They cannot, says one lawyer, withstand the sight of a woman breastfeeding her child, or a procession of orphans... .In order to gain the favour of the judges, it is enough for a woman to be attractive, according to desGlayeux.

All this is true, but it is equally true in relation to the individual mind, and there is no great difference between a separate individual and a crowd of jurors. The latter, in any case, yields to arguments as well as to the arousal of emotions. I once knew a famous lawyer from Petrograd who was able, thanks to his talent, to excite the emotions of the public and the jury, but who in celebrated trials always allowed his assistant, who based his speech exclusively on analysis and argument, to speak first.

It was this combination of analysis and argument in one speech and emotional overtones in the other that had the profoundest effect on the jury. That the crowd's inclination towards emotion is a far cry from commonplace phenomenon is proved by the well-known fact that where a single person could easily relent under the influence of a petitioner's tears, an aggregation or crowd would not be affected by those same tears, which may even create the opposite impression to what was intended.

In relation to an electoral crowd, G. Le Bon once more notes an inability to reason, a lack of critical thinking, irritability, bias, and gullibility (I.e., p. 229). But it is the spirit of parties that acts at electoral meetings, and party politics is not based on emotions or feelings, but on interests supported by certain arguments; no method of arousing emotion would be sufficient to conquer the interests of a party, expressed in its platform.

We can at least partially agree with the stand taken by G. Le Bon (I.e., p. 307) that in a crowd people are always equal and, if we are dealing with "general issues, the vote of forty academicians is no better than the vote of forty water-bearers." But we can in no way accept his arguments and assumptions pertaining to universal suffrage, which he condescendingly allows to remain in its present form only because it is a kind of contemporary religious dogma; besides, he concedes that limited voting would have no better results because it would be accompanied by the "tyranny of castes." Although G. Le Bon's opinion that the "arithmetical median of all elections in any country is an expression of the race's soul" (I.e., p 308) is quite correct, he is not right in complementing this view with the phrase: "This soul

remains almost identical from generation to generation." The diverse electoral results of various epochs in one and the same country prove that, notwithstanding its immense stability, the change of a nation's "soul" is a far from exceptional phenomenon.

Furthermore, it is worthy of attention that authors who wrote of the crowd were inclined to maintain that a collective resolution, thanks to the interaction of separate members of the crowd, will always be of lower quality and less intelligent than the decision of an individual. But this is a significant misconception. The fact is that we cannot analyze a congregation as an exclusively collective mind, since we are usually dealing with the union of the latter with the work of the individual mind. Any question is commonly prepared by a specialized speaker for a meeting, where, through criticism and discussion, the rough points of the presentation—such as overestimation of facts, exaggeration, or one-sidedness—are smoothed over and eliminated, in the manner of a lathe rounding the sharp edges of a block of wood. Sometimes, gaps in the presentation are filled out with new arguments or facts. Under such circumstances, can it be doubted that the results of this work would be better and more valuable than those of the work of an individual mind? And who does not know that the decision of a board, for example that of the supreme court or court of arbitration, is acknowledged to be far more authoritative than the opinion or command of an individual representative of one administrative power or another. It goes without saying that a board can also make erroneous dicta, which occurred, for example, when the French Academy of Sciences rejected the possibility of man's flight as "contrary to his nature," and when the Paris Academy of Medicine saw nothing more exceptional than plays of the imagination in mesmerization, which was later acknowledged by science to be the product of hypnosis, a very real phenomenon. We can likewise mention as another example the latest dictum of the Petrograd Medical Council, which requires that the same procedures and measures as are taken during operations be followed during the hypnotism of patients, which is almost equivalent to completely prohibiting the use of hypnosis in Russia. Examples of mistakes in court that have led to the conviction of innocent people are also widely known. However, these errors of the collective mind are not, in essence, related to its work, but rather to the errors and inadequate knowledge of those specialists who spoke out on the above issues, and who, thanks to their authority, were

able to convince their audience of that which was the product of their own ignorance.

Thus, the blame for these collective faults falls not on the work of the collective mind, but on the work of the individual mind that in the given case chaired a meeting and overpowered it with its authority.

It is often forgotten, when the work of individual and collective minds is juxtaposed, that the types are essentially incomparable. The fact is that when we are dealing with the work of the individual mind, we are first of all dealing with a specialist who often has excellent erudition, has the opportunity to concentrate on the given task, to make the necessary investigations, and who can spend days, months or even years on this process. Naturally, the resulting work is substantial and more or less exact, although sometimes one-sided.

On the other hand, in the collective mind we are not always dealing with such specialists, and there is no opportunity to concentrate, make investigations over long periods of time, etc. However, meetings usually contain versatile specialists, if not experts, in the given issue, as well as representatives of related fields, who analyze and discuss the issue from different viewpoints and who are usually more disinterested than separate individuals. Collective work is distinguished by a more careful attitude to the task at hand and a more comprehensive study of the subject; nevertheless, it cannot have the same depth as that of the individual mind.

In this way, it is clear that the probability of a correct decision is greatest in those cases when the work of the collective mind is combined with that of the individual mind, whose role is to prepare the issue and present it for common discussion or criticism.

This is what should be analyzed in the interaction of minds in collective meetings. It should be noted at this point that the work of the individual mind also cannot be analyzed exclusively as such. It is known that geniuses employ the results achieved by thousands of preceding minds, in the way that a good tradesman uses the knowledge and skills of a large number of past tradesmen. This is why great inventions and discoveries, which seem to be the product of an individual mind, are really the result of the activity of the collective mind of prior generations. As we know, even general concepts owe their existence to the creativity of the collective mind, i.e., to that of the crowd, not of the individual.

The results of the interaction of classes and other collectives are shown not only in the gradual mitigation of differences, but also in the level of influence of more significant collectives on the general course of events. Even the government of a country is in direct correspondence with the significance and strength of a certain class or party.

The law of class conflict states that if a given class establishes its rule in one way or another over a country, it creates its own government. If two rival classes cannot decide which of them is superior, the government can only be coalitional if there is the possibility of agreement between them. If there is no agreement, and both classes are weakened by mutual conflict, the way is cleared for the development of a dictatorship. This is what happened at the outset of Napoleon III's reign in France, when Cavaignac had to step down, thus giving the former the position of dictator.

When we are dealing with the government of a parliamentary country, the governing organ—the ministry—is upheld by the parliamentary majority, and if for some reason such a majority is not formed in its favor, the government makes compromises, attracting various parties in order to gain the support of most of the nation's representatives.

But the interaction of collectives is not limited to politics; it is manifested to no lesser extent in the areas of science and art, since here we also have separate parties of scientists or trends in art, in a word, separate camps that interact with one another, continually smoothing over their differences, sometimes even entering agreements for combined work. Once again, highest significance is achieved by the party or trend that has greater real force.

In economics we also meet with the interaction of collective groups at every step.

Interaction is first of all established between vendors and consumers or purchasers, thanks to which the setting of prices (supply and demand) occurs. The very character of the objects used by a given collective depends on the established tastes, habits, and other conditions in which the consumers live. Furthermore, interaction is manifested between rivals who each strive to place themselves ahead of the competition. In the same way, interaction between various groups of consumers is established, especially when there is a lack of goods, etc.

Finally, in international relations we are likewise dealing with the interaction of every national collective with its neighbor and with all collectives in general that it encounters in any way. Here, as in every other

case, the result of interaction is the authority of one or several collectives over all others. But in the end, the course of international life is determined by the force resultant from the interaction of all national collectives that come in contact with one another in order to establish international ties.

Notes

- ¹ See Haeckel, *Generelle Morphologie d Organismen*, 1866, p. 286. Berlin. Rd.
- ² See Sukachev, *An Introduction to the Study of Plant Communities*. Morozov, *The Forest as a Plant Community*. Vagner, "Sociology in Botanies" (in Russian), *Priroda*, 1912. Henslow, "L'écologie au point de vue de l'évolution des végétaux, *Scientia* t.XIII, Annex 87-107.
- ³ Waxweiler, *Esquisse d'une sociologie*, pp. 1-87; P. Sorokin, *System of Sociology* (in Russian), p. 10 and ff.

21

The Law of Compensation or Substitution

In this case we are dealing with a law in accordance with which any motion or any other form of energy, upon encountering an obstacle to its manifestation, is replaced by another motion or energy (also essentially a kind of movement). There is no need to dwell on the fact that this principle is the direct consequence of the principle of conservation of energy and is supported by countless examples of phenomena in the physical environment. Moreover, it is the basis of the so-called principle of the unity of energy. In chemical reactions, whenever one substance is replaced by another, we likewise meet with this law.

In the organic world, the above principle is also manifested everywhere, since, for example, the artificial inhibition of the development of roots and stems in any vegetation is compensated by the growth of new roots and stems from other parts of the plant. In the animal kingdom, the absence of an organ is compensated by the intensified function of other organs. In the functions of the nervous system we meet with the substitution of destroyed or degenerate systems and fibers by healthy systems and filaments.

In correlative activity, we also have the substitution of one action, one forgotten word, one unsatisfied demand by another. Thus, sexual needs are often replaced by artistic or religious quests and vice versa. As a matter of fact, the method of sublimation in the treatment of neuroses is based on this principle.

The same process of substitution can be observed in the activity of the collective. When its mental development is hindered, as it was, for example, in Russia under the tsarist regime, it is replaced by an astonishing rise in the popularity of mystical doctrines, gambling, and alcoholism. When the government forbids open agitation, the latter finds expression in underground propaganda. In the interests of inhibiting the latter, governments often attempt to divert a national movement by, for example, participating in a war. In order to distract a conquered nation from aggressive acts towards its conqueror, deft diplomats strive to redirect the energy of the vanquished nation by supporting its attempts to capture other nations. Thus, Bismarck, having taken Alsace and Lorraine from France, encouraged its colonial politics. Wilhelm II, not having understood the good sense of these tactics and having made a false step in provoking the Agadir incident, set the stage for the world war that would prove unfavorable for Germany, as well as for himself.

The law of compensation played a large role in the dissemination of Christianity at the end of the second and beginning of the third century. As is known, under the Antonines and especially under the Severi an astounding awakening of religious thought was manifested in Roman society, along with a thirst for belief in the exalted, spiritual, and immortal.

This was really a reaction, in the form of the replacement of social philosophy by a new doctrine, to the indulgence in sensual pleasures that was so widespread at the time. Moreover, the Roman Empire accepted into its pantheon the gods of all conquered nations, and naturally the existence of countless gods undermined belief in them and weakened the social significance of pagan religion.

Attempts had been made, in connection to the above aspiration of society towards the exalted and spiritual, to moralize ancient myths and inspire pagan religion, but these did not satisfy society. It was at this time that Christianity, with its doctrine of a single God in heaven and spirit of love and forgiveness, saw that it could make use of the reaction and quickly established its rights. The heroism of the first Christian martyrs, their unshakable belief in the righteousness of their doctrine and the incompatibility of Christianity with other religions, from which stemmed bloody persecution, completed its victory over paganism.

The fact that the realization of every great thing in social life is often connected to terrible trials, and that only sometimes, thanks to these trials and

to the suffering of its defendants, does it acquire final acknowledgement, is based on the same law of compensation.

This law also explains the remarkable contradictions that are manifested, especially during periods of insurrection, in the collective life of nations. Thus, in the most tragic days of the Terror, filled with almost incessant executions, when thousands of people died on the scaffold, all places of entertainment were full to the overflowing. Such was the situation during the Great French Revolution and during the Russian Revolution. The reason lies partly in the fact that the populace, previously deprived of all forms of pleasure as a result of poverty and oppression, having attained freedom, pursued the various entertainments that were previously inaccessible to it, and partly in the need to relieve the oppressive conditions of life. This is why all forms of diversion enjoyed full attendance while the guillotine reigned in Paris, and why the dancing-halls were full during the Russian Revolution at a time of ceaseless executions and looting.

Something analogous can be seen in the acts of the Parisian prisoners who, despite the Sword of Damocles—the guillotine—that already hung over them, indulged without any sense of moderation in amorous escapades. Here, for example, is how Cabanis and L. Nass describe life at the Conciergerie prison of Paris, based on contemporary accounts:

Having passed the first grille, of which there were at least four, you reached an area girdled by iron chains. Here, meetings with visitors were allowed; usually only women came to see the prisoners, and were greeted, of course, with the greatest affability... Here, in the words of one contemporary, husbands became lovers, and lovers redoubled their affection. It was decided without any formal agreement to no longer heed the laws of social propriety, which were "valid only when it was possible to postpone expressions of fondness to another, more convenient place and time." The tenderest of kisses were recklessly dispensed right and left. Under cover of darkness or of a commodious skirt, intimacies gratifying even the fieriest impulses were brazenly exchanged. At times, these ecstasies were interrupted by the advent of the unfortunates who had just been sentenced to execution, and who were led through that area from the hall in which the tribunal presided. For a moment, a death-like silence would reign, frightened looks would be exchanged, and then .. .then the

tender words, the kisses, the fervid declarations would recommence, everything would acquire its former aspect and take its course.

The same could be observed at the main entrance. Husbands, wives, and lovers filled the long benches that lined the walls, embracing and caressing one another with the same happiness and tranquility as if they lay on a bed of roses. Rarely, some wept and bemoaned their fate.

No less intense love affairs unfolded in other wings of the prison. During the day, inmates occupied themselves with their toilet, dressing and adorning their bodies, and in the evening love came into its own. All was employed in its favour: the evening dusk, the shadow of a passing cloud, the tiredness of the guard, of the overseers, the modesty and sleep of the other prisoners. (I.e., p. 132)

The process of communication by signs and symbols is also based on the principle of compensation or substitution. It is necessary to keep in mind that any sociality allows for the development of individuals' abilities to signal one another. Primitive signaling is the language of mime. The effect of fright experienced by one individual is a signal of danger for another, while exclamations of despair are cries for help, etc. The mime of sexual arousal is once again a sign of a certain type, leading to the corresponding stimulation of the opposite sex. The grasshopper's chirp, the bird's twitter or song, the dog's bark, the sheep's bleat, the rabbit's thump, the cow's low, the horse's neigh, the macaque's squeals and other noises—all these are essentially sound signals, lying at the foundation of animal language, completely understandable to animals of the same species and attaining, among humans, the level of complex articulate speech.

These signals are, of course, not always sound. They can also be photic, such as the glow of a firefly and the change in color of some animals' skin; mechanical, such as the movement of an ant's antennae and the expressive poses or movements of many animals, human gestures; olfactory, in the form of animals' secretions when in heat, etc. But they always serve as characteristic indicators of the animal's condition—a signal or sign of its condition to others:

When a deer throws up its head, all other deer do the same. This is a sign, which means "Listen." If the first deer cannot see the object that attracted

its attention and seemed suspicious, it emits a low sound. This is a word that means "Caution." If the deer then finds that the object is not only suspicious but also dangerous, it employs a further form of language—intonation: in place of the low sound "Caution," it makes a harsh, loud cry, signifying "Run, in order to save your life." This is the nature of language: sign or gesture, sound or word, and intonation (H. Drummond, I.e., p. 181).

It goes without saying that the higher an animal is in terms of development, the more complex its signal is. Let us take the dog as an example. C. Darwin noted from four to five variations in the barking of a dog. But that animal also expresses its condition through growling, whining, and also through the movement of its tail, ears, or entire body, its expression, etc. Here, language is already so multi-faceted that it exposes many aspects of the dog's internal condition, which could not be manifested by, for example, a bird. Monkeys have up to six vocal variations and a primitive language that serves to signify certain objects, as well as a diverse body language.

Human speech, which arose through the evolution of the primitive language of sound reflexes, gestures, and mimed movements, is the product of sociality and developed thanks to the continuous communication between people, without which signals in the form of speech could never have existed. In any case, a number of facts support the presumption that this ability to signal is not inherited, and consequently not inborn, but is acquired by experience in the form of higher or associative reflexes.¹ The fact that a person deaf from birth is also always mute but replaces the lack of speech with a profusion of gestures upholds this standpoint.²

On the other hand, someone who is blind from birth has poor gestures and mime, but excellent intonation. A person deprived from birth of communication with his kind, as was Hasper Hans, has no developed speech. Finally, someone who is born to parents of one nation and is taken in infancy to another country acquires the latter's language without retaining any trace of his parent's speech.

It should be noted that the language of gestures and mime develops under the control of sight, and the language of sounds under the control of hearing. Actually, a significant similarity is observed between the gestures and mime of deaf-mutes and those of primitive peoples, which is due to the fact that the

limited sounds produced by the former and the latter force them to highly develop gesture and movement.

Even the gestures employed for abbreviation are common to various nations. All this would be puzzling, if it were not explained by the fact that those gestures and mimed movements are reproductions of illustrative, indicative, and other movements which cannot differ from nation to nation because they correspond to the same external object.

"Mallery brought together a few foreigners from the Utah territory and one deaf-mute. The latter recounted to them the long story of a certain robbery, which contained dialogue; it turned out that they understood each other perfectly" (Ribot, *The Evolution of General Ideas*, pp. 48-49).

As was mentioned, deaf-mutes develop mime and gesture to the highest possible level, thanks to which they easily understand each other and, without being taught, use more or less identical illustrative and indicative gestures. These form a terminology and simple syntax. The latter consists of the order or sequence of the gestures one after another and their combination.

For example, let us say that the gesture for a child is the indication of small stature, and that an infant is designated by an imitation of breastfeeding, carrying or rocking. A dog is represented by barking accompanied by movements of the head and mouth. Loss is represented by the futile search for a dropped object, "forget" by a hand drawn across the forehead and a shrug of the shoulders. A hand placed on the heart demonstrates love, while hate requires the same gesture complemented by negative shaking of the head, etc.

As for syntax, according to the investigations of several authors (Scott, Taylor, Romanes, and others) it does not contain parts of speech or terms of a certain linguistic significance, but is based exclusively on the sequence of signs/gestures, with signs arranged in the order of their relative importance. To be precise, the subject comes before the predicate, the object before the verb, and the passive before the active; all unnecessary parts are eliminated. For example, the gestures for the expression "I beat somebody with a stick" would come in the following order: I-somebody-beat-stick, and for the expression "I was beaten by somebody with a stick," the sequence of gestures would be as follows: somebody-I-beat-stick. This syntax is, of course, extremely poor and can hardly be compared even to the limited syntax of the feeble-minded. Moreover, in most cases religious, moral, cosmic, and

metaphysical concepts are inaccessible to such uneducated deaf-mutes. However, the latter are not bereft of creativity (Ribot, p. 43).

Why exactly did humanity develop a primarily spoken language? I have written about the anatomical conditions that assist the development of a language of sounds in another work.³ Nonetheless, we cannot overlook the especial advantages of a spoken language over one of gestures, and the fact that the former gave a certain direction to human symbolism.

There is reason to think that, at first, man had the opportunity to use both forms of language, but that spoken language gained precedence due to the following conditions. A language of gestures requires the use of the hands, hindering their application to other employments; it is not effective in the dark, or across large distances. Furthermore, gesture and mime are of an exclusively imitative origin, as a result of which they are connected to concrete stimuli and are ill adapted to generalizations, although it cannot be denied that further development of the language of gestures may lead to the creation of generalizing signs. Support for this possibility can be found in the existing gestures for abbreviation. On the other hand, speech allows for development and subtle differentiation with respect to external sound stimuli and to the manifestation of the voice or sounds, in relation both to intonation and to the complexity of sounds, and all this with minimal expenditure of energy.

The depiction of written speech was at first derived from the language of gestures (the so-called ideographic script). M. Müller provides a number of examples that are quite edifying in this respect. The similarity to gestures is found in the isolation of terms, in their order, and in the concrete illustration of actions. He cites a letter from a citizen of the Caroline Islands to the captain of a Spanish ship: a person with outstretched hands is depicted at the top (request), below and to the left of whom are drawn the proffered objects: five large and seven smaller shells; directly to the right of this are the objects required in exchange: three large and four small fishing hooks, two pieces of iron and two axes.

Signaling has a huge significance not only in human collective life, but also in that of animals. A flock of birds calmly feeds until its sentry cries out and, raising its wings, gives the signal for all to fly from the approaching enemy.

The same is known in relation to quadrupeds that live in herds. The entire herd obeys the directions of its leader, who signals how it should behave.

Human symbolism, other than writing and drawing, takes on the following forms:

1. inarticulate sounds in the form of simple reflexes;
2. inarticulate sounds in the form of associative reflexes;
3. articulate sounds as a further development of associative reflexes, in the form of sound imitation;
4. articulate sounds as a further development of associative reflexes through prefixes, complications, and suffixes (speech);
5. reflexive mime and gesture, such as crying, laughter, etc., as a result of external excitants; defensive movements of the hands to cover the face and eyes; and removal of the hands from acute irritants;
6. mime and gesture in the form of associative reflexes;
7. pantomimical gestures as reproduction of actions (indicative, illustrative, etc.).

The significance of signal in the social conditions of human life is acquired, as in individual life, through the experience and training of members of the collective.

A cry of despair, according to general experience, signifies misfortune, while shooting, once again in accordance with the experience of all those who have come in contact with firearms, is assumed to be accompanied by grave casualties. This is enough for both one and the other, even in the absence of true misfortune or of any casualties, to have the usual consequences in a crowd, i.e., panic in the first case and answering shots in the second.

An alarm bell, as is known, is sounded in case of fire to summon assistance, whereas a church's bells call its congregation to attend the service. Thus, both serve as a signal to leave one's house—in one case, to go to the scene of a fire, and in the other to attend church. Analogous conditions can be observed in all other cases.

In human life, signals are encountered at every step.

Let us say that the training of troops is taking place. All await a certain signal. The command is given, and the military collective comes into motion. Battle also commences upon an agreed signal, in the form of a command, and all further developments are likewise in accord with signals.

Take a team of firefighters. A signal given from the tower mobilizes a mass of firefighters, and each sets about performing his task. In an instant, all is ready for departure; when the fire is reached, the words of the team, playing the role of signals, activate a mass of people. Having completed their mission, the firefighters rest until they once again hear the fire alarm, calling them to duty.

Another example is a mass of people, say, guests at a ball. For a while they wander up and down the room. But now the music, serving as a signal, commences and you see first one pair, then another separate from the crowd and begin dancing.

You are in the theatre, awaiting the commencement of the play. The public is gradually arriving, and at the appointed time the conductor appears in the orchestra. He lifts his baton, which is a signal for the musicians, and a moment of tense concentration for the audience. But now the curtain rises, the play begins. Characters come onto the stage, speak to each other and to the public. There are signals for this mass of people on the stage that are not noticeable to the audience. These signals are given by the prompter, and consist of phrases that should be pronounced at certain times, and reminders of actions that should accompany the given scene.

Observe, furthermore, how a choir sings. First of all, the precentor gives a signal in the form of a tuning fork's sound; then, at a gesture of his hand, the choir begins singing. Otherwise, the choir may sing after the wave of a conductor's baton, with the initiation of music, or, finally, upon the termination of an introductory tune. All these are essentially signals that bring on a certain form of collective action.

Let us go on to collective gymnastic exercise. What guarantees the coordination of the actions of the many people who participate in complicated gymnastics? Once again, it is a command or signal that determines the beginning of activity.

Suppose you are in a school where a mass of students are employed in collective mental labor. During break, they wander through the corridors shouting and making noise. But now the bell rings and everyone rushes to their class to take their places. Little more than three-quarters of an hour pass, and another bell announces the beginning of the next break. In the class itself, the teacher may signal the commencement of collective work such as a dictation.

While attending communal games, or even horse-races, you can once again observe that collective action is initiated by a certain signal.

If you study the collective actions of a workers' co-operative, you will likewise see that coordination of common labor is achieved by a command or by the singing of a workers' song.

In large collectives, a bell serves to arouse everyone, to summon them to meals, to their studies, and to bed. In a word, signal plays an important role in all cases that require coordinated collective activity.

In all the above instances, we are dealing with an organized collective, in which the signal itself is a previously appointed and habitual sign. Even in unorganized collectives, the signal holds an analogous significance. Thus, for a crowd that is confined to a small area, the cry of "Fire!" can serve as a signal to the development of panic. At large conferences it is often difficult to hear anything over the din of conversation, but as soon as the chairman of the meeting rises from the table to pick up the bell, everyone quiets.

Likewise, in a crowd it is enough to shout: "Stop that thief!" for a few people to take on pursuit of the said wrongdoer.

A provocational shot often serves to begin a street fight and lead to civil carnage.

The alarm-bell sends people to the scene of a fire. Church bells call them to church and prayer.

A cry of desperation sets a crowd in motion.

At large meetings, where a certain decision must be made, in other words when the collective must express a certain attitude to whatever occurrence, one word is pronounced by the chairperson and a majority of raised hands determines the nature of the meeting's resolution.

As can be seen from the given examples, signal is an indispensable condition in various instances of collective activity. It is in some cases a warning sign that stimulates collective activity, whether offensive or defensive. In other cases, it can serve as a source of coordination of collective activity, without which the latter would never be realized. Finally, it determines the termination point of common action.

This clearly outlines the significance of signal in collective activity, where it has, perhaps, even greater importance than in individual existence, since without it the lives of many people would naturally be endangered, and collective activity itself would in many cases be unrealizable.

Let us note at this point that the development of sign communication is related to the question of social heredity, thanks to which the experience of individuals of one generation is passed on to those of the next, establishing the social succession that has determined the historical development of social origins.⁴

Furthermore, the sign or signal in social life plays the role of a symbol; gestures and the spoken word are essentially symbols of surrounding phenomena or of certain interactions and actions. In the life of civilized nations, symbolism plays a very broad role, giving rise to a number of artificial symbols, often having a specific historic origin.

Thus, the regalia of tsarist rule, insignia, coats of arms, etc.—all this is nothing more than an expression of social symbol. Symbolism is manifested everywhere in social life, as, for example, in customs, in religious ceremonies, in the so-called language of flowers, and even in the style of dress or jewelry.

Some phenomena of symbolism are transferred from generation to generation in the form of habit, having lost their initial meaning.

Thus, the burning of bonfires on Midsummer's Night, being a vestige of pagan times, has lost all of its original significance. On the other hand, some remaining marriage customs have survived and retained their symbolic aspect to this day.

The ceremonial side of religion, carried out during communal worship, is full of symbolic meaning, and there is not one movement of the priest during a church service that does not symbolize one event or another from the life of Jesus and from his doctrine.

The same should be said of all communal ceremonies and rites in general, such as communal celebrations, dances, funerals, games, etc. Everywhere, we meet with social symbolism in its most variegated manifestations. As a matter of fact, court etiquette is also resplendent in symbolism, being based on the deification of the ruler and his elevation to the state of idol. N. Kareev says in a newspaper article:

Such idols, to whom divine homage was paid, incense burned, and sacrifices made—the incense of glorification and flattery, and sacrifices of reason and conscience, human pride and honour—have at one time held the highest positions in all countries.

From the times of ancient Egypt, where the pharaoh was "king and God," this tradition of pharaohism, of the deification of the ruler, spans the ages; it can be seen in the hellenistic kingdom of the Ptolemies, in the Roman Empire, and in a more recent form in Christian Europe during the period of absolutism.

Was not the monarch made the representative of God on earth? Was not his rule ascribed to God's will? Were not kings named earthly Gods? Did not Bossuet exclaim: "Kings—you are gods!" and call them "Christs" in the sense of the Lord's Anointed? Was not the official epithet "adored" applied to monarchs? Was not their person "holy"? Were not many generations raised with such attitudes towards the ruling power, and did this not become a deeply rooted tradition? (N. Kareev, *Nash Vek*, 1917)

But this deification of the ruler can also be translated to the collective power of the nation, to which J. Rousseau assigned absoluteness and infallibility in his "Social Contract." National government is likewise treated to the servility that was once given rulers of the monarchical era.

Here is how, in one newspaper article from the beginning of the Russian Revolution, the new national leadership was glorified (I take this quotation from Kareev's article):

To her majesty the Nation, proud and triumphant, wise and powerful, the heroic nation, the selfless nation, the righteous nation, the nation of clear conscience, the nation-prophet, the teacher of all nations, etc.

The Russian people were then praised as

pure and righteous, on the path to God's kingdom. Honor to you, honor to the Russian nation, mysterious and radiant as He whose blessing casts light on all your future enterprises. Honor to the nation-Saviour, honour to the nation that harbours in its womb shining revelation for the whole world. Honor to the nation that will teach and be loved by all that is disturbed, restless, and without peace or happiness on this Earth.

In the national intellect the author, as a matter of fact, observed "something approaching the image and likeness of that perfection which has been the goal of spiritual aspiration from the day of the Creation."

It could be countered that this deification represents the view of only one person. No, this text was printed in a widely read newspaper, apparently for the enlightenment of a mass of subscribers. Speeches in the same vein were made at meetings, deifying the national government or at least establishing its infallibility.

Lectures, entitled "What does 'a great noise from a northern country' signify?" were even given on the prophetic prediction in biblical times of the Russian Revolution.

This theme was addressed numerous times by one member of the order of the "Eastern Star," and attracted a huge audience. Yet the deification of national power was not limited to newspaper and lectoral eulogies.

Kerensky, embodying power as the national favorite in the first part of the Russian Revolution, did, in fact, take on the role of highest, autonomous leader of the people, before whom the same ceremonies had to be enacted as before the true, hereditary monarch who had ruled "by the grace of God." All the prerogatives of absolute power belonged to him, for at critical moments of the Russian Revolution all ministers handed their portfolios over to him, giving him the unqualified right to control their fate. It goes without saying that accommodation for the new ruler had to be the same (the Winter Palace) as that occupied by the tsars before the revolution.

Symbolism also penetrates our everyday life in the form of the admiration and flattery of those holding powerful positions, and likewise in the form of politeness towards all individuals, be they of lower social standing or higher, with the exception of family members, with whom there is no need for either etiquette or conventional politeness in view of its utter pointlessness.

The significance of symbols in social life, however, is not limited to the above, for symbolism is manifested in such reflections of social life as sculpture, architecture, art, literature, and poetry. Moreover, mime and mannerisms are full of symbolism, which is not difficult to discover through even a very superficial analysis of these phenomena.

Social unity, as any complex object, can be symbolized, usually by a banner, sign, or emblem, e.g., uniform, flag, etc. The role of the symbol may be played by a person or even a group of people, e.g., a presidium, military leadership, etc., that is an expression or representation of the given collective. That same symbol is often a factor conducive to unity, cementing ties between many members of the community by reminding them of their association to the given collective and binding them to a certain solidarity

and code of behavior, in agreement with the interests of the community. This cementing role of symbols is discussed by P. Sorokin in his *System of Sociology* (Part I, pp. 322-323).

Furthermore, are not various badges, signs, and orders, heraldry, religious and state ceremonies, dances, even games and celebrations, social symbols?

In view of the fact that symbolic signs, thanks to established customs, embody historic facts, they have so powerful an influence over people's minds as to be fetishes. The regimental banner can serve as an example. I remember how, during the World War, one of the Russian nurses happened to find, on a battlefield already taken by the Germans, a banner belonging to a regiment that had been defeated along with the other divisions under General Samsonov's command in Eastern Prussia. She kept that banner with her throughout the duration of her captivity, and, being subsequently released, smuggled it with the greatest precautions through Germany and other countries. And at the moment when she, released along with other nurses thanks to a recent agreement, crossed our border from Finland, I (being accidentally on the same train) could myself witness the effect of this fetish on the minds of the soldiers. When the nurses unfurled the rescued banner from the window of the carriage, the few soldiers who were on the platform, and who had no relation whatsoever to the destroyed regiment, reacted to the pennant as to a holy object. Some of them crossed themselves, others cried, still others kissed the ragged cloth in veneration.

It goes without saying that the nurse thought herself a heroine, and felt she should personally present the banner to the sovereign. Such is the power of a symbol that embodies the strength of a regiment as a military unit, in its historic past and in the present. The banner ceases to be an object, it is not even a simple symbol, it is something holy for which life is sacrificed without second thought, and which is an honor to carry during battle despite the fact that doing so places the standard-bearer in great danger, since he has no ability to protect himself.

A similar deification of objects can be found in religion, for example, with respect to pieces of wood (as among the Australian aborigines), rocks, mountains, plants, creatures (e.g., totemic animals, worship of bulls, crocodiles, and of the ibis in Egypt). No less significant is the regard shown for objects such as coats-of-arms, symbols of the judiciary, medals, amulets, uniforms (symbolizing military honor), icons crosses, statues of the saints,

the Papal slipper, countless relics; and this type of worship takes place in all areas of life (Sorokin, *System of Sociology: Part I*, p. 186).

The extent to which the crowd is attracted to symbolism can be seen in the establishment of new celebrations by the Convention during the great revolution in France, as a result of the appearance at the same time of the new Cult of Reason. It is interesting to note that these celebrations were seemingly a copy of ancient church ceremonies, replete in allegory, proving just how much symbolism is a necessity of the masses.

Here, for example, is the manner in which a typical celebration of the Triumph of Reason was held at the Cathedral of Notre Dame on the 20 Brumaire, year 11, or rather on the 10 December, 1793 (see Cabanis and L. Nass, I.e., pp. 332-333):

A mountain was erected at the centre of the church, covering the choir. A round portico in the Grecian style was located at its peak, bearing the inscription "philosophy" on its façade. On either side of this were busts of philosophy's apostles: Voltaire, Rousseau, Franklin, and Montesquieu. A sacred Fire of Truth burned on the slope of the mountain. To the accompaniment of music, two groups of girls in tricolour sashes, crowned with flowers and holding torches, climbed the mountain and met at the altar, where each paid homage to the holy flames. Then a beautiful woman, clothed in a white garment and red head-dress, came out of the temple. This was the embodiment of freedom, before whom all the republicans bowed to the strains of a laudatory cantata.

All customs pertaining to various social institutions are once again proof of the significance of symbolism to the masses. Even during the period of revolutionary persecution of the church, the state funerals at Petrograd on 7 May did not much differ from religious funerals. Besides, are not celebrations for whatever reason, not to mention respect for rank and various military ceremonies at times of peace, permeated by symbolism from beginning to end?

In a word, the life of the collective is full of symbolism due to the fact that the symbol facilitates the mastering of complex interactions by replacing them with comparatively simple and easily reproducible signs, often having a historical significance.

Finally, the most important actions of individual historic figures become symbols for the following generations. Prometheus' heroic act became the symbol of martyrdom for truth; Jesus' heroism is the symbol of gentleness and endurance; Svyatoslav's bravery is the symbol of courage; while the conquests of Napoleon, Suvorov, and others are symbols of military strategy. The ancients' stoicism, the epicureans' doctrine, the Cynic Diogenes' behavior—all this has turned into contemporary symbolism.

In this way, all significant heroic deeds of individuals, having outlived themselves in the history of nations, become nothing more than symbolic forms to subsequent generations. Even important events take on the status of symbols, e.g., "Sedan," "Tsushima," "Panama," etc.

We also meet with the phenomenon of compensation in national creativity in the form of so-called personification, observable in relation to one abstraction or another. In religion, we meet everywhere with personification of gods, in place of whatever abstraction. The same, obviously, can be observed in many traditional legends.

In Pareto's words:

The highest level of personification exists where a name is given to abstraction, uniformity, or emotion, transforming the latter into objective entities. Then personification progresses, turning into anthropomorphism. Through the acquisition of gender, personification becomes more concrete. Such personifications originate of their own accord, aside from the participation of consciousness.

In this way, symbolic definitions are inevitably sources of personification, and it can be enough just to name a certain abstract concept for it to be personified and anthropomorphized via substitution. Such is the personification in traditional folktales of Miserable Sorrow, Jealousy, etc. In the ancient world, the same process of personification led to the creation of such gods as Fortuna and Victoria. And according to Boudain (*Les cuites paiens dans l'empire romain: Part I*, pp. 415-416), there were many other, less famous gods and goddesses in Rome who all embodied some metaphysical concept, e.g., Aequitas, Bonus eventus, Concordia, Copia, Disciplina, Fama, Felicitas, Fides publica, Gloria, Chronos, Juventus, Libertas, Mens, Pax, Pietas, Prosperitas, Providentia, Salus generis humani, Sanctitas, Vertus.⁵ It goes without saying that the personification of one

abstraction or another achieves the pinnacle of its development when a legend is created that confines the abstraction not only to a certain personality, but also to various actions.

The fetishization and deification of people is widely known. Here, a role is played not only by position, with which an aureole of fame, power, etc. is already historically associated (e.g., the deification of tsars), but also by success in actions (e.g., the deification of an of ficer, ruler, demagogue, etc.). Examples of such deification are numerous, including the cult of Roman emperors, the cult of the holy pope, etc. There is even an example of the Zulus' deification of their king. In a letter addressed to His Majesty, the following can be read:

O Pedzuhn, king of kings, king of the heavens! What are glorious lands in comparison to our great king? What comes of the mighty forests before the great Elephant? His trunk hews the branches of trees. His breath in the face of enemies is as fire falling on dried earth. Enemies are devoured by fire before the person of the king of kings. Being himself the king of fire, he rises to the azure heavens; he sends his lightning to the clouds and forces the rain to fall from them. Mountains, forests, and you, the green fields! Hear the voice of the son of Maggoban, king of the heavens.⁶

We could give many other analogous examples of the deification of oriental kings, but the above is sufficient to prove how personalities achieve, under certain conditions, in the eyes of one collective or another, a position on the same level as that of a god.

Notes

¹ See V. Bekhterev, "The biological development of mimicry," *Vestnik Znaniia* and sep. pub.(in Russian), St. Petersburg.

² There are also experimentally ascertained facts which demonstrate that deaf animals are unable to produce [vocal] sounds. In experiments on dogs performed in our laboratory by Dr. Larionov, animals whose hearing was lost after their temporal lobes had been removed eventually lost the ability to bark.

³ V. Bekhterev, *Objective Psychology* (in Russian), St. Petersburg.

⁴ For details see the following of my works:

"The significance of social selection to the development of organisms" (in Russian), *Vestnik Znaniia*, 1912; "Hormonism and its significance in the

development of organisms" (in Russian), *Priroda*, 1916; and others.

⁵ Pareto, *Trattato di sociologia generala*, Part I.

⁶ M. Kovalevskii, *Sociology* (in Russian), Part II, pp. 194-95.

22

The Law of Dependent Relations

We will not tarry here on the significance of the law of dependent relations in the inorganic and organic worlds, where all things obviously have a certain relation to each other. This relationship is defined in the following manner:

An effect can be traced to its direct cause, which is in turn an effect, and so on. This chain of consecutive causes and effects of indefinite length—a chain in which a given phenomenon forms but one link—is in essence a linear sequence. An infinitely large number of such sequences can coexist, and these sequences may cross, so that one and the same phenomenon, if it is the product of several different phenomena, becomes an effect among a number of sequences of effects; though these may share a common origin, they will in the future remain disparate and completely disconnected.¹

We will only be occupied here with events of the social order, which make up the substantial part of our analysis.

There is not a single social movement that has not been the consequence of preceding events. The fact of the matter is that the development of social life always follows a particular sequence, just as cause generates effect.

The justification of establishing this law in the supraorganic world can be seen in the fact that wherever a social movement is delayed, external or internal reasons for this delay can be determined, as can the reasons for a developed social movement's gradual change in direction.

If the actions of separate personalities are "causal" and so determined by the successively transmitted influence of certain people on others, then it is

clear that all historical events also have a similar "causality" and succession.

In fact, it is easy to be convinced by historical analysis that contemporary events have their origins in past human actions; going successively from present events to past ones, we will quickly reach prehistoric times, and even be forced to trace contemporary events back to the most primeval epoch of human life and still farther into the depths of the centuries.

As is known, a number of authors, from Aristotle to Marx, Engels, and the latest sociologists and economists, have already discussed the question of the correlation of various phenomena in social life and have attempted to clarify the regularity and conditionality of diverse social phenomena.

One of the more productive and exact methods in this respect is the statistical method, thanks to which a number of social correlations have been determined, such as the correlation between the development of crime and social inequality, cost of living, and many other conditions; between the cost of produce and the magnitude of salaries; between the extent of dispossession of land and the growth of the working class; between birth and death rates, etc.

Other scientific methods of determining the regularity of social phenomena include the facts provided by genetic sociology, which trace the development of social life and outline the displacement of some phenomena by others. This course enables us to ascertain a particular regularity in relation to the self-sufficiency of every independent society, to the development of marriage and the family, and to the development of certain aspects of the economic life of organizations of power and legal regulations, as well as the regularities between means of subsistence and population size established by Condorcet and Malthus, between the division of labor and the stratification of society into groups and classes, between the ruling class and the organization of power. Both the statistical method and the method of genetic sociology give the opportunity to anticipate events in social life and establish the dependence of some phenomena on others.

Let us take once again the example of famine, when the behavior of the masses is subject to all the consequences inevitably following this factor. In this case, all the endeavors of the masses are directed towards the procurement of nourishment. We are dealing first of all with the possible economization of produce, with its more uniform distribution, and with the intensification of production of foodstuffs through increased development of agriculture or market-gardening and the invention of food surrogates. Since

collective concentration is determined by the same factor, famine is manifested not only in actions, but also in all conversations; everywhere, the dominant topic of discussion is the question of nutrition, the size of the food rations, etc. This preoccupation is reflected in speech reflexes, with the spoken definitions of famine, rations, and foodstuffs being discussed in all forms. Moreover, the government itself, as the leader of the collective, is subject to the aforementioned factor and interferes in personal life, imposing upon it the interests of the whole collective, for the most part with respect to the question of nourishment. Thanks to this, limiting measures are implemented on the export of produce, and means of acquiring extra foodstuffs are developed; state or social allotment of produce is implemented; communalization of farming is established; incentive measures are taken towards the intensification of agriculture and livestock raising, etc.

If all these measures prove to be insufficient, the natural consequences are the emigration of a fraction of the population and the organization of attacks on neighboring territories in the form of raids, wars, and marches, in order to seize foodstuffs in exchange for other produce and to obtain new colonies. Finally, the equal distribution of property takes place within the country not only through the forceful communalization of foodstuffs, but also through the rise of theft and burglary, and lastly through revolts by the deprived classes who wish to attain power and gain better food products. It goes without saying that all these inescapable consequences of famine can be exposed on the basis of many documented facts and depicted as numbers on the grounds of statistical material. As is known on the basis of statistics, periods of famine are always associated with limited population growth due to a lower birth and higher death rate—that unpleasant proof of social calamity—regardless of whether this heightened death rate is the result of epidemics and greater susceptibility to illness, or of an increase in the number of the murders and suicides that usually accompany the development of starvation.²

If we look at a factor of another nature, for example the sexual dissatisfaction of the collective as a result of, say, an imbalance of the sexes, then we will once again have the total dependence of the behavior of the masses on this factor, namely: intensified discussion of sexual themes among the dissatisfied segments of the population; the limitation, within reason, of sexual encounters; their replacement by surrogates such as onanism, homosexuality, and the movement of the masses in the direction of possible regulation of marital relations, e.g., polygamy in the case of an excess of

women, polyandry with an excess of men; the implementation of various measures controlling internal order, e.g., compulsory marital cohabitation, as was stipulated sometimes among convicts on lengthy prison labor, or those exiled to uninhabited areas (Sakhalin Island); the abduction of women from neighboring areas, even mass assault to this end on neighbors (witness the abduction of Sabine women in Roman history); mass rape of women during attacks on defenseless towns and villages, etc. Intellectual dissatisfaction, in its turn, also has significant social consequences, characterized by a higher demand for books in libraries and in the market; a rise in the amount of original and translated works; a greater number of research organizations (laboratories, institutes, scientific expeditions, etc.); a larger number of people entering learning institutions, etc.

Obviously, periods of great social upheaval, requiring a heightened pace of life, lead to an increased popularity of mental stimulants in the form of narcotics (alcohol, tobacco, opium, morphine, ether, cocaine, etc.)

It is evident that the existence of societies, and likewise the life of separate individuals, is determined by needs that are partly hereditary/organic, or common to all people (attraction to food, to the opposite sex, to society), and partly acquired through habit in the course of life (attraction to mental nourishment, to stimulating substances). An insufficient satisfaction of these needs decides the direction of social aspirations, which, depending on the magnitude of dissatisfaction, vary in their intensity and extent. This situation can be expressed by a particular mathematical formula. If we designate the quantity of material that completely satisfies the given demand as a , and the insufficiency of material for the satisfaction of the given demand as b , then the level of dissatisfaction should be indicated by $a - b$. The consequences of this discontent should also be numerical, and so can be similarly represented by variables. Now we can express the changes in collective reflexes under the influence of certain forces, in the form of dissatisfaction, by the corresponding mathematical formula: $a - b = \pm x \pm y \pm z \dots$. Since lack of satisfaction, being an irritant, can be signified by a particular number through the accumulation of certain facts and through the given calculation, and since numerical material on the consequences of this irritant can be obtained from statistics, we can define more precisely, with the help of mathematical computations, the interaction between the irritant and its consequences in form of collective reflexes.

Even Spencer acknowledged the regularity of phenomena in social life, comparing, as did Comte, human society to an organism, explaining why it is subject to the laws of integration and differentiation, the laws of rhythm and equilibrium, the law of struggle for survival and the law of cooperation and division of social labor. The process of evolution of any society is contained in the manifestation of these laws; as a matter of fact, Spencer studied social phenomena in terms of their development, in the manner of Aristotle. After Spencer came many other authors who supported the evolution theory, including Espinas, Letourneau, Worms, Charlet, Lilienfeld, et al.

It should be noted that Aristotle, the true founder of sociology, in his discussion of various political forms in *Politics*, elucidates the regularity of their exchange under the influence of class conflict and of the interaction between social groups. Aristotle's doctrine on society and government, as self-sufficient communication between people, is itself a very important law, on whose basis the existence of any society or state is founded. Later, the concept of the regularity of social phenomena was supported by Machiavelli and a number of more recent authors, the most important of whom are Vico, Montesquieu, Adam Smith, Turgot, Condorcet, Malthus, Saint-Simon, Comte, Quetlet, Spencer, Marx, Engels, et al.

Upon investigating the development of human societies, we meet, generally speaking, with an amazing regularity in the progression of social forms. "This regular recurrence found in the most varied forms of social life, observable in both the evolutionary and revolutionary social development of all nations, gives sociology the opportunity to discuss the regularity of the development of social life" (Takhtarev, *Sociology, its Concise History*, p. 40).

At the present time, there is no need to prove the existence of a successive tie between historical events. We must keep in mind, however, that behind historical events there are always national movements, or manifestations of the collective reflexes of the masses. In other words, historical succession presupposes the succession of national movements. But the exchange of social movements can even be seen independently from the succession of historical events in the fact that one movement is always connected to a number of preceding ones, which, in turn, are related to other, even earlier ones, etc.

Let us take the progression of social moods during the war with Germany. When the war began, a patriotic uplift occurred, uniting for a time all parties and classes in a country [Russia] that for a long period had been divided by

party conflict. Gradually, this primary impulse was replaced by a calmer, nationalist mood, which, not finding the necessary satisfaction due to a lack of victories, vented itself in a revolution that had long been anticipated in Russia, but which had been quelled through political repression. This is why the nationalist mood of the country, not having found a corresponding outlet, was released upon the disbandment of the State Duma (in which the people saw further diminution of their rights) in the form of harsh opposition to the government. Opposition cleared the way for revolution, which, having begun under the flag of bourgeois patriotism, led to the liberation of the proletariat; as a result, the class struggle of the latter was greatly facilitated, ending in domination by the proletariat and in the rise of socialist aspirations, leading to the rebirth of the International.

In this way, everything is in one common chain, all is logically connected, one social movement is the inevitable consequence of another, and is itself the cause of following movements. There is not one social movement that is not directly related to preceding ones as their effect, or to following ones as their cause, or that would stand on its own, disconnected from any other phenomena.

Just as there is logic in a chain of oral reflexes, there is logic in the sequence of actions. He who has said *a* should be prepared to say *b* as well. That same logic is manifested with immutable regularity in relation to collective aspirations and movements. In this respect, the logic of historical events is subject to the same regularity and cannot differ from the logic of language or of individual actions. Here, effect stems from cause in the same measure as a logical deduction springs from preceding premises.

All this is clarified if we recall that new social movements are, in essence, a development of previous social movements.

There is not one social movement that does not have its roots in the past, or roots that are not the inevitable consequence of even earlier social movements. This is the essence of the law of dependent relations.

Let us take as an example a large-scale social movement, for instance the Great War.

Why did such a terrifying event develop, and what brought the nations of the world into conflict?

The external aspect of this colossal social event is not complicated and can be summarized in the following way: in the summer of 1914, an Austrian archduke was assassinated at Sarajevo.

Austria, which had long extended the tendrils of its political desire over Serbia, seeing that the roots of this crime appeared to be embedded in Serbian propaganda, pronounced the famous ultimatum to the State of Serbia, which was almost equivalent to a removal of its independence.

Serbia, in its turn, applied to Russia for help through tribal and political ties.

Guided by the response received from Russia, Serbia agreed to most Austrian conditions, with the exception of those points that were incompatible with the understanding of an independent Serbia. The presentation of the ultimatum was, however, the work not only of Austria, but also of Germany. Notwithstanding the diplomatic attempts of Russia, England, and America to find a route to reconciliation, Berlin provoked Russia by publishing a decree on mobilization in the newspapers. The Russian ambassador immediately telegraphed this fact to Petersburg.

The order turned out to be apocryphal, and was consequently officially recalled. News of the correction was likewise telegraphed to Petersburg. However, the transfer of this notice was delayed by twelve hours.

The "publicized" German order to mobilize was followed by Russian mobilization directed against Austria, which had meanwhile declared war on Serbia and had mobilized its own troops to that end.

Then came the German ultimatum to Russia on demobilization, which, not having any effect, was succeeded by German declaration of war on Russia.

The Franco-German war was declared soon after, on account of the Russo-French treaty; then England joined France and Russia as a result of the violation of the Belgian neutrality. Later, Japan, Italy, Portugal and America, in that order, united forces with the allies; Turkey and Bulgaria took the part of Austria and Germany. But all this is only the external and consequently immaterial aspect of the conflict. Far more significant is the fact that German culture had for a long time been at an extraordinary level of development, as a result of a number of conditions that there is no need to discuss here. Then, after the military conquests of the 1860s, which led to the rise of Prussia in the German alliance, and especially after the war of 1870-71, Germany's internal politics were directed towards appeasing the interests of the military caste, or the so-called Prussian cadets, and of the German bourgeoisie. This led to an augmentation of Germany's imperialist interests, to economic and political rivalry with England, and to the sustainment of armed peace in Europe.

In the end, the tension endured by European countries for half a century could not but erupt into a terrible war, for which Germany had systematically prepared itself in the course of no less than forty-five years.

In this way, the Great War was the predictable result of the imperialistic militarism that arose in Europe after the war of 1870; hence the murder of an archduke in Serbia was enough to cause the spontaneous outbreak of a world war. It was impossible, in view of the natural logic of events, to localize the military conflagration through any diplomatic mediation, and states were drawn into the war one after another.

This, of course, is not all.

From the time of the Franco-Prussian war, the psychology of the German people had completely changed. The country, made great through victory, carried this triumphant mood over to its ideology, which released itself in an uncontrollable stream of aspiration to world control in politics, industry, science, etc. Colonies were rapidly seized, world markets were secured for German industry, and at the same time a number of institutions were established in Germany that answered to its dreams of world politics and its aspiration to extend its trade influence over the whole world, which could not but instigate firm and systematic opposition from England. Austria, Germany's neighbor and ally, was the first to come under the scope of German politics; Russia was to have been the next, but she defended herself in good time through an alliance with France. Then German influence spread via Austria to the Balkan peninsula, after which Kaiser Wilhelm, representing the Prussian military acquired economic and political control of Turkey, both its European and Asian territories from Asia Minor to the Persian Gulf. Finally, thanks to the Saunders mission, Germany was able to implement its own agenda at Constantinople.

This very situation contained the origin of the Great European war, for it compromised the political interests of Russia, Italy, England and France. Moreover, the humiliation of France by Germany in 1870-71 and the increasing political rivalry between Germany and England were substantial factors leading towards the confrontation between nations.

It is necessary to keep in mind that so-called interests are once again the fruit of national aspirations, prepared for decades and even centuries, and having deep roots in preceding national movements of a political, economic, and religious nature.

It should also be noted that almost the entire nineteenth century, especially its latter half, was spent in the conflict between nationalism and socialism or internationalism, and there was probably no country where nationalism was so firmly established as in Germany.

Nationalism is collective egotism. Aside from certain tribal peculiarities, nationalism is sustained by habitual lifestyle and original development of intellectual and spiritual culture.

Patriotism, a display of state egotism, arises on the basis of nationalism.

The concepts of internationalism and cosmopolitanism are in opposition to the latter, as manifestations of international and interstate altruism.

Economic dependence of one nation on another, international intellectual exchange, world trade, international industry and other ties between civilized peoples contribute to the development of internationalism and cosmopolitanism. In this respect, a large role is also played by the proletariat, which, being bereft of property and living only on salary, is not tied to a certain area and changes its living quarters depending on the location of its work. Moreover, the working class' struggle with the bourgeois and the wealthy unites the proletariat of all countries under a common objective.

Here is what Laufenberg says on the subject in his work *The Great War*, p. 6:

No matter how often France, Russia and England call this conflict with Germany a defensive war, a war that nobody wanted, no matter what purpose is assigned to it, be it the defense of threatened national domain or independence, no matter how frequently it is asserted that its goal was the well-being of nations and the guarantee of future peace, the fact that all states contrived this war with every resource at their disposal, and that capitalist development led to world conflict in incontrovertible sequence, is indisputable. World war was the inevitable consequence of a capitalist system of global economy, and involved the revolt of productive forces against the predatory capitalist and nationalist forms that had governed up to that point. Now, the near future belongs to a new economic system, one that was born of terrible suffering and upheaval.

A struggle between nationalism and internationalism was going on simultaneously with the conflict outlined above, and although that struggle is not yet concluded, it is certain that, in the end, internationalism will triumph.

Meanwhile, those states that have developed extensively but have not yet assimilated their citizens or united them culturally and economically in the manner of Russia, Austria, and Germany are destined for a reformation, in the sense of an isolation of the nations that form them.

From the aforesaid it is clear that such an immense national movement as the Great War has its roots in the distant past and even in early historic times; if we analyze these roots, then once again we will find they are related to preceding national and social movements, and so on to infinity. In a word, we are dealing with the immutable phenomenon of dependent relations in the form of historical succession, without which not one social movement could be understood, regardless of its magnitude.

But the logic of historical events extends still farther.

As a result of the colossal expenditure and sacrifice that must accompany success in contemporary warfare, it so happened that nations could not sustain the tension connected to victories, and consequently the momentous war that had induced a remarkable effort on the part of national forces began to stimulate greater patriotism in some segments of the population and dissatisfaction in others, especially in the masses, as well as the development of internationalist aspirations. The longer the war lasted, the more repression, unstable in its political forms, developed within states, and finally the masses lost patience. An underground reaction first developed in the country subject to the greatest tyranny and administrative repression, in which the masses were least accustomed to self-restraint and discipline, and which was poorer and generally not as well equipped in comparison to other countries: Russia. Moreover, its administrative mechanism had displayed obvious inadequacy prior to the war and, unable to sustain the pressure of social aspiration to democracy, was crushed with the initiation of the February Revolution. The latter soon adopted the slogan of protest against military repression and against the war itself, as a result of which the army began to deteriorate; hence Order N° 1 was simply an expression of this irrepressible social protest against military control.

In other countries, thanks to differently structured government, better systems of conducting war, and other related conditions, mass protests arose later. But they did commence, both in Italy and elsewhere. Of course, in Italy the situation was temporarily rectified thanks to active support of the allies and to the development of patriotism in the country under the influence of defeats and of a deep-seated rivalry with Austria. Even in America, a strike

movement began early in 1918. Central European countries were last to be affected by unrest, for they were momentarily the conquerors, and the noise of victory covers the murmur of the people; nevertheless, mass protests in the form of general strikes took place in 1918 first in Austro-Hungary, and then in Germany. Obviously, in this case, notwithstanding military success and acquisition of territory, a general reaction arose in the form of struggle for immediate peace, as a consequence of both famine and fatigue due to war which resulted in a catastrophe.

If we specifically analyze the development of the Russian Revolution, we also meet with complete historical regularity.

In order to understand the social process that created the Russian Revolution, it is necessary to take into account the continual drain on a country of drawn-out, unsuccessful battle, the discovery of all the sore points of past social life, the lack of technical education, the almost total absence of industry, the illiteracy of the masses, the incompetence and decay of the administration, and the utter depletion of the state's economic life.

Naturally, under such circumstances the people protested against the old form of government, being generally dissatisfied with the old order.

Moreover, during the national conflict all classes and parties—except the socio-democrats and internationalists, who were in the minority—desired and aspired towards one goal: the defense of the country from the external enemy. Yet confidence in the fact that all of the country's resources were employed for this defense, and that the latter was led by honest administrators, wavered more and more under the influence of both obvious factors and rumors that frequently grew to enormous dimensions, implicating even Defense Minister Sukhomlinov in treasonous acts. This is what led to the overthrow of the old government, as represented by a weak-willed monarch surrounded by similarly wretched minions and political charlatans. The possibility and hope of conquering the external enemy, of terminating economic ruin, and of revitalizing the country was originally embodied in the State Duma because it had, over the past ten years, led an open battle against the decaying tsarist autocracy, and so had acquired, in the eyes of many, or at least of the so-called bourgeois, the status of an organizing factor of the masses.

Of course, the lower classes were silent on the matter; but political propaganda could be found in the army, in rural areas, and at the factory, and although the dissatisfaction may have stemmed from different causes, all agreed on one point—life could not continue in the same manner.

For those classes that included the bourgeoisie and part of the intelligentsia, the deposition of the monarchy was a reaction to the inability of the old order to organize the defense of the country, and consequently signified Russian rejuvenation and the beginnings of citizens' rights and freedoms. For the proletariat, which was subject to social trends, the demise of the old order meant release from centuries-long oppression, and so the strengthening of the revolution was more valuable to it than any nationalist aspirations, which in any case were not easily understood by the uneducated masses.

The overthrow of the monarchy occurred with incredible ease, which is explained by the fact that it had almost no supporters; it had lost all defenders in the army, which was composed primarily of fresh troops that had been infected by the same socialist propaganda as had the entire urban proletariat. However, during the first days of the revolution, upon the formation of the Provisional Government, an argument arose between representatives of the bourgeoisie and of democracy—an argument that nearly brought about a total rift between the groups.

This shows that, at least with respect to the Provisional Government, rot had begun to eat away at the issue of social unity.

Consequently, soon after the onset of the February Revolution, the masses were split into two different camps—supporters of the bourgeoisie on the one hand, and supporters of democracy on the other—which were not reconciled at the Moscow State Assembly.

It should be kept in mind that the people, having remained in slavery and illiteracy for centuries and being bereft of independence in social life, as well as never having known politics, could not be expected to resolve important state questions; on the other hand, a revolutionary intelligentsia had arisen over the long period of tsarist repression, and led the struggle against the patriotic and nationalist tendencies of the bourgeoisie. Thus, any further so-called "deepening" of the revolution was really the result of conflict between one force and another. It goes without saying that the turbulent sea of impoverished, exhausted, and underdeveloped masses, whose patriotism was eradicated by a complete exclusion from matters of state and social life, followed the demagogues of an internationalist inclination, which meant that a social movement directed against the war, the bourgeoisie, and capitalism in general arose.

This is the explanation for the denationalization of the Russian Revolution, in which all socialist parties took part, and which evolved so quickly that neither the first, primarily bourgeois government, nor the following coalitional one could improve the military situation or increase material well-being with respect to land and grain, while the imperialist aims of the war naturally stimulated the people to oppose its continuation.

The famous Order N° 1; the activity of the Soviet of Worker and Soldier Deputies that had the government wrapped around its little finger; Zimmerwald's triumphant march into Russia; and above all the struggle with the bourgeois classes' imperialistic tendencies—these are the main stages of the first period of the Russian Revolution, which led it to Bolshevism and transformed it from a national uprising to a party revolution of the proletarian classes.

Alongside this process of denationalization was the gradual process of state decay and disintegration, the reasons for which lay partially in the instability of the former tsarist order. An especially large role was played in this respect by Bolshevism, which had, as the purveyor of proletarian internationalist tendencies, converted the revolution into a struggle of the working and military masses against all other classes of the population.

This Bolshevik Revolution, initiated by the coup of 25 October 1917, led the country to proletarian dictatorship and civil war. The isolation of Russia from its allies, which were branded representatives of bourgeois imperialism, the separate armistice with Germany, and subsequent separate peace talks were natural consequences of the coup.

With time, however, the masses were disillusioned even by Bolshevism, partly as a result of its failure to achieve the promised peace, but especially in view of the insufficiency of produce, of the widespread industrial and economic ruin, and the impending famine.

Obviously, at the outset of the revolution there was no one in the huge organism of the old Russian government system who could have effectuated constructive labor on the necessary scale, primarily due to the fact that the former government, as was mentioned, stifled any manifestation of sociality, making society a passive herd in which the beginnings of statehood and healthy nationalism were simultaneously extinguished.

The Russian Revolution is commonly compared to the Great French Revolution, and there are, in truth, a number of similarities between the two. There is also, however, a significant difference that has apparently been

overlooked by those who closely analyzed the French and Russian revolutionary movements, namely: the former originated in peaceful times and so could manifest colossal energy in the military respect, while the latter began in a period of taxing and prolonged warfare, which had sapped the strength of the nation, and consequently eliminated any call to arms and led, in the end, to severe protest against the war. All parties, as well as their representatives, that had supported the concept of conducting war until victory or death were immediately cast away, and even those that had spoken of continuing war, if only with democratic considerations, suffered a fiasco, since at the beginning of the revolution the pull of the land was strongest, and therefore the attraction of peace at all costs was also greatest.

Bolshevism, which had been the first to announce "peace at all costs" as its objective, naturally attracted the people's interest. At the same time, our army began disintegrating with astounding rapidity. Factors accounting for this, aside from Order N° 1, included the lack of patriotism of the Russian people.

It could even be said that patriotism was systematically eradicated in the nation. The slogan of the tsarist regime was "defense of the tsar, the faith, and the fatherland," from which it is clear that the fatherland was always the last consideration. Moreover, our schools were distinguished by an absence of courses pertaining to knowledge of the fatherland; self-government was withheld from the people, and political judgment was forbidden, contributing to the development of passivity in the general public towards the fate of the country. The majority was illiterate, crushed by poverty, and deprived of a unifying exchange of ideas because of the persecution of any discussion on political grounds.

As for the intellectuals, it should be noted that, in view of their opposition to the ruling power, the word "patriot" was to them almost equivalent to blasphemy, due to the fact that patriotic slogans were usurped by the tsar's toadies, lackeys, and their vile minions, who called themselves true patriots and true Russians. The very understanding of patriotism was embodied in the support of an absolute monarchy, which was despicable to most of the intelligentsia. As well, the vastness of the land did not suggest any immediate danger from the enemy, and so soothed the populace by providing apparent security.

In view of these factors, it is not surprising that the army did not experience particular fervor. At the beginning of the war, a dislike of the Germans was incited by media baiting, which stimulated a patriotic uplift in

the country for a time; however, inept management of the war, general chaos, lack of arms and ammunition, unjustifiable rapaciousness and unceremonious theft, and the incompetence of or betrayal by such leaders as former Defense Minister Sukhomlinov, generated an even greater decrease in the patriotic mood.

The revolution developed as a protest against the given situation, and it appeared as though the country's defense would consequently be improved. However, since the revolution arose in an area debilitated by war, it could only lead to fraternization at the front; to the slogan "Peace at all costs," without annexation on the grounds of the self-determining ability of nations; to the homily that the Germans were in fact friends, while the enemy was the bourgeoisie and "counterrevolutionary" officers, as well as the capitalists of the world, including our allies—England, France, and America. At the same time, the slogans of international law, equality, and justice were inaccessible to the understanding of the Russian people. Besides, the pull of the land was heightened under the influence of rumors of the division of property.

"What do I need with land and freedom, if I am to be killed?" was the characteristic question put to Minister Kerensky when the latter toured the front lines, pronouncing his patriotic speeches. In this query can be seen the crude, obtuse egotism that accompanies utter disinterest in the fate of one's country.

In social life, all is linked together, thanks to which every element is part of an indissoluble fabric; it is enough to yank out one thread, without having bridged the resultant fissure or rearranged the elements dependent on the lost thread, for the structured harmony of the collective to falter, leading to complete disorder. This is what causes social ruin, which is always more difficult to remedy than to prevent.

G. Le Bon, while discussing the mental preconditions to the Great French Revolution, notes (p. 134):

Complaisant humanitarianism, beginning with pastoral idylls and philosophical reasoning and ending with the guillotine, is an excellent representative of those apparently innocent ideas that quickly led the ruling classes to weakness and disorganization. They no longer believed in their own purpose and were even, as Michelet so justly observed, the enemies of their cause.

When, on the night of 4 August 1789, the nobility renounced its centuries-old rights and privileges, the revolution was already prepared. The people had but to follow the directions they were given, and, as usual, the latter were taken to extremes.

It goes without saying that the reasons for the Great French Revolution lie significantly deeper than G. Le Bon supposes, but he is right in saying that a step, once taken, inevitably has further consequences that, as the given example proves, can take on the magnitude of an event of great importance, such as a revolution.

There is a view that society's evolution, under normal conditions, is quite slow. On social spheres, G. Le Bon writes (p. 58):

Excepting those cases in which civilizations were destroyed by invaders, they have always changed very gradually. Many institutions have collapsed, many gods have fallen from their pedestals, but only after long periods of senility. Great states have disintegrated, but only following a period of gradual decline that neither society nor individual people could prevent.

It should be kept in mind, however, that the development of social life does not often proceed unimpeded. For this reason, when the evolution of state life is hindered by artificial barriers, a revolution unfolds, which causes far more drastic changes than those that would have been achieved through peaceful evolution. The history of the French and Russian revolutions provides edifying examples in this respect. The same holds true in relation to uprisings and wars.

The correlation discussed above is not limited to the behavior of certain social groups towards others, but exists between social groups and the surrounding conditions of nature, which even Bockle acknowledged. It can be said with certainty that not only products of national creativity, such as language, custom, and myth, but also any literature, art, and even social and state institutions reflect the influence of surrounding nature. Moreover, life itself and even the fate of nations correlates not only with other social collectives but also with the natural environment, which there is no need to prove. Finally, even the personality of a given people, its "spiritual" physiognomy, and partially its physical appearance, is in direct correlation

with nature. Inhabitants of the north and south, of the continental and coastal areas differ in many respects, as do natives of mountainous and flat regions.

But this by itself is insufficient proof of the influence of nature. There are facts that leave no doubt about the existence of a correlation between the behavior of large collectives and the condition of the sun, and since the condition of the sun is, in its turn, related to the condition of similar stars, it is clear that the fate of humanity is bound to that of the entire universe.

Here, we will limit ourselves to facts on the correlation of major collective movements in the human world with meteorological phenomena, the causes of which are related to the changes sustained by our Sun when it is covered by a greater or smaller number of sun-spots.

It has long been known that the periodicity of the maximum number of sunspots appearing on the sun, measured in eleven-year intervals, corresponds to the maximum magnetic storms and aurorae occurring on the earth, also every eleven years. The coincidence in this case is so complete that the section of the sun occupied by spots can be determined from the position of a magnetic needle. Aside from this, sunspots affect the average temperature on the earth's surface, precipitation, and consequently, according to the abbot Moret, are reflected on world grain and grape crops, on the flowering of lilacs and return of swallows to France.

The same periodicity, in relation to the pulsation of the sun, controls the number of tropical cyclones and polar icebergs. Obviously, all this is a product of the heat and electrical energy of the sun. If a connection can be made between the periodicity of sunspots and harvests on Earth, then the former should also be related to the origination of economic and financial crises (Prof. Jung). As a matter of fact, Moret is presently establishing the dependence of social and political phenomena on the periodicity of sunspots.

In the economic life of nations, we have the same constancy of known correlations. Trade and currency depend on one event or another in a country that affects the harvest. War in one part of a continent is reflected on world trade, etc. On the other hand, the condition of the world market can depend on an agreement between two nations, or even between two large trade firms. "World trade is such a delicate organization, that at the right moment it is enough to place a feather on the balance of the scales in order to change the flow of trade and credit and induce an unprecedented boom or crash."

It goes without saying that the status of world trade is in direct correlation with meteorological conditions, since the latter affect harvest; and since

meteorological conditions are, in turn, determined by the sun, the dependence of trade on earth must be carried over to the depths of the cosmos.

Not only trade, but also various human interactions should be placed in correlation with the condition of the sun. In this respect, the facts given by the French author Moret in his article "The Sun and Ourselves" are of particular interest.

In the words of Ya. Perelman:

The material collected in D. O. Sviatsky's article deals with this question on a broader level. He compares the chronological dates of large-scale revolutions with the years of maximum solar activity. The resultant picture of their relationship is astounding. Indeed, it cannot be passed off as simple coincidence that the years of especially high sunspot generational activity—1830, 1848, 1860, 1870, 1905 and 1917—registered on Earth not only in the form of magnetic storms, but also in the form of widespread social upheaval: the July revolution (1830); the February revolution (1848); the revolution in Italy (1860); the Paris Commune (1870); the first and second Russian revolutions (1905 and 1917).

If all this is not accidental, if the pulse of humanity really throbs in unison with the beating of the cosmic heart of our planetary system, then we can attempt to create on this basis a kind of political horoscope for the coming years.

To the question of whether the political storm currently raging in our country will spread still further, as the supporters of permanent revolution assert, or whether it will fall into decline, the astronomer can answer: "The latter is more likely than the former. The pinnacle of solar activity has passed, the thermal and electrical heart of our planetary system should soon settle down."

Our body is incomparably more sensitive than was suspected by ancient medics: we cannot see electricity, yet we can feel it when a storm is approaching.

Examples? I have collected hundreds of them throughout the period that I have been making observations. A significant part of my life has been spent in teaching, and, consequently, I have been in continual contact with

students, both young and adult. And I have found, as a general rule, that the number of punishments increases on days of strong electromagnetic disturbances. This means that sunspots stimulate not only the magnetic iron strip of a compass, but also the complex organism of a child, sensitive to external forces; unable to withstand sudden influence, the youth surrenders to various excesses in moments of general excitement.

But this is only one of many examples. All of us, young and old, must admit that the electrical condition of the atmosphere, connected to the Sun, influences our character and disposition, our mood and, in all probability, accord within our families and our parliamentary sittings, as well as the making of laws, the aggravation of relations between states, and even the declaration of wars.

What is true for separate individuals is likewise applicable to the masses, where individual will weakens and behaviour is subject to the law of large numbers.

If this is so, periods of lesser solar activity should coincide with periods of peace and tranquility within nations. Glance at the curve of the Sun's activity, and you will be convinced of the extent to which astronomy has the potential to point out dangerous shoals and deeps to the helmsman of the state.

It has long been noted that world exhibitions take place in times of decreased solar activity. Then as solar activity augments, humanity seems to succumb to the grip of a fever — conflicts are born, wars arise. It is as though a gust of insanity blows through peoples' minds.

Here is chronological reference for the preceding lines. The years of minimum solar activity—1867, 1878, 1889, 1900, and 1910— correspond to the years of world exhibitions in Paris (the first four) and in Ghent (1910). By contrast, the years of maximum solar activity coincide with periods of political turmoil and military conflict in France: 1870—Franco-Prussian war; 1881-83—Tunis, Annam, Tonkin; 1894-96—Madagascar; 1905—Morocco; 1914-17—First World War.

It is still uncertain whether we are dealing with the direct or indirect influence of the Sun's activity. D. Sviatsky notes, however, that if solar

corpuscles, which according to Arrhenius bombard our atmosphere more strongly during periods of increased solar activity, creating magnetic storms and lighting the polar aurorae, affect our meteorology and-even our grain prices, as was shown by Hershen, then should not humanity's collective technology also experience the powerful influence of our sun, thanks to which "we live, move, and exist"?³

We have quoted this interesting section in order to awaken in the reader's mind the thought that dependent relations in a social environment are not limited to the natural environment on our earth, but are significantly more extensive, reaching into the depths of the universe and tapping into its infinite amount of energy.

Notes

¹ Coumot, *Essai sur lesfondements de nos connaissances*, Paris, 1851, p. 49 and ff.

² Addendum upon reading the manuscript proofs. Recently, P. Sorokin made a presentation at the scientific conference hosted by the Institute on the Study of the Brain and Psychic Activity, entitled "The influence of hunger on people's behavior and social processes," in which the author approached the question in a manner similar to my own, but developed his thoughts in significantly greater detail. He defined the role of hunger in human behavior as "nutritaxis," which in my opinion should be replaced by "attraction to food [nutrients]." Having characterized the physiological and correlative manifestations of "nutritaxis" among separate individuals, the author went on to outline the consequences of mass starvation. The most important of these were (1) the change in the average import and export of foodstuffs into and out of the affected region; (2) the invention and alteration of means of obtaining food; (3) emigration (in peacetime and wartime); (4) aggravation of social struggle in the form of crime; (5) hungerriots, disorders and revolutions; (6) the introduction of centralization and "communalization" in the socio-political structure of society; (7) an increased death rate; (8) a decreased birth rate; (9) a change in the society's "morality, customs, and ideologies." The number of and extent to which these conditions develop in a society affected by famine depends on several other conditions (for example, whether the lacking products are supplemented by import). The general conclusion: "At the heart of many of the most significant social phenomena that, at first glance, seem to have nothing to do with nutrition, lies the independent variable in question—in other words, the amount and quality of chemical elements that enter the organisms of the members of a given society:'

³ *Bulletin of the Russian society for the promotion of natural and social science* (in Russian), Book VI, 6,1917, p. 310 and ff.

23

The Law of Individuality

The law of individuality must be placed among the most important conditions of social life, for it is in essence a universal law. Every atom, every molecule is a kind of individual in the form of a closed system of a certain type, while chemical bonding creates a colony of individuals. In this way, the inorganic world is itself development according to an individual type.

In the organic world, which includes both plant and animal kingdoms, the individual is at the foundation of the evolutionary process and of any development in general. In this way, the organic and inorganic worlds evolve in accordance with the law of individuality, in the sense that every new transformation does not destroy former individuality as a more or less closed system, but only changes its aspect.

In a word, in the constant variability of forms there is something that, rather than being destroyed, is preserved in the form of basic attributes that characterize a certain individual. For example, no matter how much a person changes throughout the aging process, he retains a definite physiognomy with its characteristic features. The same can be seen in the supraorganic world. Any profession, any party, and any society in general has its own characteristic peculiarities, which are preserved in the face of all possible alterations; similarly, no matter how much one nation or another changes, it remains a distinct collective individual. Their way of life, their form of state government could change, but the Russian people will remain Russian, with their own peculiarities; French, Italian, and other nations will continue being French, Italian, etc. It is clear, in this way, that the supraorganic and social

world is subject to a certain regularity, developing in agreement with the principle of individuality.

This thesis is also important for other reasons, since the conditions of the development of phenomena in closed systems are applicable to any individual. It is in these closed systems that, due to the interrelation of various processes, the principle of equilibrium predominates. When the process of combustion develops in a physically closed system, it is terminated of its own accord, thanks to the accumulation of carbon dioxide and a lack of oxygen. A fever developing in an organism is regulated by the production of antitoxins in the blood, on condition, of course, that the heart can withstand the higher body temperature. A parallel phenomenon can be observed among supraorganic individuals or collectives. After discontent arises for whatever reason in a social environment, it gradually subsides thanks to the opposition of various elements. There is no need to discuss this important fact in detail. It is self-explanatory.

The cornerstone of any society is the family; a number of families forms a class; several classes form a community; several communities form a tribe with one or another social structure; then come alliances of tribes, etc. Social groups in the form of clubs are created on the basis of political views; then some parties may make alliances; the latter can be united into a bloc, etc. All these are social individualities of a kind, with their own characteristic traits.

In the area of economics we have artels, cooperatives, syndicates, trusts, etc. In administrative spheres we likewise have special organizations that gradually broaden the scope of their activities. The same can be observed in the military, where troops are divided into smaller organized units. Any society, in its turn, is divisible into classes, guilds, professional unions, clubs, etc. In a word, any social environment that is undergoing the process of organization separates certain parts from itself, as a consequence of one demand or another. These parts have their own relatively isolated existence, pursuing their corporate interests; each segment obeys its own code of conduct, consequently creating its own views, habits, manners, and sometimes fashion of dress or special badge, etc. All this leads to the fact that each part of society acquires its own individual traits and form of organization; it can and must be acknowledged as being social or collective individual.

The same can be said of larger collectives, such as races.

"One of the more vivid impressions," says G. Le Bon, (*The Psychology of Nations and Masses*, p. 5),

that I have gleaned from extensive travel through various countries is that every nation possesses a spiritual structure just as invariable as its physical peculiarities, and which generates the nation's feelings, thoughts, institutions, beliefs, and arts.

The commonality of feelings, ideas, beliefs and interests, created through gradual inherited accumulation, gives the psyche of a nation great unity and stability, and at the same time guarantees immense strength, (p. 15)

Qualities of character are the inalterable basic elements that allow us to define the psychic composition of higher nations. Discoveries made by the human mind are the common property of mankind; strengths or weaknesses of character are the sole inheritance of a given nation, (p. 29)

Moral and intellectual peculiarities, whose aggregate expresses of national soul, are essentially the synthesis of a given nation's past, of the legacy of its ancestors and of the motives of its actions... This aggregate of common psychological features comprises that which is rightly called national character, (p. 10)

In the end, every Englishman, Frenchman, and German has common traits, characteristic of his nation.

The question of to what extent institutions hold educational significance for the nation is not without importance. Of course, this educational significance of institutions cannot be denied, but it is also beyond doubt that they are more often a by-product of national culture.

The influence of race can easily be demonstrated by the following example: a theatrical work that receives widespread acclaim in one country frequently has little success in another.

When a foreign institution, religion, language or art is adopted by a nation, it acquires new traits that are related to the latter's character. Much interesting information can be found pertaining to this fact in two chapters of G. Le Bon's *Psychology of Nations*, pp. 77-107. We will cite here only the concluding lines of the last of those chapters:

No matter how great religious influence is, and it is especially so in the East, the influence of race is greater still. Racial soul, guiding the fate of nations, also guides their beliefs, institutions, and art; that soul can be found in whatever characteristic of a peoples we choose to study. It is an active force that cannot be dominated by any other. It represents thousands of generations, and is the synthesis of their thought.

If we supplement what has been said with information on the type and development of industry present in various nations, on their diverse social traditions, on customs, on family structure and on peculiarities pertaining to social life, we will have all the features that characterize the individuality of every nation.

National individuality vastly facilitates our understanding of national history:

History can be analyzed as a simple account of results achieved by the psychological character of races. It develops out of that character just as fishes' gills developed as a consequence of an exclusively underwater existence. Without a preliminary knowledge of a nation's spiritual constitution, its history appears to be a chaos of events governed only by chance. When the soul of a nation is bared, however, its life is seen to be a proper and inevitable consequence of its psychological traits. In all the manifestations of a nation's life, we find that the immutable spirit of the race weaves its own fate. The power of racial soul is most obviously manifested in political institutions. (I.e., p. 48)

It is acknowledged, not without foundation, that every nation has the form of government most natural to it.

Keeping in mind that the existence of nations is manifested in the creation of sciences, beliefs, arts, material culture and in collective actions, it cannot be denied that not one of these aspects, taken separately, gives a full understanding of character; only all of them together can represent the personality of a given people, since one nation may excel in an area more than another, or may have a different set of values from another, etc., and that is what lies at the foundation of the ability to distinguish between nations.

As was already stated, any social environment divides into a number of parties, corporations, alliances, clubs, and other organizations that, created

under the influence of the demands and interests of their members, stratify the social environment, becoming separate social cells or individuals.

Any independent collective begins life with self-determination, in other words it establishes and clarifies its own purpose and goals, separating them from the objectives of other collectives. Only after that is interaction between collectives set up. In disorganized collectives, such as street crowds, this gradual self-determination obviously does not take place, but it nevertheless is sometimes realized seemingly of its own accord.

Every social cell leads its own life, existing until it satisfies the interests of its members to a certain extent, or in any case until it achieves its initial purpose. It is known that Aristotle in his *Politics* established the principle of autarchy or self-sufficiency, in accordance with which the state, or any social organization, be it a party, alliance, club, or other community, exists only as long as it remains self-sufficient, i.e., satisfies the interests of the majority of its members.

In this case, we are essentially dealing with individual self-sufficiency, which lies at the basis of the existence of any society, any collective personality, for only with the achievement of individual independence does it acquire its own *raison d'être*.

If, for example, a party loses its *raison d'être*, it is no longer self-sufficient and consequently either falls apart, allowing its members to join other parties, or undergoes corresponding transformations, supplementing and altering its program in order to become self-sufficient once again.

An economic organization also subsists only while self-sufficient in the sense that it serves the interests of its clients. As soon as it sacrifices its autarchy, however, it loses its clientele and ceases to exist. A labor organization likewise exists only to the point at which it answers to the interests of its members, and is extinguished as soon as the interests of the majority are not satisfied. Any financial organization is subject to the same principle. While it is in a position to withstand the competition of other organizations, i.e., is self-sufficient in the aforementioned sense, it continues its existence, which is terminated only when the organization cannot sustain competition or insufficiently serves the interests of its members.

A government organization, founded on the principle of protecting the life and possessions of its fellow citizens and possibly of the better satisfaction of their needs, ceases to be self-sufficient in the eyes of other states when its power becomes inadequate to execute the above functions. In order to retain

its position, the organization is forced to make alliances, or else must succumb to more powerful state organisms and is eventually swallowed up by them.

In this respect, as in others, we have a perfect analogy with the existence of any entity in the plant or animal kingdoms, which lives only while it is self-supporting, i.e., is able to satisfy its needs and withstand rivalry in the fight for survival. When this self-sufficiency is terminated for whatever reason, the individual expires. Similarly, an inorganic individual, in the form of a chemical bond, begins to deteriorate following loss of self-sufficiency. Radium is not a self-sufficient element under the present conditions of the earth's crust, and so constantly decays.

It is clear that the law of individuality, as was noted at the beginning of this chapter, is universal and can be applied not only to organic and inorganic phenomena, but also to the supraorganic or social world.

24

Conclusion

In the preceding pages, I have attempted to outline the main laws which the human society or collective, when studied objectively, is found to obey. I hasten to note that, quite possibly, some principles could have been united into still more general laws, while others, as yet unknown, will be discovered upon further investigation. Thus, for example, some prefer to analyze the law of evolution, the law of differentiation, and the law of selective integration as one body. As we know, the processes of differentiation and integration have until now been united in biology by authors such as Spencer under the law of evolution. I prefer, however, to examine each of these principles independently in order to arrive at a more detailed analysis of their phenomena. This in no way diminishes the interconnection of phenomena and dependences, for it cannot be doubted that when a law of the development of forms is established, further clarification of the progression of forms inevitably leads us to deduce that new forms are conceived through the differentiation of older ones, and through the unification of their parts into new combinations.

In this way, we seem to be dealing with a single process. But the fact of the matter is that, if we further clarify the manner in which the evolutionary process proceeds, we are forced to say that the law of continual motion and change is itself the origin or basis of the evolutionary process, since movement is connected to the variability of position—and variability of position lies at the foundation of the variation of forms, i.e., evolution. On the other hand, is not the law of historical succession also closely related to the law of evolution? The law of inertia, thanks to which forms temporarily

retain their original appearance, the law of reproduction, which lies at the basis of heredity, and the law of similarity, which controls the similar development of organs under identical conditions—are not these and other laws closely tied to the law of evolution? Yet no one would insist on their assimilation into a single law. This is why we deem it necessary to isolate the laws of differentiation and selective generalization as independent in their own right.

We consider it best to analyze laws or dependences of phenomena from their primary source. It is obvious, of course, that all laws are in a certain correlation with each other, since the world itself is a coherent and harmonious unit, in which certain phenomena are related to, or more precisely flow out of others. All that exists in the world is in a state of interaction and mobile equilibrium.¹

Thus, we do not attach especial significance to the question of whether or not certain laws should be merged into a more general law, since in the end we are dealing with a single cosmic process. It makes no difference whether we analyze the latter in greater or lesser detail, since difference of opinion could not undermine its basic principles; the social or supraorganic world should be studied just as objectively as the world of biological phenomena, without any excursions into the area of the elusive and intangible national or collective soul. When studied from a strictly objective viewpoint, in agreement with the methods of reflexology, the social world is seen as a continuation of the organic and inorganic ones, and so the energetic point of view that governs physical phenomena is also applicable to social ones. Just as the material world, be it inorganic or organic, comes down to various manifestations of energy, so should the supraorganic or social world be analyzed as a manifestation of energy in masses of people, united into a collective personality. Similarly, a separate individual's activity is simply a manifestation of energy received from without.

When a strictly objective method is applied to the study of the social or supraorganic worlds, we find that the resultant new scientific discipline—collective or social reflexology—closely corresponds to my theory of individual reflexology. In both the latter and the former, human activity is analyzed from the viewpoint of upbringing and of the development and coordination of various associative reflexes, which arise due to stimulation of the sensory organs, i.e., transformers of external energy. The fact that social reflexology resembles individual is completely natural. If a collective that

unites a group of people through common mood or interest is a collective personality, then obviously everything that relates to the separate personality should be applicable to it, with the exception of mutual induction, imitation, suggestion, and other forms of influence peculiar to the collective personality.

Naturally, the development of collective reflexes and manifestations of the collective personality follow the same path as the manifestations of separate personalities.

The strictly objective study of the personality leads us to conclude that the individual is the product of the influence of external energies and that he is himself an amalgam of energies, the primary accumulators of which are cellular elements and, partially, the nervous system and its central organs. Thus, all actions of the individual personality are externally determined, which I discuss in detail in my book *General Principles of Human Reflexology* (p. 81 and ff.), and consequently are subject to the action of a number of external causes. The dependence of the number of crimes, suicides, births, etc., on the price of grain is indisputable. According to Prussian statistics, fluctuation in the number of thefts, for example, corresponds amazingly closely to fluctuation in grain prices. Even the frequency of seemingly insignificant actions, such as errors in the writing of addresses on letters, is directly dependent on socioeconomic conditions.

My specific investigations in this area leave no doubt as to the truth of the above statements. "In the end," I write in a work dedicated to this subject,² "crime is the fatal consequence of a number of both generally predisposing and directly responsible or immediate factors." The latter include not only the character and temperament of the personality itself, which reflect its past life and the conditions of its development in terms of heredity, conception, and reproduction, but also all those conditions that act upon the given personality, such as upbringing and environment (comradeship, loss of class, economic situation), and finally one direct cause or another. These are the factors that determine why, under given socioeconomic conditions, one personality rather than another commits a crime.

But if the regularity of phenomena is established in relation to the separate personality, then it must also exist in relation to the actions of the collective, which is an aggregate of personalities as accumulators of energy.

We already know that geographic, climatic, politico-economic and other conditions determine the character and morals of social institutions.

Historical events likewise inevitably follow certain antecedent conditions; a good leader should foresee the course of social life and its outcome. "Gouverner c'est prévoir" goes the French saying, and it is a practical deduction made from observation of national movement, which, in the end, is subject to laws as immutable as those that govern the movement of celestial bodies.

It is also natural that, if the collective is an aggregate personality made up of individual accumulators of energy, the regularity of its phenomena must be manifested in the same forms as that of the individual personality. This is why analysis of external manifestations of individual and collective personalities shows a more or less complete correspondence between their respective dependences (laws); these dependences are, in turn, a reproduction and extension of their counterparts in the organic and inorganic worlds. There is no other way that matters could stand; and if this has not been determined until now, then it is solely due to the fact that the external phenomena affecting human and social activity were analyzed from a subjective viewpoint, which served as an obstacle to the merging of humanitarian knowledge with the physical and biological sciences. The cosmic process is nonetheless a manifestation of a single cosmic energy, and no matter where or in what forms the latter is found, it is demonstrated in the same interactions and is subject to the same dependences or laws.

Notes

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- 1 For greater detail on this subject, see my work: "On the basic laws of the world, in relation to the analysis of social life from the point of view of reflexology" (in Russian), in *Issues in the Study and Education of the Personality*, issues 2,3 and 4, 1921.
 - 2 V. Bekhterev, "The application of an objective psychological method to the study of crime" (in Russian), *Sbomik, posviashonniy pamiati D. A. Drila* (Collection dedicated to the memory of D. A. Dril), and sep. pub.

Index

acquisition, passion for, [190](#)
action coordination, [62](#)
activity, theory of, [6-8](#)
adaptation, [462](#)
law of, [445](#), [451-55](#)
Adler, Alfred, [279](#)
agitators, [178-79](#)
Allport, Gordon, [283](#)
altruism, [464](#). *see also* [self-sacrificing crowds](#)
Andizhan garrison, [243-44](#)
animals, [141](#), [154-56](#), [183](#), [411](#)
 experiments with, [156-57](#)
 imitation among, [141](#), [154](#), [345-46](#)
 suggestion among, [348](#)
animism, [407](#), [487-88](#)
animosity, crowd, [210-11](#)
antagonism. *see also* [hostile groups](#)
 ideological, [134](#)
antagonistic association, interaction, and collective unity, [134-35](#)
anthropomorphism, [20](#)
Apraksin Market in Petrograd, riot at, [208-9](#)
arbitration, international, [303-5](#)
archeology, [56-57](#)
Aristotle, [495](#), [514](#)

armies, 101, 105-6, 183, 226. *see also* war(s); *specific topics*
as based on coordination of movement, 267

disobeying leader, 396

art, 150, 193, 255, 299, 324, 426, 434. *see also* creativity
cyclicity in, 362
revolutions and, 359-60
science and, 408, 442

Asroy, 213

assemblies, 118, 127-28, 136, 137

associative memory, 67

associative reflex formation, 40

associative reflex theory, 24-25, 132

associative reflex(es), 6, 23, 39, 42, 181-82. *see also* *specific topics*
development of collective movements as, 181-96
direct influence via, 350
individual and collective, 181
law of, 177, 178, 194, 458

asthenic reactions, 200, 218

attack(s)
collective, 459
collective reflex of, 178

attention and concentration, 454
collective, 237-40, 334-35, 395, 421
prolonged, 153

audiences, theatre, 185

Augereau, 98

Austria, 496-97

automatism, 156

autosuggestion, 155

Avenarius, 445, 446

Batkin, 219-20

Bekhterev, Vladimir M. *see also* *specific topics*
formative years of his ideas, 3-4
methodological imperatives, 10-11

writings

On the Basic Laws of the World, 309

General Principles of Human Reflexology, 276, 291, 519

Suggestion and its Role in Social Life, 30, 142, 166, 175-76, 244, 276, 283, 349

Bentham, Jeremy, 251

Bernhardi, 33-34

biased judgments and decisions of crowds, 246, 252, 468-72

biological factors in formation of collectives, 160

biosocial hypothesis, 23-24

biosocial phenomenon, personality as, 88-89, 95

biosocial principle, 54

biosocial theory, 89

biosociology, 20, 68, 282

Bittner, V., 461-62

Blanc, Louis, 438

body movements. *see* [gestures](#); [signaling](#)

Bolsheviks, 127, 180, 228, 359, 443. *see also* [Russian Revolution\(s\)](#)

Bolshevism, 188-89, 227, 318, 358, 363, 503, 504

Bolshevist Revolution, 503

"Bonds of Freedom" campaign, 220

Boodin, 65

Bordier, 152

bourgeoisie, 300

Cabanis, 256, 257, 260, 263, 333, 488

capitalism, 300

capitalist countries, 204

Carlyle, 224

categories, 260-61

Catholicism, and French Revolution, 358-59, 365-66, 404

Cattelain, 332

causality, 257, 406, 491

change. *see* [adaptation](#); [continual motion and change](#); [economy](#); [relativity](#)

charisma, 98-99, 152

charity, [192-93](#), [207](#)
Charles XII, [100](#)
children. *see also* [parenting](#)
 in a collective, [90](#)
 interdependence with family members, [160](#)
Christianity, [288](#), [301](#), [363](#), [367](#), [404](#), [448](#), [476](#). *see also* [Catholicism](#)
Chuprov, A.A., [447](#)
Cicero, [146-47](#)
 civilizations
 historical development of, [342](#)
 life cycles of, [356-57](#)
clans, [463](#)
class struggle/conflict, [49](#), [460-61](#), [473](#), [499](#)
coalitions, [319](#)
"collective characterology," [52](#)
collective experience, interpretation of, [19](#)
collective reflexes, [175-80](#). *see also* [reflex\(es\)](#)
 evolution of, [57](#), [194](#)
collective reflexology, [25](#), [33](#), [39](#), [40](#). *see also specific topics*
 aims/tasks, [20](#), [45-46](#), [50](#)
 definition, [20](#), [47-48](#), [50](#), [51](#)
 experimentation in, [79-85](#)
 from a historical perspective, [56](#)
 individual reflexology and, [28](#), [46](#), [74](#)
collective statements, verbal, [248](#)
collective(s), [61-63](#), [143](#)
 classification, [113-14](#)
 as composite organism, [116](#)
 development and formation of, [115](#)
 factors in, [121-22](#), [160-61](#)
 during prehistoric stage of human existence, [40-41](#)
 intention and initiative in, [102](#)
 organization
 cohesiveness and, [50](#), [105-6](#), [127](#)

- constituent units/elements, 105, 186
- internal, 62-63, 116, 186
- lack of appropriate, 268
- organized vs. unorganized, 114-15
- of people related by blood, 115
- size, 63-64, 186, 223
- stupidity/lack of intelligence, 251
- training people to act as, 115
- types, 115
- unifying principle, 106
- collectivism, 464-65
 - law of, 464
- Collet, P., 255
- "collision," 140
- colors, 348-49
- commands, 183
- "common mood in a country," 223. *see also* mood
- communal ceremonies and rites, 485
- communal living and experience, 131, 142, 458. *see also* collective experience
- communal principle, 203
- communal structure, 403
- communal worship, 484
- communication
 - need for, 193
 - promoting unity, 136-39
- communist parties, 127. *see also* Bolshevism
- community, 110
- community membership, 200-201
- compensation/substitution, 475-76, 489. *see also* signaling; symbolism
 - law of, 475-77
- competition and rivalry, 313-16, 336
 - cooperation and, 6, 408-9
 - imitation and, 147, 325

between social movements, 302
Compte, A., 109, 406
Comte, Auguste, 54
concepts, 256, 259-60. *see also* ideas
"conductors," 138, 140
confidence, 34
Congress of the Party for National Freedom, 220
consciousness, 66, 67. *see also* soul/consciousness
defined, 66
society and the evolution of, 87-88
unity of, 31, 32
conservatism of crowds, 365
conspiracies and conspirators, 332
Constitutional Democratic Party, 393
contagion, 9, 141, 142, 152, 154. *see also* imitation
"mental," 146
mutual, 145
continual motion and change, 399
law of, 383-88, 399, 517
cooperation, 49, 54, 81, 203, 313, 408, 463
competition and, 6, 408-9
coordinated collective actions, 265-70
cosmopolitanism, 499
counter-imitation, 26, 374
counterrevolution(s), 343
countersuggestion, 352
court system, 169
creative activity, 30, 45, 92, 358, 426
creativity, 138, 191, 433. *see also* invention(s) and discovery(ies)
collective, 30, 245, 246, 248, 253-55, 263
imitation and, 357-58
during sleep, 276
symbolism in national, 489
criminal crowds, 125

criticism, mutual, 246, 247
cross-criticism, 246, 247
crowd psychology, 30
crowd(s), 47. *see also specific topics*
 aspects of, 124
 as bringing out the best and worst in people, 249-50
 concept and definition of, 113, 114, 117
 development, 111
 excitability, 124
 guiding and controlling, 100
 gullibility and exaggeration, 102, 123
 lack of capacity for reason and restraint, 100-101, 469, 470
 motivators, 125
 organization, 101, 114
 power, 100
 "self," 112
 variation among, 125
cruelty, 124, 214
curiosity, collective, 191-92
customs. *see traditions*
cycles
 in economy, 363
 life, of civilizations, 356-57
cyclicality, 355, 356, 362, 387. *see also rhythm; sunspots*
czar. *see tsar*
dancing, 183-84
Darwin, Charles, 6, 464
Darwinism, 457, 461-62
De la Grasserie, 47, 52, 66-67
de Roberti, Evgenii V., 4, 24, 54, 89, 412
 interpretation of collective experience, 19
deaf-mute persons, 262, 479
death sentence, 396-97. *see also executions*
debates, 247

decisions of collectives, 246, 249, 251. *see also* [biased judgments and decisions of crowds](#)

defense and attack, collective, 459
 need for, 41

defensive reflex, 200

deification, of people, 97-98, 101-2, 485-86, 488-90

democracy, 104-5. *see also* [elections](#)

dependence(s), 41-42, 287-89, 518

dependent relations, law of, 491

depersonalization, 92

developmental methodology, 10-11

differentiation, 417, 419, 424
 law of, 415-16, 418, 420, 423

discourse, 136-37

discoveries. *see* [invention\(s\) and discovery\(ies\)](#)

discussion, collective, 248, 254

dissociation, 279, 313, 314

donations, charitable, 192-93

Dragichesko, 87-88

Dragomarov, 99

dreams, and the reflexological approach, 280-81

drugs, 423-24

Drummond, H., 170, 237

Duma, 229-30, 299, 301, 338, 443

Durkheim, Emile, 65, 87, 111, 245, 256, 257, 260, 263

Durov, 156-59

"dynamic-material conductors," 138

dynamometers, 81, 82

ecology, 467

economic dependence, 499

economics, 34-38, 54, 55, 187, 299. *see also* [specific topics](#)
 interaction of collective groups in, 473

economy
 cycles in, 363

law of, [445-50](#), [454](#) "education," socially mediated, [7](#)
egoism, [169](#), [460](#)
egotism, [464-65](#)
 crowds and, [125-26](#)
elections, [203](#), [249](#), [470](#)
electoral crowds, [470](#)
Ellwood, [68](#)
empathy. *see* [understanding others](#)
energie science, [289](#), [295](#)
energy, [218](#), [275](#), [279](#), [475](#)
 among nations, [376](#)
 circulation of, [386](#)
 coordination and, [267](#), [268](#)
 dissipation of. *see* [entropy](#)
 "hidden," [5](#)
 accumulation and constant transformation of, [5-7](#)
 idea/notion of, [5](#)
 evolution and, [7](#)
 as unifying concept of all biological phenomena, [6](#)
 law of conservation of, [291-95](#)
 "nervous," [5](#)
principle of the unity of, [475](#)
energy level, stability of, [267](#)
"energy reserve," [5-6](#)
energy theory, [64](#)
England, crime in, [78](#)
enthusiasm, [232-33](#)
entropy, law of, [389-91](#)
equality
 national battles for, [302](#)
 social, [409](#)
equality and fraternity, doctrine of, [461](#)
Espinass, [154](#)
espionage, [241-43](#)

evolution, [6-7](#), [517](#). *see also* [natural selection](#)
doctrine/theory of, [460](#), [461](#)
law of, [399-401](#), [517-18](#)
exchange, processes of, [178](#)
executions, [394](#), [477](#). *see also* [death sentence](#)
reactions to, [331-33](#)
eyewitness testimony, [123](#)
"familial-cooperative groups," [462](#)
family, [205-6](#), [462-63](#), [512](#)
family relationships, [160](#), [388](#). *see also* [marriage](#)
famine/food shortages, [200-204](#), [208-11](#), [492-93](#)
fashion, [151](#)
February Revolution, [188](#), [239](#), [443](#), [502](#)
feelings, [278](#)
Ferri, E., [249](#)
Ferri, N., [460](#)
fighting, [134](#)
folie à deux, [9](#)
folk psychology, [29](#), [30](#), [45](#), [47](#)
vs. collective reflexology, [45](#)
folk reflexology, [126](#)
folk wisdom, [388](#)
Fourcroix, [365-66](#)
France, [234-35](#)
Franco-German war, [497](#), [498](#)
freedom, [95](#), [96](#), [104-5](#), [251-52](#), [294](#), [302](#)
absolute, [321-22](#)
attraction to, [212](#)
defense of, [303](#)
French Revolution, Great, [124](#), [128](#), [188](#), [213](#), [230](#), [241](#)
art and, [360](#)
Catholicism and, [358-59](#), [365-66](#)

- mental preconditions to, 505-6
- mysticism and, 269
- violence and, 397

Freud, Sigmund, 278-81

friendly vs. unfriendly moods, 217-18

Gabelli, 249

Gapon, Georgiy, 337

Garibaldi, 100

gatherings, 106, 120. *see also specific topics*

types of, 114

Gefding, 66

Geiger, L., 422

General Principles of Human Reflexology (Bekhterev), 276, 291, 519

German Revolution, 342-43

Germany, 210, 231

- and World War 1, 106, 231, 233, 304-8, 497-98

germination of seeds, 309-10

Gershel'man, S., 33

Gerver, V., 273-74, 278

gestures, 137, 141, 164-65, 232, 262, 479-81. *see also signaling*

Giddings, 100, 112

gift-giving, 192-93

Glennie, Stuart, 406

"goal," 73

goal setting, 70

god(s), 476

Grant-Allen, 460

gravitational attraction, law of, 309-12, 447

Great World War, 106, 163-64, 228, 331, 500-501

- Austria and, 496-97
- causes, 496-500
- Germany and, 106, 231, 233, 30-48, 497-98
- Laufenberg on, 499

nationalism vs. internationalism and, 499-500
Russia and, 330, 497, 498
United States and, 303-8, 497
Greeks, 94, 150
Guyau, 140
gymnastic exercises, 183
habit, 192, 372. *see also* inertia
Haekkel, 66-67, 415
hallucinations, collective, 166, 240, 243-44
hearing, 141-42
Hénault, 224
herd instinct, 132-33
herd-like existence, 121
herds, animal, 141, 175
hereditary-organic reflexes, 175, 186, 199-201, 204-5, 208. *see also*
collective reflexes; specific reflexes
heredity, 23, 88, 465. *see also* reproduction
social, 122, 426, 427
"spiritual," 425-26
hierarchy, 315
historical illusionism, law of, 288
historical succession, law of, 441-44
history, 55-56
great personalities in, 96-97
homogeneity vs. heterogeneity
of collectives, 105, 127, 128, 163, 186, 373
of crowds, 100, 101, 222, 325
hostile groups, 135. *see also* antagonism
hunger, 328. *see also* famine
hypnoid states, 155
hypnosis, 442
Bekhterev's interest in, and foundations of social life, 8-10
hypnotism, imitation and, 152-54
iconoclasy, 359

ideas, 256, 259-60

prejudice against new, 377-78

identity, individual. *see also* individuality

loss of, 92

idols

deification of popular, 97-98

former, 361-62

illusions, collective, 166, 240

imitation, 21-22, 74, 110, 141-55. *see also* similarity, law of; Tarde

among animals, 141, 154, 345-46

as basis of community, 353

in children, 146, 149

Cicero on, 146-47

as collective associative reflex, 195

competition, rivalry, and, 147, 325

creativity and, 357-58

in crowds, 84, 137

and homogeneity of environment, 325

hypnotism and, 152-54

impact on social movements, 325

in language, 436

laws of, 26, 36, 41, 149, 153, 154

of leaders and persons in positions of power, 141, 147

literature on, 145

and loss of novelty, 193

predisposition to, 152

repeated, 196

reproduction and, 344-45

significance for composite personality, 197

social stagnation based on, 192

stimulated by voice, 142

suggestion and, 348, 349

imitative reflexes, 154

immigration, 299

- imperialism, 302
- impulsive behavior, 250-51
- independence of crowds, 252
- individual aspirations, inhibition by collective, 103
- individual behavior
 - as automatic/predetermined, 93, 95-96
 - in isolation vs. in a crowd, 92
- individual differences, 133
 - innate, 163
 - "opposites attract," 133
- individual insufficiency, 133
- individualism, 95
- individuality, 90-91, 112
 - law of, 511-15
- induction, 83, 344-47. *see also* thought transmission
- industrial progress
 - imitation, leveling, and, 150
- inertia, law of, 36, 327-28, 357, 365-74, 379-81
- infanticide, 205-6, 213
- influence, 351-53
 - via associative reflexes, 350
- inheritance, 465
 - hypothesis of physiological, 465
 - social, 378, 379
- inhibited reflexes, and the experience of mind, 275-78
- inhibition
 - collective, 321
- and progress vs. regress, 357
- inhibitors, social, 394
- innovation(s), 315. *see also* invention(s)
 - "collision" interference of, 324
 - opposition to, 333-34
 - thirst for, 192
- inquisitiveness, 191

instincts, [26](#), [70](#), [132](#), [215n](#)
integration. *see* [selective integration](#)
intellectual development, [406](#). *see also* [science](#)
interaction, [110](#), [111](#), [160](#)
 direct vs. indirect, [160](#)
 law of, [467-74](#)
 psychological, [110](#)
internationalism, [499](#)
interpersonal relations, [40](#)
intuition, [71](#), [276](#)
invention(s) and discovery(ies), [74](#), [88](#), [150-51](#), [314](#), [390](#), [433-34](#), [442-43](#).
 see also [innovation\(s\)](#)
Jesus Christ, [97-98](#)
Jews, [328-29](#)
judgment, [259](#)
 collective, [246](#), [248](#)
 social, [89-90](#)
Jung, Carl Gustav, [279](#)
Juquelier, [145](#)
jurors, [469-70](#)
Kareev, N., [53](#), [485](#)
Kerensky, A.F., [350](#), [396](#), [486](#), [505](#)
 attacks on, [220](#)
 downfall, [227-28](#)
 persuasion, [165](#)
 speeches, [224](#), [227](#), [505](#)
 success and popularity, [225-27](#), [350](#), [486](#)
knowledge, [258-59](#)
 attraction to and acquisition of, [191](#)
 language and the evolution of, [168](#)
Kopel'man, A., [31-32](#)
Kornilov, General, [188](#), [189](#), [370-71](#)
Kornilov affair, [188-89](#), [227-29](#)
Koshevskii, [221](#)

Kotik, 156

Kovalevsky, M.M., 4, 104

Kronshtadt uprising, 178-79

Kropotkin, P., 133, 310, 458

labor

- combined/collective, 81-83

- division of, 54-55, 88, 133

- "epidemic" of, 320-21

Lamarck, J.B., 6, 399

land. *see* property

language(s), 255, 259. *see also* speech

- as collective creation, 261-62

- foreign, 192

- functions, 167-68

- as unifying factor, 164-69

- imitation and, 148

- origins, development, and evolution of, 22-23, 57, 170-73, 385, 388, 400, 406, 421-22, 435, 478-81

- rivalry between different linguistic forms and, 326

- signaling, 478, 481-84

Lasalle, 468

Laufenberg, 499

law(s), 89

Lazarus, 23, 29

Lazurskii, Alexander F., 4

Le Bon, Gustave, 250-53, 402, 460, 468, 470

- on French Revolution, 505, 506

- on great people in history, 96, 97

- on the Greeks, 94

- on jurors, 469-70

- on language, 169

- on marriage, 373

- on national unity, 435

- on psychology of crowds, 92, 122-25

- on race and the sexes, [418](#), [513](#)
- on religion, [380](#), [402](#)
- on social psychology, [52](#)
- on social spheres, [506](#)
- on socialism, [397](#), [444](#)
- on suggestion, [166](#)
- on uprising of crowds, [363](#)
- writings
 - The Evolution of Matter*, [386](#)
 - The Psychology of Nations and Masses*, [124](#), [356](#), [365](#), [377-78](#), [380](#), [402](#), [418](#), [435](#), [469-70](#), [512-14](#)
 - The Psychology of Socialism*, [92](#), [122](#), [363-64](#), [373](#), [397](#), [444](#), [460](#), [461](#)
- leaders, [114](#), [252](#)
 - deification and faith in, [97-98](#), [10-12](#), [485-86](#)
 - demagogical methods, [153](#)
 - interaction established by, [160](#)
- Letourneau, Ch.T., [213](#), [315](#), [342](#), [383](#), [397](#), [404-5](#)
- "leveling" effect, [146](#)
- Levins, R., [284](#)
- Lewis, [23](#)
- linguistics, [255](#). *see also* [language\(s\)](#)
- Lissagaray, [438](#)
- literature, [268](#), [324](#), [362](#), [388](#), [426](#)
- living organism. *see* [organism\(s\)](#)
- Loeb, [67](#)
- logic, [258](#), [259](#)
- logical judgment, [256](#), [259](#)
- lottery bonds, [37-38](#)
- Louis XV, [224-25](#)
- Maillard tribunal, [230](#)
- Mallery, [479](#)
- manufactured commodities, [187](#)
- manufacturing, [299](#)
- Marat, [97-98](#)

marriage, [160](#), [206](#), [373](#), [387-88](#)
Martynov, G., [102](#)
martyrdom, [213](#)
Marx, Karl, [81](#), [438](#), [460](#)
massacres, [230](#), [329](#). *see also* [pogroms](#)
Matter, Principle of Conservation of, [291](#)
McDougall, W., [26](#), [27](#), [132](#)
mechanization, [192](#)
medicine man, [168](#)
Meillard, [33](#)
Mendeleev, [287](#), [442](#)
mental phenomena, [70](#)
meteorites, [441](#)
military, [298](#), [482](#). *see also* [armies](#); [war\(s\)](#)
military operations, [420](#)
Miliukov, P.N., [220](#)
Mill, J.S., [53](#)
mime, [480](#)
mimico-somatic drive, [278](#)
mimico-somatic reflexes/reaction, [132](#), [133](#), [176-77](#), [240](#), [251](#), [266](#), [331](#), [346](#)
 feelings as, [278](#), [280](#)
 mood and, [217-32](#)
 suppression of, [332](#)
 violence and, [331-32](#)
mind-body dualism, [7-8](#)
mind-body unity, [276](#)
minorities and majorities, [103](#)
mob law, [211](#)
money, [138](#), [178](#)
 supply and circulation, [34-35](#)
monism, [66](#)
monoideism, [122](#), [153](#)
Monroe doctrine, [303](#)
mood (collective), [217-35](#), [251](#), [268](#)

- changes in, [229-31](#), [234-35](#), [361-62](#)
- crowd attitude as determined by, [124](#)
- inhibition of, [331](#)
- meaning, [235n](#)
- measurement/assessment of, [234](#), [268](#)
- revolutions and, [358](#)
- moral beliefs, reflected in art, [360](#)
- moral force/element, [33](#), [38](#)
- moral nature, movements of a, [298](#)
- moral principles, [464](#)
- moral vs. immoral behavior, [250](#)
 - religion and, [394-95](#)
- morality, social, [435](#)
- motion, [475](#). *see also* [continual motion and change](#)
- movement(s)
 - expressive, [137](#), [141](#)
 - restricted, [153](#)
- Münsterberg, [36](#), [37](#)
- mushrooms, [309-10](#)
- music, [141-42](#), [183-84](#), [232](#), [266](#), [439](#), [482](#)
- mutual aid, [310](#)
- mutual assistance, [133-35](#), [459](#), [461](#)
- mutual attraction amongst individuals, [132-33](#)
- mutual contact, [140](#)
- mysticism, [269](#), [406-7](#)
- myths, [46](#), [150](#), [400](#). *see also* [folk psychology](#)
- Napoleon Bonaparte, [33](#)
 - charisma, [98-100](#)
- Nass, L., [269](#), [333](#), [488](#)
- national characterology, [45](#), [92-93](#), [151-52](#), [378](#)
- national self-determination, [104-5](#)
- nationalism, [106-7](#), [151-52](#), [499](#)
- nations, [118](#), [121](#). *see also* [state](#)
 - battle and conflict between, [302](#), [376](#), [390](#)

- history of, 56
- imitation of, 149-50
- threatened, 319
- natural selection, 122, 417, 460. *see also* social selection
 - agentive role of, 7
 - law/theory of, 131, 399, 458
- need satisfaction, 185-87, 190, 199, 200, 266, 429-30, 448, 457-58, 462
- need(s)
 - existential, 187, 194, 195
 - mutual, 133, 186-87
- negotiation. *see* arbitration
- neopositivism, 19-21, 49, 256-57
- nervous system, 6, 452
- neural networks. *see* associative reflex
- neutrality (World War I), 303-6
- Newton, Isaac, 447
- Nicholas II, 373. *see also* tsar
- Nikonians, 134
- Nordau, Max, 249
- novelty, 193, 194
- objective study of developing phenomena, 10-11
- objective study of social phenomena, 27, 53, 518-19
- objective vs. subjective terminology, 60
- objectivism vs. subjectivism, 19-20, 66-69, 72-74, 277
- observation, collective, 237-39
- On the Basic Laws of the World* (Bekhterev), 309
- one regularity/one dependence, 41
- opposites, attraction of, 311
- organism-environment interaction, 6, 89
- organism(s)
 - as active agent in organism-environment adaptation, 6
 - vs. physical objects, 5
- pacification, crowd, 181, 182
- Paient, G., 88

paleontology, [57](#)
pan-psychism, [67](#)
panic, [175-76](#), [184](#), [185](#)
parasites and parasitical plants, [309-10](#)
parasitism, [463](#)
parental "instinct(s)", [206](#)
parenting, [207](#), [429-30](#)
 cooperative, [206](#)
 joint, [206](#)
Pareto, [489](#)
Paris Academy of Sciences, [441](#), [442](#)
Paris Commune, [438](#)
parliamentary systems and parliamentarism, [151](#), [190](#), [251-52](#), [473](#)
Parshin, A., [461-62](#)
parties. *see also specific topics*
 intense conflicts between, [299-300](#)
Party for National Freedom, [222](#)
Party for National Liberty, [393](#)
patriotic mood, [218](#), [228](#), [230-31](#), [233](#), [331](#), [394](#)
patriotism, [191](#), [233](#), [318](#), [499](#)
Pavlov, I., [68-69](#)
peace, world, [302](#)
perception(s), [256](#)
Perelman, Ya., [508](#)
perfection, [410-11](#)
periodicity, [355](#), [356](#), [358](#), [507-8](#). *see also rhythm*
Perovskaia, [323](#)
personality
 as biosocial phenomenon, [88-89](#), [95](#)
 composite, [25](#), [28](#), [39](#), [42](#), [109-11](#), [117-18](#), [120](#). *see also specific topics*
 basic laws of correlative activity of, [41-42](#)
 three forms of, [115](#)
 as highest ethical value, [112](#)
 liberation of, [103](#)

as self-expression vs. representing society, 92
social manifestations of intergenerational transmission of the, 94
suppression of individual, 95, 103, 126
personality development, 20-23, 57
 and progress, 103
 as social formation, 87, 92-95
persuasion, 165, 227, 350, 353. *see also* suggestion
 by deduction, 351
 suggestion vs., 352-53
Petrazhitskii, 71
philosophy and philosophical systems, 324-25, 362
phonetic writing, 22
physiology, social phenomenon and, 68-69
planned nature of collective work, 267
pogroms, 201, 202, 208
police violence, 210
political meetings and associations, 117-18, 122
political organization, is always accompanied by integration, 431-32
politics, great people in, 97
Polovtsev, General, 229
power, 326-27
 evolution of, 403
preconceived notions, 26
prisoners, liberating, 212
prisoners of war (POWs), 242-43
production processes, 266. *see also* manufacturing
 professional groups and assemblies, 118, 127-28, 136, 137
professions, trends toward particular, 298-99
progress, 411-12. *see also* evolution; innovation(s); invention(s)
 concept and definition of, 409, 411
propaganda, 167, 178-80, 314, 337, 369, 504, 505
property, protecting, 211
property inheritance, 204
property ownership, 403-4

protecting members of group/community, 462
protesting unity, 177-78
protests, 190
proximity, physical, 132, 140
Psycho-Neurological Institute, 3-4, 446
psychoanalytic movement, Bekhterev and, 278-81
psychological infection, 8-10
psychology, 24
 comparative, 69
 defined, 66
 "intermental," 51
public consciousness. *see* [soul/consciousness](#)
public control, importance of, 239
public opinion, 169
 language and the evolution of, 169
questionnaires, 79
race, 513
racial psychology, 52
radical parties vj. moderate trends, 370
rapaciousness, 463
rape, 213
Ravenarius, 421-22
Razvedchik, 243
reaction, law of equal and opposite, 317-19, 321, 339
 inhibition as manifestation of, 322-39
reaction-formation, 188-89
realism, 112
reason, 70, 87, 469, 488. *see also* [logical judgment](#)
 crowds' lack of capacity for, 100-101, 469, 470
reasoning
 collective, 245
 influence/persuasion through, 353
reciprocal psychology, 51
reflex(es)

- concept of, [273-75](#)
- generality of, [275](#)
- higher order. *see* [associative reflex\(es\)](#)
- offensive and defensive, [187](#)
- replaced by their symbols, [181-82](#)
- reflexological processes, from individual to group, [282-83](#)
- reflexology. *see also* [collective reflexology](#)
 - general/grand theory of, [281](#), [283](#)
 - relationship with biology and sociology, [20](#)
- regress, social, [413](#)
- regression, [280-81](#)
- regularity, [41](#)
- relativity, law of, [393-98](#)
- religion, [257](#), [288](#), [463](#), [476](#), [484](#). *see also* [Christianity](#); [deification](#)
 - revolutions and, [358-59](#), [365-66](#)
 - science vs., [323](#)
 - separation of church and state, [358](#), [364](#), [373](#)
 - symbolism in, [487-89](#)
- religious beliefs, [257](#)
 - protecting, [211](#)
 - reflected in art, [360](#)
- religious doctrines, [203](#), [377](#)
 - evolution, [400-402](#), [463](#)
- religious mood, [218](#)
- religious movements, [269](#), [298](#)
- religious persecution, [301](#), [328-29](#)
- religious societies and cults, [213](#), [269](#)
- reproduction, [205](#), [207](#). *see also* [imitation](#)
 - law of, [425-28](#)
- reproductive reflexes, [202](#), [366](#)
- repulsion, law of, [313-16](#)
- revenge, thirst for, [124](#)
- revolutionaries, repression of former oppressor by, [397](#)
- revolutionary movements, [376-77](#)

under monarchies, [323](#)
revolution(s). *see also specific revolutions*
 inevitability, [381](#)
 mysticism and, [269](#)
Rezanov, A.S., [99](#), [101](#)
rhythm, [183-84](#). *see also cyclicity; music*
 law of, [355-57](#), [360](#), [363](#), [425](#)
Ribot, [416](#), [479](#)
Richet, [156](#)
riots, [208-10](#), [212](#), [214](#)
rivalry. *see competition*
robbery, [202](#)
Rodzianko, Mr., [229-30](#)
Roman Empire, [476](#)
Roman law, [334](#)
Romans, [150](#)
Ross, [163](#)
Rossi, P., [47](#), [52](#)
Rousseau, J., [485](#)
rulers. *see leaders*
Russia. *see also specific topics*
 education funding, [78](#)
 political strike of 1905, [336-38](#)
 revolutionary process and economics in, [38](#)
 Russian Civil War, [330](#)
 Russian psychology, [10-11](#)
 Russian revolutionary movement, [300](#)
 Russian Revolution(s), [41](#), [107](#), [179-80](#), [188-90](#), [369](#), [372](#), [398](#), [436](#). *see also Kerensky; Kornilov;*
Russia, political strike of 1905
 Civil War, [231](#)
 counterrevolution during, [241](#)
 death sentence during, [396](#)
 Democratic Assembly, [189](#)

February Revolution, [188](#), [239](#), [443](#), [502](#)
French Revolution compared with, [128](#), [347](#), [401](#), [503-6](#)
law of inertia and, [369-72](#), [374-75](#)
liberation of prisoners, [212](#)
"Loan of Freedom" concert meeting, [218-21](#)
new national leadership and Russian people during, [485](#)
October coup/October Revolution, [227](#), [231](#), [443](#)
officers' council meeting, [222](#)
precipitants and development of, [182](#), [300](#), [301](#), [501-3](#)
Provisionary Government, [127](#), [502](#)
"revolt" against, [229](#)
public attitudes and mood during, [331](#)
religion during, [270](#)
riots, [208-9](#)
satisfaction of demands. *see* [need satisfaction](#)
scapegoating, [328-29](#)
Schefflin, Mr., [231](#)
Schismatics, [134](#)
Schleicher, A., [421-22](#)
science, [287-88](#), [324](#), [377](#), [426](#). *see also specific topics*
change and the evolution of, [299](#), [385](#), [387](#), [406-8](#), [412](#). *see also* [historical succession](#)
scientists, [168](#)
seeds, germination of, [309-10](#)
selection, law of, [457](#)
selective integration, [430-34](#), [436-39](#)
law of, [429](#), [431](#), [435](#), [437](#), [439](#) "self-consciousness" of society, [112](#)
self-criticism, crowds' incapacity for, [250-51](#)
self-determination, [104-5](#), [514](#)
self-expression, individual, [92-94](#), [126](#), [253](#). *see also* [creativity](#)
self-observation, [39](#)
self-preservation instinct, [190](#), [202-4](#), [212](#), [455](#)
 altruistic and collective motives and, [464](#)
 and development of civilization, [310](#)

- religion and, [211](#), [463](#)
- struggle for existence and, [459](#)
- when facing life and death issues, [208](#)
- self-sacrificing crowds, [125](#). *see also* [altruism](#)
- self-sufficiency, [135](#), [514-15](#)
 - law of, [122](#)
- Serbia, [497](#)
- sexual attraction, [205-8](#)
- sexual desire, [211](#), [213](#), [478](#), [493](#)
- sexual deviance and acting out, [213](#), [214](#), [493-94](#)
- sexual energy, [279](#)
- Shakespeare, William, [178](#)
- Shavrov, N., [464](#), [465](#)
- Sighele, S., [47](#), [52](#), [105](#), [124](#), [152](#), [249](#), [253](#)
- signaling, [478](#), [481-84](#)
- similarity, law of, [341-54](#)
- singing, [184](#), [221](#), [482](#)
- slavery, [360](#)
- sleep, [281](#)
- Sluchevskii, [124](#)
- "social affinity," [310](#)
- social communion, political vs. religious
 - concept of, [109-10](#)
- social customs and standards, [96](#)
- social development, [20-22](#), [57](#)
- social differentiation, [88](#)
- social groups
 - defined, [110](#)
 - expansion of, [88](#)
- social instinct, [121](#), [207](#), [214-15](#)
 - theory of, [131-32](#)
- social leveling, law of, [150](#)
- social life, bases of, [54](#), [55](#)
- "social mechanics," [64-65](#)

social mind, [100](#)

social movements. *see also specific movements; specific topics*

- competition/rivalry between, [302](#)
- opposing each other. *see* [reaction](#), [law of equal and opposite](#)
- origin and development of, [298](#), [300-301](#), [420](#)
- temporary breaks in, [300](#)

social necessity, [95](#)

social organization, characteristics of good, [96](#), [97](#)

social personality(ies)

- unity, [250](#)
- well-developed, [90](#)

social phenomenon, study of, [64-68](#)

social pressure, [93](#)

social psychology. *see also specific topics*

- development of, [29](#)

social-reflexological theory, Bekhterev's fate of, [283-84](#)

social selection, [122](#), [316](#), [417](#), [459](#), [465](#)

- theory of, [131](#)

social stagnation, [192](#), [369](#)

social statutes, [93](#)

social subdivisions, [93](#)

social unanimity, achievement of, [329](#)

socialism, [95](#), [204](#), [397](#), [409](#), [460](#)

sociality, [19](#), [132](#), [196](#), [457](#), [460-63](#), [477-78](#)

society(ies)

- definition, [109](#)
- as dynamic equilibrium, [116](#)
- as "ever-existing personality," [109](#)
- evolution of, [94](#)
- nature of, [111](#)
- permanent v5. accidental/short-term, [251](#)
- survival, [122](#)
- viewed as organism, [53](#)

sociology. *see also specific topics*

- biological and psychological bases of, 20
- dependence in, 41-42, 288-89
- physiological, 68
- reflexology and, 47-56
- Sokolsky rallies, 183
- solidarity, 459, 461, 487
- somnambulant states, 152-53
- Sorokin, P., 112, 160, 164, 418, 509n
 - on antagonistic interaction, 134
 - on communication, 73
 - on family, church, and state, 138-39
 - on interactive relations, 138, 160, 467
 - on mutual assistance, 133-34
 - on objectivism and subjectivism, 66, 69-70
 - on subject matter of sociology, 52-53
 - on symbols, 487
 - The System of Sociology*, 52-53, 66, 112, 139-40, 431, 467, 487
 - Tarde and, 112
- soul/consciousness, 277. *see also* consciousness
 - collective, 19, 29-32, 47, 65, 112, 470-71
- sound, 141-42, 184. *see also* speech
- Soviet psychology. *see* Russian psychology
- space and spatial organization, 257-58
- "Spartacus" uprising in Berlin, 182-83
- specialization, 88
- speech, 23, 136, 164, 353, 478-79. *see also* language(s)
 - and persuasion, 165
- speeches, 224, 246-47
 - inflammatory, 142, 178, 218-20
- speed and motive power, law of proportional relationship between, 297, 299
- Spencer, Herbert, 6, 451, 495
- "spirit," 19
- "spiritual" death of a culture, 295
- "spiritual" heredity, 425-26

- "spiritual" relationships and interdependence, 160
- "spiritual" self, 293
- "spiritual" sphere, 445
- "spiritual" unity, 134, 161
- "spiritual" wealth, 426
- "spy mania," 241-43
- state, the. *see also* nations
 - authority and power, 104
 - nature of, 119
 - personality and, 104
- State Assembly in Moscow, 222
- state integration, 431
- statistics and statistical method, 75-79, 492
- Steinthal, 29
- sthenic reaction, 187, 191, 195, 196, 200, 201, 212, 218, 348
- stimuli and stimulation, external, 138, 177, 186, 297-98
 - power of, 297
- struggle for existence, law/principle of, 459-62, 464-65, 495
 - subbotnichestvo*, 319-21
- subjectivism, 19-20, 25
- sublimation, 475
- suggestibility of crowds, 123-24
 - objective conditions of, 153
- suggestion, 153-54, 348-49. *see also* thought transmission
 - among animals, 348
 - in crowds, 101
 - experiments on, 83
 - hypnotic, 152-53
 - imitation and, 348, 349
 - Le Bon on, 166
 - mutual, 145
 - vs. persuasion, 352-53
 - resistance to, 9-10
 - social, 9-10

susceptibility to, 145
Suggestion and its Role in Social Life (Bekhterev), 30, 142, 166, 175-76, 244, 276, 283, 349
sunspots, correlation of collective movements with, 507-9
superstitions, 380, 406, 407
supervision, collective/public, 239
supply and demand, 34-37, 187-88
survival. *see* self-preservation; struggle for existence
Suvorov, 99
Sviatsky, D.O., 508, 509
Swiss revolution, 301
symbiosis, 309, 310
symbolic collective reflexes, 176
symbolism, 232, 259-60, 262, 349-50, 447, 480, 481, 484-87. *see also* signaling
synergism of crowds, 249-50
Taine, H., 97, 124, 128
Takhtarev, K.M., 55, 403-4, 431-32, 495
Tarde, G., 26, 92, 140, 152, 153, 357, 365, 400, 406, 427, 442-43
 on beliefs, 322-23
 on conservatism and inertia, 365
 on counter-imitation, 374
 on crowds, 113, 124
 on desire, aspiration, and faith, 74
 on imitation and invention, 41, 78, 146-48, 193, 357
 on intermental psychology, 51
 on language and speech, 170-72
 The Laws of Imitation, 78, 171-72, 193, 254-55, 314, 322-26, 334, 365, 378, 380
 on leaders, 99
 on nations, 121
 on politeness, 193
 on process of discovery, 254-55
 on repetition, 334

on science, [433](#)
on social inheritance, [378](#)
Social Logic, [170](#), [329](#), [406](#)
on society and social communion, [109-10](#), [116-17](#)
Sorokin and, [112](#)
on traditions, [378](#), [380](#)
on violence, [124](#)
technology imitation, leveling, and, [150](#)
telegrams and communication, [71-72](#)
teleology, [412-13](#)
telepathy. *see* [thought transmission](#)
territories, [133](#)
theatre, [185](#), [193](#), [439](#)
Thorngate, Warren, [283-84](#)
thought, [277](#). *see also* [concepts](#); [reason](#)
thought transmission, over distance, [156-59](#)
time, [257](#)
Tocqueville, Clerel de, [143](#)
Tolstoy, Leo, [293-94](#), [301](#)
touch, physical, [140](#)
trade, [506-7](#)
traditions, [46](#), [378-80](#), [448](#). *see also* [folk psychology](#)
trauma, [278](#)
tribes, [463](#)
tsar, [336-38](#), [371](#), [373](#)
tsarism, [363](#)
tsarist regime, [360-61](#), [396](#), [504](#)
Tsushima incident, [374](#)
Tugan-Baranovskii, [34](#)
Uhlans, [99](#)
understanding others, [70-73](#)
unification of members of social group, [354](#)
Union of the Russian People, [437](#)
Union of Unions, [231](#)

unity, [64](#), [115-16](#), [134-36](#). *see also* "spiritual" unity
 concept of, [135](#)
 of crowds, [100](#), [101](#), [136](#), [250](#)
 internal vs. external, [135](#)
 movements facilitating, [140-41](#)
 protesting, [177-78](#)
 "spiritual," [134](#), [161](#)
 symbolism and, [486-87](#)
uprisings, [212](#), [363-64](#)
 development of, [177-79](#), [420](#)
utilitarianism, [109](#), [412](#), [449](#)
Valden, P.I., [442](#)
Vandamme, General, [98](#)
Vavulin, N., [242](#)
Vico, [386](#)
Vigouroux, [145](#)
VIKZhel, [128](#)
violence, [327-28](#), [477](#). *see also specific topics*
voice, [141](#), [142](#)
Wagner, W., [69](#), [70](#)
Waltman, [417](#)
war(s), [102](#), [106-7](#), [134](#), [166-67](#), [184](#), [303](#). *see also* [Great World War](#);
 [Napoleon](#)
 declarations of, [230](#)
 national unification during, [319](#)
 role of moral force in winning, [33](#)
weapons, real and symbolic, [181](#), [182](#)
Western countries, Russians imitating, [149](#)
will, [442](#)
 free, [322](#)
Wilson, Woodrow, [224](#), [306-7](#)
wizards, [168](#)
word association, [279](#)
work performance, [84](#)

World War I. *see* [Great World War](#)

writing, [255](#), [443](#)

Wundt, Wilhelm, [30](#), [45](#), [66](#)

yawning, [145](#)

Zelenyi, [68](#)

Zellner, [445](#)

Zheliabov, [323](#)

Zhuk, [156](#)

zoo-reflexology, [69](#)

zoopsychology, [68-69](#)